WRITTEN PROOFS OF LEAN-CHECKED RESULTS, STAGE 1: SYMMETRIC POINT SETS

Abstract. The Erdős–Szekeres project checks some of its results in Lean 4. This document is Stage 1 of a series of written proofs of these results. It treats the Lean files listed under "Scope of Stage 1" below, which concern mainly symmetric point sets, and the foundations they use; other Lean-checked results of the project, among them further results on symmetric point sets, are left to later stages. Each section restates the Lean-checked theorems of one part, faithfully to the Lean declarations, and gives a written proof that a mathematician can check without Lean, with two exceptions. Two published theorems that the Lean code uses, Baek's theorem and the cups–caps theorem of Erdős and Szekeres, are cited and not reproved. Statements marked computer-assisted are reduced to a finite computation that is stated precisely and carried out by a machine; it is not carried out by hand, except where a section gives a proof by hand as well. The Lean checks remain the primary evidence for these theorems; the proofs here are written proofs and have not been refereed. Where a Lean theorem takes unproved propositions or satisfiability verdicts as hypotheses, it is stated with those hypotheses and is established here only as that conditional statement.

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Scope of Stage 1

Paths follow the convention of Section 1: a path that begins with `ErdosSzekeres/`, `docs/`, `experiments/` or `scripts/` is relative to the root of the project repository, and every other path is relative to its research-records directory `cloud/es7-2026-10-19/results`.

Treated here. The sections restate the Lean-checked results of the following files and give written proofs of them. They also restate declarations of the main library `ErdosSzekeres/` that these files use: the foundations (Section 1), and the formalizations of Baek's theorem and of the cups–caps theorem with their bridge files (Section 10); a few such declarations are also restated in Sections 7 and 9. These four sections list the main-library files concerned, with their hashes.

- `LAYERLEAN/LayerLemma.lean`, mainly in Sections 2 and 3;
- the subdirectories `orbit`, `merged`, `keys`, `phase`, `linked`, `capstone` and `es7corr` of `LEANEXT`, in Sections 4, 5 and 6;
- `ES8SYM/lean` and `PRIMEFOLD/compat/lean`, in Section 6;
- `SYMLEAN/SymmetricReduction.lean` and `SYMLEAN/AffineReduction.lean`, in Sections 7 and 8, and `SYMLEAN/mirror/ReductionThreeClasses.lean`, in Section 10;
- `MIRRORETV/lean`, in Sections 9 and 10;
- `C3LEAN` and `C2FINAL` (the declarations of the latter are Lean-checked according to the record), in Section 11;
- `AUDIT3/A1_reduction` and `ES7SPLIT/review/equivalence`, in Section 12.

Not restated, although they belong to these records: verbatim copies, tests and controls, `#print` and `#check` commands, the review files in `LEANEXT/review`, `SYMLEAN/review` and `MIRRORETV/review` (re-checks, independent formalizations and tests; a few of their theorems are mentioned in remarks), `MIRRORETV/lean/FarFlat.lean` (estimates for a lower-bound construction) and a few further declarations. The Coverage paragraph at the end of each section gives the details and the reasons. The Lean files of `C3L` (a generator of CNF formulas) and of `AUDIT2` (audit scripts) state no theorems and are not treated.

Left to later stages. The following Lean-checked work of the project is not treated in this document. Planned for Stage 2, the SAT encodings:

- `ENCBLEAN`: encoder B of the sixfold case of the $k = 8$ problem as a Lean program, with its soundness, and the geometric assembly in `ENCBLEAN/geo`, which derives its copy of `Sixfold66` (Definition 2.27) from the unsatisfiability of 64 cube formulas (`EncB.Geo.sixfold66_of_cubes64`; see Remark 2.30);
- `H` and `H2`: Lean versions of clause generators of the project's SAT encodings, among them mirror-symmetric and D_3-symmetric ones, with their soundness;
- `H3`, `V1-L`, `V2-L` and `V2-L2`: the derivation of the mirror statement `Mir.MirrorExclusionLine` from hypotheses asserting that certain CNF formulas are unsatisfiable (Remark 7.45).

Planned for Stage 3:

- `experiments/omission-compact-checker-2026-11-05`: the compact checker of omission certificates, with its Lean check of $\mathrm{ES}(8) \le 311$;
- the main library beyond the declarations treated here.

The Lean files of `ES8SLACK` are not treated here; written proofs of its lemmas are in `ES8SLACK/proofs.md`.

Evidence outside Lean. Some hypotheses of conditional theorems are supported in the project by SAT certificates. For these the project counts only the verdicts of the proof checker cake_lpr [11]; the checker

lrat-check of drat-trim [3] is a secondary check, whose verdicts are not counted (`WRITEUP/SYMMETRIC.md`, §3, and `LRATCHECK20261105`). None of this evidence is checked in this document.

Works cited. The sections cite published works inline, by author together with title or arXiv number, by author alone or by name; the References list them. They are: Lean 4 [2] and Mathlib [6]; textbooks for standard facts [7, 8]; the theorems of Baek [1] and of Erdős and Szekeres [5], which are cited and not reproved (Theorems 10.4 and 10.6); earlier work with priority for the fivefold case (Dumitru [4]) and for the fourfold case (Subercaseaux, Mackey, Qian and Heule [10]; see Remark 7.8); and the archive of M. Schroeder [9], from which six of the bridge files of Section 10 are adapted.

1. Conventions and foundations

Purpose of this document. This document collects human-readable proofs of results of a research project on the Erdős–Szekeres problem whose formal versions have been checked with the interactive theorem prover Lean 4 and its mathematical library Mathlib. The Lean checks remain the primary evidence for these results. The written proofs are meant to make the arguments accessible to a mathematician who does not read Lean: every statement that corresponds to a Lean declaration is stated exactly as in Lean, with the name of the declaration and its file and line, and it receives a complete proof that can be checked without Lean. There are two exceptions. A statement labelled computer-assisted is reduced to a finite computation, which is stated precisely and need not be carried out by hand; where this document also gives a complete written proof by hand, the label is qualified as "computer-assisted in Lean" (see the rigor labels below). A few published theorems that the Lean code uses (for example Baek's theorem, Theorem 10.4) are cited in the exact form in which they are used and are not reproved. A convenient variant of a Lean statement, for example one in a different generality, is labelled as a variant. These written proofs have not been refereed. Where a Lean theorem assumes propositions that are not proved in Lean (for example verdicts of SAT solvers), the statement says so and the theorem is presented as a conditional statement.

Files, commit and versions. All files belong to the project repository at commit `7e32ee259` (branch `es7-night-2026-11-01`; Lean package `erdos_szekeres`). A path that begins with `ErdosSzekeres/`, `docs/`, `experiments/` or `scripts/` is relative to the repository root, and so are the files `lean-toolchain`, `lake-manifest.json` and `lakefile.toml` at the top of the repository. The directory `ErdosSzekeres/` is the *main library*, the default build target of the package. Every other path, for example `LAYERLEAN/LayerLemma.lean:298`, is relative to the research-records directory, whose path from the repository root is `cloud/es7-2026-10-19/results`. The Lean files of the research records are not part of the build target; they are checked separately, for example with `lake env lean` run from the repository root (as `LAYERLEAN/logs/axioms_recheck_20261106/README.txt` records for a copy of `LAYERLEAN/LayerLemma.lean` followed by `#print axioms` commands), and each section names the build and axioms logs it relies on. After a section has cited a file with its full path, it may refer to it by its bare file name. A suffix `:N` is a line number; for a declaration it is the line that carries the declaration keyword (`theorem`, `lemma`, `def`, `abbrev`, `structure` or `inductive`, possibly preceded by an attribute such as `@[simp]` or a modifier such as `noncomputable` or `private`). A bare `:N` written after a file has been cited refers to line N of that file. Lean declarations are given by their full names in statement heads, for example `ErdosSzekeres.Suk.convexPosition_of_four_subsets`; in running text we sometimes omit the common prefix `ErdosSzekeres.` (so `Suk.orient_reflectY` is `ErdosSzekeres.Suk.orient_reflectY`). Names in a namespace outside `ErdosSzekeres`, such as `MirrorOdd.hasConvexNGon_of_le` in the top-level namespace `MirrorOdd`, are always written in full. Mathlib declarations are named as in Mathlib. Lean expressions are quoted in typewriter type, for example `Finset Point`. The Lean files were checked with Lean 4.24.0 (file `lean-toolchain`) and Mathlib at revision `f897ebcf` (full hash `f897ebcf72cd16f89ab4577d0c826cd14afaafc7`, file `lake-manifest.json`).

Rigor labels. We use exactly three labels; the first and the third also occur in a qualified form, defined below.

- *Lean-checked*: the statement is a Lean declaration that Lean 4.24.0 with Mathlib `f897ebcf` accepted, using no axioms other than the three standard axioms `propext`, `Classical.choice` and `Quot.sound`; for a declaration whose proof uses Lean's `native_decide`, directly or through a declaration on which it depends, the two further axioms `Lean.ofReduceBool` and `Lean.trustCompiler` that `native_decide` adds are allowed as well, and the section that cites such a declaration names them (no declaration cited in this section is of this kind). The evidence is the saved output of Lean's command `#print axioms`, for the declaration itself or for a declaration whose statement or proof uses it (this command reports every axiom in the dependency closure, which contains the declarations used in statements as well as in proofs). Each section names the build and axioms logs on which this label rests. Where a declaration is

not covered by such an output, the section says so and names the evidence used instead, which is of one of two kinds:

(a) evidence that the current version of the file that contains the declaration compiled without errors (exit status 0), namely a saved compile or build log of that file, or a project record of such a build when saved Lean output of a build that includes the file (for example an axiom audit) exists as well; together with a text search of the file and of the project files it imports that finds no `sorry` (a declaration that uses `sorry` compiles with a warning only, so the search, not the exit status, excludes it), no `admit`, no `axiom` declaration, no `native_decide`, no `+native` (as in `decide +native`, of which `native_decide` is a synonym), no `Lean.ofReduceBool` or `Lean.ofReduceNat` (the axioms through which values computed by Lean's compiler or interpreter enter a proof) and no `debug.skipKernelTC` (an option that skips the kernel's type check);

(b) a `#guard_msgs` command, compiled in a build whose log is saved, that pins the output of `#print axioms` for the declaration or for a declaration whose statement or proof uses it.

A text search of every Lean file of the repository that this document cites, and of the project files these import, finds none of the strings `+native`, `ofReduceBool`, `ofReduceNat` and `skipKernelTC`, so every application of (a) in this document remains valid, also where it reports only a search for `sorry`, `admit`, `axiom` declarations and `native_decide`. If no Lean output of any kind (compile or build log, axiom output, pinned message) is saved for a build that includes a file, and the label of its declarations rests only on the statement of a project record such as a README, we write *Lean-checked according to the record*.

- *Written proof (not refereed)*: every proof in this document. It is meant to be checkable by a mathematician without Lean; it has not been refereed.
- *Computer-assisted*: a statement that rests on a finite computation carried out by a machine. This is the case, first, when the Lean proof of the statement, or of one of its parts, is itself a finite computation: that statement or part is decided by kernel evaluation of a decision procedure (Lean's `decide` or `decide +kernel`, which add no axiom) or by compiled evaluation (Lean's `native_decide`, which adds the axioms `Lean.ofReduceBool` and `Lean.trustCompiler` and trusts the compiler), or it is obtained by evaluating generated data or by replaying a certificate produced by an external program (for example an LRAT refutation found by a SAT solver, replayed in Lean and checked by the kernel). It is the case, second, when the written proof of the statement uses a machine computation that it does not carry out by hand, for example by applying a statement of the first kind that has no written proof by hand. For a computer-assisted statement we give the mathematical reduction to the computation and describe the computation precisely; we do not claim to have carried out by hand a computation for which no written proof by hand is given. If the Lean proof of a statement is a finite computation and this document also gives a complete written proof of the statement that does not use the computation, the statement is labelled *computer-assisted in Lean*, usually with a pointer to the written proof (as in "computer-assisted in Lean; written proof by hand"); that written proof can be checked without a machine. A Lean proof that uses `decide` only for side facts inside the proof of a statement that is not itself a finite computation (such as $0 < 1$ in Fin 4, or closed instances that the written proof checks by hand) does not make the statement computer-assisted. Certificates may also be checked by programs outside Lean. For the SAT certificates of the project only the checker `cake_lpr` counts, and a verdict of it counts only if it prints the line `s VERIFIED UNSAT`; verdicts of `lrat-check` alone are not counted (`LRATCHECK20261105/README.md`; `WRITEUP/SYMMETRIC.md`, section 3). A section may report that a result of the project's records, such as an unsatisfiability verdict, is computer-assisted; this is a claim of the records, which this document does not prove, and a Lean theorem that takes such a result as a hypothesis is presented as conditional on it.

A statement, or a part of a statement, whose head names a declaration of the project in the form "Lean: name, file:line" is Lean-checked in this sense. For a definition this records only that Lean accepted the definition: a definition asserts nothing, and what this document claims for it is that the written definition restates the Lean definition. A part marked "written step" does not carry this label, even where the text names a Lean counterpart of it elsewhere, and a Mathlib declaration named in a head only identifies the Mathlib counterpart of a statement that is proved here. A Lean theorem that takes unproved propositions as hypotheses is Lean-checked only as the conditional statement it is, and the hypotheses are named. We use no other words of this kind (such as "verified" or "certified") for the results of this document.

Lean's arithmetic, written explicitly. Lean statements often use natural numbers inside real expressions and subtract natural numbers. We make both explicit. A natural number m used as a real number is written $(m : \mathbb{R})$ whenever the cast matters; for example the Lean angle `2 * Real.pi / m` with `m` a natural number is $2\pi/(m : \mathbb{R})$. Real division in Lean is total, with $x/0 = 0$; hence $2\pi/(m : \mathbb{R}) = 0$ when $m = 0$. Subtraction of natural numbers in Lean is truncated ($a - b = 0$ when $a < b$); whenever a Lean statement subtracts natural numbers we write the difference as $\max(a - b, 0)$, or as $a \dot{-} b = \max(a - b, 0)$ where a section defines this symbol, or we state the inequality $b \le a$ that makes it an ordinary difference. $\mathrm{Fin}\, n = \{0, 1, \ldots, n - 1\}$ carries its natural order. For a map f we write $f^{[n]}$ for its n-th iterate ($f^{[0]} = \mathrm{id}$, $f^{[n+1]} = f^{[n]} \circ f$), as Lean does; the recursion $f^{[n+1]} = f \circ f^{[n]}$ defines the same maps (see the proof of Lemma 1.11(c)). Mathlib's `Function.minimalPeriod` $f\ x$ is the least $n > 0$ with $f^{[n]}(x) = x$, or 0 if there is no such n.

Default notation. Points, finite sets and $|S|$ are as in Definition 1.1; orient is Definition 1.3; conv is the convex hull of Definition 1.4. For a finite set X of points, $\mathrm{hv}(X) = \{x \in X : x \notin \mathrm{conv}(X \setminus \{x\})\}$ is the set of hull vertices (Lean `hullVerts`) and $\mathrm{peel}(X, 0) = X$, $\mathrm{peel}(X, n + 1) = \mathrm{peel}(X, n) \setminus \mathrm{hv}(\mathrm{peel}(X, n))$ (Lean `peel`); these two notions are defined formally, with their Lean names, in Definitions 2.4 and 2.18. "S is K-free" means $\neg\, \mathrm{HasConvexNGon}\, K\, S$ (Definition 1.7). We write $\rho_{c,\theta}$ for the rotation $\mathrm{rotAbout}\, c\, \theta$ of Definition 1.12; rotations act counterclockwise (Lemma 1.14(e)). In this section R_θ always denotes the rotation matrix of Lemma 1.14, not a map. The notation of this paragraph and of the previous one is the default for the whole document. A later section may write the same objects differently, for example $\mathrm{rotAbout}(c, \theta)$ or $\mathrm{rotAbout}_{c,\theta}$ for $\rho_{c,\theta}$, a letter such as R with subscripts for the rotation map itself and another letter for its matrix, f^n for $f^{[n]}$, or HasConvexNGon in another typeface; such a section defines its symbols where it introduces them.

This section. We define points, the orientation function, general position, convex position, convex n-gons, K-freeness, forcing, invariance and rotations exactly as the Lean code does, and we prove the basic facts that all later sections use: the symmetries of orient and its behaviour under affine maps; the equivalence of $\mathrm{orient}(a, b, c) = 0$ with collinearity; the hull form of convex position; monotonicity; invariance of convex and general position under injective affine maps; Carathéodory's theorem in the plane; the four-point criteria for convex position; the criterion that points whose increasing triples are all counterclockwise are in convex position; and the separation of a point from a finite set. This section relies on no other section. Its sources, with the first 12 hexadecimal digits of their SHA-256 hashes, are listed in the following table, the table of sources.

file	SHA-256 (prefix)
`ErdosSzekeres/Definitions.lean`	`2425500eefd1`
`ErdosSzekeres/Orientation.lean`	`f67955ac7091`
`ErdosSzekeres/OrientationGeometry.lean`	`c7db9bbbb65e`
`ErdosSzekeres/Basic.lean`	`13667aa2d3dc`
`ErdosSzekeres/Suk/FourPoint.lean`	`f3783fe9766f`
`ErdosSzekeres/Suk/SupportingLines.lean`	`6a8e12677685`
`ErdosSzekeres/Suk/OrientationClique.lean`	`cf63d5219b47`
`ErdosSzekeres/ConvexCycleCuts.lean`	`b4ad0ba9e583`
`ErdosSzekeres/Suk/Reflection.lean`	`8b99cbf5fa2b`
`LAYERLEAN/LayerLemma.lean`	`afb8542dceb0`
`SYMLEAN/SymmetricReduction.lean`	`6d5ec90a4215`
`SYMLEAN/AffineReduction.lean`	`41cfc0b60aec`
`LEANEXT/orbit/OrbitConsistency.lean`	`9b843180ac38`
`MIRRORETV/lean/MirrorMain.lean`	`ca623e55a1b9`
`SYMLEAN/mirror/ReductionThreeClasses.lean`	`0cc83994fe8c`

Axioms evidence. In this paragraph, "line N" without a file name refers to the log, audit or report named last before it. A declaration said to be used "at" a file and line is used on that line, inside the proof named; the line is not the line of its own declaration.

- *Main library, printed theorems.* The saved axiom audit `docs/validation/axiom-audit-lower-bound-2026-10-11.txt` reports exactly the axioms `propext`, `Classical.choice`, `Quot.sound` for `ErdosSzekeres.hasConvexNGon_of_positive_cycle` (line 87), `ErdosSzekeres.Suk.convexPosition_of_four_subsets` (line 103), `ErdosSzekeres.Suk.convexPosition_of_strict_supports` (line 113), `ErdosSzekeres.Suk.convexPosition_four_of_positive` (line 114),

`ErdosSzekeres.Suk.convexPosition_of_positive_triples` (line 119), `ErdosSzekeres.HasConvexNGon.linearEquiv` (line 175) and `ErdosSzekeres.Suk.generalPosition_reflectY` (line 176).

- *Main library, declarations used by these theorems.* Since `#print axioms` reports every axiom used anywhere in the dependency closure of a declaration, these lines also cover the declarations used in those proofs: `Suk.convexPosition_of_subsets_card_le_four`, `convexPosition_iff`, `ConvexPosition.mono`, `Suk.convex_orient_positive`, `Suk.orient_affine_combination`, `Suk.orient_support_sum`, `Suk.reflectY`, `Suk.orient_reflectY`, `generalPosition_iff_orient_ne_zero` and `orient_eq_zero_iff_collinear` (`Suk.generalPosition_reflectY` uses `generalPosition_iff_orient_ne_zero` at `ErdosSzekeres/Suk/Reflection.lean:55`, and the proof of the latter uses `orient_eq_zero_iff_collinear`).
- *Main library, further theorems of the audit.* Other theorems printed in the same audit, with the same three axioms, cover further declarations through their statements or proofs:
  - line 725, `conjecturedBound_le_of_forces`, uses `ForcesConvexNGon.mono` at `ErdosSzekeres/LowerBound/Construction.lean:118`;
  - line 732, `esNumber_three` (declared at `ErdosSzekeres/LowerBound/Number.lean:40`), uses `forces_triangle`, whose proof uses `forcesConvexNGon_iff_exact` and `convexPosition_of_card_three` at `ErdosSzekeres/Basic.lean:73` and `:75`; the proof of `forcesConvexNGon_iff_exact` uses `GeneralPosition.mono` and `HasConvexNGon.mono` at `ErdosSzekeres/Basic.lean:45`;
  - line 720, `LowerBound.exists_cupCapFree`, uses `LowerBound.generalPosition_translate` at `ErdosSzekeres/LowerBound/Construction.lean:54`, whose proof uses `orient_translate` at `ErdosSzekeres/LowerBound/Placement.lean:20`;
  - line 624, `Omission.Configuration.orient_ne_of_distinct`, uses `orient_swap` and `orient_cycle` at `ErdosSzekeres/OrientationObstructions.lean:64` and `:68`;
  - line 132, `Suk.AdjacentCapFrames.glue_chain_caps`, uses `ConvexPosition.hasConvexNGon` at `ErdosSzekeres/Suk/SpikeChain.lean:140`;
  - line 49, `VisibleEdgeInsertion.convexPosition_insert_of_one_visible_edge`, uses `Suk.convex_orient_nonnegative` at `ErdosSzekeres/VisibleEdgeInsertion.lean:77`;
  - line 178, `Suk.horizontal_finite_density_bound`, uses `Suk.hasConvexNGon_of_reflectY` at `ErdosSzekeres/Suk/FiniteDensityBound.lean:66`;
  - line 88 prints `convexCycleCut_of_avoids`, whose statement contains `ConvexCycleCut`;
  - line 529 prints `Omission.BoundaryGeometry.exposed_not_in_hull` (declared at `ErdosSzekeres/Omission/BoundaryGeometry.lean:30`), which the Coverage paragraph names.

  The definitions `Point`, `orient`, `GeneralPosition`, `ConvexPosition`, `HasConvexNGon` and `ForcesConvexNGon` occur in the statements of the printed theorems above (for example in those of lines 87, 103, 176 and 725), so they are covered as well. Every main-library file named in this paragraph is unchanged since commit `7f0868fdb` (11 October 2026), which added the audit.
- *Main library, remaining declarations.* The remaining main-library declarations named in this section, `HasConvexNGon.card_le`, `Suk.convexPosition_of_negative_triples` and the three simplification lemmas `Suk.reflectY_x`, `Suk.reflectY_y`, `Suk.reflectY_reflectY` of Remark 1.28, are not covered by a printed line that we identified. For them the evidence is of kind (a) of the rigor labels: the build record `docs/VALIDATION.md:73` (on 11 October 2026 the main library, which contains their files, builds with `lake build --wfail`, which fails on any warning, in particular on the warning that Lean issues for a declaration that uses `sorry`); the audit above, saved Lean output of the same library, whose printed theorems include theorems of `ErdosSzekeres/Suk/OrientationClique.lean` and `ErdosSzekeres/Suk/Reflection.lean` and depend on declarations of `ErdosSzekeres/Basic.lean`; and a text search of their files and of the main-library files these import (`ErdosSzekeres/Suk/BaseOrder.lean`, `ErdosSzekeres/Suk/BaseChart.lean`, `ErdosSzekeres/Suk/TriangleOrder.lean`, `ErdosSzekeres/Suk/CupsCaps.lean` and the four files of `ErdosSzekeres/Vendor/ErdosSzekeresConvex/`, besides the files in the table of sources), which finds no `sorry`, `admit`, `axiom` declaration or `native_decide`, and no `implemented_by` or `extern` attribute.
- *The layer-lemma file.* The review report `LEANEXT/review/REPORT.md` (item 1) records that `LAYERLEAN/LayerLemma.lean`, in its current version (1148 lines, SHA-256 prefix `afb8542d`), compiles with exit code 0 and no messages. The re-check `LAYERLEAN/logs/axioms_recheck_20261106/out.txt` records

exit status 0 for a file whose first 1148 lines are byte-identical to `LAYERLEAN/LayerLemma.lean`, followed by 19 `#print axioms` commands; the only Lean output in it is the 19 printed axiom lines. (The log `LAYERLEAN/logs/build.log` names no file hash; it was committed with the first, 454-line version of the file and has not been updated since, and we do not use it.) The review log `LEANEXT/review/axioms.log`, which prints the axioms of the theorems of `LAYERLEAN/LayerLemma.lean`, lists `ErdosSzekeres.LayerLemma.iterate_mem` with `propext` and `Quot.sound` only (line 1) and `ErdosSzekeres.LayerLemma.rotAbout_eq_crot` (line 22), `ErdosSzekeres.LayerLemma.rotAbout_injective` (line 23), `ErdosSzekeres.LayerLemma.hullVerts_eq_orbitFS_rotAbout` (line 26; its proof uses `ErdosSzekeres.LayerLemma.rotAff` at `LAYERLEAN/LayerLemma.lean:368`) and `ErdosSzekeres.LayerLemma.rotAbout_iterate` (line 33) with the three standard axioms.

- *The symmetric and affine reduction files.* `SYMLEAN/logs/runs.tsv` records that `SYMLEAN/SymmetricReduction.lean` (SHA-256 prefix `6d5ec90a`, the current file) compiles with exit code 0 and no output (line 2), and so does `SYMLEAN/AffineReduction.lean` (prefix `41cfc0b6`, the current file; line 4). The file `SYMLEAN/AffineReductionAxioms.lean` is `SYMLEAN/AffineReduction.lean` followed by nine `#print axioms` commands and one `#check` command; it compiles with exit code 0 (`SYMLEAN/logs/runs.tsv`, line 5), and its output `SYMLEAN/logs/AffineReductionAxioms.out` lists with the three standard axioms
  - `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affine_coords` (lines 9–11), whose proof uses `ErdosSzekeres.SymReduction.hasConvexNGon_of_affine_coords_symmetry` at `SYMLEAN/AffineReduction.lean:1528`, whose proof in turn uses `ErdosSzekeres.SymReduction.injective_of_affine` at `SYMLEAN/AffineReduction.lean:1462`;
  - `ErdosSzekeres.SymReduction.generalPosition_image_affineEquiv` (lines 13–15), whose proof uses `ErdosSzekeres.SymReduction.generalPosition_image_affine` at `SYMLEAN/AffineReduction.lean:1295`, whose proof in turn uses `ErdosSzekeres.SymReduction.orient_affine` at `SYMLEAN/AffineReduction.lean:1281`;
  - `ErdosSzekeres.SymReduction.hasConvexNGon_image_affine` (line 16), whose proof uses `ErdosSzekeres.SymReduction.convexPosition_image_affine` at `SYMLEAN/AffineReduction.lean:1303`.
- *The orbit-consistency file.* `LEANEXT/orbit/logs/build.log` records exit code 0 for the build of `LEANEXT/orbit/OrbitConsistency.lean`; the log contains only the exit code, and the report `LEANEXT/orbit/REPORT.md` that accompanies these logs identifies the checked file (line 3: 429 lines, SHA-256 prefix `9b843180`, the current file). `LEANEXT/orbit/logs/axioms.log` lists the two theorems of that file named in this section, `ErdosSzekeres.OrbitConsistency.orient_rotAbout_inv` (line 1) and `ErdosSzekeres.OrbitConsistency.orient3_perm` (line 5), with the three standard axioms. From `SYMLEAN/SymmetricReduction.lean` and `LEANEXT/orbit/OrbitConsistency.lean` we otherwise cite only definitions.
- *The mirror files.* `MirrorOdd.hasConvexNGon_of_le` is used at `MIRRORETV/lean/MirrorMain.lean:2520`, in the proof of `MirrorOdd.mirror_odd_general` (declared at `MIRRORETV/lean/MirrorMain.lean:2498`), which the compile log `MIRRORETV/lean/MirrorMain.out` of that file (warnings, no errors) lists with the three standard axioms (line 858). Its verbatim copy in `SYMLEAN/mirror/ReductionThreeClasses.lean` is used in the same way, at `SYMLEAN/mirror/ReductionThreeClasses.lean:4079`, in the proof of the copy of `MirrorOdd.mirror_odd_general` (declared at `SYMLEAN/mirror/ReductionThreeClasses.lean:4057`), which line 11 of `SYMLEAN/mirror/axioms.out` lists with the three standard axioms; and `SYMLEAN/mirror/README.md` records that this file compiles with exit code 0 and no errors.
- *Text search.* A text search of the six research files of the table of sources finds no `sorry`, `admit`, `axiom` declaration or `native_decide`, and no `implemented_by` or `extern` attribute.

1.1. **Definitions.**

Definition 1.1 (Points and finite point sets; Lean: `ErdosSzekeres.Point`, `ErdosSzekeres/Definitions.lean:9`)**.** A *point* is an element of $\mathbb{R}^2$. In Lean, `Point` abbreviates the type of functions from $\operatorname{Fin} 2 = \{0, 1\}$ to $\mathbb{R}$; we write $p = (p_0, p_1)$, where $p_0 = p(0)$ is the first (x-)coordinate and $p_1 = p(1)$ the second (y-)coordinate. Addition, subtraction and multiplication by real numbers are coordinatewise.

A *finite set of points* is a Lean `Finset Point`: a finite set, without multiplicities. We use ordinary set notation for finite sets: $S \subseteq T$, $S \cup T$, $S \setminus T$, $S \setminus \{c\}$ (Lean `S.erase c`), $|S|$ (Lean `S.card`), and $f(S) = \{f(p) : p \in S\}$

for a map f (Lean `S.image f`, again a finite set, with $|f(S)| = |S|$ when f is injective). Where Lean needs a set rather than a finite set, for example as the argument of a convex hull, it uses the coercion of S to a `Set Point`, which has the same elements; we do not distinguish the two. Brace notation $\{a, b, c\}$ denotes the set with these elements; it has fewer than three elements when two of a, b, c coincide.

Remark 1.2 (Norm and topology). Mathlib gives the Lean type of points its default sup norm $\max(|p_0|, |p_1|)$. No statement of this section uses a norm or a metric; Euclidean quantities are always written out in coordinates (as in Lemma 1.40). The only topological argument of this section is the compactness argument in the proof of Lemma 1.40, which takes place in $\mathbb{R}^m$ with its usual topology. (On $\mathbb{R}^2$ the sup norm and the Euclidean norm define the same, usual, topology.)

Definition 1.3 (Orientation; Lean: `ErdosSzekeres.orient`, `ErdosSzekeres/Orientation.lean:9`)**.** Let R be a commutative ring and let $a, b, c \in R^2$ (in Lean: functions from $\mathrm{Fin}\,2$ to R). Then

$$\mathrm{orient}(a, b, c) = (b_0 - a_0)(c_1 - a_1) - (b_1 - a_1)(c_0 - a_0) \in R.$$

Unless stated otherwise $R = \mathbb{R}$; Lean also evaluates orient over $\mathbb{Q}$ and $\mathbb{Z}$ in exact certificate checks. Equivalently, $\mathrm{orient}(a, b, c) = \det\big(b - a \mid c - a\big)$ is the determinant of the 2×2 matrix with columns $b - a$ and $c - a$, that is, twice the signed area of the triangle abc. Positive orientation means counterclockwise; for example $\mathrm{orient}((0,0),(1,0),(0,1)) = 1 > 0$. For $a \neq b$ we say that c lies (strictly) to the *left* of the line through a and b directed from a to b if $\mathrm{orient}(a, b, c) > 0$, and to the *right* if $\mathrm{orient}(a, b, c) < 0$.

Definition 1.4 (Convex sets, convex hulls, affine maps; Mathlib `Convex`, `convexHull`, `AffineIndependent`, `AffineMap`, `AffineEquiv`)**.** A set $K \subseteq \mathbb{R}^2$ is *convex* if $\lambda x + \mu y \in K$ for all $x, y \in K$ and all reals $\lambda, \mu \geq 0$ with $\lambda + \mu = 1$ (Mathlib `Convex`). The *convex hull* $\mathrm{conv}\, Y$ of a set $Y \subseteq \mathbb{R}^2$ is the intersection of all convex sets that contain Y (Mathlib `convexHull`, defined as the smallest convex superset). A *convex combination* of points $y_1, \ldots, y_k$ ($k \geq 1$) is a point $\sum_{i=1}^{k} \lambda_i y_i$ with all $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. A finite set $T \subseteq \mathbb{R}^2$ is *affinely independent* if the only family of reals $(\mu_y)_{y \in T}$ with $\sum_{y \in T} \mu_y = 0$ and $\sum_{y \in T} \mu_y\, y = 0$ is the zero family (equivalent to Mathlib's `AffineIndependent` for the inclusion map of T). An *affine map* of $\mathbb{R}^2$ is a map $g(x) = Ax + v$, where A is a real 2×2 matrix (the *linear part* of g) and $v \in \mathbb{R}^2$. In Mathlib, an affine map f from the plane to itself (`AffineMap`) is a map $f \colon \mathbb{R}^2 \to \mathbb{R}^2$ together with a linear map $f_{\mathrm{lin}} \colon \mathbb{R}^2 \to \mathbb{R}^2$ (Lean `f.linear`) such that $f(w + p) = f_{\mathrm{lin}}(w) + f(p)$ for all $p, w \in \mathbb{R}^2$. These are exactly the affine maps just defined: taking $p = 0$ gives $f(x) = A_f x + f(0)$, where A_f is the matrix of f_{lin}; conversely, $g(x) = Ax + v$ satisfies $g(w + p) = Aw + g(p)$, so it is a Mathlib affine map whose linear map is $x \mapsto Ax$. A Mathlib *affine equivalence* of the plane (`AffineEquiv`) is a bijection f of $\mathbb{R}^2$ together with a linear bijection f_{lin} of $\mathbb{R}^2$ with the same property; in particular it is an injective affine map.

Definition 1.5 (General position; Lean: `ErdosSzekeres.GeneralPosition`, `ErdosSzekeres/Definitions.lean:12`)**.** A set $X \subseteq \mathbb{R}^2$ is *collinear* (Mathlib `Collinear`) if the linear span of the set of differences $\{p - q : p, q \in X\}$ (Mathlib `vectorSpan`) has dimension at most 1 (in Lean: `Module.rank` of this span is at most 1). A finite set S of points is *in general position* if for all $a, b, c \in S$ with $a \neq b$, $a \neq c$ and $b \neq c$ the set $\{a, b, c\}$ is not collinear.

Definition 1.6 (Convex position; Lean: `ErdosSzekeres.ConvexPosition`, `ErdosSzekeres/Definitions.lean:17`)**.** A finite set S of points is *in convex position* if the inclusion map of S into $\mathbb{R}^2$ is convex independent over $\mathbb{R}$ (Mathlib `ConvexIndependent`). Mathlib calls a family $p \colon \iota \to \mathbb{R}^2$ convex independent if for every set $t \subseteq \iota$ and every $x \in \iota$, $p(x) \in \mathrm{conv}\, p(t)$ implies $x \in t$. For the inclusion map of S the index set is S itself (in Lean, the type of elements of S) and p is the identity on it, so the definition says: *for every $T \subseteq S$ and every $x \in S$, if $x \in \mathrm{conv}\, T$ then $x \in T$.* Lemma 1.22 gives the familiar form: no point of S lies in the convex hull of the other points of S.

Definition 1.7 (Convex n-gons and K-free sets; Lean: `ErdosSzekeres.HasConvexNGon`, `ErdosSzekeres/Definitions.lean:21`)**.** For $n \in \mathbb{N}$ and a finite set S of points, $\mathrm{HasConvexNGon}\, n\, S$ holds if there is a finite set $T \subseteq S$ with $|T| = n$ that is in convex position. There is no emptiness condition: points of $S \setminus T$ may lie in $\mathrm{conv}\, T$. For $K \in \mathbb{N}$ we say that S is *K-free* if $\neg\, \mathrm{HasConvexNGon}\, K\, S$.

Remark 1.8. $\mathrm{HasConvexNGon}\, 0\, S$ holds for every S (Lemma 1.26(a)), so no set is 0-free, and a hypothesis "S is 0-free" is contradictory. If S is K-free and $K \leq K'$, then S is K'-free (contrapositive of Lemma 1.23(d)). If $\mathrm{HasConvexNGon}\, n\, S$, then $n \leq |S|$ (Lemma 1.24(a)).

Definition 1.9 (Forcing a convex n-gon; Lean: `ErdosSzekeres.ForcesConvexNGon`, `ErdosSzekeres/Definitions.lean:25`)**.** For $N, n \in \mathbb{N}$, ForcesConvexNGon $N\,n$ holds if every finite set S of points that is in general position and satisfies $N \le |S|$ satisfies HasConvexNGon $n\,S$.

Definition 1.10 (Invariant sets; Lean: `ErdosSzekeres.LayerLemma.Invariant`, `LAYERLEAN/LayerLemma.lean:105`, Lean: `ErdosSzekeres.SymReduction.Invariant`, `SYMLEAN/SymmetricReduction.lean:271`, Lean: `ErdosSzekeres.OrbitConsistency.Invariant`, `LEANEXT/orbit/OrbitConsistency.lean:47`)**.** For a map $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ and a finite set S of points, Invariant $f\,S$ holds if $f(p) \in S$ for every $p \in S$; that is, f maps S *into* S. For injective f this is equivalent to $f(S) = S$, and then also $f^{-1}(S) = \{p \in \mathbb{R}^2 : f(p) \in S\} = S$ (Lemma 1.11). The three cited Lean definitions are textually identical.

Lemma 1.11 (Invariance and images; (a), (b) written step; (c) Lean: `ErdosSzekeres.LayerLemma.iterate_mem`, `LAYERLEAN/LayerLemma.lean:52`)**.** *Let X be a set, let $f\colon X \to X$ be a map and let S be a finite subset of X.*

(a) If $X = \mathbb{R}^2$ and f is injective, then Invariant $f\,S$ *holds if and only if $f(S) = S$.*
(b) If $X = \mathbb{R}^2$, f is injective and Invariant $f\,S$ *holds, then $\{p \in \mathbb{R}^2 : f(p) \in S\} = S$.*
(c) If $f(y) \in S$ for every $y \in S$, then $f^{[i]}(x) \in S$ for every $x \in S$ and every $i \in \mathbb{N}$.

Part (c) is stated for an arbitrary set X, as in Lean, where X is an arbitrary type; parts (a) and (b) take $X = \mathbb{R}^2$ because Invariant *is defined for maps of the plane.*

Proof. (a) If $f(S) = S$, then $f(p) \in f(S) = S$ for every $p \in S$. Conversely, if Invariant $f\,S$ holds, then $f(S) \subseteq S$; since f is injective, $|f(S)| = |S|$, and a subset of a finite set with the same number of elements is the whole set, so $f(S) = S$. (Lean uses this argument in the step at line 165 of the proof of `ErdosSzekeres.LayerLemma.map_mem_hullVerts`, `LAYERLEAN/LayerLemma.lean:162`.)

(b) If $p \in S$ then $f(p) \in S$ by invariance. If $f(p) \in S$, then by (a) $f(p) \in f(S)$, so $f(p) = f(q)$ for some $q \in S$, and $p = q \in S$ by injectivity.

(c) Induction on i. For $i = 0$, $f^{[0]}(x) = x \in S$. If $f^{[i]}(x) \in S$, then $f^{[i+1]}(x) = f\big(f^{[i]}(x)\big) \in S$ by the hypothesis applied to $y = f^{[i]}(x)$. Here we used $f^{[i+1]} = f \circ f^{[i]}$, which follows from the definition $f^{[i+1]} = f^{[i]} \circ f$ by induction on i ($f^{[1]} = f^{[0]} \circ f = f = f \circ f^{[0]}$, and $f^{[i+2]} = f^{[i+1]} \circ f = f \circ f^{[i]} \circ f = f \circ f^{[i+1]}$; Mathlib's `Function.iterate_succ_apply'`). This is the Lean proof. □

Definition 1.12 (Rotation about a point; Lean: `ErdosSzekeres.LayerLemma.rotAbout`, `LAYERLEAN/LayerLemma.lean:298`, Lean: `ErdosSzekeres.SymReduction.rotAbout`, `SYMLEAN/SymmetricReduction.lean:400`, Lean: `ErdosSzekeres.OrbitConsistency.rotAbout`, `LEANEXT/orbit/OrbitConsistency.lean:93`)**.** For $c \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$, the map $\rho_{c,\theta} = \mathrm{rotAbout}\,c\,\theta\colon \mathbb{R}^2 \to \mathbb{R}^2$ is

$$\rho_{c,\theta}(p) = \big(c_0 + \cos\theta\,(p_0 - c_0) - \sin\theta\,(p_1 - c_1),\ c_1 + \sin\theta\,(p_0 - c_0) + \cos\theta\,(p_1 - c_1)\big).$$

In complex notation, identifying p with $p_0 + ip_1$ and c with $c_0 + ic_1$, this is $z \mapsto c + e^{i\theta}(z - c)$ (Lemma 1.14(a)): the counterclockwise rotation about c by the angle θ (Lemma 1.14(e)). The three cited Lean definitions are textually identical; a text search of the research records finds 19 definitions of `rotAbout`, all with this same body.

Remark 1.13 (Angles $2\pi/m$). In later sections the angle is usually written `2 * Real.pi / m` in Lean with `m` a natural number, that is, $\theta = 2\pi/(m : \mathbb{R})$. For $m = 0$ Lean's division gives $\theta = 0$, and $\rho_{c,0}$ is the identity (Lemma 1.14(c)). The propositions `Sixfold66` and `Sevenfold70` (Definition 2.27) write their angles as `2 * Real.pi / 6` and `2 * Real.pi / 7` with real numerals; since the real numbers 6 and 7 are the casts of the natural numbers 6 and 7, these are the angles $2\pi/(m : \mathbb{R})$ for $m = 6$ and $m = 7$.

Lemma 1.14 (Basic properties of rotations; (a) Lean: `ErdosSzekeres.LayerLemma.rotAbout_eq_crot`, `LAYERLEAN/LayerLemma.lean:302`; (b) Lean: `ErdosSzekeres.LayerLemma.rotAff`, `LAYERLEAN/LayerLemma.lean:311`; (c) Lean: `ErdosSzekeres.LayerLemma.rotAbout_iterate`, `LAYERLEAN/LayerLemma.lean:465`; (d) Lean: `ErdosSzekeres.LayerLemma.rotAbout_injective`, `LAYERLEAN/LayerLemma.lean:319`; (e), (f) written step)**.** *Let $c \in \mathbb{R}^2$ and $\theta, \alpha, \beta \in \mathbb{R}$. Write $\rho_\theta = \rho_{c,\theta}$ and $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.*

(a) For every $p \in \mathbb{R}^2$, the complex number $\rho_\theta(p)_0 + i\,\rho_\theta(p)_1$ equals $\gamma + e^{i\theta}(z - \gamma)$, where $z = p_0 + ip_1$ and $\gamma = c_0 + ic_1$. (In Lean: $\mathrm{rotAbout}\,c\,\theta = \mathrm{crot}\,(\mathrm{toC}\,c)\,(\exp(\theta i))$*, where* $\mathrm{toC}\,p = p_0 + ip_1$ *and* $\mathrm{crot}\,\gamma\,w\,p$ *is the point with complex coordinate $\gamma + w(\mathrm{toC}\,p - \gamma)$.)*
(b) $\rho_\theta(p) = c + R_\theta(p - c) = R_\theta\,p + (c - R_\theta\,c)$ for every p; thus ρ_θ is an affine map with linear part R_θ, that is, $\rho_\theta(p + v) = \rho_\theta(p) + R_\theta v$ for all $p, v \in \mathbb{R}^2$. In particular $\rho_\theta(c) = c$. (The Lean definition

`LayerLemma.rotAff` *cited for (b) packages ρ_θ as an affine map with linear part R_θ, its defining property being the identity $\rho_\theta(p+v)=\rho_\theta(p)+R_\theta v$. The matrix form $\rho_\theta(p)=c+R_\theta(p-c)=R_\theta\,p+(c-R_\theta\,c)$ of Definition 1.12 and the identity $\rho_\theta(c)=c$ are written steps here; Lean proves the latter in the step at line 330 of the proof of* `ErdosSzekeres.LayerLemma.invariant_erase_centre`, `LAYERLEAN/LayerLemma.lean:325`.*)*

(c) $\rho_\alpha\circ\rho_\beta=\rho_{\alpha+\beta}$ *and* $\rho_0=\mathrm{id}$. *Hence* $\rho_\alpha^{[j]}=\rho_{(j:\mathbb{R})\alpha}$ *for every* $j\in\mathbb{N}$. *(The Lean declaration cited for (c) is this iterate formula; the composition law and* $\rho_0=\mathrm{id}$ *are written steps here.)*

(d) ρ_θ *is a bijection of* $\mathbb{R}^2$ *with inverse* $\rho_{-\theta}$*; in particular it is injective. (The Lean declaration cited for (d) states injectivity; bijectivity and the inverse are a written step here.)*

(e) $\operatorname{orient}\big(c,p,\rho_\theta(p)\big)=\sin\theta\,\big((p_0-c_0)^2+(p_1-c_1)^2\big)$ *for every* p. *Hence, if* $0<\theta<\pi$ *and* $p\neq c$, *then* $\operatorname{orient}(c,p,\rho_\theta(p))>0$*: the rotation turns counterclockwise.*

(f) $(\rho_\theta(p)_0-c_0)^2+(\rho_\theta(p)_1-c_1)^2=(p_0-c_0)^2+(p_1-c_1)^2$ *for every* p.

Proof. We give direct proofs in coordinates; the Lean proofs of (c) and (d) pass through the complex form (a). Put $u=p-c$. By definition,

$$\rho_\theta(p)-c=\big(\cos\theta\,u_0-\sin\theta\,u_1,\ \sin\theta\,u_0+\cos\theta\,u_1\big)=R_\theta\,u.$$

(b) This is the displayed identity; $R_\theta(p-c)=R_\theta p-R_\theta c$ by linearity. Hence $\rho_\theta(p+v)=c+R_\theta(p-c)+R_\theta v=\rho_\theta(p)+R_\theta v$, and $p=c$ gives $u=0$ and $\rho_\theta(c)=c$.

(a) By Euler's formula $e^{i\theta}=\cos\theta+i\sin\theta$, and $(\cos\theta+i\sin\theta)(u_0+iu_1)=(\cos\theta\,u_0-\sin\theta\,u_1)+i(\sin\theta\,u_0+\cos\theta\,u_1)$. Adding $\gamma=c_0+ic_1$ gives the two coordinates of $\rho_\theta(p)$.

(c) By (b), $\rho_\alpha(\rho_\beta(p))=c+R_\alpha R_\beta(p-c)$. By the addition formulas $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$ and $\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$,

$$R_\alpha R_\beta=\begin{pmatrix}\cos\alpha\cos\beta-\sin\alpha\sin\beta & -(\sin\alpha\cos\beta+\cos\alpha\sin\beta)\\ \sin\alpha\cos\beta+\cos\alpha\sin\beta & \cos\alpha\cos\beta-\sin\alpha\sin\beta\end{pmatrix}=R_{\alpha+\beta},$$

so $\rho_\alpha\circ\rho_\beta=\rho_{\alpha+\beta}$. Since R_0 is the identity matrix, $\rho_0=\mathrm{id}$. Now induct on j: $\rho_\alpha^{[0]}=\mathrm{id}=\rho_0=\rho_{(0:\mathbb{R})\alpha}$, and $\rho_\alpha^{[j+1]}=\rho_\alpha^{[j]}\circ\rho_\alpha=\rho_{(j:\mathbb{R})\alpha}\circ\rho_\alpha=\rho_{(j:\mathbb{R})\alpha+\alpha}=\rho_{(j+1:\mathbb{R})\alpha}$.

(d) By (c), $\rho_{-\theta}\circ\rho_\theta=\rho_0=\mathrm{id}$ and $\rho_\theta\circ\rho_{-\theta}=\mathrm{id}$.

(e) With $u=p-c$ as above,

$$\operatorname{orient}(c,p,\rho_\theta(p))=u_0(\sin\theta\,u_0+\cos\theta\,u_1)-u_1(\cos\theta\,u_0-\sin\theta\,u_1)=\sin\theta\,(u_0^2+u_1^2).$$

If $0<\theta<\pi$ then $\sin\theta>0$, and if $p\neq c$ then $u_0^2+u_1^2>0$.

(f) $(\cos\theta\,u_0-\sin\theta\,u_1)^2+(\sin\theta\,u_0+\cos\theta\,u_1)^2=(\cos^2\theta+\sin^2\theta)(u_0^2+u_1^2)=u_0^2+u_1^2$, the cross terms cancelling. □

1.2. **Facts.**

Lemma 1.15 (Symmetries of the orientation; (a) Lean: `ErdosSzekeres.orient_swap`, `ErdosSzekeres/Orientation.lean:12`; (b) Lean: `ErdosSzekeres.orient_cycle`, `ErdosSzekeres/Orientation.lean:15`; (c) Lean: `ErdosSzekeres.orient_translate`, `ErdosSzekeres/Orientation.lean:18`; (d), (e) written step)**.** *Let R be a commutative ring and let $a,b,c,v\in R^2$.*

(a) $\operatorname{orient}(a,c,b)=-\operatorname{orient}(a,b,c)$.

(b) $\operatorname{orient}(b,c,a)=\operatorname{orient}(a,b,c)$. *Consequently* $\operatorname{orient}(a,b,c)=\operatorname{orient}(b,c,a)=\operatorname{orient}(c,a,b)$.

(c) $\operatorname{orient}(a+v,b+v,c+v)=\operatorname{orient}(a,b,c)$.

(d) For all $p_1,p_2,p_3\in R^2$ *and every permutation* π *of* $\{1,2,3\}$, $\operatorname{orient}(p_{\pi(1)},p_{\pi(2)},p_{\pi(3)})=\operatorname{sgn}(\pi)\operatorname{orient}(p_1,p_2,p_3)$.

(e) $\operatorname{orient}(a,b,c)=0$ *whenever two of a,b,c are equal. Consequently, if* $\operatorname{orient}(a,b,c)\neq 0$, *then a,b,c are pairwise distinct.*

Proof. The Lean proofs of (a)–(c) expand both sides and compare polynomials (`unfold orient; ring` for (a) and (b), `simp only [orient, Pi.add_apply]; ring` for (c)); we organise the same computation. For $x,y\in R^2$ put $[x,y]=x_0y_1-x_1y_0$. Then $[\cdot,\cdot]$ is bilinear, $[y,x]=-[x,y]$ and $[x,x]=0$. Expanding the definition (the terms a_0a_1 cancel),

$$\operatorname{orient}(a,b,c)=a_0b_1-a_1b_0+b_0c_1-b_1c_0+c_0a_1-c_1a_0=[a,b]+[b,c]+[c,a].$$

(b) $\operatorname{orient}(b,c,a)=[b,c]+[c,a]+[a,b]=\operatorname{orient}(a,b,c)$. Applying this twice gives the second statement.

(a) $\operatorname{orient}(a,c,b)=[a,c]+[c,b]+[b,a]=-[c,a]-[b,c]-[a,b]=-\operatorname{orient}(a,b,c)$.

(c) The defining formula involves only the differences $b-a$ and $c-a$, and $(b+v)-(a+v) = b-a$, $(c+v)-(a+v) = c-a$.

(d) The permutation group of $\{1,2,3\}$ has six elements. The identity and the two 3-cycles have sign $+1$; they give $\operatorname{orient}(p_1,p_2,p_3)$, $\operatorname{orient}(p_2,p_3,p_1)$ and $\operatorname{orient}(p_3,p_1,p_2)$, which are equal by (b). The three transpositions have sign -1; they give $\operatorname{orient}(p_1,p_3,p_2)$, $\operatorname{orient}(p_2,p_1,p_3)$ and $\operatorname{orient}(p_3,p_2,p_1)$. The first equals $-\operatorname{orient}(p_1,p_2,p_3)$ by (a). By (b) applied to the triple (p_2,p_1,p_3) we get $\operatorname{orient}(p_1,p_3,p_2) = \operatorname{orient}(p_2,p_1,p_3)$, and by (b) applied to (p_3,p_2,p_1) we get $\operatorname{orient}(p_2,p_1,p_3) = \operatorname{orient}(p_3,p_2,p_1)$; so all three equal $-\operatorname{orient}(p_1,p_2,p_3)$.

(e) If $a=b$ the first factors b_0-a_0 and b_1-a_1 vanish; if $a=c$ the factors c_1-a_1 and c_0-a_0 vanish; if $b=c$ the two products $(b_0-a_0)(b_1-a_1)$ and $(b_1-a_1)(b_0-a_0)$ are equal. In each case $\operatorname{orient}(a,b,c)=0$. □

Remark 1.16 (The real case of (d) in Lean). For $R=\mathbb{R}$, part (d) of Lemma 1.15 is the Lean theorem `ErdosSzekeres.OrbitConsistency.orient3_perm` (`LEANEXT/orbit/OrbitConsistency.lean:178`; Lemma 4.14). It is stated for ordered triples $t\colon \operatorname{Fin}3 \to \mathbb{R}^2$, with $\operatorname{orient}_3(t) = \operatorname{orient}(t(0),t(1),t(2))$ (Lean `orient3`, `LEANEXT/orbit/OrbitConsistency.lean:168`), and says that $\operatorname{orient}_3(t\circ\sigma) = \operatorname{sgn}(\sigma)\operatorname{orient}_3(t)$ for every permutation σ of $\operatorname{Fin}3 = \{0,1,2\}$ and every t. Put $t(i)=p_{i+1}$ and $\sigma(i)=\pi(i+1)-1$; then $\sigma = s^{-1}\circ\pi\circ s$ for the bijection $s(i)=i+1$ from $\{0,1,2\}$ to $\{1,2,3\}$, so σ has the same cycle type and hence the same sign as π, and $(t\circ\sigma)(i) = p_{\pi(i+1)}$. So $\operatorname{orient}_3(t\circ\sigma) = \operatorname{orient}(p_{\pi(1)},p_{\pi(2)},p_{\pi(3)})$ and $\operatorname{orient}_3(t) = \operatorname{orient}(p_1,p_2,p_3)$, and the Lean statement is (d) for $R=\mathbb{R}$. Parts (d) for a general commutative ring and (e) are written steps.

Lemma 1.17 (Orientation under affine maps; Lean: `ErdosSzekeres.SymReduction.orient_affine`, `SYMLEAN/AffineReduction.lean:1241`; (i) Lean: `ErdosSzekeres.OrbitConsistency.orient_rotAbout_inv`, `LEANEXT/orbit/OrbitConsistency.lean:117`; (ii) written step)**.** *Let $g\colon \mathbb{R}^2\to\mathbb{R}^2$ be a map, let A be a real 2×2 matrix and $v\in\mathbb{R}^2$, and suppose that $g(x)=Ax+v$ for every $x\in\mathbb{R}^2$. Then for all $a,b,c\in\mathbb{R}^2$*

$$\operatorname{orient}\big(g(a),g(b),g(c)\big) = \det(A)\,\operatorname{orient}(a,b,c).$$

(In Lean, g is an arbitrary map with the hypothesis that $g(p)=Lp+w$ for every p, where L is a real 2×2 matrix and $w\in\mathbb{R}^2$ (Lean's `b`), and the conclusion is $\operatorname{orient}(g(x),g(y),g(z)) = \det(L)\operatorname{orient}(x,y,z)$ for all x,y,z; this is the statement above with $L=A$ and $w=v$.) In particular:

(i) (rotations have determinant 1) for all $o\in\mathbb{R}^2$, $\theta\in\mathbb{R}$ and $a,b,d\in\mathbb{R}^2$, $\operatorname{orient}(\rho_{o,\theta}(a),\rho_{o,\theta}(b),\rho_{o,\theta}(d)) = \operatorname{orient}(a,b,d)$, because the linear part R_θ of $\rho_{o,\theta}$ (Lemma 1.14(b)) has $\det R_\theta = \cos^2\theta+\sin^2\theta = 1$;

(ii) (reflections have determinant -1) for $o\in\mathbb{R}^2$ and $\varphi\in\mathbb{R}$ let $F_\varphi = \begin{pmatrix}\cos 2\varphi & \sin 2\varphi\\ \sin 2\varphi & -\cos 2\varphi\end{pmatrix}$ and $\sigma(x) = o+F_\varphi(x-o)$, the reflection in the line through o with direction $u=(\cos\varphi,\sin\varphi)$ (indeed $F_\varphi u = u$ and $F_\varphi u^\perp = -u^\perp$ for $u^\perp=(-\sin\varphi,\cos\varphi)$). Then $\operatorname{orient}(\sigma(a),\sigma(b),\sigma(d)) = -\operatorname{orient}(a,b,d)$ for all a,b,d, because $\det F_\varphi = -\cos^2 2\varphi-\sin^2 2\varphi = -1$. For $o=0$ and $\varphi=0$, σ is the reflection $(p_0,p_1)\mapsto(p_0,-p_1)$ in the first coordinate axis.

Proof. By Definition 1.3, $\operatorname{orient}(a,b,c) = \det(b-a \mid c-a)$. Since $g(b)-g(a)=A(b-a)$ and $g(c)-g(a)=A(c-a)$, the matrix with columns $g(b)-g(a)$ and $g(c)-g(a)$ is the product $A\,(b-a \mid c-a)$, and the determinant of a product of 2×2 matrices is the product of the determinants (a polynomial identity in the eight entries, checked by expanding both sides). This proves the formula. (The Lean proof substitutes $g(p)=Lp+w$, writes $\det L = L_{00}L_{11}-L_{01}L_{10}$, expands both sides in coordinates and compares polynomials, which is the same computation.)

(i) $\rho_{o,\theta}(x) = R_\theta x + (o - R_\theta o)$ by Lemma 1.14(b), and $\det R_\theta = \cos\theta\cdot\cos\theta - (-\sin\theta)\sin\theta = 1$.

(ii) $\sigma(x) = F_\varphi x + (o - F_\varphi o)$ has linear part F_φ, and $\det F_\varphi = -\cos^2 2\varphi - \sin^2 2\varphi = -1$. Using $\cos 2\varphi\cos\varphi + \sin 2\varphi\sin\varphi = \cos\varphi$ and $\sin 2\varphi\cos\varphi - \cos 2\varphi\sin\varphi = \sin\varphi$ (the subtraction formulas for $2\varphi-\varphi$) one gets $F_\varphi u = u$ and $F_\varphi u^\perp = (\sin\varphi,-\cos\varphi) = -u^\perp$, so σ fixes the line $o+\mathbb{R}u$ pointwise and reverses the perpendicular direction. For $\varphi=0$, $F_0=\operatorname{diag}(1,-1)$. □

Remark 1.18 (The formula and its special cases in Lean). The general formula is the Lean theorem `ErdosSzekeres.SymReduction.orient_affine`, cited in the head; a verbatim copy, with the same name, is at `SYMLEAN/mirror/ReductionThreeClasses.lean:1267`. Part (i) is a separate Lean theorem, `ErdosSzekeres.OrbitConsistency.orient_rotAbout_inv`, also cited in the head (Lemma 4.7). It is stated for the copy `ErdosSzekeres.OrbitConsistency.rotAbout` of the rotation of Definition 1.12, and it calls the centre c instead of o. Part (ii) is a written step. Its case $o=0$, $\varphi=0$, the sign change under $(p_0,p_1)\mapsto(p_0,-p_1)$, is the Lean theorem `ErdosSzekeres.Suk.orient_reflectY` (`ErdosSzekeres/Suk/Reflection.lean:47`).

Lemma 1.19 (Collinearity through a point; Mathlib `collinear_iff_of_mem`; written step)**.** *Let $X \subseteq \mathbb{R}^2$ and $p^* \in X$. Then X is collinear if and only if there is $v \in \mathbb{R}^2$ such that every $p \in X$ can be written $p = p^* + r\,v$ with $r \in \mathbb{R}$.*

Proof. Let W be the linear span of $\{p-q : p, q \in X\}$. Since $p^* \in X$, W is also the span of $\{p-p^* : p \in X\}$: these vectors are among the differences, and conversely $p-q = (p-p^*)-(q-p^*)$. If X is collinear, then $\dim W \le 1$, so $W \subseteq \mathbb{R}v$ for some v (take $v = 0$ if $W = 0$, and a basis vector of W if $\dim W = 1$); then $p - p^* \in W$ gives $p = p^* + rv$ for some r. Conversely, if every $p - p^*$ lies in $\mathbb{R}v$, then $W \subseteq \mathbb{R}v$ and $\dim W \le \dim \mathbb{R}v \le 1$. □

Lemma 1.20 (Orientation and collinearity; (a) Lean: `ErdosSzekeres.orient_eq_zero_iff_collinear`, `ErdosSzekeres/OrientationGeometry.lean:36`; (b) Lean: `ErdosSzekeres.generalPosition_iff_orient_ne_zero`, `ErdosSzekeres/OrientationGeometry.lean:66`)**.** *The orientation detects collinearity, and hence general position.*

(a) For all $a, b, c \in \mathbb{R}^2$: $\operatorname{orient}(a,b,c) = 0$ *if and only if the set $\{a, b, c\}$ is collinear. This includes degenerate triples, in which two or three of the points coincide.*

(b) A finite set S of points is in general position if and only if $\operatorname{orient}(a,b,c) \neq 0$ *for all $a, b, c \in S$ with $a \neq b$, $a \neq c$ and $b \neq c$.*

Proof. We follow the Lean proof. (a) We use Lemma 1.19 with $X = \{a, b, c\}$ and $p^* = a$.

Suppose $\{a, b, c\}$ is collinear. Then there are v and reals r, s with $b = a + rv$ and $c = a + sv$, so $\operatorname{orient}(a,b,c) = (rv_0)(sv_1) - (rv_1)(sv_0) = 0$.

Suppose $\operatorname{orient}(a,b,c) = 0$. If $b = a$, put $v = c - a$; then $a = a + 0 \cdot v$, $b = a + 0 \cdot v$ and $c = a + 1 \cdot v$, so $\{a, b, c\}$ is collinear. If $b \neq a$, put $u = b - a \neq 0$ and $w = c - a$; then $u_0 w_1 - u_1 w_0 = \operatorname{orient}(a,b,c) = 0$. We find r with $w = ru$. If $u_0 \neq 0$, let $r = w_0/u_0$; then $ru_0 = w_0$ and $ru_1 = w_0 u_1/u_0 = u_0 w_1/u_0 = w_1$. If $u_0 = 0$, then $u_1 \neq 0$ and $0 = u_0 w_1 - u_1 w_0 = -u_1 w_0$ gives $w_0 = 0$; let $r = w_1/u_1$, so that $ru_0 = 0 = w_0$ and $ru_1 = w_1$. Now $a = a + 0 \cdot u$, $b = a + 1 \cdot u$ and $c = a + r\,u$, so $\{a, b, c\}$ is collinear.

(b) By Definition 1.5, S is in general position iff $\{a, b, c\}$ is not collinear for all pairwise distinct $a, b, c \in S$; by (a) this says $\operatorname{orient}(a,b,c) \neq 0$ for all such triples. □

Lemma 1.21 (Convex hulls; written step)**.** *Let $Y, Y' \subseteq \mathbb{R}^2$.*

(a) $Y \subseteq \operatorname{conv} Y$; $\operatorname{conv} Y$ is convex; and $\operatorname{conv} Y \subseteq K$ for every convex set K with $Y \subseteq K$. If $Y \subseteq Y'$ then $\operatorname{conv} Y \subseteq \operatorname{conv} Y'$. Moreover $\operatorname{conv} \emptyset = \emptyset$ and $\operatorname{conv}\{y\} = \{y\}$.

(b) $\operatorname{conv} Y$ is the set of all convex combinations of finitely many points of Y.

(c) If $F = \{y_1, \dots, y_m\}$ is a finite set of $m \ge 1$ distinct points, then $\operatorname{conv} F = \big\{\sum_{i=1}^m \lambda_i y_i : \lambda_i \ge 0,\ \sum_i \lambda_i = 1\big\}$. In particular $\operatorname{conv}\{y, z\} = \{\lambda y + (1-\lambda) z : 0 \le \lambda \le 1\}$.

(d) For $w \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$, the open half-plane $\{z \in \mathbb{R}^2 : w_0 z_0 + w_1 z_1 < \alpha\}$ is convex.

Proof. (a) An intersection of convex sets is convex, so $\operatorname{conv} Y$ is convex; it contains Y; and it is contained in each convex $K \supseteq Y$, being the intersection of all of them. If $Y \subseteq Y'$, every convex set containing Y' contains Y, so $\operatorname{conv} Y \subseteq \operatorname{conv} Y'$. The empty set and $\{y\}$ are convex (for $\{y\}$: $\lambda y + \mu y = y$ when $\lambda + \mu = 1$), which gives the last two claims.

(b) Let C be the set of convex combinations of finitely many points of Y. Every $y \in Y$ is the convex combination $1 \cdot y$, so $Y \subseteq C$. C is convex: if $x = \sum_i \lambda_i y_i$ and $x' = \sum_j \lambda'_j y'_j$ are in C and $\lambda, \mu \ge 0$ with $\lambda + \mu = 1$, then $\lambda x + \mu x' = \sum_i (\lambda\lambda_i) y_i + \sum_j (\mu\lambda'_j) y'_j$ has nonnegative coefficients with sum $\lambda + \mu = 1$. Hence $\operatorname{conv} Y \subseteq C$ by (a). Conversely, let $K \supseteq Y$ be convex; we show by induction on k that every convex combination $x = \sum_{i=1}^k \lambda_i y_i$ of points of Y lies in K. For $k = 1$, $x = y_1 \in K$. For $k+1$ terms: if $\lambda_{k+1} = 1$, all other coefficients vanish and $x = y_{k+1} \in K$; otherwise $s = \sum_{i \le k} \lambda_i = 1 - \lambda_{k+1} > 0$, the point $x' = \sum_{i \le k} (\lambda_i / s) y_i$ is a convex combination of k points of Y, so $x' \in K$, and $x = s\,x' + \lambda_{k+1} y_{k+1}$ with $s + \lambda_{k+1} = 1$ lies in K by convexity. Taking $K = \operatorname{conv} Y$ gives $C \subseteq \operatorname{conv} Y$.

(c) By (b), $\operatorname{conv} F$ is the set of convex combinations of points of F. In such a combination we may merge the terms that use the same point of F (adding their coefficients) and add the missing points of F with coefficient 0; this gives the stated form, and conversely every point of the stated form is a convex combination. For $F = \{y, z\}$ with $y \neq z$ this is the segment; for $y = z$ both sides are $\{y\}$.

(d) If $w_0 z_0 + w_1 z_1 < \alpha$ and $w_0 z'_0 + w_1 z'_1 < \alpha$, and $\lambda, \mu \ge 0$ with $\lambda + \mu = 1$, then $w_0(\lambda z_0 + \mu z'_0) + w_1(\lambda z_1 + \mu z'_1) < \lambda\alpha + \mu\alpha = \alpha$, because at least one of λ, μ is positive and the other term is at most the corresponding multiple of α. □

Lemma 1.22 (Convex position via hulls; Lean: `ErdosSzekeres.convexPosition_iff`, `ErdosSzekeres/Basic.lean:15`)**.** *A finite set S of points is in convex position if and only if $p \notin \operatorname{conv}(S \setminus \{p\})$ for every $p \in S$.*

Proof. In Lean this is Mathlib's `convexIndependent_set_iff_notMem_convexHull_diff` applied to S; we give a direct proof from the form of Definition 1.6.

Suppose S is in convex position and $p \in S$. If $p \in \operatorname{conv}(S \setminus \{p\})$, then the definition with $T = S \setminus \{p\}$ and $x = p$ gives $p \in S \setminus \{p\}$, which is absurd.

Conversely, suppose $p \notin \operatorname{conv}(S \setminus \{p\})$ for every $p \in S$. Let $T \subseteq S$ and $x \in S$ with $x \in \operatorname{conv} T$, and suppose $x \notin T$. Then $T \subseteq S \setminus \{x\}$, so $x \in \operatorname{conv} T \subseteq \operatorname{conv}(S \setminus \{x\})$ by Lemma 1.21(a), a contradiction. Hence $x \in T$, and S is in convex position. □

Lemma 1.23 (Monotonicity; (a) Lean: `ErdosSzekeres.ConvexPosition.mono`, `ErdosSzekeres/Basic.lean:11`; (b) Lean: `ErdosSzekeres.GeneralPosition.mono`, `ErdosSzekeres/Basic.lean:6`; (c) Lean: `ErdosSzekeres.HasConvexNGon.mono`, `ErdosSzekeres/Basic.lean:20`; (d) Lean: `MirrorOdd.hasConvexNGon_of_le`, `MIRRORETV/lean/MirrorMain.lean:2182`)**.** *Let s, t, S be finite sets of points and $n, n' \in \mathbb{N}$.*

(a) If s is in convex position and $t \subseteq s$, then t is in convex position.
(b) If s is in general position and $t \subseteq s$, then t is in general position.
(c) If HasConvexNGon $n\, s$ *and $s \subseteq t$, then* HasConvexNGon $n\, t$.
(d) If HasConvexNGon $n\, S$ *and $n' \le n$, then* HasConvexNGon $n'\, S$.

Proof. (a) Let $p \in t$. Then $t \setminus \{p\} \subseteq s \setminus \{p\}$, so $\operatorname{conv}(t \setminus \{p\}) \subseteq \operatorname{conv}(s \setminus \{p\})$ (Lemma 1.21(a)), and the latter does not contain p by Lemma 1.22, since $p \in s$. By Lemma 1.22 again, t is in convex position. (Lean uses Mathlib's `ConvexIndependent.mono`.)

(b) Pairwise distinct $a, b, c \in t$ are pairwise distinct points of s, so $\{a, b, c\}$ is not collinear. This is the Lean proof.

(c) A witness $T \subseteq s$ with $|T| = n$ in convex position satisfies $T \subseteq t$. This is the Lean proof.

(d) Let $T \subseteq S$ with $|T| = n$ be in convex position. Since $n' \le n = |T|$, there is $T' \subseteq T$ with $|T'| = n'$; it is in convex position by (a), and $T' \subseteq S$. This is the Lean proof (it applies Lemma 1.24(b) to T and then part (c) above). The same declaration, with the same name and proof, also appears at `SYMLEAN/mirror/ReductionThreeClasses.lean:3741`. □

Lemma 1.24 (Sizes of convex subsets; (a) Lean: `ErdosSzekeres.HasConvexNGon.card_le`, `ErdosSzekeres/Basic.lean:25`; (b) Lean: `ErdosSzekeres.ConvexPosition.hasConvexNGon`, `ErdosSzekeres/Basic.lean:31`)**.** *Let s be a finite set of points and $n \in \mathbb{N}$.*

(a) If HasConvexNGon $n\, s$, *then $n \le |s|$.*
(b) If s is in convex position and $n \le |s|$, then HasConvexNGon $n\, s$.

Proof. (a) A witness $T \subseteq s$ has $n = |T| \le |s|$. (b) Choose $T \subseteq s$ with $|T| = n$; it is in convex position by Lemma 1.23(a). Both are the Lean proofs. □

Lemma 1.25 (Forcing; (a) Lean: `ErdosSzekeres.ForcesConvexNGon.mono`, `ErdosSzekeres/Basic.lean:47`; (b) Lean: `ErdosSzekeres.forcesConvexNGon_iff_exact`, `ErdosSzekeres/Basic.lean:37`)**.** *Let $N, M, n \in \mathbb{N}$.*

(a) If ForcesConvexNGon $N\, n$ *and $N \le M$, then* ForcesConvexNGon $M\, n$.
(b) ForcesConvexNGon $N\, n$ *holds if and only if every finite set S of points in general position with $|S| = N$ satisfies* HasConvexNGon $n\, S$.

Proof. (a) If S is in general position and $M \le |S|$, then $N \le |S|$. (b) The forward direction is the case $|S| = N$. Conversely, let S be in general position with $N \le |S|$. Choose $T \subseteq S$ with $|T| = N$. By Lemma 1.23(b) T is in general position, so HasConvexNGon $n\, T$ by hypothesis, and HasConvexNGon $n\, S$ by Lemma 1.23(c). Both are the Lean proofs. □

Lemma 1.26 (At most three points; written step, the last assertion Lean: `ErdosSzekeres.convexPosition_of_card_three`, `ErdosSzekeres/Basic.lean:58`)**.** *Sets of at most three points behave as follows.*

(a) For every finite set S of points, HasConvexNGon $0\, S$ *holds.*
(b) Every finite set of at most two points is in convex position.
(c) Let a, b, c be pairwise distinct points. Then $\{a, b, c\}$ is in convex position if and only if $\{a, b, c\}$ is not collinear. In particular, if a finite set s is in general position and $|s| = 3$, then s is in convex position.

Proof. (a) $T = \emptyset$ is a subset of S with 0 elements, and it is in convex position by Lemma 1.22 (there is nothing to check).

(b) We use Lemma 1.22 and Lemma 1.21(a). For $\emptyset$ there is nothing to check; for $\{x\}$, $x \notin \operatorname{conv}\emptyset = \emptyset$; for $\{x, y\}$ with $x \neq y$, $x \notin \operatorname{conv}\{y\} = \{y\}$ and $y \notin \operatorname{conv}\{x\} = \{x\}$.

(c) Since a, b, c are pairwise distinct, $\{a, b, c\} \setminus \{a\} = \{b, c\}$, and similarly for b and c.

Suppose $\{a, b, c\}$ is not collinear. If $a \in \operatorname{conv}\{b, c\}$, then by Lemma 1.21(c) $a = \lambda b + (1 - \lambda)c$ for some $\lambda \in [0, 1]$; with $v = b - c$ we get $a = c + \lambda v$, $b = c + 1 \cdot v$ and $c = c + 0 \cdot v$, so $\{a, b, c\}$ is collinear by Lemma 1.19 (with $p^* = c$), a contradiction. The same argument with the roles of a, b, c exchanged shows $b \notin \operatorname{conv}\{a, c\}$ and $c \notin \operatorname{conv}\{a, b\}$. By Lemma 1.22, $\{a, b, c\}$ is in convex position.

Suppose $\{a, b, c\}$ is collinear. By Lemma 1.19 with $p^* = a$ there are v and reals r_b, r_c with $b = a + r_b v$ and $c = a + r_c v$; put $\tau(a) = 0$, $\tau(b) = r_b$, $\tau(c) = r_c$, so that every $x \in \{a, b, c\}$ equals $a + \tau(x)v$. Since the points are pairwise distinct, the three reals $0, r_b, r_c$ are pairwise distinct (equal parameters would give equal points). Let x be the point whose parameter $\tau(x)$ is the middle one of the three, and let y, z be the other two points, with $\tau(y) < \tau(x) < \tau(z)$. With $\lambda = (\tau(z) - \tau(x))/(\tau(z) - \tau(y)) \in (0, 1)$ we have $\tau(x) = \lambda\tau(y) + (1 - \lambda)\tau(z)$, hence $x = \lambda y + (1 - \lambda)z \in \operatorname{conv}\{y, z\} = \operatorname{conv}(\{a, b, c\} \setminus \{x\})$, and $\{a, b, c\}$ is not in convex position by Lemma 1.22.

Finally, if s is in general position and $|s| = 3$, then $s = \{a, b, c\}$ with a, b, c pairwise distinct and $\{a, b, c\}$ not collinear, so s is in convex position. (The Lean proof of this last assertion derives affine independence of a, b, c from non-collinearity and uses that affinely independent families are convex independent; we argued directly.) □

Lemma 1.27 (Injective affine maps; (a) Mathlib `AffineMap.image_convexHull`; (b′) Lean: `ErdosSzekeres.SymReduction.convexPosition_image_affine`, `SYMLEAN/AffineReduction.lean:1263`; (c′) Lean: `ErdosSzekeres.SymReduction.generalPosition_image_affine`, `SYMLEAN/AffineReduction.lean:1278`, Lean: `ErdosSzekeres.SymReduction.generalPosition_image_affineEquiv`, `SYMLEAN/AffineReduction.lean:1290`; (d′) Lean: `ErdosSzekeres.SymReduction.hasConvexNGon_image_affine`, `SYMLEAN/AffineReduction.lean:1298`; (e′) Lean: `ErdosSzekeres.SymReduction.injective_of_affine`, `SYMLEAN/AffineReduction.lean:1305`; the rest written step)**.** *Let $g\colon \mathbb{R}^2 \to \mathbb{R}^2$ be an affine map, $g(x) = Ax + v$.*

(a) For every $Y \subseteq \mathbb{R}^2$, $g(\operatorname{conv} Y) = \operatorname{conv}\big(g(Y)\big)$.

Suppose moreover that g is injective. Then $\det A \neq 0$, and g is a bijection of $\mathbb{R}^2$ whose inverse $y \mapsto A^{-1}(y - v)$ is affine. For every finite set S of points and every $n \in \mathbb{N}$ (with $g(S)$ the image set, $|g(S)| = |S|$):

(b) S is in convex position if and only if $g(S)$ is in convex position;

(c) S is in general position if and only if $g(S)$ is in general position;

(d) $\operatorname{HasConvexNGon}\, n\, S$ holds if and only if $\operatorname{HasConvexNGon}\, n\, g(S)$ holds.

The forward directions of (b)–(d), and the converse of the implication "g injective $\Rightarrow \det A \neq 0$", are Lean theorems in the following forms. Here S is a finite set of points, $n \in \mathbb{N}$, f is an affine map of $\mathbb{R}^2$ in the sense of Mathlib (`AffineMap`; Definition 1.4), and $h\colon \mathbb{R}^2 \to \mathbb{R}^2$ is a map with $h(p) = Lp + w$ for every $p \in \mathbb{R}^2$, where L is a real 2×2 matrix and $w \in \mathbb{R}^2$ (Lean's `b`).

(b′) If S is in convex position and f is injective, then $f(S)$ is in convex position.

(c′) If $\det L \neq 0$ and S is in general position, then $h(S)$ is in general position. If S is in general position and ϕ is an affine equivalence of $\mathbb{R}^2$ in the sense of Mathlib (`AffineEquiv`; Definition 1.4), then $\phi(S)$ is in general position.

(d′) If $\operatorname{HasConvexNGon}\, n\, S$ and f is injective, then $\operatorname{HasConvexNGon}\, n\, f(S)$.

(e′) If $\det L \neq 0$, then h is injective.

Proof. For $x, x' \in \mathbb{R}^2$ and reals λ, μ with $\lambda + \mu = 1$,

$$(*) \qquad g(\lambda x + \mu x') = \lambda Ax + \mu Ax' + (\lambda + \mu)v = \lambda g(x) + \mu g(x').$$

(a) By $(*)$, $g(\operatorname{conv} Y)$ is convex: any two of its points have the form $g(x), g(x')$ with $x, x' \in \operatorname{conv} Y$, and for $\lambda, \mu \geq 0$ with $\lambda + \mu = 1$, $\lambda g(x) + \mu g(x') = g(\lambda x + \mu x')$ with $\lambda x + \mu x' \in \operatorname{conv} Y$. It contains $g(Y)$, so $\operatorname{conv}(g(Y)) \subseteq g(\operatorname{conv} Y)$ by Lemma 1.21(a). Conversely, the set $K = \{x : g(x) \in \operatorname{conv}(g(Y))\}$ is convex by $(*)$ and contains Y, so $\operatorname{conv} Y \subseteq K$, that is, $g(\operatorname{conv} Y) \subseteq \operatorname{conv}(g(Y))$. (This is the proof of Mathlib's `AffineMap.image_convexHull`.)

Now let g be injective, and write $A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$. If $\det A = \alpha\delta - \beta\gamma = 0$, there is $x \neq 0$ with $Ax = 0$: if $(\alpha, \beta) \neq (0, 0)$ take $x = (\beta, -\alpha)$, for which $Ax = (\alpha\beta - \beta\alpha,\ \gamma\beta - \delta\alpha) = (0, 0)$; if $\alpha = \beta = 0$ take $x = (\delta, -\gamma)$ when $(\gamma, \delta) \neq (0, 0)$, for which $Ax = (0, \gamma\delta - \delta\gamma) = (0, 0)$, and any $x \neq 0$ when $A = 0$. Then $g(x) = v = g(0)$

with $x \neq 0$, contradicting injectivity. So $\det A \neq 0$, A has the inverse $A^{-1} = (\det A)^{-1} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}$, and g is bijective with inverse $y \mapsto A^{-1}(y - v)$, which is affine. Injectivity gives $|g(S)| = |S|$.

(b) The points of $g(S)$ are the $g(p)$ with $p \in S$, and $g(S) \setminus \{g(p)\} = g(S \setminus \{p\})$ by injectivity. By (a), $\operatorname{conv}\big(g(S) \setminus \{g(p)\}\big) = g\big(\operatorname{conv}(S \setminus \{p\})\big)$, and by injectivity $g(p)$ belongs to this set if and only if $p \in \operatorname{conv}(S \setminus \{p\})$. Lemma 1.22 applied to S and to $g(S)$ gives (b).

(c) Every triple of points of $g(S)$ has the form $(g(a), g(b), g(c))$ with $a, b, c \in S$; by injectivity $g(a), g(b), g(c)$ are pairwise distinct if and only if a, b, c are; and $\operatorname{orient}(g(a), g(b), g(c)) = \det(A) \operatorname{orient}(a, b, c)$ (Lemma 1.17) with $\det A \neq 0$, so one orientation vanishes if and only if the other does. Lemma 1.20(b) gives (c).

(d) If $T \subseteq S$, $|T| = n$, is in convex position, then $g(T) \subseteq g(S)$, $|g(T)| = n$, and $g(T)$ is in convex position by (b). Conversely, let $T' \subseteq g(S)$ with $|T'| = n$ be in convex position, and let $T = \{p \in S : g(p) \in T'\}$. Since every point of T' is $g(p)$ for some $p \in S$, we have $g(T) = T'$; hence $|T| = |T'| = n$ by injectivity, T is in convex position by (b), and $T \subseteq S$.

(b′)–(e′). By Definition 1.4, a Mathlib affine map f of $\mathbb{R}^2$ has the form $f(x) = A_f x + f(0)$, where A_f is the matrix of its linear part (in Lean: `ErdosSzekeres.SymReduction.affineMap_eq_mulVec`, `SYMLEAN/AffineReduction.lean:1284`). So (b′) and (d′) are the forward directions of (b) and (d) for $g = f$. (e′): if $h(p) = h(q)$, then $L(p - q) = (Lp + w) - (Lq + w) = h(p) - h(q) = 0$. Write $L = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$; the matrix $L^\sharp = \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}$ satisfies $L^\sharp L = (\det L)\, I$ (multiply out), so $(\det L)(p - q) = L^\sharp L(p - q) = 0$, and $p = q$ because $\det L \neq 0$. (c′): if $\det L \neq 0$, then h is an affine map with linear part L, and it is injective by (e′), so the forward direction of (c) applies to $g = h$. A Mathlib affine equivalence ϕ of $\mathbb{R}^2$ is an injective affine map (Definition 1.4), so the forward direction of (c) applies to $g = \phi$ as well. (The Lean proofs follow the same routes: `SymReduction.convexPosition_image_affine` uses `AffineMap.image_convexHull` and injectivity as in (b); `SymReduction.generalPosition_image_affine` uses `SymReduction.orient_affine`, that is, Lemma 1.17; `SymReduction.generalPosition_image_affineEquiv` derives $\det \neq 0$ for the matrix of the linear part of ϕ from the invertibility of that linear part.) □

Remark 1.28 (Special cases in the main library). Besides the general forms (b′)–(e′), the main library proves special cases of Lemma 1.27 in `ErdosSzekeres/Suk/Reflection.lean`. `ErdosSzekeres.HasConvexNGon.linearEquiv` (`ErdosSzekeres/Suk/Reflection.lean:6`) transports $\operatorname{HasConvexNGon}\, n\, S$ to $\operatorname{HasConvexNGon}\, n\, e(S)$ for a linear bijection e of the plane (Mathlib `LinearEquiv`): the forward direction of (d) for $g = e$. The reflection $\sigma(p_0, p_1) = (p_0, -p_1)$ is the linear bijection `ErdosSzekeres.Suk.reflectY` (`ErdosSzekeres/Suk/Reflection.lean:34`); the simplification lemmas `Suk.reflectY_x` and `Suk.reflectY_y` (`ErdosSzekeres/Suk/Reflection.lean:42` and `:43`) give its two coordinates, and `Suk.reflectY_reflectY` (`ErdosSzekeres/Suk/Reflection.lean:44`) says that $\sigma(\sigma(p)) = p$. `ErdosSzekeres.Suk.generalPosition_reflectY` (`ErdosSzekeres/Suk/Reflection.lean:52`) transports general position from S to $\sigma(S)$ (the forward direction of (c) for $g = \sigma$), and `ErdosSzekeres.Suk.hasConvexNGon_of_reflectY` (`ErdosSzekeres/Suk/Reflection.lean:64`) says that $\operatorname{HasConvexNGon}\, n\, \sigma(S)$ implies $\operatorname{HasConvexNGon}\, n\, S$ (the reverse direction of (d) for $g = \sigma$).

Lemma 1.29 (Carathéodory's theorem in the plane; written step)**.** *Let $Y \subseteq \mathbb{R}^2$ and $x \in \operatorname{conv} Y$. Then there is a finite, affinely independent set $Y' \subseteq Y$ with $|Y'| \leq 3$ and $x \in \operatorname{conv} Y'$.*

Proof. By Lemma 1.21(b), x is a convex combination of finitely many points of Y; if T_0 is the finite set of these points, then $x \in \operatorname{conv} T_0$ by Lemma 1.21(c). Among all finite sets $T \subseteq Y$ with $x \in \operatorname{conv} T$ choose one, $T = \{y_1, \dots, y_k\}$ with distinct y_i, of least cardinality k. Then $k \geq 1$ because $\operatorname{conv} \emptyset = \emptyset$. By Lemma 1.21(c), $x = \sum_i \lambda_i y_i$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. Every λ_j is positive: if $k = 1$ then $\lambda_1 = 1$, and if $k \geq 2$ and $\lambda_j = 0$, then x is a convex combination of the points of $T \setminus \{y_j\}$, so $x \in \operatorname{conv}(T \setminus \{y_j\})$, contradicting minimality.

T is affinely independent. Otherwise there are reals $\mu_1, \dots, \mu_k$, not all zero, with $\sum_i \mu_i = 0$ and $\sum_i \mu_i y_i = 0$. Some μ_i is positive (they sum to 0 and are not all zero). Let $t = \min\{\lambda_i/\mu_i : \mu_i > 0\} > 0$, attained at $i = j$, and put $\lambda'_i = \lambda_i - t\mu_i$. Then $\lambda'_i \geq 0$ for all i (if $\mu_i \leq 0$ then $\lambda'_i \geq \lambda_i$; if $\mu_i > 0$ then $t \leq \lambda_i/\mu_i$), $\lambda'_j = 0$, $\sum_i \lambda'_i = 1 - t \cdot 0 = 1$ and $\sum_i \lambda'_i y_i = x - t \cdot 0 = x$. So x is a convex combination of the points of $T \setminus \{y_j\}$, contradicting minimality again (note $k \geq 2$ here, since $\sum \lambda'_i = 1$ forces some $\lambda'_i \neq 0$, $i \neq j$).

Finally $k \leq 3$: if $k \geq 4$, the $k - 1 \geq 3$ vectors $y_2 - y_1, \dots, y_k - y_1$ in the two-dimensional space $\mathbb{R}^2$ are linearly dependent, $\sum_{i \geq 2} \nu_i (y_i - y_1) = 0$ with $\nu \neq 0$; then $\mu_1 = -\sum_{i \geq 2} \nu_i$ and $\mu_i = \nu_i$ $(i \geq 2)$ satisfy $\sum_i \mu_i = 0$ and $\sum_i \mu_i y_i = \sum_{i \geq 2} \nu_i (y_i - y_1) = 0$, with $\mu \neq 0$, contradicting affine independence. Take $Y' = T$.

The Lean code uses Mathlib's version of this theorem (module `Mathlib.Analysis.Convex.Caratheodory`, with `Caratheodory.minCardFinsetOfMemConvexHull`), whose proof is the same minimal-cardinality argument; the bound $|Y'| \leq 3$ is derived in Lean from affine independence as $|Y'| \leq \dim + 1 = 3$. □

Lemma 1.30 (Four-point criteria; (a) Lean: `ErdosSzekeres.Suk.convexPosition_of_subsets_card_le_four`, `ErdosSzekeres/Suk/FourPoint.lean:10`; (b) Lean: `ErdosSzekeres.Suk.convexPosition_of_four_subsets`, `ErdosSzekeres/Suk/FourPoint.lean:40`). *Let s be a finite set of points.*

(a) If every $t \subseteq s$ with $|t| \le 4$ is in convex position, then s is in convex position.

(b) If $4 \le |s|$ and every $t \subseteq s$ with $|t| = 4$ is in convex position, then s is in convex position.

No general-position assumption is made.

Proof. We follow the Lean proofs. (a) By Lemma 1.22 we must show $p \notin \operatorname{conv}(s \setminus \{p\})$ for every $p \in s$. Suppose $p \in \operatorname{conv}(s \setminus \{p\})$. By Lemma 1.29 there is a finite $Y' \subseteq s \setminus \{p\}$ with $|Y'| \le 3$ and $p \in \operatorname{conv} Y'$. Let $u = Y' \cup \{p\} \subseteq s$. Since $p \notin Y'$, $|u| = |Y'| + 1 \le 4$ and $u \setminus \{p\} = Y'$. By hypothesis u is in convex position, so $p \notin \operatorname{conv}(u \setminus \{p\}) = \operatorname{conv} Y'$ by Lemma 1.22, a contradiction.

(b) Let $t \subseteq s$ with $|t| \le 4$. Since $|s| \ge 4$, we can add $4 - |t|$ points of $s \setminus t$ to t (there are $|s| - |t| \ge 4 - |t|$ of them) and obtain u with $t \subseteq u \subseteq s$ and $|u| = 4$. By hypothesis u is in convex position, hence so is t (Lemma 1.23(a)). Now (a) applies. □

Lemma 1.31 (Half-planes of the orientation; (a) Lean: `ErdosSzekeres.Suk.orient_affine_combination`, `ErdosSzekeres/Suk/SupportingLines.lean:6`; (b) Lean: `ErdosSzekeres.Suk.convex_orient_positive`, `ErdosSzekeres/Suk/SupportingLines.lean:11`; (c) Lean: `ErdosSzekeres.Suk.convex_orient_nonnegative`, `ErdosSzekeres/Suk/SupportingLines.lean:21`). *The orientation is affine in its third argument, and its half-planes are convex.*

(a) For $p, q, a, b \in \mathbb{R}^2$ and $u, v \in \mathbb{R}$ with $u + v = 1$: $\operatorname{orient}(p, q, u\,a + v\,b) = u \operatorname{orient}(p, q, a) + v \operatorname{orient}(p, q, b)$.

(b) For $p, q \in \mathbb{R}^2$ the set $\{r \in \mathbb{R}^2 : 0 < \operatorname{orient}(p, q, r)\}$ is convex.

(c) For $p, q \in \mathbb{R}^2$ the set $\{r \in \mathbb{R}^2 : 0 \le \operatorname{orient}(p, q, r)\}$ is convex.

Proof. (a) $\operatorname{orient}(p, q, r) = (q_0 - p_0)(r_1 - p_1) - (q_1 - p_1)(r_0 - p_0)$ is a linear function of $r - p$. For $r = ua + vb$ with $u + v = 1$ we have $r - p = u(a - p) + v(b - p)$, and the claim follows. (Lean closes the same polynomial identity with the certificate $(p_1 q_0 - p_0 q_1)(u + v - 1)$.)

(b) Let $\operatorname{orient}(p, q, a) > 0$, $\operatorname{orient}(p, q, b) > 0$ and $u, v \ge 0$ with $u + v = 1$. By (a) the value at $ua + vb$ is $u \operatorname{orient}(p, q, a) + v \operatorname{orient}(p, q, b)$. If $u = 0$ then $v = 1$ and this is $\operatorname{orient}(p, q, b) > 0$; otherwise $u > 0$, the first term is positive and the second is nonnegative.

(c) By (a) the value at $ua + vb$ is a sum of two nonnegative terms. □

Lemma 1.32 (Strict supporting lines; Lean: `ErdosSzekeres.Suk.convexPosition_of_strict_supports`, `ErdosSzekeres/Suk/SupportingLines.lean:28`). *Let s be a finite set of points such that for every $p \in s$ there is $q \in \mathbb{R}^2$ with $0 < \operatorname{orient}(p, q, r)$ for every $r \in s$ with $r \ne p$. Then s is in convex position.*

Proof. We follow the Lean proof. Let $p \in s$ and choose q as in the hypothesis. The set $H = \{r : 0 < \operatorname{orient}(p, q, r)\}$ is convex (Lemma 1.31(b)) and contains $s \setminus \{p\}$, so $\operatorname{conv}(s \setminus \{p\}) \subseteq H$ by Lemma 1.21(a). But $\operatorname{orient}(p, q, p) = 0$ (Lemma 1.15(e)), so $p \notin H$ and hence $p \notin \operatorname{conv}(s \setminus \{p\})$. By Lemma 1.22, s is in convex position. □

Lemma 1.33 (A support identity; Lean: `ErdosSzekeres.Suk.orient_support_sum`, `ErdosSzekeres/Suk/SupportingLines.lean:40`). *For all $a, p, b, r \in \mathbb{R}^2$: $\operatorname{orient}(p,\ p + b - a,\ r) = \operatorname{orient}(a, p, r) + \operatorname{orient}(p, b, r)$.*

Proof. For $w \in \mathbb{R}^2$, $\operatorname{orient}(p, p + w, r) = \det(w \mid r - p)$ is linear in w. With $w = b - a = (p - a) + (b - p)$ we get $\operatorname{orient}(p, p + b - a, r) = \det(p - a \mid r - p) + \det(b - p \mid r - p)$. The second term is $\operatorname{orient}(p, b, r)$. For the first, $\operatorname{orient}(a, p, r) = \det(p - a \mid r - a) = \det(p - a \mid (r - p) + (p - a)) = \det(p - a \mid r - p)$, since a determinant with two equal columns vanishes. (The Lean proof expands both sides.) □

Lemma 1.34 (Four counterclockwise points; Lean: `ErdosSzekeres.Suk.convexPosition_four_of_positive`, `ErdosSzekeres/Suk/SupportingLines.lean:47`). *Let $a, b, c, d \in \mathbb{R}^2$ with*

$$0 < \operatorname{orient}(a, b, c), \qquad 0 < \operatorname{orient}(a, b, d), \qquad 0 < \operatorname{orient}(a, c, d), \qquad 0 < \operatorname{orient}(b, c, d).$$

Then the finite set $\{a, b, c, d\}$ is in convex position.

Proof. We follow the Lean proof, which exhibits a strict supporting line at each of the four points and applies Lemma 1.32. (The hypotheses imply, by Lemma 1.15(e), that the four points are pairwise distinct, but the argument does not need this.) Write $o_{abc} = \operatorname{orient}(a, b, c)$, $o_{abd} = \operatorname{orient}(a, b, d)$, $o_{acd} = \operatorname{orient}(a, c, d)$ and $o_{bcd} = \operatorname{orient}(b, c, d)$, all positive. By Lemma 1.15(b) every cyclic rotation of the arguments leaves orient unchanged, and by Lemma 1.15(e) orient vanishes when two arguments are equal. For $p \in \{a, b, c, d\}$ and

$r \in \{a,b,c,d\}$ with $r \neq p$, the point r is one of the three letters other than p (as a point, r differs from p), so it suffices to check the three letters other than p. Each line below applies Lemma 1.33.

- $p = a$, $q = a + b - d$: $\operatorname{orient}(a,q,r) = \operatorname{orient}(d,a,r) + \operatorname{orient}(a,b,r)$. For $r = b$: $\operatorname{orient}(d,a,b) + 0 = o_{abd}$. For $r = c$: $\operatorname{orient}(d,a,c) + o_{abc} = o_{acd} + o_{abc}$. For $r = d$: $0 + o_{abd}$.
- $p = b$, $q = b + c - a$: $\operatorname{orient}(b,q,r) = \operatorname{orient}(a,b,r) + \operatorname{orient}(b,c,r)$. For $r = a$: $0 + \operatorname{orient}(b,c,a) = o_{abc}$. For $r = c$: $o_{abc} + 0$. For $r = d$: $o_{abd} + o_{bcd}$.
- $p = c$, $q = c + d - b$: $\operatorname{orient}(c,q,r) = \operatorname{orient}(b,c,r) + \operatorname{orient}(c,d,r)$. For $r = a$: $\operatorname{orient}(b,c,a) + \operatorname{orient}(c,d,a) = o_{abc} + o_{acd}$. For $r = b$: $0 + \operatorname{orient}(c,d,b) = o_{bcd}$. For $r = d$: $o_{bcd} + 0$.
- $p = d$, $q = d + a - c$: $\operatorname{orient}(d,q,r) = \operatorname{orient}(c,d,r) + \operatorname{orient}(d,a,r)$. For $r = a$: $\operatorname{orient}(c,d,a) + 0 = o_{acd}$. For $r = b$: $\operatorname{orient}(c,d,b) + \operatorname{orient}(d,a,b) = o_{bcd} + o_{abd}$. For $r = c$: $0 + \operatorname{orient}(d,a,c) = o_{acd}$.

All twelve values are positive, so each of a,b,c,d has a strict supporting line, and Lemma 1.32 shows that $\{a,b,c,d\}$ is in convex position. □

Lemma 1.35 (Positive triples; Lean: `ErdosSzekeres.Suk.convexPosition_of_positive_triples`, `ErdosSzekeres/Suk/OrientationClique.lean:9`)**.** *Let ι be a type with a linear order $<$, let $p\colon \iota \to \mathbb{R}^2$ be injective, and let s be a finite subset of ι with $4 \le |s|$. Suppose that $0 < \operatorname{orient}(p(i),p(j),p(k))$ for all $i,j,k \in s$ with $i < j$ and $j < k$. Then the finite set $p(s) = \{p(i) : i \in s\}$ is in convex position.*

Proof. We follow the Lean proof. Since p is injective, $|p(s)| = |s| \ge 4$, so by Lemma 1.30(b) it suffices to show that every $t \subseteq p(s)$ with $|t| = 4$ is in convex position. Let $u = \{i \in s : p(i) \in t\}$. Since $t \subseteq p(s)$, $p(u) = t$, and $|u| = |t| = 4$ by injectivity. List the elements of the finite subset u of the linearly ordered set ι in increasing order, $e_0 < e_1 < e_2 < e_3$. Then $t = \{p(e_0), p(e_1), p(e_2), p(e_3)\}$, and the hypothesis applied to the increasing triples (e_0,e_1,e_2), (e_0,e_1,e_3), (e_0,e_2,e_3) and (e_1,e_2,e_3) of elements of s gives exactly the four hypotheses of Lemma 1.34 for $(a,b,c,d) = (p(e_0),p(e_1),p(e_2),p(e_3))$. Hence t is in convex position. □

Remark 1.36. (i) Only s is assumed finite; ι may be infinite, and injectivity is assumed for p on all of ι. (ii) The Lean proof discharges the order facts $0 < 1 < 2 < 3$ in $\operatorname{Fin} 4$ and the equality of the set of all elements of $\operatorname{Fin} 4$ with $\{0,1,2,3\}$ by `decide`, which the kernel evaluates without any additional axiom. These are side facts inside the proof of a statement that is not itself a finite computation, and the written proof above needs no machine computation; so the statement is not computer-assisted in the sense of the rigor labels. (iii) The hypothesis $4 \le |s|$ is needed by the Lean proof (through Lemma 1.30(b)). As a variant, not part of the Lean statement: the conclusion also holds when $|s| \le 3$, by Lemma 1.26(b) when $|p(s)| \le 2$, and when $|s| = 3$ because the three points then have nonzero orientation, hence are pairwise distinct and not collinear (Lemmas 1.15(e) and 1.20(a)), so Lemma 1.26(c) applies.

Lemma 1.37 (Negative triples; Lean: `ErdosSzekeres.Suk.convexPosition_of_negative_triples`, `ErdosSzekeres/Suk/OrientationClique.lean:33`)**.** *Let ι be a type with a linear order $<$, let $p\colon \iota \to \mathbb{R}^2$ be injective, and let s be a finite subset of ι with $4 \le |s|$. Suppose that $\operatorname{orient}(p(i),p(j),p(k)) < 0$ for all $i,j,k \in s$ with $i < j$ and $j < k$. Then $p(s)$ is in convex position.*

Proof. We follow the Lean proof, which applies Lemma 1.35 to the reversed order. Let $i \prec j$ mean $j < i$; this is a linear order on ι. If $i,j,k \in s$ with $i \prec j \prec k$, then $k < j < i$, so $\operatorname{orient}(p(k),p(j),p(i)) < 0$ by hypothesis, and by Lemma 1.15(d) (the transposition of the first and third arguments has sign -1) $\operatorname{orient}(p(i),p(j),p(k)) = -\operatorname{orient}(p(k),p(j),p(i)) > 0$. Lemma 1.35 for $(\iota, \prec)$ gives the claim. □

Lemma 1.38 (Positive cycles; Lean: `ErdosSzekeres.hasConvexNGon_of_positive_cycle`, `ErdosSzekeres/ConvexCycleCuts.lean:11`)**.** *Let $N,k \in \mathbb{N}$ with $4 \le k$. Let $p\colon \operatorname{Fin} N \to \mathbb{R}^2$ be injective, let $e\colon \operatorname{Fin} k \to \operatorname{Fin} N$ be an embedding (an injective map; Lean type `Fin k ↪ Fin N`), and let P be a finite set of points with $p(e(i)) \in P$ for every $i \in \operatorname{Fin} k$. Suppose that*

$$0 < \operatorname{orient}\big(p(e(i)),\, p(e(j)),\, p(e(l))\big) \qquad \text{for all } i,j,l \in \operatorname{Fin} k \text{ with } i < j \text{ and } j < l.$$

Then $\operatorname{HasConvexNGon}\, k\, P$.

Proof. We follow the Lean proof. The map $q = p \circ e\colon \operatorname{Fin} k \to \mathbb{R}^2$ is injective, as a composition of injective maps. Let $T = q(\operatorname{Fin} k) = \{q(0), \dots, q(k-1)\}$. Then $T \subseteq P$ by hypothesis and $|T| = k$ by injectivity. Lemma 1.35, applied to $\iota = \operatorname{Fin} k$ with its natural order, the injective map q and $s = \operatorname{Fin} k$ (so $|s| = k \ge 4$), shows that $T = q(s)$ is in convex position; its hypothesis is exactly the assumed positivity. So T witnesses $\operatorname{HasConvexNGon}\, k\, P$. □

Remark 1.39. The hypothesis concerns all increasing triples of the ordering e, not only consecutive ones. The name refers to the clause family `ConvexCycleCut` (`ErdosSzekeres/ConvexCycleCuts.lean:8`), which is used as cuts in the SMT coordinate-repair searches described in `docs/SAT.md:168–233`: `convexCycleCut_of_avoids` (`ErdosSzekeres/ConvexCycleCuts.lean:31`) is the contrapositive of Lemma 1.38: under the same hypotheses on k, p, e and P (without the positivity hypothesis), if P is k-free then some $i < j < l$ in $\operatorname{Fin} k$ has $\operatorname{orient}(p(e(i)), p(e(j)), p(e(l))) \le 0$.

Lemma 1.40 (Separation of a point from a finite set; written step)**.** *Let $F \subseteq \mathbb{R}^2$ be finite and $x \in \mathbb{R}^2$. Then $x \notin \operatorname{conv} F$ if and only if there is a linear functional $\ell\colon \mathbb{R}^2 \to \mathbb{R}$ (that is, $\ell(z) = w_0 z_0 + w_1 z_1$ for some $w \in \mathbb{R}^2$) with $\ell(x) > \ell(y)$ for every $y \in F$.*

Proof. Write $\langle w, z\rangle = w_0 z_0 + w_1 z_1$ and $\|z\|^2 = \langle z, z\rangle$ (the Euclidean norm, written out).

($\Leftarrow$, elementary.) Let $\ell(z) = \langle w, z\rangle$ satisfy $\ell(x) > \ell(y)$ for all $y \in F$. The set $H = \{z : \ell(z) < \ell(x)\}$ is convex by Lemma 1.21(d) and contains F, so $\operatorname{conv} F \subseteq H$ by Lemma 1.21(a); since $x \notin H$, $x \notin \operatorname{conv} F$. (If $F = \emptyset$ this is also clear from $\operatorname{conv}\emptyset = \emptyset$.)

($\Rightarrow$.) Suppose $x \notin \operatorname{conv} F$. If $F = \emptyset$, the functional $\ell = 0$ satisfies the condition vacuously. Otherwise write $F = \{y_1, \dots, y_m\}$ with $m \ge 1$ distinct points and let $\Delta = \{\lambda \in \mathbb{R}^m : \lambda_i \ge 0,\ \sum_i \lambda_i = 1\}$. The set Δ is closed and bounded, hence compact by the Heine–Borel theorem (a closed and bounded subset of $\mathbb{R}^m$ is compact; W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., Theorem 2.41), and the function $h(\lambda) = \|x - \sum_i \lambda_i y_i\|^2$ is a polynomial in λ, hence continuous; so h attains its minimum on the nonempty compact set Δ at some λ^* (a continuous real function on a nonempty compact metric space attains its minimum; Rudin, op. cit., Theorem 4.16). Put $q = \sum_i \lambda_i^* y_i$. By Lemma 1.21(c), $\operatorname{conv} F = \{\sum_i \lambda_i y_i : \lambda \in \Delta\}$, so $q \in \operatorname{conv} F$ and

$$\|x - q\|^2 \le \|x - z\|^2 \qquad \text{for every } z \in \operatorname{conv} F. \tag{$**$}$$

Since $x \notin \operatorname{conv} F$, $x \ne q$; put $w = x - q \ne 0$ and $\ell(z) = \langle w, z\rangle$. Let $y \in F$ and $0 < t \le 1$. The point $z_t = (1-t)q + ty$ lies in $\operatorname{conv} F$ (convexity, Lemma 1.21(a)), and $x - z_t = w - t(y - q)$, so by $(**)$

$$\|w\|^2 \le \|w - t(y-q)\|^2 = \|w\|^2 - 2t\langle w, y - q\rangle + t^2\|y - q\|^2.$$

Hence $2\langle w, y - q\rangle \le t\|y - q\|^2$ for every $t \in (0, 1]$, and letting $t \to 0$ gives $\langle w, y - q\rangle \le 0$, that is, $\ell(y) \le \ell(q)$. Finally $\ell(x) - \ell(q) = \langle w, x - q\rangle = \|w\|^2 > 0$, so $\ell(y) \le \ell(q) < \ell(x)$. □

Coverage.

- *Covered definitions, with the label of the statement.* `ErdosSzekeres.Point` (1.1), `ErdosSzekeres.orient` (1.3), `ErdosSzekeres.GeneralPosition` (1.5), `ErdosSzekeres.ConvexPosition` (1.6), `ErdosSzekeres.HasConvexNGon` (1.7), `ErdosSzekeres.ForcesConvexNGon` (1.9), the three copies of `Invariant` (1.10) and the three cited copies of `rotAbout` (1.12), together with the Mathlib notions `Convex`, `convexHull`, `AffineIndependent`, `AffineMap`, `AffineEquiv` (1.4), `Collinear` (1.5) and `ConvexIndependent` (1.6).
- *Covered facts, with the label of the statement.* `orient_swap`, `orient_cycle`, `orient_translate` (1.15); `SymReduction.orient_affine` and, for part (i), `OrbitConsistency.orient_rotAbout_inv` (1.17); `orient_eq_zero_iff_collinear`, `generalPosition_iff_orient_ne_zero` (1.20); `convexPosition_iff` (1.22); `ConvexPosition.mono`, `GeneralPosition.mono`, `HasConvexNGon.mono`, `MirrorOdd.hasConvexNGon_of_le` (1.23); `HasConvexNGon.card_le`, `ConvexPosition.hasConvexNGon` (1.24); `ForcesConvexNGon.mono`, `forcesConvexNGon_iff_exact` (1.25); `convexPosition_of_card_three` (1.26); `SymReduction.convexPosition_image_affine`, `SymReduction.generalPosition_image_affine`, `SymReduction.generalPosition_image_affineEquiv`, `SymReduction.hasConvexNGon_image_affine`, `SymReduction.injective_of_affine` (1.27, parts (b′)–(e′)); `Suk.convexPosition_of_subsets_card_le_four`, `Suk.convexPosition_of_four_subsets` (1.30); `Suk.orient_affine_combination`, `Suk.convex_orient_positive`, `Suk.convex_orient_nonnegative` (1.31); `Suk.convexPosition_of_strict_supports` (1.32); `Suk.orient_support_sum` (1.33); `Suk.convexPosition_four_of_positive` (1.34); `Suk.convexPosition_of_positive_triples` (1.35); `Suk.convexPosition_of_negative_triples` (1.37); `hasConvexNGon_of_positive_cycle` (1.38); `LayerLemma.iterate_mem` (1.11); `LayerLemma.rotAbout_eq_crot`, `LayerLemma.rotAff`, `LayerLemma.rotAbout_iterate`, `LayerLemma.rotAbout_injective` (1.14).
- *Written steps.* The following statements and parts carry no Lean tag in this section, and their proofs above are written steps. We found no declaration in the files of the table of sources that states one of them in the form given here. Several of them have Lean counterparts in another form, special cases or

variants, in the project or in Mathlib; we name some of these, here or in the remark indicated, but the research records contain many further copies and variants of such elementary facts, and we do not list them all. Naming a counterpart does not make a written step Lean-checked in the sense of the rigor labels; we give axioms evidence for such a counterpart only where we say so. The written steps are:

- 1.15(d) for a general commutative ring (its real case is `OrbitConsistency.orient3_perm`, Remark 1.16) and (e);
- part (ii) of 1.17 (its case $o = 0$, $\varphi = 0$ is `Suk.orient_reflectY`, Remark 1.18; with the line given by a unit direction vector instead of the angle φ, (ii) is stated in the research file `H2/D3Encoding/Geometry.lean` as `D3.orient_reflC` (`:87`));
- 1.19 (Mathlib's `collinear_iff_of_mem`) and 1.21;
- 1.11(a),(b) (the argument of (a) is a step of the proof of `LayerLemma.map_mem_hullVerts`);
- in 1.14, the matrix form of Definition 1.12 and the identity $\rho_\theta(c) = c$ in (b) (the latter a step of the proof of `LayerLemma.invariant_erase_centre`), the composition law and $\rho_0 = \mathrm{id}$ in (c), the bijectivity and the inverse in (d), and (e), (f) (the identity $\rho_\theta(c) = c$ and the identity of (e) are stated, for the copy of `rotAbout` in the review file `SYMLEAN/review/ReviewSanity.lean`, as `SymReduction.Review.rotAbout_center` (`:1086`) and `SymReduction.Review.orient_rotAbout` (`:1091`, which writes $(p_0 - c_0)^2 + (p_1 - c_1)^2$ as `eDist2 p c`); we do not cite them as Lean-checked, because none of the theorems whose axioms the saved output `SYMLEAN/logs/ReviewSanity.out` of that file prints uses them);
- 1.26(a)–(c) apart from its last assertion;
- in 1.27, part (a) (Mathlib's `AffineMap.image_convexHull`), the implication "injective $\Rightarrow$ $\det A \neq 0$" with the inverse, and the reverse directions of (b)–(d) (for the reflection $(p_0, p_1) \mapsto (p_0, -p_1)$, the reverse direction of (d) is `Suk.hasConvexNGon_of_reflectY`; Remark 1.28 also names the forward-direction special cases `HasConvexNGon.linearEquiv` and `Suk.generalPosition_reflectY`);
- 1.29 (Mathlib's Carathéodory module);
- 1.40 (a variant of its easy direction is the main-library theorem `Omission.BoundaryGeometry.exposed_not_in_hull`, `ErdosSzekeres/Omission/BoundaryGeometry.lean:30`: if S is a set of points, p a point and A, B, C reals with $Ap_0 + Bp_1 + C = 0$ and $Aq_0 + Bq_1 + C > 0$ for every $q \in S \setminus \{p\}$, then $p \notin \mathrm{conv}(S \setminus \{p\})$).

- *Other declarations named.* Other declarations of the project that are named in statements, remarks, proofs or this paragraph as Lean counterparts, as definitions and lemmas that these rely on, or as the location of a proof step: `OrbitConsistency.orient3_perm` and `OrbitConsistency.orient3`, `Suk.orient_reflectY`, `HasConvexNGon.linearEquiv`, `Suk.reflectY`, `Suk.reflectY_x`, `Suk.reflectY_y`, `Suk.reflectY_reflectY`, `Suk.generalPosition_reflectY`, `Suk.hasConvexNGon_of_reflectY`, `ConvexCycleCut`, `convexCycleCut_of_avoids`, `SymReduction.affineMap_eq_mulVec`, `LayerLemma.map_mem_hullVerts`, `LayerLemma.invariant_erase_centre`, `D3.orient_reflC`, `SymReduction.Review.rotAbout_center`, `SymReduction.Review.orient_rotAbout` and `Omission.BoundaryGeometry.exposed_not_in_hull`.
- *Deliberately not covered.* Of the declarations of the main-library files in the table of sources, all others are listed here: in `ErdosSzekeres/Definitions.lean`, the bound function `conjecturedBound` ($n \mapsto 2^{\max(n-2,0)} + 1$) and the open conjectures `ErdosSzekeresConjecture` and `HeptagonConjecture` (propositions that are defined but not asserted), none of which is used by the facts of this section; in `ErdosSzekeres/Orientation.lean` and `ErdosSzekeres/OrientationGeometry.lean`, `rationalToReal`, `orient_rationalToReal`, `orient_rationalToReal_ne_zero`, `RationalGeneralPosition` and `generalPosition_of_rational` (transfer of exact rational certificates to real points, outside the foundations list), and the private helper `exists_smul_of_det_zero` (absorbed into the proof of Lemma 1.20); in `ErdosSzekeres/Basic.lean`, `convexIndependent_of_affineIndependent` (a bridge to Mathlib's affine independence, used in the main library only for `convexPosition_of_card_three`, which we prove directly) and `forces_triangle`, `conjecture_case_three`, `conjecture_implies_seven`, `counterexample_refutes_conjecture` (small corollaries about the conjecture, not used by the facts of this section; `forces_triangle` is named above only in the axioms evidence); and in `ErdosSzekeres/Suk/OrientationClique.lean`, `convexPosition_insert_of_positive_triples` and `convexPosition_insert_of_negative_triples` (outside the foundations list). Of the research files in the table, only the declarations named in the statements, remarks or proofs

of this section belong to its scope; the theorems named in the axioms-evidence paragraph only as evidence, and the other declarations of these files, are treated in other sections or lie outside Stage 1. In particular, `toC`, `ofC`, `crot` and their lemmas in `LAYERLEAN/LayerLemma.lean` and its copies are implementation devices for complex coordinates, replaced here by the direct complex form in Lemma 1.14(a).

2. The layer lemma and the reduction to exactly mt points

This section gives written proofs of the declarations of the Lean file `LAYERLEAN/LayerLemma.lean` (namespace `ErdosSzekeres.LayerLemma`) that concern hull vertices, orbits of a map of finite period, the layer lemma, the peeling of convex layers, and the reduction of the sixfold and sevenfold octagon bounds to statements about sets of exactly mt points. The main results are:

- the layer lemma in abstract form (Theorem 2.8): if an injective affine map f sends a finite set S into itself, every point of S has minimal period $m > 0$ under f, and no K points of S are in convex position for some $K \le 2m$, then for every hull vertex x of S the hull vertices of S are exactly the points $x, f(x), \dots, f^{m-1}(x)$, so that, if S is nonempty, they form a single f-orbit; no general position is assumed;
- its rotation form (Theorem 2.17), the description of all convex layers (Theorem 2.22, Lemma 2.23) and of their sizes (Theorem 2.24);
- the reduction to exactly mt points (Theorem 2.26) and the sixfold and sevenfold octagon theorems (Theorems 2.28 and 2.29). The last two are *conditional*: they take the propositions `Sixfold66` and `Sevenfold70` (Definition 2.27) as hypotheses, and these propositions are not proved in Lean (Remark 2.30).

Sources. The Lean source is `LAYERLEAN/LayerLemma.lean` (1148 lines, SHA-256 `afb8542dceb0`...). This section treats its lines 1–469 and 1034–1148; the geometric lemmas of lines 471–1032 (strict nesting, the centre for even m, chains, polar rules, radius order) belong to other sections (for example Theorem 3.16 and Lemma 3.18). The notions of point, general position, convex position and `HasConvexNGon` come from `ErdosSzekeres/Definitions.lean` (SHA-256 `2425500eefd1`...) and are recalled in Definitions 1.1, 1.5, 1.6 and 1.7.

Lean evidence. The file was checked with Lean 4.24.0 and Mathlib `f897ebcf` (the repository's pins) by `lake env lean`, run from the repository root. It is not a module of the default build target, so a plain `lake build` does not re-check it. Two records show that the current file compiles. The review report `LEANEXT/review/REPORT.md` of 4 November 2026 (item 1) records exit code 0 and no messages for the reviewed file, which it identifies by its SHA-256 `afb8542d`... and its 1148 lines. The re-check `LAYERLEAN/logs/axioms_recheck_20261106/out.txt` of 6 November 2026 records exit status 0 for a file whose first 1148 lines are byte-identical to `LAYERLEAN/LayerLemma.lean`, followed by 19 `#print axioms` commands; its only Lean output is the 19 printed axiom lines. The log `LAYERLEAN/logs/build.log` (exit code 0, no messages) names no file hash; it was committed with the first version of the file (454 lines, SHA-256 `804b1d25f79a`..., commit `6429e9a70`) and was not updated by the five later commits that changed the file, so we do not use it. Axioms: `LAYERLEAN/logs/axioms.log` (28 theorems) and the re-check output just named (19 theorems) report exactly the axioms `propext`, `Classical.choice` and `Quot.sound` for every printed theorem. The review log `LEANEXT/review/axioms.log` prints 59 theorems, namely all theorems of the file that are not `@[simp]` lemmas, among them every theorem of this section except the seven `@[simp]` lemmas of Lemma 2.12(a)–(b); it reports the same three axioms for each printed theorem, except that `iterate_mem` depends only on `propext` and `Quot.sound`. This log comes from the review whose report is named above; the check file from which the axioms were printed is not kept in the record, so the identification of the printed theorems with those of this file rests on the report. The seven `@[simp]` lemmas are printed in none of the three logs; their proofs (lines 236–255) are `rfl`, extensionality (`Complex.ext`, `funext`) and, for `toC_crot`, the lemma `toC_ofC`. The file contains no `sorry`, `admit`, `axiom` or `native_decide`, and its only `decide` (line 1125) proves that 6 is even and is evaluated by the kernel. So no statement of this section depends on `sorryAx`, on `Lean.ofReduceBool` (`native_decide`) or on an axiom of the project. Every lemma and theorem below whose head names a Lean declaration is *Lean-checked* in this sense (Theorems 2.28 and 2.29 as implications from their hypotheses); for Lemma 2.14(b) the Lean declaration is the definition `rotAff`, whose affine-map property is a proof obligation discharged inside the definition and checked with it. The definitions whose heads name a Lean declaration restate Lean definitions, which the same build type-checks; being definitions, they are not Lean-checked claims. In particular, Definition 2.27 only introduces the propositions `Sixfold66` and `Sevenfold70`, which are not proved (Remark 2.30). The proofs below are

written proofs (not refereed); for each statement that corresponds to a Lean declaration we say whether its proof follows the Lean proof.

Dependencies. From the foundations section we use Definitions 1.1, 1.3, 1.5, 1.6, 1.7, 1.10, 1.12, Lemmas 1.22, 1.23, 1.20 and 1.27, and the lemma of that section titled "Convex hulls" (in (C3) below). Theorem 2.28 uses Lemma 3.18 (exclusion of the centre for even m); for the case needed we also include the short argument. Nothing else from other sections is used.

2.1. **Conventions.** Points are elements of $\mathbb{R}^2$ (Lean `Point`, defined as `Fin 2 → ℝ`), written $p = (p_0, p_1)$. We use the following conventions and standard facts.

(C1) *Finite sets.* A finite set of points is a Lean `Finset Point`; $|S|$ is its cardinality; $S \setminus \{c\}$ is Lean `S.erase c`; "S is nonempty" is Lean `S.Nonempty`.

(C2) *Iterates and minimal period.* For a set α and $f : \alpha \to \alpha$ put $f^0 = \mathrm{id}$ and $f^{i+1} = f \circ f^i$. Lean's `f^[i]` (`Nat.iterate`) is defined by the other recursion, $f^{i+1} = f^i \circ f$; an easy induction shows that both recursions define the same maps (Mathlib's `Function.iterate_succ'`), and $f^{i+j} = f^i \circ f^j$ (Mathlib's `Function.iterate_add`). The *minimal period* $\mathrm{per}_f(x)$ of $x \in \alpha$ is Mathlib's `Function.minimalPeriod f x`: the least $n \geq 1$ with $f^n(x) = x$ if such an n exists, and 0 otherwise. Directly from this definition:

(P1) $f^{\mathrm{per}_f(x)}(x) = x$ for every x (if no period exists, $f^0(x) = x$);

(P2) if $n \geq 1$ and $f^n(x) = x$, then $1 \leq \mathrm{per}_f(x) \leq n$.

(C3) *Convex hulls.* For $A \subseteq \mathbb{R}^2$, $\mathrm{conv}(A)$ is the smallest convex set containing A (Mathlib `convexHull`). It is the set of all convex combinations $\sum_{i=1}^{r} \lambda_i a_i$ with $r \geq 1$, $a_i \in A$, $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. This is part (b) of the lemma titled "Convex hulls" in the foundations section, where it is proved for the whole document (see also R. T. Rockafellar, *Convex Analysis*, Princeton 1970, Theorem 2.3). Hence: $A \subseteq \mathrm{conv}(A)$; $A \subseteq B$ implies $\mathrm{conv}(A) \subseteq \mathrm{conv}(B)$; if $A \subseteq C$ with C convex then $\mathrm{conv}(A) \subseteq C$, so $\mathrm{conv}(\mathrm{conv}(A)) = \mathrm{conv}(A)$; $\mathrm{conv}(\emptyset) = \emptyset$; a convex combination of points of a convex set C lies in C. Merging equal points and deleting zero coefficients, every point of $\mathrm{conv}(A)$ is a convex combination of pairwise distinct points of A with positive coefficients and at least one term.

(C4) *Extreme points.* A point x is an *extreme point* of $A \subseteq \mathbb{R}^2$ (Mathlib `Set.extremePoints`) if $x \in A$ and, whenever $x = ax_1 + bx_2$ with $x_1, x_2 \in A$, $a, b > 0$ and $a + b = 1$, we have $x_1 = x$. This is Mathlib's definition at the pinned revision (with x in the open segment between x_1 and x_2); exchanging the roles of x_1 and x_2 shows that then also $x_2 = x$.

(C5) *Affine maps.* A Lean affine map f, an element of the type `Point →ᵃ [ℝ] Point`, consists of a map $f : \mathbb{R}^2 \to \mathbb{R}^2$ and a linear map L with $f(v + p) = Lv + f(p)$ for all v, p; taking $p = 0$, $f(v) = Lv + b$ with $b = f(0)$, and conversely every map $p \mapsto Lp + b$ with L linear is of this form. "f is injective" refers to the underlying map on all of $\mathbb{R}^2$. If $\sum_i \lambda_i = 1$ then $f(\sum_i \lambda_i a_i) = \sum_i \lambda_i (La_i + b) = \sum_i \lambda_i f(a_i)$; hence, by (C3), $\mathrm{conv}(f(A)) \subseteq f(\mathrm{conv}(A))$ for every $A \subseteq \mathbb{R}^2$ (this inclusion is part of Lemma 1.27).

(C6) *Convex position.* Convex position, general position and `HasConvexNGon` are as in Definitions 1.6, 1.5 and 1.7: `HasConvexNGon K S` means that some $T \subseteq S$ with $|T| = K$ is in convex position, and "S is K-free" means $\neg$ `HasConvexNGon K S`. We use Lemma 1.22 (S is in convex position iff $p \notin \mathrm{conv}(S \setminus \{p\})$ for every $p \in S$) and Lemma 1.23 (monotonicity). In particular the empty set is in convex position, so `HasConvexNGon 0 S` holds for every S.

(C7) *Angles.* In Lean the angle $2\pi/m$ is `2 * Real.pi / m` with $m \in \mathbb{N}$ cast to $\mathbb{R}$; division by 0 gives 0. For $t \in \mathbb{N}$ the subtraction $t \dot{-} 1$ is truncated ($0 \dot{-} 1 = 0$).

2.2. **Orbit segments.**

Definition 2.1 (orbit segment; Lean: `ErdosSzekeres.LayerLemma.orbitFS`, `LAYERLEAN/LayerLemma.lean:48`)**.** Let α be a set (Lean: a type with decidable equality), $f : \alpha \to \alpha$, $p \in \mathbb{N}$ and $x \in \alpha$. The *orbit segment* is the finite set

$$\mathrm{orb}_p(f, x) = \{f^i(x) : \ i \in \mathbb{N},\ 0 \leq i < p\}$$

(Lean `orbitFS f p x`, the image of $\{0, \ldots, p-1\}$ under $i \mapsto f^i(x)$). It is empty for $p = 0$.

Lemma 2.2 (orbit segments; Lean: `ErdosSzekeres.LayerLemma.iterate_mem`, `LAYERLEAN/LayerLemma.lean:52`, Lean: `ErdosSzekeres.LayerLemma.mem_orbitFS_self`, `LAYERLEAN/LayerLemma.lean:59`, Lean: `ErdosSzekeres.LayerLemma.orbitFS_subset`, `LAYERLEAN/LayerLemma.lean:62`, Lean: `ErdosSzekeres.LayerLemma.orbitFS_invariant`, `LAYERLEAN/LayerLemma.lean:68`, Lean: `ErdosSzekeres.LayerLemma.card_orbitFS_of_minimalPeriod`, `LAYERLEAN/LayerLemma.lean:80`, Lean: `ErdosSzekeres.LayerLemma.mem_orbitFS_of_not_disjoint`, `LAYERLEAN/LayerLemma.lean:91`)**.** *Let α be a set and $f : \alpha \to \alpha$.*

(a) (`iterate_mem`*) If $S \subseteq \alpha$ is finite, $f(y) \in S$ for every $y \in S$, and $x \in S$, then $f^i(x) \in S$ for every $i \in \mathbb{N}$.*
(b) (`mem_orbitFS_self`*) If $0 < p$, then $x \in \mathrm{orb}_p(f, x)$ for every $x \in \alpha$.*
(c) (`orbitFS_subset`*) If $S \subseteq \alpha$ is finite, $f(y) \in S$ for every $y \in S$, and $x \in S$, then $\mathrm{orb}_p(f, x) \subseteq S$ for every $p \in \mathbb{N}$.*
(d) (`orbitFS_invariant`*) If $0 < p$ and $f^p(x) = x$, then $f(y) \in \mathrm{orb}_p(f, x)$ for every $y \in \mathrm{orb}_p(f, x)$.*
(e) (`card_orbitFS_of_minimalPeriod`*) If $\mathrm{per}_f(x) = p$, then $|\mathrm{orb}_p(f, x)| = p$.*
(f) (`mem_orbitFS_of_not_disjoint`*) If $0 < p$, $f^p(x) = x$, $f^p(y) = y$ and the sets $\mathrm{orb}_p(f, x)$ and $\mathrm{orb}_p(f, y)$ are not disjoint, then $y \in \mathrm{orb}_p(f, x)$.*

Proof. We follow the Lean proofs. (a) Induction on i: $f^0(x) = x \in S$, and if $f^i(x) \in S$ then $f^{i+1}(x) = f(f^i(x)) \in S$. (b) Take $i = 0 < p$: $f^0(x) = x$. (c) Every element of $\mathrm{orb}_p(f, x)$ is some $f^i(x)$, which lies in S by (a). (d) Let $y = f^i(x)$ with $0 \le i < p$; then $f(y) = f^{i+1}(x)$. If $i + 1 < p$ this lies in $\mathrm{orb}_p(f, x)$. Otherwise $i + 1 = p$ and $f(y) = f^p(x) = x \in \mathrm{orb}_p(f, x)$ by (b). (e) If $p = 0$, then $\mathrm{orb}_0(f, x) = \emptyset$ has 0 elements. Let $p \ge 1$. It suffices to show that $f^0(x), \dots, f^{p-1}(x)$ are pairwise distinct. By (P1), $f^p(x) = x$. Suppose $f^i(x) = f^j(x)$ with $0 \le i < j \le p - 1$. Applying f^{p-j} gives

$$f^{p-j+i}(x) = f^{p-j}(f^i(x)) = f^{p-j}(f^j(x)) = f^p(x) = x.$$

Since $j \le p-1$ and $i \le j-1$, we have $1 \le p-j+i \le p-1$, so (P2) gives $\mathrm{per}_f(x) \le p-j+i < p$, a contradiction. (f) Let z lie in both sets, say $z = f^j(y)$ with $0 \le j < p$. Then $f^{p-j}(z) = f^p(y) = y$. By (d), f maps the finite set $\mathrm{orb}_p(f, x)$ into itself, and $z \in \mathrm{orb}_p(f, x)$; so by (a), applied to this set, $y = f^{p-j}(z) \in \mathrm{orb}_p(f, x)$. □

Definition 2.3 (invariance; Lean: `ErdosSzekeres.LayerLemma.Invariant`, `LAYERLEAN/LayerLemma.lean: 105`)**.** For a map $f : \mathbb{R}^2 \to \mathbb{R}^2$ and a finite set $S \subseteq \mathbb{R}^2$, S is *f-invariant* (Lean `Invariant f S`) if $f(p) \in S$ for every $p \in S$, that is, f maps S *into* itself. This is the notion of Definition 1.10. (For injective f it implies $f(S) = S$; see the proof of Lemma 2.7.)

2.3. **Hull vertices.**

Definition 2.4 (hull vertices; Lean: `ErdosSzekeres.LayerLemma.hullVerts`, `LAYERLEAN/LayerLemma.lean: 111`)**.** For a finite set $S \subseteq \mathbb{R}^2$ let

$$\mathrm{hv}(S) = \{x \in S : \ x \notin \mathrm{conv}(S \setminus \{x\})\}$$

(Lean `hullVerts S`, the subset of S cut out by this condition). Its elements are the *hull vertices* of S.

Lemma 2.5 (membership; Lean: `ErdosSzekeres.LayerLemma.mem_hullVerts`, `LAYERLEAN/LayerLemma.lean: 114`, Lean: `ErdosSzekeres.LayerLemma.hullVerts_subset`, `LAYERLEAN/LayerLemma.lean:118`)**.** *Let S be a finite set of points. (a) For every point x: $x \in \mathrm{hv}(S)$ if and only if $x \in S$ and $x \notin \mathrm{conv}(S \setminus \{x\})$. (b) $\mathrm{hv}(S) \subseteq S$.*

Proof. As in Lean: (a) is Definition 2.4, and (b) follows from (a). □

Lemma 2.6 (hull vertices; Lean: `ErdosSzekeres.LayerLemma.convexPosition_hullVerts`, `LAYERLEAN/LayerLemma.lean:121`, Lean: `ErdosSzekeres.LayerLemma.extreme_mem_hullVerts`, `LAYERLEAN/LayerLemma.lean:130`, Lean: `ErdosSzekeres.LayerLemma.subset_convexHull_hullVerts`, `LAYERLEAN/LayerLemma.lean:146`, Lean: `ErdosSzekeres.LayerLemma.hullVerts_nonempty`, `LAYERLEAN/LayerLemma.lean:139`)**.** *Let S be a finite set of points.*

(a) (`convexPosition_hullVerts`*) $\mathrm{hv}(S)$ is in convex position.*
(b) (`extreme_mem_hullVerts`*) Every extreme point of $\mathrm{conv}(S)$ belongs to $\mathrm{hv}(S)$.*
(c) (`subset_convexHull_hullVerts`*) $S \subseteq \mathrm{conv}(\mathrm{hv}(S))$.*
(d) (`hullVerts_nonempty`*) If S is nonempty, then $\mathrm{hv}(S)$ is nonempty.*

Proof. The proof of (a) follows Lean. For (b)–(d) we give elementary proofs, different from the Lean proofs. Lean proves (b) with two Mathlib facts: the extreme points of $\mathrm{conv}(S)$ lie in S, and "x is an extreme point of a convex set C iff $x \in C$ and $C \setminus \{x\}$ is convex". It deduces (d) from (b) and the existence of an extreme point of the nonempty compact set $\mathrm{conv}(S)$ (Mathlib's Krein–Milman lemma), and (c) from (b) and the Krein–Milman theorem ($\mathrm{conv}(S)$ is the closure of the convex hull of its extreme points).

(a) By Lemma 1.22 we must show $x \notin \mathrm{conv}(\mathrm{hv}(S) \setminus \{x\})$ for every $x \in \mathrm{hv}(S)$. Since $\mathrm{hv}(S) \setminus \{x\} \subseteq S \setminus \{x\}$, (C3) gives $\mathrm{conv}(\mathrm{hv}(S) \setminus \{x\}) \subseteq \mathrm{conv}(S \setminus \{x\})$, and $x \notin \mathrm{conv}(S \setminus \{x\})$ because $x \in \mathrm{hv}(S)$.

(b) Let x be an extreme point of $C = \mathrm{conv}(S)$. We first show:

(∗) if $x = \sum_{i=1}^{r} \lambda_i s_i$ with $r \ge 1$, $s_i \in C$, $\lambda_i > 0$ and $\sum_i \lambda_i = 1$, then $s_i = x$ for every i.

If $r = 1$ then $\lambda_1 = 1$ and $s_1 = x$. Let $r \geq 2$ and fix j. Then $0 < \lambda_j < 1$, since some other coefficient is positive. The point $y_j = \sum_{i \neq j} \frac{\lambda_i}{1-\lambda_j} s_i$ is a convex combination of points of the convex set C, so $y_j \in C$ by (C3), and $x = \lambda_j s_j + (1 - \lambda_j) y_j$. By (C4), applied with $x_1 = s_j$, $x_2 = y_j$, $a = \lambda_j$, $b = 1 - \lambda_j$, we get $s_j = x$. This proves $(*)$.

Since $x \in C$, by (C3) $x = \sum_{i=1}^{r} \lambda_i s_i$ with $r \geq 1$, $s_i \in S \subseteq C$, $\lambda_i > 0$, $\sum_i \lambda_i = 1$; by $(*)$, $x = s_1 \in S$. Suppose $x \in \operatorname{conv}(S \setminus \{x\})$. Then $x = \sum_{i=1}^{r} \mu_i t_i$ with $r \geq 1$, $t_i \in S \setminus \{x\} \subseteq C$, $\mu_i > 0$, $\sum_i \mu_i = 1$, and $(*)$ gives $t_1 = x$, contradicting $t_1 \neq x$. Hence $x \notin \operatorname{conv}(S \setminus \{x\})$, and $x \in \operatorname{hv}(S)$.

(c) Strong induction on $|S|$. If $\operatorname{hv}(S) = S$, then $S \subseteq \operatorname{conv}(S) = \operatorname{conv}(\operatorname{hv}(S))$; this covers $S = \emptyset$. Otherwise choose $y \in S \setminus \operatorname{hv}(S)$; then $y \in \operatorname{conv}(S')$ where $S' = S \setminus \{y\}$. As $|S'| < |S|$, the induction hypothesis gives $S' \subseteq \operatorname{conv}(\operatorname{hv}(S'))$.

We claim $\operatorname{hv}(S') \subseteq \operatorname{hv}(S)$. Let $z \in \operatorname{hv}(S')$, so $z \in S'$ (in particular $z \in S$ and $z \neq y$) and $z \notin \operatorname{conv}(A)$, where $A = S' \setminus \{z\} = S \setminus \{y, z\}$. Suppose $z \in \operatorname{conv}(S \setminus \{z\})$. Since $S \setminus \{z\} = A \cup \{y\}$ and $S' = A \cup \{z\}$, (C3) and merging of equal points give

$$z = \sum_{a \in A} \alpha_a a + \lambda y, \qquad y = \sum_{a \in A} \beta_a a + \mu z,$$

with all $\alpha_a, \beta_a, \lambda, \mu \geq 0$, $\sum_a \alpha_a + \lambda = 1$ and $\sum_a \beta_a + \mu = 1$ (the second representation exists because $y \in \operatorname{conv}(S')$). Substituting the second into the first,

$$(1 - \lambda\mu)\, z = \sum_{a \in A} (\alpha_a + \lambda \beta_a)\, a.$$

If $\lambda\mu < 1$, the coefficients $(\alpha_a + \lambda\beta_a)/(1 - \lambda\mu)$ are nonnegative with sum $\big((1 - \lambda) + \lambda(1 - \mu)\big)/(1 - \lambda\mu) = 1$, so $z \in \operatorname{conv}(A)$, a contradiction. If $\lambda\mu = 1$, then $\lambda = \mu = 1$ because $\lambda, \mu \in [0, 1]$; so all $\alpha_a = 0$ and $z = y$, contradicting $z \neq y$. This proves the claim.

By the claim and (C3), $S' \subseteq \operatorname{conv}(\operatorname{hv}(S')) \subseteq \operatorname{conv}(\operatorname{hv}(S))$, and then $y \in \operatorname{conv}(S') \subseteq \operatorname{conv}(\operatorname{conv}(\operatorname{hv}(S))) = \operatorname{conv}(\operatorname{hv}(S))$. Hence $S = S' \cup \{y\} \subseteq \operatorname{conv}(\operatorname{hv}(S))$.

(d) If $\operatorname{hv}(S) = \emptyset$, then by (c) $S \subseteq \operatorname{conv}(\emptyset) = \emptyset$, so S is empty. □

Lemma 2.7 (affine invariance of hull vertices; Lean: `ErdosSzekeres.LayerLemma.map_mem_hullVerts`, `LAYERLEAN/LayerLemma.lean:162`)**.** *Let f be an affine map of $\mathbb{R}^2$ (Lean: f of type `Point` $\to^{\mathrm{a}}$ $[\mathbb{R}]$ `Point`) that is injective, and let S be a finite set with $f(y) \in S$ for every $y \in S$. If $x \in \operatorname{hv}(S)$, then $f(x) \in \operatorname{hv}(S)$.*

Proof. We follow the Lean proof. (1) $f(S) = S$: as f is injective, $|f(S)| = |S|$, and $f(S) \subseteq S$ with S finite. (2) $f(x) \in S$, since $x \in S$. (3) Suppose $f(x) \in \operatorname{conv}(S \setminus \{f(x)\})$. Every $y \in S$ with $y \neq f(x)$ is, by (1), of the form $y = f(z)$ with $z \in S$, and $z \neq x$ since $y \neq f(x)$; thus $S \setminus \{f(x)\} \subseteq f(S \setminus \{x\})$. By (C3) and (C5),

$$f(x) \in \operatorname{conv}\big(f(S \setminus \{x\})\big) \subseteq f\big(\operatorname{conv}(S \setminus \{x\})\big),$$

so $f(x) = f(u)$ for some $u \in \operatorname{conv}(S \setminus \{x\})$. Injectivity of f on all of $\mathbb{R}^2$ (note that u need not lie in S) gives $x = u \in \operatorname{conv}(S \setminus \{x\})$, contradicting $x \in \operatorname{hv}(S)$. By Lemma 2.5(a), (2) and (3) give $f(x) \in \operatorname{hv}(S)$. □

2.4. **The core lemma.**

Theorem 2.8 (layer lemma, abstract form; Lean: `ErdosSzekeres.LayerLemma.hullVerts_eq_orbitFS`, `LAYERLEAN/LayerLemma.lean:189`)**.** *Let f be an affine map of $\mathbb{R}^2$ (Lean: f of type `Point` $\to^{\mathrm{a}}$ $[\mathbb{R}]$ `Point`) that is injective, let S be a finite set of points, and let $m, K \in \mathbb{N}$ with $0 < m$ and $K \leq 2m$. Assume:*

(i) $f(y) \in S$ *for every* $y \in S$*;*
(ii) $\operatorname{per}_f(y) = m$ *for every* $y \in S$*;*
(iii) S *is K-free, i.e.* $\neg$ `HasConvexNGon` K S*.*

Then for every $x \in \operatorname{hv}(S)$,

$$\operatorname{hv}(S) = \operatorname{orb}_m(f, x) = \{f^i(x) : \ 0 \leq i < m\}.$$

Proof. We follow the Lean proof. Write $H = \operatorname{hv}(S)$, a finite set with $H \subseteq S$ (Lemma 2.5).

Step 1. $f(y) \in H$ for every $y \in H$, by Lemma 2.7 and (i).

Step 2. $f^m(y) = y$ for every $y \in S$, by (ii) and (P1).

Step 3. $\operatorname{orb}_m(f, x) \subseteq H$, by Lemma 2.2(c) applied to the finite set H (Step 1) and $x \in H$.

Step 4. $H \subseteq \operatorname{orb}_m(f, x)$. Let $y \in H$ and suppose $y \notin \operatorname{orb}_m(f, x)$. Both x and y lie in S, so $f^m(x) = x$ and $f^m(y) = y$ by Step 2. If $\operatorname{orb}_m(f, x)$ and $\operatorname{orb}_m(f, y)$ were not disjoint, Lemma 2.2(f) (with $p = m > 0$) would

give $y \in \mathrm{orb}_m(f,x)$; so they are disjoint. Both are contained in H, by Step 1 and Lemma 2.2(c), and each has exactly m elements by (ii) and Lemma 2.2(e). Hence

$$|H| \;\geq\; |\mathrm{orb}_m(f,x) \cup \mathrm{orb}_m(f,y)| = 2m \;\geq\; K.$$

Choose $T \subseteq H$ with $|T| = K$. By Lemma 2.6(a) and Lemma 1.23(a), T is in convex position, and $T \subseteq H \subseteq S$. So `HasConvexNGon K S` holds, contradicting (iii). Hence $y \in \mathrm{orb}_m(f,x)$.

Steps 3 and 4 give $H = \mathrm{orb}_m(f,x)$. □

Lemma 2.9 (invariance of $S \setminus \mathrm{hv}(S)$; Lean: `ErdosSzekeres.LayerLemma.invariant_sdiff_hullVerts`, `LAYERLEAN/LayerLemma.lean:217`)**.** *Let f be an affine map of $\mathbb{R}^2$ that is injective, S a finite set and $m \in \mathbb{N}$ with $0 < m$. Assume $f(y) \in S$ for every $y \in S$ and $f^m(y) = y$ for every $y \in S$. Then $f(y) \in S \setminus \mathrm{hv}(S)$ for every $y \in S \setminus \mathrm{hv}(S)$.*

Proof. We follow the Lean proof. Let $y \in S \setminus \mathrm{hv}(S)$. Then $f(y) \in S$. Suppose $f(y) \in \mathrm{hv}(S)$. By Lemma 2.7, f maps the finite set $\mathrm{hv}(S)$ into itself, so Lemma 2.2(a) gives $f^{m-1}(f(y)) \in \mathrm{hv}(S)$. As $m \geq 1$, $f^{m-1}(f(y)) = f^m(y) = y$, so $y \in \mathrm{hv}(S)$, a contradiction. Hence $f(y) \in S \setminus \mathrm{hv}(S)$. □

Remark 2.10 (on Theorem 2.8). (1) The proof uses only two geometric facts: an injective affine map that sends S into itself sends $\mathrm{hv}(S)$ into itself, hence permutes the finite set $\mathrm{hv}(S)$ (Lemma 2.7), and the hull vertices are in convex position (Lemma 2.6(a)). No general position is assumed, and f need not be a rotation. (2) If $K = 0$, hypothesis (iii) is false by (C6), and the theorem is vacuous; if $\mathrm{hv}(S) = \emptyset$, which by Lemma 2.6(d) happens only for $S = \emptyset$, there is no x. (3) Under the hypotheses of the theorem, if $S \neq \emptyset$ then $\mathrm{hv}(S)$ contains some x (Lemma 2.6(d)), and the conclusion with Lemma 2.2(e) and (ii) gives $|\mathrm{hv}(S)| = m$.

2.5. **Rotations.**

Definition 2.11 (complex coordinates; Lean: `ErdosSzekeres.LayerLemma.toC`, `LAYERLEAN/LayerLemma.lean:231`, Lean: `ErdosSzekeres.LayerLemma.ofC`, `LAYERLEAN/LayerLemma.lean:234`, Lean: `ErdosSzekeres.LayerLemma.crot`, `LAYERLEAN/LayerLemma.lean:252`)**.** For $p \in \mathbb{R}^2$ let $\mathrm{toC}(p) = p_0 + ip_1 \in \mathbb{C}$; for $z \in \mathbb{C}$ let $\mathrm{ofC}(z) = (\mathrm{Re}\, z, \mathrm{Im}\, z) \in \mathbb{R}^2$; for $\gamma, w \in \mathbb{C}$ and $p \in \mathbb{R}^2$ let

$$\mathrm{crot}_{\gamma,w}(p) = \mathrm{ofC}\big(\gamma + w\,(\mathrm{toC}(p) - \gamma)\big)$$

(Lean `crot c w p`, where $c \in \mathbb{C}$ plays the role of γ).

Lemma 2.12 (complex coordinates; Lean: `ErdosSzekeres.LayerLemma.toC_re`, `LAYERLEAN/LayerLemma.lean:236`, Lean: `ErdosSzekeres.LayerLemma.toC_im`, `LAYERLEAN/LayerLemma.lean:237`, Lean: `ErdosSzekeres.LayerLemma.ofC_zero`, `LAYERLEAN/LayerLemma.lean:238`, Lean: `ErdosSzekeres.LayerLemma.ofC_one`, `LAYERLEAN/LayerLemma.lean:239`, Lean: `ErdosSzekeres.LayerLemma.toC_ofC`, `LAYERLEAN/LayerLemma.lean:241`, Lean: `ErdosSzekeres.LayerLemma.ofC_toC`, `LAYERLEAN/LayerLemma.lean:244`, Lean: `ErdosSzekeres.LayerLemma.toC_injective`, `LAYERLEAN/LayerLemma.lean:248`, Lean: `ErdosSzekeres.LayerLemma.toC_crot`, `LAYERLEAN/LayerLemma.lean:254`, Lean: `ErdosSzekeres.LayerLemma.crot_one`, `LAYERLEAN/LayerLemma.lean:257`, Lean: `ErdosSzekeres.LayerLemma.crot_comp`, `LAYERLEAN/LayerLemma.lean:262`, Lean: `ErdosSzekeres.LayerLemma.crot_iterate`, `LAYERLEAN/LayerLemma.lean:267`, Lean: `ErdosSzekeres.LayerLemma.crot_eq_self_iff`, `LAYERLEAN/LayerLemma.lean:273`, Lean: `ErdosSzekeres.LayerLemma.crot_injective`, `LAYERLEAN/LayerLemma.lean:287`)**.** *For all $p \in \mathbb{R}^2$, $z, \gamma, w, v \in \mathbb{C}$ and $k \in \mathbb{N}$:*

- *(a) (`toC_re`, `toC_im`, `ofC_zero`, `ofC_one`, `toC_ofC`, `ofC_toC`, `toC_injective`)* $\mathrm{Re}\,\mathrm{toC}(p) = p_0$, $\mathrm{Im}\,\mathrm{toC}(p) = p_1$, $\mathrm{ofC}(z)_0 = \mathrm{Re}\, z$, $\mathrm{ofC}(z)_1 = \mathrm{Im}\, z$, $\mathrm{toC}(\mathrm{ofC}(z)) = z$ *and* $\mathrm{ofC}(\mathrm{toC}(p)) = p$*; in particular* toC *is injective;*
- *(b) (`toC_crot`)* $\mathrm{toC}(\mathrm{crot}_{\gamma,w}(p)) = \gamma + w(\mathrm{toC}(p) - \gamma)$*;*
- *(c) (`crot_one`)* $\mathrm{crot}_{\gamma,1} = \mathrm{id}$*;*
- *(d) (`crot_comp`)* $\mathrm{crot}_{\gamma,w}(\mathrm{crot}_{\gamma,v}(p)) = \mathrm{crot}_{\gamma,wv}(p)$*;*
- *(e) (`crot_iterate`)* $(\mathrm{crot}_{\gamma,w})^k = \mathrm{crot}_{\gamma,w^k}$*;*
- *(f) (`crot_eq_self_iff`) if $w \neq 1$, then* $\mathrm{crot}_{\gamma,w}(p) = p$ *if and only if* $\mathrm{toC}(p) = \gamma$*;*
- *(g) (`crot_injective`) if $w \neq 0$, then* $\mathrm{crot}_{\gamma,w}$ *is injective.*

Proof. We follow the Lean proofs. (a) The four coordinate identities are the definitions; they give $\mathrm{toC}(\mathrm{ofC}(z)) = \mathrm{Re}\, z + i\,\mathrm{Im}\, z = z$ and $\mathrm{ofC}(\mathrm{toC}(p)) = (p_0, p_1) = p$, and a map with a left inverse is injective. (b) follows from $\mathrm{toC} \circ \mathrm{ofC} = \mathrm{id}$. (c) By (b), $\mathrm{toC}(\mathrm{crot}_{\gamma,1}(p)) = \gamma + (\mathrm{toC}(p) - \gamma) = \mathrm{toC}(p)$; apply injectivity of toC. (d) By (b) twice,

toC of the left side is $\gamma + w\big((\gamma + v(\mathrm{toC}(p) - \gamma)) - \gamma\big) = \gamma + wv(\mathrm{toC}(p) - \gamma)$, which is toC of the right side; apply injectivity. (e) Induction on k: $(\mathrm{crot}_{\gamma,w})^0 = \mathrm{id} = \mathrm{crot}_{\gamma,1}$ by (c), and $(\mathrm{crot}_{\gamma,w})^{k+1}(p) = \mathrm{crot}_{\gamma,w}(\mathrm{crot}_{\gamma,w^k}(p)) = \mathrm{crot}_{\gamma,w\,w^k}(p)$ by (d). (f) Write $u = \mathrm{toC}(p)$. By (a) and (b), $\mathrm{crot}_{\gamma,w}(p) = p$ iff $\gamma + w(u - \gamma) = u$, i.e. iff $(w-1)(u-\gamma) = 0$, i.e. (as $w \neq 1$ and $\mathbb{C}$ is a field) iff $u = \gamma$. (g) If $\mathrm{crot}_{\gamma,w}(p) = \mathrm{crot}_{\gamma,w}(q)$, then by (b) $w(\mathrm{toC}(p) - \mathrm{toC}(q)) = 0$, so $\mathrm{toC}(p) = \mathrm{toC}(q)$ as $w \neq 0$, and $p = q$ by (a). □

Definition 2.13 (rotation about a point; Lean: `ErdosSzekeres.LayerLemma.rotAbout`, `LAYERLEAN/LayerLemma.lean:298`)**.** For $c \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$ let $\rho_{c,\theta} : \mathbb{R}^2 \to \mathbb{R}^2$ (Lean `rotAbout c θ`) be

$$\rho_{c,\theta}(p) = \big(c_0 + \cos\theta\,(p_0 - c_0) - \sin\theta\,(p_1 - c_1),\ \ c_1 + \sin\theta\,(p_0 - c_0) + \cos\theta\,(p_1 - c_1)\big).$$

This is the map of Definition 1.12, the counterclockwise rotation by θ about c. For $m \in \mathbb{N}$ the map $\rho_{c,2\pi/m}$ is Lean's `rotAbout c (2 * Real.pi / m)`, with the convention (C7) on the angle.

Lemma 2.14 (rotations; Lean: `ErdosSzekeres.LayerLemma.rotAbout_eq_crot`, `LAYERLEAN/LayerLemma.lean:302`, Lean: `ErdosSzekeres.LayerLemma.rotAff`, `LAYERLEAN/LayerLemma.lean:311`, Lean: `ErdosSzekeres.LayerLemma.rotAbout_injective`, `LAYERLEAN/LayerLemma.lean:319`, Lean: `ErdosSzekeres.LayerLemma.rotAbout_iterate`, `LAYERLEAN/LayerLemma.lean:465`)**.** *Let $c \in \mathbb{R}^2$, $\theta, \alpha \in \mathbb{R}$ and $j \in \mathbb{N}$.*

(a) (`rotAbout_eq_crot`) $\rho_{c,\theta} = \mathrm{crot}_{\mathrm{toC}(c),\,e^{i\theta}}$.

(b) (`rotAff`) For all $p, v \in \mathbb{R}^2$, $\rho_{c,\theta}(v + p) = A_\theta v + \rho_{c,\theta}(p)$, where A_θ is the linear map with matrix $\begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}$ acting on column vectors. So $\rho_{c,\theta}$ is an affine map with linear part A_θ; Lean's `rotAff c θ` is this affine map, and its underlying map is $\rho_{c,\theta}$.

(c) (`rotAbout_injective`) $\rho_{c,\theta}$ is injective.

(d) (`rotAbout_iterate`) $(\rho_{c,\alpha})^j = \rho_{c,j\alpha}$ (with j cast to $\mathbb{R}$).

Proof. We follow the Lean proofs. (a) Let $u = \mathrm{toC}(p) = p_0 + ip_1$, $\gamma = \mathrm{toC}(c) = c_0 + ic_1$ and $e^{i\theta} = \cos\theta + i\sin\theta$. Then

$$\gamma + e^{i\theta}(u - \gamma) = \big(c_0 + \cos\theta(p_0 - c_0) - \sin\theta(p_1 - c_1)\big) + i\big(c_1 + \sin\theta(p_0 - c_0) + \cos\theta(p_1 - c_1)\big),$$

whose real and imaginary parts are the coordinates of $\rho_{c,\theta}(p)$. Applying ofC (Lemma 2.12(a)) gives $\mathrm{crot}_{\gamma,e^{i\theta}}(p) = \rho_{c,\theta}(p)$. (b) The formula of Definition 2.13 reads $\rho_{c,\theta}(p) = A_\theta(p - c) + c$. Hence $\rho_{c,\theta}(v + p) = A_\theta(v + p - c) + c = A_\theta v + \rho_{c,\theta}(p)$ by linearity of A_θ. (c) By (a), $\rho_{c,\theta} = \mathrm{crot}_{\gamma,e^{i\theta}}$ with $e^{i\theta} \neq 0$; apply Lemma 2.12(g). (d) By (a) and Lemma 2.12(e), $(\rho_{c,\alpha})^j = \mathrm{crot}_{\gamma,(e^{i\alpha})^j}$, and $(e^{i\alpha})^j = e^{ij\alpha}$; by (a) again this is $\rho_{c,j\alpha}$. □

Lemma 2.15 (invariance of $S\backslash\{c\}$; Lean: `ErdosSzekeres.LayerLemma.invariant_erase_centre`, `LAYERLEAN/LayerLemma.lean:325`)**.** *Let S be a finite set, c a point and $\theta \in \mathbb{R}$. If S is $\rho_{c,\theta}$-invariant, then $S \setminus \{c\}$ is $\rho_{c,\theta}$-invariant.*

Proof. We follow the Lean proof. Write $\rho = \rho_{c,\theta}$. Putting $p = c$ in Definition 2.13 gives $\rho(c) = c$. Let $p \in S \setminus \{c\}$. Then $\rho(p) \in S$. If $\rho(p) = c = \rho(c)$, then $p = c$ by Lemma 2.14(c), which is false. Hence $\rho(p) \in S \setminus \{c\}$. □

Lemma 2.16 (exact period of the rotation; Lean: `ErdosSzekeres.LayerLemma.minimalPeriod_rotAbout`, `LAYERLEAN/LayerLemma.lean:336`)**.** *Let c be a point, $m \in \mathbb{N}$ with $0 < m$, and y a point with $y \neq c$. Then $\mathrm{per}_\rho(y) = m$ for $\rho = \rho_{c,2\pi/m}$.*

Proof. We follow the Lean proof. Let $\gamma = \mathrm{toC}(c)$ and $\omega = e^{2\pi i/m}$, where $2\pi/m$ is the real number of (C7). By Lemma 2.14(a), $\rho = \mathrm{crot}_{\gamma,\omega}$, and by Lemma 2.12(e), $\rho^k = \mathrm{crot}_{\gamma,\omega^k}$ for $k \in \mathbb{N}$. We have $\omega^k = e^{2\pi ik/m} = \cos(2\pi k/m) + i\sin(2\pi k/m)$. Hence $\omega^m = \cos 2\pi + i\sin 2\pi = 1$, and for $0 < k < m$ we have $0 < 2\pi k/m < 2\pi$, so $\cos(2\pi k/m) < 1$ and $\omega^k \neq 1$. (That is, ω is a primitive m-th root of unity; Lean uses Mathlib's `Complex.isPrimitiveRoot_exp`.)

By Lemma 2.12(e) and (c), $\rho^m = \mathrm{crot}_{\gamma,\omega^m} = \mathrm{crot}_{\gamma,1} = \mathrm{id}$, so $\rho^m(y) = y$ with $m \geq 1$, and (P2) gives $1 \leq q \leq m$ for $q = \mathrm{per}_\rho(y)$. Suppose $q < m$. By (P1), $\mathrm{crot}_{\gamma,\omega^q}(y) = \rho^q(y) = y$, where $\omega^q \neq 1$ because $0 < q < m$. Lemma 2.12(f) gives $\mathrm{toC}(y) = \gamma = \mathrm{toC}(c)$, hence $y = c$ by Lemma 2.12(a), a contradiction. So $q = m$. □

Theorem 2.17 (layer lemma, rotations; Lean: `ErdosSzekeres.LayerLemma.hullVerts_eq_orbitFS_rotAbout`, `LAYERLEAN/LayerLemma.lean:364`)**.** *Let $m, K \in \mathbb{N}$ with $0 < m$ and $K \leq 2m$, let S be a finite set and c a point with $c \notin S$. Assume that S is ρ-invariant for $\rho = \rho_{c,2\pi/m}$ and that S is K-free. Then for every $x \in \mathrm{hv}(S)$,*

$$\mathrm{hv}(S) = \mathrm{orb}_m(\rho, x) = \{\rho^i(x) :\ 0 \leq i < m\}.$$

Proof. As in Lean, apply Theorem 2.8 to the affine map $f = \rho$ (Lemma 2.14(b); Lean's `rotAff c (2 * Real.pi / m)`, whose underlying map is ρ), which is injective by Lemma 2.14(c). Its hypotheses hold: $0 < m$ and $K \le 2m$; (i) is the invariance of S; for (ii), every $y \in S$ satisfies $y \neq c$ because $c \notin S$, so $\mathrm{per}_\rho(y) = m$ by Lemma 2.16; (iii) is the K-freeness of S. □

2.6. **Peeling the layers.**

Definition 2.18 (peeling; Lean: `ErdosSzekeres.LayerLemma.peel`, `LAYERLEAN/LayerLemma.lean:374`)**.** For a finite set S define finite sets $\mathrm{peel}(S, n)$, $n \in \mathbb{N}$, by

$$\mathrm{peel}(S, 0) = S, \qquad \mathrm{peel}(S, n+1) = \mathrm{peel}(S, n) \setminus \mathrm{hv}(\mathrm{peel}(S, n)).$$

So $\mathrm{peel}(S, n)$ is S with its n outermost convex layers removed, and $\mathrm{hv}(\mathrm{peel}(S, n))$ is the n-th convex layer of S (counting from 0).

Lemma 2.19 (peeling, basic facts; Lean: `ErdosSzekeres.LayerLemma.peel_succ`, `LAYERLEAN/LayerLemma.lean:378`, Lean: `ErdosSzekeres.LayerLemma.peel_subset`, `LAYERLEAN/LayerLemma.lean:381`)**.** *Let S be a finite set. (a)* $\mathrm{peel}(S, n+1) = \mathrm{peel}(S, n) \setminus \mathrm{hv}(\mathrm{peel}(S, n))$ *for every $n \in \mathbb{N}$. (b)* $\mathrm{peel}(S, n) \subseteq S$ *for every $n \in \mathbb{N}$.*

Proof. As in Lean: (a) is the definition. (b) By induction: $\mathrm{peel}(S, 0) = S$, and $\mathrm{peel}(S, n+1) \subseteq \mathrm{peel}(S, n) \subseteq S$. □

Lemma 2.20 (peeling is antitone; Lean: `ErdosSzekeres.LayerLemma.peel_antitone`, `LAYERLEAN/LayerLemma.lean:460`)**.** *Let S be a finite set and $n, n' \in \mathbb{N}$ with $n \le n'$. Then* $\mathrm{peel}(S, n') \subseteq \mathrm{peel}(S, n)$*.*

Proof. Induction on $n' \ge n$, as in Lean: for $n' = n$ there is nothing to prove, and $\mathrm{peel}(S, n'+1) \subseteq \mathrm{peel}(S, n') \subseteq \mathrm{peel}(S, n)$ by Lemma 2.19(a). □

Lemma 2.21 (the peels are invariant; Lean: `ErdosSzekeres.LayerLemma.invariant_peel`, `LAYERLEAN/LayerLemma.lean:386`)**.** *Let $m \in \mathbb{N}$ with $0 < m$, let S be a finite set and c a point with $c \notin S$, and assume that S is ρ-invariant for $\rho = \rho_{c,2\pi/m}$. Then* $\mathrm{peel}(S, n)$ *is ρ-invariant for every $n \in \mathbb{N}$.*

Proof. Induction on n, as in Lean. For $n = 0$ this is the hypothesis. Assume $\mathrm{peel}(S, n)$ is ρ-invariant. Every $y \in \mathrm{peel}(S, n)$ lies in S (Lemma 2.19(b)), so $y \neq c$, so $\mathrm{per}_\rho(y) = m$ by Lemma 2.16 and $\rho^m(y) = y$ by (P1). Lemma 2.9, applied to the finite set $\mathrm{peel}(S, n)$ and the injective affine map ρ (Lemma 2.14(b),(c)), shows that ρ maps $\mathrm{peel}(S, n) \setminus \mathrm{hv}(\mathrm{peel}(S, n)) = \mathrm{peel}(S, n+1)$ into itself. □

Theorem 2.22 (every layer is one orbit; Lean: `ErdosSzekeres.LayerLemma.layers_are_orbits`, `LAYERLEAN/LayerLemma.lean:402`)**.** *Let $m, K \in \mathbb{N}$ with $0 < m$ and $K \le 2m$, let S be a finite set and c a point with $c \notin S$, and write $\rho = \rho_{c,2\pi/m}$. Assume that S is ρ-invariant and K-free, and let $n \in \mathbb{N}$ be such that* $\mathrm{peel}(S, n)$ *is nonempty. Then there is $x \in \mathrm{peel}(S, n)$ such that*

(a) $\mathrm{hv}(\mathrm{peel}(S, n)) = \mathrm{orb}_m(\rho, x)$*;*
(b) $|\mathrm{orb}_m(\rho, x)| = m$*;*
(c) $\mathrm{peel}(S, n+1) \subseteq \mathrm{conv}\big(\mathrm{hv}(\mathrm{peel}(S, n))\big)$*.*

Proof. We follow the Lean proof. By Lemma 2.6(d) there is $x \in \mathrm{hv}(\mathrm{peel}(S, n))$, and $x \in \mathrm{peel}(S, n)$ by Lemma 2.5(b). The finite set $U = \mathrm{peel}(S, n)$ satisfies the hypotheses of Theorem 2.17: $c \notin U$ since $U \subseteq S$ (Lemma 2.19(b)); U is ρ-invariant (Lemma 2.21); and U is K-free, since a K-element subset of U in convex position would be one of S (Lemma 1.23(c)). Theorem 2.17 gives (a). Since $x \in U$, $x \neq c$, so $\mathrm{per}_\rho(x) = m$ (Lemma 2.16) and (b) follows from Lemma 2.2(e). For (c), $\mathrm{peel}(S, n+1) = U \setminus \mathrm{hv}(U) \subseteq U \subseteq \mathrm{conv}(\mathrm{hv}(U))$ by Lemma 2.6(c). □

Lemma 2.23 (the layers exhaust S; Lean: `ErdosSzekeres.LayerLemma.exists_layer`, `LAYERLEAN/LayerLemma.lean:419`)**.** *Let S be a finite set of points and $y \in S$. Then there is $n \in \mathbb{N}$ with $y \in \mathrm{hv}(\mathrm{peel}(S, n))$.*

Proof. We follow the Lean proof. There are no other hypotheses: no symmetry, no general position, no K-freeness.

Claim: for every n, either $|\mathrm{peel}(S, n)| + n \le |S|$ or $\mathrm{peel}(S, n) = \emptyset$. For $n = 0$ the first alternative holds. Assume the claim for n. If $\mathrm{peel}(S, n) = \emptyset$, then $\mathrm{peel}(S, n+1) = \emptyset \setminus \mathrm{hv}(\emptyset) = \emptyset$. Otherwise the first alternative holds for n; moreover $\mathrm{hv}(\mathrm{peel}(S, n))$ is a nonempty subset of $\mathrm{peel}(S, n)$ (Lemmas 2.6(d) and 2.5(b)), so $|\mathrm{peel}(S, n+1)| \le |\mathrm{peel}(S, n)| - 1$ and $|\mathrm{peel}(S, n+1)| + (n+1) \le |\mathrm{peel}(S, n)| + n \le |S|$.

Taking $n = |S| + 1$: if $y \in \mathrm{peel}(S, |S|+1)$, this set is nonempty, so $|\mathrm{peel}(S, |S|+1)| + |S| + 1 \le |S|$, which is impossible. Hence $y \notin \mathrm{peel}(S, k)$ for some k; let k be the least such index. Since $\mathrm{peel}(S, 0) = S \ni y$, $k \neq 0$; write $k = n + 1$. By minimality $y \in \mathrm{peel}(S, n)$, and $y \notin \mathrm{peel}(S, n+1) = \mathrm{peel}(S, n) \setminus \mathrm{hv}(\mathrm{peel}(S, n))$, so $y \in \mathrm{hv}(\mathrm{peel}(S, n))$. □

Theorem 2.24 (each peel removes exactly m points; Lean: `ErdosSzekeres.LayerLemma.card_peel`, `LAYERLEAN/LayerLemma.lean:1037`)**.** *Let $m, K \in \mathbb{N}$ with $0 < m$ and $K \le 2m$, let S be a finite set and c a point with $c \notin S$, and assume that S is $\rho_{c,2\pi/m}$-invariant and K-free. Let $t \in \mathbb{N}$ with $m \cdot (t \dot{-} 1) < |S|$. Then for every $n \le t$,*

$$|\mathrm{peel}(S,n)| + m \cdot n = |S|.$$

Proof. Induction on n, as in Lean. For $n = 0$: $|\mathrm{peel}(S,0)| + 0 = |S|$. Let $n+1 \le t$ and assume $|\mathrm{peel}(S,n)| + mn = |S|$ (the case $n \le t$ of the induction hypothesis). Then $t \ge 1$ and $n \le t - 1 = t \dot{-} 1$, so $mn \le m(t \dot{-} 1) < |S|$ and $|\mathrm{peel}(S,n)| = |S| - mn > 0$: the set $\mathrm{peel}(S,n)$ is nonempty. Theorem 2.22 gives x with $\mathrm{hv}(\mathrm{peel}(S,n)) = \mathrm{orb}_m(\rho,x)$ and $|\mathrm{orb}_m(\rho,x)| = m$, so $|\mathrm{hv}(\mathrm{peel}(S,n))| = m$. As $\mathrm{hv}(\mathrm{peel}(S,n)) \subseteq \mathrm{peel}(S,n)$, we have $|\mathrm{peel}(S,n)| \ge m$ and

$$|\mathrm{peel}(S,n+1)| = |\mathrm{peel}(S,n) \setminus \mathrm{hv}(\mathrm{peel}(S,n))| = |\mathrm{peel}(S,n)| - m.$$

Hence $|\mathrm{peel}(S,n+1)| + m(n+1) = |\mathrm{peel}(S,n)| - m + mn + m = |S|$. □

The next corollary is not a Lean declaration; it assembles the preceding results into the layer decomposition of a set of exactly mt points.

Corollary 2.25 (layer decomposition; written step)**.** *Let $m, K, t \in \mathbb{N}$ with $0 < m$ and $K \le 2m$, let S be a finite set and c a point with $c \notin S$, and write $\rho = \rho_{c,2\pi/m}$. Assume that S is ρ-invariant and K-free and that $|S| = mt$. For $n < t$ put $L_n = \mathrm{hv}(\mathrm{peel}(S,n))$. Then:*

- *(a) for every $n < t$ there is $x_n \in \mathrm{peel}(S,n)$ with $L_n = \mathrm{orb}_m(\rho, x_n)$ and $|L_n| = m$;*
- *(b) for every $n \le t$, $\mathrm{peel}(S,n)$ is the disjoint union of $L_n, L_{n+1}, \dots, L_{t-1}$; in particular $\mathrm{peel}(S,t) = \emptyset$ and $S = L_0 \sqcup \cdots \sqcup L_{t-1}$;*
- *(c) $L_j \subseteq \mathrm{peel}(S, i+1) \subseteq \mathrm{conv}(L_i)$ for all $i < j < t$.*

Proof. If $t = 0$, then $S = \emptyset$; (a) and (c) are vacuous and (b) says $\mathrm{peel}(S,0) = \emptyset$. Let $t \ge 1$. Then $m(t \dot{-} 1) = mt - m < mt = |S|$, so Theorem 2.24 gives $|\mathrm{peel}(S,n)| = m(t-n)$ for $n \le t$. Thus $\mathrm{peel}(S,n) \ne \emptyset$ for $n < t$, and $\mathrm{peel}(S,t) = \emptyset$. (a) is Theorem 2.22(a),(b). For (b): for $n < t$, $L_n \subseteq \mathrm{peel}(S,n)$ and $\mathrm{peel}(S,n+1) = \mathrm{peel}(S,n) \setminus L_n$, so $\mathrm{peel}(S,n)$ is the disjoint union of L_n and $\mathrm{peel}(S,n+1)$; downward induction on n, starting from $\mathrm{peel}(S,t) = \emptyset$, gives (b). For (c): $L_j \subseteq \mathrm{peel}(S,j) \subseteq \mathrm{peel}(S,i+1)$ by Lemma 2.20 ($i+1 \le j$), and $\mathrm{peel}(S,i+1) \subseteq \mathrm{conv}(L_i)$ by Theorem 2.22(c), applicable since $\mathrm{peel}(S,i) \ne \emptyset$. □

2.7. Reduction to exactly mt points and the octagon theorems.

Theorem 2.26 (reduction to exactly mt points; Lean: `ErdosSzekeres.LayerLemma.convexNGon_of_layers`, `LAYERLEAN/LayerLemma.lean:1066`)**.** *Let $m, K, t \in \mathbb{N}$ with $0 < m$ and $K \le 2m$, and assume the premise*

- *(H) for every finite set T and every point c': if T is in general position, $c' \notin T$, T is $\rho_{c',2\pi/m}$-invariant and $|T| = m \cdot t$, then* `HasConvexNGon` K T.

Then for every finite set S and every point c: if S is in general position, S is $\rho_{c,2\pi/m}$-invariant, and $m \cdot (t \dot{-} 1) < |S \setminus \{c\}|$, then `HasConvexNGon` K S.

In Lean the bound centre in (H) is also called c; we rename it c'. The size condition is on $S \setminus \{c\}$ (Lean `S.erase c`), not on S, and $c \in S$ is allowed. The premise (H) is an explicit hypothesis of the theorem, and no theorem of `LAYERLEAN/LayerLemma.lean` proves it. In the two applications below it is supplied by the propositions `Sixfold66` and `Sevenfold70`, which are themselves hypotheses (Remark 2.30).

Proof. We follow the Lean proof. Write $\rho = \rho_{c,2\pi/m}$ and suppose that S is K-free. Put $S' = S \setminus \{c\}$. Then $c \notin S'$; S' is ρ-invariant by Lemma 2.15; and S' is K-free by Lemma 1.23(c), since $S' \subseteq S$.

Theorem 2.24 applies to S' with the given t (its size hypothesis $m(t \dot{-} 1) < |S'|$ is our hypothesis on $|S \setminus \{c\}|$), and with $n = t$ it gives $|\mathrm{peel}(S',t)| + mt = |S'|$. Let $T = S' \setminus \mathrm{peel}(S',t)$. Since $\mathrm{peel}(S',t) \subseteq S'$ (Lemma 2.19(b)), $|T| = |S'| - |\mathrm{peel}(S',t)| = mt$.

T is ρ-invariant. Let $y \in T$, so $y \in S'$ and $y \notin \mathrm{peel}(S',t)$. Then $\rho(y) \in S'$. Suppose $\rho(y) \in \mathrm{peel}(S',t)$. By Lemma 2.21 (applied to S'; $c \notin S'$) the set $\mathrm{peel}(S',t)$ is ρ-invariant, so $\rho^{m-1}(\rho(y)) \in \mathrm{peel}(S',t)$ by Lemma 2.2(a). As $y \in S'$, $y \ne c$, so $\mathrm{per}_\rho(y) = m$ (Lemma 2.16) and, by (P1) and $m \ge 1$, $\rho^{m-1}(\rho(y)) = \rho^m(y) = y$. Hence $y \in \mathrm{peel}(S',t)$, a contradiction. So $\rho(y) \in T$.

Moreover $T \subseteq S' \subseteq S$, so T is in general position (Lemma 1.23(b)), and $c \notin T$. By (H) with $c' = c$, `HasConvexNGon` K T, hence `HasConvexNGon` K S (Lemma 1.23(c)), contradicting the assumption. Hence `HasConvexNGon` K S. □

Definition 2.27 (the propositions `Sixfold66` and `Sevenfold70`; Lean: `ErdosSzekeres.LayerLemma.Sixfold66`, `LAYERLEAN/LayerLemma.lean:1102`, Lean: `ErdosSzekeres.LayerLemma.Sevenfold70`, `LAYERLEAN/LayerLemma.lean:1107`). `Sixfold66` is the proposition: for every finite set T and every point c, if T is in general position, $c \notin T$, T is $\rho_{c,2\pi/6}$-invariant and $|T| = 66$, then `HasConvexNGon` 8 T.

`Sevenfold70` is the proposition: for every finite set T and every point c, if T is in general position, $c \notin T$, T is $\rho_{c,2\pi/7}$-invariant and $|T| = 70$, then `HasConvexNGon` 8 T.

In both, the angle is written with a real numeral, `2 * Real.pi / 6` and `2 * Real.pi / 7` (Remark 2.31). Both are definitions of propositions; neither is proved in Lean (Remark 2.30).

Theorem 2.28 (sixfold octagon theorem, conditional on `Sixfold66`; Lean: `ErdosSzekeres.LayerLemma.sixfold_octagon`, `LAYERLEAN/LayerLemma.lean:1113`). *Assume the proposition* `Sixfold66` *(a hypothesis, not proved in Lean; Remark 2.30). Let S be a finite set of points and c a point such that S is in general position, S is $\rho_{c,2\pi/6}$-invariant, and $61 \le |S|$. Then* `HasConvexNGon` 8 *S: some 8 points of S are in convex position.*

Proof. We follow the Lean proof. Write $\rho = \rho_{c,2\pi/6}$; by Remark 2.31 this is $\rho_{c,2\pi/m}$ for the natural number $m = 6$.

Step 1. There is $p \in S$ with $p \neq c$: otherwise $S \subseteq \{c\}$ and $|S| \le 1 < 61$.

Step 2. $c \notin S$. This is Lemma 3.18 (Lean `centre_not_mem_of_even`) with $m = 6$, which is positive and even, applied to the ρ-invariant set S in general position and the point p of Step 1. For completeness, the argument in this case: suppose $c \in S$. By Lemma 2.14(d), $\rho^3 = \rho_{c,3\cdot 2\pi/6} = \rho_{c,\pi}$, and since $\cos\pi = -1$, $\sin\pi = 0$, Definition 2.13 gives $q := \rho^3(p) = 2c - p$, which lies in S by Lemma 2.2(a). The points p, c, q of S are pairwise distinct: $p \neq c$; $q = c$ and $q = p$ would each give $p = c$. But, by the formula of Definition 1.3,

$$\operatorname{orient}(p, c, 2c - p) = (c_0 - p_0)\cdot 2(c_1 - p_1) - (c_1 - p_1)\cdot 2(c_0 - p_0) = 0,$$

so $\{p, c, q\}$ is collinear by Lemma 1.20, contradicting general position (Definition 1.5).

Step 3. Apply Theorem 2.26 with $m = 6$, $K = 8$, $t = 11$: $0 < 6$ and $8 \le 12$. Premise (H) holds: for T and c' as there (the rotation $\rho_{c',2\pi/m}$ with $m = 6$ is $\rho_{c',2\pi/6}$, Remark 2.31), $|T| = 6 \cdot 11 = 66$, and `Sixfold66` gives `HasConvexNGon` 8 T. S is in general position and ρ-invariant. By Step 2, $S \setminus \{c\} = S$, so $6 \cdot (11 \dot{-} 1) = 60 < 61 \le |S| = |S \setminus \{c\}|$. Theorem 2.26 gives `HasConvexNGon` 8 S. □

Theorem 2.29 (sevenfold octagon theorem, conditional on `Sevenfold70`; Lean: `ErdosSzekeres.LayerLemma.sevenfold_octagon`, `LAYERLEAN/LayerLemma.lean:1136`). *Assume the proposition* `Sevenfold70` *(a hypothesis, not proved in Lean; Remark 2.30). Let S be a finite set of points and c a point such that S is in general position, S is $\rho_{c,2\pi/7}$-invariant, and $65 \le |S|$. Then* `HasConvexNGon` 8 *S. (The centre c may belong to S.)*

Proof. We follow the Lean proof. Apply Theorem 2.26 with $m = 7$, $K = 8$, $t = 10$ (Remark 2.31 for the angle): $0 < 7$ and $8 \le 14$. Premise (H) holds: for T and c' as there (the rotation is $\rho_{c',2\pi/7}$), $|T| = 7 \cdot 10 = 70$, and `Sevenfold70` gives `HasConvexNGon` 8 T. S is in general position and $\rho_{c,2\pi/7}$-invariant, and $|S \setminus \{c\}| \ge |S| - 1 \ge 64 > 63 = 7 \cdot (10 \dot{-} 1)$. Theorem 2.26 gives `HasConvexNGon` 8 S. No exclusion of the centre is needed. □

Remark 2.30 (status of `Sixfold66` and `Sevenfold70`). `Sixfold66` and `Sevenfold70` are Lean definitions of propositions (`def ... : Prop`), neither theorems nor axioms. Nothing in `LAYERLEAN/LayerLemma.lean` proves them; Theorems 2.28 and 2.29 take them as explicit hypotheses, so they do not enter the axiom lists of these theorems. Among the Lean files of the repository, the only theorems whose conclusion is `Sixfold66` are in `ENCBLEAN/geo/ES8Geo/Main.lean` and take further hypotheses (see below), and no theorem has the conclusion `Sevenfold70`. What is Lean-checked is the implication: `Sixfold66` implies that every $\rho_{c,2\pi/6}$-invariant set of at least 61 points in general position has 8 points in convex position, and `Sevenfold70` implies the same for $\rho_{c,2\pi/7}$ and at least 65 points. These octagon statements are therefore established here only conditionally.

In the project, the two propositions are supported as follows; none of this is part of the present section. The record `C6UPPER` gives the unsatisfiability of SAT encodings of octagon-free configurations of eleven nested regular hexagons in general position, by cake_lpr verdicts on cube formulas whose cover is checked there: the formula of encoder B (`C6UPPER/encb/gen_b2.py`) alone, through 64 cubes, and that of encoder A (`C7UPPER/gen.py`) with an added symmetry-breaking unit clause, through 32 cubes. The record `C7UPPER` gives the unsatisfiability of two separately implemented encodings of the same kind for ten nested regular heptagons (encoder A's with an added symmetry-breaking unit clause, encoder B's alone), each by a cake_lpr verdict. In the terms of this document these unsatisfiability results are *computer-assisted*: they rest on LRAT proofs checked by cake_lpr, outside Lean. The project's records call such a result a "certificate" (an unsatisfiability claim whose LRAT proof cake_lpr accepted; `WRITEUP/note/symmetric_extremal.tex`, lines 109–114); that is a label of the project, not one of

the labels of this document. The step from these unsatisfiability results to `Sixfold66` and `Sevenfold70` is the written soundness proof of the encodings in `WRITEUP/note/k8_soundness.tex` (a written proof, not refereed), which also covers the symmetry breaking and the cube cover. That note lists the gaps that remain, (G1)–(G4): the correspondence between the generating programs and their clauses is established by reading the source code; the passage from the formulas to `Sixfold66` and `Sevenfold70` is a written proof; the angular part of encoder A rests on binary64 floating-point arithmetic; and several of its written proofs have had no internal review. For $m = 6$ the unit `ENCBLEAN` also contains a Lean derivation of `Sixfold66`, `EncB.Geo.sixfold66_of_cubes64`; its `Sixfold66` is that of `ENCBLEAN/geo/ES8Geo/Capstone2.lean`, which contains a verbatim copy of the declarations of this file (over a byte-identical copy of `ErdosSzekeres/Definitions.lean`). Its hypothesis `EncB.CubesUnsat64` is the unsatisfiability of the 64 cube formulas built from a Lean definition of encoder B's formula (`EncB.encB`). Neither the identity of the printed texts of these formulas with the cube files that carry the cake_lpr verdicts nor the reading of those verdicts as unsatisfiability is Lean-checked; both rest on the trust items (E1)–(E6) listed under gap (G2) of that note.

The link with nested polygons is Corollary 2.25: a set T that violates `Sixfold66` is in general position, centre-free, $\rho_{c,2\pi/6}$-invariant, 8-free and has $66 = 6 \cdot 11$ points, so (with $m = 6$, $K = 8 \le 12$, $t = 11$) it is the disjoint union of eleven $\rho_{c,2\pi/6}$-orbits $L_0, \dots, L_{10}$ of 6 points each, with $L_j \subseteq \mathrm{conv}(L_i)$ for $i < j$; likewise a set violating `Sevenfold70` is the disjoint union of ten orbits of 7 points.

Remark 2.31 (real-cast numerals). In the general statements of this section the angle is `2 * Real.pi / m` with $m \in \mathbb{N}$ cast to $\mathbb{R}$ (C7). In `Sixfold66`, `Sevenfold70`, `sixfold_octagon` and `sevenfold_octagon` the angle is `2 * Real.pi / 6`, resp. `2 * Real.pi / 7`, where 6 and 7 are real numerals. The cast of the natural number 6 is the real number 6 (likewise for 7), so $\rho_{c,2\pi/6}$ is the same map in both readings; the Lean proofs bridge the two forms with `exact_mod_cast`, and the cardinality conditions $|T| = 6 \cdot 11$ and $|T| = 66$ (resp. $7 \cdot 10$ and 70) with `simpa`. Mathematically nothing changes.

Coverage. All declarations of this section's scope in `LAYERLEAN/LayerLemma.lean` are covered: `orbitFS` (Definition 2.1); `iterate_mem`, `mem_orbitFS_self`, `orbitFS_subset`, `orbitFS_invariant`, `card_orbitFS_of_minimalPeriod`, `mem_orbitFS_of_not_disjoint` (Lemma 2.2); `Invariant` (Definition 2.3); `hullVerts` (Definition 2.4); `mem_hullVerts`, `hullVerts_subset` (Lemma 2.5); `convexPosition_hullVerts`, `extreme_mem_hullVerts`, `subset_convexHull_hullVerts`, `hullVerts_nonempty` (Lemma 2.6; the proofs of (b)–(d) are elementary and differ from the Lean proofs, which use two Mathlib facts on extreme points for (b), the Krein–Milman theorem for (c) and the Krein–Milman lemma for (d)); `map_mem_hullVerts` (Lemma 2.7); `hullVerts_eq_orbitFS` (Theorem 2.8); `invariant_sdiff_hullVerts` (Lemma 2.9); `toC`, `ofC`, `crot` (Definition 2.11); `toC_re`, `toC_im`, `ofC_zero`, `ofC_one`, `toC_ofC`, `ofC_toC`, `toC_injective`, `toC_crot`, `crot_one`, `crot_comp`, `crot_iterate`, `crot_eq_self_iff`, `crot_injective` (Lemma 2.12); `rotAbout` (Definition 2.13); `rotAbout_eq_crot`, `rotAff`, `rotAbout_injective`, `rotAbout_iterate` (Lemma 2.14); `invariant_erase_centre` (Lemma 2.15); `minimalPeriod_rotAbout` (Lemma 2.16); `hullVerts_eq_orbitFS_rotAbout` (Theorem 2.17); `peel` (Definition 2.18); `peel_succ`, `peel_subset` (Lemma 2.19); `peel_antitone` (Lemma 2.20); `invariant_peel` (Lemma 2.21); `layers_are_orbits` (Theorem 2.22); `exists_layer` (Lemma 2.23); `card_peel` (Theorem 2.24); `convexNGon_of_layers` (Theorem 2.26); `Sixfold66`, `Sevenfold70` (Definition 2.27); `sixfold_octagon` (Theorem 2.28, conditional on `Sixfold66`); `sevenfold_octagon` (Theorem 2.29, conditional on `Sevenfold70`). Corollary 2.25 is a written step, not a Lean declaration. Nothing in the scope is deliberately left out. The declarations of lines 471–1032 of the same file (from `orient_rotAbout` to `dist2_le_of_later`) are outside this section's scope; `rotAbout_iterate` (line 465) is included here because it lies in the range of this section, and Step 2 of our proof of Theorem 2.28 uses it.

3. Geometry of the layers: nesting, chains and polar rules

This section proves the geometric facts about the convex layers of a rotation-invariant finite point set that the layered encoders rely on, together with the elementary identities behind them. In the statements about the layers (Lemmas 3.1 and 3.30, Theorems 3.16, 3.20 and 3.32) and in Lemma 3.18, S is a finite set of points that is mapped into itself by the counterclockwise rotation ρ by $2\pi/m$ ($m \ge 1$) about a point c; in all of them except Lemma 3.18, also $c \notin S$ and S has no K points in convex position for some $K \le 2m$. Theorem 3.28, the only other statement with a set S, involves no centre and no rotation. The section contains:

- *rotations and regular polygons* (§3.2): iterates of a rotation, the orientation of three vertices of a regular polygon (Lemmas 3.5, 3.6, 3.7, 3.12) and the position of the vertices relative to the tangent at a vertex (Lemmas 3.10, 3.11);

- *convexity* of the sets $\{q : 0 \le \mathrm{orient}(a, b, q)\}$ (closed half-planes when $a \ne b$) and of closed disks (§3.3);
- *nesting, centre and radii* (§3.4): if $m \ge 2$ and S is in general position, every point of a later layer lies strictly to the left of every edge $x \to \rho(x)$ of an earlier layer (Theorem 3.16); if $m \ge 2$ is even and S is in general position and has a point other than c, then $c \notin S$ (Lemma 3.18); the points of a layer and of all later layers are no farther from c than the vertices of that layer (Theorem 3.20);
- *fans and chains* (§3.5): a pivot followed by $n \ge 3$ chain points with positive fan and turn orientations, all in an open half-plane whose boundary line passes through the pivot, gives $n + 1$ points in convex position (Theorem 3.28; by its proof, the pivot and the chain points themselves); if $m \ge 3$ and S is in general position, all other points of the layers $\ge i$ lie strictly on the inner side of the tangent at a vertex of layer i (Lemma 3.30); hence, if moreover $K \ge 4$, no vertex of layer i is the pivot of a sequence of $K - 1$ points of the layers $\ge i$, all different from it, with positive fan and turn orientations (Theorem 3.32), which is the geometric content of the chain clauses (see the paragraph *Encoders* below);
- *polar rules* (§3.6): the orientation of three points given in polar coordinates about c (Lemma 3.35) and the angular rules A and B (Lemmas 3.36, 3.38).

Sources. Every Lean declaration that heads a statement of this section is in the file `LAYERLEAN/LayerLemma.lean` (namespace `ErdosSzekeres.LayerLemma`, 1148 lines, SHA-256 `afb8542dceb0`…), lines 465–1032; its README is `LAYERLEAN/README.md` (SHA-256 `9eaec241a1d2`…). The declarations of the same file that §3.1 recalls lie outside these lines and are restated in the section on the layer lemma. The Lean proofs of the declarations stated here use the following declarations of the main library; the foundations section restates them and proves the lemmas among them.

- The definitions `Point`, `GeneralPosition`, `ConvexPosition` and `HasConvexNGon` of `ErdosSzekeres/Definitions.lean` (SHA-256 `2425500eefd1`…).
- The orientation `ErdosSzekeres.orient` and the lemma `ErdosSzekeres.orient_cycle`, in `ErdosSzekeres/Orientation.lean`, lines 9 and 15 (SHA-256 `f67955ac7091`…).
- The lemma `ErdosSzekeres.generalPosition_iff_orient_ne_zero`, in `ErdosSzekeres/OrientationGeometry.lean:66` (SHA-256 `c7db9bbbb65e`…).
- The lemma `ErdosSzekeres.Suk.convexPosition_of_positive_triples`, in `ErdosSzekeres/Suk/OrientationClique.lean:9` (SHA-256 `cf63d5219b47`…).

The written proofs use the results listed under *Dependencies* below, among them Lemma 1.23(c) (Lean `HasConvexNGon.mono`, `ErdosSzekeres/Basic.lean:20`), which the Lean proofs do not use.

The file was checked with Lean 4.24.0 and Mathlib `f897ebcf` by `lake env lean` from the repository root; it is not part of the default `lake build` target. The log `LAYERLEAN/logs/build.log` (exit code 0, no messages) names no file hash; it was committed with the first version of the file (454 lines, commit `6429e9a70`) and was not updated by the five later commits that changed the file, so it is not a record of the present file, and we do not use it. The records for the present file are the review report `LEANEXT/review/REPORT.md` of 4 November 2026, which reports exit 0 and no messages for the reviewed file, whose SHA-256 it gives as `afb8542d`… (1148 lines), and the axiom re-check of 6 November 2026 (whose output is named below), which compiled `LAYERLEAN/logs/axioms_recheck_20261106/LayerLemmaAxioms.lean`, whose first 1148 lines are this file, with exit status 0; the files `README.txt` and `run.sh` of that directory record the toolchain and the command given above. The file contains no `sorry`, `admit`, `axiom` or `native_decide`.

Axioms. The log `LAYERLEAN/logs/axioms.log` (SHA-256 `3c4b7aac502e`…) prints the axioms of 16 of the 23 theorems of this section: all except `rotAbout_iterate`, `orient_rotAbout`, `convex_orient_nonneg`, `orient_eq_lin`, `lin_rotAbout`, `lin_orbit_pos` and `orient_next_prev_pos`. The review log `LEANEXT/review/axioms.log` of 4 November 2026 (SHA-256 `386c4f578725`…) prints all 23. It covers 59 declarations of the reviewed file (SHA-256 `afb8542d`…, the file above), namely all its `theorem` declarations except the 7 marked `@[simp]`; the file has 66 in all, and `LEANEXT/review/REPORT.md` speaks of "all 59 theorem declarations". In each of the two logs, every theorem of this section that it prints depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The re-check `LAYERLEAN/logs/axioms_recheck_20261106/out.txt` (SHA-256 `a59c3816cbe7`…) gives the same result for the 11 of them that it lists.

Rigor. Each lemma or theorem below whose head carries a Lean name is *Lean-checked*, with the axioms just listed; none of them takes an unproved proposition as a hypothesis, and none is a finite computation. Each definition whose head carries a Lean name restates that Lean definition. The proofs written here are *written proofs (not refereed)*. They follow the structure of the Lean proofs. Where Lean checks an algebraic identity or inequality with an automatic tactic (such as `ring`, `linear_combination`, `nlinarith` or `positivity`), the written proof derives it by hand; every other difference from the Lean proof is stated in a remark or in a

parenthetical note in the proof. Statements tagged "written step" are not Lean declarations, and they are proved here. Lemma 3.1 collects the opening steps of the Lean proofs of Theorems 3.16 and 3.20 and Lemma 3.30. Lemma 3.2 collects elementary facts about rotations that the Lean proofs obtain differently (through the complex form of the rotation, by direct computation, or through minimal periods). Lemma 3.22 is a general identity of which Lean checks only the three instances used, each by `ring`. Corollary 3.4 restates Lemma 3.3 for ρ and adds $\rho^m = \mathrm{id}$, which replaces a minimal-period argument of the Lean proof of Lemma 3.30.

Encoders. Several results below are the geometric facts behind parts of the project's layered encoders: Theorem 3.16 behind the nesting clauses, Theorem 3.32 behind the chain clauses, and Lemmas 3.36 and 3.38, with the weak radius order of Theorem 3.20 that rule B needs, behind the angular rules A and B, which the encoders apply to triples of points. Only the geometric facts are stated and proved here. That the encoders' programs emit exactly these clause families, and which rule they apply to which triple, are not part of `LAYERLEAN/LayerLemma.lean`; `LAYERLEAN/README.md` lists both under "What is not in Lean" and gives their status as "written proof, review and tests". For the encoders of the $k = 8$ bounds the code-to-clause step is recorded as gap (G1) of `WRITEUP/note/k8_soundness.tex`: it rests on a written proof (not refereed) by reading the code, with the Python interpreter trusted to run the programs as written. For encoder B at $m = 6$ that note also records a second route on which the step is Lean-checked for a Lean definition of the formula (`ENCBLEAN`). On that route the note lists, under gap (G2), the trust items (E1)–(E6), which are not Lean-checked: among them, that the texts printed from this definition are the 64 cube files that carry the cake_lpr verdicts is (E1), evidence from running compiled code and not a Lean theorem, and those 64 verdicts and their provenance are covered by (E2).

Dependencies. The section uses the conventions and facts of the foundations section (Definitions 1.1, 1.3, 1.5, 1.6, 1.7, 1.10, 1.12 and Lemmas 1.15, 1.20, 1.23, 1.35) and, from the section on the layer lemma, Definitions 2.4 and 2.18, Lemmas 2.6, 2.20, 2.21 and Theorem 2.17. These, except Lemma 1.35, which is used only in Step 3 of the proof of Theorem 3.28, are recalled in §3.1.

3.1. **Notation and the results used.**

Vectors. Points are elements of $\mathbb{R}^2$, written $p = (p_0, p_1)$ (Definition 1.1; Lean `Point`). For $X, Y \in \mathbb{R}^2$ put

$$\det(X,Y) = X_0Y_1 - X_1Y_0, \qquad \langle X,Y\rangle = X_0Y_0 + X_1Y_1, \qquad |X|^2 = \langle X,X\rangle, \qquad JX = (-X_1, X_0).$$

So J is the counterclockwise quarter turn, det is bilinear and antisymmetric, and $\det(X,Y) = \langle JX, Y\rangle$. In particular

$$\begin{gathered}\det(X,X) = 0, \qquad \det(X,JX) = |X|^2, \qquad \det(JX,X) = -|X|^2,\\ \det(JX,JX) = 0, \qquad \langle X, JX\rangle = 0.\end{gathered} \tag{1}$$

Orientation. We use the following facts about orient (Definition 1.3).

- (O1) $\mathrm{orient}(a,b,d) = (b_0-a_0)(d_1-a_1) - (b_1-a_1)(d_0-a_0) = \det(b-a,\, d-a)$.
- (O2) orient is invariant under cyclic permutations of its three arguments (Lemma 1.15(b)). Hence $\mathrm{orient}(a,b,o) = \mathrm{orient}(o,a,b) = \det(a-o,\, b-o)$ for all points a, b, o.
- (O3) $\mathrm{orient}(a,b,d) = 0$ whenever two of a, b, d coincide. Indeed, by (O1), $\mathrm{orient}(a,a,d) = \det(0, d-a) = 0$, $\mathrm{orient}(a,b,a) = \det(b-a, 0) = 0$ and $\mathrm{orient}(a,b,b) = \det(b-a, b-a) = 0$.
- (O4) For fixed a, b the map $q \mapsto \mathrm{orient}(a,b,q) = \det(b-a,\, q-a)$ is affine: if $u + v = 1$ then $\mathrm{orient}(a,b,uq_1 + vq_2) = u\,\mathrm{orient}(a,b,q_1) + v\,\mathrm{orient}(a,b,q_2)$, because $(uq_1 + vq_2) - a = u(q_1 - a) + v(q_2 - a)$ and det is linear in its second argument.

Rotations. For $\theta \in \mathbb{R}$ let R_θ be the matrix with rows $(\cos\theta, -\sin\theta)$ and $(\sin\theta, \cos\theta)$, so that $R_\theta X = \cos\theta\, X + \sin\theta\, JX$. Definition 1.12 reads

$$\mathrm{rotAbout}(c,\theta)(p) = c + R_\theta(p-c) = \big(c_0 + \cos\theta\,(p_0-c_0) - \sin\theta\,(p_1-c_1),\ c_1 + \sin\theta\,(p_0-c_0) + \cos\theta\,(p_1-c_1)\big).$$

For a map f and $j \in \mathbb{N}$, f^j is the j-fold iterate: $f^0 = \mathrm{id}$ and $f^{j+1} = f \circ f^j$ $(= f^j \circ f$; Lean `f^[j]`$)$. For $m \in \mathbb{N}$ the angle $2\pi/m$ is the real number $2\pi/(m : \mathbb{R})$ (Lean `2 * Real.pi / m`, with m cast to $\mathbb{R}$; Lean's convention would give 0 for $m = 0$, but every statement of this section in which $2\pi/m$ occurs assumes $m \geq 1$). Whenever c and m are given we write

$$\rho = \mathrm{rotAbout}(c, 2\pi/m).$$

A natural-number subtraction such as $m - 1$ or $K - 1$ is truncated at 0 in Lean; in every use below the minuend is at least the subtrahend, so it is the ordinary difference.

Sine and cosine. Besides the values at 0, π and 2π, the addition, subtraction and double-angle formulas, $\sin^2+\cos^2=1$, the oddness of the sine and the 2π-periodicity, we use two standard facts about the real sine and cosine (see, e.g., W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., Chapter 8; in Mathlib `Real.sin_pos_of_pos_of_lt_pi`, `Real.cos_le_one` and `Real.cos_eq_one_iff`):

(S1) $\sin\theta>0$ for $0<\theta<\pi$; hence $\sin\theta\ge 0$ for $0\le\theta\le\pi$, as $\sin 0=\sin\pi=0$;
(S2) $\cos\theta\le 1$ for every $\theta\in\mathbb{R}$, and $\cos\theta=1$ if and only if $\theta\in 2\pi\mathbb{Z}$.

Hypotheses on S. All point sets are finite (Lean `Finset Point`). "ρ maps S into S" is the Lean hypothesis `Invariant (rotAbout c (2 * Real.pi / m)) S`, i.e. $\rho(p)\in S$ for all $p\in S$ (Definition 1.10); by induction on i it implies $\rho^i(p)\in S$ for all $p\in S$ and $i\in\mathbb{N}$. "S is K-free" means $\neg$ `HasConvexNGon K S`: no K-element subset of S is in convex position (Definitions 1.6, 1.7). By Lemma 1.23(c), every subset of a K-free set is K-free. "General position" is Definition 1.5; by Lemma 1.20, S is in general position if and only if $\mathrm{orient}(a,b,d)\neq 0$ for all pairwise distinct $a,b,d\in S$.

Convexity. A set $C\subseteq\mathbb{R}^2$ is convex (Mathlib `Convex`) if $uy_1+vy_2\in C$ whenever $y_1,y_2\in C$, $u,v\ge 0$ and $u+v=1$. The convex hull $\mathrm{conv}(A)$ is the smallest convex set containing A; so $A\subseteq C$ with C convex implies $\mathrm{conv}(A)\subseteq C$ (Mathlib `convexHull_min`).

Layers. $\mathrm{hv}(X)=\{x\in X : x\notin\mathrm{conv}(X\setminus\{x\})\}\subseteq X$ (Definition 2.4), and $\mathrm{peel}(S,0)=S$, $\mathrm{peel}(S,n+1)=\mathrm{peel}(S,n)\setminus\mathrm{hv}(\mathrm{peel}(S,n))$ (Definition 2.18). For a map f, a number $k\in\mathbb{N}$ and a point x, Lean's `orbitFS f k x` (`LAYERLEAN/LayerLemma.lean:48`) is the finite set $\{f^j(x):0\le j<k\}$. We use the following results of the section on the layer lemma.

(L1) For every finite X: $X\subseteq\mathrm{conv}(\mathrm{hv}(X))$ (Lemma 2.6(c), `subset_convexHull_hullVerts`).
(L2) If $n\le n'$ then $\mathrm{peel}(S,n')\subseteq\mathrm{peel}(S,n)$ (Lemma 2.20); in particular $\mathrm{peel}(S,n)\subseteq\mathrm{peel}(S,0)=S$.
(L3) If $0<m$, $c\notin S$ and ρ maps S into S, then ρ maps $\mathrm{peel}(S,n)$ into $\mathrm{peel}(S,n)$ for every n (Lemma 2.21, `invariant_peel`).
(L4) If $0<m$, $K\le 2m$, $c\notin S$, ρ maps S into S, S is K-free and $x\in\mathrm{hv}(S)$, then $\mathrm{hv}(S)=\{\rho^j(x):0\le j<m\}$ (Theorem 2.17, `hullVerts_eq_orbitFS_rotAbout`).

The Lean proofs of Theorems 3.16 and 3.20 and of Lemma 3.30 each begin with the same few steps; we collect them once.

Lemma 3.1 (layers of a layered configuration; written step)**.** *Let $m,K\in\mathbb{N}$ with $0<m$ and $K\le 2m$, let S be a finite set and c a point with $c\notin S$, let $\rho=\mathrm{rotAbout}(c,2\pi/m)$ map S into S, and let S be K-free. Let $n\in\mathbb{N}$, put $L=\mathrm{hv}(\mathrm{peel}(S,n))$, and let $x\in L$. Then:*

(a) $L=\{\rho^j(x):0\le j<m\}$*; in particular* $\rho^j(x)\in L$ *for* $0\le j<m$*;*
(b) $L\subseteq\mathrm{peel}(S,n)\subseteq S$*, and* $x\neq c$*;*
(c) $\mathrm{peel}(S,n')\subseteq\mathrm{conv}(L)$ *for every* $n'\ge n$*;*
(d) $\mathrm{peel}(S,n')\cap L=\emptyset$ *for every* $n'>n$*.*

Proof. By (L2), $\mathrm{peel}(S,n)\subseteq S$. Hence $c\notin\mathrm{peel}(S,n)$, and $\mathrm{peel}(S,n)$ is K-free (a K-element subset of $\mathrm{peel}(S,n)$ in convex position would be one of S). By (L3), ρ maps $\mathrm{peel}(S,n)$ into itself. So (L4) applies to the finite set $\mathrm{peel}(S,n)$ and its hull vertex x, and gives (a). (b) $L\subseteq\mathrm{peel}(S,n)$ by the definition of hv, and $\mathrm{peel}(S,n)\subseteq S$; since $x\in L\subseteq S$ and $c\notin S$, $x\neq c$. (c) If $n'\ge n$ then $\mathrm{peel}(S,n')\subseteq\mathrm{peel}(S,n)\subseteq\mathrm{conv}(\mathrm{hv}(\mathrm{peel}(S,n)))=\mathrm{conv}(L)$ by (L2) and (L1). (d) If $n'>n$ then $n+1\le n'$, so $\mathrm{peel}(S,n')\subseteq\mathrm{peel}(S,n+1)=\mathrm{peel}(S,n)\setminus L$ by (L2) and the definition of peel. □

3.2. **Rotations and regular polygons.**

Lemma 3.2 (rotation facts; written step)**.** *For all points c,p and all $\alpha,\beta,\theta\in\mathbb{R}$:*

(a) $\mathrm{rotAbout}(c,0)(p)=p$*;*
(b) $\mathrm{rotAbout}(c,\alpha)\big(\mathrm{rotAbout}(c,\beta)(p)\big)=\mathrm{rotAbout}(c,\alpha+\beta)(p)$*;*
(c) $\mathrm{rotAbout}(c,2\pi)(p)=p$*;*
(d) $|\mathrm{rotAbout}(c,\theta)(p)-c|^2=|p-c|^2$*.*

Proof. Put $u=p-c$, so that $\mathrm{rotAbout}(c,\theta)(p)=c+R_\theta u$. (a), (c): $\cos 0=\cos 2\pi=1$ and $\sin 0=\sin 2\pi=0$, so $R_0u=R_{2\pi}u=u$. (b) $\mathrm{rotAbout}(c,\beta)(p)-c=R_\beta u$, hence $\mathrm{rotAbout}(c,\alpha)(\mathrm{rotAbout}(c,\beta)(p))=c+R_\alpha R_\beta u$. The product $R_\alpha R_\beta$ has rows $(\cos\alpha\cos\beta-\sin\alpha\sin\beta,\ -(\sin\alpha\cos\beta+\cos\alpha\sin\beta))$ and $(\sin\alpha\cos\beta+\cos\alpha\sin\beta,\ \cos\alpha\cos\beta-\sin\alpha\sin\beta)$, which is $R_{\alpha+\beta}$ by the addition formulas $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$ and $\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$. (d) $|R_\theta u|^2=(\cos\theta\,u_0-\sin\theta\,u_1)^2+(\sin\theta\,u_0+\cos\theta\,u_1)^2=(\cos^2\theta+\sin^2\theta)(u_0^2+u_1^2)=|u|^2$; the mixed terms cancel. □

Lemma 3.3 (iterates of the rotation about a point; Lean: `ErdosSzekeres.LayerLemma.rotAbout_iterate`, `LAYERLEAN/LayerLemma.lean:465`)**.** *For every point c, every $\alpha \in \mathbb{R}$ and every $j \in \mathbb{N}$,*

$$\big(\mathrm{rotAbout}(c,\alpha)\big)^{j} = \mathrm{rotAbout}(c,\, j\alpha)$$

as maps $\mathbb{R}^2 \to \mathbb{R}^2$, where j is cast to $\mathbb{R}$.

Proof. Induction on j. For $j = 0$ the left side is the identity by definition of the iterate, and the right side is $\mathrm{rotAbout}(c,0) = \mathrm{id}$ by Lemma 3.2(a). If the claim holds for j, then

$$\big(\mathrm{rotAbout}(c,\alpha)\big)^{j+1} = \mathrm{rotAbout}(c,\alpha) \circ \mathrm{rotAbout}(c,j\alpha) = \mathrm{rotAbout}(c,\alpha+j\alpha) = \mathrm{rotAbout}(c,(j+1)\alpha)$$

by Lemma 3.2(b). (Lean argues instead with the complex form: in the complex coordinate z, with $\gamma = c_0 + ic_1$, the map $\mathrm{rotAbout}(c,\alpha)$ is $z \mapsto \gamma + e^{i\alpha}(z-\gamma)$ (`rotAbout_eq_crot`, line 302), whose j-th iterate is $z \mapsto \gamma + (e^{i\alpha})^j(z-\gamma)$ (`crot_iterate`, line 267), and $(e^{i\alpha})^j = e^{ij\alpha}$.) □

Corollary 3.4 (ρ^m is the identity; written step)**.** *Let c be a point, $m \in \mathbb{N}$ with $0 < m$, and $\rho = \mathrm{rotAbout}(c, 2\pi/m)$. Then $\rho^j = \mathrm{rotAbout}(c, 2\pi j/m)$ for every $j \in \mathbb{N}$, and $\rho^m(p) = p$ for every point p.*

Proof. By Lemma 3.3, $\rho^j = \mathrm{rotAbout}(c, j\cdot(2\pi/m))$. For $j = m$ the angle is $m\cdot(2\pi/m) = 2\pi$ because $(m : \mathbb{R}) \neq 0$, and $\mathrm{rotAbout}(c,2\pi) = \mathrm{id}$ by Lemma 3.2(c). □

Lemma 3.5 (orientation of a regular-polygon triple; Lean: `ErdosSzekeres.LayerLemma.orient_rotAbout`, `LAYERLEAN/LayerLemma.lean:472`)**.** *For all points c, x and all $\alpha, \beta \in \mathbb{R}$,*

$$\mathrm{orient}\big(x, \mathrm{rotAbout}(c,\alpha)(x), \mathrm{rotAbout}(c,\beta)(x)\big) = \big((x_0-c_0)^2 + (x_1-c_1)^2\big)\big((\cos\alpha - 1)\sin\beta - \sin\alpha\,(\cos\beta - 1)\big).$$

Proof. Put $u = x - c$, $b = \mathrm{rotAbout}(c,\alpha)(x)$ and $d = \mathrm{rotAbout}(c,\beta)(x)$. Then $b - x = R_\alpha u - u = (\cos\alpha - 1)u + \sin\alpha\, Ju$ and $d - x = (\cos\beta - 1)u + \sin\beta\, Ju$. By (O1), bilinearity of det and (1),

$$\begin{aligned}\mathrm{orient}(x,b,d) = \det(b-x, d-x) &= (\cos\alpha-1)(\cos\beta-1)\det(u,u) + (\cos\alpha-1)\sin\beta\,\det(u,Ju)\\ &\quad + \sin\alpha\,(\cos\beta-1)\det(Ju,u) + \sin\alpha\sin\beta\,\det(Ju,Ju)\\ &= \big((\cos\alpha-1)\sin\beta - \sin\alpha\,(\cos\beta-1)\big)|u|^2,\end{aligned}$$

and $|u|^2 = (x_0-c_0)^2 + (x_1-c_1)^2$. No relation between cos and sin is used: this is a polynomial identity in $\cos\alpha, \sin\alpha, \cos\beta, \sin\beta$ and the coordinates, which is how Lean proves it (`ring`). □

Lemma 3.6 (a product formula for the orientation factor; Lean: `ErdosSzekeres.LayerLemma.trig_layer`, `LAYERLEAN/LayerLemma.lean:480`)**.** *For all $s, t \in \mathbb{R}$,*

$$(\cos 2t - 1)\sin 2s - \sin 2t\,(\cos 2s - 1) = 4\sin t\,\sin s\,\sin(s-t).$$

Proof. By the double-angle formulas, $\cos 2t - 1 = -2\sin^2 t$, $\cos 2s - 1 = -2\sin^2 s$, $\sin 2s = 2\sin s\cos s$ and $\sin 2t = 2\sin t\cos t$. Hence the left side equals

$$-4\sin^2 t\,\sin s\cos s + 4\sin t\cos t\,\sin^2 s = 4\sin t\,\sin s\,(\sin s\cos t - \cos s\sin t) = 4\sin t\,\sin s\,\sin(s-t).$$

(Lean rewrites with $\cos 2x = 2\cos^2 x - 1$ and $\sin 2x = 2\sin x\cos x$ for $x = s, t$ and with the subtraction formula for $\sin(s-t)$, and closes the remaining polynomial identity by `linear_combination` with $\sin^2 + \cos^2 = 1$ for s and for t.) □

Lemma 3.7 (vertices lie weakly left of each edge; Lean: `ErdosSzekeres.LayerLemma.orient_regular_nonneg`, `LAYERLEAN/LayerLemma.lean:488`)**.** *Let c, x be points, $m \in \mathbb{N}$ with $0 < m$, $j \in \mathbb{N}$ with $j \le m$, and $\rho = \mathrm{rotAbout}(c, 2\pi/m)$. Then*

$$0 \le \mathrm{orient}\big(x, \rho(x), \rho^j(x)\big).$$

Proof. By Lemma 3.3, $\rho^j(x) = \mathrm{rotAbout}(c,\beta)(x)$ with $\beta = j\cdot(2\pi/m)$. By Lemma 3.5 with $\alpha = 2\pi/m$,

$$\mathrm{orient}(x, \rho(x), \rho^j(x)) = |x-c|^2\,F, \qquad F = (\cos\alpha - 1)\sin\beta - \sin\alpha\,(\cos\beta - 1).$$

The first factor is ≥ 0; we show $F \ge 0$. If $j = 0$ then $\beta = 0$, so $\sin\beta = 0 = \cos\beta - 1$ and $F = 0$. Let $1 \le j \le m$ and put $t = \pi/m$. Then $\alpha = 2t$ and $\beta = 2(jt)$, so Lemma 3.6 with $s = jt$ gives

$$F = 4\sin t\,\sin(jt)\,\sin(jt - t).$$

Since $m \ge 1$, $0 < t \le \pi$; since $j \le m$, $0 \le jt = j\pi/m \le \pi$; and since $j \ge 1$, $0 \le jt - t = (j-1)t \le jt \le \pi$. By (S1) the sine is ≥ 0 on $[0,\pi]$, so all three factors are ≥ 0, and $F \ge 0$. □

Remark 3.8. The proof shows $\mathrm{orient}(x, \rho(x), \rho^j(x)) = 4|x-c|^2 \sin(\pi/m)\sin(j\pi/m)\sin((j-1)\pi/m)$ for $1 \le j \le m$: every vertex of the regular m-gon $\{\rho^j(x)\}$ lies weakly to the left of the directed edge $x \to \rho(x)$. (If moreover $x \ne c$ and $m \ge 2$, then by (S1) $\sin(\pi/m) > 0$, $\sin(j\pi/m) = 0$ only for $j = m$ and $\sin((j-1)\pi/m) = 0$ only for $j = 1$, so the value is 0 exactly for $j \in \{1, m\}$; this refinement is not used.)

Definition 3.9 (lin; Lean: `ErdosSzekeres.LayerLemma.lin`, `LAYERLEAN/LayerLemma.lean:625`)**.** For points w, a, y,

$$\mathrm{lin}(w, a, y) = w_0(y_0 - a_0) + w_1(y_1 - a_1) = \langle w,\, y - a\rangle.$$

For $w \ne 0$, $\{y : \mathrm{lin}(w, a, y) > 0\}$ is the open half-plane bounded by the line through a orthogonal to w, on the side into which w points. Note $\mathrm{lin}(w, a, a) = 0$.

Lemma 3.10 (the tangent functional on the orbit; Lean: `ErdosSzekeres.LayerLemma.lin_rotAbout`, `LAYERLEAN/LayerLemma.lean:794`)**.** *For all points c, p and every $\beta \in \mathbb{R}$,*

$$\mathrm{lin}\big(c - p,\ p,\ \mathrm{rotAbout}(c, \beta)(p)\big) = \big((p_0 - c_0)^2 + (p_1 - c_1)^2\big)(1 - \cos\beta),$$

where $c - p$ is the coordinatewise difference.

Proof. Put $u = p - c$; then $c - p = -u$ and $\mathrm{rotAbout}(c, \beta)(p) - p = R_\beta u - u = (\cos\beta - 1)u + \sin\beta\, Ju$. Hence, by (1),

$$\mathrm{lin}(c-p, p, \mathrm{rotAbout}(c,\beta)(p)) = \langle -u, (\cos\beta-1)u + \sin\beta\, Ju\rangle = (1-\cos\beta)|u|^2 - \sin\beta\,\langle u, Ju\rangle = (1-\cos\beta)|u|^2. \quad \square$$

Lemma 3.11 (other vertices lie strictly inside the tangent; Lean: `ErdosSzekeres.LayerLemma.lin_orbit_pos`, `LAYERLEAN/LayerLemma.lean:801`)**.** *Let c, p be points with $p \ne c$, let $m \in \mathbb{N}$ with $0 < m$, let $j \in \mathbb{N}$ with $0 < j$ and $j < m$, and let $\rho = \mathrm{rotAbout}(c, 2\pi/m)$. Then*

$$0 < \mathrm{lin}\big(c - p,\ p,\ \rho^j(p)\big).$$

Proof. By Lemma 3.3, $\rho^j(p) = \mathrm{rotAbout}(c, \theta)(p)$ with $\theta = j \cdot (2\pi/m)$, so by Lemma 3.10 the value is $|p - c|^2(1 - \cos\theta)$. Since $p \ne c$, one of $p_0 - c_0$, $p_1 - c_1$ is non-zero, so $|p - c|^2 > 0$. Since $0 < j < m$, $0 < \theta < 2\pi$. By (S2), $\cos\theta \le 1$, and $\cos\theta = 1$ only for $\theta \in 2\pi\mathbb{Z}$, which has no point in $(0, 2\pi)$; hence $1 - \cos\theta > 0$, and the product is positive. $\square$

Lemma 3.12 (next and previous vertex; Lean: `ErdosSzekeres.LayerLemma.orient_next_prev_pos`, `LAYERLEAN/LayerLemma.lean:830`)**.** *Let c, p be points with $p \ne c$, let $m \in \mathbb{N}$ with $3 \le m$, and let $\rho = \mathrm{rotAbout}(c, 2\pi/m)$. Then*

$$0 < \mathrm{orient}\big(p, \rho(p), \rho^{m-1}(p)\big).$$

Proof. As $m \ge 1$, the natural number $m-1$ has real cast $(m : \mathbb{R}) - 1$. By Lemma 3.3, $\rho^{m-1}(p) = \mathrm{rotAbout}(c, \beta)(p)$ with $\beta = (m - 1) \cdot (2\pi/m)$. By Lemma 3.5 with $\alpha = 2\pi/m$, the orientation equals $|p - c|^2 F$ with F as in the proof of Lemma 3.7. Put $t = \pi/m$, so $mt = \pi$, $\alpha = 2t$ and $\beta = 2((m-1)t)$. Lemma 3.6 with $s = (m-1)t$ gives

$$F = 4\sin t\, \sin((m-1)t)\, \sin((m-1)t - t) = 4\sin t\, \sin(\pi - t)\, \sin(\pi - 2t).$$

Since $m \ge 3$, $0 < t \le \pi/3$; hence t, $\pi - t$ and $\pi - 2t$ (which is $\ge \pi/3$) all lie in the open interval $(0, \pi)$, where the sine is positive by (S1). So $F > 0$; also $|p - c|^2 > 0$ as $p \ne c$. The product is positive. $\square$

3.3. **Convexity of half-planes and disks.**

Lemma 3.13 ($\{\mathrm{orient}(a, b, \cdot) \ge 0\}$ is convex; Lean: `ErdosSzekeres.LayerLemma.convex_orient_nonneg`, `LAYERLEAN/LayerLemma.lean:516`)**.** *For all points a, b the set $\{q \in \mathbb{R}^2 : 0 \le \mathrm{orient}(a, b, q)\}$ is convex.*

Proof. Let q_1, q_2 be in the set and $u, v \ge 0$ with $u + v = 1$. By (O4), $\mathrm{orient}(a, b, uq_1 + vq_2) = u\,\mathrm{orient}(a, b, q_1) + v\,\mathrm{orient}(a, b, q_2)$, a sum of two products of non-negative numbers. $\square$

For $a \ne b$ this set is the closed half-plane to the left of the directed line $a \to b$, the line included; for $a = b$ it is the whole plane, since $\mathrm{orient}(a, a, q) = 0$ by (O3).

Definition 3.14 (dist2; Lean: `ErdosSzekeres.LayerLemma.dist2`, `LAYERLEAN/LayerLemma.lean:999`)**.** For points c, y, $\mathrm{dist2}(c, y) = (y_0 - c_0)^2 + (y_1 - c_1)^2 = |y - c|^2$, the squared Euclidean distance (defined by this formula, not through a norm).

Lemma 3.15 (disks; Lean: `ErdosSzekeres.LayerLemma.convex_disk`, `LAYERLEAN/LayerLemma.lean:1002`)**.** *For every point c and every $R \in \mathbb{R}$ the set $\{y \in \mathbb{R}^2 : \mathrm{dist2}(c, y) \le R\}$ is convex.*

Proof. Let y_1, y_2 be in the set and $u, v \geq 0$ with $u + v = 1$. Put $A = y_1 - c$ and $B = y_2 - c$. Since $u + v = 1$, $(uy_1 + vy_2) - c = uA + vB$, and, using $1 - u = v$ and $1 - v = u$,

$$u|A|^2 + v|B|^2 - |uA + vB|^2 = u(1-u)|A|^2 + v(1-v)|B|^2 - 2uv\langle A, B\rangle = uv\,|A - B|^2 \geq 0.$$

Hence $\mathrm{dist2}(c, uy_1 + vy_2) = |uA + vB|^2 \leq u|A|^2 + v|B|^2 \leq uR + vR = R$. (Lean proves the same inequality with `nlinarith`, from the non-negativity of the squares of the coordinate differences of y_1, y_2 and of uv.) □

For $R \geq 0$ the set is the closed disk of radius $\sqrt{R}$ about c; for $R < 0$ it is empty.

3.4. **Strict nesting, the centre, and the radius order.**

Theorem 3.16 (strict nesting of the convex layers; Lean: `ErdosSzekeres.LayerLemma.strict_nesting`, `LAYERLEAN/LayerLemma.lean:529`)**.** *Let S be a finite set of points, c a point, and $m, K \in \mathbb{N}$ with $2 \leq m$ and $K \leq 2m$; let $\rho = \mathrm{rotAbout}(c, 2\pi/m)$. Assume that $c \notin S$, that ρ maps S into S, that S is K-free, and that S is in general position. Let $n, n' \in \mathbb{N}$ with $n < n'$, let $x \in \mathrm{hv}(\mathrm{peel}(S, n))$ and $y \in \mathrm{peel}(S, n')$. Then*

$$0 < \mathrm{orient}\big(x, \rho(x), y\big).$$

Proof. Since $m \geq 2$, also $0 < m$, so Lemma 3.1 applies to n and x. Put $L = \mathrm{hv}(\mathrm{peel}(S, n))$. By Lemma 3.1(a), $L = \{\rho^j(x) : 0 \leq j < m\}$; by (c), applied with $n' \geq n$, $y \in \mathrm{conv}(L)$; by (d), applied with $n' > n$, $y \notin L$.

Weak inequality. Let $H = \{q : 0 \leq \mathrm{orient}(x, \rho(x), q)\}$. Every element of L is $\rho^j(x)$ with $j < m$, in particular $j \leq m$, so it lies in H by Lemma 3.7. H is convex by Lemma 3.13 (with $a = x$, $b = \rho(x)$). Hence $\mathrm{conv}(L) \subseteq H$, and $y \in H$, that is, $0 \leq \mathrm{orient}(x, \rho(x), y)$.

Strict inequality. As $1 < m$, $\rho(x) = \rho^1(x) \in L$ by Lemma 3.1(a). The three points $x, \rho(x), y$ lie in S: $x, \rho(x) \in L \subseteq S$ by Lemma 3.1(b), and $y \in \mathrm{peel}(S, n') \subseteq S$ by (L2). They are pairwise distinct:

- $x \neq \rho(x)$: $x \neq c$ by Lemma 3.1(b), so Lemma 3.11 with $p = x$ and $j = 1$ ($0 < 1 < m$) gives $0 < \mathrm{lin}(c - x, x, \rho(x))$, whereas $\mathrm{lin}(c - x, x, x) = 0$;
- $y \neq x$ and $y \neq \rho(x)$, because $x, \rho(x) \in L$ and $y \notin L$.

By general position and Lemma 1.20, $\mathrm{orient}(x, \rho(x), y) \neq 0$. Together with the weak inequality, $0 < \mathrm{orient}(x, \rho(x), y)$. □

Remark 3.17. (1) Lean proves $x \neq \rho(x)$ differently: $x \neq c$ has minimal period $m \geq 2$ under ρ (Lemma 2.16), while a fixed point of ρ has minimal period 1. (2) For $m = 2$ the hypotheses of Theorem 3.16 are never all satisfied. Indeed, then $L = \{x, \rho(x)\}$ and $y \in \mathrm{conv}\{x, \rho(x)\}$. The segment $\{(1-\lambda)x + \lambda\rho(x) : 0 \leq \lambda \leq 1\}$ contains L and is convex (a convex combination of the points with parameters λ_1, λ_2 is the point with the parameter given by the same combination of λ_1, λ_2), so it contains $\mathrm{conv}\{x, \rho(x)\}$, and $y = (1-\lambda)x + \lambda\rho(x)$ with $0 \leq \lambda \leq 1$. By (O4) and (O3) $\mathrm{orient}(x, \rho(x), y) = (1-\lambda)\,\mathrm{orient}(x, \rho(x), x) + \lambda\,\mathrm{orient}(x, \rho(x), \rho(x)) = 0$, contradicting the conclusion. So for $m = 2$ the theorem says that, under its hypotheses on S, $\mathrm{peel}(S, n') = \emptyset$ for all $n' > n$ as soon as $\mathrm{hv}(\mathrm{peel}(S, n)) \neq \emptyset$. (3) The theorem is the orientation statement behind the nesting clauses of the layered encoders. That the encoders emit exactly these clauses is not part of `LAYERLEAN/LayerLemma.lean`; see the paragraph *Encoders* at the beginning of this section.

Lemma 3.18 (even orders exclude the centre; Lean: `ErdosSzekeres.LayerLemma.centre_not_mem_of_even`, `LAYERLEAN/LayerLemma.lean:582`)**.** *Let S be a finite set, c a point, and $m \in \mathbb{N}$ with $0 < m$ and m even; let $\rho = \mathrm{rotAbout}(c, 2\pi/m)$. Assume that ρ maps S into S and that S is in general position. Let p be a point with $p \in S$ and $p \neq c$. Then $c \notin S$.*

Proof. Suppose $c \in S$. Since m is even, $m = k + k$ for some $k \in \mathbb{N}$ (Mathlib's `Even`); $k \geq 1$ because $m > 0$, and the natural-number quotient $m/2$ equals k. Let $q = \rho^k(p)$ (Lean: $\rho^{m/2}(p)$). Since ρ maps S into S, $q \in S$. By Corollary 3.4, $\rho^k = \mathrm{rotAbout}(c, 2\pi k/m)$, and $2\pi k/m = 2\pi k/(2k) = \pi$. As $\cos\pi = -1$ and $\sin\pi = 0$,

$$q = \mathrm{rotAbout}(c, \pi)(p) = \big(c_0 - (p_0 - c_0),\ c_1 - (p_1 - c_1)\big) = 2c - p.$$

By (O1), $\mathrm{orient}(p, c, q) = \det(c - p,\ 2c - 2p) = 2\det(c - p,\ c - p) = 0$. The points p, c, q are pairwise distinct: $p \neq c$ by hypothesis; $q = c$ would mean $2c - p = c$, i.e. $p = c$; and $p = q$ would mean $p = 2c - p$, i.e. $p = c$. Since $p, c, q \in S$, general position and Lemma 1.20 give $\mathrm{orient}(p, c, q) \neq 0$, a contradiction. Hence $c \notin S$. □

Remark 3.19. There is no K-freeness hypothesis and no hypothesis $c \notin S$. The hypothesis that S has a point $p \neq c$ cannot be dropped: $S = \{c\}$ is mapped into itself by ρ, is in general position (it has no three distinct points), and contains c.

Theorem 3.20 (the radius order of the layers; Lean: `ErdosSzekeres.LayerLemma.dist2_le_of_later`, `LAYERLEAN/LayerLemma.lean:1012`)**.** *Let S be a finite set, c a point, and $m, K \in \mathbb{N}$ with $0 < m$ and $K \le 2m$; let $\rho = \mathrm{rotAbout}(c, 2\pi/m)$. Assume that $c \notin S$, that ρ maps S into S, and that S is K-free. Let $n, n' \in \mathbb{N}$ with $n \le n'$, let $x \in \mathrm{hv}(\mathrm{peel}(S, n))$ and $y \in \mathrm{peel}(S, n')$. Then*

$$\mathrm{dist2}(c, y) \le \mathrm{dist2}(c, x).$$

Proof. Put $L = \mathrm{hv}(\mathrm{peel}(S, n))$. By Lemma 3.1(a), $L = \{\rho^j(x) : 0 \le j < m\}$. For each j, $\rho^j(x) = \mathrm{rotAbout}(c, 2\pi j/m)(x)$ by Corollary 3.4, so $\mathrm{dist2}(c, \rho^j(x)) = \mathrm{dist2}(c, x)$ by Lemma 3.2(d). Hence L is contained in $D = \{z : \mathrm{dist2}(c, z) \le \mathrm{dist2}(c, x)\}$, which is convex by Lemma 3.15 (with $R = \mathrm{dist2}(c, x)$). So $\mathrm{conv}(L) \subseteq D$. By Lemma 3.1(c) (as $n' \ge n$), $y \in \mathrm{conv}(L) \subseteq D$. □

Remark 3.21. Only the weak inequality is stated, for $n \le n'$ (including $n = n'$), and general position is not assumed. Strictly decreasing radii are not claimed here. Rule B (Lemma 3.38) uses only weak inequalities between radii.

3.5. **Fans and the chain clauses.** The two orientation identities behind the fan lemma, and the identity behind Lemma 3.30, are all instances of one vector identity in the plane.

Lemma 3.22 (a Cramer identity; written step)**.** *For all $X, Y, Z \in \mathbb{R}^2$,*

$$\det(X, Y)\, Z + \det(Y, Z)\, X + \det(Z, X)\, Y = 0.$$

Consequently, for all $w, V \in \mathbb{R}^2$,

$$\det(X, Y)\langle w, Z\rangle + \det(Y, Z)\langle w, X\rangle + \det(Z, X)\langle w, Y\rangle = 0,$$
$$\det(X, Y)\det(V, Z) + \det(Y, Z)\det(V, X) + \det(Z, X)\det(V, Y) = 0.$$

Proof. Fix $k \in \{0, 1\}$ and consider the 3×3 matrix with rows (X_k, Y_k, Z_k), (X_0, Y_0, Z_0), (X_1, Y_1, Z_1). Two of its rows are equal, so its determinant vanishes. Expanding along the first row gives $X_k \det(Y, Z) - Y_k \det(X, Z) + Z_k \det(X, Y) = 0$, which is the k-th coordinate of the first identity because $-\det(X, Z) = \det(Z, X)$. The other two identities follow by applying the linear maps $v \mapsto \langle w, v\rangle$ and $v \mapsto \det(V, v)$. □

Lemma 3.23 (transitivity in a half-plane; Lean: `ErdosSzekeres.LayerLemma.orient_trans_halfplane`, `LAYERLEAN/LayerLemma.lean:629`)**.** *Let a, b, c, e, w be points (here c is an arbitrary point, not a centre). If*

$$0 < \mathrm{lin}(w, a, b), \quad 0 < \mathrm{lin}(w, a, c), \quad 0 < \mathrm{lin}(w, a, e), \quad 0 < \mathrm{orient}(a, b, c), \quad 0 < \mathrm{orient}(a, c, e),$$

then $0 < \mathrm{orient}(a, b, e)$.

Proof. Put $B = b - a$, $C = c - a$, $E = e - a$. By (O1) and Definition 3.9, $\mathrm{orient}(a, b, c) = \det(B, C)$, $\mathrm{orient}(a, c, e) = \det(C, E)$, $\mathrm{orient}(a, b, e) = \det(B, E)$, and $\mathrm{lin}(w, a, z) = \langle w, z - a\rangle$. The second identity of Lemma 3.22 with $(X, Y, Z) = (B, C, E)$ reads $\det(B, C)\langle w, E\rangle + \det(C, E)\langle w, B\rangle + \det(E, B)\langle w, C\rangle = 0$. Since $\det(E, B) = -\det(B, E)$, this is

$$\mathrm{orient}(a, b, e)\,\mathrm{lin}(w, a, c) = \mathrm{orient}(a, b, c)\,\mathrm{lin}(w, a, e) + \mathrm{orient}(a, c, e)\,\mathrm{lin}(w, a, b). \tag{2}$$

The right side is a sum of two products of positive numbers, hence positive. As $\mathrm{lin}(w, a, c) > 0$, it follows that $\mathrm{orient}(a, b, e) > 0$. (Lean checks identity (2) directly by `ring`, without Lemma 3.22.) □

In words: inside an open half-plane bounded by a line through a, the counterclockwise angular order around a is transitive.

Lemma 3.24 (fan step; Lean: `ErdosSzekeres.LayerLemma.fan_step`, `LAYERLEAN/LayerLemma.lean:641`)**.** *Let p, a, b, c, e be points (here c is an arbitrary point, not a centre). If*

$$0 < \mathrm{orient}(b, c, e), \quad 0 < \mathrm{orient}(p, c, e), \quad 0 < \mathrm{orient}(p, b, c), \quad 0 < \mathrm{orient}(a, b, c), \quad 0 < \mathrm{orient}(p, a, c),$$

then $0 < \mathrm{orient}(a, c, e)$.

Proof. Put $P = p - c$, $A = a - c$, $B = b - c$, $E = e - c$. By (O2) (cyclic invariance, and $\mathrm{orient}(x, y, c) = \det(x - c, y - c)$ for all x, y),

$$\mathrm{orient}(p, b, c) = \det(P, B), \qquad \mathrm{orient}(a, b, c) = \det(A, B), \qquad \mathrm{orient}(p, a, c) = \det(P, A),$$
$$\mathrm{orient}(a, c, e) = \mathrm{orient}(e, a, c) = \det(E, A), \qquad \mathrm{orient}(p, c, e) = \mathrm{orient}(e, p, c) = \det(E, P),$$
$$\mathrm{orient}(b, c, e) = \mathrm{orient}(e, b, c) = \det(E, B).$$

The third identity of Lemma 3.22 with $(X, Y, Z, V) = (P, B, A, E)$ reads

$$\det(P,B)\det(E,A) + \det(B,A)\det(E,P) + \det(A,P)\det(E,B) = 0.$$

Since $\det(B,A) = -\det(A,B)$ and $\det(A,P) = -\det(P,A)$, this is

$$\operatorname{orient}(p,b,c)\operatorname{orient}(a,c,e) = \operatorname{orient}(p,c,e)\operatorname{orient}(a,b,c) + \operatorname{orient}(b,c,e)\operatorname{orient}(p,a,c). \tag{3}$$

By the hypotheses the right side is positive, and $\operatorname{orient}(p,b,c) > 0$; hence $\operatorname{orient}(a,c,e) > 0$. (Lean checks identity (3) directly by `ring`, without Lemma 3.22.) □

Lemma 3.25 (orientation as a linear functional; Lean: `ErdosSzekeres.LayerLemma.orient_eq_lin`, `LAYERLEAN/LayerLemma.lean:653`)**.** *For all points p, x, y,*

$$\operatorname{orient}(p,x,y) = \operatorname{lin}\big((-(x_1 - p_1),\ x_0 - p_0),\ x,\ y\big).$$

Proof. By (O1), $\operatorname{orient}(p,x,y) = \det(x-p,\ y-p) = \det(x-p,\ y-x) + \det(x-p,\ x-p) = \det(x-p,\ y-x) = \langle J(x-p),\ y-x\rangle$, and $J(x-p) = (-(x_1-p_1),\ x_0-p_0)$. □

So, for $x \neq p$, $\{y : \operatorname{orient}(p,x,y) > 0\}$ is the open half-plane to the left of the directed line $p \to x$; its bounding line passes through x.

Lemma 3.26 (fan chains have all triples positive; Lean: `ErdosSzekeres.LayerLemma.fan_positive`, `LAYERLEAN/LayerLemma.lean:658`)**.** *Let p, w be points, $q : \mathbb{N} \to \mathbb{R}^2$ a sequence of points (write $q_s = q(s)$), and $n \in \mathbb{N}$. Assume*

(H) $0 < \operatorname{lin}(w,p,q_s)$ *for all* $s < n$*;*
(F) $0 < \operatorname{orient}(p,q_s,q_{s+1})$ *for all* s *with* $s+1 < n$*;*
(T) $0 < \operatorname{orient}(q_s,q_{s+1},q_{s+2})$ *for all* s *with* $s+2 < n$*.*

Then (a) $0 < \operatorname{orient}(p,q_j,q_k)$ *for all* j,k *with* $j < k$ *and* $k < n$*; and (b)* $0 < \operatorname{orient}(q_i,q_j,q_k)$ *for all* i,j,k *with* $i < j$*,* $j < k$ *and* $k < n$*.*

Proof. (a) Fix j. We show by induction on $k \geq j+1$ that $k < n$ implies $\operatorname{orient}(p,q_j,q_k) > 0$. For $k = j+1$ this is (F) with $s = j$. Let $k \geq j+1$ satisfy the claim and let $k+1 < n$. Then $k < n$, so $\operatorname{orient}(p,q_j,q_k) > 0$; (F) with $s = k$ gives $\operatorname{orient}(p,q_k,q_{k+1}) > 0$; and (H) gives $\operatorname{lin}(w,p,q_t) > 0$ for $t \in \{j,k,k+1\}$, all of which are $< n$. Lemma 3.23 with $(a,b,c,e) = (p,q_j,q_k,q_{k+1})$ gives $\operatorname{orient}(p,q_j,q_{k+1}) > 0$.

(b_0) We first show: for all $i < s$ with $s+1 < n$, $\operatorname{orient}(q_i,q_s,q_{s+1}) > 0$. Fix i and induct on $s \geq i+1$. For $s = i+1$ this is (T) with $s = i$ (note $i+2 < n$). Let $s \geq i+1$ satisfy the claim and let $s+2 < n$. Apply Lemma 3.24 with $(p,a,b,c,e) = (p,q_i,q_s,q_{s+1},q_{s+2})$; its five hypotheses hold: $\operatorname{orient}(q_s,q_{s+1},q_{s+2}) > 0$ by (T); $\operatorname{orient}(p,q_{s+1},q_{s+2}) > 0$ by (F) with $s+1$; $\operatorname{orient}(p,q_s,q_{s+1}) > 0$ by (F) with s; $\operatorname{orient}(q_i,q_s,q_{s+1}) > 0$ by the induction hypothesis (as $s+1 < n$); and $\operatorname{orient}(p,q_i,q_{s+1}) > 0$ by (a), since $i < s+1 < n$. The conclusion is $\operatorname{orient}(q_i,q_{s+1},q_{s+2}) > 0$.

(b) Fix $i < j$ and induct on $k \geq j+1$, showing that $k < n$ implies $\operatorname{orient}(q_i,q_j,q_k) > 0$. For $k = j+1$ this is (b_0) with $s = j$. Let $k \geq j+1$ satisfy the claim and let $k+1 < n$. Put $w' = (-(q_{i,1} - p_1),\ q_{i,0} - p_0)$, where $q_i = (q_{i,0}, q_{i,1})$. By Lemma 3.25, $\operatorname{lin}(w',q_i,y) = \operatorname{orient}(p,q_i,y)$ for every y, so by (a) $\operatorname{lin}(w',q_i,q_t) > 0$ for every t with $i < t < n$, in particular for $t \in \{j,k,k+1\}$. Moreover $\operatorname{orient}(q_i,q_j,q_k) > 0$ by the induction hypothesis (as $k < n$), and $\operatorname{orient}(q_i,q_k,q_{k+1}) > 0$ by (b_0) with $s = k$ (as $i < k$ and $k+1 < n$). Lemma 3.23 with $(a,b,c,e,w) = (q_i,q_j,q_k,q_{k+1},w')$ gives $\operatorname{orient}(q_i,q_j,q_{k+1}) > 0$. □

Definition 3.27 (fan points; Lean: `ErdosSzekeres.LayerLemma.fanPt`, `LAYERLEAN/LayerLemma.lean:694`)**.** For $n \in \mathbb{N}$, a point p, $q : \mathbb{N} \to \mathbb{R}^2$ and $i \in \{0, 1, \ldots, n\}$ (Lean `Fin (n + 1)`; n is an implicit argument),

$$\operatorname{fanPt}(p,q)(i) = \begin{cases} p & \text{if } i = 0, \\ q(i-1) & \text{otherwise.} \end{cases}$$

Theorem 3.28 (fan chains give convex polygons; Lean: `ErdosSzekeres.LayerLemma.hasConvexNGon_of_fan`, `LAYERLEAN/LayerLemma.lean:700`)**.** *Let S be a finite set, p, w points, $q : \mathbb{N} \to \mathbb{R}^2$ (write $q_s = q(s)$) and $n \in \mathbb{N}$ with $3 \leq n$. Assume that $p \in S$, that $q_s \in S$ for all $s < n$, and that (H), (F) and (T) of Lemma 3.26 hold:* $0 < \operatorname{lin}(w,p,q_s)$ *for all* $s < n$*;* $0 < \operatorname{orient}(p,q_s,q_{s+1})$ *for all* s *with* $s+1 < n$*; and* $0 < \operatorname{orient}(q_s,q_{s+1},q_{s+2})$ *for all* s *with* $s+2 < n$*. Then `HasConvexNGon (n + 1) S`: some $(n+1)$-element subset of S is in convex position.*

Proof. Let $x_i = \mathrm{fanPt}(p,q)(i)$ for $i \in \{0,\dots,n\}$; so $x_0 = p$ and $x_i = q_{i-1}$ for $1 \le i \le n$.

Step 1: every increasing triple is positive. Let $0 \le i < j < k \le n$. If $i = 0$, then $j, k \ge 1$ and $\mathrm{orient}(x_0, x_j, x_k) = \mathrm{orient}(p, q_{j-1}, q_{k-1}) > 0$ by Lemma 3.26(a), since $j - 1 < k - 1 < n$. If $i \ge 1$, then $\mathrm{orient}(x_i, x_j, x_k) = \mathrm{orient}(q_{i-1}, q_{j-1}, q_{k-1}) > 0$ by Lemma 3.26(b), since $i - 1 < j - 1 < k - 1 < n$.

Step 2: the points $x_0, \dots, x_n$ are pairwise distinct. Suppose $x_i = x_j$ with $i \ne j$. As $n + 1 \ge 4 > 2$, there is $k \in \{0,\dots,n\}$ different from i and j. Arrange i, j, k increasingly as $i' < j' < k'$. By Step 1, $\mathrm{orient}(x_{i'}, x_{j'}, x_{k'}) > 0$; but two of these three points are equal (namely x_i and x_j), so the orientation is 0 by (O3), a contradiction. (Lean chooses k explicitly and treats the possible orders of i, j, k one by one, using `orient_cycle` and $\mathrm{orient}(a, a, d) = 0$; the argument is the same.)

Step 3: convex position. Apply Lemma 1.35 (`Suk.convexPosition_of_positive_triples`) with the linearly ordered index set $\{0,\dots,n\}$ (Lean `Fin (n + 1)` with its usual order), the injective map $i \mapsto x_i$ (Step 2), and the full index set, which has $n + 1 \ge 4$ elements; its hypothesis is Step 1. Hence $T = \{x_0, \dots, x_n\}$ is in convex position.

Step 4. $T \subseteq S$, since $x_0 = p \in S$ and $x_i = q_{i-1} \in S$ for $1 \le i \le n$ (as $i - 1 < n$); and $|T| = n + 1$ by Step 2. Hence `HasConvexNGon (n + 1) S`. □

Remark 3.29. General position is not assumed, and the chain points are not assumed distinct: distinctness of $p, q_0, \dots, q_{n-1}$ follows from the positivity of the orientations (Step 2), and the proof shows that $\{p, q_0, \dots, q_{n-1}\}$ itself is the convex $(n + 1)$-gon.

Lemma 3.30 (inner side of the tangent in the layers; Lean: `ErdosSzekeres.LayerLemma.lin_pos_of_peel`, `LAYERLEAN/LayerLemma.lean:864`)**.** *Let S be a finite set, c a point, and $m, K \in \mathbb{N}$ with $3 \le m$ and $K \le 2m$; let $\rho = \mathrm{rotAbout}(c, 2\pi/m)$. Assume that $c \notin S$, that ρ maps S into S, that S is K-free, and that S is in general position. Let $i \in \mathbb{N}$, $p \in \mathrm{hv}(\mathrm{peel}(S, i))$, and $y \in \mathrm{peel}(S, i)$ with $y \ne p$. Then*

$$0 < \mathrm{lin}(c - p,\ p,\ y).$$

Proof. Since $m \ge 3$, $0 < m$ and Lemma 3.1 applies to i and p. Put $L = \mathrm{hv}(\mathrm{peel}(S, i))$; then $L = \{\rho^j(p) : 0 \le j < m\}$ and $p \ne c$.

Case $y \in L$. Then $y = \rho^j(p)$ for some $j < m$, and $j \ne 0$ because $\rho^0(p) = p \ne y$. So $0 < j < m$, and Lemma 3.11 gives $0 < \mathrm{lin}(c - p, p, y)$.

Case $y \notin L$. Then $y \in \mathrm{peel}(S, i) \setminus L = \mathrm{peel}(S, i + 1)$. Let $p' = \rho^{m-1}(p)$. Since $m - 1 < m$, $p' \in L$ by Lemma 3.1(a); and $\rho(p') = \rho^m(p) = p$ by Corollary 3.4. We collect five positive quantities.

- (e1) $\mathrm{orient}(p, \rho(p), y) > 0$: Theorem 3.16 with $(n, n', x) = (i, i + 1, p)$ (its hypotheses hold: $2 \le m$, $K \le 2m$, $c \notin S$, invariance, K-freeness, general position, $p \in \mathrm{hv}(\mathrm{peel}(S, i))$, $y \in \mathrm{peel}(S, i + 1)$).
- (e2) $\mathrm{orient}(p', p, y) > 0$: Theorem 3.16 with $(n, n', x) = (i, i+1, p')$ gives $\mathrm{orient}(p', \rho(p'), y) > 0$, and $\rho(p') = p$.
- (w1) $\mathrm{lin}(c - p, p, \rho(p)) > 0$: Lemma 3.11 with $j = 1$ (as $0 < 1 < m$).
- (w2) $\mathrm{lin}(c - p, p, p') > 0$: Lemma 3.11 with $j = m - 1$ (as $0 < m - 1 < m$).
- (o) $\mathrm{orient}(p, \rho(p), p') > 0$: Lemma 3.12 (as $p \ne c$ and $m \ge 3$).

Now put $B = \rho(p) - p$, $C = p' - p$, $E = y - p$ and $w = c - p$. By (O1) and (O2), $\mathrm{orient}(p, \rho(p), p') = \det(B, C)$, $\mathrm{orient}(p, \rho(p), y) = \det(B, E)$ and $\mathrm{orient}(p', p, y) = \mathrm{orient}(p, y, p') = \det(E, C)$; and $\mathrm{lin}(w, p, z) = \langle w, z - p\rangle$. The second identity of Lemma 3.22 with $(X, Y, Z) = (B, C, E)$ gives $\det(B, C)\langle w, E\rangle = -\det(C, E)\langle w, B\rangle - \det(E, B)\langle w, C\rangle = \det(E, C)\langle w, B\rangle + \det(B, E)\langle w, C\rangle$, that is,

$$\mathrm{orient}(p, \rho(p), p')\,\mathrm{lin}(c - p, p, y) = \mathrm{orient}(p', p, y)\,\mathrm{lin}(c - p, p, \rho(p)) + \mathrm{orient}(p, \rho(p), y)\,\mathrm{lin}(c - p, p, p'). \tag{4}$$

By (e2), (w1), (e1) and (w2) the right side is positive, and by (o) $\mathrm{orient}(p, \rho(p), p') > 0$. Hence $\mathrm{lin}(c - p, p, y) > 0$. □

Remark 3.31. (1) Geometrically, $\{y : \mathrm{lin}(c - p, p, y) = 0\}$ is the tangent at p to the circle about c through p, and the lemma says that every other point of the layers $\ge i$ lies strictly on the side of the centre. Identity (4) holds for arbitrary points $p, \rho(p), p', y, c$; Lean proves it by `ring`. (2) Lean obtains $\rho(p') = \rho^m(p) = p$ from the minimal period of p (Lemma 2.16); we used Corollary 3.4 instead.

Theorem 3.32 (geometric chain soundness; Lean: `ErdosSzekeres.LayerLemma.chain_clauses_sound`, `LAYERLEAN/LayerLemma.lean:914`)**.** *Let S be a finite set, c a point, and $m, K \in \mathbb{N}$ with $3 \le m$, $4 \le K$ and $K \le 2m$; let $\rho = \mathrm{rotAbout}(c, 2\pi/m)$. Assume that $c \notin S$, that ρ maps S into S, that S is K-free, and that S is in general position. Let $i \in \mathbb{N}$, $p \in \mathrm{hv}(\mathrm{peel}(S, i))$, and $q : \mathbb{N} \to \mathbb{R}^2$ (write $q_s = q(s)$) be such that*

(C1) $q_s \in \mathrm{peel}(S, i)$ for all $s < K - 1$;

(C2) $q_s \neq p$ *for all* $s < K-1$*;*
(C3) $0 < \mathrm{orient}(p, q_s, q_{s+1})$ *for all* s *with* $s+1 < K-1$*;*
(C4) $0 < \mathrm{orient}(q_s, q_{s+1}, q_{s+2})$ *for all* s *with* $s+2 < K-1$.
Then a contradiction follows (the Lean conclusion is `False`*). In other words, no such* i*,* p *and* q *exist.*

Proof. Since $K \geq 4$, $K-1 \geq 3$ is the ordinary difference. We have $p \in \mathrm{peel}(S,i)$, as $\mathrm{hv}(\mathrm{peel}(S,i)) \subseteq \mathrm{peel}(S,i)$. For every $s < K-1$, Lemma 3.30 with $y = q_s$ (which lies in $\mathrm{peel}(S,i)$ by (C1) and differs from p by (C2)) gives $0 < \mathrm{lin}(c-p, p, q_s)$. Apply Theorem 3.28 to the finite set $\mathrm{peel}(S,i)$, the pivot p, the vector $w = c-p$, the sequence q and $n = K-1 \geq 3$: its hypotheses are $p \in \mathrm{peel}(S,i)$, (C1), the half-plane inequalities just proved, (C3) and (C4). Hence $\mathrm{peel}(S,i)$ contains $(K-1)+1 = K$ points in convex position. Since $\mathrm{peel}(S,i) \subseteq S$ by (L2), so does S (Lemma 1.23(c)), contradicting the K-freeness of S. (Lean unfolds the definition of `HasConvexNGon` here instead of citing Lemma 1.23(c).) □

Remark 3.33. The chain points q_s are not assumed distinct, and they may lie in any layer $\geq i$. The theorem is the geometric statement behind the chain clauses of the layered encoders ("no pivot of layer i with $K-1$ chain points whose consecutive fan and turn orientations are positive"). That the encoders' clauses express exactly this family (including, where auxiliary chain variables are used, that their least-fixpoint values satisfy every chain clause; `LEANEXT/review/REPORT.md`, finding F5) is a code-level step outside `LAYERLEAN/LayerLemma.lean`; see the paragraph *Encoders* at the beginning of this section.

3.6. **Polar coordinates and the angular rules.**

Definition 3.34 (polar; Lean: `ErdosSzekeres.LayerLemma.polar`, `LAYERLEAN/LayerLemma.lean:932`)**.** For a point c and $r, \theta \in \mathbb{R}$,

$$\mathrm{polar}(c, r, \theta) = \big(c_0 + r\cos\theta,\ c_1 + r\sin\theta\big).$$

There is no sign condition on r.

Lemma 3.35 (orientation of three points in polar form; Lean: `ErdosSzekeres.LayerLemma.orient_polar`, `LAYERLEAN/LayerLemma.lean:935`)**.** *For every point* c *and all* $r_a, r_b, r_d, \alpha, \beta, \delta \in \mathbb{R}$*,*

$$\mathrm{orient}\big(\mathrm{polar}(c, r_a, \alpha), \mathrm{polar}(c, r_b, \beta), \mathrm{polar}(c, r_d, \delta)\big) = r_a r_b \sin(\beta-\alpha) + r_b r_d \sin(\delta-\beta) + r_d r_a \sin(\alpha-\delta).$$

Proof. Write $e(\varphi) = (\cos\varphi, \sin\varphi)$ and let a, b, d be the three points; then $a-c = A := r_a e(\alpha)$, $b-c = B := r_b e(\beta)$, $d-c = D := r_d e(\delta)$, so $b-a = B-A$ and $d-a = D-A$. By (O1) and bilinearity and antisymmetry of det,

$$\begin{aligned}\mathrm{orient}(a,b,d) = \det(B-A, D-A) &= \det(B,D) - \det(B,A) - \det(A,D) + \det(A,A)\\ &= \det(A,B) + \det(B,D) + \det(D,A).\end{aligned}$$

Finally, for all $r, s, \varphi, \psi \in \mathbb{R}$, $\det(re(\varphi), se(\psi)) = rs(\cos\varphi\sin\psi - \sin\varphi\cos\psi) = rs\sin(\psi-\varphi)$; applied to the three terms this gives the formula. (Lean expands the three sines on the right by the subtraction formula and concludes with `ring`.) □

Lemma 3.36 (rule A, centre inside; Lean: `ErdosSzekeres.LayerLemma.orient_polar_centre_inside`, `LAYERLEAN/LayerLemma.lean:943`)**.** *(Angular rule A: centre inside the triangle.) Let* c *be a point and* $r_a, r_b, r_d, \alpha, \beta, \delta \in \mathbb{R}$ *with* $0 < r_a$*,* $0 < r_b$*,* $0 < r_d$*, and*

$$0 < \beta - \alpha < \pi, \qquad 0 < \delta - \beta < \pi, \qquad 0 < \alpha - \delta + 2\pi < \pi.$$

Then $0 < \mathrm{orient}\big(\mathrm{polar}(c, r_a, \alpha), \mathrm{polar}(c, r_b, \beta), \mathrm{polar}(c, r_d, \delta)\big)$.

Proof. By Lemma 3.35 the orientation is $r_a r_b \sin(\beta-\alpha) + r_b r_d \sin(\delta-\beta) + r_d r_a \sin(\alpha-\delta)$. By (S1) the sine is positive on $(0,\pi)$, so $\sin(\beta-\alpha) > 0$, $\sin(\delta-\beta) > 0$, and, by 2π-periodicity, $\sin(\alpha-\delta) = \sin(\alpha-\delta+2\pi) > 0$. With $r_a, r_b, r_d > 0$ all three terms are positive. □

The hypotheses say $\alpha < \beta < \delta < \alpha + 2\pi$ with all three counterclockwise gaps $\beta-\alpha$, $\delta-\beta$, $\alpha+2\pi-\delta$ (which sum to 2π) in $(0,\pi)$. The name of the rule refers to the position of c inside the triangle; that geometric fact is neither stated nor used here, and only the orientation statement is proved. Lean states only the positive orientation.

Lemma 3.37 (sine of a sum below the sum of the sines; Lean: `ErdosSzekeres.LayerLemma.sin_add_lt`, `LAYERLEAN/LayerLemma.lean:956`)**.** *For all* $x, y \in \mathbb{R}$ *with* $0 < x < \pi$ *and* $0 < y < \pi$*:* $\sin(x+y) < \sin x + \sin y$.

Proof. By (S1), $\sin x > 0$ and $\sin y > 0$. Moreover $\cos x < 1$: by (S2), $\cos x \le 1$, and $\cos x = 1$ only for $x \in 2\pi\mathbb{Z}$, which has no point in $(0,\pi)$; likewise $\cos y < 1$. Hence $\sin x \cos y < \sin x$ and $\cos x \sin y < \sin y$, and $\sin(x+y) = \sin x \cos y + \cos x \sin y < \sin x + \sin y$. □

Lemma 3.38 (rule B, middle point outermost; Lean: `ErdosSzekeres.LayerLemma.orient_polar_halfplane`, `LAYERLEAN/LayerLemma.lean:974`)**.** *(Angular rule B: open half-plane, angularly middle point outermost.) Let c be a point and $r_a, r_b, r_d, \alpha, \beta, \delta \in \mathbb{R}$ with*

$$0 < r_a, \quad 0 < r_d, \quad r_a \le r_b, \quad r_d \le r_b, \quad 0 < \beta - \alpha, \quad 0 < \delta - \beta, \quad \delta - \alpha < \pi.$$

Then $0 < \operatorname{orient}\big(\operatorname{polar}(c, r_a, \alpha), \operatorname{polar}(c, r_b, \beta), \operatorname{polar}(c, r_d, \delta)\big)$.

Proof. Let a, b, d denote the three points $\operatorname{polar}(c, r_a, \alpha)$, $\operatorname{polar}(c, r_b, \beta)$, $\operatorname{polar}(c, r_d, \delta)$. Put $x = \beta - \alpha > 0$ and $y = \delta - \beta > 0$. Then $x + y = \delta - \alpha < \pi$, so $x = (x+y) - y < \pi$ and $y = (x+y) - x < \pi$. Since $\alpha - \delta = -(x+y)$ and the sine is odd, Lemma 3.35 gives

$$\begin{aligned} \operatorname{orient}(a,b,d) &= r_a r_b \sin x + r_b r_d \sin y - r_d r_a \sin(x+y) \\ &= r_a(r_b - r_d)\sin x + r_d(r_b - r_a)\sin y + r_a r_d\big(\sin x + \sin y - \sin(x+y)\big). \end{aligned}$$

Here $\sin x, \sin y > 0$ by (S1) (arguments in $(0,\pi)$), $r_a, r_d > 0$ and $r_b - r_d$, $r_b - r_a \ge 0$, so the first two terms are ≥ 0; the third is > 0 by Lemma 3.37 and $r_a r_d > 0$. Hence the orientation is positive. (Lean uses $r_a r_b \sin x \ge r_a r_d \sin x$, $r_b r_d \sin y \ge r_a r_d \sin y$ and Lemma 3.37, and concludes with `nlinarith`; this is the same estimate.) □

Remark 3.39. The radius hypotheses are weak inequalities ($r_b > 0$ follows from $0 < r_a \le r_b$). This is the form of the radius order of Theorem 3.20, which gives $\operatorname{dist2}(c,y) \le \operatorname{dist2}(c,x)$, that is, $r_y \le r_x$ for the radii $r = \sqrt{\operatorname{dist2}(c,\cdot)}$. The rules apply to points given in polar form with explicit angle representatives. No Lean statement of this file writes the points of a layered configuration in that form (a point y as $\operatorname{polar}(c, r, \theta)$ with $r^2 = \operatorname{dist2}(c,y)$, and a layer as $\{\operatorname{polar}(c, r_i, \theta_i + 2\pi k/m)\}$; `LEANEXT/review/REPORT.md`, finding F2); that step, and the choice of which rule applies to which triple, are outside these statements (see the paragraph *Encoders* at the beginning of this section).

Coverage. All 27 declarations of the scope of this section, which are all the declarations of lines 465–1032 of `LAYERLEAN/LayerLemma.lean` (23 theorems and 4 definitions), are covered: `rotAbout_iterate` (Lemma 3.3; the section on the layer lemma restates it as well), `orient_rotAbout` (Lemma 3.5), `trig_layer` (Lemma 3.6, stated just before Lemma 3.7, whose proof uses it), `orient_regular_nonneg` (Lemma 3.7), `lin` (Definition 3.9), `lin_rotAbout` (Lemma 3.10), `lin_orbit_pos` (Lemma 3.11), `orient_next_prev_pos` (Lemma 3.12), `convex_orient_nonneg` (Lemma 3.13), `dist2` (Definition 3.14), `convex_disk` (Lemma 3.15), `strict_nesting` (Theorem 3.16), `centre_not_mem_of_even` (Lemma 3.18), `dist2_le_of_later` (Theorem 3.20), `orient_trans_halfplane` (Lemma 3.23), `fan_step` (Lemma 3.24), `orient_eq_lin` (Lemma 3.25), `fan_positive` (Lemma 3.26), `fanPt` (Definition 3.27), `hasConvexNGon_of_fan` (Theorem 3.28), `lin_pos_of_peel` (Lemma 3.30), `chain_clauses_sound` (Theorem 3.32), `polar` (Definition 3.34), `orient_polar` (Lemma 3.35), `orient_polar_centre_inside` (Lemma 3.36), `sin_add_lt` (Lemma 3.37) and `orient_polar_halfplane` (Lemma 3.38); all in `LAYERLEAN/LayerLemma.lean`, namespace `ErdosSzekeres.LayerLemma`. The written steps (Lemmas 3.1, 3.2, 3.22 and Corollary 3.4) are not Lean declarations; the paragraph *Rigor* at the beginning of the section says how each relates to the Lean proofs. Deliberately not covered here:

- `peel_antitone` (line 460) and `card_peel` (line 1037, docstring at line 1036), which adjoin these lines but belong to the section on the layer lemma (Lemma 2.20, Theorem 2.24);
- the verbatim copies of these declarations in other units, which are duplicates. The following eight files contain lines 11–1147 of `LAYERLEAN/LayerLemma.lean` verbatim (checked with `diff`); a search of the uncompressed files of the repository finds no further copy:
  `LEANEXT/capstone/Capstone.lean`;
  `LEANEXT/capstone/Capstone2.lean`;
  `LEANEXT/merged/Merged.lean`;
  `LEANEXT/linked/Linked.lean`;
  `LEANEXT/es7corr/ES7Corollaries.lean`;
  `LEANEXT/review/VacuityTest.lean` (all 1148 lines, followed by the tests of the next item);
  `ENCBLEAN/geo/ES8Geo/Capstone2.lean`;
  `LAYERLEAN/logs/axioms_recheck_20261106/LayerLemmaAxioms.lean`;

- the 19 declarations of `LEANEXT/review/tests_tail.lean` (namespace `ErdosSzekeres.LayerLemma.Review`), which no section covers: the definitions `L4`, `S4`, `H6` and the theorems `mem_S4`, `card_S4`, `angle4`, `rot4`, `hc4`, `inv4`, `gp4`, `inner4`, `hno4`, `hv4`, `peel1_4`, `nest4`, `orbit4`, `layers4`, `chain4` and `inv6`. They are non-vacuity tests, not results (`LEANEXT/review/REPORT.md`, item 6). For the eight points $S_4 = \{(\pm 2, 0), (0, \pm 2), (\pm 1/2, \pm 1/2)\}$, the centre $c = 0$, $m = 4$ and $K = 8$, the theorems `hc4`, `inv4`, `gp4` and `hno4` state that $c \notin S_4$, that rotAbout$(0, 2\pi/4)$ maps S_4 into S_4, that S_4 is in general position and that S_4 is 8-free; `hv4` and `peel1_4` state that $(2, 0) \in \mathrm{hv}(S_4)$ and $(1/2, 1/2) \in \mathrm{peel}(S_4, 1)$; and `nest4`, `orbit4`, `layers4` and `chain4` apply `strict_nesting` (Theorem 3.16), `hullVerts_eq_orbitFS_rotAbout` (Theorem 2.17), `layers_are_orbits` (Theorem 2.22) and `chain_clauses_sound` (Theorem 3.32) to this instance. `inv6` states that rotAbout$(0, 2\pi/6)$ maps the set `H6` of the six points $(\cos(k\pi/3), \sin(k\pi/3))$, $0 \le k < 6$ (a regular hexagon), into itself. `LEANEXT/review/VacuityTest.lean` consists of all 1148 lines of `LAYERLEAN/LayerLemma.lean` followed by the 149 lines of `LEANEXT/review/tests_tail.lean`, as lines 1149–1297 (both parts checked with `cmp`), so the same 19 declarations stand there at lines 1160–1276. Both files end with `#print axioms`, `#check` and `#print` commands, which are not declarations;
- declarations in other namespaces that share a short name with one of the 27 (in `ErdosSzekeres.OrbitConsistency`, `PhaseClassification`, `C7` and `ErdosSzekeres.SymReduction.Review`); they are separate Lean declarations. Those of `LEANEXT/phase/PhaseClassification.lean` (`polar`, `orient_polar`, `orient_polar_centre_inside`, `sin_add_lt` and `orient_polar_halfplane`) are treated in the section on phase patterns.

4. Orbit literals, rays and phases, key order

This section gives written proofs of the Lean-checked results of three files of the `LEANEXT` unit. They are the geometric and arithmetic facts behind three features of the layered SAT encoders for point sets with m-fold rotational symmetry about a centre c.

(A) *Orbit literals* (§4.2, §4.3). Rotations about c preserve orient, and permuting an ordered triple multiplies its orientation $\mathrm{orient}_3(t) = \mathrm{orient}(t(0), t(1), t(2))$ by the sign of the permutation. Hence, if an ordered triple C of points arises from an ordered triple t by a permutation σ of the positions followed by a composition of rotations by integer multiples of $2\pi/m$, and $\mathrm{orient}_3(t) \neq 0$, then "$\mathrm{orient}_3(t) > 0$" is equivalent to "$\mathrm{orient}_3(C) > 0$, negated when $\operatorname{sgn}\sigma = -1$" (Theorems 4.18 and 4.19). This is the rule "one SAT variable per orbit of ordered triples".

(B) *Rays and phases for even $m > 0$* (§4.4, §4.5). For even $m > 0$, a finite set in general position that is mapped into itself by the rotation ρ by $2\pi/m$ about c contains the antipode $2c - x$ of each of its points, and no two of its points lie on one open ray from c (Lemma 4.23). Consequently two of its points whose directions agree modulo $2\pi/m$ lie in one ρ-orbit (Theorem 4.28). Combined with the layer lemma (under its hypotheses: c is not in the set, and the set has no K points in convex position for some $K \le 2m$), hull vertices of distinct layers never lie on one open ray from c through a rotated point and have directions that are incongruent modulo $2\pi/m$ (Theorems 4.32 and 4.33). Thus, for even $m > 0$, this distinctness of phases is obtained without any perturbation argument.

(C) *Key order* (§4.6). For $u > 0$, real numbers $Ju + \psi$ with $J \in \mathbb{Z}$ and $0 \le \psi < u$ are ordered lexicographically by (J, ψ), or by $(J, \operatorname{rank}\psi)$ for a strictly increasing rank; reduction modulo 2π into $[0, 2\pi)$ replaces J by $J \bmod N$ when $Nu = 2\pi$; and the antipode $\theta + \pi$ has an explicit cell (Theorem 4.41, Lemma 4.40).

Sources. File paths are relative to the results directory of the project, except that paths beginning with `ErdosSzekeres/` are relative to the repository root; a bare file name such as `OrbitConsistency.lean`, `Merged.lean` or `KeyOrder.lean` refers to the file of that name listed below. SHA-256 digests are abbreviated to their first 12 hexadecimal digits.

- `LEANEXT/orbit/OrbitConsistency.lean` (429 lines, SHA-256 `9b843180ac38`), namespace `ErdosSzekeres.OrbitConsistency`. The new content starts at line 114. Lines 39–112 repeat definitions and basic lemmas of `LAYERLEAN/LayerLemma.lean` as separate constants (Remark 4.2).
- `LEANEXT/merged/Merged.lean` (1385 lines, SHA-256 `245b7d65407a`), namespace `ErdosSzekeres.LayerLemma`, section 13 (lines 1149–1383). Lines 1–1147 of this file coincide line by line with lines 1–1147 of `LAYERLEAN/LayerLemma.lean` (SHA-256 `afb8542dceb0`); hull vertices, peels and the layer lemma used in §4.5 come from there. The seven declarations at lines 1160–1311 of `Merged.lean` restate

those at lines 273–427 of `OrbitConsistency.lean` over the definitions of the namespace `ErdosSzekeres.LayerLemma`; the proofs are identical and only the doc comments differ.

- `LEANEXT/keys/KeyOrder.lean` (158 lines, SHA-256 `6e34096cff52`), namespace `KeyOrder`; it imports Mathlib only.

Byte-identical copies of `OrbitConsistency.lean` and `KeyOrder.lean` (same digests) are `ENCBLEAN/geo/ES8Geo/OrbitConsistency.lean` and `ENCBLEAN/geo/ES8Geo/KeyOrder.lean`. Lines 1148–1383 of `Merged.lean` reappear verbatim as lines 1148–1383 of `LEANEXT/linked/Linked.lean` and as lines 1156–1391 of `LEANEXT/capstone/Capstone.lean` and of `LEANEXT/capstone/Capstone2.lean`; `ENCBLEAN/geo/ES8Geo/Capstone2.lean` is a byte-identical copy of the last (SHA-256 `f8c3930c2427`). The files are compiled from the repository root (`LEANEXT/README.md:6`), whose pinned toolchain is Lean 4.24.0 with Mathlib `f897ebcf`.

Axioms. `LEANEXT/orbit/logs/axioms.log` records `#print axioms` for the 15 main theorems of `OrbitConsistency.lean` (`orient_rotAbout_inv`, `RotComp.preservesOrient`, `rotComp_iterate`, `orient3_swap`, `orient3_perm`, `orient3_rot_perm`, `sign_eq_of_two_reps`, `orient3_of_common_rep`, `literal_sound`, `encoder_literal_sound`, `antipode_mem_of_even`, `same_ray_false`, `eq_iterate_of_phase_eq`, `eq_iterate_of_phase_eq_int`, `phase_ne_of_not_mem_orbit`), each depending on `[propext, Classical.choice, Quot.sound]`, with exit status 0. `LEANEXT/merged/axioms.log` records the same three axioms for `ErdosSzekeres.LayerLemma.phase_ne_of_distinct_layers`, `ErdosSzekeres.LayerLemma.not_same_ray_of_distinct_layers` and `ErdosSzekeres.LayerLemma.phase_ne_of_not_mem_orbit` (and for the copies in `Merged.lean` of three theorems of `LAYERLEAN/LayerLemma.lean`: `layers_are_orbits`, `convexNGon_of_layers` and `sevenfold_octagon`). Lines 151–158 of `KeyOrder.lean` print the axioms of its eight theorems; that output, again the same three axioms, is recorded for the byte-identical copy in `ENCBLEAN/logs/m4_build.log` (lines 3–10), not inside `LEANEXT`. The remaining theorems of the scope that carry a Lean reference below (the copied lemmas `iterate_mem` and `rotAbout_iterate` of `OrbitConsistency.lean`, both used by `antipode_mem_of_even`; the helpers `preservesOrient_id`, `PreservesOrient.comp`, `preservesOrient_rotAbout`, `rotAbout_polar`, `polar_add_int`; the other `Merged.lean` restatements; and `not_mem_hullVerts_peel_of_lt`, `not_mem_hullVerts_peel_of_ne`, `iterate_mem_hullVerts_peel`) are used by printed theorems, and `#print axioms` reports the axioms of all transitive dependencies; so their axioms are among the three listed. There are two exceptions. (i) `PreservesOrient.iterate` is used by no printed theorem, and no axiom output is recorded for it; that it compiles is recorded by the build log of the whole file, `LEANEXT/orbit/logs/build.log` (exit status 0). (ii) Lemmas 4.4 and 4.6 also cite the `LAYERLEAN` declarations `ErdosSzekeres.LayerLemma.rotAbout_iterate` and `ErdosSzekeres.LayerLemma.iterate_mem`. Their axioms are recorded in `LEANEXT/review/axioms.log` (lines 33 and 1), from the review of 4 November 2026 of `LAYERLEAN/LayerLemma.lean` (`LEANEXT/review/REPORT.md`, lines 3–4): the three axioms above for `rotAbout_iterate`, and only `propext` and `Quot.sound` for `iterate_mem`. None of the three files contains `sorry`, `admit`, an `axiom` declaration or `native_decide`. Within the scope of this section the only `decide` (kernel evaluation, no extra axiom) is the side goal $-1 \neq 1$ in $\mathbb{Z}^\times$ at `OrbitConsistency.lean:236`; the copied part of `Merged.lean` has one more, for Even(6) at line 1125.

Rigor. Every statement below that carries a Lean reference is *Lean-checked* in the sense of the records above (for a definition: it is part of the compiled file). The proofs in this section are *written proofs (not refereed)*. They follow the Lean proofs; where they take a different route, this is said in the proof (Lemma 4.4: addition formulas instead of complex exponentials; Lemma 4.13: the symmetries of orient instead of a six-case expansion; Lemma 4.31: $\rho^{[m]} = \mathrm{id}$ directly instead of the minimal period). No statement of this section takes an unproved proposition or a SAT verdict as a hypothesis.

Dependencies. From the foundations section we use the definitions of points, orient, general position, `HasConvexNGon`, invariance and rotations (Definitions 1.1, 1.3, 1.5, 1.7, 1.10, 1.12), the symmetries of orient (Lemma 1.15), the orientation test for general position (Lemma 1.20) and the monotonicity of `HasConvexNGon` (Lemma 1.23); Lemma 1.17 is cited only in a note on notation in Definition 4.1(b) and in a parenthetical comparison in the proof of Lemma 4.7, and the example in Remark 4.34 uses Lemmas 1.22 and 1.40. From the layer-lemma section we use the hull vertices and peels (Definitions 2.4 and 2.18), Lemma 2.20, Lemma 2.21 and Theorem 2.17; Lemma 2.16 is cited only in a comparison with the Lean proof (proof of Lemma 4.31). From the layer-geometry section we use the definition of polar (Definition 3.34) and, only in comments on hypotheses (Lemma 4.23 and Remark 4.34), Lemma 3.18. §4.6 uses no other section.

4.1. **Conventions and copied definitions.** Points are elements $p = (p_0, p_1)$ of $\mathbb{R}^2$ (Definition 1.1); sums, differences and scalar multiples of points are taken coordinatewise, and the Lean term `![u, v]` denotes the point (u, v). All sets of points in this section are finite (Lean `Finset Point`). $\mathrm{orient}(a, b, d) = (b_0 - a_0)(d_1 -$

$a_1) - (b_1 - a_1)(d_0 - a_0)$ (Definition 1.3); equivalently $\mathrm{orient}(a, b, d) = \det\big(b - a \mid d - a\big)$, the determinant of the 2×2 matrix with columns $b - a$ and $d - a$. For a map $f : X \to X$ the iterates are $f^{[0]} = \mathrm{id}$ and $f^{[s+1]} = f \circ f^{[s]}$ $(= f^{[s]} \circ f)$. For $m \in \mathbb{N}$, the real number $2\pi/m$ is 2π divided by m cast to $\mathbb{R}$, with Lean's convention $x/0 = 0$, so that $2\pi/0 = 0$. "m is even" means $m = k + k$ for some $k \in \mathbb{N}$ (Mathlib's `Even`); then the natural-number quotient $m/2$ equals k.

Definition 4.1 (invariance, rotations, polar points; Lean: `ErdosSzekeres.OrbitConsistency.Invariant`, `LEANEXT/orbit/OrbitConsistency.lean:47`, Lean: `ErdosSzekeres.OrbitConsistency.rotAbout`, `LEANEXT/orbit/OrbitConsistency.lean:93`, Lean: `ErdosSzekeres.OrbitConsistency.polar`, `LEANEXT/orbit/OrbitConsistency.lean:112`)**.** Let $f : \mathbb{R}^2 \to \mathbb{R}^2$, S a finite set of points, $c, p \in \mathbb{R}^2$ and $r, \theta \in \mathbb{R}$.

(a) *f maps S into S* (Lean `Invariant f S`) if $f(p) \in S$ for every $p \in S$ (Definition 1.10).

(b) The rotation by θ about c (Lean `rotAbout`; Definition 1.12) is

$$R_{c,\theta}(p) = \big(c_0 + \cos\theta\,(p_0 - c_0) - \sin\theta\,(p_1 - c_1),\ c_1 + \sin\theta\,(p_0 - c_0) + \cos\theta\,(p_1 - c_1)\big),$$

that is, $R_{c,\theta}(p) = c + M_\theta(p - c)$ with $M_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$; it turns counterclockwise. For $m \in \mathbb{N}$ we write $\rho = R_{c,2\pi/m}$ when c and m are clear. (Notation: Definition 1.12 and the layer-lemma section write $\rho_{c,\theta}$ for the map $R_{c,\theta}$, so that our ρ is their $\rho_{c,2\pi/m}$; the foundations and layer-geometry sections write R_θ for the matrix M_θ, for instance in Lemma 1.17. In this section $R_{c,\theta}$ always denotes the map, never a matrix.)

(c) $\mathrm{polar}(c, r, \theta) = (c_0 + r\cos\theta,\ c_1 + r\sin\theta)$, for every real r (Definition 3.34).

Remark 4.2 (two copies of the definitions). Lines 39–112 of `OrbitConsistency.lean` define their own `iterate_mem`, `Invariant`, `toC`, `ofC`, `crot`, `rotAbout`, `rotAbout_iterate`, `polar` and some helper lemmas, with the same declaration text as the corresponding declarations of `LAYERLEAN/LayerLemma.lean` (`iterate_mem` at line 52, `Invariant` at 105, the complex-coordinate helpers among lines 231–308, `rotAbout` at 298, `rotAbout_iterate` at 465, `polar` at 932), except that `iterate_mem` declares its type variable in its own binder (in `LAYERLEAN/LayerLemma.lean` it is a section variable) and that the doc comment of `polar` is worded differently; neither difference affects the meaning. These are distinct Lean constants with identical defining formulas; for instance, the Lean theorem `EncB.Geo.rotAbout_eq_orbitConsistency` (`ENCBLEAN/geo/ES8Geo/Orbit.lean:13`) states the equality of the two `rotAbout`, with proof `rfl` (in that build `ErdosSzekeres.LayerLemma.rotAbout` is the copy in `ENCBLEAN/geo/ES8Geo/Capstone2.lean`, whose text equals that of `LAYERLEAN/LayerLemma.lean:298`). The theorems of `OrbitConsistency.lean` are stated with the `OrbitConsistency` copies, those of `Merged.lean` with the `LayerLemma` copies (`ErdosSzekeres.LayerLemma.Invariant`, `LAYERLEAN/LayerLemma.lean:105`; `ErdosSzekeres.LayerLemma.rotAbout`, `LAYERLEAN/LayerLemma.lean:298`; `ErdosSzekeres.LayerLemma.polar`, `LAYERLEAN/LayerLemma.lean:932`). Mathematically each pair is one and the same function, so we state each result once and list both Lean names where both exist; the statement then reads verbatim for either copy.

Lemma 4.3 (composition of rotations about one centre; written step)**.** *For all $c, p \in \mathbb{R}^2$ and $\alpha, \beta \in \mathbb{R}$:*

(a) $R_{c,\alpha}(R_{c,\beta}(p)) = R_{c,\alpha+\beta}(p)$*;*

(b) $R_{c,0} = \mathrm{id}$*,* $R_{c,2\pi} = \mathrm{id}$ *and* $R_{c,\pi}(p) = (2c_0 - p_0,\ 2c_1 - p_1)$*.*

Proof. (a) Put $v = p - c$. Then $R_{c,\beta}(p) - c = (\cos\beta\, v_0 - \sin\beta\, v_1,\ \sin\beta\, v_0 + \cos\beta\, v_1)$, and applying $R_{c,\alpha}$ gives c plus the vector with coordinates

$$\cos\alpha(\cos\beta\, v_0 - \sin\beta\, v_1) - \sin\alpha(\sin\beta\, v_0 + \cos\beta\, v_1) = \cos(\alpha+\beta)\, v_0 - \sin(\alpha+\beta)\, v_1,$$
$$\sin\alpha(\cos\beta\, v_0 - \sin\beta\, v_1) + \cos\alpha(\sin\beta\, v_0 + \cos\beta\, v_1) = \sin(\alpha+\beta)\, v_0 + \cos(\alpha+\beta)\, v_1,$$

by the addition formulas $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$ and $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$. This is $R_{c,\alpha+\beta}(p) - c$. (b) Substitute $\cos 0 = \cos 2\pi = 1$, $\sin 0 = \sin 2\pi = 0$, and $\cos\pi = -1$, $\sin\pi = 0$: then $R_{c,0}(p) = R_{c,2\pi}(p) = c + (p - c) = p$ and $R_{c,\pi}(p) = c - (p - c) = 2c - p$. □

Lemma 4.4 (iterates of a rotation; Lean: `ErdosSzekeres.OrbitConsistency.rotAbout_iterate`, `LEANEXT/orbit/OrbitConsistency.lean:105`, Lean: `ErdosSzekeres.LayerLemma.rotAbout_iterate`, `LAYERLEAN/LayerLemma.lean:465`)**.** *For all $c \in \mathbb{R}^2$, $\alpha \in \mathbb{R}$ and $j \in \mathbb{N}$: $(R_{c,\alpha})^{[j]} = R_{c,j\alpha}$.*

Proof. Induction on j. For $j = 0$ both sides are the identity (Lemma 4.3(b)). If the claim holds for j, then $(R_{c,\alpha})^{[j+1]} = R_{c,\alpha} \circ (R_{c,\alpha})^{[j]} = R_{c,\alpha} \circ R_{c,j\alpha} = R_{c,\alpha+j\alpha} = R_{c,(j+1)\alpha}$ by Lemma 4.3(a). (The Lean proof writes $R_{c,\alpha}$ as $z \mapsto c + e^{i\alpha}(z - c)$ and uses $(e^{i\alpha})^j = e^{ij\alpha}$; the argument above is an equivalent elementary route.) □

Corollary 4.5 (period and half-turn; written step)**.** *Let $c \in \mathbb{R}^2$, $m \in \mathbb{N}$ with $0 < m$, and $\rho = R_{c,2\pi/m}$. Then $\rho^{[m]} = \mathrm{id}$. If moreover m is even, then $\rho^{[m/2]}(p) = (2c_0 - p_0,\ 2c_1 - p_1)$ for every $p \in \mathbb{R}^2$.*

Proof. Since $m \neq 0$, $m \cdot (2\pi/m) = 2\pi$, so $\rho^{[m]} = R_{c,2\pi} = \mathrm{id}$ by Lemmas 4.4 and 4.3(b). If $m = k + k$, then $k \geq 1$ (as $m > 0$), $m/2 = k$ and $k \cdot \frac{2\pi}{2k} = \pi$, so $\rho^{[m/2]} = R_{c,\pi}$, and Lemma 4.3(b) applies. □

Lemma 4.6 (iterates stay in a set mapped into itself; Lean: `ErdosSzekeres.OrbitConsistency.iterate_mem`, `LEANEXT/orbit/OrbitConsistency.lean:39`, Lean: `ErdosSzekeres.LayerLemma.iterate_mem`, `LAYERLEAN/LayerLemma.lean:52`)**.** *Let X be a type, $f : X \to X$, S a finite subset of X with $f(y) \in S$ for every $y \in S$, and $x \in S$. Then $f^{[i]}(x) \in S$ for every $i \in \mathbb{N}$.*

Proof. Induction on i: $f^{[0]}(x) = x \in S$, and if $f^{[i]}(x) \in S$ then $f^{[i+1]}(x) = f(f^{[i]}(x)) \in S$. □

4.2. **Rotations preserve orientation.**

Lemma 4.7 (rotation invariance of the orientation; Lean: `ErdosSzekeres.OrbitConsistency.orient_rotAbout_inv`, `LEANEXT/orbit/OrbitConsistency.lean:117`)**.** *For all $c \in \mathbb{R}^2$, $\theta \in \mathbb{R}$ and $a, b, d \in \mathbb{R}^2$,*

$$\mathrm{orient}\big(R_{c,\theta}(a),\ R_{c,\theta}(b),\ R_{c,\theta}(d)\big) = \mathrm{orient}(a, b, d).$$

There are no further hypotheses.

Proof. Write $M = M_\theta$. Since $R_{c,\theta}(p) = c + M(p - c)$, we have $R_{c,\theta}(b) - R_{c,\theta}(a) = M(b - a)$ and $R_{c,\theta}(d) - R_{c,\theta}(a) = M(d - a)$. Hence

$$\begin{aligned}\mathrm{orient}\big(R_{c,\theta}(a), R_{c,\theta}(b), R_{c,\theta}(d)\big) &= \det\big(M(b-a) \mid M(d-a)\big) = \det M \cdot \det\big(b - a \mid d - a\big)\\ &= (\cos^2\theta + \sin^2\theta)\ \mathrm{orient}(a, b, d),\end{aligned}$$

and $\cos^2\theta + \sin^2\theta = 1$. (This is the case $\det = 1$ of Lemma 1.17, item (i), which writes the map $R_{o,\theta}$ as $\rho_{o,\theta}$ and the matrix M_θ as R_θ.) The Lean proof is the same computation in expanded form: the difference of the two sides equals $\mathrm{orient}(a, b, d)\,(\sin^2\theta + \cos^2\theta - 1)$ as a polynomial identity in the coordinates and in $\cos\theta, \sin\theta$ (`linear_combination` with `Real.sin_sq_add_cos_sq`). □

Definition 4.8 (orientation-preserving maps; Lean: `ErdosSzekeres.OrbitConsistency.PreservesOrient`, `LEANEXT/orbit/OrbitConsistency.lean:124`)**.** A map $f : \mathbb{R}^2 \to \mathbb{R}^2$ *preserves orientation* if $\mathrm{orient}(f(a), f(b), f(d)) = \mathrm{orient}(a, b, d)$ for all $a, b, d \in \mathbb{R}^2$.

Definition 4.9 (compositions of rotations by multiples of $2\pi/m$; Lean: `ErdosSzekeres.OrbitConsistency.RotComp`, `LEANEXT/orbit/OrbitConsistency.lean:145`)**.** Let $c \in \mathbb{R}^2$ and $m \in \mathbb{N}$, and put $u = 2\pi/m$ (so $u = 0$ if $m = 0$). $\mathrm{RotComp}(c, m)$ is the inductively defined set of maps $\mathbb{R}^2 \to \mathbb{R}^2$ given by two rules:

(id) $\mathrm{id} \in \mathrm{RotComp}(c, m)$;
(step) if $g \in \mathrm{RotComp}(c, m)$ and $k \in \mathbb{Z}$, then $R_{c,ku} \circ g \in \mathrm{RotComp}(c, m)$.

It is the smallest set of maps that contains id and is closed under (step). Its elements are exactly the maps $R_{c,k_n u} \circ \cdots \circ R_{c,k_1 u}$ with $n \geq 0$ and $k_1, \ldots, k_n \in \mathbb{Z}$ (the identity for $n = 0$): this set of compositions contains id and is closed under (step), and each of its elements arises from id by n applications of (step). Proofs about $\mathrm{RotComp}(c, m)$ use *rule induction*: a property of maps that holds for id and passes from g to $R_{c,ku} \circ g$ holds for every element of $\mathrm{RotComp}(c, m)$.

Lemma 4.10 (orientation-preserving maps and rotation compositions; Lean: `ErdosSzekeres.OrbitConsistency.preservesOrient_id`, `LEANEXT/orbit/OrbitConsistency.lean:127`, Lean: `ErdosSzekeres.OrbitConsistency.PreservesOrient.comp`, `LEANEXT/orbit/OrbitConsistency.lean:129`, Lean: `ErdosSzekeres.OrbitConsistency.PreservesOrient.iterate`, `LEANEXT/orbit/OrbitConsistency.lean:134`, Lean: `ErdosSzekeres.OrbitConsistency.preservesOrient_rotAbout`, `LEANEXT/orbit/OrbitConsistency.lean:141`, Lean: `ErdosSzekeres.OrbitConsistency.RotComp.preservesOrient`, `LEANEXT/orbit/OrbitConsistency.lean:150`, Lean: `ErdosSzekeres.OrbitConsistency.rotComp_iterate`, `LEANEXT/orbit/OrbitConsistency.lean:157`)**.** *The following properties of orientation-preserving maps (Definition 4.8) and of the sets* $\mathrm{RotComp}(c, m)$ *(Definition 4.9) hold.*

(a) The identity preserves orientation. If f and g preserve orientation, so does $f \circ g$. If f preserves orientation, so does $f^{[n]}$ for every $n \in \mathbb{N}$. For all $c \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$ the rotation $R_{c,\theta}$ preserves orientation.
(b) Let $c \in \mathbb{R}^2$ and $m \in \mathbb{N}$ (any m, including $m = 0$). Every $g \in \mathrm{RotComp}(c, m)$ preserves orientation.
(c) Let $c \in \mathbb{R}^2$, $m \in \mathbb{N}$ (any m) and $\rho = R_{c,2\pi/m}$. Then $\rho^{[s]} \in \mathrm{RotComp}(c, m)$ for every $s \in \mathbb{N}$.

Proof. (a) For the identity there is nothing to show. If f and g preserve orientation, then for all a, b, d, $\operatorname{orient}(f(g(a)), f(g(b)), f(g(d))) = \operatorname{orient}(g(a), g(b), g(d)) = \operatorname{orient}(a, b, d)$, using first f (at the points $g(a), g(b), g(d)$) and then g. For the iterates use induction on n: $f^{[0]} = \mathrm{id}$, and $f^{[n+1]} = f^{[n]} \circ f$ is a composition of two orientation-preserving maps. Rotations preserve orientation by Lemma 4.7.

(b) Rule induction (Definition 4.9): the identity preserves orientation, and if g does, then so does $R_{c,ku} \circ g$, as a composition of two orientation-preserving maps by (a).

(c) Induction on s. $\rho^{[0]} = \mathrm{id} \in \mathrm{RotComp}(c, m)$ by (id). If $\rho^{[s]} \in \mathrm{RotComp}(c, m)$, then $\rho^{[s+1]} = \rho \circ \rho^{[s]} = R_{c,1\cdot u} \circ \rho^{[s]}$ with $u = 2\pi/m$, which lies in $\mathrm{RotComp}(c, m)$ by (step) with $k = 1$. □

For $m = 0$ we have $u = 0$ and $R_{c,0} = \mathrm{id}$, so every element of $\mathrm{RotComp}(c, 0)$ is the identity; the statements of this subsection and the next hold for every $m \in \mathbb{N}$.

4.3. **Permutations and orbit literals.**

Definition 4.11 (orientation of an ordered triple; Lean: `ErdosSzekeres.OrbitConsistency.orient3`, `LEANEXT/orbit/OrbitConsistency.lean:168`)**.** An *ordered triple* is a map $t : \{0, 1, 2\} \to \mathbb{R}^2$ (in Lean a function `Fin 3 → Point`), and $\operatorname{orient}_3(t) = \operatorname{orient}(t(0), t(1), t(2))$. For a permutation σ of $\{0, 1, 2\}$ and a map $g : \mathbb{R}^2 \to \mathbb{R}^2$, the triples $t \circ \sigma$ and $g \circ t \circ \sigma$ are $i \mapsto t(\sigma(i))$ and $i \mapsto g(t(\sigma(i)))$.

Fact 4.12 (the sign of a permutation; cited, Mathlib `Equiv.Perm.sign`)**.** *Let F be a finite set and* $\mathrm{Perm}(F)$ *the group of its permutations, with product $(\sigma\tau)(i) = \sigma(\tau(i))$. Mathlib's sign is a group homomorphism* $\operatorname{sgn} : \mathrm{Perm}(F) \to \mathbb{Z}^\times$ *with the following properties, which are all we use.*

- *(S1)* $\mathbb{Z}^\times = \{1, -1\}$ *(*`Int.units_eq_one_or`*).*
- *(S2)* $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$ *(*`Equiv.Perm.sign_mul`*); hence, with (S1),* $\operatorname{sgn}(\mathrm{id}) = 1$ *and* $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)^{-1} = \operatorname{sgn}(\sigma)$.
- *(S3)* $\operatorname{sgn}(\operatorname{swap}(x, y)) = -1$ *for $x \neq y$, where* $\operatorname{swap}(x, y)$ *exchanges x and y (*`Equiv.Perm.sign_swap`*).*
- *(S4) Every permutation of F is a product of transpositions; in the form used: if a property P of permutations holds for* id*, and $P(f)$ implies $P(\operatorname{swap}(x, y)\, f)$ for every f and all $x \neq y$, then P holds for every permutation (*`Equiv.Perm.swap_induction_on`*).*

In real formulas $\operatorname{sgn}\sigma$ *stands for the image of the unit* $\operatorname{sgn}\sigma$ *under $\mathbb{Z}^\times \to \mathbb{Z} \to \mathbb{R}$, the real number 1 or -1. This map is multiplicative, as a composite of monoid homomorphisms, so (S2) also holds for these real numbers; and it is injective because $1 \neq -1$ in $\mathbb{R}$.*

Lemma 4.13 (a transposition negates the orientation; Lean: `ErdosSzekeres.OrbitConsistency.orient3_swap`, `LEANEXT/orbit/OrbitConsistency.lean:171`)**.** *For every ordered triple t and all $x, y \in \{0, 1, 2\}$ with $x \neq y$:* $\operatorname{orient}_3(t \circ \operatorname{swap}(x, y)) = -\operatorname{orient}_3(t)$.

Proof. Since $\operatorname{swap}(x, y) = \operatorname{swap}(y, x)$, it suffices to treat $\{x, y\} = \{1, 2\}, \{0, 1\}, \{0, 2\}$. Write $t_i = t(i)$ and $o = \operatorname{orient}(t_0, t_1, t_2)$; we use $\operatorname{orient}(a, c, b) = -\operatorname{orient}(a, b, c)$ and $\operatorname{orient}(b, c, a) = \operatorname{orient}(a, b, c)$ (Lemma 1.15(a),(b)). For $\{1, 2\}$: $t \circ \operatorname{swap}(1, 2) = (t_0, t_2, t_1)$ and $\operatorname{orient}(t_0, t_2, t_1) = -o$. For $\{0, 1\}$: $t \circ \operatorname{swap}(0, 1) = (t_1, t_0, t_2)$, and the cyclic rule applied to (t_1, t_0, t_2) gives $\operatorname{orient}(t_1, t_0, t_2) = \operatorname{orient}(t_0, t_2, t_1) = -o$. For $\{0, 2\}$: $t \circ \operatorname{swap}(0, 2) = (t_2, t_1, t_0)$, and the cyclic rule applied to (t_2, t_1, t_0) gives $\operatorname{orient}(t_2, t_1, t_0) = \operatorname{orient}(t_1, t_0, t_2) = -o$ by the previous case. (The Lean proof runs through the six ordered pairs (x, y) and expands orient.) □

Lemma 4.14 (permutations multiply the orientation by the sign; Lean: `ErdosSzekeres.OrbitConsistency.orient3_perm`, `LEANEXT/orbit/OrbitConsistency.lean:178`)**.** *For every permutation σ of $\{0, 1, 2\}$ and every ordered triple t,*

$$\operatorname{orient}_3(t \circ \sigma) = \operatorname{sgn}(\sigma) \cdot \operatorname{orient}_3(t).$$

Proof. We follow the Lean proof, an induction over transpositions (Fact 4.12(S4)) for the property $P(\sigma)$: "$\operatorname{orient}_3(t \circ \sigma) = \operatorname{sgn}(\sigma)\operatorname{orient}_3(t)$ for *every* triple t". $P(\mathrm{id})$ holds since $t \circ \mathrm{id} = t$ and $\operatorname{sgn}(\mathrm{id}) = 1$. Assume $P(f)$ and $x \neq y$, and let t be any triple. Since $(\operatorname{swap}(x, y)\, f)(i) = \operatorname{swap}(x, y)(f(i))$, we have $t \circ (\operatorname{swap}(x, y)\, f) = (t \circ \operatorname{swap}(x, y)) \circ f$. By $P(f)$ for the triple $t \circ \operatorname{swap}(x, y)$ and by Lemma 4.13,

$$\operatorname{orient}_3\big(t \circ (\operatorname{swap}(x, y)\, f)\big) = \operatorname{sgn}(f)\, \operatorname{orient}_3(t \circ \operatorname{swap}(x, y)) = -\operatorname{sgn}(f)\, \operatorname{orient}_3(t),$$

and $\operatorname{sgn}(\operatorname{swap}(x, y)\, f) = \operatorname{sgn}(\operatorname{swap}(x, y))\operatorname{sgn}(f) = -\operatorname{sgn}(f)$ by (S2) and (S3). So $P(\operatorname{swap}(x, y)\, f)$ holds, and (S4) gives $P(\sigma)$ for all σ. □

Lemma 4.15 (rotations and permutations together; Lean: `ErdosSzekeres.OrbitConsistency.orient3_rot_perm`, `LEANEXT/orbit/OrbitConsistency.lean:194`)**.** *Let $c \in \mathbb{R}^2$, $m \in \mathbb{N}$, $g \in \mathrm{RotComp}(c,m)$, σ a permutation of $\{0,1,2\}$, and t, t' ordered triples with $t' = g \circ t \circ \sigma$. Then* $\mathrm{orient}_3(t') = \mathrm{sgn}(\sigma) \cdot \mathrm{orient}_3(t)$.

Proof. By Lemma 4.10(b), g preserves orientation, so $\mathrm{orient}_3(t') = \mathrm{orient}\big(g(t(\sigma(0))), g(t(\sigma(1))), g(t(\sigma(2)))\big) = \mathrm{orient}\big(t(\sigma(0)), t(\sigma(1)), t(\sigma(2))\big) = \mathrm{orient}_3(t \circ \sigma)$, and $\mathrm{orient}_3(t \circ \sigma) = \mathrm{sgn}(\sigma)\,\mathrm{orient}_3(t)$ by Lemma 4.14. □

Lemma 4.16 (two representations of one triple; Lean: `ErdosSzekeres.OrbitConsistency.sign_eq_of_two_reps`, `LEANEXT/orbit/OrbitConsistency.lean:204`, Lean: `ErdosSzekeres.OrbitConsistency.orient3_of_common_rep`, `LEANEXT/orbit/OrbitConsistency.lean:216`)**.** *Let $c \in \mathbb{R}^2$, $m \in \mathbb{N}$ and $g_1, g_2 \in \mathrm{RotComp}(c,m)$, and let σ_1, σ_2 be permutations of $\{0,1,2\}$.*

(a) (`sign_eq_of_two_reps`) If t, t' are ordered triples with $t' = g_1 \circ t \circ \sigma_1$ and $t' = g_2 \circ t \circ \sigma_2$, and $\mathrm{orient}_3(t) \neq 0$*, then* $\mathrm{sgn}\,\sigma_1 = \mathrm{sgn}\,\sigma_2$ *in $\mathbb{Z}^\times$.*

(b) (`orient3_of_common_rep`) If t_1, t_2, C are ordered triples with $C = g_1 \circ t_1 \circ \sigma_1$ and $C = g_2 \circ t_2 \circ \sigma_2$, then $\mathrm{sgn}(\sigma_1)\,\mathrm{orient}_3(t_1) = \mathrm{sgn}(\sigma_2)\,\mathrm{orient}_3(t_2)$ *in $\mathbb{R}$.*

Proof. (a) Lemma 4.15, applied to both representations, gives $\mathrm{sgn}(\sigma_1)\,\mathrm{orient}_3(t) = \mathrm{orient}_3(t') = \mathrm{sgn}(\sigma_2)\,\mathrm{orient}_3(t)$. Since $\mathrm{orient}_3(t) \neq 0$, the real numbers $\mathrm{sgn}\,\sigma_1$ and $\mathrm{sgn}\,\sigma_2$ are equal, and so are the units (Fact 4.12: the map $\mathbb{Z}^\times \to \mathbb{R}$ is injective). (b) By Lemma 4.15 both sides equal $\mathrm{orient}_3(C)$. □

Definition 4.17 (the literal of the encoder; Lean: `ErdosSzekeres.OrbitConsistency.litVal`, `LEANEXT/orbit/OrbitConsistency.lean:223`)**.** For $\varepsilon \in \mathbb{Z}^\times$ and an ordered triple C, $\mathrm{litVal}(\varepsilon, C)$ is the proposition "$0 < \mathrm{orient}_3(C)$" if $\varepsilon = 1$, and the proposition "not $0 < \mathrm{orient}_3(C)$" otherwise (that is, if $\varepsilon = -1$). It models the SAT literal "variable of the triple C, negated when the parity ε is odd", where the variable of C stands for $0 < \mathrm{orient}_3(C)$.

Theorem 4.18 (literal soundness; Lean: `ErdosSzekeres.OrbitConsistency.literal_sound`, `LEANEXT/orbit/OrbitConsistency.lean:228`)**.** *Let $c \in \mathbb{R}^2$, $m \in \mathbb{N}$, $g \in \mathrm{RotComp}(c,m)$, σ a permutation of $\{0,1,2\}$, and t, C ordered triples with $C = g \circ t \circ \sigma$ and* $\mathrm{orient}_3(t) \neq 0$*. Then*

$$0 < \mathrm{orient}_3(t) \iff \mathrm{litVal}(\mathrm{sgn}\,\sigma,\, C).$$

Proof. By Lemma 4.15, $\mathrm{orient}_3(C) = \mathrm{sgn}(\sigma)\,\mathrm{orient}_3(t)$. By (S1), $\mathrm{sgn}\,\sigma = 1$ or $\mathrm{sgn}\,\sigma = -1$. If $\mathrm{sgn}\,\sigma = 1$, then $\mathrm{litVal}(\mathrm{sgn}\,\sigma, C)$ is "$0 < \mathrm{orient}_3(C)$" and $\mathrm{orient}_3(C) = \mathrm{orient}_3(t)$, so the two sides coincide. If $\mathrm{sgn}\,\sigma = -1$, then $\mathrm{sgn}\,\sigma \neq 1$, so $\mathrm{litVal}(\mathrm{sgn}\,\sigma, C)$ is "not $0 < \mathrm{orient}_3(C)$", and $\mathrm{orient}_3(C) = -\,\mathrm{orient}_3(t)$. If $0 < \mathrm{orient}_3(t)$, then $-\,\mathrm{orient}_3(t) < 0$, so the literal holds. Conversely, if the literal holds, then $-\,\mathrm{orient}_3(t) \leq 0$, i.e. $\mathrm{orient}_3(t) \geq 0$, and $\mathrm{orient}_3(t) \neq 0$ gives $0 < \mathrm{orient}_3(t)$. □

Theorem 4.19 (literal soundness, label-level form modelled on the encoder; Lean: `ErdosSzekeres.OrbitConsistency.encoder_literal_sound`, `LEANEXT/orbit/OrbitConsistency.lean:253`)**.** *Let ι be any type (of labels), $P : \iota \to \mathbb{R}^2$ and $r : \iota \to \iota$ maps, $c \in \mathbb{R}^2$, $m \in \mathbb{N}$, and $\rho = R_{c,2\pi/m}$. Assume*

$$P(r(i)) = \rho(P(i)) \qquad \text{for every } i \in \iota.$$

Let $T, T_0, C : \{0,1,2\} \to \iota$, let σ_1, σ_2 be permutations of $\{0,1,2\}$, and $s \in \mathbb{N}$, with

$$T_0 = T \circ \sigma_1 \qquad \text{and} \qquad C = r^{[s]} \circ T_0 \circ \sigma_2.$$

If $\mathrm{orient}_3(P \circ T) \neq 0$*, then*

$$0 < \mathrm{orient}_3(P \circ T) \iff \mathrm{litVal}\big(\mathrm{sgn}\,\sigma_1 \cdot \mathrm{sgn}\,\sigma_2,\ P \circ C\big).$$

Nothing else is assumed: P need not be injective, and m is arbitrary (for $m = 0$, $\rho = R_{c,0} = \mathrm{id}$). Distinctness of the labels $T(0), T(1), T(2)$ is not a separate hypothesis; it follows from $\mathrm{orient}_3(P \circ T) \neq 0$*, because* orient *vanishes when two of its arguments are equal (Lemma 1.15(e)).*

Proof. Step 1: $P \circ r^{[s]} = \rho^{[s]} \circ P$. Induction on s: for $s = 0$ both sides are P; if it holds for s, then for every label i, $P(r^{[s+1]}(i)) = P(r(r^{[s]}(i))) = \rho(P(r^{[s]}(i))) = \rho(\rho^{[s]}(P(i))) = \rho^{[s+1]}(P(i))$.

Step 2: $P \circ C = \rho^{[s]} \circ (P \circ T) \circ (\sigma_1\sigma_2)$. For $i \in \{0,1,2\}$, $P(C(i)) = P\big(r^{[s]}(T_0(\sigma_2(i)))\big) = \rho^{[s]}\big(P(T(\sigma_1(\sigma_2(i))))\big)$ by Step 1 and $T_0 = T \circ \sigma_1$, and $\sigma_1(\sigma_2(i)) = (\sigma_1\sigma_2)(i)$.

Step 3. $\rho^{[s]} \in \mathrm{RotComp}(c,m)$ by Lemma 4.10(c). Theorem 4.18 with $g = \rho^{[s]}$, $\sigma = \sigma_1\sigma_2$, $t = P \circ T$ and the triple $P \circ C$ (Step 2), and $\mathrm{orient}_3(P \circ T) \neq 0$, gives $0 < \mathrm{orient}_3(P \circ T) \iff \mathrm{litVal}(\mathrm{sgn}(\sigma_1\sigma_2), P \circ C)$. Finally $\mathrm{sgn}(\sigma_1\sigma_2) = \mathrm{sgn}\,\sigma_1 \cdot \mathrm{sgn}\,\sigma_2$ by (S2). □

Remark 4.20 (parity by counting inversions is not modelled in Lean). The encoders compute parities by counting inversions, in Python. For a list a of three labels let $\mathrm{inv}(a)$ be the number of pairs $i < j$ of positions with $a(i) > a(j)$. The function `sortpar` of `C7UPPER/gen.py` (line 108) bubble-sorts a and flips the parity at each exchange of adjacent entries; each exchange removes exactly one inversion and the sorted list has none, so it returns $(-1)^{\mathrm{inv}(a)}$. It is applied to the ordered label triple (line 120) and to the rotations of the sorted triple (line 78); these give the parities that play the roles of $\operatorname{sgn}\sigma_1$ and $\operatorname{sgn}\sigma_2$. The function `perm_sign` of `C6UPPER/encb/gen_b2.py` (line 56) counts the inversions of its argument and also returns $(-1)^{\mathrm{inv}(a)}$; it is applied (lines 72–73) to the list $\sigma(0)\sigma(1)\sigma(2)$ of the permutation σ that sorts the rotated triple, so this encoder corresponds to $\sigma_1 = \mathrm{id}$ and $\sigma_2 = \sigma$. Theorem 4.19 is stated with Mathlib's sgn instead. The two agree on lists of three distinct labels, by a six-case check. Write a permutation τ of $\{0,1,2\}$ as the list $\tau(0)\tau(1)\tau(2)$; so $\mathrm{inv}(\tau)$ is the number of pairs $i < j$ with $\tau(i) > \tau(j)$. If σ sorts a list a of three distinct labels, i.e. $a \circ \sigma$ is increasing, then $a(i) > a(j)$ iff $\sigma^{-1}(i) > \sigma^{-1}(j)$, so $\mathrm{inv}(a) = \mathrm{inv}(\sigma^{-1})$. For the six permutations τ one has

τ	τ as a product of transpositions	$\mathrm{inv}(\tau)$	$\operatorname{sgn}\tau$ by (S2), (S3)
012	id	0	1
102	$\mathrm{swap}(0,1)$	1	-1
021	$\mathrm{swap}(1,2)$	1	-1
210	$\mathrm{swap}(0,2)$	3	-1
120	$\mathrm{swap}(0,1)\,\mathrm{swap}(1,2)$	2	1
201	$\mathrm{swap}(1,2)\,\mathrm{swap}(0,1)$	2	1

so $(-1)^{\mathrm{inv}(\tau)} = \operatorname{sgn}\tau$ for every τ. Hence `perm_sign`, applied to the list of a permutation σ, returns $(-1)^{\mathrm{inv}(\sigma)} = \operatorname{sgn}\sigma$; and `sortpar`, applied to a list a of distinct labels that σ sorts, returns $(-1)^{\mathrm{inv}(a)} = (-1)^{\mathrm{inv}(\sigma^{-1})} = \operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$ by (S2). This six-case argument, the reading of the code above and the correspondence between the generator code and the data $T, T_0, C, \sigma_1, \sigma_2, s$ of Theorem 4.19 are *not* part of any Lean statement. The Python script `LEANEXT/orbit/check_literals.py` checked (lines 36–39) that `sortpar` sorts each of the six orders of $(0,1,2)$ and returns the same parity as the script's own inversion count `perm_sign` (lines 25–31); this compares two inversion counts, not `sortpar` with Mathlib's sgn. The same script compared the literal code of `C7UPPER/gen.py` with floating-point orientations on 368,814 ordered triples with no mismatch (`LEANEXT/orbit/logs/check_literals.log`, lines 1–5). These Python checks are evidence, not proof. The design notes of the ENCBLEAN route record, in their entry for this theorem, the mismatch between sgn and the inversion count and that a six-case bridge would be needed, and give the verdict "Not used" (`ENCBLEAN/DESIGN.md:625`).

Remark 4.21 (how the theorems are used). As described in the doc comment of `encoder_literal_sound`, the encoder keeps one Boolean variable $V(C)$ per canonical representative C of an orbit of ordered label triples under the label rotation r and the permutations of the positions; $V(C)$ is meant to be $[0 < \mathrm{orient}_3(P \circ C)]$. For an ordered triple T it sorts T ($T_0 = T \circ \sigma_1$), maps the sorted triple to the representative ($C = r^{[s]} \circ T_0 \circ \sigma_2$), and uses $V(C)$, negated when $\operatorname{sgn}\sigma_1 \operatorname{sgn}\sigma_2 = -1$. Under the semiconjugacy $P \circ r = \rho \circ P$ and $\mathrm{orient}_3(P \circ T) \neq 0$ (which holds when the three points are distinct points of a set in general position, by Lemma 1.20), Theorem 4.19 says that this literal is true exactly when $\mathrm{orient}(P(T(0)), P(T(1)), P(T(2))) > 0$, whichever representation is chosen; Lemma 4.16 shows directly that the sign does not depend on the representation.

4.4. **Even order: antipodes, rays and phases.**

Lemma 4.22 (antipodes; Lean: `ErdosSzekeres.OrbitConsistency.antipode_mem_of_even`, `LEANEXT/orbit/OrbitConsistency.lean:273`, Lean: `ErdosSzekeres.LayerLemma.antipode_mem_of_even`, `LEANEXT/merged/Merged.lean:1160`)**.** *Let S be a finite set of points, $c \in \mathbb{R}^2$ and $m \in \mathbb{N}$ with m even and $0 < m$, and assume that $\rho = R_{c,2\pi/m}$ maps S into S. If $x \in S$, then the point $(2c_0 - x_0,\ 2c_1 - x_1) = 2c - x$ belongs to S.*

Proof. By Lemma 4.6, $\rho^{[m/2]}(x) \in S$, and $\rho^{[m/2]}(x) = (2c_0 - x_0, 2c_1 - x_1)$ by Corollary 4.5. □

Lemma 4.23 (no two points on one open ray from the centre; Lean: `ErdosSzekeres.OrbitConsistency.same_ray_false`, `LEANEXT/orbit/OrbitConsistency.lean:295`, Lean: `ErdosSzekeres.LayerLemma.same_ray_false`, `LEANEXT/merged/Merged.lean:1181`)**.** *Let S be a finite set of points, $c \in \mathbb{R}^2$ and $m \in \mathbb{N}$ with m even and $0 < m$; assume that $\rho = R_{c,2\pi/m}$ maps S into S and that S is in general position. Let $x, y \in S$ with $x \neq y$, and $l \in \mathbb{R}$ with $l > 0$. Then*

$$y - c \neq l\,(x - c).$$

(In Lean the equation $y - c = l\,(x - c)$ is a hypothesis and the conclusion is `False`*; this is the same statement.) The condition $c \notin S$ is not a hypothesis; it follows from the others by Lemma 3.18, since $x \neq y$ gives a point of S other than c, but the proof below does not use it.*

Proof. Suppose $y - c = l(x - c)$, i.e. $y = c + l(x - c)$. Let $q = 2c - x$; then $q \in S$ by Lemma 4.22.

(i) $x \neq c$: if $x = c$, then $y = c + l \cdot 0 = c = x$, contrary to $x \neq y$.
(ii) $x \neq q$: $x = q$ means $2(x - c) = 0$, i.e. $x = c$, excluded by (i).
(iii) $y \neq q$: $y = q$ means $c + l(x - c) = 2c - x$, i.e. $(l + 1)(x - c) = 0$; as $l + 1 > 0$, this gives $x = c$, excluded by (i).
(iv) $\operatorname{orient}(x, y, q) = 0$: $y - x = (l - 1)(x - c)$ and $q - x = -2(x - c)$, so $\operatorname{orient}(x, y, q) = \det\big(y - x \mid q - x\big) = -2(l - 1)\det\big(x - c \mid x - c\big) = 0$.

Thus x, y, q are pairwise distinct points of S with $\operatorname{orient}(x, y, q) = 0$, which contradicts general position by Lemma 1.20 (`generalPosition_iff_orient_ne_zero`, `ErdosSzekeres/OrientationGeometry.lean:66`). □

Remark 4.24 (the hypothesis $l > 0$). The proof uses $l > 0$ only through $l + 1 \neq 0$. The hypothesis cannot be replaced by $l \neq 0$: for any $x \in S$ with $x \neq c$, the antipode $y = 2c - x$ lies in S (Lemma 4.22), $y \neq x$, and $y - c = (-1)(x - c)$. Such S exist, for example $m = 2$ and $S = \{x, 2c - x\}$ with $x \neq c$ (general position holds because S has only two points).

Lemma 4.25 (rotating and shifting polar points; Lean: `ErdosSzekeres.OrbitConsistency.rotAbout_polar`, `LEANEXT/orbit/OrbitConsistency.lean:355`, Lean: `ErdosSzekeres.OrbitConsistency.polar_add_int`, `LEANEXT/orbit/OrbitConsistency.lean:383`, Lean: `ErdosSzekeres.LayerLemma.rotAbout_polar`, `LEANEXT/merged/Merged.lean:1241`, Lean: `ErdosSzekeres.LayerLemma.polar_add_int`, `LEANEXT/merged/Merged.lean:1269`)**.** *A rotation about c adds its angle to the angle of a polar point about c, and polar points are 2π-periodic in the angle:*

(a) (`rotAbout_polar`*) For all $c \in \mathbb{R}^2$ and $r, \theta, \varphi \in \mathbb{R}$: $R_{c,\varphi}(\operatorname{polar}(c, r, \theta)) = \operatorname{polar}(c, r, \theta + \varphi)$.*
(b) (`polar_add_int`*) For all $c \in \mathbb{R}^2$, $r, \theta \in \mathbb{R}$ and $n \in \mathbb{Z}$: $\operatorname{polar}(c, r, \theta + n \cdot 2\pi) = \operatorname{polar}(c, r, \theta)$.*

Proof. (a) $\operatorname{polar}(c, r, \theta) - c = (r\cos\theta, r\sin\theta)$, so $R_{c,\varphi}$ maps the point to $c + \big(r(\cos\varphi\cos\theta - \sin\varphi\sin\theta),\ r(\sin\varphi\cos\theta + \cos\varphi\sin\theta)\big) = c + \big(r\cos(\theta + \varphi),\ r\sin(\theta + \varphi)\big)$ by the addition formulas. (b) cos and sin have period 2π, so $\cos(\theta + 2\pi n) = \cos\theta$ and $\sin(\theta + 2\pi n) = \sin\theta$ for $n \in \mathbb{Z}$. □

Lemma 4.26 (equal phases modulo $2\pi/m$, natural form; Lean: `ErdosSzekeres.OrbitConsistency.eq_iterate_of_phase_eq`, `LEANEXT/orbit/OrbitConsistency.lean:363`, Lean: `ErdosSzekeres.LayerLemma.eq_iterate_of_phase_eq`, `LEANEXT/merged/Merged.lean:1249`)**.** *Let S be a finite set of points, $c \in \mathbb{R}^2$ and $m \in \mathbb{N}$ with m even and $0 < m$; assume that $\rho = R_{c,2\pi/m}$ maps S into S and that S is in general position. Let $r, r', \alpha, \beta \in \mathbb{R}$ with $r > 0$, $r' > 0$, $\operatorname{polar}(c, r, \alpha) \in S$ and $\operatorname{polar}(c, r', \beta) \in S$, and let $k \in \mathbb{N}$ with $\beta = \alpha + k \cdot (2\pi/m)$. Then*

$$\operatorname{polar}(c, r', \beta) = \rho^{[k]}\big(\operatorname{polar}(c, r, \alpha)\big).$$

Proof. Let $z = \rho^{[k]}(\operatorname{polar}(c, r, \alpha))$ and $y = \operatorname{polar}(c, r', \beta)$. By Lemma 4.4, $\rho^{[k]} = R_{c,k(2\pi/m)}$, so by Lemma 4.25(a), $z = \operatorname{polar}(c, r, \alpha + k(2\pi/m)) = \operatorname{polar}(c, r, \beta)$; and $z \in S$ by Lemma 4.6. Suppose $y \neq z$. Then

$$y - c = (r'\cos\beta,\ r'\sin\beta) = \tfrac{r'}{r}\,(r\cos\beta,\ r\sin\beta) = \tfrac{r'}{r}\,(z - c),$$

with $r'/r > 0$. So $z, y \in S$, $z \neq y$ and $y - c = l(z - c)$ with $l = r'/r > 0$, which Lemma 4.23 (applied with $x := z$) excludes. Hence $y = z$. □

Lemma 4.27 (equal phases modulo $2\pi/m$, integer form; Lean: `ErdosSzekeres.OrbitConsistency.eq_iterate_of_phase_eq_int`, `LEANEXT/orbit/OrbitConsistency.lean:389`, Lean: `ErdosSzekeres.LayerLemma.eq_iterate_of_phase_eq_int`, `LEANEXT/merged/Merged.lean:1275`)**.** *Let S be a finite set of points, $c \in \mathbb{R}^2$ and $m \in \mathbb{N}$ with m even and $0 < m$; assume that $\rho = R_{c,2\pi/m}$ maps S into S and that S is in general position. Let $r, r', \alpha, \beta \in \mathbb{R}$ with $r > 0$, $r' > 0$, $\operatorname{polar}(c, r, \alpha) \in S$ and $\operatorname{polar}(c, r', \beta) \in S$, and let $k \in \mathbb{Z}$ with $\beta = \alpha + k \cdot (2\pi/m)$. Then there is $j \in \mathbb{N}$ with*

$$\operatorname{polar}(c, r', \beta) = \rho^{[j]}\big(\operatorname{polar}(c, r, \alpha)\big).$$

(The proof takes for j the remainder of k modulo m.)

Proof. Since $m > 0$, Euclidean division of k by m gives $q = \lfloor k/m \rfloor \in \mathbb{Z}$ and a remainder $j = k - mq \in \{0, 1, \dots, m-1\}$, which we regard as a natural number (Lean: `(k % m).toNat`, with $0 \le k \bmod m$ by `Int.emod_nonneg` and $k = (k \bmod m) + m\lfloor k/m \rfloor$ by `Int.emod_add_mul_ediv`). Then $k = j + mq$, and, as $m \neq 0$, $k \cdot \frac{2\pi}{m} = j \cdot \frac{2\pi}{m} + q \cdot 2\pi$. Put $\beta' = \alpha + j \cdot (2\pi/m)$; then $\beta = \beta' + q \cdot 2\pi$, so $\mathrm{polar}(c, r', \beta) = \mathrm{polar}(c, r', \beta')$ by Lemma 4.25(b). In particular $\mathrm{polar}(c, r', \beta') \in S$, and Lemma 4.26 (with β' and the natural number j) gives $\mathrm{polar}(c, r', \beta') = \rho^{[j]}(\mathrm{polar}(c, r, \alpha))$. Hence the claim. □

Theorem 4.28 (points outside an orbit have different phases; Lean: `ErdosSzekeres.OrbitConsistency.phase_ne_of_not_mem_orbit`, `LEANEXT/orbit/OrbitConsistency.lean:420`, Lean: `ErdosSzekeres.LayerLemma.phase_ne_of_not_mem_orbit`, `LEANEXT/merged/Merged.lean:1304`)**.** *Let S be a finite set of points, $c \in \mathbb{R}^2$ and $m \in \mathbb{N}$ with m even and $0 < m$; assume that $\rho = R_{c,2\pi/m}$ maps S into S and that S is in general position. Let $r, r', \alpha, \beta \in \mathbb{R}$ with $r > 0$, $r' > 0$, $\mathrm{polar}(c, r, \alpha) \in S$, $\mathrm{polar}(c, r', \beta) \in S$, and*

$$\mathrm{polar}(c, r', \beta) \neq \rho^{[j]}\big(\mathrm{polar}(c, r, \alpha)\big) \qquad \text{for every } j \in \mathbb{N}.$$

Then $\beta \neq \alpha + k \cdot (2\pi/m)$ for every $k \in \mathbb{Z}$.

Proof. Let $k \in \mathbb{Z}$ and suppose $\beta = \alpha + k(2\pi/m)$. Lemma 4.27 gives $j \in \mathbb{N}$ with $\mathrm{polar}(c, r', \beta) = \rho^{[j]}(\mathrm{polar}(c, r, \alpha))$, contrary to the hypothesis. □

Remark 4.29 (what is and is not claimed). The doc comment of the `OrbitConsistency.lean` version of `phase_ne_of_not_mem_orbit` adds that, with the encoder's labelling (layer 0 has a point of direction 0), the phases of distinct layers differ and the phases of the layers $i \ge 1$ are nonzero. That reading needs facts that are not hypotheses of Theorem 4.28: that a point of one layer is not in the ρ-orbit of a point of another layer (this is what the proof of Theorem 4.33 below establishes, from Lemmas 4.30 and 4.31), and the normalisation that layer 0 contains a point of direction 0, together with the definition of the phases relative to it, which is not part of any Lean statement of this section.

4.5. **Distinct layers: rays and phases.** We use the hull vertices $\mathrm{hv}(X) = \{x \in X : x \notin \mathrm{conv}(X \setminus \{x\})\}$ of a finite set X (Definition 2.4) and the peels $\mathrm{peel}(S, 0) = S$, $\mathrm{peel}(S, n+1) = \mathrm{peel}(S, n) \setminus \mathrm{hv}(\mathrm{peel}(S, n))$ (Definition 2.18); the n-th layer of S is $\mathrm{hv}(\mathrm{peel}(S, n))$. We use three facts from the layer-lemma section:

- (L1) $\mathrm{hv}(X) \subseteq X$ (by definition), and $\mathrm{peel}(S, n') \subseteq \mathrm{peel}(S, n)$ for $n \le n'$ (Lemma 2.20); in particular $\mathrm{peel}(S, n) \subseteq \mathrm{peel}(S, 0) = S$.
- (L2) (Lemma 2.21) If $0 < m$, $c \notin S$ and $\rho = R_{c,2\pi/m}$ maps S into S, then ρ maps every $\mathrm{peel}(S, n)$ into itself.
- (L3) (Theorem 2.17, `hullVerts_eq_orbitFS_rotAbout`, `LAYERLEAN/LayerLemma.lean:364`) If $0 < m$, $K \le 2m$, X is a finite set of points with $c \notin X$, ρ maps X into X, X has no K points in convex position, and $x \in \mathrm{hv}(X)$, then $\mathrm{hv}(X) = \{\rho^{[a]}(x) : 0 \le a < m\}$ (Lean: `orbitFS` of ρ, m and x).

"S has no K points in convex position" is $\neg$ `HasConvexNGon K S` (Definition 1.7).

Lemma 4.30 (distinct layers are disjoint; Lean: `ErdosSzekeres.LayerLemma.not_mem_hullVerts_peel_of_lt`, `LEANEXT/merged/Merged.lean:1316`, Lean: `ErdosSzekeres.LayerLemma.not_mem_hullVerts_peel_of_ne`, `LEANEXT/merged/Merged.lean:1322`)**.** *Let S be a finite set of points (no other hypothesis).*

(a) If $i < j$ and $y \in \mathrm{hv}(\mathrm{peel}(S, j))$, then $y \notin \mathrm{hv}(\mathrm{peel}(S, i))$.

(b) If $i \neq j$, then no point y satisfies both $y \in \mathrm{hv}(\mathrm{peel}(S, i))$ and $y \in \mathrm{hv}(\mathrm{peel}(S, j))$.

Proof. (a) By (L1) and $i + 1 \le j$: $y \in \mathrm{hv}(\mathrm{peel}(S, j)) \subseteq \mathrm{peel}(S, j) \subseteq \mathrm{peel}(S, i+1) = \mathrm{peel}(S, i) \setminus \mathrm{hv}(\mathrm{peel}(S, i))$, so $y \notin \mathrm{hv}(\mathrm{peel}(S, i))$. (b) If $i < j$, apply (a) to y as a point of layer j; if $j < i$, apply (a) with i and j exchanged. □

Lemma 4.31 (each layer is closed under the rotation; Lean: `ErdosSzekeres.LayerLemma.iterate_mem_hullVerts_peel`, `LEANEXT/merged/Merged.lean:1330`)**.** *Let S be a finite set of points, $c \in \mathbb{R}^2$, and $m, K \in \mathbb{N}$ with $0 < m$ and $K \le 2m$; assume $c \notin S$, that $\rho = R_{c,2\pi/m}$ maps S into S, and that S has no K points in convex position. If $i \in \mathbb{N}$ and $x \in \mathrm{hv}(\mathrm{peel}(S, i))$, then $\rho^{[k]}(x) \in \mathrm{hv}(\mathrm{peel}(S, i))$ for every $k \in \mathbb{N}$.*

Proof. Put $X = \mathrm{peel}(S, i)$. By (L1), $X \subseteq S$; hence $c \notin X$, and X has no K points in convex position (such points would be K points of S in convex position; Lemma 1.23(c)). By (L2), ρ maps X into X. By (L3), $\mathrm{hv}(X) = O := \{\rho^{[a]}(x) : 0 \le a < m\}$. The set O is mapped into itself by ρ: for $a + 1 < m$, $\rho(\rho^{[a]}(x)) = \rho^{[a+1]}(x) \in O$, and for $a = m - 1$, $\rho(\rho^{[m-1]}(x)) = \rho^{[m]}(x) = x = \rho^{[0]}(x) \in O$, because $\rho^{[m]} = \mathrm{id}$ (Corollary 4.5). Since $x = \rho^{[0]}(x) \in O$ (as $0 < m$), Lemma 4.6 gives $\rho^{[k]}(x) \in O = \mathrm{hv}(X)$ for every k. (The Lean proof obtains $\rho^{[m]}(x) = x$ from $x \neq c$ and the minimal period, Lemma 2.16; the direct identity $\rho^{[m]} = \mathrm{id}$ used here needs neither.) □

Theorem 4.32 (distinct layers never share a ray through a rotated point; Lean: `ErdosSzekeres.LayerLemma.not_same_ray_of_distinct_layers`, `LEANEXT/merged/Merged.lean:1350`)**.** *Let S be a finite set of points, $c \in \mathbb{R}^2$, and $m, K \in \mathbb{N}$ with $2 \le m$, m even and $K \le 2m$; assume $c \notin S$, that $\rho = R_{c,2\pi/m}$ maps S into S, that S is in general position, and that S has no K points in convex position. Let $i, j \in \mathbb{N}$ with $i \ne j$, $x \in \mathrm{hv}(\mathrm{peel}(S,i))$, $y \in \mathrm{hv}(\mathrm{peel}(S,j))$, $k \in \mathbb{N}$ and $l \in \mathbb{R}$ with $l > 0$. Then*

$$y - c \ne l\,\big(\rho^{[k]}(x) - c\big).$$

(In Lean the equation is a hypothesis and the conclusion is `False`*.)*

Proof. We have $0 < m$. Put $x' = \rho^{[k]}(x)$; then $x' \in \mathrm{hv}(\mathrm{peel}(S,i))$ by Lemma 4.31. Also $x' \ne y$: otherwise y would lie in $\mathrm{hv}(\mathrm{peel}(S,i))$ and in $\mathrm{hv}(\mathrm{peel}(S,j))$, contrary to Lemma 4.30(b). By (L1), $x', y \in S$. If $y - c = l(x' - c)$, then Lemma 4.23 (with $x := x'$, using m even, $0 < m$, invariance and general position) gives a contradiction. □

Theorem 4.33 (distinct layers have phases incongruent modulo $2\pi/m$; Lean: `ErdosSzekeres.LayerLemma.phase_ne_of_distinct_layers`, `LEANEXT/merged/Merged.lean:1367`)**.** *Let S be a finite set of points, $c \in \mathbb{R}^2$, and $m, K \in \mathbb{N}$ with $2 \le m$, m even and $K \le 2m$; assume $c \notin S$, that $\rho = R_{c,2\pi/m}$ maps S into S, that S is in general position, and that S has no K points in convex position. Let $i, j \in \mathbb{N}$ with $i \ne j$, and $r, r', \alpha, \beta \in \mathbb{R}$ with $r > 0$, $r' > 0$,*

$$\mathrm{polar}(c, r, \alpha) \in \mathrm{hv}(\mathrm{peel}(S,i)) \qquad \text{and} \qquad \mathrm{polar}(c, r', \beta) \in \mathrm{hv}(\mathrm{peel}(S,j)).$$

Then $\beta - \alpha \ne k \cdot (2\pi/m)$ for every $k \in \mathbb{Z}$.

Proof. We apply Theorem 4.28. Its hypotheses hold: m is even and $0 < m$; ρ maps S into S; S is in general position; $r, r' > 0$; both points lie in S by (L1). For the orbit hypothesis, suppose $\mathrm{polar}(c, r', \beta) = \rho^{[n]}(\mathrm{polar}(c, r, \alpha))$ for some $n \in \mathbb{N}$; then this point lies in $\mathrm{hv}(\mathrm{peel}(S,i))$ by Lemma 4.31, and also in $\mathrm{hv}(\mathrm{peel}(S,j))$, contrary to Lemma 4.30(b). Now let $k \in \mathbb{Z}$ and suppose $\beta - \alpha = k(2\pi/m)$. Then $\beta = \alpha + k(2\pi/m)$, which Theorem 4.28 excludes. □

Remark 4.34 (on the hypotheses and the conclusion). (1) The hypothesis $c \notin S$ of Theorems 4.32 and 4.33 follows from the others, although Lean keeps it as a hypothesis: the two given points are distinct by Lemma 4.30(b), so S contains a point $p \ne c$, and Lemma 3.18 (`centre_not_mem_of_even`: $0 < m$, m even, ρ maps S into S, S in general position, $p \in S$, $p \ne c$) gives $c \notin S$. (2) The points of Theorem 4.33 are given in polar form with positive radii; no statement of this section produces such a form from a point. (3) The modulus $2\pi/m$ cannot be replaced by π/m. Take $m = 4$, $c = (0,0)$, $K = 8$ and $S = \{(\pm 2, 0), (0, \pm 2), (\pm\frac{1}{2}, \pm\frac{1}{2})\}$, with four outer points $(\pm 2, 0), (0, \pm 2)$ and four inner points $(\pm\frac{1}{2}, \pm\frac{1}{2})$. The rotation by $\pi/2$ maps S onto itself and $c \notin S$. S is in general position: among any three points of S two are of the same kind (outer or inner), so it suffices that the line through two points of the same kind contains no third point of S. The lines through two of the outer points are $x \pm y = \pm 2$, $x = 0$ and $y = 0$, and the lines through two of the inner points are $x = \pm\frac{1}{2}$, $y = \pm\frac{1}{2}$ and $x = \pm y$; no third point of S lies on any of them, since inner points have $|x| = |y| = \frac{1}{2}$ and outer points have one coordinate 0 and the other ± 2. For $\varepsilon_1, \varepsilon_2 \in \{1, -1\}$,

$$\left(\tfrac{\varepsilon_1}{2}, \tfrac{\varepsilon_2}{2}\right) = \tfrac{3}{8}(2\varepsilon_1, 0) + \tfrac{3}{8}(0, 2\varepsilon_2) + \tfrac{1}{8}(-2\varepsilon_1, 0) + \tfrac{1}{8}(0, -2\varepsilon_2),$$

a convex combination of the outer points; hence no inner point is a hull vertex of S, S is not in convex position (Lemma 1.22), and S (which has 8 points) has no 8 points in convex position. Each outer point is a hull vertex of S, since it is the only point of S maximising one of the linear functions $\pm x$, $\pm y$ (easy direction of Lemma 1.40); so layer 0 consists of the four outer points. Likewise each inner point is the only point of $\mathrm{peel}(S, 1)$ (the four inner points) maximising one of $\pm x \pm y$, so layer 1 consists of the inner points. With $(2, 0) = \mathrm{polar}(c, 2, 0)$ in layer 0 and $(\frac{1}{2}, \frac{1}{2}) = \mathrm{polar}(c, \frac{\sqrt{2}}{2}, \frac{\pi}{4})$ in layer 1, all hypotheses of Theorem 4.33 hold and $\beta - \alpha = \frac{\pi}{4} = 1 \cdot \frac{\pi}{m}$.

4.6. **Key order.** This subsection concerns only real numbers and integers. Throughout, intervals $[a, b) = \{y \in \mathbb{R} : a \le y < b\}$ are half-open: the left end is included and the right end is not.

Definition 4.35 (reduction modulo p, Euclidean division, lexicographic order, strict monotonicity; Mathlib)**.** We use the following notions, as defined in Mathlib and in the Lean core library.

(a) Let $p > 0$ and $a, b \in \mathbb{R}$. There is exactly one $n \in \mathbb{Z}$ with $b - np \in [a, a + p)$: for existence take $n = \lfloor (b - a)/p \rfloor$, so that $np \le b - a < np + p$; for uniqueness, if $b - np$ and $b - n'p$ both lie in $[a, a + p)$ then $|(n' - n)p| < p$, so $n = n'$. Mathlib calls this integer `toIcoDiv` and defines $\mathrm{toIcoMod}_p(a, b) = b - np$ (`toIcoMod hp a b`, where `hp` is a proof of $0 < p$). In `KeyOrder.lean`, $p = 2\pi$ (with `Real.two_pi_pos`) and $a = 0$: $\mathrm{toIcoMod}_{2\pi}(0, x)$ is the representative of x modulo 2π in $[0, 2\pi)$.

(b) For $N \in \mathbb{Z}$ with $N > 0$ and $J \in \mathbb{Z}$, Lean's integer quotient `J / N` (`Int.ediv`) is $\lfloor J/N \rfloor$, the floor of the rational number J/N, and Lean's remainder `J % N` (`Int.emod`) is $J \bmod N = J - N\lfloor J/N \rfloor$. Thus $0 \le J \bmod N \le N-1$ (`Int.emod_nonneg`, `Int.emod_lt_of_pos`) and $J = (J \bmod N) + N\lfloor J/N \rfloor$ (`Int.emod_add_mul_ediv`).
(c) On $\mathbb{Z} \times \mathbb{N}$ the lexicographic order (Lean `toLex` into $\mathbb{Z} \times_\ell \mathbb{N}$) is $(a,b) <_{\text{lex}} (a',b')$ iff $a < a'$, or $a = a'$ and $b < b'$ (`Prod.Lex.lt_iff`).
(d) A map $\text{rank} : \mathbb{R} \to \mathbb{N}$ is *strictly increasing on* $S \subseteq \mathbb{R}$ (`StrictMonoOn rank S`) if $\text{rank}(a) < \text{rank}(b)$ for all $a, b \in S$ with $a < b$.

Lemma 4.36 (characterisation of toIcoMod; written step (Mathlib `toIcoMod_eq_iff`)). *Let $p > 0$ and $a, b, y \in \mathbb{R}$. Then* $\text{toIcoMod}_p(a,b) = y$ *if and only if $a \le y < a + p$ and $b = y + zp$ for some $z \in \mathbb{Z}$. In particular* $\text{toIcoMod}_p(a,b) \in [a, a+p)$.

Proof. Let n be the integer of Definition 4.35(a). If $y = \text{toIcoMod}_p(a,b) = b - np$, then $y \in [a, a+p)$ and $b = y + np$. Conversely, if $y \in [a, a+p)$ and $b = y + zp$ with $z \in \mathbb{Z}$, then $b - zp = y \in [a, a+p)$, so $z = n$ by uniqueness, and $\text{toIcoMod}_p(a,b) = b - zp = y$. □

Lemma 4.37 (cells: order and equality; Lean: `KeyOrder.cell_lt_iff`, `LEANEXT/keys/KeyOrder.lean:23`, Lean: `KeyOrder.cell_eq_iff`, `LEANEXT/keys/KeyOrder.lean:45`). *Let $u \in \mathbb{R}$ with $u > 0$, $J_1, J_2 \in \mathbb{Z}$, and $\psi_1, \psi_2 \in \mathbb{R}$ with $0 \le \psi_1 < u$ and $0 \le \psi_2 < u$. Then*

(a) $J_1 u + \psi_1 < J_2 u + \psi_2$ if and only if $J_1 < J_2$, or $J_1 = J_2$ and $\psi_1 < \psi_2$;
(b) $J_1 u + \psi_1 = J_2 u + \psi_2$ if and only if $J_1 = J_2$ and $\psi_1 = \psi_2$.

Proof. (a) "If": when $J_1 < J_2$, we have $J_1 + 1 \le J_2$, hence $J_1 u + \psi_1 < J_1 u + u = (J_1+1)u \le J_2 u \le J_2 u + \psi_2$, using $\psi_1 < u$, $u > 0$ and $\psi_2 \ge 0$; when $J_1 = J_2$ and $\psi_1 < \psi_2$, add $J_1 u$ to $\psi_1 < \psi_2$. "Only if": assume $J_1 u + \psi_1 < J_2 u + \psi_2$. If $J_1 < J_2$ we are done. If $J_1 = J_2$, cancelling $J_1 u$ gives $\psi_1 < \psi_2$. If $J_1 > J_2$, then $J_1 - J_2 - 1 \ge 0$ and

$$(J_1 u + \psi_1) - (J_2 u + \psi_2) = (J_1 - J_2 - 1)\, u + \psi_1 + (u - \psi_2) > 0,$$

since the three terms are ≥ 0, ≥ 0 and > 0; this contradicts the assumption. (b) "If" is clear. "Only if": if $J_1 u + \psi_1 = J_2 u + \psi_2$, then neither side is smaller than the other, so by (a), applied in both orders, $J_1 \not< J_2$ and $J_2 \not< J_1$, hence $J_1 = J_2$; and then $\psi_1 \not< \psi_2$ and $\psi_2 \not< \psi_1$, hence $\psi_1 = \psi_2$. □

Lemma 4.38 (cells and ranks; Lean: `KeyOrder.cell_lt_iff_rank`, `LEANEXT/keys/KeyOrder.lean:62`, Lean: `KeyOrder.cell_lt_iff_rank_finset`, `LEANEXT/keys/KeyOrder.lean:71`). *Let $u \in \mathbb{R}$ with $u > 0$, let $S \subseteq \mathbb{R}$, and let* $\text{rank} : \mathbb{R} \to \mathbb{N}$ *be strictly increasing on S. Let $J_1, J_2 \in \mathbb{Z}$ and $\psi_1, \psi_2 \in S$ with $0 \le \psi_1 < u$ and $0 \le \psi_2 < u$. Then*

$$J_1 u + \psi_1 < J_2 u + \psi_2 \iff (J_1, \text{rank}\,\psi_1) <_{\text{lex}} (J_2, \text{rank}\,\psi_2) \quad \text{in } \mathbb{Z} \times_\ell \mathbb{N}.$$

`cell_lt_iff_rank` *is this statement for a set S (a Lean* `Set` *of reals);* `cell_lt_iff_rank_finset` *is the same statement for a finite set S (a Lean* `Finset` *of reals, with strict monotonicity on the underlying set and $\psi_1, \psi_2 \in S$).*

Proof. By Lemma 4.37(a), the left side is equivalent to "$J_1 < J_2$, or $J_1 = J_2$ and $\psi_1 < \psi_2$". By Definition 4.35(c), the right side is "$J_1 < J_2$, or $J_1 = J_2$ and $\text{rank}\,\psi_1 < \text{rank}\,\psi_2$". It remains to show $\text{rank}\,\psi_1 < \text{rank}\,\psi_2 \iff \psi_1 < \psi_2$ for $\psi_1, \psi_2 \in S$ (`StrictMonoOn.lt_iff_lt`). "If" is strict monotonicity. "Only if": if $\psi_1 \not< \psi_2$, then either $\psi_1 = \psi_2$, and then $\text{rank}\,\psi_1 = \text{rank}\,\psi_2$, or $\psi_2 < \psi_1$, and then $\text{rank}\,\psi_2 < \text{rank}\,\psi_1$; in both cases $\text{rank}\,\psi_1 \not< \text{rank}\,\psi_2$. The finite-set version is the set version applied to the set of elements of the finite set. □

Lemma 4.39 (reduction modulo 2π; Lean: `KeyOrder.reduce_mod_two_pi`, `LEANEXT/keys/KeyOrder.lean:81`). *Let $N \in \mathbb{Z}$ with $N > 0$, $u \in \mathbb{R}$ with $u > 0$ and $Nu = 2\pi$, $J \in \mathbb{Z}$, and $\psi \in \mathbb{R}$ with $0 \le \psi < u$. Then*

$$\text{toIcoMod}_{2\pi}(0,\, Ju + \psi) = (J \bmod N)\, u + \psi.$$

Proof. Let $R = J \bmod N$ and $Q = \lfloor J/N \rfloor$, so that $0 \le R \le N-1$ and $J = R + NQ$ (Definition 4.35(b)). By Lemma 4.36 (with $p = 2\pi$, $a = 0$) it suffices to show (i) $0 \le Ru + \psi$, (ii) $Ru + \psi < 2\pi$, and (iii) $Ju + \psi = (Ru + \psi) + Q \cdot 2\pi$. (i) holds as $R \ge 0$, $u > 0$ and $\psi \ge 0$. (ii): $2\pi - (Ru + \psi) = Nu - Ru - \psi = (N - 1 - R)u + (u - \psi) > 0$, since $N - 1 - R \ge 0$ and $\psi < u$. (iii): $Ju + \psi = (R + NQ)u + \psi = Ru + \psi + Q(Nu) = Ru + \psi + Q \cdot 2\pi$. □

Lemma 4.40 (cells of antipodes; Lean: `KeyOrder.antipode_odd`, `LEANEXT/keys/KeyOrder.lean:100`, Lean: `KeyOrder.antipode_even`, `LEANEXT/keys/KeyOrder.lean:115`). *Adding π to a direction and reducing modulo 2π changes its cell index as follows and keeps its phase ψ:*

(a) (`antipode_odd`*, units π/m) Let $m \in \mathbb{N}$ with $0 < m$, $J \in \mathbb{Z}$ and $\psi \in \mathbb{R}$ with $0 \le \psi < \pi/m$. Then*

$$\mathrm{toIcoMod}_{2\pi}\big(0,\ J \cdot \tfrac{\pi}{m} + \psi + \pi\big) = \big((J+m) \bmod 2m\big) \cdot \tfrac{\pi}{m} + \psi.$$

(b) (`antipode_even`*, units $2\pi/m$) Let $m \in \mathbb{N}$ with $0 < m$ and m even, $J \in \mathbb{Z}$ and $\psi \in \mathbb{R}$ with $0 \le \psi < 2\pi/m$. Then*

$$\mathrm{toIcoMod}_{2\pi}\big(0,\ J \cdot \tfrac{2\pi}{m} + \psi + \pi\big) = \big((J+m/2) \bmod m\big) \cdot \tfrac{2\pi}{m} + \psi,$$

where $m/2$ is the natural-number quotient, cast to $\mathbb{Z}$.

In (a) the integers $J+m$ and $2m$ are formed in $\mathbb{Z}$; oddness of m is not assumed (the name refers to the encoder's use of units π/m for odd m).

Proof. (a) As $m \neq 0$, $m \cdot \frac{\pi}{m} = \pi$, so $J\frac{\pi}{m} + \psi + \pi = (J+m)\frac{\pi}{m} + \psi$. Apply Lemma 4.39 with $N = 2m > 0$, $u = \pi/m > 0$ (indeed $Nu = 2m \cdot \frac{\pi}{m} = 2\pi$), the integer $J+m$, and ψ, which satisfies $0 \le \psi < u$. (b) Write $m = k + k$ with $k \in \mathbb{N}$; then $k \ge 1$ and $m/2 = k$. Since $k \cdot \frac{2\pi}{2k} = \pi$, we get $J\frac{2\pi}{m} + \psi + \pi = (J+k)\frac{2\pi}{m} + \psi$. Apply Lemma 4.39 with $N = m > 0$, $u = 2\pi/m > 0$ (so $Nu = 2\pi$), the integer $J + k = J + m/2$, and ψ, which satisfies $0 \le \psi < u$. □

Theorem 4.41 (sorting reduced directions by integer keys; Lean: `KeyOrder.reduced_lt_iff_key`, `LEANEXT/keys/KeyOrder.lean:139`)**.** *Let $N \in \mathbb{Z}$ with $N > 0$, $u \in \mathbb{R}$ with $u > 0$ and $Nu = 2\pi$, $S \subseteq \mathbb{R}$, and $\mathrm{rank} : \mathbb{R} \to \mathbb{N}$ strictly increasing on S. Let $J_1, J_2 \in \mathbb{Z}$ and $\psi_1, \psi_2 \in S$ with $0 \le \psi_1 < u$ and $0 \le \psi_2 < u$. Then*

$$\mathrm{toIcoMod}_{2\pi}(0, J_1 u + \psi_1) < \mathrm{toIcoMod}_{2\pi}(0, J_2 u + \psi_2) \iff (J_1 \bmod N, \mathrm{rank}\,\psi_1) <_{\mathrm{lex}} (J_2 \bmod N, \mathrm{rank}\,\psi_2).$$

Proof. By Lemma 4.39, applied twice, $\mathrm{toIcoMod}_{2\pi}(0, J_1 u + \psi_1) = (J_1 \bmod N)u + \psi_1$ and $\mathrm{toIcoMod}_{2\pi}(0, J_2 u + \psi_2) = (J_2 \bmod N)u + \psi_2$. Lemma 4.38, applied to the integers $J_1 \bmod N$ and $J_2 \bmod N$ and to ψ_1, ψ_2 (with the same u, S and rank), gives the claim. □

Remark 4.42 (half-open intervals). (1) The strict bound $\psi < u$ cannot be weakened to $\psi \le u$: for $J_1 = 0$, $\psi_1 = u$, $J_2 = 1$, $\psi_2 = 0$ we get $J_1 u + \psi_1 = u = J_2 u + \psi_2$, so the left side of Lemma 4.37(a) is false while its right side $(0 < 1)$ is true, and Lemma 4.37(b) fails as well. (2) The reduction lands in the half-open interval $[0, 2\pi)$: for instance $J = N$ and $\psi = 0$ give $\mathrm{toIcoMod}_{2\pi}(0, 2\pi) = 0 = (N \bmod N)u + 0$, in accordance with Lemma 4.39.

Remark 4.43 (what remains outside Lean). The statements of this subsection are about real numbers already written as $Ju + \psi$ with $0 \le \psi < u$, and about a rank map assumed strictly increasing on the phases. Three steps are not part of any Lean statement here: that the encoder's directions have this form (encoder B computes the cell index as $J = 2k + h_i$, reduced modulo $2m$, for odd m, and as $J = k$ for even m; `C6UPPER/encb/gen_b2.py`, lines 106–109); that its ranks (which are local ranks) are strictly increasing on the phases; and that sorting the keys gives the circular order of the actual points. The first two are listed as hypotheses in `LEANEXT/keys/REPORT.md` (lines 22–25; see also `ENCBLEAN/DESIGN.md:626`), the third as outside Lean in `LEANEXT/README.md` (line 25). (Lines 16–17 of `LEANEXT/keys/REPORT.md` call `reduced_lt_iff_key` the step "sorting keys gives the circular order" in `C6UPPER/encb/gen_b2.py`; Theorem 4.41 states that step only for real numbers already in the form above, with a rank as above.) For even $m \ge 2$, under the hypotheses of Theorem 4.33 on the point set (it does not contain c, ρ maps it into itself, it is in general position, and it has no K points in convex position for some $K \le 2m$), that theorem gives, for hull vertices of distinct layers given in polar form with positive radii, directions that are incongruent modulo $2\pi/m$; once these directions are written as $Ju + \psi$ with $u = 2\pi/m$ and $0 \le \psi < u$, it follows that their phases ψ differ. This distinctness of the phases of distinct layers is what a strictly increasing rank needs.

4.7. **Coverage.** *Covered.* From `LEANEXT/orbit/OrbitConsistency.lean`: `orient_rotAbout_inv` (Lemma 4.7); `PreservesOrient` (Definition 4.8); `RotComp` (Definition 4.9); `preservesOrient_id`, `PreservesOrient.comp`, `PreservesOrient.iterate`, `preservesOrient_rotAbout`, `RotComp.preservesOrient`, `rotComp_iterate` (Lemma 4.10); `orient3` (Definition 4.11); `orient3_swap` (Lemma 4.13); `orient3_perm` (Lemma 4.14); `orient3_rot_perm` (Lemma 4.15); `sign_eq_of_two_reps`, `orient3_of_common_rep` (Lemma 4.16); `litVal` (Definition 4.17); `literal_sound` (Theorem 4.18); `encoder_literal_sound` (Theorem 4.19); `antipode_mem_of_even` (Lemma 4.22); `same_ray_false` (Lemma 4.23); `rotAbout_polar`, `polar_add_int` (Lemma 4.25); `eq_iterate_of_phase_eq` (Lemma 4.26); `eq_iterate_of_phase_eq_int` (Lemma 4.27); `phase_ne_of_not_mem_orbit` (Theorem 4.28). Of the copied block (lines 39–112): `Invariant`, `rotAbout`, `polar` (Definition 4.1), `rotAbout_iterate` (Lemma 4.4) and `iterate_mem` (Lemma 4.6). From `LEANEXT/merged/Merged.lean`, section 13: the seven restatements `antipode_mem_of_even`, `same_ray_false`, `rotAbout_polar`, `eq_iterate_of_phase_eq`, `polar_add_int`, `eq_iterate_of_phase_eq_int`, `phase_ne_`

`of_not_mem_orbit` in the namespace `ErdosSzekeres.LayerLemma` (same labels as their originals); `not_mem_hullVerts_peel_of_lt`, `not_mem_hullVerts_peel_of_ne` (Lemma 4.30); `iterate_mem_hullVerts_peel` (Lemma 4.31); `not_same_ray_of_distinct_layers` (Theorem 4.32); `phase_ne_of_distinct_layers` (Theorem 4.33). From `LEANEXT/keys/KeyOrder.lean`: `cell_lt_iff`, `cell_eq_iff` (Lemma 4.37); `cell_lt_iff_rank`, `cell_lt_iff_rank_finset` (Lemma 4.38); `reduce_mod_two_pi` (Lemma 4.39); `antipode_odd`, `antipode_even` (Lemma 4.40); `reduced_lt_iff_key` (Theorem 4.41).

Not covered, deliberately. (i) The complex-coordinate helpers of the copied block of `OrbitConsistency.lean` (lines 49–90 and 97–103: `toC`, `ofC`, `toC_re`, `toC_im`, `ofC_zero`, `ofC_one`, `toC_ofC`, `ofC_toC`, `toC_injective`, `crot`, `toC_crot`, `crot_one`, `crot_comp`, `crot_iterate`, `rotAbout_eq_crot`): copies of declarations among lines 231–308 of `LAYERLEAN/LayerLemma.lean`, which belong to the layer-lemma unit; our proofs do not need them (the written proof of Lemma 4.4 uses the addition formulas). (ii) Lines 1–1147 of `Merged.lean`: a copy of `LAYERLEAN/LayerLemma.lean`, treated in the layer-lemma and layer-geometry sections. (iii) Lines 151–158 of `KeyOrder.lean`: `#print axioms` commands, not results. (iv) The verbatim copies listed under "Sources" (section 13 of `Merged.lean` in `LEANEXT/linked/Linked.lean`, `LEANEXT/capstone/Capstone.lean`, `LEANEXT/capstone/Capstone2.lean` and `ENCBLEAN/geo/ES8Geo/Capstone2.lean`; the byte-identical `ENCBLEAN/geo/ES8Geo/OrbitConsistency.lean` and `ENCBLEAN/geo/ES8Geo/KeyOrder.lean`), which are the same declarations. (v) The Python scripts and logs of `LEANEXT/orbit` (evidence, cited in Remark 4.20) and the mutant logs (audit only).

5. Phase patterns and angular rules

Fix a centre $c \in \mathbb{R}^2$ and three directions $\alpha < \beta < \delta < \alpha + 2\pi$, read counterclockwise around c. Together with the three antipodal directions $\alpha + \pi$, $\beta + \pi$, $\delta + \pi$ they give six points on the circle of directions, distinct unless one of the gaps $\beta - \alpha$, $\delta - \beta$, $\alpha + 2\pi - \delta$ equals π, in which case a direction and an antipode coincide (Remark 5.7). This section gives written proofs of the Lean-checked results that classify the cyclic pattern of these points, expressed by the arc predicates of Definition 5.6 (which are defined in every case), and connect it with the sign of orient for three points in polar form about c.

- The pattern alternates (every open arc between consecutive directions contains an antipode; pattern $PAPAPA$) if and only if all three gaps are smaller than π (Theorem 5.12).
- A run PPP with middle point β (no antipode in the arcs (α, β) and (β, δ)) is equivalent to $\delta - \alpha < \pi$, provided $\alpha + 2\pi - \delta \neq \pi$ (Theorem 5.15).
- If no gap equals π, the pattern alternates or has a run, and alternation excludes every run (Theorem 5.19).
- Rule A: an alternating triple at positive radii turns counterclockwise (Theorem 5.21). Rule B: a triple with a run turns counterclockwise if the gap not adjacent to the middle point of the run differs from π, and the other two radii are positive and at most the radius of the middle point (Theorem 5.22). The two rules combined (Theorem 5.23), and a version for arbitrary real directions (Theorem 5.26).
- The dichotomy for cyclic Boolean words of length 6 (Theorem 5.28; computer-assisted in Lean, with a written proof by hand here).
- Rule B on the convex layers of a rotationally symmetric set, where the radius condition follows from the layer indices (Theorem 5.30).

Sources. File paths are relative to the results directory `cloud/es7-2026-10-19/results` of the repository. The Lean sources are:

- `LEANEXT/phase/PhaseClassification.lean` (545 lines, SHA-256 `490efcd0a7e4`...), namespace `PhaseClassification` (`LEANEXT/phase/REPORT.md` gives 540 lines for this file, with the same hash prefix);
- `LEANEXT/linked/Linked.lean` (1416 lines, SHA-256 `f1e18c119276`...); its new lines are 1385–1416 (`dist2_polar` and `orient_pos_of_layers`, namespace `ErdosSzekeres.LayerLemma`);
- `LEANEXT/capstone/Capstone.lean` (2004 lines, SHA-256 `dc382395ca55`...); its new lines are 1966–2004 (namespace `ErdosSzekeres.Capstone`);
- `LEANEXT/capstone/Capstone2.lean` (2132 lines, SHA-256 `f8c3930c2427`...); its new lines are 2004–2132.

The following relations were checked with `diff`. Lines 40–102 of `PhaseClassification.lean` repeat lines 931–996 of `LAYERLEAN/LayerLemma.lean` (`polar`, `orient_polar`, `orient_polar_centre_inside`, `sin_add_lt`, `orient_polar_halfplane`) as new constants of the namespace `PhaseClassification`; only the docstrings differ. Lines 1–1147 of `Linked.lean` are lines 1–1147 of `LAYERLEAN/LayerLemma.lean`, and lines 1–1384 of `Linked.lean` are

those of LEANEXT/merged/Merged.lean. Capstone.lean consists of an 18-line header, lines 11–1416 of Linked.lean (as its lines 19–1424), a blank line, lines 6–545 of PhaseClassification.lean (as its lines 1426–1965) and the new lines 1966–2004. Lines 1–2003 of Capstone2.lean are those of Capstone.lean. Hence a declaration at line L of PhaseClassification.lean reappears at line $L + 1420$ of Capstone.lean and Capstone2.lean, and one at line $L \geq 11$ of Linked.lean reappears at line $L + 8$. We cite every declaration at its first occurrence. Capstone2.lean is byte-identical to the copy ENCBLEAN/geo/ES8Geo/Capstone2.lean that the ENCBLEAN geometric package imports (hash list ENCBLEAN/geo/SOURCES.sha256). Further Lean files cited below, with the first 12 hex digits of their SHA-256: LAYERLEAN/LayerLemma.lean (afb8542dceb0), LEANEXT/merged/Merged.lean (245b7d65407a), ENCBLEAN/geo/ES8Geo/Audit.lean (39f214bbf30b), ENCBLEAN/geo/ES8Geo/Labelling.lean (489ba89f8d6f), ENCBLEAN/geo/ES8Geo/ArcBridge.lean (46966b48a101), ENCBLEAN/geo/ES8Geo/ModelOfLabelling.lean (5b532f49cc11) and ENCBLEAN/EncB/SoundWord.lean (c1bdb4175ee5). The files were built with Lean 4.24.0 and, except ENCBLEAN/EncB/SoundWord.lean (part of a package without Mathlib), with Mathlib f897ebcf; these are the repository's pins.

Rigor and axioms. Every definition, lemma, proposition, theorem and corollary below whose head or one of whose parts carries a Lean reference is *Lean-checked*; the proofs given here are written proofs (not refereed). Remarks are commentary and, even when they name Lean declarations, are not Lean-checked as a whole; the mathematical claims they add to the Lean statements are written steps and are not Lean-checked. The log LEANEXT/phase/logs/axioms.log records the output of #print axioms for 16 theorems of PhaseClassification.lean (alternating_iff_gaps_lt_pi, alternating_exactly_one, runMidB_iff, gap_classification, phase_trichotomy, orient_pos_of_alternating, orient_pos_of_runMidA, orient_pos_of_runMidB, orient_pos_of_runMidD, orient_pos_of_phase, orient_sign_of_phase, centre_inside_of_alternating, halfplane_of_runMidB, exists_polar, polar_toIcoMod, word_dichotomy); each depends only on propext, Classical.choice and Quot.sound. Every other declaration of that file covered here is used in the statement or the proof of one of these 16 theorems, so its axioms are among these three. For the new declarations of Linked.lean, Capstone.lean and Capstone2.lean no axiom output is saved in LEANEXT; LEANEXT/linked/REPORT.md and LEANEXT/capstone/REPORT.md state the same three axioms. Of these declarations, dist2_polar, Capstone.polar_eq, Capstone.exists_angle, Capstone.sqrt_dist2_pos and Capstone.ne_of_mem (with Capstone.polar_zero, used in the proof of sqrt_dist2_pos) are used, together with PhaseClassification.orient_pos_of_alternating, PhaseClassification.orient_pos_of_runMidB and PhaseClassification.polar_add_int_two_pi, through the byte-identical copy of Capstone2.lean, in proofs in ENCBLEAN/geo/ES8Geo/Labelling.lean and ENCBLEAN/geo/ES8Geo/ArcBridge.lean on which the declarations EncB.Geo.labelling_of_counterexample (in Labelling.lean) and EncB.Geo.modelOfLabelling (in ENCBLEAN/geo/ES8Geo/ModelOfLabelling.lean) depend. The axioms of these two declarations are pinned to the same three by #guard_msgs in ENCBLEAN/geo/ES8Geo/Audit.lean, lines 45–49. The package compiles the copy from the same sources, against the same imports, as the original file: besides the byte-identical copy and the common Lean and Mathlib pins (see Sources), the ten repository modules that Capstone2.lean needs (its imports ErdosSzekeres.Definitions, ErdosSzekeres.Orientation, ErdosSzekeres.OrientationGeometry and ErdosSzekeres.Suk.OrientationClique, with the repository modules they import) are compiled there from verbatim copies, whose SHA-256 hashes are listed in ENCBLEAN/geo/VENDOR.sha256 and agree with those of the repository files. So the axioms of the declarations used are among these three as well. The remaining new declarations (orient_pos_of_layers, the six layer rules of ErdosSzekeres.Capstone and Capstone.orient_pos_of_alternating_polar) are not used in the ENCBLEAN geometric package outside the copy itself, and no #print axioms output is saved for them; besides the axiom statements of the two reports, the evidence for them is the clean builds stated in these reports and the ENCBLEAN build log ENCBLEAN/logs/m4_build.log, a successful build of the package that imports the byte-identical copy of Capstone2.lean as a module, in which only the KeyOrder module prints messages (its axiom lines). A text search of the four files finds no sorry, admit, axiom declaration or native_decide; among the declarations of this section, decide occurs only in the proof of word_dichotomy.

Dependencies. We use Definitions 1.1, 1.3, 1.12, 1.10 and 1.7, and Lemma 1.15. From the layer sections we use the hull vertices hv and the peeling peel (Definitions 2.4 and 2.18), the constant LayerLemma.polar (Definition 3.34), its polar formula and angular rules (Lemmas 3.35, 3.36, 3.37 and 3.38), and the radius order of the layers (Theorem 3.20).

5.1. **Conventions and polar points.** Points are elements of $\mathbb{R}^2$, written $p = (p_0, p_1)$ (Lean Point, Definition 1.1); two points are equal when both coordinates agree. We use $\mathrm{orient}(a, b, d) = (b_0 - a_0)(d_1 - a_1) - (b_1 - a_1)(d_0 - a_0)$ (Definition 1.3), which is positive when a, b, d turn counterclockwise, and from

Lemma 1.15: $\mathrm{orient}(a,d,b) = -\mathrm{orient}(a,b,d)$ (`orient_swap`), $\mathrm{orient}(b,d,a) = \mathrm{orient}(a,b,d)$ (`orient_cycle`) and $\mathrm{orient}(a+v,b+v,d+v) = \mathrm{orient}(a,b,d)$ (`orient_translate`). Angles are arbitrary real numbers. The number $\pi > 0$ is Lean's `Real.pi`, and an integer k is identified with its image in $\mathbb{R}$ (Lean writes this cast explicitly).

Reduction modulo 2π. For $\alpha, \theta \in \mathbb{R}$ we write $\mathrm{toIcoMod}(\alpha,\theta)$ for Mathlib's `toIcoMod` with period 2π (its positivity argument is `Real.two_pi_pos`), base point α and argument θ. By definition $\mathrm{toIcoMod}(\alpha,\theta) = \theta - 2\pi n$, where $n = \mathrm{toIcoDiv}(\alpha,\theta)$ is the unique integer with $\theta - 2\pi n \in [\alpha, \alpha+2\pi)$, namely $n = \lfloor(\theta-\alpha)/(2\pi)\rfloor$. Only two properties are used (Mathlib `toIcoMod_mem_Ico` and `toIcoMod_add_toIcoDiv_zsmul`):

(T1) $\alpha \le \mathrm{toIcoMod}(\alpha,\theta) < \alpha + 2\pi$;
(T2) $\theta = \mathrm{toIcoMod}(\alpha,\theta) + n \cdot 2\pi$ with the integer $n = \mathrm{toIcoDiv}(\alpha,\theta)$.

Definition 5.1 (polar points and squared distance; Lean: `PhaseClassification.polar`, `LEANEXT/phase/PhaseClassification.lean:41`, Lean: `ErdosSzekeres.LayerLemma.dist2`, `LAYERLEAN/LayerLemma.lean:999`)**.** For $c \in \mathbb{R}^2$ and $r, \theta \in \mathbb{R}$,

$$\mathrm{polar}(c, r, \theta) = (c_0 + r\cos\theta,\ c_1 + r\sin\theta).$$

No sign condition is imposed on r. For $c, y \in \mathbb{R}^2$, $\mathrm{dist2}(c,y) = (y_0 - c_0)^2 + (y_1 - c_1)^2$.

The constant `ErdosSzekeres.LayerLemma.polar` (Definition 3.34; `LAYERLEAN/LayerLemma.lean:932`, repeated at `LEANEXT/linked/Linked.lean:932` and `LEANEXT/capstone/Capstone2.lean:940`) is a different Lean constant given by the same formula; Lemma 5.5(a) records that the two agree. We write polar for both; every statement below that involves polar says, directly or through the convention of its subsection, which of the two constants its Lean version uses. The definition of dist2 is repeated at `LEANEXT/linked/Linked.lean:999` and `LEANEXT/capstone/Capstone2.lean:1007`.

5.2. **The polar orientation formula and the two rules.**

Lemma 5.2 (polar formula, rule A, rule B; Lean: `PhaseClassification.orient_polar`, `LEANEXT/phase/PhaseClassification.lean:44` for (a), and the declarations cited in (b)–(e))**.** *Let $c \in \mathbb{R}^2$; all other letters denote real numbers, and* polar *is* `PhaseClassification.polar`*.*

(a) $\mathrm{orient}(\mathrm{polar}(c,r_a,\alpha), \mathrm{polar}(c,r_b,\beta), \mathrm{polar}(c,r_d,\delta)) = r_a r_b \sin(\beta-\alpha) + r_b r_d \sin(\delta-\beta) + r_d r_a \sin(\alpha-\delta)$.

(b) (Lean: `PhaseClassification.orient_polar_centre`, `LEANEXT/phase/PhaseClassification.lean:407`.) $\mathrm{orient}(\mathrm{polar}(c,r_a,\alpha), \mathrm{polar}(c,r_b,\beta), c) = r_a r_b \sin(\beta-\alpha)$.

(c) (Rule A; Lean: `PhaseClassification.orient_polar_centre_inside`, `LEANEXT/phase/PhaseClassification.lean:51`.) *If* $0 < r_a$, $0 < r_b$, $0 < r_d$, $0 < \beta-\alpha$, $\beta-\alpha < \pi$, $0 < \delta-\beta$, $\delta-\beta < \pi$, $0 < \alpha - \delta + 2\pi$ *and* $\alpha - \delta + 2\pi < \pi$, *then* $0 < \mathrm{orient}(\mathrm{polar}(c,r_a,\alpha), \mathrm{polar}(c,r_b,\beta), \mathrm{polar}(c,r_d,\delta))$.

(d) (Lean: `PhaseClassification.sin_add_lt`, `LEANEXT/phase/PhaseClassification.lean:64`.) *If* $0 < x$, $x < \pi$, $0 < y$ *and* $y < \pi$, *then* $\sin(x+y) < \sin x + \sin y$.

(e) (Rule B; Lean: `PhaseClassification.orient_polar_halfplane`, `LEANEXT/phase/PhaseClassification.lean:80`.) *If* $0 < r_a$, $0 < r_d$, $r_a \le r_b$, $r_d \le r_b$, $0 < \beta-\alpha$, $0 < \delta-\beta$ *and* $\delta - \alpha < \pi$, *then* $0 < \mathrm{orient}(\mathrm{polar}(c,r_a,\alpha), \mathrm{polar}(c,r_b,\beta), \mathrm{polar}(c,r_d,\delta))$.

Parts (a), (c) and (e) are the statements of Lemmas 3.35, 3.36 and 3.38 with the constant `PhaseClassification.polar` in place of `LayerLemma.polar`, and part (d) is the statement of Lemma 3.37; the Lean proofs are the same. For completeness we repeat the short arguments.

Proof. (a) Put $u = (r_a\cos\alpha, r_a\sin\alpha)$, $v = (r_b\cos\beta, r_b\sin\beta)$, $w = (r_d\cos\delta, r_d\sin\delta)$, so that the three points are $c+u$, $c+v$, $c+w$, and write $[x,y] = x_0y_1 - x_1y_0$ for $x, y \in \mathbb{R}^2$. By translation invariance, $\mathrm{orient}(c+u,c+v,c+w) = \mathrm{orient}(u,v,w) = [v-u, w-u]$, and by bilinearity and antisymmetry of $[\cdot,\cdot]$,

$$[v-u, w-u] = [v,w] - [v,u] - [u,w] + [u,u] = [u,v] + [v,w] + [w,u].$$

Finally $[u,v] = r_a r_b(\cos\alpha\sin\beta - \sin\alpha\cos\beta) = r_a r_b \sin(\beta-\alpha)$, and in the same way $[v,w] = r_b r_d \sin(\delta-\beta)$ and $[w,u] = r_d r_a \sin(\alpha-\delta)$.

(b) For every δ, $\mathrm{polar}(c,0,\delta) = c$. Hence (b) is (a) with $r_d = 0$. (Lean expands the definitions directly; the computation is the same.)

(c) By (a) the orientation is $r_a r_b \sin(\beta-\alpha) + r_b r_d \sin(\delta-\beta) + r_d r_a \sin(\alpha-\delta)$. The sine is positive on $(0,\pi)$, so $\sin(\beta-\alpha) > 0$, $\sin(\delta-\beta) > 0$ and $\sin(\alpha-\delta) = \sin(\alpha-\delta+2\pi) > 0$. All radii are positive, so all three terms are positive.

(d) By the addition formula, $\sin x + \sin y - \sin(x+y) = \sin x\,(1-\cos y) + \sin y\,(1-\cos x)$. Here $\sin x > 0$ and $\sin y > 0$. Moreover $\cos x < 1$ and $\cos y < 1$, because cos is strictly decreasing on $[0,\pi]$ with $\cos 0 = 1$. So both products are positive.

(e) Put $x = \beta - \alpha > 0$ and $y = \delta - \beta > 0$. Then $x + y = \delta - \alpha < \pi$, so $x, y \in (0,\pi)$, and $\alpha - \delta = -(x+y)$. By (a), the orientation O in (e) satisfies

$$\begin{aligned} O &= r_a r_b \sin x + r_b r_d \sin y - r_d r_a \sin(x+y) \\ &= r_a(r_b - r_d)\sin x + r_d(r_b - r_a)\sin y + r_a r_d\big(\sin x + \sin y - \sin(x+y)\big). \end{aligned}$$

The first two terms are ≥ 0 (as $r_a, r_d > 0$, $r_b - r_d \geq 0$, $r_b - r_a \geq 0$, $\sin x, \sin y > 0$), and the last is > 0 by (d) and $r_a r_d > 0$. (Positivity of r_b is not assumed; it follows from $r_b \geq r_a > 0$.) □

5.3. **Periodicity and polar forms.**

Lemma 5.3 (periodicity; Lean: `PhaseClassification.polar_add_int_two_pi`, `LEANEXT/phase/PhaseClassification.lean:106`, Lean: `PhaseClassification.polar_add_two_pi`, `LEANEXT/phase/PhaseClassification.lean:111`, Lean: `PhaseClassification.polar_toIcoMod`, `LEANEXT/phase/PhaseClassification.lean:485`). *Let $c \in \mathbb{R}^2$ and $r, \theta, \alpha \in \mathbb{R}$, and let* polar *be* `PhaseClassification.polar`.

(a) For every $k \in \mathbb{Z}$, $\mathrm{polar}(c, r, \theta + k \cdot 2\pi) = \mathrm{polar}(c, r, \theta)$.
(b) $\mathrm{polar}(c, r, \theta + 2\pi) = \mathrm{polar}(c, r, \theta)$.
(c) $\mathrm{polar}(c, r, \mathrm{toIcoMod}(\alpha, \theta)) = \mathrm{polar}(c, r, \theta)$.

Proof. (a) $\cos(\theta + 2\pi k) = \cos\theta$ and $\sin(\theta + 2\pi k) = \sin\theta$ for $k \in \mathbb{Z}$. (b) is (a) with $k = 1$. (c) By (T2), $\theta = \mathrm{toIcoMod}(\alpha, \theta) + n \cdot 2\pi$ with $n \in \mathbb{Z}$, so by (a) $\mathrm{polar}(c, r, \theta) = \mathrm{polar}(c, r, \mathrm{toIcoMod}(\alpha, \theta) + n \cdot 2\pi) = \mathrm{polar}(c, r, \mathrm{toIcoMod}(\alpha, \theta))$. □

Lemma 5.4 (every point other than the centre has a polar form; Lean: `PhaseClassification.exists_polar`, `LEANEXT/phase/PhaseClassification.lean:459`). *For all $c, p \in \mathbb{R}^2$ with $p \neq c$ there are $r, \theta \in \mathbb{R}$ with $0 < r$ and $p = \mathrm{polar}(c, r, \theta)$, where* polar *is* `PhaseClassification.polar`.

Proof. Let $u = p - c \neq 0$ and $r = \sqrt{u_0^2 + u_1^2} > 0$, and put $t = u_0/r \in [-1, 1]$; note $1 - t^2 = u_1^2/r^2$. Recall that $\arccos t \in [0, \pi]$, $\cos(\arccos t) = t$ and $\sin(\arccos t) = \sqrt{1 - t^2} = |u_1|/r$. If $u_1 \geq 0$, let $\theta = \arccos t$; then $\cos\theta = u_0/r$ and $\sin\theta = |u_1|/r = u_1/r$. If $u_1 < 0$, let $\theta = -\arccos t$; then $\cos\theta = t = u_0/r$ and $\sin\theta = -|u_1|/r = u_1/r$. Hence $p = c + u = (c_0 + r\cos\theta, c_1 + r\sin\theta) = \mathrm{polar}(c, r, \theta)$. (The Lean proof takes $r = |z|$ and $\theta = \arg z$ for the complex number $z = u_0 + iu_1$, using $|z|\cos(\arg z) = \mathrm{Re}\, z$ and $|z|\sin(\arg z) = \mathrm{Im}\, z$; this is the same polar representation.) □

Lemma 5.5 (helpers for layered points; Lean: `ErdosSzekeres.Capstone.polar_eq`, `LEANEXT/capstone/Capstone.lean:2001`, Lean: `ErdosSzekeres.Capstone.polar_zero`, `LEANEXT/capstone/Capstone2.lean:2051`, Lean: `ErdosSzekeres.LayerLemma.dist2_polar`, `LEANEXT/linked/Linked.lean:1390`, Lean: `ErdosSzekeres.Capstone.exists_angle`, `LEANEXT/capstone/Capstone2.lean:2056`, Lean: `ErdosSzekeres.Capstone.sqrt_dist2_pos`, `LEANEXT/capstone/Capstone2.lean:2064`, Lean: `ErdosSzekeres.Capstone.ne_of_mem`, `LEANEXT/capstone/Capstone2.lean:2073`). *Let $c, p \in \mathbb{R}^2$ and $r, \theta \in \mathbb{R}$. In* (b)–(e), polar *is the constant* `LayerLemma.polar`.

(a) $\mathtt{LayerLemma.polar}(c, r, \theta) = \mathtt{PhaseClassification.polar}(c, r, \theta)$.
(b) $\mathrm{polar}(c, 0, \theta) = c$.
(c) $\mathrm{dist2}(c, \mathrm{polar}(c, r, \theta)) = r^2$.
(d) If $p \neq c$, there is $\theta \in \mathbb{R}$ with $p = \mathrm{polar}\big(c, \sqrt{\mathrm{dist2}(c, p)}, \theta\big)$.
(e) If $p \neq c$, then $0 < \sqrt{\mathrm{dist2}(c, p)}$.
(f) If S is a finite set of points with $c \notin S$ and $p \in S$, then $p \neq c$.

Proof. (a) Both constants are defined by the formula of Definition 5.1 (in Lean the proof is `rfl`). (b) $c_0 + 0 \cdot \cos\theta = c_0$ and $c_1 + 0 \cdot \sin\theta = c_1$. (c) $(r\cos\theta)^2 + (r\sin\theta)^2 = r^2(\cos^2\theta + \sin^2\theta) = r^2$. (d) By Lemma 5.4 and (a) there are $r > 0$ and θ with $p = \mathrm{polar}(c, r, \theta)$. By (c), $\mathrm{dist2}(c, p) = r^2$, and $\sqrt{r^2} = r$ because $r \geq 0$. Hence $p = \mathrm{polar}(c, \sqrt{\mathrm{dist2}(c, p)}, \theta)$. (e) The square root is ≥ 0. If it were 0, then by (d) and (b) $p = \mathrm{polar}(c, 0, \theta) = c$, a contradiction. (f) If $p = c$, then $c = p \in S$, contrary to $c \notin S$. □

5.4. **Arcs and phase patterns.**

Definition 5.6 (arcs and phase patterns; Lean: `PhaseClassification.inArc`, `LEANEXT/phase/PhaseClassification.lean:117`, Lean: `PhaseClassification.arcHasAntipode`, `LEANEXT/phase/PhaseClassification.lean:120`, Lean: `PhaseClassification.Alternating`, `LEANEXT/phase/PhaseClassification.lean:126`, Lean: `PhaseClassification.RunMidB`, `LEANEXT/phase/PhaseClassification.lean:132`, Lean: `PhaseClassification.RunMidD`, `LEANEXT/phase/PhaseClassification.lean:136`, Lean: `PhaseClassification.RunMidA`, `LEANEXT/phase/PhaseClassification.lean:140`). For $s, g, \theta \in \mathbb{R}$,

$$\mathrm{inArc}(s,g,\theta) \ :\Longleftrightarrow\ \exists k \in \mathbb{Z} :\ s < \theta + 2\pi k \ \text{ and } \ \theta + 2\pi k < s + g.$$

For $\alpha, \beta, \delta, s, g \in \mathbb{R}$,

$$\mathrm{arcHasAntipode}(\alpha,\beta,\delta,s,g) \ :\Longleftrightarrow\ \mathrm{inArc}(s,g,\alpha+\pi) \vee \mathrm{inArc}(s,g,\beta+\pi) \vee \mathrm{inArc}(s,g,\delta+\pi).$$

For $\alpha, \beta, \delta \in \mathbb{R}$ abbreviate (this notation is not a Lean name)

$$H_1 = \mathrm{arcHasAntipode}(\alpha,\beta,\delta,\alpha,\beta-\alpha),$$
$$H_2 = \mathrm{arcHasAntipode}(\alpha,\beta,\delta,\beta,\delta-\beta),$$
$$H_3 = \mathrm{arcHasAntipode}(\alpha,\beta,\delta,\delta,\alpha+2\pi-\delta),$$

and write $H_j(\alpha,\beta,\delta)$ when the triple has to be shown. Then

$$\mathrm{Alternating}(\alpha,\beta,\delta) :\Longleftrightarrow H_1 \wedge H_2 \wedge H_3, \qquad \mathrm{RunMidB}(\alpha,\beta,\delta) :\Longleftrightarrow \neg H_1 \wedge \neg H_2,$$
$$\mathrm{RunMidD}(\alpha,\beta,\delta) :\Longleftrightarrow \neg H_2 \wedge \neg H_3, \qquad \mathrm{RunMidA}(\alpha,\beta,\delta) :\Longleftrightarrow \neg H_3 \wedge \neg H_1.$$

Remark 5.7 (reading the definitions). For $0 < g \le 2\pi$, $\mathrm{inArc}(s,g,\theta)$ says that the direction θ, taken modulo 2π, lies on the open counterclockwise arc from s to $s+g$. When $\alpha < \beta < \delta < \alpha + 2\pi$, the arcs of H_1, H_2, H_3 are the open arcs (α,β), (β,δ), $(\delta,\alpha+2\pi)$ between counterclockwise-consecutive directions (for H_3, $s+g = \delta + (\alpha+2\pi-\delta) = \alpha+2\pi$), and H_j says that the j-th arc contains the direction of an antipode. We use the gaps $g_1 = \beta-\alpha$, $g_2 = \delta-\beta$, $g_3 = \alpha+2\pi-\delta$; they satisfy $g_1+g_2+g_3 = 2\pi$, and $g_1, g_2, g_3 > 0$ when $\alpha < \beta < \delta < \alpha+2\pi$. Note $g_1+g_2 = \delta-\alpha$, $g_2+g_3 = \alpha+2\pi-\beta$ and $g_3+g_1 = \beta+2\pi-\delta$. If $\alpha < \beta < \delta < \alpha+2\pi$ and no gap equals π, the six directions $\alpha, \beta, \delta, \alpha+\pi, \beta+\pi, \delta+\pi$ are distinct modulo 2π (a direction and an antipode coincide modulo 2π exactly when $\beta-\alpha$, $\delta-\beta$ or $\delta-\alpha$ equals π, that is, when g_1, g_2 or g_3 equals π). In that case, writing P for a direction and A for an antipode, Alternating is the cyclic pattern $PAPAPA$, and RunMidB (resp. RunMidD, RunMidA) is a cyclic run PPP whose middle point is β (resp. δ, α). The definitions themselves assume no order; every result below states the order it uses.

Lemma 5.8 (membership and non-membership; Lean: `PhaseClassification.inArc_of`, `LEANEXT/phase/PhaseClassification.lean:143`, Lean: `PhaseClassification.not_inArc`, `LEANEXT/phase/PhaseClassification.lean:147`). *Let $s, g, \theta \in \mathbb{R}$.*

(a) For every $k \in \mathbb{Z}$: if $s < \theta + 2\pi k$ and $\theta + 2\pi k < s + g$, then $\mathrm{inArc}(s,g,\theta)$.
(b) For every $m \in \mathbb{Z}$: if $s + g \le \theta + 2\pi m$ and $\theta + 2\pi m \le s + 2\pi$, then $\neg\,\mathrm{inArc}(s,g,\theta)$.

Proof. (a) is the definition, with witness k. (b) Suppose $k \in \mathbb{Z}$ satisfies $s < \theta + 2\pi k < s + g$. If $k \ge m$, then $\theta + 2\pi k \ge \theta + 2\pi m \ge s + g$ (as $\pi > 0$), contradicting $\theta + 2\pi k < s + g$. If $k \le m-1$, then $\theta + 2\pi k \le \theta + 2\pi m - 2\pi \le s$, contradicting $s < \theta + 2\pi k$. (The Lean proof makes the same case split on the sign of $k - m$.) □

Lemma 5.9 (shift invariance and relabelling; written step). *Let $s, g, \theta, \alpha, \beta, \delta \in \mathbb{R}$ and $n \in \mathbb{Z}$.*

(a) $\mathrm{inArc}(s,g,\theta+2\pi n) \iff \mathrm{inArc}(s,g,\theta)$.
(b) $\mathrm{inArc}(s+2\pi n,g,\theta) \iff \mathrm{inArc}(s,g,\theta)$.
(c) $\mathrm{RunMidD}(\alpha,\beta,\delta) \iff \mathrm{RunMidB}(\beta,\delta,\alpha+2\pi)$.
(d) $\mathrm{RunMidA}(\alpha,\beta,\delta) \iff \mathrm{RunMidB}(\delta,\alpha+2\pi,\beta+2\pi)$.

Proof. (a) If $s < \theta + 2\pi n + 2\pi k < s + g$, then $k' = n + k$ witnesses $\mathrm{inArc}(s,g,\theta)$; conversely a witness k' for θ gives the witness $k' - n$ for $\theta + 2\pi n$. (b) $s + 2\pi n < \theta + 2\pi k < s + 2\pi n + g$ is equivalent to $s < \theta + 2\pi(k-n) < s + g$.

(c) For all s, g, $\mathrm{arcHasAntipode}(\beta,\delta,\alpha+2\pi,s,g)$ is the disjunction of $\mathrm{inArc}(s,g,\beta+\pi)$, $\mathrm{inArc}(s,g,\delta+\pi)$ and $\mathrm{inArc}(s,g,\alpha+\pi+2\pi)$; by (a) with $n=1$ the last one is equivalent to $\mathrm{inArc}(s,g,\alpha+\pi)$. Hence $\mathrm{arcHasAntipode}(\beta,\delta,\alpha+2\pi,s,g) \iff \mathrm{arcHasAntipode}(\alpha,\beta,\delta,s,g)$ for all s, g. By definition, $\mathrm{RunMidB}(\beta,\delta,\alpha+2\pi)$ is the conjunction of $\neg\,\mathrm{arcHasAntipode}(\beta,\delta,\alpha+2\pi,\beta,\delta-\beta)$ and $\neg\,\mathrm{arcHasAntipode}(\beta,\delta,\alpha+2\pi,\delta,(\alpha+2\pi)-\delta)$, which are therefore equivalent to $\neg H_2$ and $\neg H_3$ (of (α,β,δ)). So it is equivalent to $\mathrm{RunMidD}(\alpha,\beta,\delta)$.

(d) As in (c), arcHasAntipode$(\delta, \alpha+2\pi, \beta+2\pi, s, g) \iff$ arcHasAntipode$(\alpha, \beta, \delta, s, g)$ for all s, g (the antipodes $\alpha+2\pi+\pi$, $\beta+2\pi+\pi$ are shifted by 2π). RunMidB$(\delta, \alpha+2\pi, \beta+2\pi)$ is the conjunction of $\neg$ arcHasAntipode$(\delta, \alpha+2\pi, \beta+2\pi, \delta, (\alpha+2\pi)-\delta)$, equivalent to $\neg H_3$, and of $\neg$ arcHasAntipode$(\delta, \alpha+2\pi, \beta+2\pi, \alpha+2\pi, (\beta+2\pi)-(\alpha+2\pi))$. Since $(\beta+2\pi)-(\alpha+2\pi) = \beta-\alpha$ and, by (b) with $n=1$, inArc$(\alpha+2\pi, \beta-\alpha, \theta) \iff$ inArc$(\alpha, \beta-\alpha, \theta)$ for every θ, the second conjunct is equivalent to $\neg H_1$. So the conjunction is equivalent to $\neg H_3 \wedge \neg H_1 = \text{RunMidA}(\alpha, \beta, \delta)$.

The Lean proofs of `orient_pos_of_runMidD` and `orient_pos_of_runMidA` (`LEANEXT/phase/PhaseClassification.lean`, lines 323–335 and 346–362) prove the implications from left to right inside the proofs, disjunct by disjunct, keeping the witness k or changing it by $+1$ or -1, as (a) and (b) do above. □

Lemma 5.10 (empty arcs; Lean: `PhaseClassification.arc1_empty`, `LEANEXT/phase/PhaseClassification.lean:161`, Lean: `PhaseClassification.arc2_empty`, `LEANEXT/phase/PhaseClassification.lean:169`, Lean: `PhaseClassification.arc2_empty'`, `LEANEXT/phase/PhaseClassification.lean:177`, Lean: `PhaseClassification.arc3_empty`, `LEANEXT/phase/PhaseClassification.lean:185`, Lean: `PhaseClassification.arc3_empty'`, `LEANEXT/phase/PhaseClassification.lean:193`, Lean: `PhaseClassification.arc1_empty'`, `LEANEXT/phase/PhaseClassification.lean:201`). *Let $\alpha, \beta, \delta \in \mathbb{R}$, with H_1, H_2, H_3 as in Definition 5.6.*

(a) (`arc1_empty`) *If $\alpha < \beta$, $\beta < \delta$ and $\delta - \alpha \le \pi$, then $\neg H_1$.*
(b) (`arc2_empty`) *If $\alpha < \beta$, $\beta < \delta$ and $\delta - \alpha \le \pi$, then $\neg H_2$.*
(c) (`arc2_empty'`) *If $\beta < \delta$, $\delta < \alpha + 2\pi$ and $\alpha + 2\pi - \beta \le \pi$, then $\neg H_2$.*
(d) (`arc3_empty`) *If $\beta < \delta$, $\delta < \alpha + 2\pi$ and $\alpha + 2\pi - \beta \le \pi$, then $\neg H_3$.*
(e) (`arc3_empty'`) *If $\alpha < \beta$, $\delta < \alpha + 2\pi$ and $\beta + 2\pi - \delta \le \pi$, then $\neg H_3$.*
(f) (`arc1_empty'`) *If $\alpha < \beta$, $\delta < \alpha + 2\pi$ and $\beta + 2\pi - \delta \le \pi$, then $\neg H_1$.*

Proof. In each part we apply Lemma 5.8(b) to each of the three antipodes $\theta \in \{\alpha+\pi, \beta+\pi, \delta+\pi\}$ with the integer m indicated, and verify the two inequalities $s+g \le \theta+2\pi m \le s+2\pi$ for the arc in question; this excludes all three disjuncts of arcHasAntipode. Throughout we use $\pi > 0$. (The Lean proof uses the same integers m.)

(a) Arc H_1: $s = \alpha$, $s+g = \beta$, $s+2\pi = \alpha+2\pi$. From $\beta < \delta$ and $\delta - \alpha \le \pi$ we get $\beta < \alpha+\pi$ and $\beta - \alpha < \pi$. $\theta = \alpha+\pi$, $m=0$: $\beta \le \alpha+\pi \le \alpha+2\pi$. $\theta = \beta+\pi$, $m=0$: $\beta \le \beta+\pi$, and $\beta+\pi \le \alpha+2\pi$ as $\beta - \alpha < \pi$. $\theta = \delta+\pi$, $m=0$: $\beta \le \delta+\pi$ as $\beta < \delta$, and $\delta+\pi \le \alpha+2\pi$ as $\delta - \alpha \le \pi$.

(b) Arc H_2: $s = \beta$, $s+g = \delta$, $s+2\pi = \beta+2\pi$. From $\alpha < \beta$ and $\delta - \alpha \le \pi$ we get $\delta - \beta < \pi$. $\theta = \alpha+\pi$, $m=0$: $\delta \le \alpha+\pi$, and $\alpha+\pi \le \beta+2\pi$ as $\alpha < \beta$. $\theta = \beta+\pi$, $m=0$: $\delta \le \beta+\pi$ as $\delta - \beta < \pi$, and $\beta+\pi \le \beta+2\pi$. $\theta = \delta+\pi$, $m=0$: $\delta \le \delta+\pi$, and $\delta+\pi \le \beta+2\pi$ as $\delta - \beta < \pi$.

(c) Arc H_2 as in (b). The hypothesis says $\alpha+\pi \le \beta$; with $\delta < \alpha+2\pi$ this gives $\delta < \beta+\pi$. $\theta = \alpha+\pi$, $m=1$: $\delta \le \alpha+3\pi$ as $\delta < \alpha+2\pi$, and $\alpha+3\pi \le \beta+2\pi$ as $\alpha+\pi \le \beta$. $\theta = \beta+\pi$, $m=0$: $\delta \le \beta+\pi$, and $\beta+\pi \le \beta+2\pi$. $\theta = \delta+\pi$, $m=0$: $\delta \le \delta+\pi$, and $\delta+\pi \le \beta+2\pi$ as $\delta < \beta+\pi$.

(d) Arc H_3: $s = \delta$, $s+g = \alpha+2\pi$, $s+2\pi = \delta+2\pi$. The hypothesis says $\alpha+\pi \le \beta$; with $\beta < \delta$ this gives $\alpha+\pi < \delta$. $\theta = \alpha+\pi$, $m=1$: $\alpha+2\pi \le \alpha+3\pi$, and $\alpha+3\pi \le \delta+2\pi$ as $\alpha+\pi < \delta$. $\theta = \beta+\pi$, $m=0$: $\alpha+2\pi \le \beta+\pi$ as $\alpha+\pi \le \beta$, and $\beta+\pi \le \delta+2\pi$ as $\beta < \delta$. $\theta = \delta+\pi$, $m=0$: $\alpha+2\pi \le \delta+\pi$ as $\alpha+\pi < \delta$, and $\delta+\pi \le \delta+2\pi$.

(e) Arc H_3 as in (d). The hypothesis says $\beta+\pi \le \delta$; with $\alpha < \beta$ this gives $\alpha+\pi < \delta$. $\theta = \alpha+\pi$, $m=1$: $\alpha+2\pi \le \alpha+3\pi$, and $\alpha+3\pi \le \delta+2\pi$ as $\alpha+\pi < \delta$. $\theta = \beta+\pi$, $m=1$: $\alpha+2\pi \le \beta+3\pi$ as $\alpha < \beta$, and $\beta+3\pi \le \delta+2\pi$ as $\beta+\pi \le \delta$. $\theta = \delta+\pi$, $m=0$: $\alpha+2\pi \le \delta+\pi$ as $\alpha+\pi < \delta$, and $\delta+\pi \le \delta+2\pi$.

(f) Arc H_1 as in (a). The hypothesis says $\beta+\pi \le \delta$; with $\delta < \alpha+2\pi$ this gives $\beta < \alpha+\pi$ and $\beta+\pi < \alpha+2\pi$. $\theta = \alpha+\pi$, $m=0$: $\beta \le \alpha+\pi \le \alpha+2\pi$. $\theta = \beta+\pi$, $m=0$: $\beta \le \beta+\pi$, and $\beta+\pi \le \alpha+2\pi$. $\theta = \delta+\pi$, $m=-1$: $\beta \le \delta-\pi$ as $\beta+\pi \le \delta$, and $\delta-\pi \le \alpha+2\pi$ as $\delta < \alpha+2\pi$. □

Lemma 5.11 (a long arc meets an antipode; Lean: `PhaseClassification.arc12_nonempty`, `LEANEXT/phase/PhaseClassification.lean:229`). *Let $\alpha, \beta, \delta \in \mathbb{R}$ with $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha+2\pi$ and $\pi < \delta - \alpha$. Then $H_1 \vee H_2$.*

Proof. We give a three-case proof; the Lean proof uses a finer case split (on $\delta - \beta$ against π first, then on $\beta - \alpha$), each case again exhibiting one antipode with an explicit witness k.

Case $\beta - \alpha < \pi$: the antipode $\alpha+\pi$ lies in the arc of H_2 with $k=0$, since $\beta < \alpha+\pi$ and $\alpha+\pi < \delta$ (the latter is $\pi < \delta - \alpha$). So H_2.

Case $\beta - \alpha > \pi$: $\alpha+\pi$ lies in the arc of H_1 with $k=0$, since $\alpha < \alpha+\pi < \beta$. So H_1.

Case $\beta-\alpha=\pi$: the antipode $\delta+\pi$ lies in the arc of H_1 with $k=-1$: indeed $\alpha<\delta-\pi$ is $\pi<\delta-\alpha$, and $\delta-\pi<\beta=\alpha+\pi$ is $\delta<\alpha+2\pi$. So H_1. □

5.5. **Alternation and runs.**

Theorem 5.12 (alternation iff all gaps are $<\pi$; Lean: `PhaseClassification.alternating_iff_gaps_lt_pi`, `LEANEXT/phase/PhaseClassification.lean:210`). *Let $\alpha,\beta,\delta\in\mathbb{R}$ with $\alpha<\beta$, $\beta<\delta$ and $\delta<\alpha+2\pi$. Then*

$$\mathrm{Alternating}(\alpha,\beta,\delta)\iff\big(\beta-\alpha<\pi\ \wedge\ \delta-\beta<\pi\ \wedge\ \alpha+2\pi-\delta<\pi\big).$$

Proof. ($\Rightarrow$) Assume $H_1\wedge H_2\wedge H_3$. If $\beta-\alpha\geq\pi$, then $\alpha+2\pi-\beta\leq\pi$, and Lemma 5.10(c) (with $\beta<\delta<\alpha+2\pi$) gives $\neg H_2$, a contradiction. If $\delta-\beta\geq\pi$, then $\beta+2\pi-\delta\leq\pi$, and Lemma 5.10(e) (with $\alpha<\beta$, $\delta<\alpha+2\pi$) gives $\neg H_3$. If $\alpha+2\pi-\delta\geq\pi$, then $\delta-\alpha\leq\pi$, and Lemma 5.10(a) gives $\neg H_1$. Hence all three gaps are $<\pi$.

($\Leftarrow$) Assume $g_1,g_2,g_3<\pi$. By Lemma 5.8(a): H_1 holds through $\delta+\pi$ with $k=-1$, since $\alpha<\delta-\pi$ (that is $g_3<\pi$) and $\delta-\pi<\beta$ (that is $g_2<\pi$); H_2 holds through $\alpha+\pi$ with $k=0$, since $\beta<\alpha+\pi$ ($g_1<\pi$) and $\alpha+\pi<\delta$ ($g_3<\pi$); H_3 holds through $\beta+\pi$ with $k=0$, since $\delta<\beta+\pi$ ($g_2<\pi$) and $\beta+\pi<\delta+(\alpha+2\pi-\delta)=\alpha+2\pi$ ($g_1<\pi$). □

Remark 5.13 (a gap equal to π). No genericity is assumed: a gap may equal π, and then alternation fails. For example, if $\beta-\alpha=\pi$, the antipode $\alpha+\pi$ is the endpoint β of the open arc (β,δ), and Lemma 5.10(c) shows that this arc contains no antipode.

Proposition 5.14 (each arc of an alternating pattern holds exactly the opposite antipode; Lean: `PhaseClassification.alternating_exactly_one`, `LEANEXT/phase/PhaseClassification.lean:386`). *Let $\alpha,\beta,\delta\in\mathbb{R}$ with $\alpha<\beta$, $\beta<\delta$, $\delta<\alpha+2\pi$ and* $\mathrm{Alternating}(\alpha,\beta,\delta)$. *Then*

(i) $\mathrm{inArc}(\alpha,\beta-\alpha,\delta+\pi)$, $\neg\,\mathrm{inArc}(\alpha,\beta-\alpha,\alpha+\pi)$ *and* $\neg\,\mathrm{inArc}(\alpha,\beta-\alpha,\beta+\pi)$;
(ii) $\mathrm{inArc}(\beta,\delta-\beta,\alpha+\pi)$, $\neg\,\mathrm{inArc}(\beta,\delta-\beta,\beta+\pi)$ *and* $\neg\,\mathrm{inArc}(\beta,\delta-\beta,\delta+\pi)$;
(iii) $\mathrm{inArc}(\delta,\alpha+2\pi-\delta,\beta+\pi)$, $\neg\,\mathrm{inArc}(\delta,\alpha+2\pi-\delta,\alpha+\pi)$ *and* $\neg\,\mathrm{inArc}(\delta,\alpha+2\pi-\delta,\delta+\pi)$.

Proof. By Theorem 5.12, $g_1,g_2,g_3<\pi$. The three memberships were shown in the proof of Theorem 5.12, ($\Leftarrow$). For the non-memberships we use Lemma 5.8(b):

- arc (α,β), i.e. $s+g=\beta$, $s+2\pi=\alpha+2\pi$: for $\alpha+\pi$ with $m=0$, $\beta\leq\alpha+\pi$ ($g_1<\pi$) and $\alpha+\pi\leq\alpha+2\pi$; for $\beta+\pi$ with $m=0$, $\beta\leq\beta+\pi$ and $\beta+\pi\leq\alpha+2\pi$ ($g_1<\pi$);
- arc (β,δ), i.e. $s+g=\delta$, $s+2\pi=\beta+2\pi$: for $\beta+\pi$ with $m=0$, $\delta\leq\beta+\pi$ ($g_2<\pi$) and $\beta+\pi\leq\beta+2\pi$; for $\delta+\pi$ with $m=0$, $\delta\leq\delta+\pi$ and $\delta+\pi\leq\beta+2\pi$ ($g_2<\pi$);
- arc $(\delta,\alpha+2\pi)$, i.e. $s+g=\alpha+2\pi$, $s+2\pi=\delta+2\pi$: for $\alpha+\pi$ with $m=1$, $\alpha+2\pi\leq\alpha+3\pi$ and $\alpha+3\pi\leq\delta+2\pi$ ($g_3<\pi$); for $\delta+\pi$ with $m=0$, $\alpha+2\pi\leq\delta+\pi$ ($g_3<\pi$) and $\delta+\pi\leq\delta+2\pi$. □

Theorem 5.15 (a run with middle β iff an open half-plane; Lean: `PhaseClassification.runMidB_iff`, `LEANEXT/phase/PhaseClassification.lean:251`). *Let $\alpha,\beta,\delta\in\mathbb{R}$ with $\alpha<\beta$, $\beta<\delta$, $\delta<\alpha+2\pi$ and $\alpha+2\pi-\delta\neq\pi$. Then*

$$\mathrm{RunMidB}(\alpha,\beta,\delta)\iff\delta-\alpha<\pi.$$

Proof. ($\Leftarrow$) If $\delta-\alpha<\pi$, then $\delta-\alpha\leq\pi$, and Lemma 5.10(a),(b) give $\neg H_1$ and $\neg H_2$. ($\Rightarrow$) Assume $\neg H_1\wedge\neg H_2$ and, for a contradiction, $\delta-\alpha\geq\pi$. Since $\alpha+2\pi-\delta\neq\pi$ we have $\delta-\alpha\neq\pi$, hence $\pi<\delta-\alpha$, and Lemma 5.11 gives $H_1\vee H_2$, a contradiction. □

Remark 5.16 (the gap hypothesis is needed). The direction ($\Rightarrow$) fails without $\alpha+2\pi-\delta\neq\pi$. Take any $\alpha<\beta<\delta$ with $\delta-\alpha=\pi$, for instance $(\alpha,\beta,\delta)=(0,\pi/2,\pi)$. Then $\delta<\alpha+2\pi$, and $\mathrm{RunMidB}(\alpha,\beta,\delta)$ holds by Lemma 5.10(a),(b) (their hypothesis is $\delta-\alpha\leq\pi$), but $\delta-\alpha<\pi$ is false. Geometrically, points at positive radii in the directions α and $\delta=\alpha+\pi$ lie on opposite rays from c, so no open half-plane bounded by a line through c contains both (compare Proposition 5.25). The direction ($\Leftarrow$) does not use the gap hypothesis.

5.6. **Classification under genericity.**

Proposition 5.17 (gap classification; Lean: `PhaseClassification.gap_classification`, `LEANEXT/phase/PhaseClassification.lean:268`). *Let $\alpha,\beta,\delta\in\mathbb{R}$ with $\alpha<\beta$, $\beta<\delta$, $\delta<\alpha+2\pi$, $\beta-\alpha\neq\pi$, $\delta-\beta\neq\pi$ and $\alpha+2\pi-\delta\neq\pi$. Then*

$$\big(\beta-\alpha<\pi\wedge\delta-\beta<\pi\wedge\alpha+2\pi-\delta<\pi\big)\ \vee\ \delta-\alpha<\pi\ \vee\ \alpha+2\pi-\beta<\pi\ \vee\ \beta+2\pi-\delta<\pi.$$

Proof. The gaps g_1, g_2, g_3 are positive and sum to 2π, and none equals π, so each is $< \pi$ or $> \pi$. If all three are $< \pi$, the first disjunct holds. Two gaps cannot both exceed π: their sum would exceed 2π, whereas it equals 2π minus the (positive) third gap. If exactly one gap exceeds π, the other two sum to less than π: if $g_3 > \pi$ then $\delta - \alpha = g_1 + g_2 = 2\pi - g_3 < \pi$; if $g_1 > \pi$ then $\alpha + 2\pi - \beta = g_2 + g_3 < \pi$; if $g_2 > \pi$ then $\beta + 2\pi - \delta = g_3 + g_1 < \pi$. (The Lean proof runs through the same 2^3 sign cases, closing the impossible ones by linear arithmetic.) □

Remark 5.18 (boundary cases and exclusivity). (1) Genericity cannot be dropped here: for $(\alpha, \beta, \delta) = (0, \pi, 3\pi/2)$ all hypotheses hold except $\beta - \alpha \neq \pi$ (here $g_1 = \pi$, $g_2 = g_3 = \pi/2$), and the four disjuncts read $\pi < \pi$ (first conjunct of the first disjunct), $3\pi/2 < \pi$, $\pi < \pi$ and $3\pi/2 < \pi$, all false.

(2) The docstring of `gap_classification` speaks of exactly one case; the Lean statement is only the disjunction above. The disjuncts are in fact mutually exclusive whenever $\alpha < \beta < \delta < \alpha + 2\pi$ (a written step, not part of the Lean statement): if all gaps are $< \pi$, then each sum of two gaps is 2π minus the third, hence $> \pi$; and two of the last three disjuncts together give, after adding, $2\pi + g_j < 2\pi$ for some j, which contradicts $g_j > 0$.

Theorem 5.19 (phase trichotomy; Lean: `PhaseClassification.phase_trichotomy`, `LEANEXT/phase/PhaseClassification.lean:285`). *Let $\alpha, \beta, \delta \in \mathbb{R}$ with $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\beta - \alpha \neq \pi$, $\delta - \beta \neq \pi$ and $\alpha + 2\pi - \delta \neq \pi$. Then*

(i) $\mathrm{Alternating}(\alpha, \beta, \delta) \vee \mathrm{RunMidA}(\alpha, \beta, \delta) \vee \mathrm{RunMidB}(\alpha, \beta, \delta) \vee \mathrm{RunMidD}(\alpha, \beta, \delta)$, *and*

(ii) $\mathrm{Alternating}(\alpha, \beta, \delta) \Rightarrow \big(\neg\,\mathrm{RunMidA}(\alpha, \beta, \delta) \wedge \neg\,\mathrm{RunMidB}(\alpha, \beta, \delta) \wedge \neg\,\mathrm{RunMidD}(\alpha, \beta, \delta)\big)$.

Proof. (i) Apply Proposition 5.17. If all gaps are $< \pi$, Theorem 5.12 gives Alternating. If $\delta - \alpha < \pi$, Lemma 5.10(a),(b) give $\neg H_1 \wedge \neg H_2$, i.e. RunMidB. If $\alpha + 2\pi - \beta < \pi$, Lemma 5.10(c),(d) give $\neg H_2 \wedge \neg H_3$, i.e. RunMidD. If $\beta + 2\pi - \delta < \pi$, Lemma 5.10(e),(f) give $\neg H_3 \wedge \neg H_1$, i.e. RunMidA. (Each strict inequality implies the weak one required by the lemma.)

(ii) Alternating is $H_1 \wedge H_2 \wedge H_3$. Its conjunct H_3 contradicts the first conjunct $\neg H_3$ of RunMidA, H_1 contradicts the first conjunct $\neg H_1$ of RunMidB, and H_2 contradicts the first conjunct $\neg H_2$ of RunMidD. This part uses none of the hypotheses. □

Remark 5.20 (what Theorem 5.19 does not state). The Lean statement is the conjunction of the disjunction (i) and the implication (ii). It does *not* state that the three runs exclude each other. They do (a written step, not Lean-checked): if $\alpha < \beta < \delta < \alpha + 2\pi$, then $H_1 \vee H_2 \vee H_3$. Indeed $\alpha < \alpha + \pi < \alpha + 2\pi$. If $\alpha + \pi < \beta$, the antipode $\alpha + \pi$ gives H_1 (with $k = 0$); if $\beta < \alpha + \pi < \delta$, it gives H_2; if $\delta < \alpha + \pi$, it gives H_3, since $\alpha + \pi < \alpha + 2\pi$. If $\alpha + \pi = \beta$, then $\delta \in (\alpha + \pi, \alpha + 2\pi)$, so $\delta - \pi \in (\alpha, \beta)$, and $\delta + \pi$ gives H_1 with $k = -1$. If $\alpha + \pi = \delta$, then $\beta \in (\alpha, \alpha + \pi)$, so $\beta + \pi \in (\delta, \alpha + 2\pi)$, and $\beta + \pi$ gives H_3 with $k = 0$. Any two of RunMidA, RunMidB, RunMidD together give $\neg H_1 \wedge \neg H_2 \wedge \neg H_3$, which is therefore impossible. Hence, under the hypotheses of Theorem 5.19, exactly one of the four patterns holds.

5.7. **The rules in pattern form.** In this subsection $c \in \mathbb{R}^2$, $r_a, r_b, r_d, \alpha, \beta, \delta \in \mathbb{R}$, polar is `PhaseClassification.polar`, and

$$a = \mathrm{polar}(c, r_a, \alpha), \qquad b = \mathrm{polar}(c, r_b, \beta), \qquad d = \mathrm{polar}(c, r_d, \delta).$$

Theorem 5.21 (rule A from the pattern; Lean: `PhaseClassification.orient_pos_of_alternating`, `LEANEXT/phase/PhaseClassification.lean:301`). *If $0 < r_a$, $0 < r_b$, $0 < r_d$, $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$ and* $\mathrm{Alternating}(\alpha, \beta, \delta)$, *then* $0 < \mathrm{orient}(a, b, d)$.

Proof. By Theorem 5.12, $\beta - \alpha < \pi$, $\delta - \beta < \pi$ and $\alpha + 2\pi - \delta < \pi$; the ordering gives $0 < \beta - \alpha$, $0 < \delta - \beta$ and $0 < \alpha - \delta + 2\pi$. Lemma 5.2(c) applies. □

Theorem 5.22 (rule B from a run; Lean: `PhaseClassification.orient_pos_of_runMidB`, `LEANEXT/phase/PhaseClassification.lean:310`, Lean: `PhaseClassification.orient_pos_of_runMidD`, `LEANEXT/phase/PhaseClassification.lean:318`, Lean: `PhaseClassification.orient_pos_of_runMidA`, `LEANEXT/phase/PhaseClassification.lean:342`). *Let c, $r_a, r_b, r_d, \alpha, \beta, \delta$ and a, b, d be as above.*

(B) *If $0 < r_a$, $0 < r_d$, $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\alpha + 2\pi - \delta \neq \pi$,* $\mathrm{RunMidB}(\alpha, \beta, \delta)$, *$r_a \leq r_b$ and $r_d \leq r_b$, then* $0 < \mathrm{orient}(a, b, d)$.

(D) *If $0 < r_a$, $0 < r_b$, $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\beta - \alpha \neq \pi$,* $\mathrm{RunMidD}(\alpha, \beta, \delta)$, *$r_b \leq r_d$ and $r_a \leq r_d$, then* $0 < \mathrm{orient}(a, b, d)$.

(A) *If $0 < r_b$, $0 < r_d$, $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\delta - \beta \neq \pi$,* $\mathrm{RunMidA}(\alpha, \beta, \delta)$, *$r_b \leq r_a$ and $r_d \leq r_a$, then* $0 < \mathrm{orient}(a, b, d)$.

In each case the radius of the middle point of the run is not assumed positive.

Proof. (B) Theorem 5.15 turns RunMidB(α,β,δ) into $\delta-\alpha<\pi$. With $0<\beta-\alpha$ and $0<\delta-\beta$ from the ordering, Lemma 5.2(e) gives the claim.

(D) Put $(\alpha',\beta',\delta')=(\beta,\delta,\alpha+2\pi)$. Then $\alpha'<\beta'$ ($\beta<\delta$), $\beta'<\delta'$ ($\delta<\alpha+2\pi$) and $\delta'<\alpha'+2\pi$ ($\alpha+2\pi<\beta+2\pi$, from $\alpha<\beta$). Further $\alpha'+2\pi-\delta'=\beta-\alpha\neq\pi$, and RunMidB$(\alpha',\beta',\delta')$ holds by Lemma 5.9(c). Apply (B) to these directions with the radii (r_b,r_d,r_a) in place of (r_a,r_b,r_d); its radius hypotheses $0<r_b$, $0<r_a$, $r_b\le r_d$, $r_a\le r_d$ are among ours. It gives $0<\mathrm{orient}(\mathrm{polar}(c,r_b,\beta),\mathrm{polar}(c,r_d,\delta),\mathrm{polar}(c,r_a,\alpha+2\pi))$. By Lemma 5.3(b) the last point is a, so $0<\mathrm{orient}(b,d,a)=\mathrm{orient}(a,b,d)$ by cyclic invariance.

(A) Put $(\alpha',\beta',\delta')=(\delta,\alpha+2\pi,\beta+2\pi)$. Then $\delta<\alpha+2\pi$, $\alpha+2\pi<\beta+2\pi$ and $\beta+2\pi<\delta+2\pi$, so $\alpha'<\beta'<\delta'<\alpha'+2\pi$; further $\alpha'+2\pi-\delta'=\delta-\beta\neq\pi$, and RunMidB$(\alpha',\beta',\delta')$ holds by Lemma 5.9(d). Apply (B) with the radii (r_d,r_a,r_b) in place of (r_a,r_b,r_d); its radius hypotheses $0<r_d$, $0<r_b$, $r_d\le r_a$, $r_b\le r_a$ are among ours. It gives $0<\mathrm{orient}(\mathrm{polar}(c,r_d,\delta),\mathrm{polar}(c,r_a,\alpha+2\pi),\mathrm{polar}(c,r_b,\beta+2\pi))=\mathrm{orient}(d,a,b)$ by Lemma 5.3(b), and $\mathrm{orient}(d,a,b)=\mathrm{orient}(a,b,d)$ by cyclic invariance. □

Theorem 5.23 (sign from the phase pattern, sorted directions; Lean: `PhaseClassification.orient_pos_of_phase`, `LEANEXT/phase/PhaseClassification.lean:372`)**.** *Assume* $0<r_a$, $0<r_b$, $0<r_d$, $\alpha<\beta$, $\beta<\delta$, $\delta<\alpha+2\pi$, $\beta-\alpha\neq\pi$, $\delta-\beta\neq\pi$, $\alpha+2\pi-\delta\neq\pi$, *and*

- RunMidA$(\alpha,\beta,\delta)\Rightarrow(r_b\le r_a\wedge r_d\le r_a)$,
- RunMidB$(\alpha,\beta,\delta)\Rightarrow(r_a\le r_b\wedge r_d\le r_b)$,
- RunMidD$(\alpha,\beta,\delta)\Rightarrow(r_b\le r_d\wedge r_a\le r_d)$.

Then $0<\mathrm{orient}(a,b,d)$.

Proof. By Theorem 5.19(i) one of four cases holds. If Alternating(α,β,δ), apply Theorem 5.21. If RunMidA, the first implication gives $r_b\le r_a$ and $r_d\le r_a$, and Theorem 5.22(A) applies (with $0<r_b$, $0<r_d$, $\delta-\beta\neq\pi$). If RunMidB, the second gives $r_a\le r_b$, $r_d\le r_b$, and (B) applies (with $0<r_a$, $0<r_d$, $\alpha+2\pi-\delta\neq\pi$). If RunMidD, the third gives $r_b\le r_d$, $r_a\le r_d$, and (D) applies (with $0<r_a$, $0<r_b$, $\beta-\alpha\neq\pi$). □

The two patterns have a direct geometric meaning.

Proposition 5.24 (alternation puts the centre inside the triangle; Lean: `PhaseClassification.centre_inside_of_alternating`, `LEANEXT/phase/PhaseClassification.lean:414`)**.** *If* $0<r_a$, $0<r_b$, $0<r_d$, $\alpha<\beta$, $\beta<\delta$, $\delta<\alpha+2\pi$ *and* Alternating(α,β,δ), *then*

$$0<\mathrm{orient}(a,b,c),\qquad 0<\mathrm{orient}(b,d,c),\qquad 0<\mathrm{orient}(d,a,c).$$

Proof. By Theorem 5.12, $\beta-\alpha$, $\delta-\beta$ and $\alpha+2\pi-\delta$ lie in $(0,\pi)$. By Lemma 5.2(b), applied to the pairs (a,b), (b,d) and (d,a),

$$\mathrm{orient}(a,b,c)=r_ar_b\sin(\beta-\alpha),\quad \mathrm{orient}(b,d,c)=r_br_d\sin(\delta-\beta),\quad \mathrm{orient}(d,a,c)=r_dr_a\sin(\alpha-\delta),$$

and $\sin(\alpha-\delta)=\sin(\alpha+2\pi-\delta)$. All three sines and all radii are positive. □

Proposition 5.25 (a run with middle β lies in an open half-plane through the centre; Lean: `PhaseClassification.halfplane_of_runMidB`, `LEANEXT/phase/PhaseClassification.lean:431`)**.** *If* $0<r_a$, $0<r_b$, $0<r_d$, $\alpha<\beta$, $\beta<\delta$, $\delta<\alpha+2\pi$, $\alpha+2\pi-\delta\neq\pi$ *and* RunMidB(α,β,δ), *then there is* $\varphi\in\mathbb{R}$ *such that for every* $p\in\{a,b,d\}$,

$$0<(p_0-c_0)\cos\varphi+(p_1-c_1)\sin\varphi.$$

Proof. By Theorem 5.15, $\delta-\alpha<\pi$. Take $\varphi=(\alpha+\delta)/2$. For $p=\mathrm{polar}(c,r,\theta)$,

$$(p_0-c_0)\cos\varphi+(p_1-c_1)\sin\varphi=r(\cos\theta\cos\varphi+\sin\theta\sin\varphi)=r\cos(\theta-\varphi).$$

For $\theta\in\{\alpha,\beta,\delta\}\subseteq[\alpha,\delta]$ we have $|\theta-\varphi|\le(\delta-\alpha)/2<\pi/2$, so $\cos(\theta-\varphi)>0$; the radius $r\in\{r_a,r_b,r_d\}$ is positive. □

5.8. **Arbitrary real directions.**

Theorem 5.26 (sign for arbitrary directions; Lean: `PhaseClassification.orient_sign_of_phase`, `LEANEXT/phase/PhaseClassification.lean:496`)**.** *Let* $c\in\mathbb{R}^2$ *and* $r_a,r_b,r_d,\alpha,\beta,\delta\in\mathbb{R}$ *with* $0<r_a$, $0<r_b$, $0<r_d$, *and let* $a=\mathrm{polar}(c,r_a,\alpha)$, $b=\mathrm{polar}(c,r_b,\beta)$, $d=\mathrm{polar}(c,r_d,\delta)$, *where* polar *is* `PhaseClassification.polar`. *Put* $\beta'=\mathrm{toIcoMod}(\alpha,\beta)$ *and* $\delta'=\mathrm{toIcoMod}(\alpha,\delta)$, *and assume* $\beta'\neq\alpha$ *and* $\delta'\neq\alpha$. *Then both of the following hold.*

(i) If $\beta'<\delta'$, $\beta'-\alpha\neq\pi$, $\delta'-\beta'\neq\pi$, $\alpha+2\pi-\delta'\neq\pi$, RunMidA$(\alpha,\beta',\delta')\Rightarrow(r_b\le r_a\wedge r_d\le r_a)$, RunMidB$(\alpha,\beta',\delta')\Rightarrow(r_a\le r_b\wedge r_d\le r_b)$ *and* RunMidD$(\alpha,\beta',\delta')\Rightarrow(r_b\le r_d\wedge r_a\le r_d)$, *then* $0<\mathrm{orient}(a,b,d)$.

(ii) *If* $\delta' < \beta'$, $\delta' - \alpha \neq \pi$, $\beta' - \delta' \neq \pi$, $\alpha + 2\pi - \beta' \neq \pi$, $\mathrm{RunMidA}(\alpha, \delta', \beta') \Rightarrow (r_d \leq r_a \wedge r_b \leq r_a)$, $\mathrm{RunMidB}(\alpha, \delta', \beta') \Rightarrow (r_a \leq r_d \wedge r_b \leq r_d)$ *and* $\mathrm{RunMidD}(\alpha, \delta', \beta') \Rightarrow (r_d \leq r_b \wedge r_a \leq r_b)$, *then* $\mathrm{orient}(a, b, d) < 0$.

Proof. By (T1), $\alpha \leq \beta' < \alpha + 2\pi$ and $\alpha \leq \delta' < \alpha + 2\pi$; with $\beta' \neq \alpha$ and $\delta' \neq \alpha$ we get $\alpha < \beta'$ and $\alpha < \delta'$. By Lemma 5.3(c), $b = \mathrm{polar}(c, r_b, \beta')$ and $d = \mathrm{polar}(c, r_d, \delta')$.

(i) The directions $(\alpha, \beta', \delta')$ satisfy $\alpha < \beta' < \delta' < \alpha + 2\pi$, and the remaining hypotheses of (i) are exactly the gap conditions and radius implications of Theorem 5.23 for these directions and the radii (r_a, r_b, r_d). Hence $0 < \mathrm{orient}(a, b, d)$.

(ii) Apply Theorem 5.23 to the directions $(\alpha, \delta', \beta')$, which satisfy $\alpha < \delta' < \beta' < \alpha + 2\pi$, and the radii (r_a, r_d, r_b) in place of (r_a, r_b, r_d). Its gap conditions are $\delta' - \alpha \neq \pi$, $\beta' - \delta' \neq \pi$, $\alpha + 2\pi - \beta' \neq \pi$, and its radius implications, with r_b and r_d interchanged, are exactly the three implications of (ii). Hence $0 < \mathrm{orient}(a, \mathrm{polar}(c, r_d, \delta'), \mathrm{polar}(c, r_b, \beta')) = \mathrm{orient}(a, d, b) = -\mathrm{orient}(a, b, d)$, so $\mathrm{orient}(a, b, d) < 0$. □

Remark 5.27 (scope of Theorem 5.26). The case $\beta' = \delta'$ (the points b and d on one ray from c) is not covered, and $\beta' = \alpha$ or $\delta' = \alpha$ is excluded by hypothesis. In Lean this theorem is not combined with the layer theorems of Subsection 5.10 (`LEANEXT/capstone/REPORT.md`, open issues).

5.9. **The discrete dichotomy of cyclic words.** Let $\mathrm{Fin}\,6 = \{0, 1, \ldots, 5\}$ with addition modulo 6, and $\mathrm{Bool} = \{\mathrm{true}, \mathrm{false}\}$ with negation $\neg$. A Boolean value x used as a statement means $x = \mathrm{true}$.

Theorem 5.28 (discrete dichotomy, computer-assisted in Lean; written proof by hand; Lean: `PhaseClassification.word_dichotomy`, `LEANEXT/phase/PhaseClassification.lean:539`). *For every* $s\colon \mathrm{Fin}\,6 \to \mathrm{Bool}$ *such that* $s(i+3) = \neg s(i)$ *for all* $i \in \mathrm{Fin}\,6$*:*

(i) $\big(\forall i:\ s(i+1) = \neg s(i)\big)\ \vee\ \big(\exists i:\ s(i) \wedge s(i+1) \wedge s(i+2)\big)$, *and*

(ii) $\neg\Big(\big(\forall i:\ s(i+1) = \neg s(i)\big) \wedge \big(\exists i:\ s(i) \wedge s(i+1) \wedge s(i+2)\big)\Big)$.

How Lean checks it. The statement quantifies over the finite type of functions $\mathrm{Fin}\,6 \to \mathrm{Bool}$, which has $2^6 = 64$ elements, and inside over $\mathrm{Fin}\,6$; all atoms are equalities of Booleans. Lean therefore has a decision procedure for it. For each of the 64 functions s, the procedure decides the implication from the premise (the six equalities $s(i+3) = \neg s(i)$) to the conclusion (i) ∧ (ii), whose atoms are the six equalities $s(i+1) = \neg s(i)$ of the alternation condition and the six triples $s(i), s(i+1), s(i+2)$ of the run condition. Lean's decision procedure for an implication evaluates the premise first and the conclusion only when the premise holds, so the conclusion is evaluated only for the 8 functions listed in the proof below. The tactic `decide` evaluates the procedure during elaboration, and the Lean kernel evaluates it again, to `true`, when it type-checks the resulting proof term (`of_decide_eq_true` applied to `Eq.refl true`). No `native_decide` is used, so no `Lean.ofReduceBool` axiom enters; `LEANEXT/phase/logs/axioms.log` records the three standard axioms. Mutant M7 of the mutation script `LEANEXT/phase/mutants.py` replaces the premise by `True`; the mutation log `LEANEXT/phase/logs/mutants.log`, which calls this mutant "drop antipodal premise in `word_dichotomy`", records that it fails to compile with the error "Tactic `decide` proved that the proposition" (the script logs only the first line of the first error message; Lean's message continues with the proposition and "is false"). The mutated statement is indeed false: for example, the word $PPAPPA$ (with $P = \mathrm{true}$, $A = \mathrm{false}$; it violates the premise, as $s(3) = s(0)$) is neither alternating nor contains a cyclic PPP.

Written proof (by hand). Let s satisfy the premise and put $x = s(0)$, $y = s(1)$, $z = s(2)$. The premise gives $s(3) = \neg x$, $s(4) = \neg y$, $s(5) = \neg z$, so $s = (x, y, z, \neg x, \neg y, \neg z)$ and there are exactly eight such words. Conversely each of these eight words satisfies the premise, since $s(i+3) = \neg s(i)$ for $i = 0, 1, 2$ gives it for $i = 3, 4, 5$ as well. In the notation $P = \mathrm{true}$, $A = \mathrm{false}$, and listing positions $0, \ldots, 5$, they are:

(x, y, z)	word	property
(P, P, P)	$PPPAAA$	run PPP at $i = 0$
(P, P, A)	$PPAAAP$	run at $i = 5$ (positions $5, 0, 1$)
(P, A, P)	$PAPAPA$	alternating
(P, A, A)	$PAAAPP$	run at $i = 4$ (positions $4, 5, 0$)
(A, P, P)	$APPPAA$	run at $i = 1$
(A, P, A)	$APAPAP$	alternating
(A, A, P)	$AAPPPA$	run at $i = 2$
(A, A, A)	$AAAPPP$	run at $i = 3$

This proves (i). For (ii): if $s(i+1) = \neg s(i)$ for all i, then $s(i)$ and $s(i+1)$ are never both true, so there is no cyclic run. (A shorter argument for (i): if s is not alternating, then $s(i+1) = s(i)$ for some i, and then also $s(i+4) = \neg s(i+1) = \neg s(i) = s(i+3)$; one of the equal pairs $\{i, i+1\}$, $\{i+3, i+4\}$ consists of two P's, say at $j, j+1$. If $s(j+2) = P$ there is a run at j; otherwise $s(j-1) = s(j+5) = \neg s(j+2) = P$ and there is a run at $j-1$.) □

Remark 5.29 (scope of Theorem 5.28). This is a statement about Boolean words only. The layered encoders read the six directions $\alpha, \beta, \delta, \alpha+\pi, \beta+\pi, \delta+\pi$, sorted counterclockwise, as such a word (P for a direction, A for an antipode). That this sorted word satisfies $s(i+3) = \neg s(i)$, and that a cyclic PPP in it is RunMidA, RunMidB or RunMidD of the middle point, is not proved in `LEANEXT` (`LEANEXT/phase/REPORT.md`, "Not in Lean", item 2). For encoder B, the Lean version of the encoder in the ENCBLEAN package has a Lean-checked counterpart of the second link that does not use Theorem 5.28: `EncB.wordSpec` (`ENCBLEAN/EncB/SoundWord.lean:222`) derives the rule-B pattern `EncB.BlockCCW` from three cyclically consecutive points of the encoder's sorted word, and, in `ENCBLEAN/geo/ES8Geo/ArcBridge.lean`, `EncB.Geo.Arc.arcFrame` (line 393) identifies, for a polar labelling of the layers, the arc tests of this pattern with the tests arcHasAntipode for true directions $\alpha < \beta < \delta < \alpha + 2\pi$, from which `EncB.Geo.labelling_ruleB` (line 473) obtains RunMidB(α, β, δ) and applies Theorem 5.22(B). These three declarations depend only on `propext`, `Classical.choice` and `Quot.sound`: for `wordSpec` by the `#guard_msgs` pin that follows it (`ENCBLEAN/EncB/SoundWord.lean`, lines 228–229; the module is built in `ENCBLEAN/logs/m3_build.log`, line 9, and `ENCBLEAN/logs/axioms.log`, line 446, gives the same list), and for `arcFrame` and `labelling_ruleB` by their `#print axioms` output in `ENCBLEAN/logs/m8_fix_build.log`, lines 35 and 29, and by the pin of `EncB.Geo.modelOfLabelling` (which uses `labelling_ruleB`) described under Rigor and axioms. The case split for real directions that is Lean-checked here is Theorem 5.19.

5.10. **Rule B on the layers of a symmetric set.** In this subsection S is a finite set of points, $c \in \mathbb{R}^2$, $m, K \in \mathbb{N}$, and $\rho = \mathrm{rotAbout}(c, 2\pi/m)$ is the counterclockwise rotation about c by $2\pi/m$ (Definition 1.12; m is cast to $\mathbb{R}$). We say that *the layer hypotheses* (L) *hold* if

$$0 < m, \qquad K \le 2m, \qquad c \notin S, \qquad \mathrm{Invariant}(\rho, S), \qquad \neg\,\mathrm{HasConvexNGon}(K, S),$$

where Invariant(ρ, S) means $\rho(p) \in S$ for all $p \in S$ (Definition 1.10) and HasConvexNGon(K, S) means that some K-element subset of S is in convex position (Definition 1.7). In this subsection polar is `LayerLemma.polar`; by Lemma 5.5(a) it is the same function as `PhaseClassification.polar`, so the results of the previous subsections apply to it.

We use two facts from the layer sections. First, $\mathrm{peel}(S, n) \subseteq S$ and $\mathrm{hv}(T) \subseteq T$ for all n and all finite T (Lean `peel_subset`, `hullVerts_subset`): $\mathrm{hv}(T)$ is defined as a subset of T (Definition 2.4), and $\mathrm{peel}(S, n+1) = \mathrm{peel}(S, n) \setminus \mathrm{hv}(\mathrm{peel}(S, n)) \subseteq \mathrm{peel}(S, n)$ (Definition 2.18), so induction on n gives the first inclusion. Second, the radius order of the layers (Theorem 3.20; Lean `ErdosSzekeres.LayerLemma.dist2_le_of_later`, `LAYERLEAN/LayerLemma.lean:1012`, repeated at `LEANEXT/linked/Linked.lean:1012` and `LEANEXT/capstone/Capstone2.lean:1020`): *if* (L) *holds,* $n \le n'$, $x \in \mathrm{hv}(\mathrm{peel}(S, n))$ *and* $y \in \mathrm{peel}(S, n')$, *then* $\mathrm{dist2}(c, y) \le \mathrm{dist2}(c, x)$. We also use the elementary fact that $0 < r$, $0 < r'$ and $r^2 \le r'^2$ imply $r \le r'$ (otherwise $r^2 - r'^2 = (r - r')(r + r') > 0$).

Theorem 5.30 (rule B on the convex layers; Lean: `ErdosSzekeres.LayerLemma.orient_pos_of_layers`, `LEANEXT/linked/Linked.lean:1398` for (a), and the six declarations of `ErdosSzekeres.Capstone` cited in (b)–(e))**.** *Assume* (L). *Let* $n_a, n_b, n_d \in \mathbb{N}$ *and* $a, b, d \in \mathbb{R}^2$. *For real* $r_a, r_b, r_d, \alpha, \beta, \delta$ *let* (P) *denote the conditions*

$$0 < r_a, \quad 0 < r_b, \quad 0 < r_d, \quad a = \mathrm{polar}(c, r_a, \alpha), \quad b = \mathrm{polar}(c, r_b, \beta), \quad d = \mathrm{polar}(c, r_d, \delta),$$

and for real α, β, δ *let* (P$\surd$) *denote the conditions*

$$a = \mathrm{polar}\big(c, \sqrt{\mathrm{dist2}(c, a)}, \alpha\big), \quad b = \mathrm{polar}\big(c, \sqrt{\mathrm{dist2}(c, b)}, \beta\big), \quad d = \mathrm{polar}\big(c, \sqrt{\mathrm{dist2}(c, d)}, \delta\big).$$

(a) (`orient_pos_of_layers`) *If* $a \in \mathrm{peel}(S, n_a)$, $b \in \mathrm{hv}(\mathrm{peel}(S, n_b))$, $d \in \mathrm{peel}(S, n_d)$, $n_b \le n_a$, $n_b \le n_d$, *and* $r_a, r_b, r_d, \alpha, \beta, \delta \in \mathbb{R}$ *satisfy* (P), $0 < \beta - \alpha$, $0 < \delta - \beta$ *and* $\delta - \alpha < \pi$, *then* $0 < \mathrm{orient}(a, b, d)$.

(b) (Lean: `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidB`, `LEANEXT/capstone/Capstone.lean:1977`.) *If* $a \in \mathrm{peel}(S, n_a)$, $b \in \mathrm{hv}(\mathrm{peel}(S, n_b))$, $d \in \mathrm{peel}(S, n_d)$, $n_b \le n_a$, $n_b \le n_d$, *and* $r_a, r_b, r_d, \alpha, \beta, \delta \in \mathbb{R}$ *satisfy* (P), $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\alpha + 2\pi - \delta \ne \pi$ *and* RunMidB(α, β, δ), *then* $0 < \mathrm{orient}(a, b, d)$.

(c) (Lean: `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidD`, `LEANEXT/capstone/Capstone2.lean:2011`.) *If* $a \in \mathrm{peel}(S, n_a)$, $b \in \mathrm{peel}(S, n_b)$, $d \in \mathrm{hv}(\mathrm{peel}(S, n_d))$, $n_d \le n_a$, $n_d \le n_b$, *and* $r_a, r_b, r_d, \alpha, \beta, \delta \in \mathbb{R}$ *satisfy* (P), $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\beta - \alpha \ne \pi$ *and* RunMidD(α, β, δ), *then* $0 < \mathrm{orient}(a, b, d)$.

(d) (Lean: `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidA`, `LEANEXT/capstone/Capstone2.lean:2031`.) *If* $a \in \mathrm{hv}(\mathrm{peel}(S, n_a))$, $b \in \mathrm{peel}(S, n_b)$, $d \in \mathrm{peel}(S, n_d)$, $n_a \le n_b$, $n_a \le n_d$, *and* $r_a, r_b, r_d, \alpha, \beta, \delta \in \mathbb{R}$ *satisfy* (P), $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\delta - \beta \ne \pi$ *and* $\mathrm{RunMidA}(\alpha, \beta, \delta)$, *then* $0 < \mathrm{orient}(a, b, d)$.

(e) (The `_intrinsic` versions.) *Let* $\alpha, \beta, \delta \in \mathbb{R}$. *Then* $0 < \mathrm{orient}(a, b, d)$ *in each of the following three cases:*

(e1) (Lean: `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidB_intrinsic`, `LEANEXT/capstone/Capstone2.lean:2082`.) $a \in \mathrm{peel}(S, n_a)$, $b \in \mathrm{hv}(\mathrm{peel}(S, n_b))$, $d \in \mathrm{peel}(S, n_d)$, $n_b \le n_a$, $n_b \le n_d$, $(\mathrm{P}\surd)$, $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\alpha + 2\pi - \delta \ne \pi$ *and* $\mathrm{RunMidB}(\alpha, \beta, \delta)$;

(e2) (Lean: `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidD_intrinsic`, `LEANEXT/capstone/Capstone2.lean:2098`.) $a \in \mathrm{peel}(S, n_a)$, $b \in \mathrm{peel}(S, n_b)$, $d \in \mathrm{hv}(\mathrm{peel}(S, n_d))$, $n_d \le n_a$, $n_d \le n_b$, $(\mathrm{P}\surd)$, $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\beta - \alpha \ne \pi$ *and* $\mathrm{RunMidD}(\alpha, \beta, \delta)$;

(e3) (Lean: `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidA_intrinsic`, `LEANEXT/capstone/Capstone2.lean:2115`.) $a \in \mathrm{hv}(\mathrm{peel}(S, n_a))$, $b \in \mathrm{peel}(S, n_b)$, $d \in \mathrm{peel}(S, n_d)$, $n_a \le n_b$, $n_a \le n_d$, $(\mathrm{P}\surd)$, $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$, $\delta - \beta \ne \pi$ *and* $\mathrm{RunMidA}(\alpha, \beta, \delta)$.

General position of S is not assumed.

Proof. (a) Apply the radius order with $(n, n', x, y) = (n_b, n_a, b, a)$ and with (n_b, n_d, b, d): this gives $\mathrm{dist2}(c, a) \le \mathrm{dist2}(c, b)$ and $\mathrm{dist2}(c, d) \le \mathrm{dist2}(c, b)$. By (P) and Lemma 5.5(c) these read $r_a^2 \le r_b^2$ and $r_d^2 \le r_b^2$, hence $r_a \le r_b$ and $r_d \le r_b$ as all radii are positive. Rule B, Lemma 5.2(e) (in Lean the `LayerLemma` copy, Lemma 3.38), with $0 < r_a$, $0 < r_d$, $0 < \beta - \alpha$, $0 < \delta - \beta$ and $\delta - \alpha < \pi$ gives $0 < \mathrm{orient}(a, b, d)$.

(b) By Theorem 5.15 (with $\alpha < \beta < \delta < \alpha + 2\pi$ and $\alpha + 2\pi - \delta \ne \pi$), $\mathrm{RunMidB}(\alpha, \beta, \delta)$ gives $\delta - \alpha < \pi$. Together with $0 < \beta - \alpha$ and $0 < \delta - \beta$, part (a) applies.

(c) The radius order with (n_d, n_a, d, a) and (n_d, n_b, d, b) gives $\mathrm{dist2}(c, a) \le \mathrm{dist2}(c, d)$ and $\mathrm{dist2}(c, b) \le \mathrm{dist2}(c, d)$, so $r_a^2 \le r_d^2$ and $r_b^2 \le r_d^2$, hence $r_a \le r_d$ and $r_b \le r_d$. Theorem 5.22(D), with $0 < r_a$, $0 < r_b$, the ordering, $\beta - \alpha \ne \pi$, $\mathrm{RunMidD}(\alpha, \beta, \delta)$, $r_b \le r_d$ and $r_a \le r_d$, gives $0 < \mathrm{orient}(a, b, d)$.

(d) The radius order with (n_a, n_b, a, b) and (n_a, n_d, a, d) gives $r_b^2 \le r_a^2$ and $r_d^2 \le r_a^2$, hence $r_b \le r_a$ and $r_d \le r_a$. Theorem 5.22(A), with $0 < r_b$, $0 < r_d$, the ordering, $\delta - \beta \ne \pi$, $\mathrm{RunMidA}(\alpha, \beta, \delta)$, $r_b \le r_a$ and $r_d \le r_a$, gives $0 < \mathrm{orient}(a, b, d)$.

(e) In each case $a, b, d \in S$: a point of $\mathrm{peel}(S, n)$ lies in S, and a point of $\mathrm{hv}(\mathrm{peel}(S, n))$ lies in $\mathrm{peel}(S, n) \subseteq S$. As $c \notin S$, Lemma 5.5(f) gives $a, b, d \ne c$, and Lemma 5.5(e) gives $0 < \sqrt{\mathrm{dist2}(c, a)}$, $0 < \sqrt{\mathrm{dist2}(c, b)}$ and $0 < \sqrt{\mathrm{dist2}(c, d)}$. So (P) holds with $r_a = \sqrt{\mathrm{dist2}(c, a)}$, $r_b = \sqrt{\mathrm{dist2}(c, b)}$, $r_d = \sqrt{\mathrm{dist2}(c, d)}$, and (e1), (e2), (e3) follow from (b), (c), (d) respectively. □

Corollary 5.31 (rule A restated; Lean: `ErdosSzekeres.Capstone.orient_pos_of_alternating_polar`, `LEANEXT/capstone/Capstone.lean:1992`)**.** *Let* $c \in \mathbb{R}^2$ *and* $r_a, r_b, r_d, \alpha, \beta, \delta \in \mathbb{R}$ *with* $0 < r_a$, $0 < r_b$, $0 < r_d$, $\alpha < \beta$, $\beta < \delta$, $\delta < \alpha + 2\pi$ *and* $\mathrm{Alternating}(\alpha, \beta, \delta)$. *Then, with* polar *the constant* `LayerLemma.polar`, $0 < \mathrm{orient}(\mathrm{polar}(c, r_a, \alpha), \mathrm{polar}(c, r_b, \beta), \mathrm{polar}(c, r_d, \delta))$. *No layer hypotheses are involved.*

Proof. By Lemma 5.5(a) this is Theorem 5.21. (The Lean proof repeats the proof of Theorem 5.21 with the `LayerLemma` copy of rule A, Lemma 3.36.) □

Remark 5.32 (what the layer theorems assume). In Theorem 5.30 the polar forms (with radius variables in (a)–(d), with the intrinsic radii $\sqrt{\mathrm{dist2}(c, \cdot)}$ in (e)) and the conditions on the given directions are hypotheses: in (a) the conditions $0 < \beta - \alpha$, $0 < \delta - \beta$ and $\delta - \alpha < \pi$; in (b)–(e) the strict order $\alpha < \beta < \delta < \alpha + 2\pi$, the gap condition $\ne \pi$ and the pattern. They are not derived from the symmetry or from general position. Directions with $(\mathrm{P}\surd)$ exist for every point other than c (Lemma 5.5(d)), but choosing them in sorted order and excluding gaps equal to π is left to the user of these theorems; the reduction of arbitrary directions in Theorem 5.26 is not combined with the layer hypotheses in Lean. The middle point (the point b of the middle direction in (a), the middle point of the run in (b)–(e)) is required to be a hull vertex of a layer whose index is at most the indices of the layers of the other two points, and the conclusion is about the fixed order (a, b, d).

Coverage. All declarations of `LEANEXT/phase/PhaseClassification.lean` are covered: `polar` (Definition 5.1); `orient_polar`, `orient_polar_centre`, `orient_polar_centre_inside`, `sin_add_lt`, `orient_polar_halfplane` (Lemma 5.2; except `orient_polar_centre`, these are the copies of lines 931–996 of `LAYERLEAN/LayerLemma.lean`, whose originals also have written proofs in the section on layer geometry); `polar_add_int_two_pi`, `polar_add_two_pi`, `polar_toIcoMod` (Lemma 5.3); `exists_polar` (Lemma 5.4); `inArc`, `arcHasAntipode`, `Alternating`, `RunMidB`, `RunMidD`, `RunMidA` (Definition 5.6); `inArc_of`, `not_inArc`

(Lemma 5.8); `arc1_empty`, `arc2_empty`, `arc2_empty'`, `arc3_empty`, `arc3_empty'`, `arc1_empty'` (Lemma 5.10); `arc12_nonempty` (Lemma 5.11); `alternating_iff_gaps_lt_pi` (Theorem 5.12); `alternating_exactly_one` (Proposition 5.14); `runMidB_iff` (Theorem 5.15); `gap_classification` (Proposition 5.17); `phase_trichotomy` (Theorem 5.19); `orient_pos_of_alternating` (Theorem 5.21); `orient_pos_of_runMidB`, `orient_pos_of_runMidD`, `orient_pos_of_runMidA` (Theorem 5.22); `orient_pos_of_phase` (Theorem 5.23); `centre_inside_of_alternating` (Proposition 5.24); `halfplane_of_runMidB` (Proposition 5.25); `orient_sign_of_phase` (Theorem 5.26); `word_dichotomy` (Theorem 5.28, computer-assisted in Lean; written proof by hand). The new declarations of `LEANEXT/linked/Linked.lean` are covered: `dist2_polar` (Lemma 5.5(c)) and `orient_pos_of_layers` (Theorem 5.30(a)); so are those of `LEANEXT/capstone/Capstone.lean`: `orient_pos_of_layers_runMidB` (Theorem 5.30(b)), `orient_pos_of_alternating_polar` (Corollary 5.31) and `polar_eq` (Lemma 5.5(a)); and those of `LEANEXT/capstone/Capstone2.lean`: `orient_pos_of_layers_runMidD`, `orient_pos_of_layers_runMidA` (Theorem 5.30(c),(d)), `polar_zero`, `exists_angle`, `sqrt_dist2_pos`, `ne_of_mem` (Lemma 5.5(b),(d),(e),(f)) and the three `_intrinsic` theorems (Theorem 5.30(e)). Deliberately not covered separately: lines 1–1384 of `Linked.lean` and the corresponding lines of `Capstone.lean` and `Capstone2.lean` (copies of `LAYERLEAN/LayerLemma.lean` and of `LEANEXT/merged/Merged.lean`, treated in the sections on the layer lemma, layer geometry and orbit literals), and lines 1426–1965 of `Capstone.lean` and `Capstone2.lean` (verbatim copies of `PhaseClassification.lean`, whose statements and proofs are those covered above, at line offset +1420). The helper Lemma 5.9 is a written step that makes explicit two `have` steps of the Lean proofs of `orient_pos_of_runMidD` and `orient_pos_of_runMidA`; Remarks 5.13, 5.16, 5.18 and 5.20 contain further written steps (with proofs) that are not Lean statements.

6. Small corollaries for $k = 7$ and $k = 8$

This section gives written proofs for a group of short Lean files about rotationally symmetric point sets.

(1) *Conditional $ES(7)$ corollaries* (Subsection 6.1). If the case classes Fivefold35 and Fourfold32 hold (they are hypotheses of the Lean theorems), then every finite set of points in general position that is invariant under the rotation by $2\pi/5$ about some point and has at least 32 points, and every such set for the rotation by $2\pi/4$ with at least 29 points, has 7 points in convex position.

(2) *Rotations of order at least* 8 (Subsection 6.2). A subset of $\mathbb{C}$ that is mapped into itself by $z \mapsto e^{2\pi i/m}z$ with $m \geq 8$ and contains no 8 points in convex position is contained in $\{0\}$. Also the threshold arithmetic for $m = 4, 5, 6, 7$: for all $t, e \in \mathbb{N}$ with $e \leq 1$, $65 \leq 4t \iff t \geq 17$, $65 \leq 5t + e \iff t \geq 13$, $65 \leq 6t \iff t \geq 11$ and $65 \leq 7t + e \iff t \geq 10$.

(3) *Orientations in inverse-radius polar form* (Subsection 6.3): the sign of the orientation of three points $(\cos\varphi_k, \sin\varphi_k)/u_k$, and its consequences for layered configurations and for nested regular m-gons.

(4) *Size compatibility* (Subsections 6.4 and 6.5): the orbit count $|S| \equiv [0 \in S] \pmod m$ for finite sets invariant under an element of exact order m, its converse under explicit hypotheses, the classification of the m with $2^{m-1} \bmod m \in \{0, 1\}$, and the consequence for m-fold symmetric sets of 2^{m-1} points.

Sources. Paths are relative to `cloud/es7-2026-10-19/results`. We computed the SHA-256 of each file; the table gives the first 12 hexadecimal digits. The line ranges are the scope of this section; the Coverage paragraph at the end lists what in them is not covered.

Lean file (lines in scope)	SHA-256	axioms log
`LEANEXT/es7corr/ES7Corollaries.lean` (1148–1233)	`5883c8ebac55`	`LEANEXT/es7corr/axioms.log`
`ES8SYM/lean/C6check.lean` (1–123)	`92f15b5cc907`	`ES8SYM/lean/C6check.out`
`ES8SYM/lean/PolarOrient.lean` (1–132)	`6e54f5f83a19`	`ES8SYM/lean/PolarOrient.out`
`PRIMEFOLD/compat/lean/PCompatCore.lean` (1–291)	`ebc6aacefadb`	`PRIMEFOLD/compat/lean/PCompatCore.out`
`PRIMEFOLD/compat/lean/L0Check.lean` (1–152)	`05a67c3b8b16`	`PRIMEFOLD/compat/lean/L0Check.out`

According to the records, all five files were compiled with the repository's pins, Lean 4.24.0 and Mathlib `f897ebcf`; nothing was re-run for this document. The hashes of `LEANEXT/es7corr/ES7Corollaries.lean` and of the two Lean files in `PRIMEFOLD/compat/lean` agree with the prefixes recorded in `LEANEXT/es7corr/REPORT.md` and `PRIMEFOLD/README.md`. For the two `ES8SYM` files, `ES8SYM/README.md` (lines 258–260) records the prefixes `8c090168` and `4d42e2fb` for the compiled files and states that the present copies differ from them only in four comment lines; we could not check this, so the logs of these two files refer to the compiled versions.

Axioms. The logs list the following theorems, each depending on at most the standard axioms `propext`, `Classical.choice` and `Quot.sound`: the five $ES(7)$ corollaries, Theorems 6.3–6.7 (`LEANEXT/es7corr/axioms.log`); `RefuteC6.part_a` and `RefuteC6.thr_m4`–`thr_m7` (`ES8SYM/lean/C6check.out`); the six theorems `C7.`

`orient_polar`, `orient_polar_cross`, `orient_layered`, `orient_layered_scale`, `nested_regular_orient`, `orient_centre` (`ES8SYM/lean/PolarOrient.out`); `PCompat.dvd_card`, `card_mod`, `exists_iff`, `compat_even`, `compat_odd`, `classification`, and `PCompat.ex341`, which depends on `propext` only (`PRIMEFOLD/compat/lean/PCompatCore.out`); `L0Check.L0_rot` (`PRIMEFOLD/compat/lean/L0Check.out`). Every other declaration covered here occurs in the statement or the proof of one of these, so the same logs cover it, except `RefuteC6.m4_with_centre_counter`, which has no axiom line (it is a closed arithmetic statement whose Lean proof is `omega`; the compile log of its file shows no error). No file uses `sorry`, `native_decide` or new axioms; `PCompat.ex341` is checked by kernel evaluation (`decide`).

Rigor and dependencies. The theorems with a Lean reference are Lean-checked, with the axioms above; for the two `ES8SYM` files this rests on the logs of the compiled versions, which according to the records differ from the present copies only in comment lines (see Sources). The proofs below are written proofs (not refereed). Statements tagged "written step" are not declarations of the five Lean files: Fact 6.14 restates Mathlib lemmas, and Corollary 6.18 has no Lean counterpart. The theorems of Subsection 6.1 are conditional on the propositions Fivefold35 and Fourfold32, and Lemma 6.32 and Theorem 6.35(c) take hypotheses on r that Lean does not discharge (Remark 6.37). We use Theorem 2.26 (reduction to exactly mt points) and Lemma 3.18 (even orders exclude the centre), and the conventions of Definitions 1.1, 1.3, 1.5, 1.6, 1.7, 1.10 and 1.12; remarks also refer to Lemma 2.16, Theorems 2.22 and 2.24, Lemma 3.35, Lemmas 4.7, 4.14 and 4.23, Theorem 4.28 and Definition 7.7. The other subsections are self-contained apart from standard facts, which we state where they are used.

6.1. **Conditional $ES(7)$ corollaries for fivefold and fourfold symmetry.** The source is Section 11 of `LEANEXT/es7corr/ES7Corollaries.lean` (lines 1148–1233, namespace `ErdosSzekeres.LayerLemma`). Lines 1–1147 of this file coincide line by line with lines 1–1147 of `LAYERLEAN/LayerLemma.lean` (we checked this with `diff`; the latter file has one more line, its closing `end`). Hence the declarations `Invariant`, `rotAbout`, `centre_not_mem_of_even` and `convexNGon_of_layers` used below are verbatim copies of those of `LAYERLEAN/LayerLemma.lean`, at the same lines (105, 298, 582, 1066).

Conventions. S is a finite set of points (Lean `Finset Point`) and c a point. rotAbout(c,θ) is the counterclockwise rotation by θ about c (Definition 1.12). "S is invariant under f" means $f(p)\in S$ for every $p\in S$ (Definition 1.10). General position is Definition 1.5, and HasConvexNGon(n,S) means that some n-element subset of S is in convex position (Definition 1.7; no emptiness condition). We write $S\setminus\{c\}$ for Lean's `S.erase c`. In the statements below the angles are $2\pi/5$ and $2\pi/4$ with real numerals 5 and 4; in Theorem 2.26 and Lemma 3.18 (items (R) and (C) below) the angle is $2\pi/m$ with $m\in\mathbb{N}$ cast to $\mathbb{R}$. For $m=5$ and $m=4$ these are the same real numbers (the Lean proofs pass between the two forms with the cast-normalizing tactic `exact_mod_cast`).

Definition 6.1 (the case classes; Lean: `ErdosSzekeres.LayerLemma.Fivefold35`, `LEANEXT/es7corr/ES7Corollaries.lean:1157`, Lean: `ErdosSzekeres.LayerLemma.Fourfold32`, `LEANEXT/es7corr/ES7Corollaries.lean:1164`)**.** Fivefold35 is the proposition: for every finite set S of points and every point c, if S is in general position, $|S|=35$, $c\notin S$, and S is invariant under rotAbout$(c,2\pi/5)$, then HasConvexNGon$(7,S)$.

Fourfold32 is the proposition: for every finite set S of points and every point c, if S is in general position, $|S|=32$, $c\notin S$, and S is invariant under rotAbout$(c,2\pi/4)$, then HasConvexNGon$(7,S)$.

Remark 6.2 (status of the case classes). No Lean theorem proves Fivefold35 or Fourfold32. The theorems of this subsection take them as hypotheses and are therefore conditional. In the project they are supported by computer-assisted results outside Lean, which also rest on written arguments. For SAT certificates the project counts only the verdicts of the proof checker cake_lpr (`WRITEUP/SYMMETRIC.md`, Section 3). The records also show acceptances by a second checker, lrat-check, which the project does not count: it is a secondary check only, and its unpatched upstream build is unsound on the empty clause (`E2/logs/lrat_check_soundness.md`; `LRATCHECK20261105`).

- Fivefold35: `SYM5` (`SYM5/README.md`). Two formulas for the 35-point fivefold case, an angle form and a layer form of one SAT encoder (not independent encodings: audit `AUDIT3`, finding A4-3), are recorded as unsatisfiable; the LRAT proofs of both were accepted by cake_lpr. The passage from these verdicts to Fivefold35 uses the soundness of the encoder (recorded as a written proof, reviewed in the audit, `AUDIT3`, A4), a normal-form lemma for the labelling (for the layer form, the layer lemma), and the centre-perturbation lemma, which makes the centre generic (`C2ENC/perturbation.md`, recorded there as a complete written proof, not Lean-checked, refereed within the project on the day it was written:

`C2ENC/referee/NOTES.md`, Section 4). `C2ENC/class_table.md` summarizes this chain for the constant of the same text in `SYMLEAN` (see below).

- Fourfold32: `C4CUT` (`C4CUT/README.md`). The layered formula for the 32-point fourfold case with six realizability cuts, each justified by a Grassmann–Plücker certificate checked in exact rational arithmetic, is recorded as unsatisfiable; its LRAT proof was accepted by cake_lpr. The final run does not record which checker builds it used (`AUDIT3`, A4-5, which concludes that the claim is carried by the cake_lpr verdict); the replay `REPLAY`, with hash-checked files and checker binaries, accepted the proof again with cake_lpr. The passage to Fourfold32 uses the soundness of the layered encoder (recorded in `C4CUT/README.md` as a written proof, refereed within the project: the review of 1 November 2026, `C4CUT/review/README.md`), the layer lemma, and the sector normal form: each of the other orbits has exactly one point in the open quadrant between the rays from the centre through two consecutive points of the outermost orbit. The audit found the argument for this not written down (`AUDIT3`, A4-1); `C4CUT/README.md`, Section 1, now gives it in one sentence: a point of another orbit on a bounding ray would be collinear with two antipodal points of the outermost orbit.

The reviews, audits and refereeing named in this list were all made within the project (internal reviews); none of them is an external peer review. Apart from the layer lemma and the ray fact for even orders behind the sector normal form, none of these written arguments is checked in Lean. Both have Lean-checked forms. For the layer lemma it is Theorem 2.22. For the ray fact it is Lemma 4.23 together with Theorem 4.28 (`LEANEXT/orbit/OrbitConsistency.lean`): for even $m > 0$, no two points of a finite set in general position that is invariant under $\mathrm{rotAbout}(c, 2\pi/m)$ lie on one open ray from c; and if x and y are points of such a set, both different from c, and y is not in the orbit of x under that rotation, then the directions of x and y at c differ by no integer multiple of $2\pi/m$. The remaining step, that each of the other orbits has exactly one point in the open quadrant, is a counting argument and is not in Lean. The rotation invariance of orientations and their sign rule under permutations, on which the encoders' single variable per orbit of triples rests, are Lean-checked too (Lemmas 4.7 and 4.14). Nothing in Lean connects any of these Lean-checked forms to the formulas, and no Lean theorem links a formula or a certificate to these propositions. Propositions with the same text occur in `SYMLEAN/SymmetricReduction.lean:477` and `SYMLEAN/SymmetricReduction.lean:496` (Definition 7.7); they are different Lean constants (namespace `ErdosSzekeres.SymReduction`, built on that file's own copies of `Invariant` and `rotAbout`), and no Lean theorem identifies the two versions. The docstring of `Fourfold32`, copied from `SYMLEAN`, says that it is used only by a variant theorem of `SYMLEAN`; in `LEANEXT/es7corr/ES7Corollaries.lean` it is the hypothesis of Theorems 6.6 and 6.7.

We use two Lean-checked results whose written proofs are in other sections, in the following form.

(R) Theorem 2.26 (`convexNGon_of_layers`, `LAYERLEAN/LayerLemma.lean:1066`). Let $m, K, t \in \mathbb{N}$ with $0 < m$ and $K \le 2m$, and assume: *for every finite set T and every point c', if T is in general position, $c' \notin T$, T is invariant under* $\mathrm{rotAbout}(c', 2\pi/m)$ *and* $|T| = m \cdot t$, *then* $\mathrm{HasConvexNGon}(K, T)$. Then for every finite set S and every point c such that S is in general position, S is invariant under $\mathrm{rotAbout}(c, 2\pi/m)$ and $m \cdot (t \dot{-} 1) < |S \setminus \{c\}|$, we have $\mathrm{HasConvexNGon}(K, S)$. Here $t \dot{-} 1 = \max(t-1, 0)$ is Lean's truncated subtraction on $\mathbb{N}$ (the notation of the layer-lemma section).

(C) Lemma 3.18 (`centre_not_mem_of_even`, `LAYERLEAN/LayerLemma.lean:582`). Let $m \in \mathbb{N}$, let S be a finite set of points and c a point. If $0 < m$, m is even, S is invariant under $\mathrm{rotAbout}(c, 2\pi/m)$ and in general position, and some $p \in S$ satisfies $p \ne c$, then $c \notin S$. (In this section, for a map f and $n \in \mathbb{N}$, f^n is the n-fold composite of f, written $f^{[n]}$ in the foundations section; so $f^0 = \mathrm{id}$. The power $\mathrm{rotAbout}(c, 2\pi/m)^{m/2}$ is the half-turn $p \mapsto 2c - p$, and p, c, $2c - p$ would be three distinct collinear points of S.)

In each of the following five theorems the case class is a hypothesis (Definition 6.1 and Remark 6.2).

Theorem 6.3 (fivefold, layer form; Lean: `ErdosSzekeres.LayerLemma.fivefold_heptagon_of_erase`, `LEANEXT/es7corr/ES7Corollaries.lean:1170`)**.** *Assume* Fivefold35. *Let S be a finite set of points and c a point such that S is in general position, S is invariant under* $\mathrm{rotAbout}(c, 2\pi/5)$, *and* $30 < |S \setminus \{c\}|$. *Then* $\mathrm{HasConvexNGon}(7, S)$.

Proof. We follow the Lean proof. Apply (R) with $m = 5$, $K = 7$, $t = 7$; here $0 < 5$ and $K = 7 \le 10 = 2m$. The assumption of (R) holds: let T be finite and c' a point such that T is in general position, $c' \notin T$, T is invariant under $\mathrm{rotAbout}(c', 2\pi/5)$ and $|T| = 5 \cdot 7 = 35$. These are the four hypotheses of Fivefold35 for T and c' (listed there in a different order), so Fivefold35 gives $\mathrm{HasConvexNGon}(7, T)$. The size condition of (R) is $5 \cdot (7 \dot{-} 1) = 5 \cdot 6 = 30 < |S \setminus \{c\}|$, which is our hypothesis. So (R) gives $\mathrm{HasConvexNGon}(7, S)$. □

Theorem 6.4 (fivefold; Lean: `ErdosSzekeres.LayerLemma.fivefold_heptagon`, `LEANEXT/es7corr/ES7Corollaries.lean:1183`)**.** *Assume* Fivefold35. *Let S be a finite set of points and c a point such that S is in general position, S is invariant under* $\mathrm{rotAbout}(c, 2\pi/5)$*, and* $32 \le |S|$*. Then* $\mathrm{HasConvexNGon}(7, S)$.

Proof. As in Lean: removing one element lowers the cardinality by at most one, so $|S \setminus \{c\}| \ge |S| - 1 \ge 31 > 30$, and Theorem 6.3 applies. □

Theorem 6.5 (centre-free; Lean: `ErdosSzekeres.LayerLemma.fivefold_heptagon_of_centre_not_mem`, `LEANEXT/es7corr/ES7Corollaries.lean:1193`)**.** *Assume* Fivefold35. *Let S be a finite set of points and c a point such that S is in general position, S is invariant under* $\mathrm{rotAbout}(c, 2\pi/5)$*, $c \notin S$, and* $31 \le |S|$*. Then* $\mathrm{HasConvexNGon}(7, S)$.

Proof. As in Lean: since $c \notin S$, $S \setminus \{c\} = S$, so $|S \setminus \{c\}| = |S| \ge 31 > 30$, and Theorem 6.3 applies. □

Theorem 6.6 (fourfold, layer form; Lean: `ErdosSzekeres.LayerLemma.fourfold_heptagon_of_erase`, `LEANEXT/es7corr/ES7Corollaries.lean:1203`)**.** *Assume* Fourfold32. *Let S be a finite set of points and c a point such that S is in general position, S is invariant under* $\mathrm{rotAbout}(c, 2\pi/4)$*, and* $28 < |S \setminus \{c\}|$*. Then* $\mathrm{HasConvexNGon}(7, S)$.

Proof. We follow the Lean proof. Apply (R) with $m = 4$, $K = 7$, $t = 8$; here $0 < 4$ and $K = 7 \le 8 = 2m$. The assumption of (R) holds by Fourfold32, exactly as in the proof of Theorem 6.3, since $4 \cdot 8 = 32$. The size condition of (R) is $4 \cdot (8 \dot{-} 1) = 4 \cdot 7 = 28 < |S \setminus \{c\}|$, our hypothesis. □

Theorem 6.7 (fourfold; Lean: `ErdosSzekeres.LayerLemma.fourfold_heptagon`, `LEANEXT/es7corr/ES7Corollaries.lean:1216`)**.** *Assume* Fourfold32. *Let S be a finite set of points and c a point such that S is in general position, S is invariant under* $\mathrm{rotAbout}(c, 2\pi/4)$*, and* $29 \le |S|$*. Then* $\mathrm{HasConvexNGon}(7, S)$.

Proof. We follow the Lean proof. Since $|S| \ge 29 > 1$, S is not contained in $\{c\}$, so some $p \in S$ satisfies $p \ne c$. Apply (C) with $m = 4$: $0 < 4$, 4 is even, S is in general position, and S is invariant under $\mathrm{rotAbout}(c, 2\pi/4)$, which is the rotation $\mathrm{rotAbout}(c, 2\pi/m)$ of (C) for $m = 4$ (the cast angle $2\pi/m$ equals $2\pi/4$ for $m = 4$; see the Conventions). So $c \notin S$. Hence $S \setminus \{c\} = S$ and $|S \setminus \{c\}| = |S| \ge 29 > 28$, and Theorem 6.6 applies. □

Remark 6.8 (what the theorems give). Under Fivefold35, Theorem 6.4 says that a finite set in general position that is invariant under $\mathrm{rotAbout}(c, 2\pi/5)$ for some c and has no 7 points in convex position has at most 31 points; under Fourfold32, Theorem 6.7 gives at most 28 points for $\mathrm{rotAbout}(c, 2\pi/4)$. These are the upper bounds $N(C_5) \le 31$ and $N(C_4) \le 28$ of the symmetric $ES(7)$ note (`WRITEUP/note/symmetric_es7.tex`, the proposition labelled `prop:C`), where they are derived from the 35-point fivefold and the 32-point fourfold case by orbit counting (the lemma labelled `lem:sizes`). A set with symmetry C_s in the sense of the note is invariant under a rotation ρ of order s about some point c, that is $\rho = \mathrm{rotAbout}(c, 2\pi k/s)$ for an integer k with $\gcd(k, s) = 1$. Choose $j \in \mathbb{N}$ with $jk \equiv 1 \pmod{s}$. By the complex form of Definition 1.12, $\rho^j = \mathrm{rotAbout}(c, 2\pi jk/s) = \mathrm{rotAbout}(c, 2\pi/s)$, since the two angles differ by a multiple of 2π; and a set mapped into itself by ρ is mapped into itself by ρ^j. So the two theorems apply to every finite set in general position with symmetry C_5, respectively C_4. The matching lower-bound examples `SYMMAX/c5_31points.txt` and `SYMMAX/c4_28points.txt` rest on exact computations outside Lean, which were not repeated for this document. The logs `LEANEXT/es7corr/mutant31.log` and `LEANEXT/es7corr/mutant28.log` record that the Lean proofs fail at their final arithmetic step when the thresholds 32 and 29 are lowered to 31 and 28; this audits the proof scripts only.

Remark 6.9 (a route without the layer lemma). Theorems 6.3 and 6.6 also follow by orbit counting, the argument of the symmetric $ES(7)$ note. Let $\rho = \mathrm{rotAbout}(c, 2\pi/m)$ with $m \in \{4, 5\}$. The map ρ fixes c and is injective, so it maps $S' = S \setminus \{c\}$ into itself, hence onto itself (S' is finite). Every $y \ne c$ has minimal period m under ρ (Lemma 2.16), so the ρ-orbits $\{y, \rho y, \dots, \rho^{m-1} y\}$ of the points $y \in S'$ have exactly m elements and partition S'. If $m(t-1) < |S'|$, with $t = 7$ for $m = 5$ and $t = 8$ for $m = 4$, then S' consists of at least t orbits; the union T of t of them has mt points (35, respectively 32), satisfies $c \notin T$, is in general position (as a subset of S) and is invariant under ρ. The case class applies to T, and 7 points of T in convex position lie in S. The Lean proofs use Theorem 2.26 instead, which takes for T the t outermost convex layers of S'.

6.2. **Rotations of order at least eight.** The source is `ES8SYM/lean/C6check.lean` (namespace `RefuteC6`; it imports Mathlib only, not the project library). It is the Lean check of a complex form of part (a) of Lemma 2 of `ES8SYM/proofs/layers.md` (the planar form is Corollary 6.18, a written step) and of the arithmetic of part (b) of that lemma.

Conventions. In this subsection points are complex numbers. $\mathbb{C}$ is a real vector space of dimension 2, $|z|$ is the modulus (the Euclidean norm), and for $r \in \mathbb{R}$ we write $B_r = \{z \in \mathbb{C} : |z| \le r\}$ (Lean `closedBall 0 r`); the circle $\{z \in \mathbb{C} : |z| = r\}$ is Lean's `sphere 0 r`, and both sets are empty if $r < 0$. conv is the convex hull over $\mathbb{R}$, the smallest convex set containing the given set. The following notions are Mathlib's.

- An *extreme point* of $A \subseteq \mathbb{C}$ (Mathlib `Set.extremePoints`) is a point $x \in A$ such that whenever $x_1, x_2 \in A$, $a, b \in \mathbb{R}$, $a > 0$, $b > 0$, $a + b = 1$ and $x = ax_1 + bx_2$, we have $x_1 = x$.
- A family $f : \iota \to \mathbb{C}$ is *convex independent* over $\mathbb{R}$ (Mathlib `ConvexIndependent`) if for every set $s \subseteq \iota$ and every $x \in \iota$, $f(x) \in \operatorname{conv} f(s)$ implies $x \in s$.
- $\zeta \in \mathbb{C}$ is a *primitive m-th root of unity* (Mathlib `IsPrimitiveRoot`) if $\zeta^m = 1$ and, for every $l \in \mathbb{N}$, $\zeta^l = 1$ implies $m \mid l$.

$\operatorname{Fin} 8 = \{0, 1, \dots, 7\}$. A finite set of points is in convex position (Definition 1.6) if the inclusion family of the set is convex independent; Corollary 6.18 makes the passage between injective families in $\mathbb{C}$ and finite sets of points precise.

Lemma 6.10 (points of a circle are extreme points of the disc; Lean: `RefuteC6.sphere_subset_extremePoints`, `ES8SYM/lean/C6check.lean:24`)**.** *For every $r \in \mathbb{R}$, every $x \in \mathbb{C}$ with $|x| = r$ is an extreme point of B_r.*

Proof. We follow the Lean proof, which uses the strict convexity of the norm of $\mathbb{C}$; we make it explicit. For $x_1, x_2 \in \mathbb{C}$ and real a, b with $a + b = 1$,

$$|ax_1 + bx_2|^2 = a|x_1|^2 + b|x_2|^2 - ab\,|x_1 - x_2|^2.$$

Indeed, with $\langle z, w\rangle = \operatorname{Re}(z\bar{w})$ the left side is $a^2|x_1|^2 + b^2|x_2|^2 + 2ab\langle x_1, x_2\rangle$, the right side is $(a - ab)|x_1|^2 + (b - ab)|x_2|^2 + 2ab\langle x_1, x_2\rangle$, and $a - ab = a(1 - b) = a^2$, $b - ab = b^2$.

Let $|x| = r$. Then $x \in B_r$. Let $x_1, x_2 \in B_r$ and $a, b > 0$ with $a + b = 1$ and $x = ax_1 + bx_2$. If $x_1 = x_2$, then $x = (a + b)x_1 = x_1$. Suppose $x_1 \neq x_2$. Since $0 \le |x_1| \le r$ we have $r \ge 0$ and $|x_1|^2 \le r^2$, and likewise $|x_2|^2 \le r^2$; also $ab|x_1 - x_2|^2 > 0$. The identity gives $r^2 = |x|^2 < ar^2 + br^2 = r^2$, a contradiction. Hence $x_1 = x$. □

Lemma 6.11 (injective families on a circle are convex independent; Lean: `RefuteC6.convexIndependent_of_norm_eq`, `ES8SYM/lean/C6check.lean:43`)**.** *Let ι be any type, $f : \iota \to \mathbb{C}$ injective, and $r \in \mathbb{R}$ with $|f(i)| = r$ for every $i \in \iota$. Then f is convex independent over $\mathbb{R}$.*

Proof. Step 1. *If x is an extreme point of a convex set $A \subseteq \mathbb{C}$, then $A \setminus \{x\}$ is convex.* Let $y, z \in A \setminus \{x\}$, $a, b \ge 0$ real with $a + b = 1$, and $w = ay + bz$; then $w \in A$. If $w = x$: for $b = 0$ we get $w = y \neq x$, for $a = 0$ we get $w = z \neq x$, and for $a, b > 0$ extremality gives $y = x$; each case is a contradiction. So $w \in A \setminus \{x\}$.

Step 2. The disc B_r is convex: for $y, z \in B_r$ and $a, b \ge 0$ with $a + b = 1$, $|ay + bz| \le a|y| + b|z| \le r$. Let $s \subseteq \iota$ and $x \in \iota$ with $f(x) \in \operatorname{conv} f(s)$, and suppose $x \notin s$. For $j \in s$ we have $|f(j)| = r$, so $f(j) \in B_r$, and $j \neq x$, so $f(j) \neq f(x)$ by injectivity. Thus $f(s) \subseteq B_r \setminus \{f(x)\}$. Since $|f(x)| = r$, $f(x)$ is an extreme point of B_r (Lemma 6.10), so $B_r \setminus \{f(x)\}$ is convex by Step 1, and therefore contains $\operatorname{conv} f(s)$. This contradicts $f(x) \in \operatorname{conv} f(s)$. Hence $x \in s$.

(The Lean proof uses Mathlib's reformulation of convex independence, "$f(i) \notin \operatorname{conv} f(s \setminus \{i\})$ for all i, s", with the same two steps.) □

Definition 6.12 (the generator; Lean: `RefuteC6.zeta`, `ES8SYM/lean/C6check.lean:60`)**.** For $m \in \mathbb{N}$, $\zeta_m = \exp(2\pi i/m) \in \mathbb{C}$, with m cast to $\mathbb{C}$. (For $m = 0$, Lean's division gives $2\pi i/0 = 0$ and $\zeta_0 = 1$; this case is not used.) Rotation by $2\pi/m$ about 0 is $z \mapsto \zeta_m z$.

Lemma 6.13 (primitive root; Lean: `RefuteC6.zeta_prim`, `ES8SYM/lean/C6check.lean:62`)**.** *For $m \in \mathbb{N}$ with $m \neq 0$, ζ_m is a primitive m-th root of unity.*

Proof. This is Mathlib's `Complex.isPrimitiveRoot_exp`, a standard fact. Proof: $\exp(w) = 1$ if and only if $w \in 2\pi i\mathbb{Z}$ (periodicity of the exponential function), and $\zeta_m^l = \exp(2\pi i l/m)$ for $l \in \mathbb{N}$. Hence $\zeta_m^l = 1$ if and only if $l/m \in \mathbb{Z}$, that is $m \mid l$; in particular $\zeta_m^m = 1$. □

Fact 6.14 (properties of primitive roots; written step)**.** *Let $m \ge 1$ and let $\zeta \in \mathbb{C}$ be a primitive m-th root of unity. Then* (i) *$\zeta \neq 0$;* (ii) *$|\zeta| = 1$;* (iii) *if $i, j < m$ and $\zeta^i = \zeta^j$, then $i = j$;* (iv) *$\zeta^j \neq 1$ for $0 < j < m$.*

Proof. (i) $\zeta^m = 1 \neq 0$ and $m \ge 1$. (ii) $|\zeta|^m = |\zeta^m| = 1$ with $|\zeta| \ge 0$, and $x \mapsto x^m$ is strictly increasing on $[0, \infty)$, so $|\zeta| = 1$. (iii) Say $i \le j$. Then $\zeta^i\zeta^{j-i} = \zeta^j = \zeta^i$, and $\zeta^i \neq 0$ by (i), so $\zeta^{j-i} = 1$, hence $m \mid j - i$; as $0 \le j - i < m$, $j = i$. (iv) $\zeta^j = 1$ would give $m \mid j$, impossible for $0 < j < m$. (In Mathlib these are `IsPrimitiveRoot.ne_zero`, `IsPrimitiveRoot.norm'_eq_one`, `IsPrimitiveRoot.pow_inj` and `IsPrimitiveRoot.pow_ne_one_of_pos_of_lt`.) □

Lemma 6.15 (eight points of an orbit; Lean: `RefuteC6.orbit_injective`, `ES8SYM/lean/C6check.lean:66`, Lean: `RefuteC6.orbit_convexIndependent`, `ES8SYM/lean/C6check.lean:75`)**.** *Let $m \in \mathbb{N}$ with $8 \le m$ and $p \in \mathbb{C}$ with $p \neq 0$, and let $F : \operatorname{Fin} 8 \to \mathbb{C}$, $F(k) = p\,\zeta_m^k$. Then* (a) *F is injective, and* (b) *F is convex independent over $\mathbb{R}$.*

Proof. Since $m \neq 0$, ζ_m is a primitive m-th root of unity (Lemma 6.13). (a) If $p\zeta_m^j = p\zeta_m^k$, then $\zeta_m^j = \zeta_m^k$ because $p \neq 0$, and $j, k < 8 \le m$, so $j = k$ by Fact 6.14(iii). (b) $|F(k)| = |p|\,|\zeta_m|^k = |p|$ by Fact 6.14(ii). Apply Lemma 6.11 with $r = |p|$ and (a). □

Theorem 6.16 (order at least 8; Lean: `RefuteC6.part_a`, `ES8SYM/lean/C6check.lean:85`)**.** *Let $m \in \mathbb{N}$ with $8 \le m$, and let $S \subseteq \mathbb{C}$ be any set (not necessarily finite) such that $\exp(2\pi i/m)\, z \in S$ for every $z \in S$. Assume that for every injective $f : \operatorname{Fin} 8 \to \mathbb{C}$ with $f(k) \in S$ for all k, f is not convex independent over $\mathbb{R}$. Then $S \subseteq \{0\}$.*

Proof. We follow the Lean proof. Let $p \in S$ and suppose $p \neq 0$. By induction, $p\zeta_m^k \in S$ for every $k \in \mathbb{N}$: $p\zeta_m^0 = p \in S$, and if $p\zeta_m^k \in S$, then $\exp(2\pi i/m)\,(p\zeta_m^k) = p\zeta_m^{k+1} \in S$ by the hypothesis (note $\exp(2\pi i/m) = \zeta_m$ by definition). Hence the family F of Lemma 6.15 takes values in S; it is injective and convex independent by that lemma, contradicting the assumption. So $p = 0$. □

Remark 6.17. The theorem uses no finiteness and no general position, and only the forward invariance $\zeta_m S \subseteq S$. Its conclusion says $S = \emptyset$ or $S = \{0\}$; both sets satisfy the hypotheses, since an injective family of 8 values needs 8 distinct points. The planar statement used in the project (`ES8SYM/proofs/layers.md`, Lemma 2(a), stated there for sets in general position; the row "order ≥ 8", labelled Lean-checked, of the $k = 8$ tables in `ES8SYM/README.md` and `WRITEUP/note/symmetric_extremal.tex`) follows from the following corollary, which does not assume general position. Lemma 2(a) is stated for the rotation by $2\pi/m$, and the corollary applies to it directly. The tables concern any rotation ρ of order m (`WRITEUP/note/symmetric_extremal.tex`, line 338), that is $\rho = \operatorname{rotAbout}(c, 2\pi k/m)$ with $\gcd(k, m) = 1$; as in Remark 6.8, the power ρ^j with $j \in \mathbb{N}$ and $jk \equiv 1 \pmod m$ is $\operatorname{rotAbout}(c, 2\pi/m)$ and maps every ρ-invariant set into itself, so the corollary applies to ρ^j. Lean checks only the complex form, Theorem 6.16; the passage to the plane is not formalized in Lean, and a written proof follows.

Corollary 6.18 (planar form of Theorem 6.16; written step)**.** *Let $m \in \mathbb{N}$ with $m \ge 8$, let c be a point, and let S be a finite set of points that is invariant under $\operatorname{rotAbout}(c, 2\pi/m)$ and satisfies $\neg\operatorname{HasConvexNGon}(8, S)$. Then $S \subseteq \{c\}$.*

Proof. Let $\varphi : \mathbb{R}^2 \to \mathbb{C}$, $\varphi(p) = (p_0 - c_0) + i(p_1 - c_1)$. It is a bijection and an affine map of real vector spaces, with inverse $\psi(z) = (c_0 + \operatorname{Re} z,\ c_1 + \operatorname{Im} z)$. By the complex form of Definition 1.12, $\varphi(\operatorname{rotAbout}(c, \theta)\, p) = e^{i\theta}\varphi(p)$ for all $\theta \in \mathbb{R}$ and all points p. We also use the standard fact that an affine map α between real vector spaces satisfies $\alpha(\operatorname{conv} Y) = \operatorname{conv}\alpha(Y)$, because $\operatorname{conv} Y$ is the set of finite convex combinations of points of Y and α preserves convex combinations. Let $\tilde{S} = \varphi(S) \subseteq \mathbb{C}$.

Invariance. Let $z = \varphi(p)$ with $p \in S$. With $\theta = 2\pi/m$ we have $e^{i\theta} = \exp(2\pi i/m)$, so $\exp(2\pi i/m)\, z = \varphi(\operatorname{rotAbout}(c, 2\pi/m)\, p) \in \tilde{S}$, because $\operatorname{rotAbout}(c, 2\pi/m)\, p \in S$.

No convex independent 8-family. Let $f : \operatorname{Fin} 8 \to \mathbb{C}$ be injective with values in $\tilde{S}$, and suppose f is convex independent. Put $g = \psi \circ f$; it is injective with values in S, so $T = g(\operatorname{Fin} 8)$ is an 8-element subset of S. We show that T is in convex position, i.e. (Definition 1.6) that for every $U \subseteq T$ and every $x \in T$, $x \in \operatorname{conv} U$ implies $x \in U$. Write $x = g(k)$ and $s = \{j \in \operatorname{Fin} 8 : g(j) \in U\}$; then $U = g(s)$, and $\varphi \circ g = f$. From $x \in \operatorname{conv} U$ we get $f(k) = \varphi(x) \in \varphi(\operatorname{conv} U) = \operatorname{conv}\varphi(U) = \operatorname{conv} f(s)$, so $k \in s$ by convex independence of f, that is $x = g(k) \in U$. Hence $\operatorname{HasConvexNGon}(8, S)$, a contradiction.

So $\tilde{S}$ satisfies the hypotheses of Theorem 6.16, and $\tilde{S} \subseteq \{0\}$. Since $\varphi(p) = 0$ only for $p = c$, $S \subseteq \{c\}$. □

Proposition 6.19 (threshold arithmetic; Lean: `RefuteC6.thr_m4`, `ES8SYM/lean/C6check.lean:108`, Lean: `RefuteC6.thr_m5`, `ES8SYM/lean/C6check.lean:109`, Lean: `RefuteC6.thr_m6`, `ES8SYM/lean/C6check.lean:110`, Lean: `RefuteC6.thr_m7`, `ES8SYM/lean/C6check.lean:111`, Lean: `RefuteC6.m4_with_centre_counter`, `ES8SYM/lean/C6check.lean:115`)**.** *In $\mathbb{N}$:*

(i) (`thr_m4`) *for every t:* $65 \le 4t \iff 17 \le t$;
(ii) (`thr_m5`) *for all t, e with $e \le 1$:* $65 \le 5t + e \iff 13 \le t$;
(iii) (`thr_m6`) *for every t:* $65 \le 6t \iff 11 \le t$;
(iv) (`thr_m7`) *for all t, e with $e \le 1$:* $65 \le 7t + e \iff 10 \le t$;
(v) (`m4_with_centre_counter`) $65 \le 4 \cdot 16 + 1$ *and* $\neg(17 \le 16)$.

(*In Lean the variable e is called* `c`.)

Proof. (i) If $t \ge 17$ then $4t \ge 68$; if $t \le 16$ then $4t \le 64$. (ii) If $t \ge 13$ then $5t + e \ge 65$; if $t \le 12$ then $5t + e \le 60 + 1 = 61$. (iii) If $t \ge 11$ then $6t \ge 66$; if $t \le 10$ then $6t \le 60$. (iv) If $t \ge 10$ then $7t + e \ge 70$; if $t \le 9$ then $7t + e \le 63 + 1 = 64$. (v) $4 \cdot 16 + 1 = 65$, and $17 > 16$. (Lean proves each line with `omega`, a decision procedure for linear arithmetic over $\mathbb{N}$ and $\mathbb{Z}$ whose proof term is checked by the kernel.) □

Remark 6.20 (the geometric part of Lemma 2(b), not in this file). In `ES8SYM/proofs/layers.md`, Lemma 2(b), the proposition is applied to a finite set S in general position, invariant under $\mathrm{rotAbout}(c, 2\pi/m)$ with $m \in \{4, 5, 6, 7\}$ and without 8 points in convex position. The set $S \setminus \{c\}$ is again invariant (the rotation is injective and fixes c), and the layer results applied to it (Theorems 2.22 and 2.24, with $K = 8 \le 2m$) give $|S \setminus \{c\}| = mt$, where t is the number of convex layers of $S \setminus \{c\}$. If moreover $S \not\subseteq \{c\}$ (for instance if $|S| \ge 2$), then $c \notin S$ for even m (Lemma 3.18). So $|S| = mt + e$ with $e = [c \in S] \le 1$, and $e = 0$ for $m = 4, 6$ when $|S| \ge 2$. Then Proposition 6.19 turns $|S| \ge 65$ into $t \ge 17, 13, 11, 10$. This geometric input is not part of `ES8SYM/lean/C6check.lean`, so Lemma 2(b) as a whole is not Lean-checked there. Item (v) records the arithmetic behind the remark of `ES8SYM/proofs/layers.md` (lines 41–42) that general position is needed: with the centre allowed for $m = 4$, 16 orbits and the centre would already give 65 points.

6.3. **Orientations in inverse-radius polar form.** The source is `ES8SYM/lean/PolarOrient.lean` (namespace `C7`; Mathlib only). It is the Lean check of Lemma 1 of `ES8SYM/proofs/layers.md` (lines 15–23) and of part of the consequence stated after it (lines 25–29; see Remark 6.30(iii)). For $x \in \mathbb{R}$, $\operatorname{sign} x \in \{-1, 0, 1\}$ is Mathlib's `SignType.sign`: 1 if $x > 0$, 0 if $x = 0$, -1 if $x < 0$. We use $\operatorname{sign}(xy) = \operatorname{sign}(x)\operatorname{sign}(y)$ and $\operatorname{sign}(x) = 1$ for $x > 0$; hence $\operatorname{sign}(xP) = \operatorname{sign}(x)$ whenever $P > 0$. For $p \in \mathbb{R} \times \mathbb{R}$ we write $p = (p_1, p_2)$ (Lean `p.1`, `p.2`).

Definition 6.21 (cross product, inverse-radius points, layered coefficients; Lean: `C7.cross`, `ES8SYM/lean/PolarOrient.lean:45`, Lean: `C7.pt`, `ES8SYM/lean/PolarOrient.lean:48`, Lean: `C7.coef`, `ES8SYM/lean/PolarOrient.lean:63`, Lean: `C7.phase`, `ES8SYM/lean/PolarOrient.lean:94`, Lean: `C7.ngonPt`, `ES8SYM/lean/PolarOrient.lean:97`)**.** The following objects are used in this subsection.

(i) For $p, q, r \in \mathbb{R} \times \mathbb{R}$: $\operatorname{cross}(p, q, r) = (q_1 - p_1)(r_2 - p_2) - (q_2 - p_2)(r_1 - p_1)$.

(ii) For $\varphi, u \in \mathbb{R}$: $\mathrm{pt}(\varphi, u) = (\cos\varphi/u,\ \sin\varphi/u) \in \mathbb{R} \times \mathbb{R}$, the point with direction φ and inverse radius u (Lean's division gives $x/0 = 0$; only $u > 0$ is used).

(iii) For a type ι, $t \in \mathbb{N}$, $L : \iota \to \operatorname{Fin} t$, $\varphi : \iota \to \mathbb{R}$ and $i, j, k \in \iota$, the vector $\mathrm{coef}_{L,\varphi}(i, j, k) \in \mathbb{R}^{\operatorname{Fin} t}$ is

$$\mathrm{coef}_{L,\varphi}(i, j, k) = \sin(\varphi_k - \varphi_j)\, e_{L(i)} + \sin(\varphi_i - \varphi_k)\, e_{L(j)} + \sin(\varphi_j - \varphi_i)\, e_{L(k)},$$

where $\varphi_i = \varphi(i)$ and e_o is the o-th unit vector (Lean `Pi.single`); when two of $L(i), L(j), L(k)$ coincide, the coefficients add.

(iv) For $m, t \in \mathbb{N}$, $\theta : \operatorname{Fin} t \to \mathbb{R}$ and $(o, k) \in \operatorname{Fin} t \times \operatorname{Fin} m$: $\mathrm{phase}_{m,\theta}(o, k) = \theta_o + 2\pi k/m$, computed in $\mathbb{R}$ as $(2\pi k)/m$ with k and m cast to $\mathbb{R}$.

(v) For moreover $r : \operatorname{Fin} t \to \mathbb{R}$: $\mathrm{ngonPt}_{m,\theta,r}(o, k) = \big(r_o \cos \mathrm{phase}_{m,\theta}(o, k),\ r_o \sin \mathrm{phase}_{m,\theta}(o, k)\big)$, vertex k of the regular m-gon with centre 0, radius r_o and phase θ_o.

For the rest of this subsection, for real $\varphi_1, \varphi_2, \varphi_3, u_1, u_2, u_3$ let

$$D = \det \begin{pmatrix} \cos\varphi_1/u_1 & \sin\varphi_1/u_1 & 1 \\ \cos\varphi_2/u_2 & \sin\varphi_2/u_2 & 1 \\ \cos\varphi_3/u_3 & \sin\varphi_3/u_3 & 1 \end{pmatrix}, \qquad \Sigma = u_1 \sin(\varphi_3 - \varphi_2) + u_2 \sin(\varphi_1 - \varphi_3) + u_3 \sin(\varphi_2 - \varphi_1).$$

Lemma 6.22 (the value identity; Lean: `C7.det_mul_eq`, `ES8SYM/lean/PolarOrient.lean:24`)**.** *For all real $\varphi_1, \varphi_2, \varphi_3, u_1, u_2, u_3$ with $u_1 \ne 0$, $u_2 \ne 0$, $u_3 \ne 0$: $D \cdot (u_1 u_2 u_3) = \Sigma$.*

Proof. Write $x_k = \cos\varphi_k/u_k$ and $y_k = \sin\varphi_k/u_k$. Expanding the determinant along its third column,

$$D = (x_2 y_3 - x_3 y_2) + (x_3 y_1 - x_1 y_3) + (x_1 y_2 - x_2 y_1).$$

For indices $a \ne b$, the subtraction formula $\sin(\beta - \alpha) = \sin\beta\cos\alpha - \cos\beta\sin\alpha$ gives $x_a y_b - x_b y_a = (\cos\varphi_a \sin\varphi_b - \cos\varphi_b \sin\varphi_a)/(u_a u_b) = \sin(\varphi_b - \varphi_a)/(u_a u_b)$. Multiplying by $u_1 u_2 u_3$ turns the three terms into $u_1 \sin(\varphi_3 - \varphi_2)$, $u_2 \sin(\varphi_1 - \varphi_3)$ and $u_3 \sin(\varphi_2 - \varphi_1)$. (The Lean proof expands the determinant by `Matrix.det_fin_three`, applies the subtraction formula, clears denominators and normalizes; it is the same computation.) □

Lemma 6.23 (polar orientation formula; Lean: `C7.orient_polar`, `ES8SYM/lean/PolarOrient.lean:37`)**.** *For all real $\varphi_1, \varphi_2, \varphi_3, u_1, u_2, u_3$ with $u_1 > 0$, $u_2 > 0$, $u_3 > 0$: $\operatorname{sign} D = \operatorname{sign} \Sigma$.*

Proof. Let $P = u_1u_2u_3 > 0$. By Lemma 6.22, $\operatorname{sign}\Sigma = \operatorname{sign}(DP) = \operatorname{sign} D$. □

Lemma 6.24 (cross-product form; Lean: `C7.orient_polar_cross`, `ES8SYM/lean/PolarOrient.lean:50`)**.** *For all real $\varphi_1, \varphi_2, \varphi_3, u_1, u_2, u_3$ with $u_1 > 0$, $u_2 > 0$, $u_3 > 0$:*

$$\operatorname{sign}\operatorname{cross}\big(\operatorname{pt}(\varphi_1, u_1), \operatorname{pt}(\varphi_2, u_2), \operatorname{pt}(\varphi_3, u_3)\big) = \operatorname{sign}\Sigma.$$

Proof. With x_k, y_k as before, $\operatorname{pt}(\varphi_k, u_k) = (x_k, y_k)$, and expanding,

$$(x_2 - x_1)(y_3 - y_1) - (y_2 - y_1)(x_3 - x_1) = (x_2y_3 - x_3y_2) + (x_3y_1 - x_1y_3) + (x_1y_2 - x_2y_1),$$

the same three-term expression as D. So $\operatorname{cross}(\dots)\cdot u_1u_2u_3 = \Sigma$ by the computation in the proof of Lemma 6.22, and the sign argument of Lemma 6.23 applies. □

Proposition 6.25 (layered configurations; Lean: `C7.orient_layered`, `ES8SYM/lean/PolarOrient.lean:68`)**.** *Let ι be a type, $t \in \mathbb{N}$, $L : \iota \to \operatorname{Fin} t$, $\varphi : \iota \to \mathbb{R}$, and $u : \operatorname{Fin} t \to \mathbb{R}$ with $u(o) > 0$ for every o. Then for all $i, j, k \in \iota$ (not necessarily distinct):*

$$\operatorname{sign}\operatorname{cross}\big(\operatorname{pt}(\varphi_i, u(L(i))), \operatorname{pt}(\varphi_j, u(L(j))), \operatorname{pt}(\varphi_k, u(L(k)))\big) = \operatorname{sign}\sum_{o \in \operatorname{Fin} t} \operatorname{coef}_{L,\varphi}(i,j,k)_o \, u(o).$$

Proof. For $a \in \operatorname{Fin} t$ and $v \in \mathbb{R}$, $\sum_o (v\, e_a)_o \, u(o) = v\, u(a)$, and the sum is linear in the coefficient vector, so

$$\sum_o \operatorname{coef}_{L,\varphi}(i,j,k)_o \, u(o) = u(L(i))\sin(\varphi_k - \varphi_j) + u(L(j))\sin(\varphi_i - \varphi_k) + u(L(k))\sin(\varphi_j - \varphi_i),$$

also when some of $L(i), L(j), L(k)$ coincide. This is Σ for $(\varphi_1, \varphi_2, \varphi_3) = (\varphi_i, \varphi_j, \varphi_k)$ and $(u_1, u_2, u_3) = (u(L(i)), u(L(j)), u(L(k)))$, all positive; apply Lemma 6.24. □

Corollary 6.26 (scaling; Lean: `C7.orient_layered_scale`, `ES8SYM/lean/PolarOrient.lean:80`)**.** *In the setting of Proposition 6.25, let $\lambda \in \mathbb{R}$ with $\lambda > 0$. Then for all $i, j, k \in \iota$,*

$$\begin{aligned}\operatorname{sign}\operatorname{cross}\big(\operatorname{pt}(\varphi_i, \lambda u(L(i))), \operatorname{pt}(\varphi_j, \lambda u(L(j))), \operatorname{pt}(\varphi_k, \lambda u(L(k)))\big)& \\ = \operatorname{sign}\operatorname{cross}\big(\operatorname{pt}(\varphi_i, u(L(i))), \operatorname{pt}(\varphi_j, u(L(j))), \operatorname{pt}(\varphi_k, u(L(k)))\big).&\end{aligned}$$

Proof. The function $o \mapsto \lambda u(o)$ is positive, so Proposition 6.25 applies to it and to u. With $c_o = \operatorname{coef}_{L,\varphi}(i,j,k)_o$, the left side is $\operatorname{sign}\sum_o c_o \lambda u(o) = \operatorname{sign}\big(\lambda \sum_o c_o u(o)\big) = \operatorname{sign}\sum_o c_o u(o)$, which is the right side. (Directly: $\operatorname{pt}(\varphi, \lambda u) = \lambda^{-1}\operatorname{pt}(\varphi, u)$ and cross is homogeneous of degree 2.) □

Lemma 6.27 (polygon vertices in inverse-radius form; Lean: `C7.ngonPt_eq_pt`, `ES8SYM/lean/PolarOrient.lean:100`)**.** *Let $m, t \in \mathbb{N}$, $\theta, r : \operatorname{Fin} t \to \mathbb{R}$ with $r_o > 0$ for every o, and $p \in \operatorname{Fin} t \times \operatorname{Fin} m$. Then $\operatorname{ngonPt}_{m,\theta,r}(p) = \operatorname{pt}\big(\operatorname{phase}_{m,\theta}(p),\, 1/r_{p_1}\big)$, where p_1 is the first component of p.*

Proof. For $r_o \neq 0$ and every real ϕ, $\cos\phi/(1/r_o) = r_o\cos\phi$ and $\sin\phi/(1/r_o) = r_o\sin\phi$. □

Theorem 6.28 (nested regular m-gons; Lean: `C7.nested_regular_orient`, `ES8SYM/lean/PolarOrient.lean:108`)**.** *Let $m, t \in \mathbb{N}$ and $\theta, r : \operatorname{Fin} t \to \mathbb{R}$ with $r_o > 0$ for every o. Then for all $a, b, c \in \operatorname{Fin} t \times \operatorname{Fin} m$:*

$$\operatorname{sign}\operatorname{cross}\big(\operatorname{ngonPt}_{m,\theta,r}(a), \operatorname{ngonPt}_{m,\theta,r}(b), \operatorname{ngonPt}_{m,\theta,r}(c)\big) = \operatorname{sign}\sum_{o \in \operatorname{Fin} t} \operatorname{coef}_{\pi_1, \operatorname{phase}_{m,\theta}}(a,b,c)_o \cdot \frac{1}{r_o},$$

where $\pi_1 : \operatorname{Fin} t \times \operatorname{Fin} m \to \operatorname{Fin} t$ is the first projection (Lean `Prod.fst`*). The coefficient vector depends only on m and θ, not on r.*

Proof. By Lemma 6.27, $\operatorname{ngonPt}_{m,\theta,r}(a) = \operatorname{pt}(\operatorname{phase}_{m,\theta}(a), u(\pi_1(a)))$ with $u(o) = 1/r_o > 0$, and likewise for b and c. Apply Proposition 6.25 with $\iota = \operatorname{Fin} t \times \operatorname{Fin} m$, $L = \pi_1$, $\varphi = \operatorname{phase}_{m,\theta}$ and this u. □

Lemma 6.29 (triples with the centre; Lean: `C7.orient_centre`, `ES8SYM/lean/PolarOrient.lean:116`)**.** *For all real $\varphi_1, \varphi_2, u_1, u_2$ with $u_1 > 0$ and $u_2 > 0$:* $\operatorname{sign}\operatorname{cross}\big(\operatorname{pt}(\varphi_1, u_1), \operatorname{pt}(\varphi_2, u_2), (0,0)\big) = \operatorname{sign}\sin(\varphi_2 - \varphi_1)$.

Proof. With x_k, y_k as before, $\operatorname{cross}((x_1,y_1),(x_2,y_2),(0,0)) = (x_2 - x_1)(-y_1) - (y_2 - y_1)(-x_1) = x_1y_2 - x_2y_1 = \sin(\varphi_2 - \varphi_1)/(u_1u_2)$. Multiply by $u_1u_2 > 0$ and argue as in Lemma 6.23. □

Remark 6.30 (scope of the polar lemmas). (i) If $\hat{p}$ denotes the point with coordinates $\hat{p}_0 = p_1$, $\hat{p}_1 = p_2$, then $\mathrm{cross}(p,q,r) = \mathrm{orient}(\hat{p},\hat{q},\hat{r})$ by the formulas (Definition 1.3), so a positive value means counterclockwise; Lean has no link between `C7.cross` and `ErdosSzekeres.orient`. By the expansions in the proofs of Lemmas 6.22 and 6.24, $\mathrm{cross}(p,q,r)$ is the determinant of the matrix with rows $(p_1,p_2,1)$, $(q_1,q_2,1)$, $(r_1,r_2,1)$. (ii) Lemma 6.23 is not the theorem `orient_polar` of `LAYERLEAN/LayerLemma.lean:935` (Lemma 3.35), which is a radius form about a centre c: $\mathrm{orient}(\mathrm{polar}(c,r_a,\alpha),\mathrm{polar}(c,r_b,\beta),\mathrm{polar}(c,r_d,\delta)) = r_a r_b \sin(\beta-\alpha) + r_b r_d \sin(\delta-\beta) + r_d r_a \sin(\alpha-\delta)$, where $\mathrm{polar}(c,r,\alpha) = (c_0 + r\cos\alpha,\, c_1 + r\sin\alpha)$. For $u \neq 0$ the point $\mathrm{pt}(\varphi,u)$ corresponds as in (i) to $\mathrm{polar}(0,1/u,\varphi)$, and for $r_a = 1/u_1$, $r_b = 1/u_2$, $r_d = 1/u_3$ this right side times $u_1u_2u_3$ is Σ with $(\varphi_1,\varphi_2,\varphi_3) = (\alpha,\beta,\delta)$, so the two agree in substance, but they are distinct declarations; the name `orient_polar` in `WRITEUP/P1/p1.tex` and `WRITEUP/note/k8_soundness.tex` refers to the radius form. (iii) Theorem 6.28 assumes only $r_o > 0$: no nesting, no order of the radii, no distinctness of the points and no general position. It says that for fixed phases every orientation sign is the sign of a linear form in $(1/r_o)_o$ whose coefficients depend only on θ and m; Corollary 6.26 adds that the signs do not change when all inverse radii are multiplied by the same $\lambda > 0$. Lemma 6.29 treats triples that contain the centre $(0,0)$, which have no finite inverse radius. The further statement of `ES8SYM/proofs/layers.md` (lines 26–27) that for fixed phases the order type is constant on the cells of a central hyperplane arrangement is not formalized as such.

6.4. **Size compatibility in a group with zero.** The source is `PRIMEFOLD/compat/lean/PCompatCore.lean` (namespace `PCompat`). Here the plane is replaced by an arbitrary commutative group with zero K (Mathlib `CommGroupWithZero`: a commutative monoid with zero in which $0 \neq 1$, with an inverse map such that $aa^{-1} = 1$ for every $a \neq 0$ and $0^{-1} = 0$; the structure also carries a division and integer powers, which are not used here), with decidable equality (Lean needs it to form finite sets by image, difference, erasure and insertion, and for the test $0 \in S$ in Theorem 6.35(b); mathematically it is no restriction). In the application $K = \mathbb{C}$, the centre is 0 and the rotation is $z \mapsto \zeta z$; a rotation about another centre reduces to this by a translation. We use only these consequences of the axioms: $0 \cdot a = 0$; if $a \neq 0$ and $ab = ac$, then $b = c$ (multiply by a^{-1}); if $a \neq 0$ and $b \neq 0$, then $ab \neq 0$ (otherwise $b = a^{-1}(ab) = 0$); hence $a^j \neq 0$ for $a \neq 0$. For a finite $T \subseteq K$, "T is ζ-invariant" means $\zeta z \in T$ for every $z \in T$. We write $[P]$ for 1 if P holds and 0 otherwise, and $N \bmod m$ for the remainder (Lean `N % m`). The hypotheses on $m \in \mathbb{N}$, $\zeta \in K$ and $r : \mathbb{N} \to K$ are:

- (O) $0 < m$, $\zeta^m = 1$, and $\zeta^j \neq 1$ for every j with $0 < j < m$ (ζ has exact order m);
- (R0) $r_i \neq 0$ for every $i \in \mathbb{N}$;
- (RD) for all $i,i',j,j' \in \mathbb{N}$ with $j < m$ and $j' < m$: if $\zeta^j r_i = \zeta^{j'} r_{i'}$, then $i = i'$ (the orbits of the r_i are pairwise disjoint).

Lemma 6.31 (powers; Lean: `PCompat.ne_zero_of_order`, `PRIMEFOLD/compat/lean/PCompatCore.lean:36`, Lean: `PCompat.pow_inj_of_order`, `PRIMEFOLD/compat/lean/PCompatCore.lean:43`, Lean: `PCompat.pow_mul_mem`, `PRIMEFOLD/compat/lean/PCompatCore.lean:58`)**.** *Let K be a commutative group with zero, $m \in \mathbb{N}$ and $\zeta \in K$.*

(a) If $0 < m$ and $\zeta^m = 1$, then $\zeta \neq 0$.
(b) If (O) *holds, $i < m$, $j < m$ and $\zeta^i = \zeta^j$, then $i = j$.*
(c) If $T \subseteq K$ is finite and ζ-invariant and $z \in T$, then $\zeta^j z \in T$ for every $j \in \mathbb{N}$.

Proof. (a) If $\zeta = 0$, then $\zeta^m = 0$ since $m > 0$, so $0 = 1$, which is false in K. (b) By (a), $\zeta \neq 0$. Suppose $i < j$. Then $\zeta^i\zeta^{j-i} = \zeta^j = \zeta^i = \zeta^i \cdot 1$, and $\zeta^i \neq 0$, so $\zeta^{j-i} = 1$ with $0 < j - i < m$, contradicting (O). The case $j < i$ is symmetric. Hence $i = j$. (c) Induction on j: $\zeta^0 z = z \in T$, and $\zeta^{j+1}z = \zeta(\zeta^j z) \in T$ by invariance. □

Lemma 6.32 (the converse constructions; Lean: `PCompat.exists_card0`, `PRIMEFOLD/compat/lean/PCompatCore.lean:129`, Lean: `PCompat.exists_card1`, `PRIMEFOLD/compat/lean/PCompatCore.lean:160`)**.** *Let K be a commutative group with zero, $m \in \mathbb{N}$, $\zeta \in K$ and $r : \mathbb{N} \to K$, and assume* (O), (R0) *and* (RD)*. Then for every $t \in \mathbb{N}$:*

(a) there is a finite ζ-invariant $S \subseteq K$ with $0 \notin S$ and $|S| = mt$;
(b) there is a finite ζ-invariant $S \subseteq K$ with $|S| = mt + 1$.

Proof. (a) Let $P = \{0,\dots,m-1\} \times \{0,\dots,t-1\}$ and $S = \{\zeta^j r_i : (j,i) \in P\}$. *Invariance:* for $(j,i) \in P$, if $j + 1 < m$ then $\zeta \cdot \zeta^j r_i = \zeta^{j+1} r_i$ with $(j+1,i) \in P$; if $j + 1 = m$ then $\zeta \cdot \zeta^j r_i = \zeta^m r_i = r_i = \zeta^0 r_i$ with $(0,i) \in P$ (as $0 < m$). $0 \notin S$: $\zeta \neq 0$ by Lemma 6.31(a), so $\zeta^j \neq 0$, and $r_i \neq 0$ by (R0), so $\zeta^j r_i \neq 0$. *Cardinality:* the map $(j,i) \mapsto \zeta^j r_i$ is injective on P: if $\zeta^j r_i = \zeta^{j'} r_{i'}$ with $(j,i),(j',i') \in P$, then $i = i'$ by (RD), then $\zeta^j = \zeta^{j'}$ by cancelling $r_i \neq 0$, then $j = j'$ by Lemma 6.31(b). Hence $|S| = |P| = mt$.

(b) Take S from (a) and $S' = S \cup \{0\}$. It is ζ-invariant ($\zeta \cdot 0 = 0$), and $|S'| = mt + 1$ because $0 \notin S$. □

Definition 6.33 (size condition; Lean: `PCompat.Compat`, `PRIMEFOLD/compat/lean/PCompatCore.lean:205`)**.** For $m, N \in \mathbb{N}$, $\mathrm{Compat}(m, N)$ means: $N \bmod m = 0$ or $N \bmod m = 1 \bmod m$. (For every $m \geq 1$ this is the same condition as $N \bmod m \in \{0, 1\}$: for $m \geq 2$, $1 \bmod m = 1$, and for $m = 1$ both conditions hold, since $N \bmod 1 = 0$. The form $1 \bmod m$ matches the right side of Theorem 6.35(b). For $m = 0$, Lean's convention $N \bmod 0 = N$ applies, but $m = 0$ is not used.)

Proposition 6.34 (parity; Lean: `PCompat.compat_even`, `PRIMEFOLD/compat/lean/PCompatCore.lean:208`, Lean: `PCompat.compat_odd`, `PRIMEFOLD/compat/lean/PCompatCore.lean:235`)**.** *Both parts are statements about natural numbers.*

(a) For every even $m \in \mathbb{N}$ with $2 \leq m$: $\mathrm{Compat}(m, 2^{m-1})$ *holds if and only if $m = 2^a$ for some $a \geq 1$.*

(b) For every odd $m \in \mathbb{N}$ with $3 \leq m$: $\mathrm{Compat}(m, 2^{m-1})$ *holds if and only if* $2^{m-1} \bmod m = 1$.

Proof. We use the standard fact (unique factorization; Mathlib `Nat.dvd_prime_pow`) that the divisors of 2^n are exactly $2^0, 2^1, \ldots, 2^n$. In both parts $m \geq 2$, so $1 \bmod m = 1$ and $\mathrm{Compat}(m, 2^{m-1})$ says $2^{m-1} \bmod m \in \{0, 1\}$.

(a) Suppose $2^{m-1} \bmod m = 0$. Then $m \mid 2^{m-1}$, so $m = 2^a$ with $a \leq m - 1$, and $a \neq 0$ because $m \geq 2$. Suppose $2^{m-1} \bmod m = 1$. Write $2^{m-1} = qm + 1$; since $2 \mid m$, 2^{m-1} is odd. But $m - 1 \geq 1$, so 2^{m-1} is even, a contradiction. Conversely, let $m = 2^a$ with $a \geq 1$. Since $a < 2^a = m$, we have $a \leq m - 1$, so $m = 2^a \mid 2^{m-1}$ and $2^{m-1} \bmod m = 0$.

(b) Suppose $2^{m-1} \bmod m = 0$. Then $m = 2^a$ for some a; $a = 0$ gives $m = 1 < 3$, and $a \geq 1$ makes m even; both are impossible. So $\mathrm{Compat}(m, 2^{m-1})$ holds exactly when $2^{m-1} \bmod m = 1$. □

Theorem 6.35 (compatibility; Lean: `PCompat.dvd_card`, `PRIMEFOLD/compat/lean/PCompatCore.lean:67`, Lean: `PCompat.card_mod`, `PRIMEFOLD/compat/lean/PCompatCore.lean:112`, Lean: `PCompat.exists_iff`, `PRIMEFOLD/compat/lean/PCompatCore.lean:174`, Lean: `PCompat.classification`, `PRIMEFOLD/compat/lean/PCompatCore.lean:253`)**.** *Let $m \in \mathbb{N}$, and in* (a)–(c) *let K be a commutative group with zero and $\zeta \in K$.*

(a) (`dvd_card`) *If* (O) *holds, then every finite ζ-invariant $T \subseteq K$ with $0 \notin T$ satisfies $m \mid |T|$.*

(b) (`card_mod`) *If* (O) *holds, then every finite ζ-invariant $S \subseteq K$ satisfies $|S| \bmod m = [0 \in S] \bmod m$.*

(c) (`exists_iff`) *Let $r : \mathbb{N} \to K$. If* (O), (R0) *and* (RD) *hold, then for every $N \in \mathbb{N}$: there is a finite ζ-invariant $S \subseteq K$ with $|S| = N$ if and only if $N \bmod m = 0$ or $N \bmod m = 1 \bmod m$ (that is,* $\mathrm{Compat}(m, N)$*).*

(d) (`classification`; this part does not involve K and ζ) *If $1 \leq m$, then* $\mathrm{Compat}(m, 2^{m-1})$ *holds if and only if $m = 1$, or $m = 2^a$ for some $a \geq 1$, or (m is odd, $3 \leq m$ and $2^{m-1} \bmod m = 1$).*

Proof. We follow the Lean proofs.

(a) By Lemma 6.31(a), $\zeta \neq 0$. We use strong induction on $|T|$. If $T = \emptyset$, then $m \mid 0$. Otherwise pick $z \in T$; $z \neq 0$ because $0 \notin T$. Let $O = \{\zeta^j z : 0 \leq j < m\}$. By Lemma 6.31(c), $O \subseteq T$. The map $j \mapsto \zeta^j z$ is injective on $\{0, \ldots, m-1\}$: if $\zeta^i z = \zeta^j z$, then $\zeta^i = \zeta^j$ (cancel $z \neq 0$) and $i = j$ by Lemma 6.31(b). So $|O| = m$, and $|T| = |T \setminus O| + m$. The set $T \setminus O$ omits 0 and is ζ-invariant: let $w \in T \setminus O$; then $\zeta w \in T$. Suppose $\zeta w \in O$, say $\zeta w = \zeta^j z$ with $j < m$. If $j = 0$, then $\zeta^{m-1} z = \zeta^{m-1} \zeta w = \zeta^m w = w$, so $w \in O$ (as $m - 1 < m$). If $j \geq 1$, then $\zeta w = \zeta \cdot \zeta^{j-1} z$, so $w = \zeta^{j-1} z \in O$ (cancel $\zeta \neq 0$). Both contradict $w \notin O$, so $\zeta w \in T \setminus O$. Since $z \in O$, $|T \setminus O| < |T|$, and the induction hypothesis gives $m \mid |T \setminus O|$. Hence $m \mid |T|$.

(b) Let $T_0 = S \setminus \{0\}$. Then $0 \notin T_0$, and T_0 is ζ-invariant: for $z \in T_0$, $\zeta z \in S$ and $\zeta z \neq 0$ since $\zeta \neq 0$ and $z \neq 0$. By (a), $|T_0| = mt$ for some t. If $0 \in S$, then $|S| = mt + 1$ and $|S| \bmod m = 1 \bmod m = [0 \in S] \bmod m$. If $0 \notin S$, then $S = T_0$ and $|S| \bmod m = 0 = [0 \in S] \bmod m$.

(c) ($\Rightarrow$) Let S be finite, ζ-invariant, with $|S| = N$. By (b), $N \bmod m = 1 \bmod m$ if $0 \in S$, and $N \bmod m = 0 \bmod m = 0$ if $0 \notin S$. ($\Leftarrow$) Write $N = m\lfloor N/m \rfloor + (N \bmod m)$. If $N \bmod m = 0$, Lemma 6.32(a) with $t = \lfloor N/m \rfloor$ gives a ζ-invariant S with $|S| = N$. Otherwise $N \bmod m = 1 \bmod m$. Then $m \neq 1$ (for $m = 1$ every remainder is 0), so $m > 1$, $1 \bmod m = 1$, $N = m\lfloor N/m \rfloor + 1$, and Lemma 6.32(b) with $t = \lfloor N/m \rfloor$ gives the set.

(d) If m is even, then $m \geq 2$, and by Proposition 6.34(a) $\mathrm{Compat}(m, 2^{m-1})$ holds iff $m = 2^a$ with $a \geq 1$; the other two alternatives of the right side are false for even m. If m is odd, then $m = 1$ or $m \geq 3$. For $m = 1$ both sides hold: $2^0 \bmod 1 = 0$, and $m = 1$. For odd $m \geq 3$, by Proposition 6.34(b) the left side holds iff $2^{m-1} \bmod m = 1$; on the right side $m = 1$ is false and $m = 2^a$ with $a \geq 1$ is impossible (m is odd), so the right side holds iff $2^{m-1} \bmod m = 1$. □

Proposition 6.36 ($m = 341$; computer-assisted in Lean; written proof by hand; Lean: `PCompat.ex341`, `PRIMEFOLD/compat/lean/PCompatCore.lean:277`)**.** $\mathrm{Compat}(341, 2^{341-1})$*;* $341 = 11 \cdot 31$*;* 341 *is not prime;* $2^{10} \bmod 11 = 1$*;* $2^{10} \bmod 31 = 1$*;* $2^{10} \bmod 341 = 1$.

Proof. This is a finite computation with closed numerals. Lean checks the five numerical claims by kernel evaluation (`decide`, with `exponentiation.threshold` raised to 400; no `native_decide`) and non-primality by Mathlib's `Nat.not_prime_mul` with $11 \neq 1$ and $31 \neq 1$; the log lists only `propext`. The computation is small enough to do by hand, as follows. $11 \cdot 31 = 341$, and 11 is a divisor with $1 < 11 < 341$, so 341 is not prime. Further $2^{10} = 1024 = 93 \cdot 11 + 1 = 33 \cdot 31 + 1 = 3 \cdot 341 + 1$, which gives the three remainders. Finally $2^{340} = (2^{10})^{34} \equiv 1^{34} = 1 \pmod{341}$, so $2^{340} \bmod 341 = 1 = 1 \bmod 341$ and $\mathrm{Compat}(341, 2^{340})$ holds by its second alternative. □

Remark 6.37 (what is and is not formalized). (i) For $K = \mathbb{C}$ and $\zeta = \zeta_m = e^{2\pi i/m}$ with $m \geq 1$, (O) holds by Lemma 6.13 and Fact 6.14(iv). (ii) In Theorem 6.35(c) the hypotheses (R0) and (RD) on r are not discharged in Lean for $\mathbb{C}$. They hold for $K = \mathbb{C}$, $\zeta = \zeta_m$ and $r_i = i + 1$ (a written step): (R0) is clear, and $|\zeta_m^j r_i| = |\zeta_m|^j (i+1) = i + 1$ by Fact 6.14(ii), so $\zeta_m^j r_i = \zeta_m^{j'} r_{i'}$ forces $i + 1 = i' + 1$. The sets of Lemma 6.32 need not be in general position (for $m = 2$ and $r_i = i + 1$ they lie on the real axis); the existence of size-compatible sets in general position (`WRITEUP/note/symmetric_extremal.tex`, the proposition labelled `prop:compat`) is not formalized. (iii) Fermat's little theorem is not formalized: the third alternative of Theorem 6.35(d) is the arithmetic condition $2^{m-1} \bmod m = 1$; by Fermat's little theorem it holds for every odd prime m, and for odd composite m it says that m is a base-2 Fermat pseudoprime, as for $m = 341$.

6.5. **Symmetric sets of 2^{m-1} points.** The source is `PRIMEFOLD/compat/lean/L0Check.lean` (namespace `L0Check`), an independent check of Lemma 0 of `PRIMEFOLD/cluster/PROOFS.md`. The plane is $\mathbb{C}$, the centre is 0, and the rotation by $2\pi/m$ is multiplication by a primitive m-th root of unity. Mathlib's base-2 Fermat pseudoprimes are defined by

$$\mathrm{FermatPsp}(m, 2) \iff m \mid 2^{m-1} - 1,\ m \text{ is not prime, and } 1 < m$$

(`Nat.FermatPsp`, with natural subtraction; $2^{m-1} \geq 1$, so $2^{m-1} - 1$ is the usual difference).

Lemma 6.38 (arithmetic core; Lean: `L0Check.arith`, `PRIMEFOLD/compat/lean/L0Check.lean:21`)**.** *Let $m, t, e \in \mathbb{N}$ with $2 \leq m$, $e \leq 1$ and $mt + e = 2^{m-1}$. Then either (m is odd, $e = 1$, and m is prime or $\mathrm{FermatPsp}(m, 2)$), or ($e = 0$ and $m = 2^j$ for some $j \in \mathbb{N}$).*

Proof. m even. Then mt is even, and 2^{m-1} is even because $m - 1 \geq 1$. Hence $e = 2^{m-1} - mt$ is even, and $e \leq 1$ gives $e = 0$. Then $mt = 2^{m-1}$, so $m \mid 2^{m-1}$ and $m = 2^j$ for some j (the divisors of 2^n are powers of 2). The second alternative holds.

m odd. If $e \neq 1$, then $e = 0$, $m \mid 2^{m-1}$ and $m = 2^j$; $j = 0$ gives $m = 1 < 2$, and $j \geq 1$ makes m even; both are impossible. So $e = 1$ and $mt + 1 = 2^{m-1}$, hence $m \mid mt = 2^{m-1} - 1$. If m is prime, the first alternative holds; otherwise $m \mid 2^{m-1} - 1$, m is not prime and $1 < m$, which is $\mathrm{FermatPsp}(m, 2)$. □

Lemma 6.39 (orbit counting in $\mathbb{C}$; Lean: `L0Check.orbit_dvd`, `PRIMEFOLD/compat/lean/L0Check.lean:60`)**.** *Let $m \in \mathbb{N}$ with $0 < m$, and let $\zeta \in \mathbb{C}$ be a primitive m-th root of unity. Then every finite $T \subseteq \mathbb{C}$ with $0 \notin T$ and $\zeta z \in T$ for all $z \in T$ satisfies $m \mid |T|$.*

Proof. $\mathbb{C}$ is a commutative group with zero, $\zeta^m = 1$, and $\zeta^j \neq 1$ for $0 < j < m$ by Fact 6.14(iv); so (O) holds and Theorem 6.35(a) applies. (The Lean file proves this statement again, independently, by the same strong induction, using Mathlib's `IsPrimitiveRoot.pow_inj` for the size of an orbit.) □

Theorem 6.40 (sets of 2^{m-1} points; Lean: `L0Check.L0`, `PRIMEFOLD/compat/lean/L0Check.lean:122`, Lean: `L0Check.L0_rot`, `PRIMEFOLD/compat/lean/L0Check.lean:144`)**.** *Let $m \in \mathbb{N}$ with $2 \leq m$.*

(a) (`L0`) *Let $\zeta \in \mathbb{C}$ be a primitive m-th root of unity, and let $S \subseteq \mathbb{C}$ be finite with $\zeta z \in S$ for all $z \in S$ and $|S| = 2^{m-1}$. Then either (m is odd, $0 \in S$, and m is prime or $\mathrm{FermatPsp}(m, 2)$), or ($0 \notin S$ and $m = 2^j$ for some $j \in \mathbb{N}$).*

(b) (`L0_rot`) *Let $S \subseteq \mathbb{C}$ be finite with $\exp(2\pi i/m)\, z \in S$ for all $z \in S$ and $|S| = 2^{m-1}$. Then the same conclusion holds.*

Proof. (a) By Fact 6.14(i), $\zeta \neq 0$. The set $T = S \setminus \{0\}$ omits 0 and is invariant: for $z \in T$, $\zeta z \in S$ and $\zeta z \neq 0$. By Lemma 6.39, $|T| = mt$ for some $t \in \mathbb{N}$. If $0 \in S$, then $mt + 1 = |S| = 2^{m-1}$, and Lemma 6.38 with $e = 1$ gives the first alternative of the lemma (the second needs $e = 0$); together with $0 \in S$ this is the first alternative of the theorem. If $0 \notin S$, then $T = S$ and $mt + 0 = 2^{m-1}$; Lemma 6.38 with $e = 0$ excludes its first alternative (which needs $e = 1$), so $m = 2^j$ for some j, and with $0 \notin S$ this is the second alternative of the theorem.

(b) $\exp(2\pi i/m) = \zeta_m$ is a primitive m-th root of unity by Lemma 6.13 ($m \geq 2 > 0$); apply (a). □

Remark 6.41 (on the hypotheses). No general-position or convexity hypothesis is assumed. For even m the centre is excluded by parity, as in the proof of Lemma 6.38: $|S| = 2^{m-1}$ is even and $|S \setminus \{0\}|$ is a multiple of m. (Lemma 0 of `PRIMEFOLD/cluster/PROOFS.md` excludes the centre by general position instead; the Lean statement does not need it.) For odd m the theorem only says that m is prime or a base-2 Fermat pseudoprime; that every odd prime actually satisfies $2^{m-1} \equiv 1 \pmod m$ is Fermat's little theorem, which is not used and not formalized here. For a rotation about a centre $c \neq 0$, translate by $-c$ first. For $k = 8$, i.e. $m = 7$ and $2^{m-1} = 64 = 7 \cdot 9 + 1$: a finite $S \subseteq \mathbb{C}$ with $e^{2\pi i/7} S \subseteq S$ and $|S| = 64$ contains 0, by part (b), since 7 is odd and not a power of 2.

Coverage. Covered, by label:

- `LEANEXT/es7corr/ES7Corollaries.lean`, Section 11: `Fivefold35` and `Fourfold32` (Definition 6.1); `fivefold_heptagon_of_erase` (Theorem 6.3); `fivefold_heptagon` (Theorem 6.4); `fivefold_heptagon_of_centre_not_mem` (Theorem 6.5); `fourfold_heptagon_of_erase` (Theorem 6.6); `fourfold_heptagon` (Theorem 6.7). The three fivefold theorems are conditional on `Fivefold35`, the two fourfold theorems on `Fourfold32`.
- `ES8SYM/lean/C6check.lean`: `sphere_subset_extremePoints` (Lemma 6.10); `convexIndependent_of_norm_eq` (Lemma 6.11); `zeta` (Definition 6.12); `zeta_prim` (Lemma 6.13); `orbit_injective` and `orbit_convexIndependent` (Lemma 6.15); `part_a` (Theorem 6.16); `thr_m4`–`thr_m7` and `m4_with_centre_counter` (Proposition 6.19).
- `ES8SYM/lean/PolarOrient.lean`: `cross`, `pt`, `coef`, `phase`, `ngonPt` (Definition 6.21); `det_mul_eq` (Lemma 6.22); `orient_polar` (Lemma 6.23); `orient_polar_cross` (Lemma 6.24); `orient_layered` (Proposition 6.25); `orient_layered_scale` (Corollary 6.26); `ngonPt_eq_pt` (Lemma 6.27); `nested_regular_orient` (Theorem 6.28); `orient_centre` (Lemma 6.29).
- `PRIMEFOLD/compat/lean/PCompatCore.lean`: `ne_zero_of_order`, `pow_inj_of_order`, `pow_mul_mem` (Lemma 6.31); `exists_card0`, `exists_card1` (Lemma 6.32); `Compat` (Definition 6.33); `compat_even`, `compat_odd` (Proposition 6.34); `dvd_card`, `card_mod`, `exists_iff`, `classification` (Theorem 6.35); `ex341` (Proposition 6.36, a finite computation).
- `PRIMEFOLD/compat/lean/L0Check.lean`: `arith` (Lemma 6.38); `orbit_dvd` (Lemma 6.39); `L0`, `L0_rot` (Theorem 6.40).

Not covered, deliberately: lines 1–1147 of `LEANEXT/es7corr/ES7Corollaries.lean` (outside the scope of this section), a verbatim copy of lines 1–1147 of `LAYERLEAN/LayerLemma.lean`, which belongs to the layer-lemma sections (we use Theorem 2.26 and Lemma 3.18 from there); the `#print axioms` commands at the ends of the four other files (`ES8SYM/lean/C6check.lean`, lines 119–123; `ES8SYM/lean/PolarOrient.lean`, lines 127–132; `PRIMEFOLD/compat/lean/PCompatCore.lean`, lines 285–291; `PRIMEFOLD/compat/lean/L0Check.lean`, line 152), which produce the logs cited above and are not results. Out of scope for this section: the Lean files of `ES8SLACK` (`ES8SLACK/review/C6_early_vertex/EarlyVertex.lean`, `ES8SLACK/review/C2_identity/Identity.lean`, and the two files that according to `ES8SLACK/README.md` have not been compiled, `ES8SLACK/review/C1_certificates/Composition311.lean` and `ES8SLACK/review/C3_residual/TemplateRowsC3.lean`); written proofs of the `ES8SLACK` lemmas are in `ES8SLACK/proofs.md`. Fact 6.14, Corollary 6.18 and the example in Remark 6.37(ii) are written steps and not declarations of the five Lean files; only the Fact has Lean counterparts, the Mathlib lemmas named in its proof.

7. Reduction of symmetric sets to case classes

This section gives written proofs of the declarations of `SYMLEAN/SymmetricReduction.lean`. Its main result, Theorem 7.39 (`heptagon_of_nontrivial_isometry`), is a conditional statement. Five propositions about point sets of an exact size with one prescribed symmetry, the *case classes* `HalfTurn34`, `Threefold33`, `Fivefold35`, `MirrorOneAxis33` and `MirrorNoAxis34` (Definition 7.7), imply the following: every finite set of at least 33 points in general position that is mapped into itself by a Euclidean isometry of the plane other than the identity contains 7 points in convex position. The classes are *hypotheses* of the theorem. Nothing in this section proves them; Remark 7.8 says which project records supply them and with which rigor label. The section also proves the variant Theorem 7.41 (`halfTurn_of_symmetric_counterexample`), in which a fourfold class `Fourfold32` replaces `HalfTurn34`, and the bridge Lemma 7.44 to a mirror statement of the directory `H3`.

The proof of Theorem 7.39 has three parts.

(1) Classification (Theorem 7.21). A Euclidean isometry $g \neq \mathrm{id}$ of the plane that maps a finite set with at least two points into itself is a rotation $z \mapsto c + a(z - c)$ with $a \neq 1$ a root of unity, or a reflection in a line. Translations and glide reflections are excluded because the set is finite.
(2) Rotations. A power of g is the half-turn, or a rotation whose factor is a primitive p-th root of unity for an odd prime p (Lemma 7.29). Whole orbits are selected (Lemma 7.16) to reach the exact sizes of the classes: 17 antipodal pairs; 11 orbits of size 3; 7 orbits of size 5. For $p \geq 7$ one orbit consists of p concyclic points, which are in convex position (Lemma 7.23).
(3) Reflections. At most two points lie on the mirror and the others come in pairs; 17 pairs, or 16 pairs and one point of the mirror, are kept.

Sources. Paths of research records are relative to `cloud/es7-2026-10-19/results`; paths of the main library are relative to the repository root. The first 12 hex digits of the SHA-256 of each Lean source used are:

- `SYMLEAN/SymmetricReduction.lean` (1059 lines; every declaration of this file is covered here): `6d5ec90a4215`;
- `SYMLEAN/SymmetricReductionAxioms.lean`: `7bf809cbdc77`. Its lines 1–1059 are byte-identical to the previous file; two `#print axioms` commands follow;
- the eight repository modules that the file imports, directly or through other imports:
  - `ErdosSzekeres/Definitions.lean`: `2425500eefd1`;
  - `ErdosSzekeres/Basic.lean`: `13667aa2d3dc`;
  - `ErdosSzekeres/Orientation.lean`: `f67955ac7091`;
  - `ErdosSzekeres/OrientationGeometry.lean`: `c7db9bbbb65e`;
  - `ErdosSzekeres/Suk/SupportingLines.lean`: `6a8e12677685`;
  - `ErdosSzekeres/Suk/BaseOrder.lean`: `d21becd29e84`;
  - `ErdosSzekeres/Suk/BaseChart.lean`: `08d0e856f005`;
  - `ErdosSzekeres/Suk/TriangleOrder.lean`: `6447cd4833d4`.

  The last three are imported only through `ErdosSzekeres/Suk/SupportingLines.lean`, and this section cites nothing from them.

The files were checked with Lean 4.24.0 and Mathlib `f897ebcf`.

Axioms and rigor.

- The compile log `SYMLEAN/logs/SymmetricReduction.out` is empty, and the exit code is 0 (`SYMLEAN/logs/runs.tsv`). So no declaration of the file uses `sorry`.
- The log `SYMLEAN/logs/SymmetricReductionAxioms.out` shows that the two theorems `heptagon_of_nontrivial_isometry` and `halfTurn_of_symmetric_counterexample` depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.
- The bridge lemma `mirrorOneAxis33_of_exclusionLine` is used in the proof of the review theorem `Review.mirrorOneAxis33_iff_exclusionLine` at `SYMLEAN/review/ReviewSanity.lean:1629`. That file's lines 1–1059 are byte-identical to `SYMLEAN/SymmetricReduction.lean`, and its log `SYMLEAN/logs/ReviewSanity.out` shows the same three axioms for this theorem.
- The file and the eight imported repository modules listed above contain no `axiom` declaration, no `sorry`, no `decide` and no `native_decide` (the word "axiom" occurs there only in comments). A declaration of the file that the printed theorems do not use (such as `IsEucIsometry.injective`) is covered by the empty compile log and by this scan: it uses no `sorry`, and neither the file nor these modules add an axiom. No statement of this section is a finite computation.

Every statement below whose head carries a Lean reference is *Lean-checked* in this sense. The proofs written here are *written proofs (not refereed)*. Unless a proof says otherwise, it follows the Lean proof step by step. Statements tagged "written step" are introduced only for the written proofs; most of them are standard facts that the Lean proofs take from Mathlib, and each is proved here.

Dependencies. The section uses the foundations: Definitions 1.1, 1.3, 1.5, 1.6, 1.7, 1.10 and 1.12, and Lemmas 1.20, 1.22 and 1.23. Lemma 7.22 is the lemma titled "Strict supporting lines" in the foundations section (the same declaration of the main library), repeated here with its short proof. Remarks 7.8 and 7.40 refer to Theorems 11.98 and 11.82 (bookkeeping of two case splits; both theorems are computer-assisted, as their Lean proofs rest on kernel evaluation, and Theorem 11.98 is Lean-checked according to the record, Theorem 11.82 Lean-checked), to Theorem 10.39, which gives written proofs of the two mirror classes, to Theorem 10.40 and to Theorem 8.28; no proof of this section uses them.

7.1. **Setting and definitions.**

Conventions. Points are elements of $\mathbb{R}^2$, written $p = (p_0, p_1)$ (Definition 1.1; Lean type `Point`). Point sets are finite (Lean `Finset Point`), and $|S|$ is the number of elements of S. The function orient is that of Definition 1.3: $\mathrm{orient}(a,b,c) = (b_0-a_0)(c_1-a_1) - (b_1-a_1)(c_0-a_0)$. General position, convex position and `HasConvexNGon n S` are as in Definitions 1.5, 1.6 and 1.7. We use them in the following forms.

- S is in general position if and only if $\mathrm{orient}(a,b,c) \neq 0$ for all pairwise distinct $a,b,c \in S$ (Lemma 1.20).
- S is in convex position if and only if $p \notin \mathrm{conv}(S \setminus \{p\})$ for every $p \in S$ (Lemma 1.22).
- `HasConvexNGon n S` means: some $T \subseteq S$ with $|T| = n$ is in convex position.
- Monotonicity (Lemma 1.23): subsets of a set in general position are in general position; subsets of a set in convex position are in convex position; `HasConvexNGon n T` and $T \subseteq S$ imply `HasConvexNGon n S`.
- If T is in convex position and $n \leq |T|$, then `HasConvexNGon n T`: any n-element subset of T is in convex position by monotonicity (Lean `ConvexPosition.hasConvexNGon`, `ErdosSzekeres/Basic.lean:31`).

For a map $f\colon X \to X$ the iterates are $f^{[0]} = \mathrm{id}$ and $f^{[k+1]} = f \circ f^{[k]}$; then also $f^{[k+1]} = f^{[k]} \circ f$ and $f^{[i+j]} = f^{[i]} \circ f^{[j]}$. Natural numbers include 0, and every natural number divides 0. For $z \in \mathbb{C}$, $\bar{z}$ is the complex conjugate (Mathlib `starRingEnd`) and $|z|$ the modulus; Mathlib's $\mathrm{normSq}(z) = (\mathrm{Re}\, z)^2 + (\mathrm{Im}\, z)^2$ equals $|z|^2$ and is multiplicative. We write $\langle z, w\rangle = \mathrm{Re}(z\bar{w}) = \mathrm{Re}\, z\, \mathrm{Re}\, w + \mathrm{Im}\, z\, \mathrm{Im}\, w$ for the Euclidean inner product of $\mathbb{C} = \mathbb{R}^2$, so that $|z|^2 = \langle z, z\rangle$.

Lean names in the text are read relative to the namespace `ErdosSzekeres.SymReduction`: a name n stands for the first of `ErdosSzekeres.SymReduction.`n, `ErdosSzekeres.`n and n that exists. So `orbit_select` is `ErdosSzekeres.SymReduction.orbit_select`, `Review.` and `VariantChecks.` denote sub-namespaces of `ErdosSzekeres.SymReduction`, `Point` is `ErdosSzekeres.Point`, `Suk.` denotes `ErdosSzekeres.Suk`, and the names of Mathlib and those with the prefixes `Mir.` (the Lean package in `H3`) and `V2L.` (the Lean package in `V2-L`) are full names. Statement heads give full names.

Definition 7.1 (Squared distance, Euclidean isometry; Lean: `ErdosSzekeres.SymReduction.eDist2`, `SYMLEAN/SymmetricReduction.lean:265`, Lean: `ErdosSzekeres.SymReduction.IsEucIsometry`, `SYMLEAN/SymmetricReduction.lean:268`)**.** For $p, q \in \mathbb{R}^2$ put $\mathrm{eDist2}(p,q) = (p_0-q_0)^2 + (p_1-q_1)^2$. A map $g\colon \mathbb{R}^2 \to \mathbb{R}^2$ is a *Euclidean isometry* (Lean `IsEucIsometry g`) if

$$\mathrm{eDist2}(g(p), g(q)) = \mathrm{eDist2}(p,q) \qquad \text{for all } p, q \in \mathbb{R}^2.$$

No further property of g is assumed: g is not assumed continuous, surjective or affine, but it is defined on the whole plane. That g, read in complex coordinates, has the form $z \mapsto az + b$ or $z \mapsto a\bar{z} + b$ with $|a| = 1$ follows from Lemma 7.20(b) and Theorem 7.21(a).

Remark 7.2 (Why an explicit Euclidean distance). Mathlib equips the type `Point` of functions $\{0,1\} \to \mathbb{R}$ with the product (sup) metric $\mathrm{dist}(p,q) = \max(|p_0-q_0|, |p_1-q_1|)$; for instance $\mathrm{dist}((0,0),(3,4)) = 4$. Mathlib's predicate `Isometry` for maps of `Point` therefore means preservation of the sup distance, which a rotation need not have. For example, the rotation ρ by $2\pi/3$ about a point c fixes c and maps $c + (1,0)$, at sup distance 1 from c, to $c + (-\frac{1}{2}, \frac{\sqrt{3}}{2})$, at sup distance $\frac{\sqrt{3}}{2}$ from c. So ρ does not preserve the sup distance, and a theorem stated with Mathlib's `Isometry` would not cover the rotations used in `Threefold33`. This is why the file states isometries with the explicit squared Euclidean distance eDist2. General position, convex position and `HasConvexNGon` are affine notions (Mathlib `Collinear` and `ConvexIndependent`) and involve no metric. Lean checks of such sup-metric facts are in the review file `SYMLEAN/review/SupMetric.lean`, which is outside the scope of this section.

Definition 7.3 (Invariance; Lean: `ErdosSzekeres.SymReduction.Invariant`, `SYMLEAN/SymmetricReduction.lean:271`)**.** For $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ and a finite set S, $\mathrm{Invariant}(f, S)$ means $f(p) \in S$ for every $p \in S$.

This is the notion of Definition 1.10: f maps S *into* S; equality $f(S) = S$ is not required. (It follows when f is injective, because S is finite, but it is never used.)

Definition 7.4 (Complex coordinates; Lean: `ErdosSzekeres.SymReduction.toC`, `SYMLEAN/SymmetricReduction.lean:274`, Lean: `ErdosSzekeres.SymReduction.ofC`, `SYMLEAN/SymmetricReduction.lean:277`)**.** For $p \in \mathbb{R}^2$ let $\mathrm{toC}(p) = p_0 + ip_1 \in \mathbb{C}$, and for $z \in \mathbb{C}$ let $\mathrm{ofC}(z) = (\mathrm{Re}\, z, \mathrm{Im}\, z) \in \mathbb{R}^2$. In proofs we abbreviate $\hat{p} = \mathrm{toC}(p)$.

Lemma 7.5 (Coordinates; Lean: `ErdosSzekeres.SymReduction.toC_re`, `SYMLEAN/SymmetricReduction.lean:279`, Lean: `ErdosSzekeres.SymReduction.toC_im`, `SYMLEAN/SymmetricReduction.lean:280`,

Lean: `ErdosSzekeres.SymReduction.ofC_zero`, `SYMLEAN/SymmetricReduction.lean:281`, Lean: `ErdosSzekeres.SymReduction.ofC_one`, `SYMLEAN/SymmetricReduction.lean:282`, Lean: `ErdosSzekeres.SymReduction.toC_ofC`, `SYMLEAN/SymmetricReduction.lean:284`, Lean: `ErdosSzekeres.SymReduction.ofC_toC`, `SYMLEAN/SymmetricReduction.lean:287`, Lean: `ErdosSzekeres.SymReduction.toC_injective`, `SYMLEAN/SymmetricReduction.lean:291`, Lean: `ErdosSzekeres.SymReduction.eDist2_eq_normSq`, `SYMLEAN/SymmetricReduction.lean:294`, Lean: `ErdosSzekeres.SymReduction.eDist2_pos_of_ne`, `SYMLEAN/SymmetricReduction.lean:297`, Lean: `ErdosSzekeres.SymReduction.point_ext`, `SYMLEAN/SymmetricReduction.lean:502`). *Let $p, q \in \mathbb{R}^2$ and $z \in \mathbb{C}$.*

(a) (`toC_re`, `toC_im`, `ofC_zero`, `ofC_one`*)* $\operatorname{Re}\mathrm{toC}(p) = p_0$, $\operatorname{Im}\mathrm{toC}(p) = p_1$, $\mathrm{ofC}(z)_0 = \operatorname{Re} z$ *and* $\mathrm{ofC}(z)_1 = \operatorname{Im} z$.

(b) (`point_ext`*) If $p_0 = q_0$ and $p_1 = q_1$, then $p = q$.*

(c) (`toC_ofC`, `ofC_toC`, `toC_injective`*)* $\mathrm{toC}(\mathrm{ofC}(z)) = z$ *and* $\mathrm{ofC}(\mathrm{toC}(p)) = p$*; the map* toC *is injective.*

(d) (`eDist2_eq_normSq`*)* $\mathrm{eDist2}(p, q) = \mathrm{normSq}(\mathrm{toC}(p) - \mathrm{toC}(q))$.

(e) (`eDist2_pos_of_ne`*) If $p \neq q$, then $0 < \mathrm{eDist2}(p, q)$.*

Proof. (a) holds by the definitions. (b) A point is a function on $\{0, 1\}$, determined by its two values. (c) By (a), $\mathrm{toC}(\mathrm{ofC}(z)) = \operatorname{Re} z + i \operatorname{Im} z = z$, and $\mathrm{ofC}(\mathrm{toC}(p))$ has coordinates p_0, p_1, so it equals p by (b). If $\mathrm{toC}(p) = \mathrm{toC}(q)$, then $p = \mathrm{ofC}(\mathrm{toC}(p)) = \mathrm{ofC}(\mathrm{toC}(q)) = q$. (d) $\mathrm{toC}(p) - \mathrm{toC}(q) = (p_0 - q_0) + i(p_1 - q_1)$, whose normSq is $(p_0 - q_0)^2 + (p_1 - q_1)^2$. (e) If $p \neq q$, then $\hat{p} - \hat{q} \neq 0$ by (c), and a nonzero complex number has positive normSq; conclude by (d). □

Definition 7.6 (Rotations, half-turn, line reflection; Lean: `ErdosSzekeres.SymReduction.crot`, `SYMLEAN/SymmetricReduction.lean:349`, Lean: `ErdosSzekeres.SymReduction.rotAbout`, `SYMLEAN/SymmetricReduction.lean:400`, Lean: `ErdosSzekeres.SymReduction.halfTurn`, `SYMLEAN/SymmetricReduction.lean:413`, Lean: `ErdosSzekeres.SymReduction.reflLine`, `SYMLEAN/SymmetricReduction.lean:423`). Let $p \in \mathbb{R}^2$.

(a) For $c, w \in \mathbb{C}$ (Lean `crot c w`): $\mathrm{crot}_{c,w}(p) = \mathrm{ofC}\big(c + w(\mathrm{toC}(p) - c)\big)$.

(b) For $c \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$ (Lean `rotAbout`; the formula of Definition 1.12):

$$\mathrm{rotAbout}_{c,\theta}(p) = \big(c_0 + \cos\theta\,(p_0 - c_0) - \sin\theta\,(p_1 - c_1),\ c_1 + \sin\theta\,(p_0 - c_0) + \cos\theta\,(p_1 - c_1)\big).$$

(c) For $c \in \mathbb{R}^2$ (Lean `halfTurn c`): $\mathrm{halfTurn}_c(p) = (2c_0 - p_0,\ 2c_1 - p_1)$.

(d) For $a, u \in \mathbb{R}^2$ (Lean `reflLine a u`): with $d = (p_0 - a_0)u_0 + (p_1 - a_1)u_1$,

$$\mathrm{reflLine}_{a,u}(p) = \big(a_0 + 2du_0 - (p_0 - a_0),\ a_1 + 2du_1 - (p_1 - a_1)\big).$$

For $|w| = 1$, $\mathrm{crot}_{c,w}$ is the rotation about the point $\mathrm{ofC}(c)$ through the angle θ with $w = e^{i\theta}$ (Lemma 7.18(a)), and $\mathrm{rotAbout}_{c,\theta}$ is the counterclockwise rotation by θ about c. For $u_0^2 + u_1^2 = 1$, $\mathrm{reflLine}_{a,u}$ is the reflection in the line $a + \mathbb{R}u$: by Lemma 7.19(a) it is $z \mapsto \hat{a} + \hat{u}^{\,2}\,\overline{(z - \hat{a})}$ in complex coordinates, and as $\hat{u}\,\overline{\hat{u}} = 1$, this map fixes $\hat{a} + t\hat{u}$ and sends $\hat{a} + it\hat{u}$ to $\hat{a} - it\hat{u}$ for every $t \in \mathbb{R}$. For $u = 0$ it is the half-turn about a. For every other u it is not even a Euclidean isometry: with $s = u_0^2 + u_1^2 \notin \{0, 1\}$, it fixes a and sends $a + u$ to $a + (2s - 1)u$, and $\mathrm{eDist2}(a + (2s - 1)u, a) = (2s - 1)^2 s \neq s = \mathrm{eDist2}(a + u, a)$. So every statement below that uses it as a reflection assumes $u_0^2 + u_1^2 = 1$. The definition has the same text as `Mir.reflLine` in `H3/MirrorEncoding/Line.lean:18`.

Definition 7.7 (The case classes; Lean: `ErdosSzekeres.SymReduction.HalfTurn34`, `SYMLEAN/SymmetricReduction.lean:467`, Lean: `ErdosSzekeres.SymReduction.Threefold33`, `SYMLEAN/SymmetricReduction.lean:472`, Lean: `ErdosSzekeres.SymReduction.Fivefold35`, `SYMLEAN/SymmetricReduction.lean:477`, Lean: `ErdosSzekeres.SymReduction.MirrorOneAxis33`, `SYMLEAN/SymmetricReduction.lean:482`, Lean: `ErdosSzekeres.SymReduction.MirrorNoAxis34`, `SYMLEAN/SymmetricReduction.lean:488`, Lean: `ErdosSzekeres.SymReduction.Fourfold32`, `SYMLEAN/SymmetricReduction.lean:496`). The following six propositions are defined (Lean: definitions of type `Prop`). In each, S ranges over all finite subsets of $\mathbb{R}^2$, and c, a, u range over all points of $\mathbb{R}^2$.

(1) `HalfTurn34`: for all S and c, if S is in general position, $|S| = 34$, $c \notin S$ and $\mathrm{Invariant}(\mathrm{halfTurn}_c, S)$, then `HasConvexNGon 7 S`.

(2) `Threefold33`: for all S and c, if S is in general position, $|S| = 33$, $c \notin S$ and $\mathrm{Invariant}(\mathrm{rotAbout}_{c,2\pi/3}, S)$, then `HasConvexNGon 7 S`.

(3) `Fivefold35`: for all S and c, if S is in general position, $|S| = 35$, $c \notin S$ and $\mathrm{Invariant}(\mathrm{rotAbout}_{c,2\pi/5}, S)$, then `HasConvexNGon 7 S`.

(4) `MirrorOneAxis33`: for all S, a and u, if $u_0^2 + u_1^2 = 1$, S is in general position, $|S| = 33$, Invariant(reflLine$_{a,u}, S$), and there is $x \in S$ with reflLine$_{a,u}(x) = x$ such that every $y \in S$ with reflLine$_{a,u}(y) = y$ equals x, then `HasConvexNGon 7 S`.
(5) `MirrorNoAxis34`: for all S, a and u, if $u_0^2 + u_1^2 = 1$, S is in general position, $|S| = 34$, Invariant(reflLine$_{a,u}, S$), and reflLine$_{a,u}(p) \neq p$ for every $p \in S$, then `HasConvexNGon 7 S`.
(6) `Fourfold32`: for all S and c, if S is in general position, $|S| = 32$, $c \notin S$ and Invariant(rotAbout$_{c,2\pi/4}, S$), then `HasConvexNGon 7 S`.

The angles are the real numbers $2\pi/3$, $2\pi/5$ and $2\pi/4$ (Lean `2 * Real.pi / 3` etc.); the rotations are counterclockwise. In `MirrorOneAxis33` the last hypothesis says that exactly one point of S is fixed by the reflection; in `MirrorNoAxis34` no point of S is fixed. For $u_0^2 + u_1^2 = 1$ the fixed points of reflLine$_{a,u}$ are exactly the points of the mirror $a + \mathbb{R}u$. Indeed, with d as in Definition 7.6(d), the equation reflLine$_{a,u}(p) = p$ says $2du_0 = 2(p_0 - a_0)$ and $2du_1 = 2(p_1 - a_1)$, so a fixed point is $p = a + du$; and for $p = a + tu$ one has $d = t(u_0^2 + u_1^2) = t$ and reflLine$_{a,u}(p) = a + 2tu - tu = p$.

These are definitions, not assertions. No class is proved in this section. Theorem 7.39 takes the first five as hypotheses, and Theorem 7.41 takes `Threefold33`, `Fourfold32`, `Fivefold35` and the two mirror classes. Remark 7.46 records which hypotheses of the classes are implied by the others.

Remark 7.8 (Where the project supplies the classes). None of the following is checked in this section; it follows `C2ENC/class_table.md`, §5 and §9 of `SYMLEAN/README.md`, §3 of `WRITEUP/SYMMETRIC.md` and the records named below. For no class is the step from a SAT certificate to the Lean class Lean-checked. On the certificate routes of `HalfTurn34`, `Threefold33`, `Fivefold35` and `Fourfold32`, and for the `SYMZ` certificates of the two mirror classes, that step rests on written proofs (the soundness of the encoders and the lemmas named below) and on the review of the class definitions in `SYMLEAN/README.md` §6, which compared each class with its certificate by reading. On the route through `H3` and `V2-L` to `MirrorOneAxis33`, the soundness of the encodings and the reduction of `Mir.MirrorExclusionLine` to CNF facts are Lean-checked. The cube covers in that reduction are checked by kernel evaluation (computer-assisted), with one exception: the cover of the top two levels of the hull-pair case split of sub-case (ii), where the point of S on the mirror lies in the convex hull of the other points (`WRITEUP/MIRROR.md` §4). That cover is itself given by $1 + 121$ of those CNF facts. What remains outside Lean on this route is described in Remark 7.45. The project trusts one proof checker, `cake_lpr`: where a certificate below is said to be accepted by both checkers, `cake_lpr` accepted it, and `lrat-check` is only a secondary check (`WRITEUP/SYMMETRIC.md` §3, revised on 6 November 2026). The trust statement in §9 of `SYMLEAN/README.md`, which this remark follows on other points, predates that revision and still lists both checkers as trusted.

- `HalfTurn34`: claimed in `C2FINAL` as computer-assisted. It rests on SAT certificates (`C2C34`: hull cover and 120 hexagonal hulls, proofs retained; `C2P`–`C2P5`: parallelogram subcases, most of which are verdicts of both checkers recorded at solve time for proofs that were then deleted; re-checks in `C2RECHECK`); on the written soundness proof of the encoder, `C2ENC/proof.md`; on the reductions A and B, which pair equivalent parallelogram cases and subcases (written proofs, reviewed in `C2P/review` and checked at the CNF level in `AUDIT3`); and on the bookkeeping of the subcase orbits, Lean-checked according to the record (`C2FINAL/README.md` §6), by kernel evaluation, and so computer-assisted (Theorem 11.98; that the listed subcases are refuted rests on the records). A table of the evidence is in `C2ENC/class_table.md` §1.
- `Threefold33`: claimed in the records as computer-assisted. It rests on `C3FINAL`, a composition of 116,883 pieces whose proofs were not retained, so that their verdicts are recorded ones (a random sample of 2,120 pieces was re-solved in `C3SPOT`, and the 204 indispensable σ-cube pieces were re-solved with retained proofs in `C3PIECES`); on the cover of the 910 roots, Lean-checked by kernel evaluation and so computer-assisted (Theorem 11.82; that each leaf names a record with recorded checker verdicts is checked outside Lean); on the written normal form `C3-S/soundness.md` and the written Lemma H6M (`C3H6M/soundness_h6m.md`, checked adversarially in `C3AUDIT2`); on the written exhaustiveness of the layer splits and σ-splits (Lemmas 1–4 and the cover theorem of `C3-S/soundness.md`, its version for the reduced cut in `C3H6M/soundness_h6m.md`, and rules 2 and 3 of `C3U2/sigma_split.md`); on the encoding `C3a`; and on the written centre-perturbation lemma `C2ENC/perturbation.md`. The perturbation lemma is needed because the class admits centres that lie on a line through two points of different orbits, which the encoder excludes.
- `Fivefold35`: claimed in the records as computer-assisted (`SYM5/README.md`). It rests on `SYM5`: two normal forms of one encoder (an angle form and a layer form; not independent encodings), each with a

retained proof that both checkers accepted; either proof would suffice for the class without the other, given the written proofs listed next. It rests further on written proofs of the encoder's soundness and of the two normal forms, and on the same perturbation lemma. An earlier computation reported by B. Dumitru (arXiv:2512.24061, §7), which has priority, also implies the fivefold bound; no proof files of that run are public (`SYM5/README.md`).

- `MirrorOneAxis33` and `MirrorNoAxis34`: Lean-checked in `SYMLEAN/mirror` (`mirrorOneAxis33_proved`, `mirrorNoAxis34_proved`), with no SAT input, from the repository's formalization of Baek's theorem (J. Baek, arXiv:2206.04260); written proofs are in Theorem 10.39. The file `SYMLEAN/mirror/ReductionThreeClasses.lean` does not import `SYMLEAN/SymmetricReduction.lean`. Its lines 38–1084 are byte-identical to lines 12–1058 of `SYMLEAN/SymmetricReduction.lean`, so the classes defined in `SYMLEAN/mirror/ReductionThreeClasses.lean` are copies of those of Definition 7.7, with the same full names and the same text. Both classes are also supplied by SAT certificates in `SYMZ` (Theorems 2a and 2b of `WRITEUP/SYMMETRIC.md`, claimed in the records as computer-assisted), and `MirrorOneAxis33` also, conditionally, by the Lean chain of `H3` and `V2-L` (Remark 7.45).
- `Fourfold32` (among the statements of this section, a hypothesis only of Theorem 7.41 and of Lemma 7.37, which its proof uses): claimed in `C4CUT`, which states it without a rigor label. It rests on a final certificate accepted by both checkers, on six cuts checked by exact computation and on a written proof of the encoder's soundness. The statement is due to Subercaseaux, Mackey, Qian and Heule (arXiv:2506.00224, §6.2).

7.2. **Finite dynamics.**

Lemma 7.9 (Iterates stay in an invariant set; Lean: `ErdosSzekeres.SymReduction.iterate_mem`, `SYMLEAN/SymmetricReduction.lean:69`, Lean: `ErdosSzekeres.SymReduction.Invariant.iterate`, `SYMLEAN/SymmetricReduction.lean:309`)**.** *Iterates of a map stay in a set that the map sends into itself.*

(a) (`iterate_mem`*) Let X be a set, $f\colon X \to X$, and $S \subseteq X$ finite with $f(y) \in S$ for every $y \in S$. If $x \in S$, then $f^{[i]}(x) \in S$ for every $i \in \mathbb{N}$.*

(b) (`Invariant.iterate`*) If $f\colon \mathbb{R}^2 \to \mathbb{R}^2$, $S \subseteq \mathbb{R}^2$ is finite and* Invariant(f, S)*, then* Invariant$(f^{[k]}, S)$ *for every $k \in \mathbb{N}$.*

Proof. (a) Induction on i: $f^{[0]}(x) = x \in S$, and $f^{[i+1]}(x) = f(f^{[i]}(x)) \in S$ because $f^{[i]}(x) \in S$. (b) is (a) applied to each $x \in S$. □

Lemma 7.10 (Pigeonhole for iterates; written step)**.** *Let $S \subseteq \mathbb{R}^2$ be finite, $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ with* Invariant(f, S)*, and $q \in S$. Then there are $i < j$ in $\mathbb{N}$ with $f^{[i]}(q) = f^{[j]}(q)$.*

Proof. By Lemma 7.9, the $|S| + 1$ points $f^{[k]}(q)$, $0 \le k \le |S|$, lie in S. By the pigeonhole principle two of them with different indices coincide; call the smaller index i and the larger j. (In Lean this is the local statement `pigeon` in the proof of `isometry_structure`, obtained from Mathlib's `Set.Finite.exists_lt_map_eq_of_forall_mem`.) □

Definition 7.11 (Minimal period; written step)**.** Let X be a set, $f\colon X \to X$ and $x \in X$. If $f^{[n]}(x) = x$ for some $n > 0$, then minimalPeriod(f, x) is the least such n; otherwise minimalPeriod$(f, x) = 0$. (This is Mathlib's `Function.minimalPeriod`.)

Lemma 7.12 (Minimal periods; written step)**.** *Let $f\colon X \to X$, $x \in X$ and $d =$* minimalPeriod(f, x)*.*

(MP1) If $f^{[n]}(x) = x$ for some $n \in \mathbb{N}$, then $d \mid n$. (Mathlib `Function.IsPeriodicPt.minimalPeriod_dvd`*.)*

(MP2) $d = 1$ if and only if $f(x) = x$. (Mathlib `Function.minimalPeriod_eq_one_iff_isFixedPt`*.)*

(MP3) $f^{[d]}(x) = x$. (Mathlib `Function.iterate_minimalPeriod`*.)*

(MP4) If $i < j < d$, then $f^{[i]}(x) \ne f^{[j]}(x)$. (Mathlib `Function.iterate_injOn_Iio_minimalPeriod`*.)*

Proof. (MP3) If some $n > 0$ has $f^{[n]}(x) = x$, this holds for the least one, d. Otherwise $d = 0$ and $f^{[0]}(x) = x$.

(MP1) If $n = 0$ there is nothing to prove. Let $n > 0$. Then $d > 0$ is the least positive m with $f^{[m]}(x) = x$. By induction on t, $f^{[td]}(x) = x$ for all $t \in \mathbb{N}$. Write $n = td + r$ with $0 \le r < d$. Then $x = f^{[n]}(x) = f^{[r]}(f^{[td]}(x)) = f^{[r]}(x)$. If $r > 0$, this contradicts the minimality of d; so $r = 0$ and $d \mid n$.

(MP2) If $f(x) = x$, then 1 is the least positive integer and $f^{[1]}(x) = x$, so $d = 1$. Conversely, if $d = 1 \ne 0$, then by definition $f^{[1]}(x) = x$.

(MP4) Suppose $f^{[i]}(x) = f^{[j]}(x)$ with $i < j < d$. Then $d > 0$, and applying $f^{[d-j]}$ to both sides gives $f^{[d-j+i]}(x) = f^{[d]}(x) = x$ by (MP3). Since $j < d$ and $i < j$, we have $0 < d - j + i < d$, which contradicts the minimality of d. □

Definition 7.13 (Forward orbit segment; Lean: `ErdosSzekeres.SymReduction.orbitFS`, `SYMLEAN/SymmetricReduction.lean:65`)**.** For $f\colon X \to X$, $p \in \mathbb{N}$ and $x \in X$, let $\mathrm{orbitFS}(f,p,x) = \{f^{[i]}(x) : 0 \le i < p\}$, a finite set (Lean: the image of $\{0,\dots,p-1\}$ under $i \mapsto f^{[i]}(x)$; Lean needs decidable equality on X to form it).

Lemma 7.14 (Orbit segments; Lean: `ErdosSzekeres.SymReduction.card_orbitFS_of_minimalPeriod`, `SYMLEAN/SymmetricReduction.lean:111`, Lean: `ErdosSzekeres.SymReduction.mem_orbitFS_self`, `SYMLEAN/SymmetricReduction.lean:90`, Lean: `ErdosSzekeres.SymReduction.orbitFS_subset`, `SYMLEAN/SymmetricReduction.lean:93`, Lean: `ErdosSzekeres.SymReduction.orbitFS_invariant`, `SYMLEAN/SymmetricReduction.lean:99`, Lean: `ErdosSzekeres.SymReduction.card_orbitFS`, `SYMLEAN/SymmetricReduction.lean:76`)**.** *Let X be a set, $f\colon X \to X$, $p \in \mathbb{N}$ and $x \in X$, and write $O = \mathrm{orbitFS}(f,p,x)$.*

(a) (`mem_orbitFS_self`) If $0 < p$, then $x \in O$.
(b) (`orbitFS_subset`) If $S \subseteq X$ is finite, $f(y) \in S$ for every $y \in S$, and $x \in S$, then $O \subseteq S$.
(c) (`orbitFS_invariant`) If $0 < p$ and $f^{[p]}(x) = x$, then $f(y) \in O$ for every $y \in O$.
(d) (`card_orbitFS`) If p is prime, $f^{[p]}(x) = x$ and $f(x) \neq x$, then $|O| = p$.
(e) (`card_orbitFS_of_minimalPeriod`) If $\mathrm{minimalPeriod}(f,x) = p$, then $|O| = p$.

Proof. (a) $x = f^{[0]}(x)$ and $0 < p$. (b) Each $f^{[i]}(x)$ lies in S by Lemma 7.9(a). (c) Let $y = f^{[i]}(x)$ with $i < p$. Then $f(y) = f^{[i+1]}(x)$. If $i + 1 < p$, this lies in O. Otherwise $i + 1 = p$ and $f(y) = f^{[p]}(x) = x \in O$ by (a). (e) By (MP4), $i \mapsto f^{[i]}(x)$ is injective on $\{0,\dots,p-1\}$, so O has p elements. (This also holds for $p = 0$, when $O = \varnothing$.) (d) By (MP1), $d = \mathrm{minimalPeriod}(f,x)$ divides the prime p, so $d = 1$ or $d = p$. By (MP2), $d \neq 1$ because $f(x) \neq x$. Hence $d = p$, and (e) applies. □

The next statement contains the induction shared by the two orbit selection lemmas. The Lean file carries out the same induction twice, once in each lemma; we do it once.

Lemma 7.15 (Orbit selection, common form; written step)**.** *Let X be a set, $f\colon X \to X$ and $p \in \mathbb{N}$ with $0 < p$. Let $S \subseteq X$ be finite, and assume: $f(y) \in S$ for every $y \in S$; f is injective on S; and every $y \in S$ satisfies $f^{[p]}(y) = y$ and $|\mathrm{orbitFS}(f,p,y)| = p$. Then $p \mid |S|$, and for every $k \in \mathbb{N}$ with $kp \le |S|$ there is $T \subseteq S$ with $f(y) \in T$ for every $y \in T$ and $|T| = kp$.*

Proof. Strong induction on $|S|$. The hypotheses on single points pass to subsets of S, and so does injectivity.

If $S = \varnothing$, then $p \mid 0$; and $kp \le 0$ forces $kp = 0$, so $T = \varnothing$ works.

Let $S \neq \varnothing$ and pick $x \in S$. Put $O = \mathrm{orbitFS}(f,p,x)$. By Lemma 7.14 and the hypotheses, $O \subseteq S$, $|O| = p$, $x \in O$, and $f(y) \in O$ for every $y \in O$. Let $S' = S \setminus O$; then $|S'| = |S| - p < |S|$.

S' is mapped into itself. Let $y \in S'$. Then $f(y) \in S$. Suppose $f(y) \in O$, say $f(y) = f^{[i]}(x)$ with $i < p$. If $i = 0$, then $f(y) = x = f^{[p]}(x) = f(f^{[p-1]}(x))$ (here $p \ge 1$). Both y and $f^{[p-1]}(x)$ lie in S (Lemma 7.9(a)), so injectivity on S gives $y = f^{[p-1]}(x) \in O$. If $i \ge 1$, then $f(y) = f(f^{[i-1]}(x))$ and injectivity gives $y = f^{[i-1]}(x) \in O$. Both contradict $y \notin O$. Hence $f(y) \in S'$.

So S' satisfies all the hypotheses, and by induction $p \mid |S'|$, and S' has f-invariant subsets of every size $k'p \le |S'|$. Since $|S| = |S'| + p$, we get $p \mid |S|$. Let $kp \le |S|$. If $k = 0$, take $T = \varnothing$. If $k = k' + 1$, then $k'p \le |S| - p = |S'|$. Take an f-invariant $T' \subseteq S'$ with $|T'| = k'p$, and put $T = T' \cup O \subseteq S$. The union of two sets that f maps into themselves is mapped into itself, and $T' \cap O = \varnothing$ because $T' \subseteq S \setminus O$. So $|T| = k'p + p = kp$. □

Lemma 7.16 (Orbit selection; Lean: `ErdosSzekeres.SymReduction.orbit_select_minimalPeriod`, `SYMLEAN/SymmetricReduction.lean:123`, Lean: `ErdosSzekeres.SymReduction.orbit_select`, `SYMLEAN/SymmetricReduction.lean:194`)**.** *Let X be a set and $f\colon X \to X$.*

(a) (`orbit_select_minimalPeriod`) Let $p \in \mathbb{N}$ with $0 < p$. For every finite $S \subseteq X$: if $f(y) \in S$ for every $y \in S$, f is injective on S, and $\mathrm{minimalPeriod}(f,y) = p$ for every $y \in S$, then $p \mid |S|$, and for every $k \in \mathbb{N}$ with $kp \le |S|$ there is $T \subseteq S$ with $f(y) \in T$ for every $y \in T$ and $|T| = kp$.
(b) (`orbit_select`) Let p be a prime. For every finite $S \subseteq X$: if $f(y) \in S$ for every $y \in S$, f is injective on S, $f^{[p]}(y) = y$ for every $y \in S$, and $f(y) \neq y$ for every $y \in S$, then $p \mid |S|$, and for every $k \in \mathbb{N}$ with $kp \le |S|$ there is $T \subseteq S$ with $f(y) \in T$ for every $y \in T$ and $|T| = kp$.

Proof. Both follow from Lemma 7.15; we check its hypotheses. (a) For $y \in S$, $f^{[p]}(y) = y$ by (MP3) and $|\mathrm{orbitFS}(f,p,y)| = p$ by Lemma 7.14(e). (b) $p > 0$ because p is prime, and $|\mathrm{orbitFS}(f,p,y)| = p$ for $y \in S$ by Lemma 7.14(d). □

In (b) the injectivity hypothesis is redundant: if $f(y) = f(y')$ with $y, y' \in S$, then applying $f^{[p-1]}$ gives $y = f^{[p]}(y) = f^{[p]}(y') = y'$. It is part of the Lean statement, and it is available in all applications.

7.3. **Rotations and reflections in coordinates.**

Lemma 7.17 (The maps crot; Lean: `ErdosSzekeres.SymReduction.toC_crot`, `SYMLEAN/SymmetricReduction.lean:351`, Lean: `ErdosSzekeres.SymReduction.crot_one`, `SYMLEAN/SymmetricReduction.lean:354`, Lean: `ErdosSzekeres.SymReduction.crot_comp`, `SYMLEAN/SymmetricReduction.lean:359`, Lean: `ErdosSzekeres.SymReduction.crot_iterate`, `SYMLEAN/SymmetricReduction.lean:364`, Lean: `ErdosSzekeres.SymReduction.crot_eq_self_iff`, `SYMLEAN/SymmetricReduction.lean:370`, Lean: `ErdosSzekeres.SymReduction.crot_injective`, `SYMLEAN/SymmetricReduction.lean:384`, Lean: `ErdosSzekeres.SymReduction.eDist2_crot_center`, `SYMLEAN/SymmetricReduction.lean:394`). *Let $c, w, v \in \mathbb{C}$ and $p \in \mathbb{R}^2$.*

(a) (`toC_crot`) $\mathrm{toC}(\mathrm{crot}_{c,w}(p)) = c + w(\mathrm{toC}(p) - c)$.
(b) (`crot_one`) $\mathrm{crot}_{c,1} = \mathrm{id}$.
(c) (`crot_comp`) $\mathrm{crot}_{c,w}(\mathrm{crot}_{c,v}(p)) = \mathrm{crot}_{c,wv}(p)$.
(d) (`crot_iterate`) $(\mathrm{crot}_{c,w})^{[k]} = \mathrm{crot}_{c,w^k}$ *for every* $k \in \mathbb{N}$.
(e) (`crot_eq_self_iff`) If $w \neq 1$, *then* $\mathrm{crot}_{c,w}(p) = p$ *if and only if* $\mathrm{toC}(p) = c$.
(f) (`crot_injective`) If $w \neq 0$, *then* $\mathrm{crot}_{c,w}$ *is injective.*
(g) (`eDist2_crot_center`) If $\mathrm{normSq}(w) = 1$, *then* $\mathrm{eDist2}(\mathrm{crot}_{c,w}(p), \mathrm{ofC}(c)) = \mathrm{eDist2}(p, \mathrm{ofC}(c))$.

Proof. Write $z = \hat{p}$. Equalities of points are proved by applying the injective map toC (Lemma 7.5(c)). (a) is $\mathrm{toC}(\mathrm{ofC}(\zeta)) = \zeta$. (b) $c + 1 \cdot (z - c) = z$. (c) By (a), $c + w\big((c + v(z-c)) - c\big) = c + wv(z-c)$. (d) Induction on k. For $k = 0$ both sides are the identity, by (b). If it holds for k, then $(\mathrm{crot}_{c,w})^{[k+1]}(p) = \mathrm{crot}_{c,w}(\mathrm{crot}_{c,w^k}(p)) = \mathrm{crot}_{c,w\cdot w^k}(p) = \mathrm{crot}_{c,w^{k+1}}(p)$ by (c). (e) By (a), $\mathrm{toC}(\mathrm{crot}_{c,w}(p)) - z = (w-1)(z-c)$. For $w \neq 1$ this vanishes if and only if $z = c$. (f) If $\mathrm{crot}_{c,w}(p) = \mathrm{crot}_{c,w}(q)$, then by (a) $w(\hat{p} - \hat{q}) = 0$, so $\hat{p} = \hat{q}$ and $p = q$. (g) By Lemma 7.5(c),(d) and (a), the left side is $\mathrm{normSq}(c + w(z-c) - c) = \mathrm{normSq}(w)\,\mathrm{normSq}(z-c) = \mathrm{normSq}(z-c)$, which is the right side. □

Lemma 7.18 (Rotations and the half-turn as crot; Lean: `ErdosSzekeres.SymReduction.rotAbout_eq_crot`, `SYMLEAN/SymmetricReduction.lean:404`, Lean: `ErdosSzekeres.SymReduction.halfTurn_eq_crot`, `SYMLEAN/SymmetricReduction.lean:415`). *For $c \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$:*

(a) (`rotAbout_eq_crot`) $\mathrm{rotAbout}_{c,\theta} = \mathrm{crot}_{\mathrm{toC}(c),\,\exp(\theta i)}$ *(equality of maps of $\mathbb{R}^2$);*
(b) (`halfTurn_eq_crot`) $\mathrm{halfTurn}_c = \mathrm{crot}_{\mathrm{toC}(c),\,-1}$.

Proof. Let $p \in \mathbb{R}^2$, $x = p_0 - c_0$ and $y = p_1 - c_1$, so that $\hat{p} - \hat{c} = x + iy$. (a) By Euler's formula $\exp(\theta i) = \cos\theta + i\sin\theta$, hence $\exp(\theta i)(x+iy) = (x\cos\theta - y\sin\theta) + i(x\sin\theta + y\cos\theta)$. Adding $\hat{c} = c_0 + ic_1$ gives $\mathrm{toC}(\mathrm{rotAbout}_{c,\theta}(p))$; conclude by Lemma 7.17(a) and the injectivity of toC. (b) $\hat{c} + (-1)(\hat{p} - \hat{c}) = 2\hat{c} - \hat{p} = \mathrm{toC}(\mathrm{halfTurn}_c(p))$. □

Lemma 7.19 (Line reflections; Lean: `ErdosSzekeres.SymReduction.toC_reflLine`, `SYMLEAN/SymmetricReduction.lean:427`, Lean: `ErdosSzekeres.SymReduction.reflLine_involutive`, `SYMLEAN/SymmetricReduction.lean:435`, Lean: `ErdosSzekeres.SymReduction.cross_of_reflLine_fixed`, `SYMLEAN/SymmetricReduction.lean:447`, Lean: `ErdosSzekeres.SymReduction.orient_eq_zero_of_cross`, `SYMLEAN/SymmetricReduction.lean:455`). *Let $a, u, p, q, r \in \mathbb{R}^2$.*

(a) (`toC_reflLine`) If $u_0^2 + u_1^2 = 1$, *then* $\mathrm{toC}(\mathrm{reflLine}_{a,u}(p)) = \hat{a} + \hat{u}^{\,2}\,\overline{(\hat{p} - \hat{a})}$.
(b) (`reflLine_involutive`) If $u_0^2 + u_1^2 = 1$, *then* $\mathrm{reflLine}_{a,u}(\mathrm{reflLine}_{a,u}(p)) = p$.
(c) (`cross_of_reflLine_fixed`) If $\mathrm{reflLine}_{a,u}(p) = p$, *then* $(p_0 - a_0)u_1 - (p_1 - a_1)u_0 = 0$. *(No hypothesis on u.)*
(d) (`orient_eq_zero_of_cross`) If $u_0^2 + u_1^2 = 1$ *and* $(z_0 - a_0)u_1 - (z_1 - a_1)u_0 = 0$ *for each* $z \in \{p, q, r\}$, *then* $\mathrm{orient}(p,q,r) = 0$.

Proof. Let $x = p_0 - a_0$, $y = p_1 - a_1$ and $d = xu_0 + yu_1$, so that $\mathrm{reflLine}_{a,u}(p) - a = (2du_0 - x,\ 2du_1 - y)$.

(a) Expanding, $\hat{u}^{\,2}\,\overline{(x+iy)} = (u_0^2 - u_1^2 + 2iu_0u_1)(x - iy) = \big((u_0^2 - u_1^2)x + 2u_0u_1y\big) + i\big(2u_0u_1x - (u_0^2 - u_1^2)y\big)$. On the other hand $2du_0 - x = (2u_0^2 - 1)x + 2u_0u_1y$ and $2du_1 - y = 2u_0u_1x + (2u_1^2 - 1)y$. Since $u_0^2 + u_1^2 = 1$, we

have $2u_0^2-1=u_0^2-u_1^2$ and $2u_1^2-1=-(u_0^2-u_1^2)$, so the two expressions agree. (In other words, the real and imaginary parts of the difference are $x(u_0^2+u_1^2-1)$ and $y(u_0^2+u_1^2-1)$.)

(b) Put $\zeta=\hat{p}-\hat{a}$; note $\hat{u}\,\overline{\hat{u}}=u_0^2+u_1^2=1$. By (a), applied twice, $\mathrm{toC}(\mathrm{reflLine}_{a,u}(\mathrm{reflLine}_{a,u}(p)))-\hat{a}=\hat{u}^{\,2}\,\overline{\hat{u}}^{\,2}\,\overline{\zeta}=(\hat{u}\,\overline{\hat{u}})^2\zeta=\zeta$. Hence the two points have the same toC, and they are equal.

(c) The fixed-point equations read $a_0+2du_0-x=p_0=a_0+x$ and $a_1+2du_1-y=a_1+y$, that is, $du_0=x$ and $du_1=y$. Hence $xu_1-yu_0=du_0u_1-du_1u_0=0$.

(d) Write $v\times w=v_0w_1-v_1w_0$ and $v\cdot w=v_0w_0+v_1w_1$ for $v,w\in\mathbb{R}^2$, so that $\mathrm{orient}(p,q,r)=(q-p)\times(r-p)$. For all $v,w,u\in\mathbb{R}^2$ one has the identity

$$(v\times w)(u_0^2+u_1^2)=(w\cdot u)(v\times u)-(v\cdot u)(w\times u),$$

checked by expanding both sides. Take $v=q-p$ and $w=r-p$. The hypotheses say $(z-a)\times u=0$ for $z=p,q,r$, so $v\times u=(q-a)\times u-(p-a)\times u=0$, and likewise $w\times u=0$. Hence $\mathrm{orient}(p,q,r)\cdot 1=0$. □

7.4. **Isometries of the plane.**

Lemma 7.20 (Isometries in coordinates; Lean: `ErdosSzekeres.SymReduction.IsEucIsometry.injective`, `SYMLEAN/SymmetricReduction.lean:301`, Lean: `ErdosSzekeres.SymReduction.isometry_toC_conj`, `SYMLEAN/SymmetricReduction.lean:313`). *Let $g\colon\mathbb{R}^2\to\mathbb{R}^2$ be a Euclidean isometry.*

(a) (`IsEucIsometry.injective`) g is injective.

(b) (`isometry_toC_conj`) The map $G\colon\mathbb{C}\to\mathbb{C}$, $G(z)=\mathrm{toC}(g(\mathrm{ofC}(z)))$, is an isometry of $\mathbb{C}$: $|G(z)-G(w)|=|z-w|$ for all $z,w\in\mathbb{C}$. (Lean: Mathlib's `Isometry G`; for the metric space $\mathbb{C}$, whose distance is $|z-w|$, this is equivalent to the displayed equation.)

Proof. (a) If $g(p)=g(q)$, then $\mathrm{eDist2}(p,q)=\mathrm{eDist2}(g(p),g(q))=0$, so $p=q$ by Lemma 7.5(e). (b) By Lemma 7.5(c),(d),

$$|G(z)-G(w)|^2=\mathrm{eDist2}\big(g(\mathrm{ofC}\,z),g(\mathrm{ofC}\,w)\big)=\mathrm{eDist2}(\mathrm{ofC}\,z,\mathrm{ofC}\,w)=|z-w|^2,$$

and both moduli are nonnegative. □

Theorem 7.21 (Isometries of the plane; Lean: `ErdosSzekeres.SymReduction.isometry_complex_cases`, `SYMLEAN/SymmetricReduction.lean:321`, Lean: `ErdosSzekeres.SymReduction.isometry_structure`, `SYMLEAN/SymmetricReduction.lean:869`). *Part (a) describes the isometries of $\mathbb{C}$; part (b) the Euclidean isometries of the plane that map a finite set with at least two points into itself.*

(a) (`isometry_complex_cases`) Let $G\colon\mathbb{C}\to\mathbb{C}$ be an isometry: $|G(z)-G(w)|=|z-w|$ for all $z,w\in\mathbb{C}$. Then there are $a,b\in\mathbb{C}$ with $|a|=1$ such that either $G(z)=az+b$ for all $z\in\mathbb{C}$, or $G(z)=a\bar{z}+b$ for all $z\in\mathbb{C}$.

(b) (`isometry_structure`) Let $S\subseteq\mathbb{R}^2$ be finite with $1<|S|$, and let $g\colon\mathbb{R}^2\to\mathbb{R}^2$ be a Euclidean isometry with $g\neq\mathrm{id}$ (as maps of $\mathbb{R}^2$) and $\mathrm{Invariant}(g,S)$. Then one of the following holds:

(i) there are $c,a\in\mathbb{C}$ with $g=\mathrm{crot}_{c,a}$, $a\neq1$, and $a^m=1$ for some $m\in\mathbb{N}$ with $0<m$;

(ii) there are $A,u\in\mathbb{R}^2$ with $u_0^2+u_1^2=1$ and $g=\mathrm{reflLine}_{A,u}$.

Proof. (a). This proof differs from the Lean proof. The Lean proof applies Mathlib's construction of an affine isometry from an isometry between real normed affine spaces with strictly convex codomain (`Isometry.affineIsometryOfStrictConvexSpace`, a definition whose underlying map is the given isometry; it is a result of Mazur–Ulam type that does not need surjectivity), and then Mathlib's theorem describing the real-linear isometric equivalences of $\mathbb{C}$ (`linear_isometry_complex`). We give a self-contained argument.

Let $b=G(0)$ and $H(z)=G(z)-b$. Then $H(0)=0$, $|H(z)-H(w)|=|z-w|$, and so $|H(z)|=|H(z)-H(0)|=|z|$. For all z,w we have $|z-w|^2=|z|^2-2\langle z,w\rangle+|w|^2$, and the same for $H(z),H(w)$. Hence

$$2\langle H(z),H(w)\rangle=|H(z)|^2+|H(w)|^2-|H(z)-H(w)|^2=|z|^2+|w|^2-|z-w|^2=2\langle z,w\rangle,$$

so H preserves the inner product. Put $a=H(1)$ and $e=H(i)$. Then $|a|=|1|=1$, $|e|=1$ and $\langle e,a\rangle=\langle i,1\rangle=0$. Let $v=e\bar{a}$. Then $|v|=1$ and $\mathrm{Re}\,v=\langle e,a\rangle=0$, so $v=i$ or $v=-i$; and $e=e\bar{a}a=va$, so $e=ia$ or $e=-ia$.

Next, every $\xi\in\mathbb{C}$ satisfies $\xi=\langle\xi,a\rangle a+\langle\xi,e\rangle e$. Indeed, write $\bar{a}\xi=\alpha+i\beta$ with $\alpha,\beta\in\mathbb{R}$; then $\xi=a\bar{a}\xi=\alpha a+\beta(ia)$, $\langle\xi,a\rangle=\mathrm{Re}(\xi\bar{a})=\alpha$ and $\langle\xi,ia\rangle=\mathrm{Re}(-i\xi\bar{a})=\beta$; and $\langle\xi,e\rangle e=\langle\xi,ia\rangle\,ia$ for both choices $e=\pm ia$.

Now let $z=x+iy$ with $x,y\in\mathbb{R}$. Since H preserves the inner product, $\langle H(z),a\rangle=\langle H(z),H(1)\rangle=\langle z,1\rangle=x$ and $\langle H(z),e\rangle=\langle z,i\rangle=y$. Hence $H(z)=xa+ye$. If $e=ia$, this is $a(x+iy)=az$; if $e=-ia$, it is $a(x-iy)=a\bar{z}$. The choice between the two cases does not depend on z. So $G(z)=az+b$ for all z, or $G(z)=a\bar{z}+b$ for all z, and $|a|=1$.

(b). This part follows the Lean proof. Let $G(z) = \mathrm{toC}(g(\mathrm{ofC}\, z))$. By Lemma 7.20(b) and part (a) there are $a, b \in \mathbb{C}$ with $|a| = 1$ such that either (R) $G(z) = az + b$ for all z, or (C) $G(z) = a\bar{z} + b$ for all z. Since $\mathrm{ofC}(\hat{p}) = p$, we have $\mathrm{toC}(g(p)) = G(\hat{p})$ for every $p \in \mathbb{R}^2$. The set S is nonempty because $|S| > 1$.

Case (R), $a = 1$. Then $\mathrm{toC}(g(p)) = \hat{p} + b$ for all p, and by induction $\mathrm{toC}(g^{[k]}(q)) = \hat{q} + kb$ for all $k \in \mathbb{N}$ and $q \in \mathbb{R}^2$. Take $q \in S$. By Lemma 7.10 there are $i < j$ with $g^{[i]}(q) = g^{[j]}(q)$, so $\hat{q} + ib = \hat{q} + jb$ and $(j - i)b = 0$. Since $j - i$ is a nonzero integer, $b = 0$. Then $\mathrm{toC}(g(p)) = \hat{p}$ for all p, so $g = \mathrm{id}$ by the injectivity of toC. This contradicts $g \neq \mathrm{id}$; so this case does not occur.

Case (R), $a \neq 1$. Put $c = b/(1 - a)$. For every p, $c + a(\hat{p} - c) = a\hat{p} + (1 - a)c = a\hat{p} + b = \mathrm{toC}(g(p))$, so $g = \mathrm{crot}_{c,a}$ by Lemma 7.17(a). Also $a \neq 0$ because $|a| = 1$. Since $|S| \geq 2$, some $q \in S$ differs from $\mathrm{ofC}(c)$. By Lemma 7.10, applied to $g = \mathrm{crot}_{c,a}$, there are $i < j$ with $(\mathrm{crot}_{c,a})^{[i]}(q) = (\mathrm{crot}_{c,a})^{[j]}(q)$. By Lemma 7.17(d),(a), $c + a^i(\hat{q} - c) = c + a^j(\hat{q} - c)$. Here $\hat{q} - c \neq 0$, since otherwise $q = \mathrm{ofC}(\hat{q}) = \mathrm{ofC}(c)$. Hence $a^i = a^j = a^i a^{j-i}$, and as $a^i \neq 0$, $a^{j-i} = 1$. With $m = j - i > 0$ we obtain alternative (i).

Case (C). We prove alternative (ii); the hypothesis $g \neq \mathrm{id}$ is not needed here. Since $|a| = 1$, $a\bar{a} = 1$. For every p,

$$\mathrm{toC}(g(g(p))) = a\,\overline{(a\bar{\hat{p}} + b)} + b = a\bar{a}\hat{p} + a\bar{b} + b = \hat{p} + t, \qquad t := a\bar{b} + b.$$

By induction, $\mathrm{toC}((g \circ g)^{[k]}(p)) = \hat{p} + kt$ for all k and p. The map $g \circ g$ maps S into S. Take $q \in S$; Lemma 7.10 applied to $g \circ g$ gives $i < j$ with $(j - i)t = 0$, hence $t = 0$, that is, $a\bar{b} = -b$. Choose $u \in \mathbb{C}$ with $u^2 = a$ (every complex number has a square root; the Lean proof takes the principal value $a^{1/2}$). Then $|u|^2 = |a| = 1$, so the point $U = \mathrm{ofC}(u)$ satisfies $U_0^2 + U_1^2 = 1$. Let $A = \mathrm{ofC}(b/2)$. By Lemma 7.19(a), for every p,

$$\mathrm{toC}(\mathrm{reflLine}_{A,U}(p)) = \tfrac{b}{2} + u^2\,\overline{\left(\hat{p} - \tfrac{b}{2}\right)} = a\bar{\hat{p}} + \tfrac{b}{2} - \tfrac{a\bar{b}}{2} = a\bar{\hat{p}} + \tfrac{b}{2} + \tfrac{b}{2} = a\bar{\hat{p}} + b = \mathrm{toC}(g(p)).$$

Hence $g = \mathrm{reflLine}_{A,U}$, which is alternative (ii). □

In (b), the pigeonhole steps are where finiteness enters: they exclude translations ($a = 1$) and glide reflections ($t \neq 0$), and they make a a root of unity.

7.5. **Elementary geometric lemmas.**

Lemma 7.22 (Strict supporting lines; Lean: `ErdosSzekeres.Suk.convexPosition_of_strict_supports`, `ErdosSzekeres/Suk/SupportingLines.lean:28`)**.** *Let $s \subseteq \mathbb{R}^2$ be finite. If for every $p \in s$ there is $q \in \mathbb{R}^2$ with* $\mathrm{orient}(p, q, r) > 0$ *for every $r \in s$ with $r \neq p$, then s is in convex position.*

Proof. This is the lemma titled "Strict supporting lines" in the foundations section, the same declaration of the main library; the short proof is repeated here because Lemma 7.23 uses it. By Lemma 1.22 it suffices to show $p \notin \mathrm{conv}(s \setminus \{p\})$ for every $p \in s$. Fix p and a point q as in the hypothesis, and let $\varphi(x) = \mathrm{orient}(p, q, x) = (q_0 - p_0)(x_1 - p_1) - (q_1 - p_1)(x_0 - p_0)$. This is an affine function of x: $\varphi(\lambda x + \mu y) = \lambda\varphi(x) + \mu\varphi(y)$ whenever $\lambda + \mu = 1$. So the open half-plane $H = \{x : \varphi(x) > 0\}$ is convex: for $x, y \in H$ and $\lambda, \mu \geq 0$ with $\lambda + \mu = 1$, both terms $\lambda\varphi(x)$ and $\mu\varphi(y)$ are ≥ 0 and at least one is > 0. By hypothesis $s \setminus \{p\} \subseteq H$, hence $\mathrm{conv}(s \setminus \{p\}) \subseteq H$. But $\varphi(p) = \mathrm{orient}(p, q, p) = 0$, so $p \notin H$, and therefore $p \notin \mathrm{conv}(s \setminus \{p\})$. □

Lemma 7.23 (Concyclic points; Lean: `ErdosSzekeres.SymReduction.convexPosition_of_concyclic`, `SYMLEAN/SymmetricReduction.lean:509`)**.** *Let $s \subseteq \mathbb{R}^2$ be finite, $c \in \mathbb{R}^2$ and $r \in \mathbb{R}$, and assume* $\mathrm{eDist2}(p, c) = r$ *for every $p \in s$. Then s is in convex position.*

No general position is assumed, and r is arbitrary (for $r = 0$ the set has at most one point).

Proof. We apply Lemma 7.22 with the tangent direction. For $p \in s$ let $q = \big(p_0 - (p_1 - c_1),\ p_1 + (p_0 - c_0)\big)$, that is, p plus the vector $p - c$ turned by $+90°$. For every $x \in \mathbb{R}^2$, expanding gives

$$\mathrm{orient}(p, q, x) = -\big[(p_0 - c_0)(x_0 - p_0) + (p_1 - c_1)(x_1 - p_1)\big] = \tfrac{1}{2}\big(\mathrm{eDist2}(x, p) + \mathrm{eDist2}(p, c) - \mathrm{eDist2}(x, c)\big).$$

(The second equality is $|x - c|^2 = |x - p|^2 + 2\langle x - p, p - c\rangle + |p - c|^2$.) For $x \in s$ we have $\mathrm{eDist2}(x, c) = r = \mathrm{eDist2}(p, c)$, so $\mathrm{orient}(p, q, x) = \frac{1}{2}\,\mathrm{eDist2}(x, p)$, which is positive when $x \neq p$ (Lemma 7.5(e)). Lemma 7.22 gives the claim. □

Corollary 7.24 (A concyclic subset gives convex polygons; Lean: `ErdosSzekeres.SymReduction.hasConvexNGon_of_concyclic`, `SYMLEAN/SymmetricReduction.lean:522`)**.** *Let $T \subseteq S$ be finite sets of points, $c \in \mathbb{R}^2$ and $r \in \mathbb{R}$ with* $\mathrm{eDist2}(p, c) = r$ *for every $p \in T$, and let $n \in \mathbb{N}$ with $n \leq |T|$. Then* ***`HasConvexNGon n S`****.*

Proof. T is in convex position by Lemma 7.23, hence `HasConvexNGon n T` since $n \leq |T|$, hence `HasConvexNGon n S` by monotonicity (Lemma 1.23). □

Lemma 7.25 (The centre of a half-turn; Lean: `ErdosSzekeres.SymReduction.center_not_mem_of_halfTurn`, `SYMLEAN/SymmetricReduction.lean:528`)**.** *Let S be in general position with $2 \le |S|$, let $c \in \mathbb{R}^2$, and assume* $\mathrm{Invariant}(\mathrm{halfTurn}_c, S)$. *Then $c \notin S$.*

Proof. Suppose $c \in S$. Since $|S| \ge 2$, there is $q \in S$ with $q \ne c$, and $q' = \mathrm{halfTurn}_c(q) = 2c - q$ lies in S. We have $q' \ne q$, since $2c - q = q$ would give $q = c$, and $q' \ne c$, since $2c - q = c$ would give $q = c$. So q, c, q' are pairwise distinct points of S. But

$$\mathrm{orient}(q, c, 2c - q) = (c_0 - q_0) \cdot 2(c_1 - q_1) - (c_1 - q_1) \cdot 2(c_0 - q_0) = 0,$$

contradicting general position (Lemma 1.20). □

7.6. **Roots of unity.**

Definition 7.26 (Primitive roots of unity, order; written step)**.** For $\zeta \in \mathbb{C}$ and $k \in \mathbb{N}$, ζ is a *primitive k-th root of unity* (Mathlib `IsPrimitiveRoot`) if $\zeta^k = 1$ and every $l \in \mathbb{N}$ with $\zeta^l = 1$ is a multiple of k. If $\zeta^m = 1$ for some $m > 0$, the *order* $\mathrm{ord}(\zeta)$ (Mathlib `orderOf`) is the least such m; otherwise it is 0.

Lemma 7.27 (Facts on roots of unity; written step)**.** *Let $\zeta, \xi \in \mathbb{C}$ and $n, k, d \in \mathbb{N}$.*

(P1) If $\zeta^m = 1$ for some $m > 0$, then $\mathrm{ord}(\zeta) > 0$, ζ is a primitive $\mathrm{ord}(\zeta)$-th root of unity, and $\mathrm{ord}(\zeta) = 1$ if and only if $\zeta = 1$. (Mathlib `IsPrimitiveRoot.orderOf`, `orderOf_eq_one_iff`.*)*

(P2) If ζ is a primitive n-th root of unity, $n > 0$ and $n = k\,d$, then ζ^k is a primitive d-th root of unity. (`IsPrimitiveRoot.pow`.*)*

(P3) A primitive 2nd root of unity equals -1. (`IsPrimitiveRoot.eq_neg_one_of_two_right`.*)*

(P4) If ζ is a primitive k-th root of unity and $k > 1$, then $\zeta^k = 1$, $\zeta \ne 1$ and $\zeta \ne 0$. (`IsPrimitiveRoot.pow_eq_one`, `IsPrimitiveRoot.ne_one`, `IsPrimitiveRoot.ne_zero`.*)*

(P5) If ζ is a primitive k-th root of unity, $k > 0$ and $\xi^k = 1$, then $\xi = \zeta^e$ for some $e < k$. (`IsPrimitiveRoot.eq_pow_of_pow_eq_one`.*)*

(P6) If $n > 0$, then $\exp(2\pi i/n)$ is a primitive n-th root of unity. (`Complex.isPrimitiveRoot_exp`.*)*

Proof. (P1) Put $s = \mathrm{ord}(\zeta)$. Since $\zeta^m = 1$ with $m > 0$, Definition 7.26 makes s the least positive integer with $\zeta^s = 1$; in particular $s > 0$. If $\zeta^l = 1$, write $l = ts + r$ with $0 \le r < s$; then $1 = \zeta^l = (\zeta^s)^t \zeta^r = \zeta^r$, so $r = 0$ by minimality, and $s \mid l$. So ζ is a primitive s-th root of unity. Finally $s = 1$ if and only if $\zeta^1 = 1$, that is, $\zeta = 1$.

(P2) $(\zeta^k)^d = \zeta^n = 1$. If $(\zeta^k)^l = 1$, then $\zeta^{kl} = 1$, so $kd = n \mid kl$; since $n > 0$ we have $k > 0$, and so $d \mid l$.

(P3) Let ζ be a primitive 2nd root. Then $\zeta^2 = 1$, and $\zeta \ne 1$ because $\zeta^1 = 1$ would give $2 \mid 1$. From $(\zeta - 1)(\zeta + 1) = \zeta^2 - 1 = 0$ we get $\zeta = -1$.

(P4) $\zeta^k = 1$ by definition. If $\zeta = 1$, then $\zeta^1 = 1$ gives $k \mid 1$, so $k = 1$. If $\zeta = 0$, then $\zeta^k = 0 \ne 1$ since $k \ge 1$.

(P5) The numbers $\zeta^0, \dots, \zeta^{k-1}$ are pairwise distinct: if $\zeta^i = \zeta^j$ with $i < j < k$, then $\zeta^i \ne 0$ (as $\zeta^k = 1$), so $\zeta^{j-i} = 1$ and $k \mid j - i$ with $0 < j - i < k$, which is impossible. They are k distinct roots of the polynomial $X^k - 1$, which has degree k and therefore at most k roots in $\mathbb{C}$. Since ξ is also a root, it is one of them.

(P6) Let $\zeta = \exp(2\pi i/n)$. For $l \in \mathbb{N}$, $\zeta^l = \exp(2\pi i l/n)$, and $\exp(w) = 1$ holds exactly when $w \in 2\pi i\mathbb{Z}$. So $\zeta^l = 1$ if and only if $l/n \in \mathbb{Z}$, that is, $n \mid l$. In particular $\zeta^n = 1$. □

Lemma 7.28 (The angles $2\pi/3$, $2\pi/4$, $2\pi/5$; Lean: `ErdosSzekeres.SymReduction.isPrimitiveRoot_rot3`, `SYMLEAN/SymmetricReduction.lean:584`, Lean: `ErdosSzekeres.SymReduction.isPrimitiveRoot_rot5`, `SYMLEAN/SymmetricReduction.lean:591`, Lean: `ErdosSzekeres.SymReduction.isPrimitiveRoot_rot4`, `SYMLEAN/SymmetricReduction.lean:598`)**.** *For $n \in \{3, 5, 4\}$, $\exp\big((2\pi/n)\,i\big)$ is a primitive n-th root of unity. Here $2\pi/n$ is the real number $2\pi/n$ (Lean* `2 * Real.pi / n`*), viewed as a complex number.*

Proof. $(2\pi/n)i = 2\pi i/n$; apply (P6). □

Lemma 7.29 (Even order or an odd prime; Lean: `ErdosSzekeres.SymReduction.root_cases`, `SYMLEAN/SymmetricReduction.lean:552`)**.** *Let $a \in \mathbb{C}$ and $m \in \mathbb{N}$ with $0 < m$, $a^m = 1$ and $a \ne 1$. Then either $a^k = -1$ for some $k \in \mathbb{N}$, or there are $p, k \in \mathbb{N}$ such that p is prime, $p \ne 2$, and a^k is a primitive p-th root of unity.*

Proof. Let $n = \mathrm{ord}(a)$. By (P1), $n > 0$, a is a primitive n-th root of unity, and $n \ne 1$ because $a \ne 1$. If $2 \mid n$, write $n = 2k = k \cdot 2$; by (P2), a^k is a primitive 2nd root of unity, so $a^k = -1$ by (P3). Otherwise let p be the least prime factor of n (it exists since $n > 1$). Then $p \ne 2$ because n is odd, and $n = (n/p)\,p$, so $a^{n/p}$ is a primitive p-th root of unity by (P2). □

In the proof the first alternative is taken for *every* even order (so orders $6, 10, 12, \dots$ lead to the half-turn), and in the second p is the least prime factor of the order; the statement itself does not record this.

Lemma 7.30 (Odd primes; Lean: `ErdosSzekeres.SymReduction.prime_cases`, `SYMLEAN/SymmetricReduction.lean:572`)**.** *If p is prime and $p \neq 2$, then $p = 3$, $p = 5$ or $7 \leq p$.*

Proof. $p \geq 2$ and $p \neq 2$, so $p \in \{3, 4, 5, 6\}$ or $p \geq 7$; and $4 = 2 \cdot 2$ and $6 = 2 \cdot 3$ are not prime. □

Lemma 7.31 (Order 2, a multiple of 4, or an odd prime; Lean: `ErdosSzekeres.SymReduction.root_cases2`, `SYMLEAN/SymmetricReduction.lean:607`)**.** *Let $a \in \mathbb{C}$ and $m \in \mathbb{N}$ with $0 < m$, $a^m = 1$ and $a \neq 1$. Then $a = -1$, or a^k is a primitive 4th root of unity for some $k \in \mathbb{N}$, or there are $p, k \in \mathbb{N}$ such that p is prime, $p \neq 2$, and a^k is a primitive p-th root of unity.*

Proof. Let $n = \mathrm{ord}(a)$; as before, $n > 1$ and a is a primitive n-th root of unity. If some prime $p \neq 2$ divides n, then $a^{n/p}$ is a primitive p-th root of unity by (P2). Otherwise 2 is the only prime factor of n. If $4 \mid n$, then $a^{n/4}$ is a primitive 4th root of unity by (P2). Otherwise n is even (its least prime factor is 2), say $n = 2m'$ with $m' \geq 1$. If $m' > 1$, a prime factor of m' divides n, so it equals 2, and then $4 \mid n$, which is excluded. So $m' = 1$, $n = 2$, and $a = -1$ by (P3). □

7.7. **The case lemmas.**

Lemma 7.32 (Half-turn invariant sets; Lean: `ErdosSzekeres.SymReduction.halfTurn_facts`, `SYMLEAN/SymmetricReduction.lean:647`)**.** *Let S be in general position with $2 \leq |S|$, let $c \in \mathbb{R}^2$, and assume* $\mathrm{Invariant}(\mathrm{halfTurn}_c, S)$*. Then $c \notin S$, $2 \mid |S|$, and for every $k \in \mathbb{N}$ with $2k \leq |S|$ there is $T \subseteq S$ with* $\mathrm{Invariant}(\mathrm{halfTurn}_c, T)$ *and $|T| = 2k$.*

Proof. $c \notin S$ by Lemma 7.25. Let $f = \mathrm{halfTurn}_c = \mathrm{crot}_{\hat{c},-1}$ (Lemma 7.18(b)). By Lemma 7.17(d),(b), $f^{[2]} = \mathrm{crot}_{\hat{c},1} = \mathrm{id}$. If $f(y) = y$ for some $y \in S$, then $\hat{y} = \hat{c}$ by Lemma 7.17(e) (as $-1 \neq 1$), so $y = c \in S$, a contradiction. f is injective by Lemma 7.17(f). Lemma 7.16(b) with the prime $p = 2$ gives the claims. □

Lemma 7.33 (Case of a half-turn; Lean: `ErdosSzekeres.SymReduction.case_halfTurn`, `SYMLEAN/SymmetricReduction.lean:670`; conditional on `HalfTurn34`)**.** *Assume* `HalfTurn34` *(a hypothesis). Let S be in general position with $33 \leq |S|$, let $c \in \mathbb{R}^2$, and assume* $\mathrm{Invariant}(\mathrm{halfTurn}_c, S)$*. Then* ***`HasConvexNGon 7 S`****.*

Proof. By Lemma 7.32, $c \notin S$ and $|S|$ is even; with $|S| \geq 33$ this gives $|S| \geq 34 = 17 \cdot 2$. So there is $T \subseteq S$, invariant under $\mathrm{halfTurn}_c$, with $|T| = 34$. T is in general position (monotonicity) and $c \notin T$. `HalfTurn34` applied to T and c gives `HasConvexNGon 7 T`, hence `HasConvexNGon 7 S` by monotonicity. □

Lemma 7.34 (Rotations of prime order off the centre; Lean: `ErdosSzekeres.SymReduction.rot_erase_facts`, `SYMLEAN/SymmetricReduction.lean:676`)**.** *Let S be finite, $c, w \in \mathbb{C}$, and p a prime with $w^p = 1$ and $w \neq 1$; assume* $\mathrm{Invariant}(\mathrm{crot}_{c,w}, S)$*. Let $S' = S \setminus \{\mathrm{ofC}(c)\}$. Then $p \mid |S'|$, and for every $k \in \mathbb{N}$ with $kp \leq |S'|$ there is $T \subseteq S$ with* $\mathrm{ofC}(c) \notin T$*,* $\mathrm{Invariant}(\mathrm{crot}_{c,w}, T)$ *and $|T| = kp$.*

Proof. Let $f = \mathrm{crot}_{c,w}$. Since $p > 0$ and $w^p = 1$, $w \neq 0$, so f is injective (Lemma 7.17(f)). The centre is fixed: $\mathrm{toC}(f(\mathrm{ofC}\, c)) = c + w(c - c) = c$, so $f(\mathrm{ofC}\, c) = \mathrm{ofC}\, c$. *$S'$ is mapped into itself:* for $y \in S'$, $f(y) \in S$, and $f(y) = \mathrm{ofC}(c) = f(\mathrm{ofC}\, c)$ would give $y = \mathrm{ofC}(c)$ by injectivity. Further $f^{[p]} = \mathrm{crot}_{c,w^p} = \mathrm{crot}_{c,1} = \mathrm{id}$ (Lemma 7.17(d),(b)). If $f(y) = y$ for some $y \in S'$, then $\hat{y} = c$ by Lemma 7.17(e), so $y = \mathrm{ofC}(c) \notin S'$, a contradiction. Lemma 7.16(b) applied to S' gives $p \mid |S'|$ and, for $kp \leq |S'|$, an f-invariant $T \subseteq S'$ with $|T| = kp$; then $T \subseteq S$ and $\mathrm{ofC}(c) \notin T$. □

Lemma 7.35 (Rotations of prime order at least 7; Lean: `ErdosSzekeres.SymReduction.case_rot_big`, `SYMLEAN/SymmetricReduction.lean:711`)**.** *Let S be finite, $c, w \in \mathbb{C}$, and p a prime with $7 \leq p$, $w^p = 1$ and $w \neq 1$. Assume* $\mathrm{Invariant}(\mathrm{crot}_{c,w}, S)$ *and $2 \leq |S|$. Then* ***`HasConvexNGon 7 S`****.*

No general position is assumed.

Proof. Let $f = \mathrm{crot}_{c,w}$. Since $|S| \geq 2$, some $q \in S$ differs from $\mathrm{ofC}(c)$. We have $\mathrm{normSq}(w)^p = \mathrm{normSq}(w^p) = 1$ and $\mathrm{normSq}(w) \geq 0$; as $t \mapsto t^p$ is strictly increasing on $[0, \infty)$, $\mathrm{normSq}(w) = 1$. Let $O = \mathrm{orbitFS}(f, p, q)$. Then $O \subseteq S$ (Lemma 7.14(b)); $f^{[p]}(q) = \mathrm{crot}_{c,w^p}(q) = q$; and $f(q) \neq q$, since otherwise $\hat{q} = c$ by Lemma 7.17(e) and $q = \mathrm{ofC}(c)$. By Lemma 7.14(d), $|O| = p$. Every $x \in O$ has the form $x = f^{[i]}(q) = \mathrm{crot}_{c,w^i}(q)$ with $\mathrm{normSq}(w^i) = \mathrm{normSq}(w)^i = 1$, so $\mathrm{eDist2}(x, \mathrm{ofC}\, c) = \mathrm{eDist2}(q, \mathrm{ofC}\, c)$ by Lemma 7.17(g). Corollary 7.24 with $T = O$, centre $\mathrm{ofC}(c)$, radius $\mathrm{eDist2}(q, \mathrm{ofC}\, c)$ and $n = 7 \leq p = |O|$ gives `HasConvexNGon 7 S`. □

Lemma 7.36 (Case of an odd prime order; Lean: `ErdosSzekeres.SymReduction.case_odd_prime`, `SYMLEAN/SymmetricReduction.lean:736`; conditional on `Threefold33` and `Fivefold35`). *Assume* `Threefold33` *and* `Fivefold35` *(hypotheses). Let S be in general position with $33 \le |S|$, let $c, w \in \mathbb{C}$, let p be a prime with $p \ne 2$, let w be a primitive p-th root of unity, and assume* $\mathrm{Invariant}(\mathrm{crot}_{c,w}, S)$. *Then* ***`HasConvexNGon 7 S`****.*

Proof. By (P4), $w^p = 1$ and $w \ne 1$. Let $S' = S \setminus \{\mathrm{ofC}(c)\}$; then $|S'| \ge |S| - 1 \ge 32$. By Lemma 7.30, $p = 3$, $p = 5$ or $p \ge 7$.

$p = 3$. By Lemma 7.34, $3 \mid |S'|$; with $|S'| \ge 32$ this gives $|S'| \ge 33 = 11 \cdot 3$. So there is $T \subseteq S$ with $\mathrm{ofC}(c) \notin T$, $\mathrm{Invariant}(\mathrm{crot}_{c,w}, T)$ and $|T| = 33$. Let $\zeta = \exp((2\pi/3)i)$. By Lemma 7.28, $\zeta^3 = 1$, so by (P5) $\zeta = w^e$ for some e. By Lemmas 7.18(a), 7.5(c) and 7.17(d),

$$\mathrm{rotAbout}_{\mathrm{ofC}(c),2\pi/3} = \mathrm{crot}_{\mathrm{toC}(\mathrm{ofC}\, c),\,\zeta} = \mathrm{crot}_{c,w^e} = (\mathrm{crot}_{c,w})^{[e]},$$

so $\mathrm{Invariant}(\mathrm{rotAbout}_{\mathrm{ofC}(c),2\pi/3}, T)$ by Lemma 7.9(b). T is in general position (monotonicity), $|T| = 33$ and $\mathrm{ofC}(c) \notin T$. `Threefold33`, applied to T and the centre $\mathrm{ofC}(c)$, gives `HasConvexNGon 7 T`, hence `HasConvexNGon 7 S`.

$p = 5$. In the same way, $5 \mid |S'|$ and $|S'| \ge 32$ give $|S'| \ge 35 = 7 \cdot 5$; take $T \subseteq S$ with $\mathrm{ofC}(c) \notin T$, invariant under $\mathrm{crot}_{c,w}$, with $|T| = 35$; with $\zeta = \exp((2\pi/5)i) = w^e$, T is invariant under $\mathrm{rotAbout}_{\mathrm{ofC}(c),2\pi/5}$; and `Fivefold35` gives `HasConvexNGon 7 T`, hence `HasConvexNGon 7 S`.

$p \ge 7$. Lemma 7.35 applies, since $|S| \ge 2$. □

Lemma 7.37 (Case of a primitive fourth root; Lean: `ErdosSzekeres.SymReduction.case_fourfold`, `SYMLEAN/SymmetricReduction.lean:763`; conditional on `Fourfold32`). *Assume* `Fourfold32` *(a hypothesis). Let S be in general position with $33 \le |S|$, let $c, w \in \mathbb{C}$, let w be a primitive 4th root of unity, and assume* $\mathrm{Invariant}(\mathrm{crot}_{c,w}, S)$. *Then* ***`HasConvexNGon 7 S`****.*

Proof. By (P4), $w^4 = 1$, $w \ne 1$ and $w \ne 0$; by (P2) with $4 = 2 \cdot 2$ and (P3), $w^2 = -1$. Let $f = \mathrm{crot}_{c,w}$. By Lemmas 7.9(b) and 7.17(d), S is invariant under $f^{[2]} = \mathrm{crot}_{c,-1}$, which equals $\mathrm{halfTurn}_{\mathrm{ofC}(c)}$ by Lemma 7.18(b) and $\mathrm{toC}(\mathrm{ofC}\, c) = c$. By Lemma 7.25, $\mathrm{ofC}(c) \notin S$. Hence $\hat{y} \ne c$ for every $y \in S$.

Every $y \in S$ has minimal period 4 under f. First, $f^{[4]}(y) = \mathrm{crot}_{c,w^4}(y) = y$, so by (MP1) $d = \mathrm{minimalPeriod}(f, y)$ divides 4, and $d \in \{1, 2, 4\}$. If $d = 1$, then $f(y) = y$ by (MP2), and $\hat{y} = c$ by Lemma 7.17(e), which is impossible. If $d = 2$, then $f^{[2]}(y) = y$ by (MP3), that is, $\mathrm{crot}_{c,-1}(y) = y$, and again $\hat{y} = c$ (as $-1 \ne 1$). So $d = 4$.

f is injective (Lemma 7.17(f)). Lemma 7.16(a) with $p = 4$ gives $4 \mid |S|$ and invariant subsets of every size $4k \le |S|$. Since $8 \cdot 4 = 32 \le |S|$, there is $T \subseteq S$ with $\mathrm{Invariant}(f, T)$ and $|T| = 32$. (Also $|S| \ge 36$, but this is not needed.) Let $\zeta = \exp((2\pi/4)i)$; by Lemma 7.28 and (P5), $\zeta = w^e$ for some e, and as in the proof of Lemma 7.36, $\mathrm{rotAbout}_{\mathrm{ofC}(c),2\pi/4} = f^{[e]}$, so T is invariant under it. T is in general position, $|T| = 32$, and $\mathrm{ofC}(c) \notin T$ because $T \subseteq S$. `Fourfold32` gives `HasConvexNGon 7 T`, hence `HasConvexNGon 7 S`. □

Lemma 7.38 (Case of a reflection; Lean: `ErdosSzekeres.SymReduction.case_mirror`, `SYMLEAN/SymmetricReduction.lean:811`; conditional on the two mirror classes). *Assume* `MirrorOneAxis33` *and* `MirrorNoAxis34` *(hypotheses). Let $a, u \in \mathbb{R}^2$ with $u_0^2 + u_1^2 = 1$, let S be in general position with $33 \le |S|$, and assume* $\mathrm{Invariant}(\mathrm{reflLine}_{a,u}, S)$. *Then* ***`HasConvexNGon 7 S`****.*

Proof. Let $\sigma = \mathrm{reflLine}_{a,u}$, $F = \{p \in S : \sigma(p) = p\}$ and $M = \{p \in S : \sigma(p) \ne p\}$, so that $|F| + |M| = |S|$.

$|F| \le 2$. Otherwise F contains three pairwise distinct points x, y, z. By Lemma 7.19(c), each $t \in \{x, y, z\}$ satisfies $(t_0 - a_0)u_1 - (t_1 - a_1)u_0 = 0$, so $\mathrm{orient}(x, y, z) = 0$ by Lemma 7.19(d), contradicting general position.

M is mapped into itself. By Lemma 7.19(b), $\sigma \circ \sigma = \mathrm{id}$; in particular σ is injective. Let $y \in M$. Then $\sigma(y) \in S$, and if $\sigma(\sigma(y)) = \sigma(y)$ then $y = \sigma(y)$, which is false; so $\sigma(y) \in M$. Moreover $\sigma^{[2]}(y) = y$ and $\sigma(y) \ne y$ for $y \in M$. Lemma 7.16(b) with the prime 2 gives: $2 \mid |M|$, and for every k with $2k \le |M|$ there is $T \subseteq M$, mapped into itself by σ, with $|T| = 2k$.

Case $|F| = 0$. Then $|M| = |S| \ge 33$ and $|M|$ is even, so $|M| \ge 34$. Take $T \subseteq M$ invariant with $|T| = 34$. $T \subseteq S$ is in general position, and no point of T is fixed by σ (as $T \subseteq M$). `MirrorNoAxis34` applied to T, a, u gives `HasConvexNGon 7 T`, hence `HasConvexNGon 7 S`.

Case $|F| \ge 1$. Pick $x \in F$. Since $|F| \le 2$, $|M| \ge 31$, and $|M|$ is even, so $|M| \ge 32$. Take $T \subseteq M$ invariant with $|T| = 32$, and put $T' = T \cup \{x\} \subseteq S$. Then $x \notin T$ (points of T are not fixed), so $|T'| = 33$. T' is invariant: $\sigma(x) = x$, and T is invariant. The point $x \in T'$ is fixed, and every fixed $y \in T'$ equals x, because no point of T is fixed. T' is in general position. `MirrorOneAxis33` applied to T', a, u gives `HasConvexNGon 7 T'`, hence `HasConvexNGon 7 S`. □

7.8. **The reduction theorems.**

Theorem 7.39 (Main reduction; Lean: `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_isometry`, `SYMLEAN/SymmetricReduction.lean:997`; conditional on five case classes)**.** *Assume the propositions* `HalfTurn34`*,* `Threefold33`*,* `Fivefold35`*,* `MirrorOneAxis33` *and* `MirrorNoAxis34` *of Definition 7.7 (hypotheses; none is proved in this section). Let S be a finite set of points and $g\colon \mathbb{R}^2 \to \mathbb{R}^2$ a map. If S is in general position, $33 \le |S|$, g is a Euclidean isometry, $g \neq \mathrm{id}$ and $\mathrm{Invariant}(g, S)$, then* ***`HasConvexNGon 7 S`****.*

Proof. Since $|S| \ge 33 > 1$, Theorem 7.21(b) applies.

Rotation: $g = \mathrm{crot}_{c,a}$ with $a \neq 1$ and $a^m = 1$, $m > 0$. Then $\mathrm{Invariant}(\mathrm{crot}_{c,a}, S)$, and for every k, $\mathrm{Invariant}(\mathrm{crot}_{c,a^k}, S)$ by Lemmas 7.9(b) and 7.17(d). By Lemma 7.29 there are two cases.

- $a^k = -1$ for some k. Then S is invariant under $\mathrm{crot}_{c,-1} = \mathrm{halfTurn}_{\mathrm{ofC}(c)}$ (Lemma 7.18(b) and $\mathrm{toC}(\mathrm{ofC}\, c) = c$), and Lemma 7.33 gives the claim.
- a^k is a primitive p-th root of unity for a prime $p \neq 2$. Then Lemma 7.36, applied with $w = a^k$, gives the claim.

Reflection: $g = \mathrm{reflLine}_{A,u}$ with $u_0^2 + u_1^2 = 1$. Lemma 7.38 gives the claim. □

Remark 7.40 (Status of Theorem 7.39). The theorem is Lean-checked as an implication: the five classes are explicit hypotheses (Lean binders `h2 h3 h5 hm1 hm0`), not axioms, and the axioms log is `SYMLEAN/logs/SymmetricReductionAxioms.out`. The unconditional statement, Theorem 7.39 without its five hypotheses, is claimed in the project as computer-assisted (Theorem A of `WRITEUP/SYMMETRIC.md`, §0, in its form for Euclidean isometries), on the evidence summarized in Remark 7.8; it is not Lean-checked as a whole. In this document the theorem is established only as the stated implication. Theorem 10.40 restates the theorem with the two mirror classes discharged, so that only `HalfTurn34`, `Threefold33` and `Fivefold35` remain hypotheses; it is stated in `SYMLEAN/mirror/ReductionThreeClasses.lean`, for the byte-identical copies of the definitions described in Remark 7.8. The symmetry g must be a Euclidean isometry of the whole plane; a distance-preserving permutation of S alone is not treated. Affine symmetries are treated in Theorem 8.28.

Theorem 7.41 (Variant; Lean: `ErdosSzekeres.SymReduction.halfTurn_of_symmetric_counterexample`, `SYMLEAN/SymmetricReduction.lean:1024`; conditional on five case classes)**.** *Assume the propositions* `Threefold33`*,* `Fourfold32`*,* `Fivefold35`*,* `MirrorOneAxis33` *and* `MirrorNoAxis34` *of Definition 7.7 (hypotheses; none is proved in this section). Let S be a finite set of points and $g\colon \mathbb{R}^2 \to \mathbb{R}^2$ a map. Assume that S is in general position, $33 \le |S|$, g is a Euclidean isometry, $g \neq \mathrm{id}$, $\mathrm{Invariant}(g, S)$, and $\neg$* ***`HasConvexNGon 7 S`****. Then there is $c \in \mathbb{R}^2$ such that $g = \mathrm{halfTurn}_c$ (as maps of $\mathbb{R}^2$), $c \notin S$, $2 \mid |S|$ and $34 \le |S|$.*

Proof. By Theorem 7.21(b), as $|S| > 1$, g is a rotation or a reflection.

Rotation: $g = \mathrm{crot}_{c,a}$ with $a \neq 1$ and $a^m = 1$, $m > 0$. By Lemma 7.31 there are three cases.

- $a = -1$. Then $g = \mathrm{crot}_{c,-1} = \mathrm{halfTurn}_{\mathrm{ofC}(c)}$ (Lemma 7.18(b) and $\mathrm{toC}(\mathrm{ofC}\, c) = c$). By Lemma 7.32, $\mathrm{ofC}(c) \notin S$ and $2 \mid |S|$; with $|S| \ge 33$ this gives $|S| \ge 34$. So the point $\mathrm{ofC}(c)$ has all the required properties.
- a^k is a primitive 4th root of unity. S is invariant under crot_{c,a^k} (Lemmas 7.9(b) and 7.17(d)), and Lemma 7.37 gives `HasConvexNGon 7 S`, contradicting the hypothesis.
- a^k is a primitive p-th root of unity for a prime $p \neq 2$. Likewise Lemma 7.36 gives a contradiction.

Reflection. Lemma 7.38 gives `HasConvexNGon 7 S`, a contradiction. □

Remark 7.42 (Scope of Theorem 7.41). The theorem concerns one symmetry at a time and Euclidean isometries only. Corollary 3 of `WRITEUP/SYMMETRIC.md` has the same per-symmetry form (every symmetry of such a set is a half-turn about a point outside the set), with the case theorems and the fourfold exclusion as assumptions, but there a symmetry may also be an affine bijection; the theorem is the isometry case of that corollary. A group form, stated as the corollary labelled `cor:only` in `WRITEUP/note/symmetric_es7.tex`, is Lean-checked for Euclidean isometries, with the same five classes as hypotheses, in the review file `SYMLEAN/review/VariantChecks.lean` (`VariantChecks.group_form`, line 1168): for S as in the theorem, the Euclidean isometries that map S into itself are the identity and at most one half-turn, whose centre is not in S. The evidence is the log `SYMLEAN/logs/VariantChecks.out`, which prints the axioms `propext`, `Classical.choice` and `Quot.sound` for `VariantChecks.group_form`, and the row `review/VariantChecks.lean` of `SYMLEAN/logs/runs.tsv`, which records exit code 0 for the file with SHA-256 prefix `4b823bb6`; the file at the commit used here has the prefix `4b823bb6cf85`. The group form follows from the theorem. Let g and g' be Euclidean isometries other than the identity that map S into itself. The theorem gives $g = \mathrm{halfTurn}_c$ and $g' = \mathrm{halfTurn}_d$ with $c, d \notin S$. If $c \neq d$, then $g \circ g'$ is the

translation $p \mapsto p + 2(c - d)$. It is a Euclidean isometry other than the identity that maps S into itself, so by the theorem it equals $\mathrm{halfTurn}_e$ for some $e \in \mathbb{R}^2$; evaluating at $p = e$ gives $e + 2(c - d) = e$, so $c = d$, a contradiction. The group form is not restated in this document as a numbered statement, and no statement of this document uses it. The variant does not use `HalfTurn34`; it uses `Fourfold32`, which rests on `C4CUT` (Remark 7.8).

Definition 7.43 (The `H3` mirror statement; Lean: `ErdosSzekeres.SymReduction.MirrorExclusionLineCopy`, `SYMLEAN/SymmetricReduction.lean:1052`)**.** `MirrorExclusionLineCopy` is the proposition: for every finite $S \subseteq \mathbb{R}^2$ and all $a, u \in \mathbb{R}^2$, if $u_0^2 + u_1^2 = 1$, S is in general position, $|S| = 33$, and $\mathrm{reflLine}_{a,u}(q) \in S$ for every $q \in S$, then `HasConvexNGon 7 S`.

Lemma 7.44 (Bridge; Lean: `ErdosSzekeres.SymReduction.mirrorOneAxis33_of_exclusionLine`, `SYMLEAN/SymmetricReduction.lean:1056`)**.** `MirrorExclusionLineCopy` *implies* `MirrorOneAxis33`.

Proof. Let S, a, u satisfy the hypotheses of `MirrorOneAxis33`. The condition $\mathrm{Invariant}(\mathrm{reflLine}_{a,u}, S)$ is, by definition, the condition $\mathrm{reflLine}_{a,u}(q) \in S$ for every $q \in S$. So all hypotheses of `MirrorExclusionLineCopy` hold (the hypothesis about the fixed point is not used), and it gives `HasConvexNGon 7 S`. □

Remark 7.45 (About the bridge). `MirrorExclusionLineCopy` has the same text as `Mir.MirrorExclusionLine` (`H3/MirrorEncoding/Line.lean:94`), and `reflLine` has the same text as `Mir.reflLine`. They are different Lean constants of separately compiled developments, and no Lean theorem in `SYMLEAN` identifies them; `SYMLEAN/README.md` §5 reports a separate check of both identities by `rfl`, whose file is not kept in `SYMLEAN`. In `H3`, `Mir.mirrorExclusionLine_of_unsat` and `Mir.mirrorExclusionLine_of_cubes` (`H3/MirrorEncoding/Line.lean:139` and `:147`) derive `Mir.MirrorExclusionLine` from hypotheses saying that certain CNF formulas are unsatisfiable (the second also from two hypotheses that the Boolean cover check `coverCheck` of a cube list returns `true`). In `V2-L`, `V2L.mirrorExclusionLine_final` (`V2-L/V2L/FinalTree.lean:28`; `V2-L/README.md` §9.1 and §9.6) derives it from hypotheses that all carry recorded verdicts. It is Lean-checked with four hypotheses, `h1`, `hroot`, `hup` and `hleaves`, which together assert $1{,}198 + 1 + 121 + 74{,}000$ facts of the form "this CNF formula is unsatisfiable"; the $1 + 121$ facts of `hroot` and `hup` are the cover obligations of the top two levels of the hull-pair case split of sub-case (ii) (Remark 7.8; `WRITEUP/MIRROR.md` §4). Each fact has a recorded `cake_lpr` verdict on a file whose proof was not retained, and the step from a verdict on a file to the Lean hypothesis is made outside Lean (`WRITEUP/MIRROR.md` §6). So Lemma 7.44 supplies `MirrorOneAxis33` only conditionally on those CNF facts and on the identification of the two texts; the Lean proof without SAT input is in `SYMLEAN/mirror` (Remark 7.8). The converse implication also holds, and is Lean-checked in the review file `SYMLEAN/review/ReviewSanity.lean` (`Review.exclusionLine_of_mirrorOneAxis33`, line 1593, and the equivalence `Review.mirrorOneAxis33_iff_exclusionLine`, line 1629, whose axioms the log `SYMLEAN/logs/ReviewSanity.out` prints): for 33 points the fixed-point hypothesis of `MirrorOneAxis33` is automatic. Indeed, let S, a, u satisfy the hypotheses of `MirrorExclusionLineCopy`. In the notation of the proof of Lemma 7.38, whose first two steps use only these hypotheses, $|F| \le 2$, $|M|$ is even and $|F| + |M| = 33$, so $|F| = 1$. The point of F is then the only point of S fixed by $\mathrm{reflLine}_{a,u}$, and `MirrorOneAxis33` gives `HasConvexNGon 7 S`. Neither review theorem is restated in this document as a numbered statement, and no statement of this document uses them.

Remark 7.46 (Hypotheses of the classes that follow from the others). Some hypotheses of the classes are implied by the others; this is not used anywhere.

- In `HalfTurn34`, $c \notin S$ follows from general position and invariance (Lemma 7.25).
- In `Fourfold32`, $c \notin S$ follows as well: by Lemma 7.18(a) the rotation is $f = \mathrm{crot}_{\hat{c},\zeta}$ with $\zeta = \exp((2\pi/4)i)$, a primitive 4th root of unity (Lemma 7.28), so $\zeta^2 = -1$ by (P2) and (P3). By Lemmas 7.17(d) and 7.18(b), $f^{[2]} = \mathrm{crot}_{\hat{c},-1} = \mathrm{halfTurn}_c$; S is invariant under it (Lemma 7.9(b)), and Lemma 7.25 applies.
- In `Threefold33` and `Fivefold35`, $c \notin S$ follows from the size. Let $p \in \{3, 5\}$ and $w = \exp((2\pi/p)i)$, so that $w^p = 1$ and $w \neq 1$ (Lemma 7.28 and (P4)), and the rotation is $\mathrm{crot}_{\hat{c},w}$ (Lemma 7.18(a)). If $c \in S$, then by Lemma 7.34 (with $\mathrm{ofC}(\hat{c}) = c$), p divides $|(S \setminus \{c\})|$, which is 32 for $p = 3$ and 34 for $p = 5$; but $3 \nmid 32$ and $5 \nmid 34$.
- In `MirrorOneAxis33`, the fixed-point hypothesis follows from $|S| = 33$ (Remark 7.45). In `MirrorNoAxis34` the hypothesis that no point is fixed is not implied in this way: for 34 points the same argument gives only $|F| \in \{0, 2\}$.

Dropping one of the implied hypotheses above gives an equivalent class. In general, adding a hypothesis to a class can only make it weaker (the class without the hypothesis implies the class with it), that is, easier to supply.

Coverage. All 72 declarations of `SYMLEAN/SymmetricReduction.lean` are covered.

- Definitions: `eDist2`, `IsEucIsometry` (Definition 7.1); `Invariant` (Definition 7.3); `toC`, `ofC` (Definition 7.4); `crot`, `rotAbout`, `halfTurn`, `reflLine` (Definition 7.6); `orbitFS` (Definition 7.13); the six classes `HalfTurn34`, `Threefold33`, `Fivefold35`, `MirrorOneAxis33`, `MirrorNoAxis34`, `Fourfold32` (Definition 7.7); `MirrorExclusionLineCopy` (Definition 7.43).
- Finite dynamics: `iterate_mem`, `Invariant.iterate` (Lemma 7.9); `mem_orbitFS_self`, `orbitFS_subset`, `orbitFS_invariant`, `card_orbitFS`, `card_orbitFS_of_minimalPeriod` (Lemma 7.14); `orbit_select_minimalPeriod`, `orbit_select` (Lemma 7.16).
- Coordinates and maps: `toC_re`, `toC_im`, `ofC_zero`, `ofC_one`, `toC_ofC`, `ofC_toC`, `toC_injective`, `eDist2_eq_normSq`, `eDist2_pos_of_ne`, `point_ext` (Lemma 7.5); `toC_crot`, `crot_one`, `crot_comp`, `crot_iterate`, `crot_eq_self_iff`, `crot_injective`, `eDist2_crot_center` (Lemma 7.17); `rotAbout_eq_crot`, `halfTurn_eq_crot` (Lemma 7.18); `toC_reflLine`, `reflLine_involutive`, `cross_of_reflLine_fixed`, `orient_eq_zero_of_cross` (Lemma 7.19).
- Isometries: `IsEucIsometry.injective`, `isometry_toC_conj` (Lemma 7.20); `isometry_complex_cases`, `isometry_structure` (Theorem 7.21).
- Geometry: `convexPosition_of_concyclic` (Lemma 7.23), `hasConvexNGon_of_concyclic` (Corollary 7.24), `center_not_mem_of_halfTurn` (Lemma 7.25); also the library lemma `Suk.convexPosition_of_strict_supports` (Lemma 7.22). This is the lemma titled "Strict supporting lines" in the foundations section; the short proof is repeated here.
- Roots of unity: `root_cases` (Lemma 7.29); `prime_cases` (Lemma 7.30); `root_cases2` (Lemma 7.31); and `isPrimitiveRoot_rot3`, `isPrimitiveRoot_rot5`, `isPrimitiveRoot_rot4` (Lemma 7.28).
- Cases: `halfTurn_facts` (Lemma 7.32); `case_halfTurn` (Lemma 7.33, conditional on `HalfTurn34`); `rot_erase_facts` (Lemma 7.34); `case_rot_big` (Lemma 7.35); `case_odd_prime` (Lemma 7.36, conditional on `Threefold33` and `Fivefold35`); `case_fourfold` (Lemma 7.37, conditional on `Fourfold32`); `case_mirror` (Lemma 7.38, conditional on `MirrorOneAxis33` and `MirrorNoAxis34`).
- Main results: `heptagon_of_nontrivial_isometry` (Theorem 7.39, conditional on `HalfTurn34`, `Threefold33`, `Fivefold35`, `MirrorOneAxis33` and `MirrorNoAxis34`); `halfTurn_of_symmetric_counterexample` (Theorem 7.41, conditional on the same classes with `Fourfold32` in place of `HalfTurn34`); `mirrorOneAxis33_of_exclusionLine` (Lemma 7.44, an implication between two propositions, so conditional on its hypothesis `MirrorExclusionLineCopy`).

The local statement `pigeon` in the proof of `isometry_structure` is Lemma 7.10. The Mathlib facts on minimal periods and on roots of unity that the Lean proofs use are restated and proved in Lemmas 7.12 and 7.27; the Mathlib results on isometries that the Lean proof uses (the construction `Isometry.affineIsometryOfStrictConvexSpace` and the theorem `linear_isometry_complex`) are replaced by the self-contained proof of Theorem 7.21(a); other standard facts (the pigeonhole principle, least prime factors, square roots of complex numbers, elementary facts on finite sets) are stated where they are used. Not covered, deliberately: `SYMLEAN/SymmetricReductionAxioms.lean` (a verbatim copy with two `#print axioms` lines); lines 1–1058 of `SYMLEAN/AffineReduction.lean` (a verbatim copy; its affine part belongs to the section on affine symmetries); the review files `SYMLEAN/review/ReviewSanity.lean`, `SYMLEAN/review/VariantChecks.lean` and `SYMLEAN/review/SupMetric.lean` (review and audit files, cited only in the paragraph on axioms and rigor and in Remarks 7.2, 7.42 and 7.45); and `SYMLEAN/mirror` (the section on mirror bounds). Two results of the review files go beyond audit checks and are not restated as numbered statements: the group form `VariantChecks.group_form` (`SYMLEAN/review/VariantChecks.lean:1168`, conditional on the five classes of Theorem 7.41), and the converse bridge `Review.exclusionLine_of_mirrorOneAxis33` (`SYMLEAN/review/ReviewSanity.lean:1593`, an implication between two propositions, so conditional on its hypothesis `MirrorOneAxis33`) with the equivalence `Review.mirrorOneAxis33_iff_exclusionLine` of the two propositions (line 1629 of the same file), which asserts neither of them. The reason is that they lie in review files and that no statement of this document uses them. Remarks 7.42 and 7.45 state them, name their logs and give short written arguments.

8. Affine symmetries

This section extends the reduction of Theorem 7.39 from Euclidean isometries to affine symmetries. Its central statement is a transfer principle (Theorem 8.25). Let $N \geq 3$ and n be natural numbers, and suppose that every finite set in general position with at least N points that is mapped into itself by a Euclidean isometry other than the identity contains n points in convex position. Then the same holds with an injective affine map $p \mapsto Lp + b$ in place of the isometry. The Euclidean statement is a *hypothesis* of the transfer (a parameter of the theorem),

so the transfer is an implication and asserts neither statement on its own. Corollary 8.27 restates the transfer for Mathlib affine maps, for affine equivalences and for maps given in coordinates. With $N = 33$, $n = 7$ and the Euclidean statement supplied by Theorem 7.39, the transfer gives five forms of the statement "a set of at least 33 points in general position with a non-trivial affine symmetry contains 7 points in convex position" (Theorem 8.28). Like Theorem 7.39, these five theorems take five of the six case classes of Definition 7.7 (all except `Fourfold32`) as hypotheses, so they are conditional (Remark 8.29).

The idea is as follows. The symmetry $g(p) = Lp+b$ permutes the finite set S, and S contains three non-collinear points, because it is in general position and has at least $N \geq 3$ points. So some power g^m with $m \geq 1$ fixes these three points, and then its linear part satisfies $L^m = 1$. The averaged quadratic form $z \mapsto \sum_{k<m} \mathrm{nsq}(L^k z)$, where nsq is the squared Euclidean norm, is L-invariant and positive definite. Writing it as $\mathrm{nsq}(Tz)$ with an explicit upper triangular matrix T (a Cholesky factor) turns g into the Euclidean isometry $T \circ g \circ T^{-1}$ of the set $T(S)$. Finally, T preserves general position and cardinality, and T^{-1} carries n points of $T(S)$ in convex position back to n points of S in convex position.

Sources. All declarations of this section are in the namespace `ErdosSzekeres.SymReduction` of the file `SYMLEAN/AffineReduction.lean` (SHA-256 `41cfc0b60aec...`), lines 1059–1533 (its §§9–10).

- Lines 1–1058 of that file are byte-identical to lines 1–1058 of `SYMLEAN/SymmetricReduction.lean` (SHA-256 `6d5ec90a4215...`). Their declarations, among them `eDist2`, `IsEucIsometry`, `Invariant`, the case classes and `heptagon_of_nontrivial_isometry` (line 997 in both files), are treated in the section of Theorem 7.39; the theorems of this section use the copies in `SYMLEAN/AffineReduction.lean`.
- The repository definitions `Point`, `GeneralPosition`, `ConvexPosition` and `HasConvexNGon` are in `ErdosSzekeres/Definitions.lean` (SHA-256 `2425500eefd1...`), and `orient` is in `ErdosSzekeres/Orientation.lean:9` (SHA-256 `f67955ac7091...`).
- Two reformulations of these definitions are used. The first is `convexPosition_iff`, at `ErdosSzekeres/Basic.lean:15` (file SHA-256 `13667aa2d3dc...`). The second is `generalPosition_iff_orient_ne_zero`, at `ErdosSzekeres/OrientationGeometry.lean:66` (file SHA-256 `c7db9bbbb65e...`).
- Toolchain: Lean 4.24.0, Mathlib `f897ebcf`.

Rigor and axioms. Every statement below that carries a Lean reference is *Lean-checked*. The evidence is as follows.

- The file `SYMLEAN/AffineReductionAxioms.lean` (SHA-256 `f6cb9372b174...`) consists of the 1533 lines of `SYMLEAN/AffineReduction.lean`, followed by a blank line, nine `#print axioms` lines and one `#check` line.
- The run table `SYMLEAN/logs/runs.tsv` records that `SYMLEAN/AffineReduction.lean` and `SYMLEAN/AffineReductionAxioms.lean` were each compiled on 1 November 2026 with exit code 0.
- The recorded output `SYMLEAN/logs/AffineReductionAxioms.out` (SHA-256 `38d0b6f59398...`) of the axioms file states that each of the following nine theorems depends on exactly the axioms `propext`, `Classical.choice` and `Quot.sound`: `heptagon_of_nontrivial_affineEquiv`, `heptagon_of_nontrivial_affineEquiv_image`, `heptagon_of_nontrivial_affineMap`, `heptagon_of_nontrivial_affine`, `heptagon_of_nontrivial_affine_coords`, `hasConvexNGon_of_affine_symmetry`, `generalPosition_image_affineEquiv`, `hasConvexNGon_image_affine` and `heptagon_of_nontrivial_isometry`.
- Every other declaration of this section is used, directly or through other declarations, in the proof of one of these nine theorems. So it depends on no further axiom. For `lin2_fst` and `lin2_snd` (Lemma 8.11(a)) this inference needs a supplement. Their Lean proofs are `rfl`, and they are used only inside `simp only` (`SYMLEAN/AffineReduction.lean`, lines 1180, 1187, 1193 and 1197), which may apply such lemmas as definitional rewrites without recording them in the proof term. So they need not occur in the proof terms from which `#print axioms` collects the axioms of the nine theorems. Apart from equality and `rfl`, which use no axiom, their statements and proofs mention only `lin2` applied to points and the coordinates, additions and multiplications in the definition of `lin2`. The axioms check of `hasConvexNGon_of_affine_symmetry` reaches all of these, because its proof applies the map T of Proposition 8.24(b), a value of `lin2`, to points (Lean lines 1373–1376), and the check also follows definitions. So these two lemmas depend on no further axiom either.
- The compilation log `SYMLEAN/logs/AffineReduction.out` of `SYMLEAN/AffineReduction.lean` is empty (no message). So no declaration of the file uses `sorry`.

- The five case classes are definitions of type `Prop` that occur as explicit premises, not axioms. The output of the `#check` line, in the same log as the axioms, displays them as premises of `heptagon_of_nontrivial_affineEquiv`.
- No proof uses `decide` or `native_decide`. Nothing in this section is a computation.

The proofs below are *written proofs (not refereed)*. They follow the Lean proofs step by step unless a proof says otherwise. Statements tagged "written step" are not Lean declarations.

Dependencies. From the foundations: Definitions 1.1, 1.3, 1.5, 1.6, 1.7 and 1.10; Lemma 1.20 (general position in terms of orient); and Lemma 1.22 (convex position in terms of convex hulls). The lemmas below re-prove, in the exact form used here, the facts summarised in Lemmas 1.17 and 1.27. From the reduction for isometries: Definition 7.1 (`eDist2` and `IsEucIsometry`), Definition 7.7 (the case classes) and Theorem 7.39.

8.1. **Conventions.**

Points, matrices, iterates. Points are elements of $\mathbb{R}^2$ (Lean `Point`, the functions from the index type `Fin 2` to the reals), written $p = (p_0, p_1)$. Points and vectors are the same objects, and $p \pm q$ and λp are taken coordinatewise.

- A *matrix* is a real 2×2 matrix $M = (M_{ij})_{i,j\in\{0,1\}}$ (Lean: a `Matrix` with rows and columns indexed by `Fin 2` and real entries).
- Mz is the matrix–vector product, $(Mz)_i = \sum_{j\in\{0,1\}} M_{ij} z_j$ (Lean `Matrix.mulVec`). It is linear in z, and $M(M'z) = (MM')z$ for matrices M, M'.
- M^k is the k-th matrix power, with $M^0 = 1$, the identity matrix, and $M^{k+1} = M^k M = M M^k$.
- $\det M = M_{00}M_{11} - M_{01}M_{10}$ (Mathlib `Matrix.det_fin_two`).
- For a map $g : \mathbb{R}^2 \to \mathbb{R}^2$, g^k is the k-fold iterate (Lean `g^[k]`, from `Nat.iterate`): $g^0 = \mathrm{id}$ and $g^{k+1} = g \circ g^k$. (The definition of `Nat.iterate` recurses on the other side, $g^{k+1} = g^k \circ g$. The form used here, $g^{k+1}(p) = g(g^k(p))$, is the lemma `Function.iterate_succ_apply'`, which the Lean proof of Lemma 8.13 uses.)
- "$g \neq \mathrm{id}$" is always an inequality of maps of the whole plane.

Finite sets. A finite set is a Lean `Finset` of points, and $|S|$ is its cardinality. For a finite set S and a map f, $f(S) = \{f(p) : p \in S\}$ is again a finite set (Lean `Finset.image`); if f is injective, then $|f(S)| = |S|$. "f maps S into S" means `Invariant f S`: $f(p) \in S$ for all $p \in S$ (Definition 1.10; Lean `SYMLEAN/AffineReduction.lean:271`). General position, convex position and `HasConvexNGon` are as in Definitions 1.5, 1.6 and 1.7. By Lemma 1.20, a finite set S is in general position if and only if $\mathrm{orient}(a, b, c) \neq 0$ for all pairwise distinct $a, b, c \in S$. By Lemma 1.22, S is in convex position if and only if $p \notin \mathrm{conv}(S \setminus \{p\})$ for every $p \in S$. Here $\mathrm{conv}(A)$ is the convex hull, the smallest convex set containing A. Finally, `HasConvexNGon` n S means that some $U \subseteq S$ with $|U| = n$ is in convex position.

Euclidean isometries. $\mathrm{eDist2}(p, q) = (p_0 - q_0)^2 + (p_1 - q_1)^2$, and a map $h : \mathbb{R}^2 \to \mathbb{R}^2$ is a *Euclidean isometry* (`IsEucIsometry` h) if $\mathrm{eDist2}(h(p), h(q)) = \mathrm{eDist2}(p, q)$ for *all* $p, q \in \mathbb{R}^2$ (Definition 7.1; Lean `SYMLEAN/AffineReduction.lean:265` and `:268`). The squared distance is written out because the metric that Mathlib puts on `Point` is the sup metric, not the Euclidean one.

Affine maps (Mathlib).

- An *affine map* f of $\mathbb{R}^2$ (Mathlib `AffineMap`) is a map $f : \mathbb{R}^2 \to \mathbb{R}^2$ together with a linear map $f_{\mathrm{lin}} : \mathbb{R}^2 \to \mathbb{R}^2$ (Lean `f.linear`) such that $f(v + p) = f_{\mathrm{lin}}(v) + f(p)$ for all $v, p \in \mathbb{R}^2$. The same letter denotes the map. A linear map φ is an affine map with $\varphi_{\mathrm{lin}} = \varphi$ (Lean `LinearMap.toAffineMap`).
- An *affine equivalence* f of $\mathbb{R}^2$ (Mathlib `AffineEquiv`) is a bijection f of $\mathbb{R}^2$ together with a linear bijection f_{lin} with the same property. Its underlying affine map (`AffineEquiv.toAffineMap`) has the same map and the same linear part.
- refl denotes the identity affine equivalence of $\mathbb{R}^2$, whose map is id (Lean `AffineEquiv.refl`, for the reals and `Point`).
- Two affine equivalences with the same map are equal (`AffineEquiv.ext`), because the linear part is determined by the map: $f_{\mathrm{lin}}(v) = f(v) - f(0)$, by the defining property with $p = 0$. Hence, for an affine equivalence g, $g \neq \mathrm{refl}$ holds if and only if $g \neq \mathrm{id}$ as maps.
- For a linear map φ of $\mathbb{R}^2$, M_φ is the matrix whose j-th column is $\varphi(e_j)$, where $e_0 = (1, 0)$ and $e_1 = (0, 1)$ (Lean `LinearMap.toMatrix'`). Then $M_\varphi z = \varphi(z)$ for every z, since $\varphi(z) = z_0\varphi(e_0) + z_1\varphi(e_1)$ (`LinearMap.toMatrix'_mulVec`). The matrix of a composition is the product of the matrices.

8.2. **The averaged quadratic form.**

Definition 8.1 (squared norm; Lean: `ErdosSzekeres.SymReduction.nsq`, `SYMLEAN/AffineReduction.lean:1088`)**.** For $v \in \mathbb{R}^2$, $\mathrm{nsq}(v) = v_0^2 + v_1^2$.

Lemma 8.2 (coordinates of Az; Lean: `ErdosSzekeres.SymReduction.mulVec2_fst`, `SYMLEAN/AffineReduction.lean:1097`, Lean: `ErdosSzekeres.SymReduction.mulVec2_snd`, `SYMLEAN/AffineReduction.lean:1101`)**.** *For every matrix A and every $z \in \mathbb{R}^2$: $(Az)_0 = A_{00}z_0 + A_{01}z_1$* (`mulVec2_fst`) *and $(Az)_1 = A_{10}z_0 + A_{11}z_1$* (`mulVec2_snd`)*.*

Proof. This is the definition $(Az)_i = \sum_{j\in\{0,1\}} A_{ij}z_j$, written out for $i = 0$ and $i = 1$. □

Lemma 8.3 (elementary facts on nsq; Lean: `ErdosSzekeres.SymReduction.nsq_nonneg`, `SYMLEAN/AffineReduction.lean:1090`, Lean: `ErdosSzekeres.SymReduction.eDist2_eq_nsq`, `SYMLEAN/AffineReduction.lean:1094`)**.** *The following hold.*

(a) (`nsq_nonneg`) $0 \le \mathrm{nsq}(v)$ *for every $v \in \mathbb{R}^2$.*
(b) (`eDist2_eq_nsq`) $\mathrm{eDist2}(p, q) = \mathrm{nsq}(p - q)$ *for all $p, q \in \mathbb{R}^2$.*

Proof. (a) $\mathrm{nsq}(v)$ is a sum of two squares. (b) $(p - q)_i = p_i - q_i$, so $\mathrm{nsq}(p - q) = (p_0 - q_0)^2 + (p_1 - q_1)^2 = \mathrm{eDist2}(p, q)$. □

Definition 8.4 (Gram entries; Lean: `ErdosSzekeres.SymReduction.gA`, `SYMLEAN/AffineReduction.lean:1106`, Lean: `ErdosSzekeres.SymReduction.gB`, `SYMLEAN/AffineReduction.lean:1107`, Lean: `ErdosSzekeres.SymReduction.gC`, `SYMLEAN/AffineReduction.lean:1108`)**.** For a matrix A,

$$g_A(A) = A_{00}^2 + A_{10}^2, \qquad g_B(A) = A_{00}A_{01} + A_{10}A_{11}, \qquad g_C(A) = A_{01}^2 + A_{11}^2$$

(Lean `gA`, `gB`, `gC`). These are the entries of the Gram matrix of the two columns of A.

Lemma 8.5 (squared norm of Az; Lean: `ErdosSzekeres.SymReduction.nsq_mulVec`, `SYMLEAN/AffineReduction.lean:1110`)**.** *For every matrix A and every $z \in \mathbb{R}^2$,* $\mathrm{nsq}(Az) = g_A(A)\, z_0^2 + 2\, g_B(A)\, z_0z_1 + g_C(A)\, z_1^2$.

Proof. By Lemma 8.2, $\mathrm{nsq}(Az) = (A_{00}z_0 + A_{01}z_1)^2 + (A_{10}z_0 + A_{11}z_1)^2$. Expanding gives $(A_{00}^2 + A_{10}^2)z_0^2 + 2(A_{00}A_{01} + A_{10}A_{11})z_0z_1 + (A_{01}^2 + A_{11}^2)z_1^2$. □

Definition 8.6 (the averaged form; Lean: `ErdosSzekeres.SymReduction.avgQ`, `SYMLEAN/AffineReduction.lean:1116`, Lean: `ErdosSzekeres.SymReduction.qA`, `SYMLEAN/AffineReduction.lean:1120`, Lean: `ErdosSzekeres.SymReduction.qB`, `SYMLEAN/AffineReduction.lean:1121`, Lean: `ErdosSzekeres.SymReduction.qC`, `SYMLEAN/AffineReduction.lean:1122`)**.** For a matrix L, $n \in \mathbb{N}$ and $z \in \mathbb{R}^2$,

$$\mathrm{avgQ}(L, n, z) = \sum_{k=0}^{n-1} \mathrm{nsq}(L^k z),$$

$$q_A(L, n) = \sum_{k=0}^{n-1} g_A(L^k), \qquad q_B(L, n) = \sum_{k=0}^{n-1} g_B(L^k), \qquad q_C(L, n) = \sum_{k=0}^{n-1} g_C(L^k)$$

(Lean `avgQ`, `qA`, `qB`, `qC`; the sums run over `Finset.range` n, and they are 0 for $n = 0$).

Lemma 8.7 (coefficients of the averaged form; Lean: `ErdosSzekeres.SymReduction.avgQ_eq`, `SYMLEAN/AffineReduction.lean:1124`)**.** *For every matrix L, every $n \in \mathbb{N}$ and every $z \in \mathbb{R}^2$,*

$$\mathrm{avgQ}(L, n, z) = q_A(L, n)\, z_0^2 + 2\, q_B(L, n)\, z_0z_1 + q_C(L, n)\, z_1^2.$$

Proof. Apply Lemma 8.5 to $A = L^k$ for each $k < n$ and add the n equalities. On the left this gives $\mathrm{avgQ}(L, n, z)$. On the right, the coefficients of z_0^2, $2z_0z_1$ and z_1^2 add up to $q_A(L, n)$, $q_B(L, n)$ and $q_C(L, n)$. (Lean argues by induction on n; each induction step is the addition of one such equality.) □

Lemma 8.8 (positivity of the averaged form; Lean: `ErdosSzekeres.SymReduction.avgQ_ge`, `SYMLEAN/AffineReduction.lean:1133`, Lean: `ErdosSzekeres.SymReduction.qA_pos`, `SYMLEAN/AffineReduction.lean:1152`, Lean: `ErdosSzekeres.SymReduction.qdisc_pos`, `SYMLEAN/AffineReduction.lean:1158`)**.** *Let L be a matrix and $n \in \mathbb{N}$ with $0 < n$. Then:*

(a) (`avgQ_ge`) $\mathrm{nsq}(z) \le \mathrm{avgQ}(L, n, z)$ *for every $z \in \mathbb{R}^2$;*
(b) (`qA_pos`) $0 < q_A(L, n)$*;*

(c) (`qdisc_pos`) $0 < q_A(L,n)\,q_C(L,n) - q_B(L,n)^2$.

Proof. (a) Since $n > 0$, the sum defining $\mathrm{avgQ}(L,n,z)$ contains the term $k = 0$, which is $\mathrm{nsq}(L^0 z) = \mathrm{nsq}(z)$. The other terms are ≥ 0 by Lemma 8.3(a).

Write $\alpha = q_A(L,n)$, $\beta = q_B(L,n)$, $\gamma = q_C(L,n)$.

(b) Apply (a) to $z = (1,0)$ and use Lemma 8.7: $1 = \mathrm{nsq}(1,0) \leq \mathrm{avgQ}(L,n,(1,0)) = \alpha$. Hence $\alpha \geq 1 > 0$.

(c) Apply (a) to $z = (-\beta, \alpha)$ and use Lemma 8.7:

$$\alpha^2 + \beta^2 = \mathrm{nsq}(-\beta,\alpha) \leq \alpha\beta^2 + 2\beta(-\beta)\alpha + \gamma\alpha^2 = \alpha(\alpha\gamma - \beta^2).$$

Hence $\alpha(\alpha\gamma - \beta^2) \geq \alpha^2 + \beta^2 \geq \alpha^2 > 0$, where the last step uses (b). Since $\alpha > 0$, we get $\alpha\gamma - \beta^2 > 0$. □

Lemma 8.9 (invariance of the averaged form; Lean: `ErdosSzekeres.SymReduction.avgQ_mulVec`, `SYMLEAN/AffineReduction.lean:1139`)**.** *Let L be a matrix and $n \in \mathbb{N}$ with $L^n = 1$. Then $\mathrm{avgQ}(L,n,Lz) = \mathrm{avgQ}(L,n,z)$ for every $z \in \mathbb{R}^2$.*

Proof. Put $f(k) = \mathrm{nsq}(L^k z)$ for $k \in \mathbb{N}$. Since $L^k(Lz) = (L^k L)z = L^{k+1}z$, we have $\mathrm{avgQ}(L,n,Lz) = \sum_{k=0}^{n-1} f(k+1)$. Splitting off the last term, respectively the first term, of $\sum_{k=0}^{n} f(k)$ gives

$$\sum_{k=0}^{n-1} f(k) + f(n) = \sum_{k=0}^{n} f(k) = \sum_{k=0}^{n-1} f(k+1) + f(0).$$

Here $f(n) = \mathrm{nsq}(L^n z) = \mathrm{nsq}(z) = \mathrm{nsq}(L^0 z) = f(0)$, because $L^n = 1 = L^0$. Hence $\sum_{k=0}^{n-1} f(k+1) = \sum_{k=0}^{n-1} f(k) = \mathrm{avgQ}(L,n,z)$. No positivity of n is needed; for $n = 0$ both sides are 0. □

8.3. **Linear maps with explicit matrices; the Cholesky factor.**

Definition 8.10 (the linear map lin2; Lean: `ErdosSzekeres.SymReduction.lin2`, `SYMLEAN/AffineReduction.lean:1167`)**.** For $a,b,c,d \in \mathbb{R}$, $\mathrm{lin2}(a,b,c,d) : \mathbb{R}^2 \to \mathbb{R}^2$ is the $\mathbb{R}$-linear map $p \mapsto (ap_0 + bp_1,\ cp_0 + dp_1)$, that is, multiplication by the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. In Lean it is a bundled linear map; its additivity and homogeneity are discharged inside the definition by coordinate computations.

Lemma 8.11 (coordinates and orientation for lin2; Lean: `ErdosSzekeres.SymReduction.lin2_fst`, `SYMLEAN/AffineReduction.lean:1174`, Lean: `ErdosSzekeres.SymReduction.lin2_snd`, `SYMLEAN/AffineReduction.lean:1175`, Lean: `ErdosSzekeres.SymReduction.orient_lin2`, `SYMLEAN/AffineReduction.lean:1177`)**.** *Let $a,b,c,d \in \mathbb{R}$ and $\Lambda = \mathrm{lin2}(a,b,c,d)$.*

(a) (`lin2_fst`, `lin2_snd`) $(\Lambda p)_0 = ap_0 + bp_1$ *and* $(\Lambda p)_1 = cp_0 + dp_1$ *for every $p \in \mathbb{R}^2$.*

(b) (`orient_lin2`) $\mathrm{orient}(\Lambda x, \Lambda y, \Lambda z) = (ad - bc)\,\mathrm{orient}(x,y,z)$ *for all $x,y,z \in \mathbb{R}^2$.*

Proof. (a) This is the definition. (b) By definition, $\mathrm{orient}(X,Y,Z) = (Y-X)_0(Z-X)_1 - (Y-X)_1(Z-X)_0$. Put $u = y - x$ and $w = z - x$. By linearity, $\Lambda y - \Lambda x = \Lambda u$ and $\Lambda z - \Lambda x = \Lambda w$. Hence, by (a),

$$\mathrm{orient}(\Lambda x, \Lambda y, \Lambda z) = (au_0 + bu_1)(cw_0 + dw_1) - (cu_0 + du_1)(aw_0 + bw_1) = (ad - bc)(u_0 w_1 - u_1 w_0),$$

and $u_0 w_1 - u_1 w_0 = \mathrm{orient}(x,y,z)$. This is the case $v = 0$ of Lemma 1.17. □

Lemma 8.12 (the Cholesky factor and its inverse; Lean: `ErdosSzekeres.SymReduction.nsq_lin2_chol`, `SYMLEAN/AffineReduction.lean:1184`, Lean: `ErdosSzekeres.SymReduction.lin2_chol_inv_left`, `SYMLEAN/AffineReduction.lean:1191`, Lean: `ErdosSzekeres.SymReduction.lin2_chol_inv_right`, `SYMLEAN/AffineReduction.lean:1195`)**.** *Let $s,t,\beta \in \mathbb{R}$. Put $T = \mathrm{lin2}(s,\beta/s,0,t)$ and $T' = \mathrm{lin2}(1/s, -\beta/(s^2 t), 0, 1/t)$. (Division follows the Lean convention $x/0 = 0$, so T and T' are defined for all s,t,β; each part below states the non-vanishing it assumes.)*

(a) (`nsq_lin2_chol`) *If $s \neq 0$, then for every $z \in \mathbb{R}^2$,*

$$\mathrm{nsq}(Tz) = s^2 z_0^2 + 2\beta z_0 z_1 + \Big(\frac{\beta^2}{s^2} + t^2\Big) z_1^2.$$

(b) (`lin2_chol_inv_left`) *If $s \neq 0$ and $t \neq 0$, then $T'(Tx) = x$ for every $x \in \mathbb{R}^2$.*

(c) (`lin2_chol_inv_right`) *If $s \neq 0$ and $t \neq 0$, then $T(T'x) = x$ for every $x \in \mathbb{R}^2$.*

Proof. By Lemma 8.11(a), $Tz = (sz_0 + (\beta/s)z_1,\ tz_1)$ and $T'x = (x_0/s - \beta x_1/(s^2 t),\ x_1/t)$.

(a) $\mathrm{nsq}(Tz) = s^2 z_0^2 + 2s(\beta/s)z_0 z_1 + (\beta/s)^2 z_1^2 + t^2 z_1^2$, and $s(\beta/s) = \beta$ because $s \neq 0$.

(b) The first coordinate of $T'(Tx)$ is $\frac{1}{s}\big(sx_0 + \frac{\beta}{s}x_1\big) - \frac{\beta}{s^2 t}\,tx_1 = x_0 + \frac{\beta x_1}{s^2} - \frac{\beta x_1}{s^2} = x_0$. The second is $\frac{1}{t}\,tx_1 = x_1$.

(c) The first coordinate of $T(T'x)$ is $s\big(\frac{x_0}{s} - \frac{\beta x_1}{s^2 t}\big) + \frac{\beta}{s}\cdot\frac{x_1}{t} = x_0 - \frac{\beta x_1}{st} + \frac{\beta x_1}{st} = x_0$. The second is $t\cdot\frac{x_1}{t} = x_1$. □

8.4. **Affine maps in matrix form.** In this subsection, $g : \mathbb{R}^2 \to \mathbb{R}^2$ is a map, L a matrix and $b \in \mathbb{R}^2$; "$g(p) = Lp + b$ for all p" is the Lean hypothesis `hg`.

Lemma 8.13 (differences of iterates; Lean: `ErdosSzekeres.SymReduction.iterate_sub_of_affine`, `SYMLEAN/AffineReduction.lean:1199`)**.** *If $g(p) = Lp + b$ for all $p \in \mathbb{R}^2$, then $g^k(p) - g^k(q) = L^k(p - q)$ for all $k \in \mathbb{N}$ and $p, q \in \mathbb{R}^2$.*

Proof. Induction on k. For $k = 0$ both sides equal $p - q$, as $L^0 = 1$. If the claim holds for k, then

$$g^{k+1}(p) - g^{k+1}(q) = \big(Lg^k(p) + b\big) - \big(Lg^k(q) + b\big) = L\big(g^k(p) - g^k(q)\big) = L\big(L^k(p - q)\big) = L^{k+1}(p - q). \qquad \square$$

Lemma 8.14 (periodic points; Lean: `ErdosSzekeres.SymReduction.exists_periodic_of_injective`, `SYMLEAN/AffineReduction.lean:1208`)**.** *Let $g : \mathbb{R}^2 \to \mathbb{R}^2$ be injective, let S be a finite set that g maps into itself, and let $q \in S$. Then there is $m \in \mathbb{N}$ with $0 < m$ and $g^m(q) = q$.*

Proof. By induction on i, $g^i(q) \in S$ for every $i \in \mathbb{N}$: $g^0(q) = q \in S$, and $g^{i+1}(q) = g(g^i(q)) \in S$ because g maps S into itself. (In Lean this is `iterate_mem`, `SYMLEAN/AffineReduction.lean:69`.) Since $\mathbb{N}$ is infinite and S is finite, the map $i \mapsto g^i(q)$ from $\mathbb{N}$ to S is not injective (pigeonhole). So there are $i < j$ with $g^i(q) = g^j(q)$. Put $m = j - i > 0$. Then $g^i(g^m(q)) = g^{i+m}(q) = g^j(q) = g^i(q)$. The map g^i is injective, as a composition of injective maps, so $g^m(q) = q$. $\square$

Lemma 8.15 (a matrix fixing two independent vectors; Lean: `ErdosSzekeres.SymReduction.matrix_eq_one_of_fix`, `SYMLEAN/AffineReduction.lean:1217`)**.** *Let M be a matrix and $u, w \in \mathbb{R}^2$ with $u_0w_1 - u_1w_0 \neq 0$, $Mu = u$ and $Mw = w$. Then $M = 1$.*

Proof. Put $\delta = u_0w_1 - u_1w_0$. By Lemma 8.2, the hypotheses say that the four quantities

$$E_1 = M_{00}u_0 + M_{01}u_1 - u_0, \quad E_2 = M_{10}u_0 + M_{11}u_1 - u_1, \quad E_3 = M_{00}w_0 + M_{01}w_1 - w_0, \quad E_4 = M_{10}w_0 + M_{11}w_1 - w_1$$

vanish. Expanding gives the polynomial identities

$$w_1E_1 - u_1E_3 = (M_{00} - 1)\delta, \quad -w_0E_1 + u_0E_3 = M_{01}\delta, \quad w_1E_2 - u_1E_4 = M_{10}\delta, \quad -w_0E_2 + u_0E_4 = (M_{11} - 1)\delta.$$

So all four right-hand sides vanish. As $\delta \neq 0$, we get $M_{00} = 1$, $M_{01} = 0$, $M_{10} = 0$ and $M_{11} = 1$, that is, $M = 1$. $\square$

Lemma 8.16 (orientation under affine maps; Lean: `ErdosSzekeres.SymReduction.orient_affine`, `SYMLEAN/AffineReduction.lean:1241`)**.** *If $g(p) = Lp + b$ for all $p \in \mathbb{R}^2$, then* $\operatorname{orient}(g(x), g(y), g(z)) = \det(L)\operatorname{orient}(x, y, z)$ *for all $x, y, z \in \mathbb{R}^2$.*

Proof. We have $g(y) - g(x) = L(y - x)$ and $g(z) - g(x) = L(z - x)$, since the translation b cancels. So the computation in the proof of Lemma 8.11(b), with the entries $L_{00}, L_{01}, L_{10}, L_{11}$ in place of the four parameters of lin2, gives the factor $L_{00}L_{11} - L_{01}L_{10} = \det L$. This is Lemma 1.17. $\square$

Lemma 8.17 (injectivity; Lean: `ErdosSzekeres.SymReduction.injective_of_affine`, `SYMLEAN/AffineReduction.lean:1305`)**.** *If $g(p) = Lp + b$ for all $p \in \mathbb{R}^2$ and $\det L \neq 0$, then g is injective.*

Proof. Let $g(p) = g(q)$ and put $v = p - q$. Then $Lv = Lp - Lq = (Lp + b) - (Lq + b) = g(p) - g(q) = 0$. So $h_0 := L_{00}v_0 + L_{01}v_1 = 0$ and $h_1 := L_{10}v_0 + L_{11}v_1 = 0$ by Lemma 8.2. Then

$$(\det L)\, v_0 = L_{11}h_0 - L_{01}h_1 = 0, \qquad (\det L)\, v_1 = L_{00}h_1 - L_{10}h_0 = 0,$$

so $v = 0$ because $\det L \neq 0$. Hence $p = q$. $\square$

Lemma 8.18 (matrix form of a Mathlib affine map; Lean: `ErdosSzekeres.SymReduction.affineMap_eq_mulVec`, `SYMLEAN/AffineReduction.lean:1284`)**.** *Let f be an affine map of $\mathbb{R}^2$ (Mathlib* `AffineMap`*) and let M_f be the matrix of its linear part f_{lin} (Lean* `LinearMap.toMatrix'` *applied to* `f.linear`*). Then $f(p) = M_fp + f(0)$ for every $p \in \mathbb{R}^2$.*

Proof. By the defining property of an affine map with $v = p$ and base point 0, $f(p) = f(p + 0) = f_{\mathrm{lin}}(p) + f(0)$ (in Mathlib, `AffineMap.decomp`). By the conventions, $f_{\mathrm{lin}}(p) = M_fp$. $\square$

8.5. **Transport of general position and convex position.**

Lemma 8.19 (general position under orientation-scaling maps; Lean: `ErdosSzekeres.SymReduction.generalPosition_image_of_orient`, `SYMLEAN/AffineReduction.lean:1249`)**.** *Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be a map and $D \in \mathbb{R}$ with $D \neq 0$ and* $\operatorname{orient}(f(x), f(y), f(z)) = D \operatorname{orient}(x, y, z)$ *for all $x, y, z \in \mathbb{R}^2$. If a finite set S is in general position, then so is $f(S)$.*

Proof. By Lemma 1.20 it suffices to show $\operatorname{orient}(X, Y, Z) \neq 0$ for pairwise distinct $X, Y, Z \in f(S)$. Write $X = f(x)$, $Y = f(y)$, $Z = f(z)$ with $x, y, z \in S$. Then x, y, z are pairwise distinct: for instance $x = y$ would give $X = Y$. By Lemma 1.20 again, $\operatorname{orient}(x, y, z) \neq 0$, and so $\operatorname{orient}(X, Y, Z) = D \operatorname{orient}(x, y, z) \neq 0$. Injectivity of f is not assumed and not needed. □

Lemma 8.20 (general position under affine maps; Lean: `ErdosSzekeres.SymReduction.generalPosition_image_affine`, `SYMLEAN/AffineReduction.lean:1278`, Lean: `ErdosSzekeres.SymReduction.generalPosition_image_affineEquiv`, `SYMLEAN/AffineReduction.lean:1290`)**.** *The following hold.*

(a) (`generalPosition_image_affine`) *Let $g : \mathbb{R}^2 \to \mathbb{R}^2$, L a matrix and $b \in \mathbb{R}^2$ with $g(p) = Lp + b$ for all p, and $\det L \neq 0$. If a finite set S is in general position, then so is $g(S)$.*

(b) (`generalPosition_image_affineEquiv`) *If a finite set S is in general position and f is an affine equivalence of $\mathbb{R}^2$ (Mathlib* `AffineEquiv`*), then $f(S)$ is in general position.*

Proof. (a) By Lemma 8.16, the map g satisfies the hypothesis of Lemma 8.19 with $D = \det L \neq 0$.

(b) By Lemma 8.18, applied to the underlying affine map, $f(p) = Mp + f(0)$ for all p, where M is the matrix of f_{lin}. The linear bijection f_{lin} has a linear inverse ψ. The product MM_ψ is the matrix of $f_{\mathrm{lin}} \circ \psi = \mathrm{id}$, so $MM_\psi = 1$ and $\det M \cdot \det M_\psi = 1$; hence $\det M \neq 0$. (Lean uses `LinearMap.det_toMatrix'` and `LinearEquiv.isUnit_det'`.) Now apply (a). □

Lemma 8.21 (convex position under injective affine maps; Lean: `ErdosSzekeres.SymReduction.convexPosition_image_affine`, `SYMLEAN/AffineReduction.lean:1263`)**.** *Let S be a finite set in convex position and let f be an affine map of $\mathbb{R}^2$ (Mathlib* `AffineMap`*) that is injective. Then $f(S)$ is in convex position.*

Proof. Step 1: $f(\operatorname{conv} A) = \operatorname{conv}(f(A))$ for every $A \subseteq \mathbb{R}^2$ (in Mathlib, `AffineMap.image_convexHull`). By Lemma 8.18, $f(p) = Mp + c$ with $M = M_f$ and $c = f(0)$. So for $x, y \in \mathbb{R}^2$ and $\lambda \in [0, 1]$,

$$f\big((1-\lambda)x + \lambda y\big) = (1-\lambda)Mx + \lambda My + \big((1-\lambda) + \lambda\big)c = (1-\lambda)f(x) + \lambda f(y).$$

Hence images and preimages of convex sets under f are convex. Let C be convex. If $X = f(x)$ and $Y = f(y)$ with $x, y \in C$, then $(1-\lambda)x + \lambda y \in C$, so $(1-\lambda)X + \lambda Y = f((1-\lambda)x + \lambda y) \in f(C)$. Let C' be convex. If $f(x), f(y) \in C'$, then $f((1-\lambda)x + \lambda y) = (1-\lambda)f(x) + \lambda f(y) \in C'$, so $(1-\lambda)x + \lambda y \in f^{-1}(C')$. Now $f(\operatorname{conv} A)$ is a convex set containing $f(A)$, so it contains $\operatorname{conv}(f(A))$. And $f^{-1}(\operatorname{conv}(f(A)))$ is a convex set containing A, so it contains $\operatorname{conv} A$; that is, $f(\operatorname{conv} A) \subseteq \operatorname{conv}(f(A))$.

Step 2. By Lemma 1.22 we must show $P \notin \operatorname{conv}(f(S) \setminus \{P\})$ for every $P \in f(S)$. Write $P = f(q)$ with $q \in S$. Since f is injective, $f(S) \setminus \{f(q)\} = f(S \setminus \{q\})$. Indeed, if $r \in S$ and $f(r) \neq f(q)$ then $r \neq q$; conversely, if $r \in S$ and $r \neq q$ then $f(r) \neq f(q)$. Suppose $f(q) \in \operatorname{conv}(f(S) \setminus \{f(q)\})$. By this identity and Step 1, this set is $\operatorname{conv}(f(S \setminus \{q\})) = f(\operatorname{conv}(S \setminus \{q\}))$. So $f(q) = f(r)$ for some $r \in \operatorname{conv}(S \setminus \{q\})$. Injectivity gives $q = r \in \operatorname{conv}(S \setminus \{q\})$, which contradicts the convex position of S (Lemma 1.22). □

Lemma 8.22 (convex n-gons under injective affine maps; Lean: `ErdosSzekeres.SymReduction.hasConvexNGon_image_affine`, `SYMLEAN/AffineReduction.lean:1298`)**.** *Let $n \in \mathbb{N}$, let S be a finite set with* `HasConvexNGon` *n S, and let f be an injective affine map of $\mathbb{R}^2$ (Mathlib* `AffineMap`*). Then* `HasConvexNGon` *n holds for $f(S)$.*

Proof. Take $U \subseteq S$ with $|U| = n$ in convex position. Then $f(U) \subseteq f(S)$, $|f(U)| = |U| = n$ since f is injective, and $f(U)$ is in convex position by Lemma 8.21. □

8.6. **The transfer theorem.**

Definition 8.23 (the Euclidean statement $\mathcal{E}(N, n)$; written step)**.** For $N, n \in \mathbb{N}$, $\mathcal{E}(N, n)$ is the following proposition. *For every finite set $S' \subseteq \mathbb{R}^2$ and every map $h : \mathbb{R}^2 \to \mathbb{R}^2$: if S' is in general position, $N \leq |S'|$, h is a Euclidean isometry (that is,* $\mathrm{eDist2}(h(p), h(q)) = \mathrm{eDist2}(p, q)$ *for all $p, q \in \mathbb{R}^2$), $h \neq \mathrm{id}$, and $h(p) \in S'$ for all $p \in S'$, then* `HasConvexNGon` *n S'.* This is the type of the hypothesis `hE` of Theorem 8.25 and Corollary 8.27, up to two changes of notation: the bound variables, `S` and `g` in Lean, are called S' and h here, and `IsEucIsometry`

and `Invariant` are unfolded. The quantifiers, the five premises and their order, and the conclusion are those of Lean. It has no name in Lean.

The next proposition collects the steps of the proof of Theorem 8.25 that construct the Euclidean isometry. It is stated separately because the remark on affine symmetries in `WRITEUP/P1/p1.tex` (Remark `rem:affine`) uses this construction.

Proposition 8.24 (averaging an affine symmetry; written step)**.** *Let S be a finite set in general position with $|S| \geq 3$. Let L be a matrix, $b \in \mathbb{R}^2$ and $g : \mathbb{R}^2 \to \mathbb{R}^2$ with $g(p) = Lp + b$ for all p. Assume that g is injective and maps S into itself.*

(a) *There is an integer $m \geq 1$ with $L^m = 1$ and $g^m = \mathrm{id}$.*

(b) *Fix any integer $m \geq 1$ with $L^m = 1$, and put $\alpha = q_A(L,m)$, $\beta = q_B(L,m)$, $\gamma = q_C(L,m)$. Then $\alpha > 0$ and $\alpha\gamma - \beta^2 > 0$. Put*

$$s = \sqrt{\alpha}, \qquad t = \sqrt{(\alpha\gamma - \beta^2)/\alpha}, \qquad T = \mathrm{lin2}(s, \beta/s, 0, t), \qquad T' = \mathrm{lin2}(1/s,\, -\beta/(s^2 t),\, 0,\, 1/t).$$

Then $s > 0$ and $t > 0$. The maps T and T' are mutually inverse linear bijections of $\mathbb{R}^2$. Moreover $\mathrm{orient}(Tx, Ty, Tz) = st \cdot \mathrm{orient}(x, y, z)$ for all x, y, z, with $st > 0$, and $\mathrm{nsq}(Tz) = \mathrm{avgQ}(L, m, z)$ for every z.

(c) *With m, T, T' as in* (b)*, the map $h = T \circ g \circ T'$ is a Euclidean isometry, and h maps $T(S)$ into $T(S)$.*

(d) *$h^k = T \circ g^k \circ T'$ for every $k \in \mathbb{N}$. Hence $h^k = \mathrm{id}$ if and only if $g^k = \mathrm{id}$. In particular, $h \neq \mathrm{id}$ if and only if $g \neq \mathrm{id}$, and h and g have the same order (the least $k \geq 1$ with $g^k = \mathrm{id}$, which exists by* (a)*).*

Proof. (a) Since $|S| \geq 3$, S contains pairwise distinct points p_0, p_1, p_2. By general position and Lemma 1.20, $\delta := \mathrm{orient}(p_0, p_1, p_2) \neq 0$. By Lemma 8.14 there are $m_0, m_1, m_2 \geq 1$ with $g^{m_i}(p_i) = p_i$. Put $m = m_0 m_1 m_2 \geq 1$. For each i, $m = r_i m_i$ for an integer $r_i \geq 1$. By induction on r, $g^{r m_i}(p_i) = p_i$: for $r = 0$ this is $g^0(p_i) = p_i$, and $g^{(r+1)m_i}(p_i) = g^{m_i}(g^{r m_i}(p_i)) = g^{m_i}(p_i) = p_i$. So $g^m(p_i) = p_i$ for $i = 0, 1, 2$.

Put $u = p_1 - p_0$ and $w = p_2 - p_0$. By Lemma 8.13, $L^m u = g^m(p_1) - g^m(p_0) = p_1 - p_0 = u$, and likewise $L^m w = w$. By the definition of orient, $u_0 w_1 - u_1 w_0 = \mathrm{orient}(p_0, p_1, p_2) = \delta \neq 0$. So Lemma 8.15 gives $L^m = 1$.

Finally, for every $p \in \mathbb{R}^2$, Lemma 8.13 gives $g^m(p) - g^m(p_0) = L^m(p - p_0) = p - p_0$. Since $g^m(p_0) = p_0$, this gives $g^m(p) = p$. So $g^m = \mathrm{id}$. (The Lean proof derives $L^m = 1$ in the same way, as its step `hLm`. It does not derive or use $g^m = \mathrm{id}$.)

(b) By Lemma 8.8(b),(c) with $n = m \geq 1$, $\alpha > 0$ and $\alpha\gamma - \beta^2 > 0$. So $(\alpha\gamma - \beta^2)/\alpha > 0$; hence $s > 0$, $t > 0$, $s^2 = \alpha$ and $t^2 = (\alpha\gamma - \beta^2)/\alpha$. As $s \neq 0$ and $t \neq 0$, Lemma 8.12(b),(c) gives $T'(Tx) = x$ and $T(T'x) = x$ for all x. So T and T' are mutually inverse bijections; they are linear by Definition 8.10. By Lemma 8.11(b), applied to $T = \mathrm{lin2}(s, \beta/s, 0, t)$, the orientation factor is $st - (\beta/s) \cdot 0 = st > 0$. Finally, by Lemma 8.12(a), $\mathrm{nsq}(Tz) = s^2 z_0^2 + 2\beta z_0 z_1 + (\beta^2/s^2 + t^2) z_1^2$. Here $s^2 = \alpha$ and

$$\frac{\beta^2}{s^2} + t^2 = \frac{\beta^2}{\alpha} + \frac{\alpha\gamma - \beta^2}{\alpha} = \gamma.$$

So $\mathrm{nsq}(Tz) = \alpha z_0^2 + 2\beta z_0 z_1 + \gamma z_1^2 = \mathrm{avgQ}(L, m, z)$ by Lemma 8.7.

(c) Let $x, y \in \mathbb{R}^2$ and put $w = T'x - T'y$. Since $g(T'x) - g(T'y) = (LT'x + b) - (LT'y + b) = Lw$ and T is linear,

$$\begin{aligned}
\mathrm{eDist2}(h(x), h(y)) &= \mathrm{nsq}\big(T(g(T'x)) - T(g(T'y))\big) && \text{Lemma 8.3(b)} \\
&= \mathrm{nsq}\big(T(Lw)\big) = \mathrm{avgQ}(L, m, Lw) && \text{linearity of } T\text{; part (b)} \\
&= \mathrm{avgQ}(L, m, w) = \mathrm{nsq}(Tw) && \text{Lemma 8.9 } (L^m = 1)\text{; part (b)} \\
&= \mathrm{nsq}\big(T(T'x) - T(T'y)\big) = \mathrm{nsq}(x - y) && \text{linearity of } T\text{; } TT' = \mathrm{id} \\
&= \mathrm{eDist2}(x, y) && \text{Lemma 8.3(b).}
\end{aligned}$$

So h is a Euclidean isometry. If $X \in T(S)$, write $X = Tp$ with $p \in S$. Then $h(X) = T(g(T'(Tp))) = T(g(p)) \in T(S)$, because $g(p) \in S$.

(d) Induction on k. For $k = 0$: $h^0 = \mathrm{id} = T \circ T' = T \circ g^0 \circ T'$. If $h^k = T \circ g^k \circ T'$, then $h^{k+1} = h \circ h^k = T \circ g \circ (T' \circ T) \circ g^k \circ T' = T \circ g^{k+1} \circ T'$, because $T' \circ T = \mathrm{id}$.

If $g^k = \mathrm{id}$, then $h^k = T \circ T' = \mathrm{id}$. If $h^k = \mathrm{id}$, then for every p, $T(g^k(p)) = T(g^k(T'(Tp))) = h^k(Tp) = Tp$, so $g^k(p) = p$ because T is injective. The case $k = 1$ gives $h \neq \mathrm{id}$ if and only if $g \neq \mathrm{id}$. Since $h^k = \mathrm{id} \Leftrightarrow g^k = \mathrm{id}$ for every k, the least $k \geq 1$ with this property is the same for g and h.

Relation to Lean. Parts (b) and (c) are steps of the Lean proof of `hasConvexNGon_of_affine_symmetry`, where T' is called `Ti`. These steps are: the positivity of α, of $\alpha\gamma-\beta^2$, of s and of t; the inverse identities `hTiT` and `hTTi`, and the injectivity of T and T' derived from them (`hTinj`, `hTiinj`); the identity `hnT`; the orientation factor $st-(\beta/s)\cdot 0$ of T, which Lean uses only inside the step `hgp'` (general position of $T(S)$, through `generalPosition_image_of_orient` and `orient_lin2`); and the steps `hiso` and `hinv'`. Of part (d), only the implication $g\neq\mathrm{id}\Rightarrow h\neq\mathrm{id}$ is in Lean (step `hne'`). □

Theorem 8.25 (affine-to-Euclidean transfer; Lean: `ErdosSzekeres.SymReduction.hasConvexNGon_of_affine_symmetry`, `SYMLEAN/AffineReduction.lean:1330`)**.** *Let $N,n\in\mathbb{N}$. Assume*

(hE) *for every finite set $S'\subseteq\mathbb{R}^2$ and every map $h:\mathbb{R}^2\to\mathbb{R}^2$: if S' is in general position, $N\le|S'|$, h is a Euclidean isometry* (`IsEucIsometry` *h:* $\mathrm{eDist2}(h(p),h(q))=\mathrm{eDist2}(p,q)$ *for all $p,q\in\mathbb{R}^2$), $h\neq\mathrm{id}$, and $h(p)\in S'$ for all $p\in S'$, then* `HasConvexNGon` *n S'*

(that is, assume $\mathcal{E}(N,n)$ of Definition 8.23*), and assume $3\le N$. Let S be a finite set, $g:\mathbb{R}^2\to\mathbb{R}^2$ a map, L a matrix and $b\in\mathbb{R}^2$, such that $g(p)=Lp+b$ for every $p\in\mathbb{R}^2$, g is injective, S is in general position, $N\le|S|$, $g\neq\mathrm{id}$, and $g(p)\in S$ for every $p\in S$. Then* `HasConvexNGon` *n S: some n points of S are in convex position.*

Proof. Since $|S|\ge N\ge 3$, Proposition 8.24 applies. Fix $m\ge 1$ with $L^m=1$ (part (a)), and let T, T' and $h=T\circ g\circ T'$ be as in parts (b) and (c). We check the hypotheses of (hE) for $S'=T(S)$ and h.

- $T(S)$ is in general position, by Lemma 8.19 with $f=T$ and $D=st\neq 0$ (Proposition 8.24(b)).
- $N\le|S|=|T(S)|$, because T is injective ($T'\circ T=\mathrm{id}$).
- h is a Euclidean isometry and maps $T(S)$ into $T(S)$ (Proposition 8.24(c)).
- $h\neq\mathrm{id}$, by Proposition 8.24(d), since $g\neq\mathrm{id}$.

So (hE) gives $U\subseteq T(S)$ with $|U|=n$ in convex position. Put $V=T'(U)$.

- $V\subseteq S$: every $y\in V$ is $y=T'(x)$ with $x\in U\subseteq T(S)$, so $x=Tp$ with $p\in S$, and $y=T'(Tp)=p\in S$.
- $|V|=|U|=n$, because T' is injective ($T\circ T'=\mathrm{id}$).
- V is in convex position, by Lemma 8.21 applied to T'. This is a linear map, hence an affine map (Lean `Ti.toAffineMap`), and it is injective.

Hence `HasConvexNGon` n S. □

Remark 8.26 (the hypothesis (hE), and what the statement forgets). The following three points concern the hypotheses of Theorem 8.25 and what its statement does not record.

(1) The hypothesis (hE) is a parameter of Theorem 8.25: the Euclidean statement $\mathcal{E}(N,n)$ for the *same* N and n. The theorem is the implication from $\mathcal{E}(N,n)$ to the corresponding statement for injective affine maps. It involves no unproved claim of the project.

(2) (hE) is used once, for the set $T(S)$ and the isometry $h=T\circ g\circ T'$. Since (hE) quantifies over *all* Euclidean isometries other than the identity, the statement does not record which isometry h is. The written Proposition 8.24(d) adds that h has the same order as g. This is not part of any Lean statement. An order-preserving form of the transfer, whose Euclidean hypothesis concerns only isometries of the order of g, is not formalized. The remark on affine symmetries in `WRITEUP/P1/p1.tex` (Remark `rem:affine`) needs such a form, and says that it is not formalized.

(3) The hypothesis $3\le N$ is used only to find three distinct points of S. Injectivity of g is a separate hypothesis. It follows from $\det L\neq 0$ (Lemma 8.17), and that is how Corollary 8.27(c) supplies it.

Corollary 8.27 (the transfer for other presentations of the map; Lean: `ErdosSzekeres.SymReduction.hasConvexNGon_of_affineMap_symmetry`, `SYMLEAN/AffineReduction.lean:1423`, Lean: `ErdosSzekeres.SymReduction.hasConvexNGon_of_affineEquiv_symmetry`, `SYMLEAN/AffineReduction.lean:1433`, Lean: `ErdosSzekeres.SymReduction.hasConvexNGon_of_affine_coords_symmetry`, `SYMLEAN/AffineReduction.lean:1444`)**.** *Let $N,n\in\mathbb{N}$. Assume* (hE)*, exactly as in Theorem* 8.25 *(the statement $\mathcal{E}(N,n)$), and assume $3\le N$. Let S be a finite set in general position with $N\le|S|$.*

(a) (`hasConvexNGon_of_affineMap_symmetry`) *If f is an affine map of $\mathbb{R}^2$ (Mathlib* `AffineMap`*) that is injective, $f\neq\mathrm{id}$ as maps, and $f(p)\in S$ for all $p\in S$, then* `HasConvexNGon` *n S.*

(b) (`hasConvexNGon_of_affineEquiv_symmetry`) *If f is an affine equivalence of $\mathbb{R}^2$ (Mathlib* `AffineEquiv`*) with $f\neq\mathrm{refl}$ and $f(p)\in S$ for all $p\in S$, then* `HasConvexNGon` *n S.*

(c) (`hasConvexNGon_of_affine_coords_symmetry`) *If $g:\mathbb{R}^2\to\mathbb{R}^2$ is a map and $a,b,c,d,e,f\in\mathbb{R}$ (here f is a real number) satisfy $ad-bc\neq 0$, $g(p)=(ap_0+bp_1+e,\ cp_0+dp_1+f)$ for every $p\in\mathbb{R}^2$, $g\neq\mathrm{id}$, and $g(p)\in S$ for all $p\in S$, then* `HasConvexNGon` *n S.*

Proof. (a) By Lemma 8.18, $f(p) = M_f p + f(0)$ for every p. Apply Theorem 8.25 with $g = f$, $L = M_f$ and $b = f(0)$; the remaining hypotheses are those of (a).

(b) The underlying affine map of f has the same map, which is injective because f is a bijection. It differs from id: if the map f were id, then f and refl would have the same map, so $f =$ refl by `AffineEquiv.ext`, a contradiction. Now apply (a).

(c) Let $L = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $b' = (e, f)$. By Lemma 8.2, $Lp + b' = (ap_0 + bp_1 + e,\ cp_0 + dp_1 + f) = g(p)$ for every p. Moreover $\det L = ad - bc \neq 0$ (Mathlib `Matrix.det_fin_two_of`), so g is injective by Lemma 8.17. Apply Theorem 8.25. □

8.7. **Affine symmetries and convex heptagons.**

Theorem 8.28 (convex heptagons from affine symmetries, conditional on the five case classes; Lean: `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affineEquiv`, `SYMLEAN/AffineReduction.lean:1501`, Lean: `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affineEquiv_image`, `SYMLEAN/AffineReduction.lean:1510`, Lean: `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affineMap`, `SYMLEAN/AffineReduction.lean:1489`, Lean: `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affine`, `SYMLEAN/AffineReduction.lean:1477`, Lean: `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affine_coords`, `SYMLEAN/AffineReduction.lean:1521`). *Assume, as hypotheses, the five propositions* `HalfTurn34`, `Threefold33`, `Fivefold35`, `MirrorOneAxis33` *and* `MirrorNoAxis34` *of Definition* 7.7. *Let $S \subseteq \mathbb{R}^2$ be a finite set in general position with $33 \leq |S|$.*

(a) (`heptagon_of_nontrivial_affineEquiv`) *If g is an affine equivalence of $\mathbb{R}^2$ (Mathlib* `AffineEquiv`*) with $g \neq$* refl *and $g(p) \in S$ for all $p \in S$, then* `HasConvexNGon 7 S`.

(b) (`heptagon_of_nontrivial_affineEquiv_image`) *If g is an affine equivalence of $\mathbb{R}^2$ with $g \neq$* refl *and $g(S) \subseteq S$ (an inclusion of sets: the image of S under g is contained in S), then* `HasConvexNGon 7 S`.

(c) (`heptagon_of_nontrivial_affineMap`) *If f is an affine map of $\mathbb{R}^2$ (Mathlib* `AffineMap`*) that is injective, $f \neq$ id as maps, and $f(p) \in S$ for all $p \in S$, then* `HasConvexNGon 7 S`.

(d) (`heptagon_of_nontrivial_affine`) *If $g : \mathbb{R}^2 \to \mathbb{R}^2$ is a map, L a matrix and $b \in \mathbb{R}^2$, with $g(p) = Lp + b$ for every p, g injective, $g \neq$ id, and $g(p) \in S$ for all $p \in S$, then* `HasConvexNGon 7 S`.

(e) (`heptagon_of_nontrivial_affine_coords`) *If $g : \mathbb{R}^2 \to \mathbb{R}^2$ is a map and $a, b, c, d, e, f \in \mathbb{R}$ (in this part b and f are real numbers) satisfy $ad - bc \neq 0$, $g(p) = (ap_0 + bp_1 + e,\ cp_0 + dp_1 + f)$ for every $p \in \mathbb{R}^2$, $g \neq$ id, and $g(p) \in S$ for all $p \in S$, then* `HasConvexNGon 7 S`.

In each case, `HasConvexNGon 7 S` *means that some* 7 *points of S are in convex position.*

Proof. By Theorem 7.39 (`heptagon_of_nontrivial_isometry`, applied to the five hypotheses), the following holds. *For every finite set S' and every map $h : \mathbb{R}^2 \to \mathbb{R}^2$: if S' is in general position, $33 \leq |S'|$, h is a Euclidean isometry, $h \neq$ id and h maps S' into S', then* `HasConvexNGon 7 S'`. This is exactly $\mathcal{E}(33, 7)$, the hypothesis (hE) with $N = 33$ and $n = 7$. Moreover $3 \leq 33$.

(a) Apply Corollary 8.27(b).

(b) For $p \in S$ we have $g(p) \in g(S) \subseteq S$, so g maps S into S. Now apply (a).

(c) Apply Corollary 8.27(a).

(d) Apply Theorem 8.25.

(e) Apply Corollary 8.27(c).

The Lean proofs are the same applications, each a two-line term. For (a), (c), (d) and (e) they pass `heptagon_of_nontrivial_isometry`, applied to the five hypotheses `h2`, `h3`, `h5`, `hm1` and `hm0`, as `hE`, and prove $3 \leq 33$ by `norm_num`. For (b) the Lean proof passes no `hE`: it applies `heptagon_of_nontrivial_affineEquiv`, that is (a), after turning the inclusion into maps-into. □

Remark 8.29 (the five hypotheses). Theorem 8.28 is a conditional statement. Its five hypotheses enter only through Theorem 7.39. In Lean they are definitions of type `Prop` that occur as explicit premises (`h2`, `h3`, `h5`, `hm1`, `hm0`), not axioms. In short, each class says that a finite set in general position contains 7 points in convex position if it has one of the following: exactly 34 points and a half-turn symmetry whose centre is not in the set; exactly 33 points and a symmetry under the rotation by $2\pi/3$ about a centre not in the set; exactly 35 points and a symmetry under the rotation by $2\pi/5$ about a centre not in the set; exactly 33 points, a reflection symmetry in a line, and exactly one point of the set on that line; exactly 34 points, a reflection symmetry, and no point of the set on the line. See Definition 7.7 for the exact statements. In the project they are supplied as follows (`SYMLEAN/README.md`, §§5, 6 and 9; `WRITEUP/SYMMETRIC.md`, §§1, 3 and 4).

- `MirrorOneAxis33` and `MirrorNoAxis34` are Lean-checked theorems: `mirrorOneAxis33_proved` (line 4274) and `mirrorNoAxis34_proved` (line 4282) of `SYMLEAN/mirror/ReductionThreeClasses.lean` (Theorem 10.39). These two theorems are about the copies of the class definitions in that file. It does not import `SYMLEAN/AffineReduction.lean`; lines 38–1559 of `SYMLEAN/mirror/ReductionThreeClasses.lean` are byte-identical to lines 12–1533 of `SYMLEAN/AffineReduction.lean`. So the two theorems discharge the premises `hm1` and `hm0` of Theorem 8.28 only through the identity of the two texts. Their axioms log `SYMLEAN/mirror/axioms.out` lists only `propext`, `Classical.choice` and `Quot.sound`. They are obtained without SAT input from the repository's Lean formalization of Baek's theorem (Theorem 10.4) and the mirror reduction. The same file restates part (a) with only the three rotation classes as hypotheses (`heptagon_of_nontrivial_affineEquiv_3`, line 4303; Theorem 10.40). Both mirror classes also have SAT certificates (record `SYMZ`).
- `HalfTurn34`, `Threefold33` and `Fivefold35` are not Lean-checked. They are computer-assisted claims of the records `C2FINAL` (with `C2C34` and `C2P`–`C2P5`), `C3FINAL` and `SYM5`. They rest on four kinds of evidence.
  - *SAT certificates.* Every certificate that these claims use, including those of the pieces of their case splits, carries a recorded acceptance by the proof checker `cake_lpr`. Since a revision of 6 November 2026 this is the only checker in the project's trust base; the second checker, `lrat-check`, is a secondary check only (`WRITEUP/SYMMETRIC.md`, §3; `SYMLEAN/README.md`, §9, which predates that revision, still names both). For the parallelogram pieces of `HalfTurn34` and the 116,883 pieces of `Threefold33`, the proof files were deleted after checking, so for most of these pieces only the recorded verdicts remain. Some of these pieces were re-solved (`C2FINAL`, §§4, 5 and 9; `C2RECHECK`; `C3SPOT`; `C3PIECES`).
  - *Written proofs, reviewed or audited within the project* (listed in `WRITEUP/SYMMETRIC.md`, §3). Among them are the soundness proofs of the encoders; the reductions A and B of the parallelogram cases of `HalfTurn34`; the normal form of `Threefold33` (`C3-S/soundness.md`) and Lemma 5 and Lemma H6M of its case split; and, for `Threefold33` and `Fivefold35`, a perturbation argument for centres that lie on a line through two points of different orbits.
  - *Exact computations* that compose the pieces (`C2FINAL`, `C3FINAL`). Two bookkeeping steps are computer-assisted because their finite checks are kernel evaluations (`decide +kernel`): the cover of the 910 threefold roots, which is Lean-checked (Theorem 11.82), and the closure of the central parallelogram orbits, which is Lean-checked according to the record (`C2FINAL/README.md`, §6; Theorem 11.98). Lean does not see the records: that every leaf named by the keys of the certificates of Theorem 11.82 and every subcase listed in the data of Theorem 11.98 has a record with the checker verdicts and a CNF hash equal to that of the rebuilt formula is an exact computation outside Lean.
  - *The match between the classes and the formulas.* No formal link joins a Lean class to its CNF formulas. That each class states what its certificates prove was checked by reading (`SYMLEAN/README.md`, §6).

  The records list the remaining caveats and open checks (`WRITEUP/SYMMETRIC.md`, §4).

The unconditional statement, Theorem 8.28 without its five hypotheses, is claimed in the project as computer-assisted (Theorem A of `WRITEUP/SYMMETRIC.md`, §0, in its form for affine bijections), with the evidence listed above; it is not Lean-checked as a whole. This document establishes it only conditionally: as Theorem 8.28 and, for part (a) with three hypotheses, as Theorem 10.40.

Remark 8.30 (what the statements of this section do not say). The statements of this section are limited in four ways.

(1) The symmetry is a map of the whole plane: an affine equivalence, an injective affine map, or an injective map $p \mapsto Lp + b$. A permutation of S that is not the restriction of such a map is not treated.
(2) One symmetry is treated at a time. Nothing is stated about the full symmetry group of S.
(3) The variant without `HalfTurn34` (Theorem 7.41) has no affine version in the Lean files (`SYMLEAN/README.md`, §9).
(4) The order of the symmetry is not tracked (Remark 8.26(2)).

Coverage. This section covers every declaration of `SYMLEAN/AffineReduction.lean`, lines 1059–1533:

- `nsq` (Definition 8.1); `mulVec2_fst` and `mulVec2_snd` (Lemma 8.2); `nsq_nonneg` and `eDist2_eq_nsq` (Lemma 8.3);

- `gA`, `gB` and `gC` (Definition 8.4); `nsq_mulVec` (Lemma 8.5);
- `avgQ`, `qA`, `qB` and `qC` (Definition 8.6); `avgQ_eq` (Lemma 8.7); `avgQ_ge`, `qA_pos` and `qdisc_pos` (Lemma 8.8); `avgQ_mulVec` (Lemma 8.9);
- `lin2` (Definition 8.10); `lin2_fst`, `lin2_snd` and `orient_lin2` (Lemma 8.11); `nsq_lin2_chol`, `lin2_chol_inv_left` and `lin2_chol_inv_right` (Lemma 8.12);
- `iterate_sub_of_affine` (Lemma 8.13); `exists_periodic_of_injective` (Lemma 8.14); `matrix_eq_one_of_fix` (Lemma 8.15); `orient_affine` (Lemma 8.16); `injective_of_affine` (Lemma 8.17); `affineMap_eq_mulVec` (Lemma 8.18);
- `generalPosition_image_of_orient` (Lemma 8.19); `generalPosition_image_affine` and `generalPosition_image_affineEquiv` (Lemma 8.20); `convexPosition_image_affine` (Lemma 8.21); `hasConvexNGon_image_affine` (Lemma 8.22);
- `hasConvexNGon_of_affine_symmetry` (Theorem 8.25, conditional on (hE)); `hasConvexNGon_of_affineMap_symmetry`, `hasConvexNGon_of_affineEquiv_symmetry` and `hasConvexNGon_of_affine_coords_symmetry` (Corollary 8.27, conditional on (hE));
- `heptagon_of_nontrivial_affineEquiv`, `heptagon_of_nontrivial_affineEquiv_image`, `heptagon_of_nontrivial_affineMap`, `heptagon_of_nontrivial_affine` and `heptagon_of_nontrivial_affine_coords` (Theorem 8.28, conditional on the case classes `HalfTurn34`, `Threefold33`, `Fivefold35`, `MirrorOneAxis33` and `MirrorNoAxis34`).

Definition 8.23 and Proposition 8.24 are written steps, not Lean declarations.

Deliberately not covered here:

- Lines 1–1058 of `SYMLEAN/AffineReduction.lean`. They are a verbatim duplicate of lines 1–1058 of `SYMLEAN/SymmetricReduction.lean` and are treated in the section of Theorem 7.39. Of them, `iterate_mem` (line 69) is re-derived inside the proof of Lemma 8.14, and `point_ext` (line 502: two points with equal coordinates are equal) is used tacitly.
- `SYMLEAN/AffineReductionAxioms.lean`. It is the same file followed by a blank line, nine `#print axioms` lines and one `#check` line, and it is used here only as evidence for the axioms.
- The verbatim copy of lines 12–1533 inside `SYMLEAN/mirror/ReductionThreeClasses.lean` (the declarations of this section sit there at their line number plus 26). The own content of that file is treated in the section of Theorem 10.40.

9. The mirror lemma and the mirror reduction (Theorem A of MIRRORETV)

This section gives written proofs for the part of the Lean development `MirrorOdd` (file `MIRRORETV/lean/MirrorMain.lean`) that concerns point sets symmetric under a reflection: the mirror lemma, the reduction of a mirror-symmetric set to an x-sorted configuration ("Theorem A" of `MIRRORETV/README.md`, in its Lean form), the transport from an arbitrary mirror line to the x-axis, and the bounds for $k = 4, 5, 6$. In this section "Theorem A" always means this mirror reduction. The main theorem of `WRITEUP/SYMMETRIC.md`, also called Theorem A there, is a different statement, about sets of at least 33 points with a symmetry; this document states it only in conditional forms, whose hypotheses are case classes (Theorems 7.39, 8.28 and 10.40).

- *Mirror lemma* (Theorems 9.8 and 9.10). Let $c_0, \dots, c_{m-1}$ have strictly increasing abscissae, and let every triple c_i, c_j, c_l with $i < j < l$ be clockwise. If $m \geq 2$ and all c_i lie strictly above the x-axis, then the points c_i and their mirror images $\mathrm{mir}(c_i)$ contain a convex $2m$-gon. If $m \geq 3$ and the first (or the last) point lies on the axis while the others lie strictly above it, they contain a convex $(2m-1)$-gon. As in Lean, the proof lists the points in an explicit cyclic order and checks that *every* triple of the list, taken in list order, is counterclockwise.
- *Theorem A* (Theorems 9.17, 9.18 and 9.23). Let $j \geq 2$, and let M be a natural number such that every x-sorted configuration with no $(j+1)$-cap and no convex $(2j+1)$-gon has at most M points. Then every finite set in general position that is symmetric under the reflection in a line and has no convex $(2j+1)$-gon has at most $2M$ points. In the even case (j-caps, convex $2j$-gons) the bound is $2M+2$. Here M is an *assumed* upper bound: the bound on configurations is a hypothesis of these theorems, and it is not proved in this section.
- *Small k* (Theorem 9.25). Without further hypotheses, a finite set in general position that is symmetric in a line and has no convex k-gon has at most 4, 8, 12 points for $k = 4, 5, 6$.

Sources. The files are the following (SHA-256, first 12 hexadecimal digits).

- The declarations treated here: `MIRRORETV/lean/MirrorMain.lean` (2677 lines, `ca623e55a1b9...`, namespace `MirrorOdd`) and, for configurations and caps, `ErdosSzekeres/Omission/Realization.lean` (`ae4970e0808e...`).
- The positive-cycle lemma (Lemma 1.38): `ErdosSzekeres/ConvexCycleCuts.lean` (`b4ad0ba9e583...`).
- The basic definitions recalled in the foundations section: `ErdosSzekeres/Definitions.lean` (`2425500eefd1...`), `ErdosSzekeres/Basic.lean` (`13667aa2d3dc...`), `ErdosSzekeres/Orientation.lean` (`f67955ac7091...`) and `ErdosSzekeres/OrientationGeometry.lean` (`c7db9bbbb65e...`).

Lines 22–1506 of `MIRRORETV/lean/MirrorMain.lean` hold five positive-cycle helpers and nine small polygons (Lemma 9.12 and Proposition 9.13). The file header (lines 4–10) describes the polygons for caps of four points, says that every orientation inequality is closed by `linarith` with a certificate found offline by a script `gen_certs.py`, and lists `MirrorPolygons.lean` and `small_polygons.lean`, generated by a script `gen_lean.py`, among the source files of the assembled file; the comment at line 993 marks the polygons for caps of two and three points as generated. Neither script is included in `MIRRORETV`, and the header lists the other source files without saying how they were written. Toolchain: Lean 4.24.0, Mathlib `f897ebcf`. The text of `MIRRORETV/lean/MirrorMain.lean`, lines 3–2677, occurs verbatim as lines 1562–4236 of `SYMLEAN/mirror/ReductionThreeClasses.lean` (`0cc83994fe8c...`): a declaration at line L of `MIRRORETV/lean/MirrorMain.lean` is at line $L + 1559$ there. That copy has its own record: lines 1–16 of `SYMLEAN/mirror/axioms.out` print the same three axioms for the sixteen theorems listed in the Rigor paragraph below, and `SYMLEAN/mirror/README.md` reports a compile without errors. With this record in place of `MIRRORETV/lean/MirrorMain.out`, everything below applies to both copies. `MIRRORETV/lean/FarFlat.lean` (`08a79f881eea...`) and the four Lean files of `MIRRORETV/review/` are not used here (see the Coverage paragraph).

Rigor. Every statement below that carries a Lean reference is *Lean-checked*, except the two parts marked as written steps (part (e) of Lemma 9.2 and part (b) of Lemma 9.7); Lemma 9.16 is a written step and carries no Lean reference. The recorded compile log `MIRRORETV/lean/MirrorMain.out`, lines 848–863, prints for the sixteen theorems `mirror_four`, `mirror_five`, `mirror_six`, `mirror_seven`, `mirror_eight`, their five `_line` versions, `mirror_odd_general`, `mirror_even_general`, `mirror_odd_general_line`, `mirror_even_general_line`, `mirror_seven'` and `mirror_eight'` (all in `MirrorOdd`) exactly the axioms `propext`, `Classical.choice` and `Quot.sound`; the log reports no error, only 121 linter warnings about unused simp arguments. Every other declaration treated in this section is used, directly or indirectly, by one of these sixteen theorems, so it depends on no further axiom. The file contains no `sorry`, `admit`, `native_decide` or `axiom` declaration; in the declarations treated here `decide` only compares explicit elements of finite index types (kernel evaluation). This section relies on the recorded log; Lean was not re-run for it. The proofs below are written proofs (not refereed). Theorems 9.17 and 9.18 and parts (b), (c) of Theorem 9.23 take a parametric hypothesis (hM): an assumed bound M, not an unproved claim of the project; several instances of (hM) are discharged by Lean-checked theorems (Remark 9.19).

Dependencies. We use the conventions and facts of the foundations section: Definitions 1.1, 1.3, 1.5, 1.6, 1.7 and 1.10, and Lemmas 1.15, 1.17, 1.20, 1.22, 1.23, 1.27 and 1.38. The bounds for $k = 7, 8, 9$, which combine the results of this section with the repository's Lean formalization of Baek's theorem (for $k = 9$ also with the vendored cups–caps theorem), are Theorems 10.31, 10.33 and 10.35.

Conventions. Points are $p = (p_0, p_1) \in \mathbb{R}^2$ (Lean `Point`, Definition 1.1); p_0 is the abscissa and p_1 the ordinate. For $m \in \mathbb{N}$ we write $[m] = \{0, 1, \dots, m-1\}$ with its usual order; it stands for Lean's `Fin m`. All point sets are finite (Lean `Finset Point`). A finite set $X \subset \mathbb{R}^2$ *has a convex n-gon* if `HasConvexNGon` n X holds (Definition 1.7): some $T \subseteq X$ with $|T| = n$ is in convex position. A finite set T is *x-injective* if $p \mapsto p_0$ is injective on T (Lean `Set.InjOn (fun p => p 0) T`). In Lean, $2m-1$, $m-1$ and $K-1$ in the statements below are truncated subtractions of natural numbers; wherever they occur, the hypotheses ($m \geq 3$, $K \geq 4$) make them equal to the ordinary differences.

9.1. **Reflection, configurations and sorting.**

Definition 9.1 (reflection, upper part, axis part; Lean: `MirrorOdd.mir`, `MIRRORETV/lean/MirrorMain.lean:16`, Lean: `MirrorOdd.Rset`, `MIRRORETV/lean/MirrorMain.lean:1598`, Lean: `MirrorOdd.Aset`, `MIRRORETV/lean/MirrorMain.lean:1599`)**.** For $p \in \mathbb{R}^2$ let $\mathrm{mir}(p) = (p_0, -p_1)$, the reflection in the x-axis. For a finite set $S \subset \mathbb{R}^2$ let

$$R(S) = \{p \in S : p_1 > 0\}, \qquad A(S) = \{p \in S : p_1 = 0\}$$

(Lean `Rset` and `Aset`). In all theorems below, symmetry of S in the x-axis is the hypothesis "$\mathrm{mir}(p) \in S$ for every $p \in S$" ("maps into", as in Definition 1.10). Since mir is injective and S is finite, this is equivalent to $\mathrm{mir}(S) = S$, but only the stated form is used.

Lemma 9.2 (identities for mir and orient; Lean: `MirrorOdd.mir_zero`, `MIRRORETV/lean/MirrorMain.lean:18`, Lean: `MirrorOdd.mir_one`, `MIRRORETV/lean/MirrorMain.lean:19`, Lean: `MirrorOdd.mir_mir`, `MIRRORETV/lean/MirrorMain.lean:1543`, Lean: `MirrorOdd.orient_mir3`, `MIRRORETV/lean/MirrorMain.lean:2187`, Lean: `MirrorOdd.orient_mir2`, `MIRRORETV/lean/MirrorMain.lean:2190`, Lean: `MirrorOdd.orient_cycl`, `MIRRORETV/lean/MirrorMain.lean:2193`, Lean: `MirrorOdd.orient_swap12`, `MIRRORETV/lean/MirrorMain.lean:2196`, Lean: `MirrorOdd.orient_rev`, `MIRRORETV/lean/MirrorMain.lean:2199`). *For all $p, a, b, d \in \mathbb{R}^2$:*

(a) $\mathrm{mir}(p)_0 = p_0$, $\mathrm{mir}(p)_1 = -p_1$ *and* $\mathrm{mir}(\mathrm{mir}(p)) = p$ *(*`mir_zero`, `mir_one`, `mir_mir`*);*

(b) $\mathrm{orient}(\mathrm{mir}\, a, \mathrm{mir}\, b, \mathrm{mir}\, d) = -\mathrm{orient}(a, b, d)$ *(*`orient_mir3`*);*

(c) $\mathrm{orient}(\mathrm{mir}\, a, \mathrm{mir}\, b, d) = -\mathrm{orient}(a, b, \mathrm{mir}\, d)$ *(*`orient_mir2`*);*

(d) $\mathrm{orient}(a, b, d) = \mathrm{orient}(b, d, a)$, $\mathrm{orient}(a, b, d) = -\mathrm{orient}(b, a, d)$ *and* $\mathrm{orient}(a, b, d) = -\mathrm{orient}(d, b, a)$ *(*`orient_cycl`, `orient_swap12`, `orient_rev`*);*

(e) (written step; not a separate Lean declaration) $\mathrm{orient}(a, b, \mathrm{mir}\, d) = \mathrm{orient}(a, b, d) - 2d_1(b_0 - a_0)$.

Proof. (a) is the definition of mir. For (b) and (e) expand $\mathrm{orient}(u, v, w) = (v_0 - u_0)(w_1 - u_1) - (v_1 - u_1)(w_0 - u_0)$ (Definition 1.3):

$$\mathrm{orient}(\mathrm{mir}\, a, \mathrm{mir}\, b, \mathrm{mir}\, d) = (b_0 - a_0)(-d_1 + a_1) - (-b_1 + a_1)(d_0 - a_0) = -\mathrm{orient}(a, b, d),$$

$$\mathrm{orient}(a, b, \mathrm{mir}\, d) = (b_0 - a_0)(-d_1 - a_1) - (b_1 - a_1)(d_0 - a_0) = \mathrm{orient}(a, b, d) - 2d_1(b_0 - a_0).$$

(c) By (a), $d = \mathrm{mir}(\mathrm{mir}\, d)$, so (b) applied to $a, b, \mathrm{mir}\, d$ gives $\mathrm{orient}(\mathrm{mir}\, a, \mathrm{mir}\, b, d) = \mathrm{orient}(\mathrm{mir}\, a, \mathrm{mir}\, b, \mathrm{mir}(\mathrm{mir}\, d)) = -\mathrm{orient}(a, b, \mathrm{mir}\, d)$. (d) These are the cases of Lemma 1.15(d) for the cyclic permutation $(a, b, d) \mapsto (b, d, a)$ (sign $+1$), the transposition of the first two arguments and the transposition of the first and the third argument (sign -1). Each can also be checked by expanding. □

Definition 9.3 (configurations, caps and sorting; Lean: `ErdosSzekeres.Omission.Configuration`, `ErdosSzekeres/Omission/Realization.lean:9`, Lean: `ErdosSzekeres.Omission.Configuration.HasCap`, `ErdosSzekeres/Omission/Realization.lean:64`, Lean: `MirrorOdd.xset`, `MIRRORETV/lean/MirrorMain.lean:1549`, Lean: `MirrorOdd.emb`, `MIRRORETV/lean/MirrorMain.lean:1553`, Lean: `MirrorOdd.sortPt`, `MIRRORETV/lean/MirrorMain.lean:1558`, Lean: `MirrorOdd.cfg`, `MIRRORETV/lean/MirrorMain.lean:1584`). The following notions concern finite sequences of points sorted by abscissa.

(a) Let $n \in \mathbb{N}$. A *configuration of n points* is a map $C : [n] \to \mathbb{R}^2$, $i \mapsto C_i$ (Lean field `point`), such that $i \mapsto (C_i)_0$ is strictly increasing (field `xOrder`) and $\mathrm{orient}(C_i, C_j, C_k) \neq 0$ for all $i < j < k$ in $[n]$ (field `generalPosition`). Its *point set* is $\mathrm{pts}(C) = \{C_i : i \in [n]\}$ (Lean `Finset.univ.image c.point`).

(b) Let $k \in \mathbb{N}$. A configuration C of n points *has a k-cap* (Lean `c.HasCap k`) if there is a strictly increasing map $f : [k] \to [n]$ such that $\mathrm{orient}(C_{f(i)}, C_{f(j)}, C_{f(l)}) < 0$ for *all* $i < j < l$ in $[k]$.

(c) For every finite set $T \subset \mathbb{R}^2$ let $\mathrm{xset}(T) = \{p_0 : p \in T\} \subset \mathbb{R}$. From now on let T also be x-injective. By Lemma 9.5(a) the set $\mathrm{xset}(T)$ has $|T|$ elements, so there is a unique strictly increasing map $\mathrm{emb}(T) : [|T|] \to \mathbb{R}$ with image $\mathrm{xset}(T)$ (Mathlib `Finset.orderEmbOfFin`); $\mathrm{emb}(T)(i)$ is the $(i+1)$-st smallest abscissa of a point of T. For $i \in [|T|]$, $\mathrm{sortPt}(T)(i)$ is a point $p \in T$ with $p_0 = \mathrm{emb}(T)(i)$. Such a point exists by Lemma 9.5(b), and Lean selects one with the axiom of choice (`Classical.choose`). By x-injectivity it is unique, but this is not used. (Parts (a) and (b) of Lemma 9.5 concern only xset and emb, so these forward references are not circular.)

(d) If, in addition, T is in general position, $\mathrm{cfg}(T)$ is the configuration of $|T|$ points $i \mapsto \mathrm{sortPt}(T)(i)$. Its two defining properties are Lemma 9.5(e) and (h).

Remark 9.4. Caps in Definition 9.3(b) are *all-triples* caps: every increasing triple of the selected points is clockwise. The written sources `MIRRORETV/README.md` and `MIRRORETV/PROOFS.md` define caps by consecutive triples; for points with increasing abscissae the two notions agree (`MIRRORETV/PROOFS.md`, Lemma 2.1(b)), but only the all-triples notion occurs in this section. For $k \leq 2$ the condition on triples is vacuous, so a configuration of n points has a k-cap if and only if $k \leq n$.

Lemma 9.5 (sorting; Lean: `MirrorOdd.xset_card`, `MIRRORETV/lean/MirrorMain.lean:1551`, Lean: `MirrorOdd.emb_mem`, `MIRRORETV/lean/MirrorMain.lean:1555`, Lean: `MirrorOdd.sortPt_mem`, `MIRRORETV/lean/MirrorMain.lean:1560`, Lean: `MirrorOdd.sortPt_x`, `MIRRORETV/lean/MirrorMain.lean:1563`, Lean: `MirrorOdd.sortPt_strictMono`, `MIRRORETV/lean/MirrorMain.lean:1566`, Lean: `MirrorOdd.sortPt_injective`, `MIRRORETV/lean/MirrorMain.lean:1572`, Lean: `MirrorOdd.sortPt_image`, `MIRRORETV/lean/MirrorMain.lean:1576`). *Let $T \subset \mathbb{R}^2$ be finite and x-injective, and write $s_i = \mathrm{sortPt}(T)(i)$ for $i \in [|T|]$.*

(a) $|\mathrm{xset}(T)| = |T|$ *(*`xset_card`*).*
(b) For every $i \in [|T|]$ *there is* $p \in T$ *with* $p_0 = \mathrm{emb}(T)(i)$ *(*`emb_mem`*).*
(c) $s_i \in T$ *for every* i *(*`sortPt_mem`*).*
(d) $(s_i)_0 = \mathrm{emb}(T)(i)$ *for every* i *(*`sortPt_x`*).*
(e) $i \mapsto (s_i)_0$ *is strictly increasing on* $[|T|]$ *(*`sortPt_strictMono`*).*
(f) $i \mapsto s_i$ *is injective (*`sortPt_injective`*).*
(g) $\{s_i : i \in [|T|]\} = T$ *(*`sortPt_image`*).*
(h) (the field `generalPosition` *of* `cfg`*,* `MIRRORETV/lean/MirrorMain.lean:1587`*) If* T *is in general position, then* $\mathrm{orient}(s_i, s_j, s_k) \neq 0$ *for all* $i < j < k$ *in* $[|T|]$*.*

Consequently, if T *is in general position,* $\mathrm{cfg}(T)$ *is a configuration of* $|T|$ *points with* $\mathrm{pts}(\mathrm{cfg}(T)) = T$*.*

Proof. (a) The map $p \mapsto p_0$ is injective on T, so its image $\mathrm{xset}(T)$ has $|T|$ elements. (b) $\mathrm{emb}(T)(i)$ lies in the image $\mathrm{xset}(T)$ of $\mathrm{emb}(T)$, so it equals p_0 for some $p \in T$. (c), (d) These are the two properties of the point chosen in Definition 9.3(c). (e) $\mathrm{emb}(T)$ is strictly increasing; use (d). (f) If $s_i = s_j$, then $(s_i)_0 = (s_j)_0$, so $i = j$ by (e). (g) By (c) the set $I = \{s_i : i \in [|T|]\}$ is contained in T, and by (f) it has $|T|$ elements. A subset of the finite set T with $|T|$ elements is T itself. (h) By (f) the points s_i, s_j, s_k are pairwise distinct, and by (c) they lie in T. Since T is in general position, $\mathrm{orient}(s_i, s_j, s_k) \neq 0$ by Lemma 1.20. Finally, (e) and (h) are the two conditions of Definition 9.3(a) for $i \mapsto s_i$, and (g) says that the point set is T. □

Lemma 9.6 (two facts on convex polygons; Lean: `MirrorOdd.hasConvexNGon_of_le`, `MIRRORETV/lean/MirrorMain.lean:2182`, Lean: `MirrorOdd.hasConvexNGon_affineMap`, `MIRRORETV/lean/MirrorMain.lean:1859`)**.** *Let* $S \subset \mathbb{R}^2$ *be finite.*

(a) (`hasConvexNGon_of_le`*) If* S *has a convex* m*-gon and* $n \leq m$*, then* S *has a convex* n*-gon.*
(b) (`hasConvexNGon_affineMap`*) If* S *has a convex* n*-gon and* $f : \mathbb{R}^2 \to \mathbb{R}^2$ *is an injective affine map, then* $f(S)$ *has a convex* n*-gon.*

Proof. (a) This is Lemma 1.23(d). Directly: take $T \subseteq S$ with $|T| = m$ in convex position and choose $T' \subseteq T$ with $|T'| = n$. Then $T' \subseteq S$, and T' is in convex position by Lemma 1.23(a).

(b) Take $T \subseteq S$ with $|T| = n$ in convex position. Then $f(T) \subseteq f(S)$ and $|f(T)| = n$ because f is injective. An affine map preserves affine combinations: $f(\sum_i \lambda_i y_i) = \sum_i \lambda_i f(y_i)$ whenever $\sum_i \lambda_i = 1$. Since the convex hull of a set consists of the convex combinations of its finite subsets, $f(\mathrm{conv}\, Y) = \mathrm{conv}\, f(Y)$ for every $Y \subseteq \mathbb{R}^2$. Let $q \in f(T)$, say $q = f(p)$ with $p \in T$. By injectivity, $f(T) \setminus \{f(p)\} = f(T \setminus \{p\})$. If $f(p) \in \mathrm{conv}\, f(T \setminus \{p\}) = f(\mathrm{conv}(T \setminus \{p\}))$, then $f(p) = f(x)$ for some $x \in \mathrm{conv}(T \setminus \{p\})$, so $p = x \in \mathrm{conv}(T \setminus \{p\})$ by injectivity, which contradicts the convex position of T (Lemma 1.22). Hence no point of $f(T)$ lies in the convex hull of the other points of $f(T)$, and $f(T)$ is in convex position by Lemma 1.22. (This is also contained in Lemma 1.27.) □

9.2. **The mirror lemma.**

Lemma 9.7 (cap facts; Lean: `MirrorOdd.cap_below_lines`, `MIRRORETV/lean/MirrorMain.lean:2204`)**.** *Both parts concern points that lie on or above the* x*-axis.*

(a) (`cap_below_lines`*) Let* $m \in \mathbb{N}$ *and let* $c : [m] \to \mathbb{R}^2$ *be such that* $i \mapsto (c_i)_0$ *is strictly increasing,* $(c_i)_1 \geq 0$ *for all* $i \in [m]$*, and* $\mathrm{orient}(c_i, c_j, c_l) < 0$ *for all* $i < j < l$ *in* $[m]$*. Then for all* $i, j, l \in [m]$ *with* $i < j$ *and* $(c_l)_1 > 0$*,*

$$\mathrm{orient}(c_i, c_j, \mathrm{mir}(c_l)) < 0.$$

(b) (Written step. It is not a separate Lean declaration: the same argument is carried out inline in the proof of `mirror_odd_general`*,* `MIRRORETV/lean/MirrorMain.lean:2542`*–2557, and as the local claim* `interior` *in* `mirror_five`*, line 2060, and in* `mirror_seven_of`*, line 1716.) Let* $p, a, q \in \mathbb{R}^2$ *with* $p_0 < a_0 < q_0$*,* $p_1 \geq 0$*,* $q_1 \geq 0$ *and* $a_1 = 0$*. Then*

$$\mathrm{orient}(p, a, q) = (a_0 - p_0)\, q_1 + p_1\, (q_0 - a_0) \;\geq\; 0.$$

In particular $\mathrm{orient}(p, a, q) < 0$ *is impossible: a point on the axis cannot be the middle point of a clockwise triple with increasing abscissae whose other two points lie on or above the axis.*

In words, (a) says that the mirror image of a cap point lying strictly above the axis lies strictly to the right of every directed chord $c_i \to c_j$ of the cap ($i < j$), that is, strictly below the line through c_i and c_j.

Proof. (a) Write $x_k = (c_k)_0$ and $y_k = (c_k)_1$. By Lemma 9.2(e),

$$(*) \qquad \operatorname{orient}(c_i, c_j, \operatorname{mir} c_l) = \operatorname{orient}(c_i, c_j, c_l) - 2y_l(x_j - x_i),$$

and $2y_l(x_j - x_i) > 0$ because $y_l > 0$ and $x_j > x_i$. We distinguish the position of l.

- $l < i$: $\operatorname{orient}(c_i, c_j, c_l) = \operatorname{orient}(c_l, c_i, c_j)$ by Lemma 9.2(d), and this is negative by the cap hypothesis for $l < i < j$. So $(*)$ is negative.
- $l = i$ or $l = j$: $\operatorname{orient}(c_i, c_j, c_l) = 0$ because two arguments coincide, so $(*)$ equals $-2y_l(x_j - x_i) < 0$.
- $i < l < j$: expanding directly,

$$\operatorname{orient}(c_i, c_j, \operatorname{mir} c_l) = -(x_j - x_i)\, y_l - y_i\,(x_j - x_l) - y_j\,(x_l - x_i).$$

  The first term is negative, and the other two are ≤ 0 because $y_i, y_j \ge 0$ and $x_i < x_l < x_j$. (The cap hypothesis is not needed in this case.)
- $l > j$: $\operatorname{orient}(c_i, c_j, c_l) < 0$ by the cap hypothesis for $i < j < l$, so $(*)$ is negative.

(b) Expanding with $a_1 = 0$: $\operatorname{orient}(p, a, q) = (a_0 - p_0)(q_1 - p_1) + p_1(q_0 - p_0) = (a_0 - p_0)q_1 + p_1(q_0 - a_0)$, and both products are ≥ 0. □

Theorem 9.8 (mirror lemma, no point on the axis; Lean: `MirrorOdd.cap_mirror_polygon`, `MIRRORETV/lean/MirrorMain.lean:2233`)**.** *Let $m \in \mathbb{N}$ with $2 \le m$, and let $c : [m] \to \mathbb{R}^2$ be such that $i \mapsto (c_i)_0$ is strictly increasing, $(c_i)_1 > 0$ for all $i \in [m]$, and $\operatorname{orient}(c_i, c_j, c_l) < 0$ for all $i < j < l$ in $[m]$. Let $P \subset \mathbb{R}^2$ be a finite set with $c_i \in P$ and $\operatorname{mir}(c_i) \in P$ for all $i \in [m]$. Then P has a convex $2m$-gon.*

Proof. We follow the Lean proof. Write $x_i = (c_i)_0$ and $y_i = (c_i)_1 > 0$. Define $q : [2m] \to \mathbb{R}^2$ by

$$q_r = \operatorname{mir}(c_r) \quad (0 \le r \le m-1), \qquad q_r = c_{2m-1-r} \quad (m \le r \le 2m-1).$$

Thus q lists $\operatorname{mir} c_0, \dots, \operatorname{mir} c_{m-1}$ (the reflected cap, from left to right) followed by $c_{m-1}, \dots, c_0$ (the cap, from right to left). For $m \le r \le 2m-1$ the index $2m-1-r$ lies in $[m]$, and $r \mapsto 2m-1-r$ is strictly decreasing. Since all $y_i > 0$, the hypotheses of Lemma 9.7(a) hold for c.

Claim 1: $\operatorname{orient}(q_r, q_s, q_t) > 0$ *for all $r < s < t$ in $[2m]$.* We distinguish how many of r, s, t are smaller than m.

- Case 1, $t < m$. By Lemma 9.2(b), $\operatorname{orient}(\operatorname{mir} c_r, \operatorname{mir} c_s, \operatorname{mir} c_t) = -\operatorname{orient}(c_r, c_s, c_t)$, which is positive by the cap hypothesis for $r < s < t$.
- Case 2, $s < m \le t$. Put $u = 2m-1-t \in [m]$, so $q_t = c_u$. By Lemma 9.2(c), $\operatorname{orient}(\operatorname{mir} c_r, \operatorname{mir} c_s, c_u) = -\operatorname{orient}(c_r, c_s, \operatorname{mir} c_u)$, and $\operatorname{orient}(c_r, c_s, \operatorname{mir} c_u) < 0$ by Lemma 9.7(a) with $(i, j, l) = (r, s, u)$, since $r < s$ and $y_u > 0$.
- Case 3, $r < m \le s$. Put $u' = 2m-1-s$ and $u = 2m-1-t$. Then $u < u'$ (because $s < t$), $q_s = c_{u'}$ and $q_t = c_u$. By the first two identities of Lemma 9.2(d),

$$\operatorname{orient}(\operatorname{mir} c_r, c_{u'}, c_u) = \operatorname{orient}(c_{u'}, c_u, \operatorname{mir} c_r) = -\operatorname{orient}(c_u, c_{u'}, \operatorname{mir} c_r),$$

  and $\operatorname{orient}(c_u, c_{u'}, \operatorname{mir} c_r) < 0$ by Lemma 9.7(a) with $(i, j, l) = (u, u', r)$, since $u < u'$ and $y_r > 0$.
- Case 4, $m \le r$. Put $\alpha = 2m-1-r$, $\beta = 2m-1-s$ and $\delta = 2m-1-t$; then $\delta < \beta < \alpha$, $q_r = c_\alpha$, $q_s = c_\beta$ and $q_t = c_\delta$. By the third identity of Lemma 9.2(d), $\operatorname{orient}(c_\alpha, c_\beta, c_\delta) = -\operatorname{orient}(c_\delta, c_\beta, c_\alpha)$, which is positive by the cap hypothesis for $\delta < \beta < \alpha$.

Claim 2: q is injective. Let $r \ne s$ in $[2m]$. If $r, s < m$ and $q_r = q_s$, then comparing abscissae gives $x_r = x_s$ (Lemma 9.2(a)), so $r = s$ because $i \mapsto x_i$ is strictly increasing; a contradiction. If $r < m \le s$, then $(q_r)_1 = -y_r < 0 < y_{2m-1-s} = (q_s)_1$, so $q_r \ne q_s$; the case $s < m \le r$ is symmetric. If $r, s \ge m$ and $q_r = q_s$, then $x_{2m-1-r} = x_{2m-1-s}$, so $2m-1-r = 2m-1-s$ and $r = s$; a contradiction.

Conclusion. Every q_r lies in P, being some $\operatorname{mir}(c_i)$ or some c_i. Apply Lemma 1.38 with $N = k = 2m$ (note $2m \ge 4$ because $m \ge 2$), the injective map $p = q$ (Claim 2), e the identity of $[2m]$ and the finite set P; its positivity hypothesis is Claim 1. Hence P has a convex $2m$-gon. □

Remark 9.9. (1) Claim 1 checks the sign of *every* triple q_r, q_s, q_t with $r < s < t$, that is, of every triple of the list taken in list order, which is what Lemma 1.38 requires. It would not suffice to check that the closed polygon $q_0 q_1 \cdots q_{2m-1}$ turns the same way at every vertex: such a polygon can wind around twice, and then a vertex can lie inside the convex hull of the others (`MIRRORETV/PROOFS.md`, remark after Lemma 3.1, gives a five-point example). (2) `MIRRORETV/PROOFS.md`, Lemma 3.1, gives a written argument by a different method (a region argument) for a version with caps C defined by consecutive triples: if the interior points of C have positive ordinates and its endpoints have nonnegative ordinates (positive if C has only one point), then $C \cup \operatorname{mir}(C)$ is in

convex position. That argument is not used here; the Lean statements are Theorems 9.8 and 9.10, and these are what we prove.

Theorem 9.10 (mirror lemma, one endpoint on the axis; Lean: `MirrorOdd.axis_left_polygon`, `MIRRORETV/lean/MirrorMain.lean:2309`, Lean: `MirrorOdd.axis_right_polygon`, `MIRRORETV/lean/MirrorMain.lean:2403`)**.** *Let $m \in \mathbb{N}$ with $3 \le m$, and let $c : [m] \to \mathbb{R}^2$ be such that $i \mapsto (c_i)_0$ is strictly increasing and* $\operatorname{orient}(c_i, c_j, c_l) < 0$ *for all $i < j < l$ in $[m]$. Let $P \subset \mathbb{R}^2$ be a finite set with $c_i \in P$ and* $\operatorname{mir}(c_i) \in P$ *for all $i \in [m]$.*

(a) (`axis_left_polygon`) If $(c_0)_1 = 0$ and $(c_i)_1 > 0$ for every $i \in [m]$ with $0 < i$, then P has a convex $(2m-1)$-gon.

(b) (`axis_right_polygon`) If $(c_{m-1})_1 = 0$ and $(c_i)_1 > 0$ for every $i \in [m]$ with $i < m-1$, then P has a convex $(2m-1)$-gon.

Proof. We follow the Lean proofs. Write $x_i = (c_i)_0$ and $y_i = (c_i)_1$. In both parts $y_i \ge 0$ for all i, so Lemma 9.7(a) applies to c; and $2m-1 \ge 5$.

(a) Here $\operatorname{mir}(c_0) = c_0$ because $(c_0)_1 = 0$. Define $q : [2m-1] \to \mathbb{R}^2$ by

$$q_r = \operatorname{mir}(c_r) \quad (0 \le r \le m-1), \qquad q_r = c_{2m-1-r} \quad (m \le r \le 2m-2).$$

So q lists $\operatorname{mir} c_0 = c_0, \operatorname{mir} c_1, \dots, \operatorname{mir} c_{m-1}$ and then $c_{m-1}, \dots, c_1$. For $m \le r \le 2m-2$ the index $2m-1-r$ lies in $\{1, \dots, m-1\}$, hence $y_{2m-1-r} > 0$.

Claim 1: $\operatorname{orient}(q_r, q_s, q_t) > 0$ *for all $r < s < t$ in $[2m-1]$.*

- Case 1 ($t < m$) and Case 4 ($m \le r$) are exactly as in the proof of Theorem 9.8; they use only the cap hypothesis (in Case 4 the indices are $2m-1-r > 2m-1-s > 2m-1-t$).
- Case 2, $s < m \le t$. With $u = 2m-1-t \in \{1, \dots, m-1\}$ we have $q_t = c_u$ and $y_u > 0$, and Case 2 of the proof of Theorem 9.8 applies verbatim.
- Case 3, $r < m \le s$. Put $u' = 2m-1-s$ and $u = 2m-1-t$, so $1 \le u < u' \le m-1$, $q_s = c_{u'}$, $q_t = c_u$.
  - Case 3a, $r = 0$. Then $q_r = \operatorname{mir} c_0 = c_0$, and by Lemma 1.15(a) (exchange of the last two arguments) $\operatorname{orient}(c_0, c_{u'}, c_u) = -\operatorname{orient}(c_0, c_u, c_{u'})$, which is positive by the cap hypothesis for $0 < u < u'$.
  - Case 3b, $r \ge 1$. Then $y_r > 0$, and Case 3 of the proof of Theorem 9.8 applies verbatim: $\operatorname{orient}(\operatorname{mir} c_r, c_{u'}, c_u) = -\operatorname{orient}(c_u, c_{u'}, \operatorname{mir} c_r) > 0$.

Claim 2: q is injective. Two indices $r \ne s$ that are both $< m$, or both $\ge m$, are separated by the abscissae as in Theorem 9.8. If $r < m \le s$, then $(q_s)_1 = y_{2m-1-s} > 0$ because $2m-1-s \ge 1$, while $(q_r)_1 = -y_r \le 0$. The case $s < m \le r$ is symmetric.

Conclusion. Every q_r lies in P, and Lemma 1.38 with $N = k = 2m-1 \ge 4$, $p = q$ and e the identity gives a convex $(2m-1)$-gon in P.

(b) Here $\operatorname{mir}(c_{m-1}) = c_{m-1}$. Define $q : [2m-1] \to \mathbb{R}^2$ by

$$q_r = \operatorname{mir}(c_r) \quad (0 \le r \le m-1), \qquad q_r = c_{2m-2-r} \quad (m \le r \le 2m-2).$$

So q lists $\operatorname{mir} c_0, \dots, \operatorname{mir} c_{m-2}, \operatorname{mir} c_{m-1} = c_{m-1}$ and then $c_{m-2}, \dots, c_0$. For $m \le r \le 2m-2$ the index $2m-2-r$ lies in $\{0, \dots, m-2\}$, hence $y_{2m-2-r} > 0$.

Claim 1.

- Cases 1 and 4 are as in Theorem 9.8, with $2m-2-\cdot$ in place of $2m-1-\cdot$.
- Case 2, $s < m \le t$. With $u = 2m-2-t \le m-2$ we have $q_t = c_u$ and $y_u > 0$, and Case 2 of Theorem 9.8 applies.
- Case 3, $r < m \le s$. Put $u' = 2m-2-s$ and $u = 2m-2-t$, so $0 \le u < u' \le m-2$.
  - Case 3a, $r = m-1$. Then $q_r = c_{m-1}$, and by the third identity of Lemma 9.2(d), $\operatorname{orient}(c_{m-1}, c_{u'}, c_u) = -\operatorname{orient}(c_u, c_{u'}, c_{m-1})$, which is positive by the cap hypothesis for $u < u' < m-1$.
  - Case 3b, $r < m-1$. Then $y_r > 0$, and Case 3 of Theorem 9.8 applies.

Claim 2. Two indices $r \ne s$ that are both $< m$, or both $\ge m$, are separated by the abscissae as in (a). If $r < m \le s$, then $(q_s)_1 = y_{2m-2-s} > 0$ because $2m-2-s \le m-2$, while $(q_r)_1 = -y_r \le 0$. The case $s < m \le r$ is symmetric.

Conclusion as in (a). □

Remark 9.11. The hypothesis $m \geq 3$ makes the list long enough for Lemma 1.38, which needs at least four points. In (a) the point c_0 occurs in the list only once, as $\operatorname{mir} c_0 = c_0$; in (b) the same holds for c_{m-1}. This is why one point is lost compared with Theorem 9.8.

9.3. **The generated small polygons.** Lines 22–1506 of `MIRRORETV/lean/MirrorMain.lean` contain five helper lemmas cyc_k ($k = 4, \dots, 8$) and nine lemmas on small mirror polygons. In Lean each polygon lemma is proved directly: the pairwise distinctness of its k vertices by comparing one coordinate, and each of the $\binom{k}{3}$ orientation inequalities, after expanding orient and mir, by linear arithmetic (`linarith`) from the hypotheses (with the cap orientations expanded) and auxiliary products $(c_i)_1 (u_0 - v_0) > 0$, where u and v are two of the lemma's points with $v_0 < u_0$ (in the axis lemmas one of them can be the axis point a), with certificates found offline; then cyc_k is applied. These Lean proofs do not use the general lemmas, which come later in the file. Our written proofs are different: we derive the nine lemmas from Theorems 9.8 and 9.10.

Lemma 9.12 (positive cycles of length 4 to 8; Lean: `MirrorOdd.cyc4`, `MIRRORETV/lean/MirrorMain.lean:995`, Lean: `MirrorOdd.cyc5`, `MIRRORETV/lean/MirrorMain.lean:1026`, Lean: `MirrorOdd.cyc6`, `MIRRORETV/lean/MirrorMain.lean:1068`, Lean: `MirrorOdd.cyc7`, `MIRRORETV/lean/MirrorMain.lean:22`, Lean: `MirrorOdd.cyc8`, `MIRRORETV/lean/MirrorMain.lean:103`)**.** *Let $k \in \{4, 5, 6, 7, 8\}$, let $q_0, \dots, q_{k-1} \in \mathbb{R}^2$, and let $P \subset \mathbb{R}^2$ be finite. Suppose that $q_r \in P$ for all $r \in [k]$, that $q_r \neq q_s$ for all $r < s$ in $[k]$, and that $\operatorname{orient}(q_r, q_s, q_t) > 0$ for all $r < s < t$ in $[k]$. Then P has a convex k-gon. (The Lean lemma cyc_k lists these k memberships, $\binom{k}{2}$ inequalities $q_r \neq q_s$ and $\binom{k}{3}$ orientation inequalities as separate hypotheses.)*

Proof. The map $r \mapsto q_r$ on $[k]$ is injective by the distinctness hypotheses. Lemma 1.38 with $N = k \geq 4$, $p = q$, e the identity of $[k]$ and the set P gives the claim; its remaining hypotheses are exactly the memberships and the orientation inequalities. This is also the Lean proof. □

Proposition 9.13 (generated small mirror polygons; Lean: `MirrorOdd.cap_mirror_quadrilateral`, `MIRRORETV/lean/MirrorMain.lean:1126`, Lean: `MirrorOdd.cap_mirror_pentagon`, `MIRRORETV/lean/MirrorMain.lean:1282`, Lean: `MirrorOdd.cap_mirror_hexagon`, `MIRRORETV/lean/MirrorMain.lean:1168`, Lean: `MirrorOdd.cap_mirror_heptagon`, `MIRRORETV/lean/MirrorMain.lean:212`, Lean: `MirrorOdd.cap_mirror_octagon`, `MIRRORETV/lean/MirrorMain.lean:392`, Lean: `MirrorOdd.axis_left_pentagon`, `MIRRORETV/lean/MirrorMain.lean:1358`, Lean: `MirrorOdd.axis_right_pentagon`, `MIRRORETV/lean/MirrorMain.lean:1433`, Lean: `MirrorOdd.axis_left_heptagon`, `MIRRORETV/lean/MirrorMain.lean:639`, Lean: `MirrorOdd.axis_right_heptagon`, `MIRRORETV/lean/MirrorMain.lean:816`)**.** *Let $a, c_0, c_1, c_2, c_3 \in \mathbb{R}^2$ and let $P \subset \mathbb{R}^2$ be finite. Write $x_i = (c_i)_0$ and $y_i = (c_i)_1$.*

- *(a) (`cap_mirror_quadrilateral`) If $x_0 < x_1$, $y_0 > 0$, $y_1 > 0$, and $\operatorname{mir} c_0, \operatorname{mir} c_1, c_1, c_0 \in P$, then P has a convex 4-gon.*
- *(b) (`cap_mirror_pentagon`) If $x_0 < x_1 < x_2$, $y_0, y_1, y_2 > 0$, $\operatorname{orient}(c_0, c_1, c_2) < 0$, and $\operatorname{mir} c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, c_2, c_1 \in P$, then P has a convex 5-gon.*
- *(c) (`cap_mirror_hexagon`) If $x_0 < x_1 < x_2$, $y_0, y_1, y_2 > 0$, $\operatorname{orient}(c_0, c_1, c_2) < 0$, and $\operatorname{mir} c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, c_2, c_1, c_0 \in P$, then P has a convex 6-gon.*
- *(d) (`cap_mirror_heptagon`) If $x_0 < x_1 < x_2 < x_3$, $y_0, y_1, y_2, y_3 > 0$, the four orientations $\operatorname{orient}(c_0, c_1, c_2)$, $\operatorname{orient}(c_0, c_1, c_3)$, $\operatorname{orient}(c_0, c_2, c_3)$, $\operatorname{orient}(c_1, c_2, c_3)$ are < 0, and $c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, \operatorname{mir} c_3, c_3, c_2, c_1 \in P$, then P has a convex 7-gon.*
- *(e) (`cap_mirror_octagon`) Under the hypotheses on $c_0, \dots, c_3$ of (d), if $\operatorname{mir} c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, \operatorname{mir} c_3, c_3, c_2, c_1, c_0 \in P$, then P has a convex 8-gon.*
- *(f) (`axis_left_pentagon`) If $a_1 = 0$, $a_0 < x_0 < x_1$, $y_0, y_1 > 0$, $\operatorname{orient}(a, c_0, c_1) < 0$, and $a, \operatorname{mir} c_0, \operatorname{mir} c_1, c_1, c_0 \in P$, then P has a convex 5-gon.*
- *(g) (`axis_right_pentagon`) If $a_1 = 0$, $x_0 < x_1 < a_0$, $y_0, y_1 > 0$, $\operatorname{orient}(c_0, c_1, a) < 0$, and $\operatorname{mir} c_0, \operatorname{mir} c_1, a, c_1, c_0 \in P$, then P has a convex 5-gon.*
- *(h) (`axis_left_heptagon`) If $a_1 = 0$, $a_0 < x_0 < x_1 < x_2$, $y_0, y_1, y_2 > 0$, the four orientations $\operatorname{orient}(a, c_0, c_1)$, $\operatorname{orient}(a, c_0, c_2)$, $\operatorname{orient}(a, c_1, c_2)$, $\operatorname{orient}(c_0, c_1, c_2)$ are < 0, and $a, \operatorname{mir} c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, c_2, c_1, c_0 \in P$, then P has a convex 7-gon.*
- *(i) (`axis_right_heptagon`) If $a_1 = 0$, $x_0 < x_1 < x_2 < a_0$, $y_0, y_1, y_2 > 0$, the four orientations $\operatorname{orient}(c_0, c_1, c_2)$, $\operatorname{orient}(c_0, c_1, a)$, $\operatorname{orient}(c_0, c_2, a)$, $\operatorname{orient}(c_1, c_2, a)$ are < 0, and $\operatorname{mir} c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, a, c_2, c_1, c_0 \in P$, then P has a convex 7-gon.*

In each item the memberships are listed in the vertex order that the Lean proof passes to cyc_k (Lemma 9.12).

Proof. We use the general theorems; this differs from the generated Lean proofs described above.

(a), (c), (e): apply Theorem 9.8 with $m = 2, 3, 4$ to $c = (c_0, \dots, c_{m-1})$ and to P. The abscissae increase strictly because consecutive ones do, and all $y_i > 0$. The cap hypothesis concerns the triples $i < j < l$ in $[m]$: there are none for $m = 2$, only $(0, 1, 2)$ for $m = 3$, and $(0, 1, 2)$, $(0, 1, 3)$, $(0, 2, 3)$, $(1, 2, 3)$ for $m = 4$; these are exactly the listed orientation hypotheses. P contains all c_i and $\operatorname{mir} c_i$. The conclusion is a convex $2m$-gon, that is, a 4-, 6- or 8-gon.

(f), (h): note that $\operatorname{mir}(a) = (a_0, -0) = a$. Apply Theorem 9.10(a) with $m = 3$ to (a, c_0, c_1), respectively with $m = 4$ to (a, c_0, c_1, c_2). The abscissae increase strictly, the first point a has ordinate 0 and the others have positive ordinates. For $m = 3$ the only triple is (a, c_0, c_1); for $m = 4$ the triples $(0, 1, 2)$, $(0, 1, 3)$, $(0, 2, 3)$, $(1, 2, 3)$ of (a, c_0, c_1, c_2) are (a, c_0, c_1), (a, c_0, c_2), (a, c_1, c_2), (c_0, c_1, c_2). These are the listed orientation hypotheses. P contains the listed points and their mirror images (recall $\operatorname{mir} a = a$). The conclusion is a convex $(2m - 1)$-gon, that is, a 5- or 7-gon.

(g), (i): the same with Theorem 9.10(b), applied to (c_0, c_1, a), respectively (c_0, c_1, c_2, a), whose last point a has ordinate 0. For $m = 4$ the four triples are (c_0, c_1, c_2), (c_0, c_1, a), (c_0, c_2, a), (c_1, c_2, a).

(b): let $Q = \{\operatorname{mir} c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, c_2, c_1, c_0\}$, a set with at most 6 elements. By (c) applied to the finite set Q in place of P, there is $T \subseteq Q$ with $|T| = 6$ in convex position. As $|Q| \le 6$, we get $T = Q$: the six listed points are pairwise distinct and Q is in convex position. Then $Q \setminus \{c_0\} = \{\operatorname{mir} c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, c_2, c_1\}$ has 5 elements, is in convex position (Lemma 1.23(a)), and is contained in P by hypothesis. So P has a convex 5-gon.

(d): in the same way, (e) applied to the set Q of its eight points shows that these eight points are pairwise distinct and in convex position. The set $Q \setminus \{\operatorname{mir} c_0\} = \{c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, \operatorname{mir} c_3, c_3, c_2, c_1\}$ has 7 elements, is in convex position and is contained in P. □

In Lean, items (a), (b), (c), (f), (g) are used in the proofs of Theorem 9.25, and items (d), (e), (h), (i) in those of the bounds for $k = 7, 8$ (Theorems 10.31 and 10.33).

9.4. **Counting and distinct abscissae.**

Lemma 9.14 (counting; Lean: `MirrorOdd.card_split`, `MIRRORETV/lean/MirrorMain.lean:1601`, Lean: `MirrorOdd.card_A_le`, `MIRRORETV/lean/MirrorMain.lean:1636`)**.** *Let $S \subset \mathbb{R}^2$ be finite.*

(a) (`card_split`) If $\operatorname{mir}(p) \in S$ for every $p \in S$, then $|S| = 2|R(S)| + |A(S)|$. (General position is not assumed.)

(b) (`card_A_le`) If S is in general position, then $|A(S)| \le 2$.

Proof. (a) Let $N = \{p \in S : p_1 < 0\}$. By trichotomy of p_1, S is the disjoint union of $R(S)$, $A(S)$ and N. We show $\operatorname{mir}(R(S)) = N$. If $q \in R(S)$, then $\operatorname{mir} q \in S$ by hypothesis and $(\operatorname{mir} q)_1 = -q_1 < 0$, so $\operatorname{mir} q \in N$. Conversely, if $p \in N$, then $\operatorname{mir} p \in S$ and $(\operatorname{mir} p)_1 = -p_1 > 0$, so $\operatorname{mir} p \in R(S)$, and $p = \operatorname{mir}(\operatorname{mir} p)$ by Lemma 9.2(a). Since mir is injective (it is its own inverse), $|N| = |R(S)|$, and therefore $|S| = |R(S)| + |A(S)| + |N| = 2|R(S)| + |A(S)|$.

(b) Suppose that $A(S)$ contains three pairwise distinct points a, b, d. Since $a_1 = b_1 = d_1 = 0$, $\operatorname{orient}(a, b, d) = (b_0 - a_0)(d_1 - a_1) - (b_1 - a_1)(d_0 - a_0) = 0$, which contradicts general position (Lemma 1.20). Hence $|A(S)| \le 2$. □

Lemma 9.15 (distinct abscissae; Lean: `MirrorOdd.injOn_x`, `MIRRORETV/lean/MirrorMain.lean:1647`)**.** *Let $S \subset \mathbb{R}^2$ be finite and in general position, with $\operatorname{mir}(p) \in S$ for every $p \in S$. Let $T \subseteq S$ be a set such that $p_1 \ge 0$ for every $p \in T$, and such that $p = q$ whenever $p, q \in T$ and $p_1 = q_1 = 0$ (at most one point of T lies on the axis). Then T is x-injective.*

Proof. First, if $p, q \in T$, $p_1 > 0$ and $p_0 = q_0$, then $p = q$. Indeed, suppose $p \ne q$. The points p, q and $\operatorname{mir} p$ lie in S (as $T \subseteq S$ and S is mir-symmetric) and are pairwise distinct: $p \ne \operatorname{mir} p$ because $p_1 > 0 > -p_1$, and $q \ne \operatorname{mir} p$ because $q_1 \ge 0 > -p_1$. But $\operatorname{orient}(p, q, \operatorname{mir} p) = (q_0 - p_0)(-p_1 - p_1) - (q_1 - p_1)(p_0 - p_0) = 0$ because $q_0 = p_0$, which contradicts general position (Lemma 1.20).

Now let $p, q \in T$ with $p_0 = q_0$. If $p_1 > 0$, then $p = q$ by the first step. Otherwise $p_1 = 0$. If $q_1 > 0$, the first step with the roles of p and q exchanged gives $q = p$. Otherwise $q_1 = 0$ as well, and $p = q$ by the hypothesis on axis points. □

Lemma 9.16 (caps in the closed upper half; written step)**.** *Let $S \subset \mathbb{R}^2$ be finite and in general position, with $\operatorname{mir}(p) \in S$ for every $p \in S$. Let $T \subseteq S$ satisfy $p_1 \ge 0$ for every $p \in T$, and $p = q$ whenever $p, q \in T$ and $p_1 = q_1 = 0$. Then T is x-injective and in general position, so $C = \operatorname{cfg}(T)$ is a configuration of $|T|$ points with $\operatorname{pts}(C) = T$. Let $m \ge 2$ and suppose that C has an m-cap.*

(a) If $p_1 > 0$ for every $p \in T$, then S has a convex $2m$-gon.

(b) If $m \geq 3$, then S has a convex $(2m-1)$-gon.

(This lemma isolates the cap argument that the Lean proofs of Theorems 9.17 and 9.18 and of parts (b) and (c) of Theorem 9.25 carry out inside their local claims `keyT` *(odd case and $k = 5$) and* `hR` *(even case and $k = 6$).)*

Proof. T is x-injective by Lemma 9.15 and in general position by Lemma 1.23(b); the statements about C are Lemma 9.5. Let $f : [m] \to [|T|]$ be strictly increasing with $\mathrm{orient}(C_{f(i)}, C_{f(i')}, C_{f(i'')}) < 0$ for all $i < i' < i''$ in $[m]$, and put $c_i = C_{f(i)}$ for $i \in [m]$. Then $i \mapsto (c_i)_0$ is strictly increasing (a composition of strictly increasing maps); $\mathrm{orient}(c_i, c_{i'}, c_{i''}) < 0$ for all $i < i' < i''$; $c_i \in T \subseteq S$, hence $\mathrm{mir}(c_i) \in S$; and $(c_i)_1 \geq 0$.

(a) Here $(c_i)_1 > 0$ for all i, because $c_i \in T$. Theorem 9.8 with $P = S$ gives a convex $2m$-gon in S.

(b) If $(c_i)_1 > 0$ for all i, Theorem 9.8 gives a convex $2m$-gon in S, hence a convex $(2m-1)$-gon by Lemma 9.6(a). Otherwise there is i_0 with $(c_{i_0})_1 \leq 0$, hence $(c_{i_0})_1 = 0$. For $i \neq i_0$ we have $(c_i)_1 > 0$: otherwise $(c_i)_1 = 0$, the hypothesis on axis points gives $c_i = c_{i_0}$, so $(c_i)_0 = (c_{i_0})_0$ and $i = i_0$. There are three cases.

- $i_0 = 0$: Theorem 9.10(a) with $P = S$ gives a convex $(2m-1)$-gon in S.
- $i_0 = m-1$: Theorem 9.10(b) with $P = S$ gives a convex $(2m-1)$-gon in S.
- $0 < i_0 < m-1$: put $p = c_{i_0-1}$, $a = c_{i_0}$ and $q = c_{i_0+1}$. Then $p_0 < a_0 < q_0$, $p_1, q_1 \geq 0$ and $a_1 = 0$, so $\mathrm{orient}(p, a, q) \geq 0$ by Lemma 9.7(b); but $\mathrm{orient}(p, a, q) < 0$ by the cap property for $i_0 - 1 < i_0 < i_0 + 1$. So this case does not occur. □

9.5. **The mirror reduction.**

Theorem 9.17 (Theorem A, odd case; Lean: `MirrorOdd.mirror_odd_general`, `MIRRORETV/lean/MirrorMain.lean:2498`)**.** *Let $j, M \in \mathbb{N}$ with $2 \leq j$, and assume the following parametric hypothesis on M, which is part of the statement (Remark 9.19):*

(hM) for every $n \in \mathbb{N}$ and every configuration C of n points: if C has no $(j+1)$-cap and $\mathrm{pts}(C)$ has no convex $(2j+1)$-gon, then $n \leq M$.

Let $S \subset \mathbb{R}^2$ be finite and in general position, with $\mathrm{mir}(p) \in S$ for every $p \in S$, and suppose that S has no convex $(2j+1)$-gon. Then $|S| \leq 2M$.

Proof. We follow the Lean proof.

Step 1. Let $T \subseteq S$ be such that $p_1 \geq 0$ for all $p \in T$, and $p = q$ whenever $p, q \in T$ and $p_1 = q_1 = 0$. Then $|T| \leq M$. Indeed, let $C = \mathrm{cfg}(T)$, a configuration of $|T|$ points with $\mathrm{pts}(C) = T$ (Lemma 9.16). If C had a $(j+1)$-cap, Lemma 9.16(b) with $m = j + 1 \geq 3$ would give a convex $(2(j+1)-1)$-gon, that is, a convex $(2j+1)$-gon, in S; so C has no $(j+1)$-cap. If $\mathrm{pts}(C) = T$ had a convex $(2j+1)$-gon, so would $S \supseteq T$ (Lemma 1.23(c)). Hypothesis (hM) with $n = |T|$ gives $|T| \leq M$.

Step 2. The set $T = R(S)$ satisfies the assumptions of Step 1: all its points have positive ordinate, so none of them lies on the axis. Hence $|R(S)| \leq M$.

Step 3. If $A(S) = \emptyset$, then $|S| = 2|R(S)| \leq 2M$ by Lemma 9.14(a). Otherwise pick $a \in A(S)$ and let $T = R(S) \cup \{a\}$. Then $a \notin R(S)$ (as $a_1 = 0$), $T \subseteq S$, $p_1 \geq 0$ on T, and a is the only point of T on the axis. Step 1 gives $|R(S)| + 1 = |T| \leq M$. By Lemma 9.14(a) and (b), $|S| = 2|R(S)| + |A(S)| \leq 2|R(S)| + 2 \leq 2M$. □

Theorem 9.18 (Theorem A, even case; Lean: `MirrorOdd.mirror_even_general`, `MIRRORETV/lean/MirrorMain.lean:2599`)**.** *Let $j, M \in \mathbb{N}$ with $2 \leq j$, and assume the following parametric hypothesis on M, which is part of the statement (Remark 9.19):*

(hM) for every $n \in \mathbb{N}$ and every configuration C of n points: if C has no j-cap and $\mathrm{pts}(C)$ has no convex $2j$-gon, then $n \leq M$.

Let $S \subset \mathbb{R}^2$ be finite and in general position, with $\mathrm{mir}(p) \in S$ for every $p \in S$, and suppose that S has no convex $2j$-gon. Then $|S| \leq 2M + 2$.

Proof. We follow the Lean proof. The set $T = R(S)$ satisfies the assumptions of Lemma 9.16, with $p_1 > 0$ for all $p \in T$. Let $C = \mathrm{cfg}(R(S))$. If C had a j-cap, Lemma 9.16(a) with $m = j \geq 2$ would give a convex $2j$-gon in S; so C has no j-cap. The set $\mathrm{pts}(C) = R(S) \subseteq S$ has no convex $2j$-gon (Lemma 1.23(c)). By (hM), $|R(S)| \leq M$, and by Lemma 9.14, $|S| = 2|R(S)| + |A(S)| \leq 2M + 2$. □

Remark 9.19 (the hypothesis (hM), and indexing). We comment on the hypothesis of Theorems 9.17 and 9.18 and of parts (b), (c) of Theorem 9.23.

(1) *(hM) is a parametric hypothesis.* M is any natural number for which (hM) holds; it is not defined as a maximum. Theorems 9.17, 9.18 and 9.23(b),(c) are implications from (hM) to a bound on $|S|$. Thus (hM) is an assumption on the parameter M, not an unproved claim of the project: for a given pair

(j, M) the bound holds as soon as (hM) is proved for that pair. In the project, (hM) is discharged by Lean-checked theorems in the following instances. Odd case: $(j, M) = (3, 16)$ by `MirrorOdd.capacity16` (in `MirrorOdd.mirror_seven'`, `MIRRORETV/lean/MirrorMain.lean:2631`) and $(j, M) = (4, 113)$ by `C12Check.capacity113` (in `C12Check.mirror_nine`, `MIRRORETV/lean/MirrorNine_append.lean:131`). Even case: $(j, M) = (4, 22)$ by `MirrorOdd.capacity22g` (in `MirrorOdd.mirror_eight'`, `MIRRORETV/lean/MirrorMain.lean:2636`). The axioms of `mirror_seven'` and `mirror_eight'` are recorded in `MIRRORETV/lean/MirrorMain.out` (lines 860–861), those of `C12Check.mirror_nine` in `MIRRORETV/lean/MirrorNine.out` (line 881; that log checks `MIRRORETV/lean/MirrorNine_append.lean` appended to `MIRRORETV/lean/MirrorMain.lean`); in each case they are the three standard axioms. These instances rest on the repository's Lean formalization of Baek's theorem (the vendored theorem `main`, Theorem 10.4), and the instance $(4, 113)$ also on the vendored cups–caps theorem `Config.cap_cup`. They are treated in the section of Theorems 10.31, 10.33 and 10.35: the three capacity bounds appear there as statements used in the proofs of these theorems, `mirror_seven'` and `mirror_eight'` in remarks after Theorems 10.31 and 10.33, and `C12Check.mirror_nine` is part (b) of Theorem 10.35. For small j: Lemma 9.24(b) is exactly (hM) for the odd case $(j, M) = (2, 4)$ (with $K = 5$) and for the even case $(j, M) = (3, 5)$ (with $K = 6$); and (hM) for the even case $(j, M) = (2, 1)$ holds because any two points of a configuration form a 2-cap. So the bounds 8, 12 and 4 of Theorem 9.25 also follow from Theorems 9.17 and 9.18; the Lean proofs of Theorem 9.25 do not take this route.

(2) *Relation to the Erdős–Tuza–Valtr function.* `MIRRORETV/README.md` and `MIRRORETV/PROOFS.md` state Theorem A with Baek's number $N(n, a, b)$ in the place of M: the largest size of an x-distinct set in general position with no n points in convex position, no a-cap and no b-cup, where caps and cups are defined by consecutive triples. In their own index j' (see (3)) the odd case uses $N(2j'-1, j', 2j'-1)$ and the even case $N(2j', j', 2j')$; in the indexing of Theorems 9.17 and 9.18 these are $N(2j+1, j+1, 2j+1)$ (with $j' = j+1$) and $N(2j, j, 2j)$ (with $j' = j$). These numbers are finite: by the cups–caps theorem of Erdős and Szekeres (1935), for $a, b \geq 2$ every set in the class that defines $N(n, a, b)$ has at most $\binom{a+b-4}{a-2}$ points (`MIRRORETV/PROOFS.md`, §1), and the two numbers above have $a, b \geq j \geq 2$. That (hM) holds for M equal to the corresponding number then follows, since the point set of a configuration of n points is an x-distinct set of n points in general position, from two facts stated in writing in `MIRRORETV/PROOFS.md`: consecutive clockwise triples imply that all increasing triples are clockwise (Lemma 2.1(b)), and a cup is in convex position (§1; see also Lemma 2.3). No declaration of `MirrorOdd` states this, and it is not used in this section. A separate Lean formalization of Theorem A in this form, written by a review with its own definitions and importing only Mathlib, is `MIRRORETV/review/MirrorA.lean` (for example `theoremA_odd_bound`, whose hypothesis is an upper bound on the sets of the class that defines $N(2j'-1, j', 2j'-1)$). It is not restated in this document; the Coverage paragraph at the end of this section describes it and gives the reason.

(3) *Index shift.* In Theorem 9.17, $k = 2j+1$ and the caps have $j+1$ points. The written sources state the odd case as $k = 2j'-1$ with j'-caps, so the Lean j equals $j'-1$; and $2 \leq j$ means $k \geq 5$, so the case $k = 3$ is not covered by Theorem 9.17. In Theorem 9.18, $k = 2j$ with j-caps, the same j as in the written sources.

(4) In the odd case, Step 3 adds an axis point to $R(S)$ at no cost: by Lemma 9.16(b) a $(j+1)$-cap through it would still give a convex $(2j+1)$-gon. This yields $2M$ instead of $2M+2$. In the even case a j-cap with an endpoint on the axis gives only $2j-1$ points, and the (at most two) axis points are simply counted.

9.6. **Arbitrary mirror lines.**

Definition 9.20 (line reflection and frame; Lean: `MirrorOdd.lineRefl`, `MIRRORETV/lean/MirrorMain.lean:1879`, Lean: `MirrorOdd.frame`, `MIRRORETV/lean/MirrorMain.lean:1883`, Lean: `MirrorOdd.unframeLin`, `MIRRORETV/lean/MirrorMain.lean:1886`, Lean: `MirrorOdd.unframe`, `MIRRORETV/lean/MirrorMain.lean:1892`)**.** Let $n_0, n_1, c \in \mathbb{R}$. For $p, q \in \mathbb{R}^2$ put

$$\begin{aligned}
\mathrm{lineRefl}(p) &= \big(p_0 - 2(n_0p_0 + n_1p_1 - c)\,n_0,\quad p_1 - 2(n_0p_0 + n_1p_1 - c)\,n_1\big),\\
\mathrm{frame}(p) &= \big(n_1(p_0 - cn_0) - n_0(p_1 - cn_1),\quad n_0(p_0 - cn_0) + n_1(p_1 - cn_1)\big),\\
\mathrm{unframeLin}(q) &= \big(n_1q_0 + n_0q_1,\quad -n_0q_0 + n_1q_1\big),\\
\mathrm{unframe}(q) &= \big(n_1q_0 + n_0q_1 + cn_0,\quad -n_0q_0 + n_1q_1 + cn_1\big).
\end{aligned}$$

These maps depend on n_0, n_1, c (Lean `lineRefl`, `frame` and `unframe` take the arguments n_0, n_1, c; `unframeLin` takes n_0, n_1). The map unframeLin is linear, and unframe is the affine map with linear part unframeLin and translation part (cn_0, cn_1).

Remark 9.21. Let $n_0^2 + n_1^2 = 1$ and $\ell = \{p : n_0p_0 + n_1p_1 = c\}$; every line in $\mathbb{R}^2$ has this form. With $\lambda(p) = n_0p_0 + n_1p_1 - c$ we have $\mathrm{lineRefl}(p) = p - 2\lambda(p)(n_0, n_1)$, so $\lambda(\mathrm{lineRefl}(p)) = -\lambda(p)$ while $n_1p_0 - n_0p_1$ is unchanged: lineRefl is the orthogonal reflection in ℓ. Moreover $\mathrm{frame}(p) = (n_1p_0 - n_0p_1,\ \lambda(p))$ is a rotation by the matrix $\left(\begin{smallmatrix} n_1 & -n_0 \\ n_0 & n_1 \end{smallmatrix}\right)$ followed by a translation, and it maps ℓ onto the x-axis. This explains the definitions; only Lemma 9.22 is used below.

Lemma 9.22 (frame identities; Lean: `MirrorOdd.unframe_apply`, `MIRRORETV/lean/MirrorMain.lean:1897`, Lean: `MirrorOdd.unframe_frame`, `MIRRORETV/lean/MirrorMain.lean:1900`, Lean: `MirrorOdd.frame_unframe`, `MIRRORETV/lean/MirrorMain.lean:1907`, Lean: `MirrorOdd.frame_lineRefl`, `MIRRORETV/lean/MirrorMain.lean:1914`, Lean: `MirrorOdd.orient_frame`, `MIRRORETV/lean/MirrorMain.lean:1920`, Lean: `MirrorOdd.frame_injective`, `MIRRORETV/lean/MirrorMain.lean:1925`, Lean: `MirrorOdd.unframe_injective`, `MIRRORETV/lean/MirrorMain.lean:1930`). *Let $n_0, n_1, c \in \mathbb{R}$. For every $q \in \mathbb{R}^2$,* $\mathrm{unframe}(q)$ *is given by the formula of Definition 9.20 (*`unframe_apply`*; no hypothesis). If $n_0^2 + n_1^2 = 1$, then for all $p, q, a, b, d \in \mathbb{R}^2$:*

(a) $\mathrm{unframe}(\mathrm{frame}(p)) = p$ *(*`unframe_frame`*);*
(b) $\mathrm{frame}(\mathrm{unframe}(q)) = q$ *(*`frame_unframe`*);*
(c) $\mathrm{frame}(\mathrm{lineRefl}(p)) = \mathrm{mir}(\mathrm{frame}(p))$ *(*`frame_lineRefl`*);*
(d) $\mathrm{orient}(\mathrm{frame}(a), \mathrm{frame}(b), \mathrm{frame}(d)) = \mathrm{orient}(a, b, d)$ *(*`orient_frame`*);*
(e) frame *and* unframe *are injective (*`frame_injective`*,* `unframe_injective`*).*

Proof. The formula for unframe holds by definition (in Lean, `unframe_apply` is proved by `rfl`). Put $s = n_0^2 + n_1^2$ and $\lambda = n_0p_0 + n_1p_1 - c$. Expanding the definition gives, with no hypothesis on n_0, n_1, c,

$$\mathrm{frame}(p) = \big(n_1p_0 - n_0p_1,\ n_0p_0 + n_1p_1 - cs\big) = Rp - (0, cs), \qquad R = \begin{pmatrix} n_1 & -n_0 \\ n_0 & n_1 \end{pmatrix}.$$

We prove four polynomial identities, valid for all n_0, n_1, c.

- $\mathrm{unframe}(\mathrm{frame}(p))$ has first coordinate $n_1(n_1p_0 - n_0p_1) + n_0(n_0p_0 + n_1p_1 - cs) + cn_0 = sp_0 - csn_0 + cn_0 = p_0 + (s-1)(p_0 - cn_0)$, and second coordinate $-n_0(n_1p_0 - n_0p_1) + n_1(n_0p_0 + n_1p_1 - cs) + cn_1 = sp_1 - csn_1 + cn_1 = p_1 + (s-1)(p_1 - cn_1)$.
- With $X = n_1q_0 + n_0q_1 + cn_0$ and $Y = -n_0q_0 + n_1q_1 + cn_1$, we get $\mathrm{frame}(\mathrm{unframe}(q)) = (n_1X - n_0Y,\ n_0X + n_1Y - cs) = (sq_0,\ sq_1 + cs - cs) = q + (s-1)q$.
- $\mathrm{lineRefl}(p) = p - 2\lambda\,(n_0, n_1)$ and $R\,(n_0, n_1)^{\mathsf{T}} = (0, s)^{\mathsf{T}}$, so $\mathrm{frame}(\mathrm{lineRefl}(p)) = \mathrm{frame}(p) - 2\lambda\,(0, s)$. Its first coordinate equals that of $\mathrm{frame}(p)$, which is also that of $\mathrm{mir}(\mathrm{frame}(p))$. Its second coordinate minus that of $\mathrm{mir}(\mathrm{frame}(p))$ is $2(n_0p_0 + n_1p_1 - cs) - 2\lambda s = -2(s-1)(n_0p_0 + n_1p_1)$.
- frame is the affine map $x \mapsto Rx - (0, cs)$ with $\det R = s$, so $\mathrm{orient}(\mathrm{frame}(a), \mathrm{frame}(b), \mathrm{frame}(d)) = s\ \mathrm{orient}(a, b, d)$ by Lemma 1.17.

With $s = 1$ these identities give (a)–(d). (They are the certificates of the Lean proofs: the first coordinate in (c) is closed by `ring`, and every other coordinate, and (d), by `linear_combination` with the multiples $p_0 - cn_0$, $p_1 - cn_1$, q_0, q_1, $-2(n_0p_0 + n_1p_1)$ and $\mathrm{orient}(a, b, d)$ of the hypothesis $s = 1$.) (e) By (a), unframe is a left inverse of frame, so frame is injective; by (b), frame is a left inverse of unframe, so unframe is injective. □

Theorem 9.23 (transport, and Theorem A for an arbitrary mirror line; Lean: `MirrorOdd.transport`, `MIRRORETV/lean/MirrorMain.lean:1937`, Lean: `MirrorOdd.mirror_odd_general_line`, `MIRRORETV/lean/MirrorMain.lean:2641`, Lean: `MirrorOdd.mirror_even_general_line`, `MIRRORETV/lean/MirrorMain.lean:2650`). *Part (a) moves a set that is symmetric in a line to a set that is symmetric in the x-axis; parts (b) and (c) are Theorems 9.17 and 9.18 for an arbitrary mirror line.*

(a) (`transport`*) Let $n_0, n_1, c \in \mathbb{R}$ with $n_0^2 + n_1^2 = 1$, let $S \subset \mathbb{R}^2$ be finite, and let $m \in \mathbb{N}$. Suppose that S is in general position, that* $\mathrm{lineRefl}(p) \in S$ *for every $p \in S$, and that S has no convex m-gon. Let $S' = \mathrm{frame}(S)$. Then (i) S' is in general position; (ii)* $\mathrm{mir}(q) \in S'$ *for every $q \in S'$; (iii) S' has no convex m-gon; and (iv) $|S'| = |S|$.*

(b) (`mirror_odd_general_line`*) Let $j, M \in \mathbb{N}$ with $2 \le j$, and assume the parametric hypothesis (hM) of Theorem 9.17. Let $n_0, n_1, c \in \mathbb{R}$ with $n_0^2 + n_1^2 = 1$, and let $S \subset \mathbb{R}^2$ be finite and in general position, with* $\mathrm{lineRefl}(p) \in S$ *for every $p \in S$ and with no convex $(2j+1)$-gon. Then $|S| \le 2M$.*

(c) (`mirror_even_general_line`*) Let $j, M \in \mathbb{N}$ with $2 \le j$, and assume the parametric hypothesis (hM) of Theorem 9.18. Let $n_0, n_1, c \in \mathbb{R}$ with $n_0^2 + n_1^2 = 1$, and let $S \subset \mathbb{R}^2$ be finite and in general position, with* $\mathrm{lineRefl}(p) \in S$ *for every $p \in S$ and with no convex $2j$-gon. Then $|S| \le 2M + 2$.*

Here lineRefl *and* frame *are the maps of Definition 9.20 for the given n_0, n_1, c.*

Proof. (a) (i) Let $a', b', d' \in S'$ be pairwise distinct, say $a' = \mathrm{frame}(a)$, $b' = \mathrm{frame}(b)$, $d' = \mathrm{frame}(d)$ with $a, b, d \in S$. Then a, b, d are pairwise distinct (equal points have equal images), so $\mathrm{orient}(a, b, d) \ne 0$ by Lemma 1.20, and $\mathrm{orient}(a', b', d') = \mathrm{orient}(a, b, d) \ne 0$ by Lemma 9.22(d). By Lemma 1.20, S' is in general position. (ii) If $q = \mathrm{frame}(p)$ with $p \in S$, then $\mathrm{mir}(q) = \mathrm{frame}(\mathrm{lineRefl}(p))$ by Lemma 9.22(c), and $\mathrm{lineRefl}(p) \in S$. (iii) Suppose that S' has a convex m-gon. The map unframe is affine and injective (Lemma 9.22(e)), so $\mathrm{unframe}(S')$ has a convex m-gon by Lemma 9.6(b). But $\mathrm{unframe}(S') = \{\mathrm{unframe}(\mathrm{frame}(p)) : p \in S\} = S$ by Lemma 9.22(a), a contradiction. (iv) frame is injective (Lemma 9.22(e)).

(b) By (a) with $m = 2j + 1$, the finite set $S' = \mathrm{frame}(S)$ is in general position, is mir-symmetric, has no convex $(2j + 1)$-gon, and $|S'| = |S|$. Theorem 9.17 (with the same j, M and (hM)) gives $|S| = |S'| \le 2M$.

(c) The same with $m = 2j$ and Theorem 9.18. This is also the structure of the Lean proofs. □

9.7. **Small k.**

Lemma 9.24 (configurations without a 3-cap; Lean: `MirrorOdd.hasCap3_of`, `MIRRORETV/lean/MirrorMain.lean:1991`, Lean: `MirrorOdd.capacity_cup`, `MIRRORETV/lean/MirrorMain.lean:2002`)**.** *Both parts concern configurations in the sense of Definition 9.3.*

(a) (`hasCap3_of`*) Let C be a configuration of n points, and let $a < b < d$ in $[n]$ with* $\mathrm{orient}(C_a, C_b, C_d) < 0$. *Then C has a 3-cap.*

(b) (`capacity_cup`*) Let $n, K \in \mathbb{N}$ with $4 \le K$, and let C be a configuration of n points that has no 3-cap and such that* $\mathrm{pts}(C)$ *has no convex K-gon. Then $n \le K - 1$.*

Proof. (a) The map $f : [3] \to [n]$ with $f(0) = a$, $f(1) = b$, $f(2) = d$ is strictly increasing, and the only triple $i < j < l$ in $[3]$ is $(0, 1, 2)$, for which the required inequality is the hypothesis.

(b) Suppose $n > K - 1$, that is, $n \ge K$. The map $i \mapsto C_i$ is injective because the abscissae increase strictly. Let $e : [K] \to [n]$ be the inclusion $i \mapsto i$ (Lean `Fin.castLEEmb`). For $i < j < l$ in $[K]$ we have $\mathrm{orient}(C_i, C_j, C_l) \ne 0$ (Definition 9.3(a)), and $\mathrm{orient}(C_i, C_j, C_l) < 0$ is impossible because by (a) it would give a 3-cap; hence $\mathrm{orient}(C_i, C_j, C_l) > 0$. Lemma 1.38 with $N = n$, $k = K \ge 4$, $p = C$, this e and the set $\mathrm{pts}(C)$ (which contains every $C_{e(i)}$) gives a convex K-gon in $\mathrm{pts}(C)$, a contradiction. □

Theorem 9.25 ($k = 4, 5, 6$; Lean: `MirrorOdd.mirror_four`, `MIRRORETV/lean/MirrorMain.lean:2019`, Lean: `MirrorOdd.mirror_five`, `MIRRORETV/lean/MirrorMain.lean:2040`, Lean: `MirrorOdd.mirror_six`, `MIRRORETV/lean/MirrorMain.lean:2129`, Lean: `MirrorOdd.mirror_four_line`, `MIRRORETV/lean/MirrorMain.lean:2159`, Lean: `MirrorOdd.mirror_five_line`, `MIRRORETV/lean/MirrorMain.lean:2165`, Lean: `MirrorOdd.mirror_six_line`, `MIRRORETV/lean/MirrorMain.lean:2171`)**.** *In (a)–(c) let $S \subset \mathbb{R}^2$ be finite and in general position, with* $\mathrm{mir}(p) \in S$ *for every $p \in S$. In (d)–(f) let $n_0, n_1, c \in \mathbb{R}$ with $n_0^2 + n_1^2 = 1$, and let $S \subset \mathbb{R}^2$ be finite and in general position, with* $\mathrm{lineRefl}(p) \in S$ *for every $p \in S$.*

(a) (`mirror_four`*) If S has no convex 4-gon, then $|S| \le 4$.*
(b) (`mirror_five`*) If S has no convex 5-gon, then $|S| \le 8$.*
(c) (`mirror_six`*) If S has no convex 6-gon, then $|S| \le 12$.*
(d) (`mirror_four_line`*) If S has no convex 4-gon, then $|S| \le 4$.*
(e) (`mirror_five_line`*) If S has no convex 5-gon, then $|S| \le 8$.*
(f) (`mirror_six_line`*) If S has no convex 6-gon, then $|S| \le 12$.*

These statements have no hypotheses beyond the displayed ones.

Proof. We follow the Lean proofs.

(a) First, $|R(S)| \le 1$. Otherwise there are $p \ne q$ in $R(S)$. Lemma 9.15, applied with $T = R(S)$ (all its points have positive ordinate, so the condition on axis points is vacuous), gives $p_0 \ne q_0$; exchanging the names if necessary, $p_0 < q_0$. Proposition 9.13(a) with $(c_0, c_1) = (p, q)$ and $P = S$ applies ($p_1, q_1 > 0$, and $\mathrm{mir}\, p, \mathrm{mir}\, q, q, p \in S$) and gives a convex 4-gon in S, a contradiction. By Lemma 9.14, $|S| = 2|R(S)| + |A(S)| \le 2 + 2 = 4$.

(b) *Step 1:* if $T \subseteq S$ has $p_1 \ge 0$ for all $p \in T$ and at most one point on the axis, then $|T| \le 4$. Let $C = \mathrm{cfg}(T)$ (Lemma 9.16). C has no 3-cap, because otherwise Lemma 9.16(b) with $m = 3$ gives a convex 5-gon in S. (The Lean proof makes the same case distinction as the proof of Lemma 9.16(b) for the cap c_0, c_1, c_2, but applies Proposition 9.13(b) when the three points lie strictly above the axis; Proposition 9.13(g), with the points a, c_0, c_1

of the Proposition taken to be c_2, c_0, c_1, when c_2 lies on the axis; and Proposition 9.13(f), with a, c_0, c_1 taken to be c_0, c_1, c_2, when c_0 lies on the axis. The case that c_1 lies on the axis is excluded by Lemma 9.7(b).) Moreover $\mathrm{pts}(C) = T \subseteq S$ has no convex 5-gon. Lemma 9.24(b) with $K = 5$ gives $|T| \le 4$. *Step 2:* $|R(S)| \le 4$ by Step 1. *Step 3:* if $A(S) = \emptyset$, then $|S| = 2|R(S)| \le 8$. Otherwise, for $a \in A(S)$, Step 1 applied to $R(S) \cup \{a\}$ (as in Step 3 of Theorem 9.17) gives $|R(S)| + 1 \le 4$, and $|S| \le 2 \cdot 3 + 2 = 8$ by Lemma 9.14.

(c) Let $C = \mathrm{cfg}(R(S))$ (Lemma 9.16 with $T = R(S)$). Suppose C has a 3-cap, given by f, and put $c_i = C_{f(i)}$; then $(c_0)_0 < (c_1)_0 < (c_2)_0$, all $(c_i)_1 > 0$, $\mathrm{orient}(c_0, c_1, c_2) < 0$, and S contains c_i and $\mathrm{mir}\, c_i$. Then Proposition 9.13(c) with $P = S$ (equivalently, Lemma 9.16(a) with $m = 3$) gives a convex 6-gon in S, a contradiction. Also $\mathrm{pts}(C) = R(S) \subseteq S$ has no convex 6-gon. Lemma 9.24(b) with $K = 6$ gives $|R(S)| \le 5$, so $|S| \le 2 \cdot 5 + 2 = 12$ by Lemma 9.14.

(d)–(f) By Theorem 9.23(a) with $m = 4, 5, 6$, the set $S' = \mathrm{frame}(S)$ satisfies the hypotheses of (a)–(c), respectively, and $|S'| = |S|$. □

Coverage. Covered, all in namespace `MirrorOdd` of `MIRRORETV/lean/MirrorMain.lean` unless stated otherwise: `mir`, `Rset`, `Aset` (Definition 9.1); `mir_zero`, `mir_one`, `mir_mir`, `orient_mir3`, `orient_mir2`, `orient_cycl`, `orient_swap12`, `orient_rev` (Lemma 9.2); `ErdosSzekeres.Omission.Configuration` and `Configuration.HasCap` (`ErdosSzekeres/Omission/Realization.lean`), `xset`, `emb`, `sortPt`, `cfg` (Definition 9.3); `xset_card`, `emb_mem`, `sortPt_mem`, `sortPt_x`, `sortPt_strictMono`, `sortPt_injective`, `sortPt_image` (Lemma 9.5); `hasConvexNGon_of_le`, `hasConvexNGon_affineMap` (Lemma 9.6); `cap_below_lines` (Lemma 9.7); `cap_mirror_polygon` (Theorem 9.8); `axis_left_polygon`, `axis_right_polygon` (Theorem 9.10); `cyc4`–`cyc8` (Lemma 9.12); `cap_mirror_quadrilateral`, `cap_mirror_pentagon`, `cap_mirror_hexagon`, `cap_mirror_heptagon`, `cap_mirror_octagon`, `axis_left_pentagon`, `axis_right_pentagon`, `axis_left_heptagon`, `axis_right_heptagon` (Proposition 9.13); `card_split`, `card_A_le` (Lemma 9.14); `injOn_x` (Lemma 9.15); `mirror_odd_general` (Theorem 9.17); `mirror_even_general` (Theorem 9.18); `lineRefl`, `frame`, `unframeLin`, `unframe` (Definition 9.20); `unframe_apply`, `unframe_frame`, `frame_unframe`, `frame_lineRefl`, `orient_frame`, `frame_injective`, `unframe_injective` (Lemma 9.22); `transport`, `mirror_odd_general_line`, `mirror_even_general_line` (Theorem 9.23); `hasCap3_of`, `capacity_cup` (Lemma 9.24); `mirror_four`, `mirror_five`, `mirror_six` and their `_line` versions (Theorem 9.25). Three items are written steps with no Lean declaration of their own: Lemma 9.16 isolates the cap argument of the local claims `keyT` and `hR`; part (b) of Lemma 9.7 states the local claim `interior` and the inline interior-point argument of the Lean proofs; and part (e) of Lemma 9.2 is an identity that the Lean proof of `cap_below_lines` uses implicitly when it expands orient and closes each case by `nlinarith`.

Not covered here, deliberately, from `MIRRORETV/lean/MirrorMain.lean`:

- `hasNGon_heptagon` (line 1512), `capacity16` (1529), `mirror_seven_of` (1676), `mirror_seven` (1803), `capacity22g` (1812), `mirror_eight` (1820), `mirror_seven_line` (1969) and `mirror_eight_line` (1977). `hasNGon_heptagon` is the bridge, used by `capacity16`, from a 7-gon of the combinatorial model of a configuration (`colorConfig`) to a convex 7-gon of its point set. The others rest on Baek's theorem (the vendored theorem `main`): `capacity16` applies it directly, `capacity22g` through `geometric_capacity22`, and the remaining five through these two. They are treated in the section of Theorems 10.31 and 10.33: `mirror_seven`, `mirror_eight` and their `_line` versions are these two theorems, and the other four are statements used in their proofs.
- `mirror_seven'` (2631) and `mirror_eight'` (2636): consistency re-derivations of the bounds 32 and 46. They are Theorem 9.17 with $(j, M) = (3, 16)$ and Theorem 9.18 with $(j, M) = (4, 22)$, with (hM) supplied by `capacity16` and `capacity22g` (see Remark 9.19); the bounds themselves are Theorems 10.31 and 10.33.
- The `#print axioms` lines 2662–2677: audit commands; their output is the axioms evidence cited at the beginning of this section.

Not covered here, deliberately, from the other Lean files of `MIRRORETV`:

- The five declarations of `MIRRORETV/lean/MirrorNine_append.lean` (namespace `C12Check`). They are treated in the section of Theorem 10.35, and `config_capacity113`, `mirror_nine` and `mirror_nine_line` are that theorem.
- The independent formalizations `MIRRORETV/review/MirrorA.lean` and `MIRRORETV/review/MirrorCap.lean`, written by reviews. Each imports only Mathlib and has its own definitions, and no Lean file of the project imports either of them. No statement or proof of this document uses them. Their content overlaps that of Theorems 9.8, 9.10, 9.17, 9.18 and 9.23, but in another formulation: points in

$\mathbb{R} \times \mathbb{R}$, caps and cups defined by consecutive triples, and an upper bound on the class that defines $N(n, a, b)$ (Remark 9.19(2)) in place of (hM). Beyond this overlap they contain the odd case $k = 3$, which Theorem 9.17 excludes (Remark 9.19(3)); the mirror lemma for caps with both endpoints on the axis; the convex position of caps and of cups; and the coincidence statement `MirrorCap.coincidences`.

- `MirrorA.lean` (891 lines, `1a8a62cb4385...`; namespace `MirrorA`; 16 definitions, 55 theorems and lemmas, and one `example`). The definitions are `orient`, the reflection `mu` in the x-axis and its linear form `muL`, `ConvexPos`, the caps and cups `IsCapSeq` and `IsCupSeq`, the set `V` of the points of a cap and their mirror images, `GenPos`, `XDistinct`, `HasGon`, `HasCap`, `HasCup`, the class `InClass` that defines $N(n, a, b)$, `MirrorSym`, `refl`, the reflection in the line through a point with a given unit direction, and `isoT`, the rigid motion that maps this line onto the x-axis. The main results are `lemma31` (282: a cap, in the sense of `IsCapSeq`, all of whose points have ordinates ≥ 0, together with its mirror image, is in convex position; the review's form of `MIRRORETV/PROOFS.md`, Lemma 3.1), `cap_convexPos` (352) and `cup_convexPos` (417) (the points of a cap, resp. of a cup, are in convex position), `theoremA_even` (593) and `theoremA_odd` (615) (in the indexing of the written sources, for $j \geq 2$: if S is symmetric in the x-axis, in general position, and has no $2j$, resp. $2j - 1$, points in convex position, then some set X in the class of $N(2j, j, 2j)$, resp. $N(2j - 1, j, 2j - 1)$, satisfies $|S| \leq 2|X| + 2$, resp. $|S| \leq 2|X|$), `theoremA_even_bound` (685) and `theoremA_odd_bound` (694) (the bounds $2N_b + 2$ and $2N_b$ from an upper bound N_b on that class), `transport` (771; the transport from an arbitrary mirror line to the x-axis by the rigid motion `isoT`), and `theoremA_even_line` (810) and `theoremA_odd_line` (821) (the two bounds for the reflection in an arbitrary line). With $j = 2$ the odd case is $k = 3$. The other 45 lemmas are auxiliary, among them three checks of the definition of `refl` (`refl_refl`, `refl_fix_line`, `refl_normal`); the `example` (852) checks that `ConvexPos` is not vacuous. The recorded log `MIRRORETV/review/MirrorA.out` prints the three standard axioms for eight of these theorems.
- `MirrorCap.lean` (354 lines, `a5c6dadb5ac6...`; namespace `MirrorCap`; 16 declarations): the definitions `orient`, `sl` (the slope of an edge) and `V` (the points of the cap and their mirror images), ten auxiliary lemmas and theorems (`edge_eq`, `orient_eq_slopes`, `slope_dec_of_orient`, `sl_anti`, `x_mono`, `chord_right`, `chord_left`, `y_nonneg`, `exposed_c`, `exposed_mu`), and `convex_position` (292), `convexIndependent` (316) and `coincidences` (327). For a cap $c_0, \ldots, c_{m-1}$ given by coordinates (strictly increasing abscissae, consecutive triples clockwise) whose interior points have positive ordinates and whose endpoints have nonnegative ordinates (positive if $m = 1$), `convex_position` says that no point of V lies in the convex hull of the other points of V (the form of `MIRRORETV/PROOFS.md`, Lemma 3.1), and `convexIndependent` says the same in Mathlib's terms. From the monotonicity of the abscissae and the positivity of the interior ordinates, `coincidences` derives that $c_i = c_k$ or $\operatorname{mir} c_i = \operatorname{mir} c_k$ forces $i = k$, and that $c_i = \operatorname{mir} c_k$ forces $i = k$, $(c_i)_1 = 0$ and $i \in \{0, m - 1\}$; by the file's comment, this gives $|V| = 2m - e$, where e is the number of endpoints of the cap on the axis. The recorded log `MIRRORETV/review/MirrorCap.out` prints the three standard axioms for these three theorems.

- The far-flat files `MIRRORETV/lean/FarFlat.lean` (`08a79f881eea...`) and `MIRRORETV/review/FarFlatExtra.lean` (144 lines, `d30a9b874ba5...`). Both import only Mathlib, and no Lean file of the project imports them. They hold estimates for the far-flat lower-bound construction (Theorem B of `MIRRORETV/README.md`). This document treats no part of Theorem B, so no section restates either file.
  - `FarFlat.lean`: the root-namespace definition `orient` on six reals (line 12), `orient_wzq_neg` (15), `orient_uqz_neg` (34), `orient_cyclic` (55), `orient_sum` (60), `inside_triangle` (67) and `line_misses` (89): three orientation estimates (`orient_wzq_neg`, `orient_uqz_neg`, `line_misses`), two identities (`orient_cyclic`, `orient_sum`) and a barycentric representation (`inside_triangle`). The recorded log `MIRRORETV/lean/FarFlat.out` prints the axioms of five of these six theorems (all but `orient_cyclic`), and they are the standard three.
  - `FarFlatExtra.lean` (namespace `FFX`): the definition `orient` on six reals (line 20), `orient_sum` (22), `bary_of_neg` (28), `bary_of_pos` (44), `gp_cross_axis` (60), `a1_inside` (69) and `lemma51_core` (91). Here `orient_sum` and `bary_of_neg` have the same statements as `orient_sum` and `inside_triangle` of `FarFlat.lean`; `bary_of_pos` is the barycentric representation for three positive orientations; `gp_cross_axis` is a general-position estimate for a point above the axis, a point below it and a point of the axis to the right of both; `a1_inside` is the counting step of Theorem B; and `lemma51_core` is the composed core of Lemma 5.1 of `MIRRORETV/PROOFS.md`. The

recorded log `MIRRORETV/review/FarFlatExtra.out` prints the axioms of five of these six theorems (all but `orient_sum`), the standard three, and four warnings about unused variables.

- `MIRRORETV/review/ReplayRepo.lean` (118 lines, `20e479e8b3ab...`) contains no definition or theorem, only a command `#replay_repo`, defined by `elab` at line 51 and run at line 118. By its header comment, it imports only library modules, reads the compiled repository modules without importing them, and sends the repository constants in the dependency closure of those that `MIRRORETV/lean/MirrorMain.lean` references directly to Lean's kernel again. The recorded log `MIRRORETV/review/ReplayRepo.out` reports 497 such constants, none of them an axiom, and a successful replay. It is an audit and states no theorem, so there is nothing to restate.

10. Mirror bounds via Baek's theorem and the three-class reduction

What this section proves. Let $S \subset \mathbb{R}^2$ be a finite set in general position that is mapped into itself by the reflection in a line. We prove that $|S| \leq 32$ if S contains no 7 points in convex position, $|S| \leq 46$ if it contains no 8 points in convex position, and $|S| \leq 226$ if it contains no 9 points in convex position (Theorems 10.31, 10.33 and 10.35; each has a version for the x-axis and a version for an arbitrary line). The bounds for 7 and 8 rest on Baek's theorem on 4-caps, the bound for 9 in addition on the cups–caps theorem of Erdős and Szekeres. These two theorems are *cited*, in the exact form in which the Lean code uses them (Theorems 10.4 and 10.6). Everything between their abstract, purely combinatorial statements and the geometric notions is proved here in full (written proofs, not refereed): the bridge from Baek's configurations to points of the plane (Subsection 10.2), the capacity bounds 16 and 22 (Subsection 10.3) and 113 (Subsection 10.7) and the mirror reductions (Subsections 10.4–10.7); the latter use the general mirror lemma and the general reductions of Theorems 9.8, 9.10, 9.17 and 9.23, which are proved elsewhere in this document. Finally (Subsection 10.8) we identify the two parametrisations of line reflections that occur in the Lean files of this section (Theorem 10.38), prove the two mirror case classes of Definition 7.7 (Theorem 10.39), and restate the reduction Theorem 7.39 and part (a) of Theorem 8.28 (the `AffineEquiv` form) with the three remaining case classes as their only case-class hypotheses (Theorem 10.40, which is *conditional* on these three statements; they are not proved in Lean or in this document, see Remark 10.41).

Sources. Paths beginning with `ErdosSzekeres/` or `docs/` are relative to the repository root; all other paths (such as `MIRRORETV/...`, `SYMLEAN/...` and the record directories named in Remark 10.41) are relative to the results directory of the project. A file name without a directory, such as `MirrorMain.lean`, is short for the file of that name whose full path is given in this section. The declarations treated here are in the following files (SHA-256, first 12 hexadecimal digits):

- `MIRRORETV/lean/MirrorMain.lean` (`ca623e55a1b9`), namespace `MirrorOdd`;
- `MIRRORETV/lean/MirrorNine_append.lean` (`c2afca380211`), namespace `C12Check`; this file is not standalone: it was compiled appended to `MirrorMain.lean`; line numbers below refer to the file itself;
- `SYMLEAN/mirror/ReductionThreeClasses.lean` (`0cc83994fe8c`), called RTC below; its Part C (lines 4238–4322) is treated in Subsection 10.8;
- the vendored formalization of Baek's theorem (adapted from Baek's Lean code, Apache-2.0 licence):
  - `ErdosSzekeres/Vendor/ETV/Main/Main.lean` (`83dd0a214204`),
  - `ErdosSzekeres/Vendor/ETV/Main/CapCup.lean` (`fb8fbefac260`),
  - `ErdosSzekeres/Vendor/ETV/Config/Defs.lean` (`ce2ccf7c1d28`),
  - `ErdosSzekeres/Vendor/ETV/Config/Lemmas.lean` (`5395e27a82cc`),
  - `ErdosSzekeres/Vendor/ETV/Lib/List/Defs.lean` (`46d750763528`);
- the bridge files: `ErdosSzekeres/Omission/Realization.lean` (`ae4970e0808e`), which is the project's own code (it has no adaptation header; `docs/THIRD_PARTY.md` says that its connections were added in the project, using the strategy of M. Schroeder's archive [9]), and six files adapted from that archive (MIT licence):
  - `ErdosSzekeres/Omission/ActualRanks.lean` (`3788f5149b3a`),
  - `ErdosSzekeres/Omission/PlanarChains.lean` (`21161d053dc0`),
  - `ErdosSzekeres/Omission/ListGeometry.lean` (`1a03c283176b`),
  - `ErdosSzekeres/Omission/BoundaryGeometry.lean` (`f55a711e7aa9`),
  - `ErdosSzekeres/Omission/BaekGeometry.lean` (`db8ac3c30dc0`),
  - `ErdosSzekeres/Omission/Baek.lean` (`3433c3c2124d`).

Lean version and axioms. The files were compiled with Lean 4.24.0 and Mathlib `f897ebcf`. The recorded `#print axioms` outputs report exactly the three standard axioms `propext`, `Classical.choice`, `Quot.sound` for the following declarations.

- `MIRRORETV/lean/MirrorMain.out` (`c6024155067e`), lines 848–863: the 16 main theorems of `MirrorOdd`, among them `mirror_seven` (line 848), `mirror_eight` (849), `mirror_seven_line` (850), `mirror_eight_line` (851), `mirror_odd_general` (858) and `mirror_odd_general_line` (862). The log records no error.
- `MIRRORETV/lean/MirrorNine.out` (`976fda168e70`), lines 879–882: `C12Check.config_capacity113`, `C12Check.capacity113`, `C12Check.mirror_nine`, `C12Check.mirror_nine_line`. This log records the compilation of the concatenation of `MirrorMain.lean` and `MirrorNine_append.lean`; in it, the lines of the appended file are shifted by 2677.
- `SYMLEAN/mirror/axioms.out` (`ab92257c6f17`): the six Part C theorems `reflLine_eq_lineRefl`, `card_le_32_of_reflLine`, `mirrorOneAxis33_proved`, `mirrorNoAxis34_proved`, `heptagon_of_nontrivial_isometry_3`, `heptagon_of_nontrivial_affineEquiv_3`, and the 16 `MirrorOdd` theorems in their copy inside RTC. This file is a filtered excerpt of the compile output: some lines are repeated, and of the five `#check` commands of RTC (lines 4312–4316) only the two for the `_3` theorems have output in it (lines 17–20, the first one again at lines 39–40), truncated after the binders `HalfTurn34` and `Threefold33`.

Since `#print axioms` reports the axioms of the whole dependency cone, these lines also cover the declarations that are used but not printed individually. Every other Lean declaration named in the head of a statement below is used by one of the printed theorems, for example `capacity16`, `capacity22g`, `hasNGon_heptagon`, the bridge lemmas of Subsection 10.2, and the vendored theorems `main` and `Config.cap_cup`. In the three files of this section the tactic `decide` occurs only on comparisons of small indices and on the binomial values $\binom{6}{2}=15$, $\binom{8}{2}=28$, $\binom{9}{3}=84$, which the kernel evaluates; no `native_decide`, `sorry` or added axiom occurs in them or in the vendored and bridge files listed above. The logs were produced by the project and were not re-run for this document.

Copies of the same code. RTC lines 1562–4236 are byte-identical to `MIRRORETV/lean/MirrorMain.lean` lines 3–2677, so line L of `MirrorMain.lean` is line $L+1559$ of RTC; we cite `MirrorOdd` declarations by their lines in `MirrorMain.lean`. RTC lines 38–1559 are byte-identical to `SYMLEAN/AffineReduction.lean` lines 12–1533, and RTC lines 38–1084 to `SYMLEAN/SymmetricReduction.lean` lines 12–1058 (so a line L of `SymmetricReduction.lean` is line $L+26$ of RTC). These identities can be checked with `sed` and `diff`.

Dependencies and rigour. We use the foundations (Definitions 1.1, 1.3, 1.5, 1.6, 1.7, 1.10 and Lemmas 1.15, 1.17, 1.20, 1.22, 1.23, 1.27, 1.40), the mirror lemma and the general reductions of Definition 9.3 and Theorems 9.8, 9.10, 9.17, 9.23, and in Subsection 10.8 Definitions 7.1, 7.7 and Theorems 7.39, 8.28. Three remarks refer to further results that no proof of this section uses: Remark 10.25 (on the Lean proofs of Lemma 10.24) to Lemma 1.38, Remark 10.34 (on the second Lean proof `mirror_eight'` of Theorem 10.33(a)) to Theorem 9.18, and Remark 10.41 (on the hypotheses of Theorem 10.40) to Theorems 2.22, 3.16, 11.98 and 11.82. Every Lean declaration stated below is *Lean-checked* (axioms evidence above); the proofs given here are *written proofs (not refereed)*. Theorems 10.4 and 10.6 are cited published theorems; their vendored formalizations are Lean-checked as part of the dependency cone above, but this document does not reprove them. Theorem 10.40 takes as hypotheses three propositions that are not proved in Lean or in this document.

Conventions. Points are $p=(p_0,p_1)\in\mathbb{R}^2$ (Lean `Point`, Definition 1.1); we also write $x(p):=p_0$ and $y(p):=p_1$, which avoids double subscripts for points that carry an index. orient is as in Definition 1.3. We write $[n]:=\{0,1,\dots,n-1\}$ (Lean `Finset.range n`). Where $[m]$ is the index set of a configuration of m points, or the domain or codomain of a cap map (such as $f\colon[k]\to[n]$ in Subsection 10.2), it is the same set viewed as Lean's `Fin m`, as in Definition 9.3. "S contains k points in convex position" means $\mathsf{HasConvexNGon}(k,S)$ of Definition 1.7: some k-element subset of S is in convex position (Definition 1.6); no emptiness is required. We use without comment the identities $\mathrm{orient}(a,c,b)=-\mathrm{orient}(a,b,c)$, $\mathrm{orient}(b,c,a)=\mathrm{orient}(a,b,c)$, $\mathrm{orient}(b,a,c)=-\mathrm{orient}(a,b,c)$, $\mathrm{orient}(a,b,a)=\mathrm{orient}(a,b,b)=0$ (Lemma 1.15; the last two are immediate from the formula of Definition 1.3), and the fact that $\mathrm{orient}(a,b,w)$ is an affine function of w for fixed a,b (it equals $(x(b)-x(a))(y(w)-y(a))-(y(b)-y(a))(x(w)-x(a))$).

10.1. Baek's configurations and the two cited theorems.

Definition 10.1 (configurations, caps, cups and gons;
Lean: `Config`, `ErdosSzekeres/Vendor/ETV/Config/Defs.lean:20`,
Lean: `Config.Cap3`, `ErdosSzekeres/Vendor/ETV/Config/Defs.lean:30`,

Lean: `Config.Cap`, `ErdosSzekeres/Vendor/ETV/Config/Defs.lean:38`,
Lean: `Config.Cup`, `ErdosSzekeres/Vendor/ETV/Config/Defs.lean:41`,
Lean: `Config.Gon`, `ErdosSzekeres/Vendor/ETV/Config/Defs.lean:45`,
Lean: `Config.NGon`, `ErdosSzekeres/Vendor/ETV/Config/Defs.lean:57`,
Lean: `Config.HasNCap`, `ErdosSzekeres/Vendor/ETV/Config/Defs.lean:60`,
Lean: `Config.HasNCup`, `ErdosSzekeres/Vendor/ETV/Config/Defs.lean:63`,
Lean: `Config.HasNGon`, `ErdosSzekeres/Vendor/ETV/Config/Defs.lean:66`). Let $(\alpha, <)$ be a linearly ordered set. A *configuration* C on α is a ternary relation $\mathrm{Cup3}_C$ on α (in Lean a structure with the field `Cup3` and a decidability witness, which plays no mathematical role). Put $\mathrm{Cap3}_C(a,b,c) :\Leftrightarrow \neg\,\mathrm{Cup3}_C(a,b,c)$ (`Config.Cap3`). Let $l = (l_0, \dots, l_{r-1})$ be a finite list of elements of α, of length $|l| = r$.

(a) l is a *C-cap* if $l_0 < l_1 < \cdots < l_{r-1}$ and $\mathrm{Cap3}_C(l_i, l_{i+1}, l_{i+2})$ for every i with $i+2 < r$; l is a *C-cup* if $l_0 < l_1 < \cdots < l_{r-1}$ and $\mathrm{Cup3}_C(l_i, l_{i+1}, l_{i+2})$ for every i with $i+2 < r$. Only consecutive triples are constrained; every strictly increasing list of length at most 2 is both a C-cap and a C-cup.

(b) A pair (l, l') of lists is a *C-gon* if $|l| \geq 2$, l is a C-cap, $|l'| \geq 2$, l' is a C-cup, and l and l' have the same first entry and the same last entry. It is an *n-gon of C* if moreover $|l| + |l'| = n + 2$.

(c) For a finite set $S \subseteq \alpha$, $\mathsf{HasNCap}_C(n, S)$ means that some C-cap of length n has all its entries in S; $\mathsf{HasNCup}_C(n, S)$ means the same for C-cups; and $\mathsf{HasNGon}_C(n, S)$ means that some n-gon (l, l') of C has all entries of l and of l' in S.

Remark 10.2. In Lean, "strictly increasing" is `List.IsChain` for the relation $<$ (consecutive entries increase, hence all entries are in increasing order), the condition on consecutive triples is `List.Chain3'` (`ErdosSzekeres/Vendor/ETV/Lib/List/Defs.lean:46`), "same first entry" is the equality `l.head? = l'.head?` of the optional first entries and "same last entry" is `l.getLast? = l'.getLast?`; since both lists are nonempty these are the stated equalities. "All entries in S" is `List.In` (`ErdosSzekeres/Vendor/ETV/Lib/List/Defs.lean:22`). An n-gon (l, l') consists of n entries if the two lists share only their end entries; this is the reason for the normalisation $|l| + |l'| = n + 2$. Nothing else is required of a gon (Baek calls such gons *weak*): for an arbitrary relation $\mathrm{Cup3}_C$ there is no geometric content. The geometric meaning enters only in Subsection 10.2.

Lemma 10.3 (monotonicity, and cups as gons;
Lean: `hasNCap_supset`, `ErdosSzekeres/Vendor/ETV/Config/Lemmas.lean:293`,
Lean: `hasNGon_supset`, `ErdosSzekeres/Vendor/ETV/Config/Lemmas.lean:305`,
Lean: `ncup_is_ngon`, `ErdosSzekeres/Vendor/ETV/Config/Lemmas.lean:275`). *Let C be a configuration on a linearly ordered set α and $n \in \mathbb{N}$.*

(a) If $S_1 \subseteq S_2 \subseteq \alpha$ are finite, then $\mathsf{HasNCap}_C(n, S_1)$ implies $\mathsf{HasNCap}_C(n, S_2)$, and $\mathsf{HasNGon}_C(n, S_1)$ implies $\mathsf{HasNGon}_C(n, S_2)$.

(b) If $S \subseteq \alpha$ is finite, $2 \leq n$ and $\mathsf{HasNCup}_C(n, S)$, then $\mathsf{HasNGon}_C(n, S)$.

Proof. (a) The same lists are witnesses, since their entries lie in $S_1 \subseteq S_2$.

(b) Let $l = (l_0, \dots, l_{n-1})$ be a C-cup with all entries in S. As $n \geq 2$ and l is strictly increasing, $l_0 < l_{n-1}$. The list (l_0, l_{n-1}) is strictly increasing and has no consecutive triple, so it is a C-cap of length 2; its entries lie in S; it has the same first entry l_0 and the same last entry l_{n-1} as l; and $2 + n = n + 2$. Hence $((l_0, l_{n-1}), l)$ is an n-gon of C with all entries in S. □

Theorem 10.4 (Baek's theorem, cited;
Lean: `main`, `ErdosSzekeres/Vendor/ETV/Main/Main.lean:87`). *Let $(\gamma, <)$ be a linearly ordered set, $n \in \mathbb{N}$, C a configuration on γ, and $S \subseteq \gamma$ a finite set with $\binom{n+2}{2} + 2 \leq |S|$. Then $\mathsf{HasNCap}_C(4, S)$ or $\mathsf{HasNGon}_C(n+3, S)$.*

Remark 10.5 (source, status and index shift). (1) *Source.* J. Baek, *On the Erdős–Tuza–Valtr conjecture*, Eur. J. Combin. 124 (2025) 104085; arXiv:2206.04260. Its geometric main theorem (Theorem 1.6 in the arXiv numbering, quoted in `MIRRORETV/PROOFS.md`, proof of Corollary 7.2) says: for every $N \geq 3$, every set of $\binom{N-1}{2} + 2$ points in the plane in general position with pairwise distinct x-coordinates contains a 4-cap or N points in convex position. Theorem 10.4 is a combinatorial statement, for an arbitrary ternary relation and weak gons. It is the theorem `main` of the vendored file `ErdosSzekeres/Vendor/ETV/Main/Main.lean`, which is adapted from Baek's own Lean code (according to its header, the changes concern only imports and compatibility with Lean/Mathlib 4.24.0); the vendored file is compiled with the repository, so the axioms evidence above covers it. `MIRRORETV/README.md`, §8, states that Baek's Theorem 2.7 holds for all colourings with weak gons; we have not compared the wording of that theorem with Theorem 10.4. In this document Theorem 10.4 is cited, not reproved.

(2) *Index shift.* The Lean theorem `main` at n is Baek's statement for gons of size $N = n + 3$: indeed $\binom{n+2}{2} + 2 = \binom{N-1}{2} + 2$. We use $n = 4$ ($N = 7$, threshold $\binom{6}{2} + 2 = 17$), $n = 5$ ($N = 8$, threshold $\binom{7}{2} + 2 = 23$) and $n = 6$ ($N = 9$, threshold $\binom{8}{2} + 2 = 30$). Equivalently: if a finite set S contains no 4-cap and no N-gon of C, then $|S| \le \binom{N-1}{2} + 1$, which is 16, 22, 29 for $N = 7, 8, 9$.

(3) *Geometric meaning.* For the configuration C_c of a point configuration c of n points (Definition 10.9), C_c-caps and C_c-cups with entries in $[n]$ are geometric caps and cups (Lemma 10.13), and for $N \in \{7, 8, 9\}$ every N-gon of C_c with entries in $[n]$ yields N of the points of c in convex position (Proposition 10.16; its proof does not use the value of N).

Theorem 10.6 (cups–caps theorem, cited;
Lean: `Config.cap_cup`, `ErdosSzekeres/Vendor/ETV/Main/CapCup.lean:46`). *Let C be a configuration on a linearly ordered set α, let $a, b \in \mathbb{N}$, and let $S \subseteq \alpha$ be finite with $\binom{a+b}{a} < |S|$. Then $\mathsf{HasNCap}_C(a + 2, S)$ or $\mathsf{HasNCup}_C(b + 2, S)$.*

Remark 10.7. This is the cups–caps theorem of P. Erdős and G. Szekeres, *A combinatorial problem in geometry*, Compositio Math. 2 (1935) 463–470. Its classical proof (induction on $a + b$; split S into the first entries of $(a + 1)$-caps, that is, of caps one entry shorter than the target $(a + 2)$-cap, and the rest) uses only consecutive triples, hence holds for every configuration; the vendored file contains a Lean proof of this form, which we cite. The proof of Theorem 10.35(a) below repeats its induction step once (target 5-cap, split by the first entries of 4-caps).

Lemma 10.8 (Baek's theorem for 8-gons;
Lean: `ErdosSzekeres.Omission.Baek.four_cap_or_eight_gon`, `ErdosSzekeres/Omission/Baek.lean:12`,
Lean: `ErdosSzekeres.Omission.Baek.capacity22`, `ErdosSzekeres/Omission/Baek.lean:16`). *Let C be a configuration on a linearly ordered set α and $S \subseteq \alpha$ finite.*

(a) If $23 \le |S|$, then $\mathsf{HasNCap}_C(4, S)$ or $\mathsf{HasNGon}_C(8, S)$.
(b) If $\neg\mathsf{HasNCap}_C(4, S)$ and $\neg\mathsf{HasNGon}_C(8, S)$, then $|S| \le 22$.

Proof. (a) is Theorem 10.4 with $n = 5$, because $\binom{7}{2} + 2 = 23$. (b) If $|S| \ge 23$, then (a) contradicts the two hypotheses; hence $|S| \le 22$. □

10.2. **From configurations to points of the plane.** We use the x-sorted point configurations of Definition 9.3. A *configuration of n points* (Lean `ErdosSzekeres.Omission.Configuration`, `ErdosSzekeres/Omission/Realization.lean:9`) is a map $c\colon [n] \to \mathbb{R}^2$, $i \mapsto c_i$, with $x(c_i) < x(c_j)$ for all $i < j$ and $\mathrm{orient}(c_i, c_j, c_k) \neq 0$ for all $i < j < k$. For $k \in \mathbb{N}$, $c.\mathsf{HasCap}(k)$ (`Configuration.HasCap`, `ErdosSzekeres/Omission/Realization.lean:64`) means that there is a strictly increasing $f\colon [k] \to [n]$ with $\mathrm{orient}(c_{f(i)}, c_{f(j)}, c_{f(l)}) < 0$ for *all* $i < j < l$ in $[k]$. We write $\mathcal{P}(c) := \{c_0, \dots, c_{n-1}\}$ for the set of points of c (Lean `Finset.univ.image c.point`; it is the point set $\mathrm{pts}(c)$ of Definition 9.3(a)); since x is strictly increasing along c, $|\mathcal{P}(c)| = n$. (A configuration of n points is not a configuration in the sense of Definition 10.1; the link between the two notions is the configuration C_c of Definition 10.9.)

Definition 10.9 (the configuration C_c of a point configuration;
Lean: `ErdosSzekeres.Omission.Configuration.cup`, `ErdosSzekeres/Omission/Realization.lean:33`,
Lean: `ErdosSzekeres.Omission.ActualRanks.colorConfig`, `ErdosSzekeres/Omission/ActualRanks.lean:28`,
Lean: `ErdosSzekeres.Omission.ListGeometry.pointAt`, `ErdosSzekeres/Omission/ListGeometry.lean:13`). Let c be a configuration of n points. For $i, j, k \in \mathbb{N}$ put $c.\mathrm{cup}(i, j, k) = \mathrm{true}$ if $i, j, k < n$ and $\mathrm{orient}(c_i, c_j, c_k) > 0$, and $c.\mathrm{cup}(i, j, k) = \mathrm{false}$ otherwise. Let C_c be the configuration on $(\mathbb{N}, <)$ with $\mathrm{Cup3}_{C_c}(i, j, k)$ if and only if $c.\mathrm{cup}(i, j, k) = \mathrm{true}$ (Lean `colorConfig c.cup true`). Finally $\mathrm{pointAt}(c, i) := c_i$ for $i < n$ and $\mathrm{pointAt}(c, i) := (0, 0)$ for $i \ge n$; for indices $i < n$, which are the only ones occurring below, we simply write c_i.

Lemma 10.10 (signs of the colouring;
Lean: `ErdosSzekeres.Omission.Configuration.cup_true_iff`, `ErdosSzekeres/Omission/Realization.lean:39`,
Lean: `ErdosSzekeres.Omission.ListGeometry.cup_sign`, `ErdosSzekeres/Omission/ListGeometry.lean:19`,
Lean: `ErdosSzekeres.Omission.ListGeometry.cap_sign`, `ErdosSzekeres/Omission/ListGeometry.`

`lean:24`). *Let c be a configuration of n points and $a, b, d \in \mathbb{N}$ with $a, b, d < n$. Then $c.\mathrm{cup}(a, b, d) = \mathrm{true}$, that is $\mathrm{Cup3}_{C_c}(a, b, d)$, holds if and only if $\mathrm{orient}(c_a, c_b, c_d) > 0$. If moreover $a < b < d$, then $c.\mathrm{cup}(a, b, d) \neq \mathrm{true}$, that is $\mathrm{Cap3}_{C_c}(a, b, d)$, holds if and only if $\mathrm{orient}(c_a, c_b, c_d) < 0$.*

Proof. The first claim is Definition 10.9, since $a, b, d < n$. If $a < b < d$, then $\mathrm{orient}(c_a, c_b, c_d) \neq 0$ because c is a configuration; so $\mathrm{Cap3}_{C_c}(a, b, d)$, which is $\neg(\mathrm{orient}(c_a, c_b, c_d) > 0)$, is equivalent to $\mathrm{orient}(c_a, c_b, c_d) < 0$. (In Lean, `cup_true_iff` states the first claim with indices i, j, k in `Fin n` and the points `c.point i`; `cup_sign` and `cap_sign` state the two claims for natural numbers a, b, d, with the bounds $a, b, d < n$ as hypotheses and the points $\mathrm{pointAt}(c, a)$, which equal c_a for $a < n$. The companion `Configuration.cup_false_iff` (`ErdosSzekeres/Omission/Realization.lean:44`), the second claim with indices in `Fin n` and the equation $c.\mathrm{cup}(i, j, k) = \mathrm{false}$, is not used on the proof paths of this section and is not covered here.) □

Lemma 10.11 (four x-sorted points;
Lean: `ErdosSzekeres.Omission.PlanarChains.four_point_identity_left`, `ErdosSzekeres/Omission/PlanarChains.lean:15`,
Lean: `ErdosSzekeres.Omission.PlanarChains.four_point_identity_right`, `ErdosSzekeres/Omission/PlanarChains.lean:19`,
Lean: `ErdosSzekeres.Omission.PlanarChains.positive_four`, `ErdosSzekeres/Omission/PlanarChains.lean:23`,
Lean: `ErdosSzekeres.Omission.PlanarChains.negative_four`, `ErdosSzekeres/Omission/PlanarChains.lean:35`). *For all $a, b, c, d \in \mathbb{R}^2$,*

$$(x(c) - x(b))\ \mathrm{orient}(a, b, d) = (x(d) - x(b))\ \mathrm{orient}(a, b, c) + (x(b) - x(a))\ \mathrm{orient}(b, c, d), \tag{5}$$

$$(x(c) - x(b))\ \mathrm{orient}(a, c, d) = (x(d) - x(c))\ \mathrm{orient}(a, b, c) + (x(c) - x(a))\ \mathrm{orient}(b, c, d). \tag{6}$$

Let moreover $x(a) < x(b) < x(c) < x(d)$. If $\mathrm{orient}(a, b, c) > 0$ and $\mathrm{orient}(b, c, d) > 0$, then $\mathrm{orient}(a, b, d) > 0$ and $\mathrm{orient}(a, c, d) > 0$. If $\mathrm{orient}(a, b, c) < 0$ and $\mathrm{orient}(b, c, d) < 0$, then $\mathrm{orient}(a, b, d) < 0$ and $\mathrm{orient}(a, c, d) < 0$.

Proof. Both identities are polynomial identities in the eight coordinates; expanding both sides gives the same polynomial (we checked this symbolically; it is also the Lean proof, `unfold orient; ring`). Under $x(a) < x(b) < x(c) < x(d)$ the coefficients $x(d) - x(b)$, $x(b) - x(a)$, $x(d) - x(c)$, $x(c) - x(a)$ on the right are positive, so each right-hand side is positive (negative) when $\mathrm{orient}(a, b, c)$ and $\mathrm{orient}(b, c, d)$ are both positive (negative). Dividing by $x(c) - x(b) > 0$ gives the claims. □

Lemma 10.12 (closure of chain conditions;
Lean: `ErdosSzekeres.Omission.PlanarChains.chain3_all`, `ErdosSzekeres/Omission/PlanarChains.lean:58`). *Let $(\alpha, <)$ be a linearly ordered set, R a ternary relation on α, and $D \subseteq \alpha$ such that for all $a, b, c, d \in D$ with $a < b < c < d$,*

$$R(a, b, c) \text{ and } R(b, c, d) \implies R(a, b, d) \text{ and } R(a, c, d). \tag{7}$$

Let $l = (l_0, \ldots, l_{r-1})$ be a list of elements of D with $l_0 < l_1 < \cdots < l_{r-1}$ and $R(l_i, l_{i+1}, l_{i+2})$ for all i with $i + 2 < r$. Then $R(a, b, c)$ holds for all entries a, b, c of l with $a < b < c$ (Lean: `OrderedTriples R l`).

Proof. Since l is strictly increasing, the claim says $R(l_i, l_j, l_k)$ for all $0 \le i < j < k < r$. We prove by downward induction on i the statement $Q(i)$: $R(l_i, l_j, l_k)$ for all j, k with $i < j < k < r$. For $i \ge r - 2$ there is nothing to prove. Let $i \le r - 3$ and assume $Q(i')$ for all $i' > i$.

Step A: $R(l_i, l_{i+1}, l_k)$ for $i + 2 \le k < r$. Induction on k. For $k = i + 2$ this is a hypothesis on consecutive triples. For $k > i + 2$ we have $R(l_i, l_{i+1}, l_{k-1})$ by the inner induction hypothesis and $R(l_{i+1}, l_{k-1}, l_k)$ by $Q(i + 1)$; the first conclusion of (7) for $l_i < l_{i+1} < l_{k-1} < l_k$ gives $R(l_i, l_{i+1}, l_k)$.

Step B: $R(l_i, l_j, l_k)$ for $i + 1 < j < k < r$. By Step A, $R(l_i, l_{i+1}, l_j)$; by $Q(i + 1)$, $R(l_{i+1}, l_j, l_k)$; the second conclusion of (7) for $l_i < l_{i+1} < l_j < l_k$ gives $R(l_i, l_j, l_k)$.

Steps A ($j = i + 1$) and B ($j > i + 1$) give $Q(i)$. All points used lie in D. (The Lean proof is an induction on the list, with the same two steps.) □

Lemma 10.13 (C_c-caps and C_c-cups are geometric;
Lean: `ErdosSzekeres.Omission.ListGeometry.cup_list_all`, `ErdosSzekeres/Omission/ListGeometry.lean:39`,
Lean: `ErdosSzekeres.Omission.ListGeometry.cap_list_all`, `ErdosSzekeres/Omission/ListGeometry.lean:53`,

Lean: `ErdosSzekeres.Omission.ListGeometry.hasCap_of_list`, `ErdosSzekeres/Omission/ListGeometry.lean:85`). *Let c be a configuration of n points and l a list all of whose entries are $< n$.*

(a) If l is a C_c-cup, then $\operatorname{orient}(c_a, c_b, c_d) > 0$ *for all entries $a < b < d$ of l.*

(b) If l is a C_c-cap, then $\operatorname{orient}(c_a, c_b, c_d) < 0$ *for all entries $a < b < d$ of l.*

(c) For every $k \in \mathbb{N}$, $\mathsf{HasNCap}_{C_c}(k, [n])$ *implies* $c.\mathsf{HasCap}(k)$.

Proof. (b) Apply Lemma 10.12 with $\alpha = \mathbb{N}$, $D = [n]$ and $R(a,b,d) :\Leftrightarrow \mathrm{Cap3}_{C_c}(a,b,d)$. Hypothesis (7): let $a < b < d < e$ be in $[n]$ with $R(a,b,d)$ and $R(b,d,e)$. By Lemma 10.10, $\operatorname{orient}(c_a, c_b, c_d) < 0$ and $\operatorname{orient}(c_b, c_d, c_e) < 0$; since c is a configuration, $x(c_a) < x(c_b) < x(c_d) < x(c_e)$; Lemma 10.11 gives $\operatorname{orient}(c_a, c_b, c_e) < 0$ and $\operatorname{orient}(c_a, c_d, c_e) < 0$, that is $R(a,b,e)$ and $R(a,d,e)$ by Lemma 10.10. The list l is strictly increasing with entries in $[n]$, and its consecutive triples satisfy $\mathrm{Cap3}_{C_c} = R$ because l is a C_c-cap. Lemma 10.12 gives $R(a,b,d)$ for all entries $a < b < d$ of l, i.e. $\operatorname{orient}(c_a, c_b, c_d) < 0$ by Lemma 10.10.

(a) The same argument with $R = \mathrm{Cup3}_{C_c}$ and the positive half of Lemma 10.11.

(c) Let $l = (l_0, \dots, l_{k-1})$ be a C_c-cap of length k with entries in $[n]$, and put $f(i) := l_i$. Then $f\colon [k] \to [n]$ is strictly increasing because l is, and for $i < j < m$ in $[k]$, (b) applied to the entries $l_i < l_j < l_m$ gives $\operatorname{orient}(c_{f(i)}, c_{f(j)}, c_{f(m)}) < 0$. Hence $c.\mathsf{HasCap}(k)$. □

The next lemma is the geometric heart of the bridge: a geometric cup and a geometric cap with the same two end points form a convex polygon. (This is the direction "⇐" of `MIRRORETV/PROOFS.md`, Lemma 2.3.)

Lemma 10.14 (cup and cap with common ends;
Lean: `ErdosSzekeres.Omission.BoundaryGeometry.boundary_polygon`, `ErdosSzekeres/Omission/BoundaryGeometry.lean:403`,
Lean: `ErdosSzekeres.Omission.BoundaryGeometry.Boundary.card`, `ErdosSzekeres/Omission/BoundaryGeometry.lean:359`,
Lean: `ErdosSzekeres.Omission.BoundaryGeometry.boundary_convex_position`, `ErdosSzekeres/Omission/BoundaryGeometry.lean:317`). *Let $s, t \in \mathbb{N}$. Let $u_0, \dots, u_{s+1} \in \mathbb{R}^2$ satisfy* $x(u_0) < x(u_1) < \cdots < x(u_{s+1})$ *and* $\operatorname{orient}(u_i, u_j, u_m) > 0$ *for all $i < j < m$, and let $v_0, \dots, v_{t+1} \in \mathbb{R}^2$ satisfy* $x(v_0) < \cdots < x(v_{t+1})$ *and* $\operatorname{orient}(v_i, v_j, v_m) < 0$ *for all $i < j < m$. Assume $u_0 = v_0$ and $u_{s+1} = v_{t+1}$, and let $B := \{u_0, \dots, u_{s+1}\} \cup \{v_0, \dots, v_{t+1}\}$. Then*

(a) $|B| = s + t + 2$;

(b) $p \notin \operatorname{conv}(B \setminus \{p\})$ for every $p \in B$; equivalently (Lemma 1.22), B is in convex position;

(c) for every finite $S \subset \mathbb{R}^2$ with $B \subseteq S$, S contains $s + t + 2$ points in convex position.

Remark 10.15 (the Lean encoding). Lean stores this data as a `Boundary s t` (`ErdosSzekeres/Omission/BoundaryGeometry.lean:120`): the lower arc `lower` is (u_i), a `CupArc s` (strictly increasing x, all ordered triples positive, line 56), and the upper arc is stored through its mirror image `upperReflected` $= (\rho(v_j))$, a `CupArc t`, where $\rho(p) = (p_0, -p_1)$ is `BoundaryGeometry.reflect` (line 69); the fields `first_eq` and `last_eq` say $u_0 = \rho(\rho(v_0)) = v_0$ and $u_{s+1} = v_{t+1}$, and `Boundary.points` is B. Since ρ preserves x and $\operatorname{orient}(\rho a, \rho b, \rho d) = -\operatorname{orient}(a,b,d)$, $(\rho(v_j))$ is a `CupArc` if and only if (v_j) has strictly increasing x and all ordered triples negative. So Lemma 10.14 is `boundary_polygon` (part (c)), `Boundary.card` (part (a)) and `boundary_convex_position` (part (b)) after the change of variables $v_j = \rho(w_j)$, where (w_j) is `upperReflected`. This change of variables is a bijection between the data (u_i), (v_j) of the lemma and the values of type `Boundary s t`, and `Boundary.card` counts the finite set `Boundary.finite`, which is B; so the lemma is the Lean statement in this encoding, not a variant of it.

Proof of Lemma 10.14. Put $L := u_0 = v_0$, $R := u_{s+1} = v_{t+1}$, $U := \{u_0, \dots, u_{s+1}\}$ and $V := \{v_0, \dots, v_{t+1}\}$. The u_i are pairwise distinct and the v_j are pairwise distinct (their x-coordinates are), so $|U| = s + 2$, $|V| = t + 2$, and $x(L) < x(R)$, in particular $L \neq R$. The proof follows the Lean proof.

Step 1 (the chord LR). For $0 < i < s+1$, $\operatorname{orient}(L, R, u_i) = -\operatorname{orient}(u_0, u_i, u_{s+1}) < 0$; for $i \in \{0, s+1\}$, $\operatorname{orient}(L, R, u_i) = 0$. For $0 < j < t+1$, $\operatorname{orient}(L, R, v_j) = -\operatorname{orient}(v_0, v_j, v_{t+1}) > 0$; for $j \in \{0, t+1\}$, $\operatorname{orient}(L, R, v_j) = 0$. (Lean: `chord_negative`, `chord_nonpositive`.)

Step 2 (only the ends are shared; part (a)). If $u_i = v_j$, then $\operatorname{orient}(L, R, u_i) = \operatorname{orient}(L, R, v_j) \geq 0$ by Step 1, so $i \in \{0, s+1\}$ by Step 1 again, i.e. $u_i \in \{L, R\}$. Hence $U \cap V = \{L, R\}$ (both L and R lie in $U \cap V$), and $|B| = |U| + |V| - |U \cap V| = (s+2) + (t+2) - 2 = s + t + 2$. (Lean: `shared_only_endpoints`, `Boundary.card`.)

Step 3 (the lower edges support B). We claim: for $0 \leq i \leq s$ and $w \in B$,

$$\operatorname{orient}(u_i, u_{i+1}, w) \geq 0, \quad \text{with strict inequality if } w \notin \{u_i, u_{i+1}\}. \tag{8}$$

If $w = u_j$: for $j \in \{i, i+1\}$ the orientation is 0; for $j < i$ it equals $\operatorname{orient}(u_j, u_i, u_{i+1}) > 0$ (cyclic invariance); for $j > i+1$ it is $\operatorname{orient}(u_i, u_{i+1}, u_j) > 0$. If $w = v_j$ with $j \in \{0, t+1\}$, then $w \in \{L, R\} \subseteq U$ and the previous case applies. If $w = v_j$ with $0 < j < t+1$, we use the identity (valid for all points, checked symbolically; Lean `chord_identity`, line 64)

$$\begin{aligned}(x(R) - x(L))\operatorname{orient}(a,b,w) = (x(R) - x(w))\operatorname{orient}(a,b,L) + (x(w) - x(L))\operatorname{orient}(a,b,R)\\ + (x(b) - x(a))\operatorname{orient}(L,R,w)\end{aligned} \tag{9}$$

with $a = u_i$, $b = u_{i+1}$. Here $x(L) < x(w) < x(R)$ because the $x(v_j)$ increase, $\operatorname{orient}(u_i, u_{i+1}, L) \geq 0$ and $\operatorname{orient}(u_i, u_{i+1}, R) \geq 0$ by the case $w \in U$ (as $L = u_0$, $R = u_{s+1}$), $x(u_{i+1}) - x(u_i) > 0$, and $\operatorname{orient}(L, R, w) > 0$ by Step 1. So the right-hand side of (9) is positive, and dividing by $x(R) - x(L) > 0$ gives $\operatorname{orient}(u_i, u_{i+1}, w) > 0$. (In this last case $w \notin U$ by Step 2.) (Lean: `edge_nonnegative`, `edge_positive`, `cross_edge_positive`, `lower_edge_support`.)

Step 4 (the upper edges support B). We claim: for $0 \leq j \leq t$ and $w \in B$,

$$\operatorname{orient}(v_{j+1}, v_j, w) \geq 0, \quad \text{with strict inequality if } w \notin \{v_j, v_{j+1}\}. \tag{10}$$

Let $\rho(p) = (p_0, -p_1)$; then $x(\rho p) = x(p)$ and $\operatorname{orient}(\rho a, \rho b, \rho d) = -\operatorname{orient}(a, b, d)$ (Lemma 1.17, $\det \rho = -1$). The points $\rho(v_0), \dots, \rho(v_{t+1})$ therefore satisfy the hypotheses on the u's (with t in place of s), the points $\rho(u_0), \dots, \rho(u_{s+1})$ satisfy the hypotheses on the v's, and $\rho(v_0) = \rho(u_0)$, $\rho(v_{t+1}) = \rho(u_{s+1})$; the set of all these points is $\rho(B)$. Step 3 for this reflected data, applied to the edge $(\rho v_j, \rho v_{j+1})$ and the point $\rho w \in \rho(B)$, gives $\operatorname{orient}(\rho v_j, \rho v_{j+1}, \rho w) \geq 0$, strictly if $\rho w \notin \{\rho v_j, \rho v_{j+1}\}$, i.e. (as ρ is injective) if $w \notin \{v_j, v_{j+1}\}$. Finally $\operatorname{orient}(\rho v_j, \rho v_{j+1}, \rho w) = -\operatorname{orient}(v_j, v_{j+1}, w) = \operatorname{orient}(v_{j+1}, v_j, w)$. (Lean: `Boundary.flip`, `upper_edge_support`.)

Step 5 (exposing functions; part (b)). We use the following elementary fact (the easy direction of Lemma 1.40; Lean `exposed_not_in_hull`, line 30): *if $g\colon \mathbb{R}^2 \to \mathbb{R}$ is affine, $g(p) = 0$ and $g(w) > 0$ for every w in a finite set F, then $p \notin \operatorname{conv} F$.* Indeed every point of $\operatorname{conv} F$ is $\sum_k \lambda_k w_k$ with $w_k \in F$, $\lambda_k \geq 0$, $\sum_k \lambda_k = 1$, and $g(\sum_k \lambda_k w_k) = \sum_k \lambda_k g(w_k) > 0$. So it suffices to find, for every $p \in B$, an affine g with $g(p) = 0$ and $g > 0$ on $B \setminus \{p\}$. Every $p \in B$ is L, R, some u_i with $0 < i < s+1$, or some v_j with $0 < j < t+1$.

- $p = L$: $g(w) = x(w) - x(L)$. Every $w \in B \setminus \{L\}$ is u_i or v_j with $i \geq 1$ or $j \geq 1$, so $x(w) > x(L)$. (Lean `boundary_first_exposed`.)
- $p = R$: $g(w) = x(R) - x(w)$, symmetrically. (Lean `boundary_last_exposed`.)
- $p = u_i$, $0 < i < s+1$: $g(w) = \operatorname{orient}(u_{i-1}, u_i, w) + \operatorname{orient}(u_i, u_{i+1}, w)$, which is affine with $g(u_i) = 0$. Let $w \in B \setminus \{u_i\}$. If $w = u_{i-1}$, the first term is 0 and the second is > 0 by (8), since $u_{i-1} \notin \{u_i, u_{i+1}\}$. If $w \neq u_{i-1}$, the first term is > 0 and the second ≥ 0 by (8). (Lean `two_edges_expose`, `lower_interior_exposed`.)
- $p = v_j$, $0 < j < t+1$: $g(w) = \operatorname{orient}(v_{j+1}, v_j, w) + \operatorname{orient}(v_j, v_{j-1}, w)$, affine with $g(v_j) = 0$. For $w \in B \setminus \{v_j\}$: if $w = v_{j+1}$ the first term is 0 and the second is > 0 by (10) (since $v_{j+1} \notin \{v_{j-1}, v_j\}$); otherwise the first term is > 0 and the second ≥ 0. (Lean `upper_interior_exposed`.)

So $p \notin \operatorname{conv}(B \setminus \{p\})$ for every $p \in B$, which is (b); by Lemma 1.22, B is in convex position.

Step 6 (part (c)). $B \subseteq S$, $|B| = s + t + 2$ and B is in convex position, so B witnesses $\mathsf{HasConvexNGon}(s + t + 2, S)$. □

Proposition 10.16 (gons of C_c are convex polygons;
Lean: `MirrorOdd.hasNGon_heptagon`, `MIRRORETV/lean/MirrorMain.lean:1512`,
Lean: `ErdosSzekeres.Omission.BaekGeometry.hasNGon_octagon`, `ErdosSzekeres/Omission/BaekGeometry.lean:94`,
Lean: `C12Check.hasNGon_nonagon`, `MIRRORETV/lean/MirrorNine_append.lean:99`). *Let c be a configuration of n points and $m \in \{7, 8, 9\}$. If $\mathsf{HasNGon}_{C_c}(m, [n])$, then $\mathcal{P}(c)$ contains m points in convex position. (For $m = 7, 8, 9$ this is* `hasNGon_heptagon`, `hasNGon_octagon`, `hasNGon_nonagon` *respectively.)*

Proof. The three Lean proofs are identical up to the number m, which plays no role. Let (l, l') be an m-gon of C_c with all entries in $[n]$: l is a C_c-cap, l' a C_c-cup, $|l|, |l'| \geq 2$, the first entries agree, the last entries agree, and $|l| + |l'| = m + 2$. Put $s := |l'| - 2$ and $t := |l| - 2$ (Lean calls these numbers `l` and `k`), so that $s + t + 2 = m$, and put $u_i := c_{l'_i}$ $(0 \leq i \leq s+1)$ and $v_j := c_{l_j}$ $(0 \leq j \leq t+1)$. The lists are strictly increasing and c is x-sorted, so $x(u_0) < \cdots < x(u_{s+1})$ and $x(v_0) < \cdots < x(v_{t+1})$. By Lemma 10.13(a), $\operatorname{orient}(u_i, u_j, u_k) > 0$ for $i < j < k$; by Lemma 10.13(b), $\operatorname{orient}(v_i, v_j, v_k) < 0$ for $i < j < k$. Equal first entries give $u_0 = v_0$ and equal last entries give $u_{s+1} = v_{t+1}$. All u_i and v_j lie in $\mathcal{P}(c)$. Lemma 10.14(c) with $S = \mathcal{P}(c)$ gives $s + t + 2 = m$ points of $\mathcal{P}(c)$

in convex position. (In Lean, `boundaryOfGon`, `ErdosSzekeres/Omission/BaekGeometry.lean:62`, builds the `Boundary` from the two lists, and `boundaryOfGon_inside`, line 77, says that its points lie in $\mathcal{P}(c)$.) □

10.3. **The capacity bounds 16 and 22.**

Proposition 10.17 (Baek at $N = 7$ for point configurations;
Lean: `MirrorOdd.capacity16`, `MIRRORETV/lean/MirrorMain.lean:1529`)**.** *Let c be a configuration of n points with $\neg\, c.\mathsf{HasCap}(4)$ such that $\mathcal{P}(c)$ does not contain 7 points in convex position. Then $n \le 16$.*

Proof. Suppose $n \ge 17$. Then $\binom{4+2}{2} + 2 = 17 \le n = |[n]|$ (the value $\binom{6}{2} = 15$ is evaluated by `decide`), so Theorem 10.4 with the value 4 for its parameter (called n there), $C = C_c$ and $S = [n]$ gives $\mathsf{HasNCap}_{C_c}(4, [n])$ or $\mathsf{HasNGon}_{C_c}(7, [n])$. In the first case $c.\mathsf{HasCap}(4)$ by Lemma 10.13(c); in the second, $\mathcal{P}(c)$ contains 7 points in convex position by Proposition 10.16. Both contradict the hypotheses. □

Lemma 10.18 (Baek at $N = 8$ on index sets;
Lean: `ErdosSzekeres.Omission.BaekGeometry.geometric_capacity22`, `ErdosSzekeres/Omission/BaekGeometry.lean:111`)**.** *Let c be a configuration of n points and $S \subseteq [n]$ such that $\neg\mathsf{HasNCap}_{C_c}(4, S)$ and $\mathcal{P}(c)$ does not contain 8 points in convex position. Then $|S| \le 22$.*

Proof. By Lemma 10.8(b) it suffices to show $\neg\mathsf{HasNGon}_{C_c}(8, S)$. An 8-gon of C_c with entries in S is one with entries in $[n]$ (Lemma 10.3(a)), so it would give 8 points of $\mathcal{P}(c)$ in convex position by Proposition 10.16. □

Proposition 10.19 (Baek at $N = 8$ for point configurations;
Lean: `MirrorOdd.capacity22g`, `MIRRORETV/lean/MirrorMain.lean:1812`)**.** *Let c be a configuration of n points with $\neg\, c.\mathsf{HasCap}(4)$ such that $\mathcal{P}(c)$ does not contain 8 points in convex position. Then $n \le 22$.*

Proof. Apply Lemma 10.18 with $S = [n]$: a C_c-cap of length 4 in $[n]$ would give $c.\mathsf{HasCap}(4)$ by Lemma 10.13(c). Hence $n = |[n]| \le 22$. □

10.4. **Mirror-symmetric sets: notation and reduction lemmas.** The Lean declarations stated in this subsection, and those of Definition 10.27 and Lemmas 10.28, 10.29 and 10.30 below, are also stated (and, where they are lemmas, proved) in the section of Theorem 9.17; we repeat them here, with proofs, so that this section is self-contained.

Definition 10.20 (the reflection in the x-axis; upper and axis parts;
Lean: `MirrorOdd.mir`, `MIRRORETV/lean/MirrorMain.lean:16`,
Lean: `MirrorOdd.Rset`, `MIRRORETV/lean/MirrorMain.lean:1598`,
Lean: `MirrorOdd.Aset`, `MIRRORETV/lean/MirrorMain.lean:1599`)**.** $\mathrm{mir}(p) := (p_0, -p_1)$. For a finite $S \subset \mathbb{R}^2$, $R(S) := \{p \in S : y(p) > 0\}$ and $A(S) := \{p \in S : y(p) = 0\}$. We say that S is mir-*symmetric* if $\mathrm{mir}(p) \in S$ for all $p \in S$ (this is the hypothesis $\forall p \in S,\ \mathrm{mir}(p) \in S$, named `sym`, of the Lean statements; for finite S it is equivalent to $\mathrm{mir}(S) = S$, because mir is injective).

Note that $\mathrm{mir}(\mathrm{mir}(p)) = p$ (`MirrorOdd.mir_mir`, line 1543), that $\mathrm{mir}(a) = a$ if $y(a) = 0$, and that $\mathrm{orient}(\mathrm{mir}\, a, \mathrm{mir}\, b, \mathrm{mir}\, d) = -\,\mathrm{orient}(a, b, d)$ (Lemma 1.17).

Lemma 10.21 (splitting a mirror-symmetric set;
Lean: `MirrorOdd.card_split`, `MIRRORETV/lean/MirrorMain.lean:1601`,
Lean: `MirrorOdd.card_A_le`, `MIRRORETV/lean/MirrorMain.lean:1636`,
Lean: `MirrorOdd.injOn_x`, `MIRRORETV/lean/MirrorMain.lean:1647`)**.** *Let $S \subset \mathbb{R}^2$ be finite.*

(a) If S is mir-*symmetric, then $|S| = 2|R(S)| + |A(S)|$.*
(b) If S is in general position, then $|A(S)| \le 2$.
(c) Let S be in general position and mir-*symmetric, and let $T \subseteq S$ be such that $y(p) \ge 0$ for all $p \in T$ and $p = q$ for all $p, q \in T$ with $y(p) = 0 = y(q)$. Then $p \mapsto x(p)$ is injective on T.*

Proof. (a) Let $N := \{p \in S : y(p) < 0\}$; S is the disjoint union of $R(S)$, $A(S)$ and N. For $p \in R(S)$, $\mathrm{mir}(p) \in S$ and $y(\mathrm{mir}\, p) = -y(p) < 0$, so mir maps $R(S)$ into N; likewise it maps N into $R(S)$. As $\mathrm{mir} \circ \mathrm{mir} = \mathrm{id}$, these two maps are mutually inverse, so $|N| = |R(S)|$ and $|S| = |R(S)| + |A(S)| + |N| = 2|R(S)| + |A(S)|$.

(b) If $A(S)$ had three distinct points a, b, d, then $\mathrm{orient}(a, b, d) = (x(b) - x(a)) \cdot 0 - 0 \cdot (x(d) - x(a)) = 0$, contradicting general position (Lemma 1.20).

(c) First let $p, q \in T$ with $y(p) > 0$ and $x(p) = x(q)$, and suppose $p \ne q$. Then $p \ne \mathrm{mir}(p)$ (as $y(p) \ne -y(p)$) and $q \ne \mathrm{mir}(p)$ (as $y(q) \ge 0 > -y(p)$), so $p, q, \mathrm{mir}(p)$ are three pairwise distinct points of S. Since $x(q) = x(p) =$

$x(\operatorname{mir} p)$, $\operatorname{orient}(p, q, \operatorname{mir} p) = (x(q) - x(p))(-2y(p)) - (y(q) - y(p)) \cdot 0 = 0$, contradicting general position. So $p = q$. Now let $p, q \in T$ with $x(p) = x(q)$. If $y(p) > 0$ or $y(q) > 0$, the first part (with the roles of p, q exchanged if needed) gives $p = q$; otherwise $y(p) = y(q) = 0$ and $p = q$ by hypothesis. □

Lemma 10.22 (sorting by abscissa;
Lean: `MirrorOdd.cfg`, `MIRRORETV/lean/MirrorMain.lean:1584`,
Lean: `MirrorOdd.sortPt_image`, `MIRRORETV/lean/MirrorMain.lean:1576`). *Let $T \subset \mathbb{R}^2$ be finite and in general position, and let $p \mapsto x(p)$ be injective on T. Then there is a configuration* $\operatorname{cfg}(T)$ *of $|T|$ points (Lean `cfg T hx hgp`, with points `sortPt T hx`) all of whose points lie in T, and $\mathcal{P}(\operatorname{cfg}(T)) = T$.*

Proof. This is the sorting of Definition 9.3. As x is injective on T, the set $\{x(p) : p \in T\}$ has $|T|$ elements $\xi_0 < \cdots < \xi_{|T|-1}$ (Lean `Finset.orderEmbOfFin`); let t_i be the point of T with $x(t_i) = \xi_i$. Then $x(t_i)$ is strictly increasing in i, so $i \mapsto t_i$ is injective and its image is a subset of T with $|T|$ elements, i.e. all of T. For $i < j < k$ the points t_i, t_j, t_k are pairwise distinct points of T, so $\operatorname{orient}(t_i, t_j, t_k) \neq 0$ by general position (Lemma 1.20). Hence (t_i) is a configuration of $|T|$ points with $\mathcal{P} = T$. □

Lemma 10.23 (an axis point is not an interior point of a cap; written step). *Let $p, a, q \in \mathbb{R}^2$ with $x(p) < x(a) < x(q)$, $y(p) \geq 0$, $y(q) \geq 0$ and $y(a) = 0$. Then* $\operatorname{orient}(p, a, q) \geq 0$.

Proof. If $y(a) = 0$, then $\operatorname{orient}(p, a, q) = (x(a) - x(p))\, y(q) + y(p)\, (x(q) - x(a))$ (expand; checked symbolically), and both products are ≥ 0. (In Lean this is the local claim `interior` inside the proof of `mirror_seven_of`.) □

The next lemma consists of four machine-generated Lean statements. We state them exactly and derive them from the general mirror lemma of Theorems 9.8 and 9.10; this is a different proof from the Lean one (see Remark 10.25).

Lemma 10.24 (mirror heptagons and octagon;
Lean: `MirrorOdd.cap_mirror_heptagon`, `MIRRORETV/lean/MirrorMain.lean:212`,
Lean: `MirrorOdd.cap_mirror_octagon`, `MIRRORETV/lean/MirrorMain.lean:392`,
Lean: `MirrorOdd.axis_left_heptagon`, `MIRRORETV/lean/MirrorMain.lean:639`,
Lean: `MirrorOdd.axis_right_heptagon`, `MIRRORETV/lean/MirrorMain.lean:816`). *Let $P \subset \mathbb{R}^2$ be finite.*

(a) Let $c_0, c_1, c_2, c_3 \in \mathbb{R}^2$ with $x(c_0) < x(c_1) < x(c_2) < x(c_3)$, $y(c_0), y(c_1), y(c_2), y(c_3) > 0$ and $\operatorname{orient}(c_0, c_1, c_2)$, $\operatorname{orient}(c_0, c_1, c_3)$, $\operatorname{orient}(c_0, c_2, c_3)$, $\operatorname{orient}(c_1, c_2, c_3) < 0$. *If P contains c_0,* $\operatorname{mir} c_1$, $\operatorname{mir} c_2$, $\operatorname{mir} c_3$, c_3, c_2, c_1, *then P contains 7 points in convex position. (`cap_mirror_heptagon`)*

(b) Under the same hypotheses on $c_0, \dots, c_3$: if P contains $\operatorname{mir} c_0$, $\operatorname{mir} c_1$, $\operatorname{mir} c_2$, $\operatorname{mir} c_3$, c_3, c_2, c_1, c_0, *then P contains 8 points in convex position. (`cap_mirror_octagon`)*

(c) Let $a, c_0, c_1, c_2 \in \mathbb{R}^2$ with $y(a) = 0$, $x(a) < x(c_0) < x(c_1) < x(c_2)$, $y(c_0), y(c_1), y(c_2) > 0$ and $\operatorname{orient}(a, c_0, c_1)$, $\operatorname{orient}(a, c_0, c_2)$, $\operatorname{orient}(a, c_1, c_2)$, $\operatorname{orient}(c_0, c_1, c_2) < 0$. *If P contains a,* $\operatorname{mir} c_0$, $\operatorname{mir} c_1$, $\operatorname{mir} c_2$, c_2, c_1, c_0, *then P contains 7 points in convex position. (`axis_left_heptagon`)*

(d) Let $a, c_0, c_1, c_2 \in \mathbb{R}^2$ with $y(a) = 0$, $x(c_0) < x(c_1) < x(c_2) < x(a)$, $y(c_0), y(c_1), y(c_2) > 0$ and $\operatorname{orient}(c_0, c_1, c_2)$, $\operatorname{orient}(c_0, c_1, a)$, $\operatorname{orient}(c_0, c_2, a)$, $\operatorname{orient}(c_1, c_2, a) < 0$. *If P contains* $\operatorname{mir} c_0$, $\operatorname{mir} c_1$, $\operatorname{mir} c_2$, a, c_2, c_1, c_0, *then P contains 7 points in convex position. (`axis_right_heptagon`)*

Proof. We use the Lean statements of the general mirror lemma. Theorem 9.8 (`cap_mirror_polygon`, `MIRRORETV/lean/MirrorMain.lean:2233`) says: if $m \geq 2$, $(e_0, \dots, e_{m-1})$ has strictly increasing x, $y(e_i) > 0$ for all i, $\operatorname{orient}(e_i, e_j, e_l) < 0$ for all $i < j < l$, and the finite set P contains every e_i and every $\operatorname{mir} e_i$, then P contains $2m$ points in convex position. Theorem 9.10 (`axis_left_polygon`, line 2309; `axis_right_polygon`, line 2403) says: if $m \geq 3$, $(e_0, \dots, e_{m-1})$ has strictly increasing x, $\operatorname{orient}(e_i, e_j, e_l) < 0$ for all $i < j < l$, P contains every e_i and every $\operatorname{mir} e_i$, and either $y(e_0) = 0$ and $y(e_i) > 0$ for $i > 0$ (left), or $y(e_{m-1}) = 0$ and $y(e_i) > 0$ for $i < m - 1$ (right), then P contains $2m - 1$ points in convex position.

(b) Apply the first statement with $m = 4$ and $e = (c_0, c_1, c_2, c_3)$: the x-coordinates increase, all $y(c_i) > 0$, the four triples $i < j < l$ of $\{0, 1, 2, 3\}$ are exactly the four given negative orientations, and P contains all c_i and all $\operatorname{mir} c_i$. So P contains $2 \cdot 4 = 8$ points in convex position.

(a) Let $P' := P \cup \{\operatorname{mir} c_0\}$. By (b) applied to P' there is $T \subseteq P'$ with $|T| = 8$ in convex position. Then $T' := T \setminus \{\operatorname{mir} c_0\} \subseteq P$, $|T'| \geq 7$, and T' is in convex position (Lemma 1.23(a)). So P contains $|T'|$ points in convex position, hence 7 (Lemma 1.23(d)).

(c) Apply the left form of the second statement with $m = 4$ and $e = (a, c_0, c_1, c_2)$: the x-coordinates increase, $y(e_0) = y(a) = 0$, $y(e_i) > 0$ for $i = 1, 2, 3$, the four triples of $\{0, 1, 2, 3\}$ are the four given negative orientations,

and P contains $e_0 = a$, $\operatorname{mir} e_0 = \operatorname{mir} a = a$ (as $y(a) = 0$), and c_i, $\operatorname{mir} c_i$ for $i = 0, 1, 2$. So P contains $2 \cdot 4 - 1 = 7$ points in convex position.

(d) Apply the right form with $m = 4$ and $e = (c_0, c_1, c_2, a)$: $y(e_3) = y(a) = 0$, $y(e_i) > 0$ for $i < 3$, the four triples are (c_0, c_1, c_2), (c_0, c_1, a), (c_0, c_2, a), (c_1, c_2, a), and $\operatorname{mir} a = a \in P$. So P contains 7 points in convex position. □

Remark 10.25 (the Lean proofs of Lemma 10.24). In Lean these four statements are proved by generated code (`MIRRORETV/lean/MirrorMain.lean`, lines 22–991, including the wrappers `cyc7` and `cyc8`; the rest of the generated block, up to line 1506, contains the smaller mirror polygons used for $k = 4, 5, 6$ and their wrappers): the listed points are taken in the cyclic order in which they appear in the hypotheses (for example $c_0, \operatorname{mir} c_1, \operatorname{mir} c_2, \operatorname{mir} c_3, c_3, c_2, c_1$ in (a)); pairwise distinctness is read off from x- or y-coordinates; each of the $\binom{7}{3} = 35$ (for (b): $\binom{8}{3} = 56$) orientation inequalities $\operatorname{orient}(q_i, q_j, q_k) > 0$, $i < j < k$, is proved by `linarith`, with hints found offline, from the hypotheses and from products of a positive ordinate and a positive difference of abscissae, $y(c_i)\,(x(q) - x(q')) > 0$ with q, q' two of the four given points and $x(q') < x(q)$ (for example $y(c_0)\,(x(c_0) - x(a))$ in (c)); and the positive-cycle criterion of Lemma 1.38 (through the wrappers `cyc7`, `cyc8`) concludes. The general lemmas used above (Theorems 9.8 and 9.10) are proved in Lean by the same method in general form. Statements (b), (c) and (d) are special cases of them ($m = 4$), and (a) follows from (b) applied to $P \cup \{\operatorname{mir} c_0\}$ by deleting the point $\operatorname{mir} c_0$; so nothing is lost by the derivation above.

10.5. **Seven points: the bound 32.**

Lemma 10.26 (the reduction for $k = 7$;
Lean: `MirrorOdd.mirror_seven_of`, `MIRRORETV/lean/MirrorMain.lean:1676`)**.** *Let (Hcap), (HL) and (HR) denote the statements of Lemma 10.24(a), (c) and (d) respectively, each universally quantified over the four points ($c_0, \dots, c_3$, resp. a, c_0, c_1, c_2) and over all finite $P \subset \mathbb{R}^2$. Assume (Hcap), (HL) and (HR). Let $S \subset \mathbb{R}^2$ be finite, in general position and* mir*-symmetric, and suppose that S does not contain 7 points in convex position. Then $|S| \le 32$.*

Proof. Key claim. Let $T \subseteq S$ with $y(p) \ge 0$ for all $p \in T$ and with $p = q$ for all $p, q \in T$ such that $y(p) = 0 = y(q)$ (at most one point of T on the axis). Then $|T| \le 16$.

By Lemma 10.21(c), x is injective on T; T is in general position (Lemma 1.23(b)). Let $c := \operatorname{cfg}(T)$ be the configuration of $|T|$ points of Lemma 10.22, so $\mathcal{P}(c) = T$. We apply Proposition 10.17 to c. Its second hypothesis holds because $\mathcal{P}(c) = T \subseteq S$ and S does not contain 7 points in convex position (Lemma 1.23(c)). For the first, suppose $c.\mathsf{HasCap}(4)$, with $f\colon [4] \to [|T|]$ strictly increasing, and put $d_i := c_{f(i)}$. Then $d_0, \dots, d_3 \in T \subseteq S$, $x(d_0) < x(d_1) < x(d_2) < x(d_3)$, $\operatorname{orient}(d_i, d_j, d_l) < 0$ for all $i < j < l$, $\operatorname{mir}(d_i) \in S$ by symmetry, and $y(d_i) \ge 0$. We distinguish cases.

- $y(d_0), y(d_1), y(d_2), y(d_3) > 0$. Then (Hcap) with $P = S$ shows that S contains 7 points in convex position: contradiction.
- $y(d_0) > 0$, $y(d_3) > 0$, $y(d_1) > 0$, $y(d_2) = 0$. Lemma 10.23 for (d_1, d_2, d_3) gives $\operatorname{orient}(d_1, d_2, d_3) \ge 0$: contradiction.
- $y(d_0) > 0$, $y(d_3) > 0$, $y(d_1) = 0$. Lemma 10.23 for (d_0, d_1, d_2) gives $\operatorname{orient}(d_0, d_1, d_2) \ge 0$: contradiction.
- $y(d_0) > 0$, $y(d_3) = 0$. If $y(d_1) = 0$, then $d_1 = d_3$ (at most one axis point in T), contradicting $x(d_1) < x(d_3)$; so $y(d_1) > 0$, and likewise $y(d_2) > 0$. Then (HR) with $a = d_3$ and $(c_0, c_1, c_2) = (d_0, d_1, d_2)$, and $P = S$, gives 7 points of S in convex position: contradiction. (The hypotheses of (HR) are $y(d_3) = 0$, the order $x(d_0) < x(d_1) < x(d_2) < x(d_3)$, the signs $y(d_i) > 0$ for $i < 3$, the four negative orientations, and $\operatorname{mir} d_0, \operatorname{mir} d_1, \operatorname{mir} d_2, d_3, d_2, d_1, d_0 \in S$.)
- $y(d_0) = 0$. As before, $y(d_i) > 0$ for $i = 1, 2, 3$ (otherwise $d_i = d_0$, contradicting $x(d_0) < x(d_i)$). Then (HL) with $a = d_0$ and $(c_0, c_1, c_2) = (d_1, d_2, d_3)$, and $P = S$ (which contains $d_0, \operatorname{mir} d_1, \operatorname{mir} d_2, \operatorname{mir} d_3, d_3, d_2, d_1$), gives a contradiction.

These cases are exhaustive (since every $y(d_i) \ge 0$). Hence $\neg\, c.\mathsf{HasCap}(4)$, and Proposition 10.17 gives $|T| = |\mathcal{P}(c)| \le 16$.

Counting. Write $R = R(S)$ and $A = A(S)$. By Lemma 10.21(a),(b), $|S| = 2|R| + |A|$ and $|A| \le 2$. The key claim applies to $T = R$ (all $y > 0$, no axis point), so $|R| \le 16$. If $A = \emptyset$, then $|S| = 2|R| \le 32$. Otherwise pick $a \in A$; then $a \notin R$, and $T = R \cup \{a\} \subseteq S$ has all $y \ge 0$ and exactly one axis point, so the key claim gives $|R| + 1 \le 16$, and $|S| = 2|R| + |A| \le 2 \cdot 15 + 2 = 32$. □

Definition 10.27 (reflection in a line given by a unit normal; the frame;
Lean: `MirrorOdd.lineRefl`, `MIRRORETV/lean/MirrorMain.lean:1879`,

Lean: `MirrorOdd.frame`, `MIRRORETV/lean/MirrorMain.lean:1883`,
Lean: `MirrorOdd.unframe`, `MIRRORETV/lean/MirrorMain.lean:1892`). For $n_0, n_1, c \in \mathbb{R}$ and $p, q \in \mathbb{R}^2$ put

$$\begin{aligned}
\mathrm{lineRefl}_{n_0,n_1,c}(p) &:= \big(p_0 - 2(n_0p_0 + n_1p_1 - c)\,n_0,\ p_1 - 2(n_0p_0 + n_1p_1 - c)\,n_1\big),\\
\mathrm{frame}_{n_0,n_1,c}(p) &:= \big(n_1(p_0 - cn_0) - n_0(p_1 - cn_1),\ n_0(p_0 - cn_0) + n_1(p_1 - cn_1)\big),\\
\mathrm{unframe}_{n_0,n_1,c}(q) &:= \big(n_1q_0 + n_0q_1 + cn_0,\ -n_0q_0 + n_1q_1 + cn_1\big).
\end{aligned}$$

unframe is an affine map (Lean: a Mathlib `AffineMap` of `Point`) with linear part $(q_0, q_1) \mapsto (n_1q_0 + n_0q_1, -n_0q_0 + n_1q_1)$ (`MirrorOdd.unframeLin`, line 1886). When $n_0^2 + n_1^2 = 1$, lineRefl is the Euclidean reflection in the line $\{p : n_0p_0 + n_1p_1 = c\}$ (every line has this form), and frame is a rotation composed with a translation which takes this line onto the x-axis; no condition on n_0, n_1 is built into the definitions.

Lemma 10.28 (identities of the frame;
Lean: `MirrorOdd.frame_lineRefl`, `MIRRORETV/lean/MirrorMain.lean:1914`,
Lean: `MirrorOdd.orient_frame`, `MIRRORETV/lean/MirrorMain.lean:1920`,
Lean: `MirrorOdd.unframe_frame`, `MIRRORETV/lean/MirrorMain.lean:1900`,
Lean: `MirrorOdd.frame_unframe`, `MIRRORETV/lean/MirrorMain.lean:1907`,
Lean: `MirrorOdd.frame_injective`, `MIRRORETV/lean/MirrorMain.lean:1925`,
Lean: `MirrorOdd.unframe_injective`, `MIRRORETV/lean/MirrorMain.lean:1930`). *Let $n_0^2 + n_1^2 = 1$ and $c \in \mathbb{R}$; write* frame, unframe, lineRefl *for the maps with these parameters. Then for all $p, q, a, b, d \in \mathbb{R}^2$:*

(a) $\mathrm{frame}(\mathrm{lineRefl}(p)) = \mathrm{mir}(\mathrm{frame}(p))$*;*
(b) $\mathrm{orient}(\mathrm{frame}(a), \mathrm{frame}(b), \mathrm{frame}(d)) = \mathrm{orient}(a, b, d)$*;*
(c) $\mathrm{unframe}(\mathrm{frame}(p)) = p$ *and* $\mathrm{frame}(\mathrm{unframe}(q)) = q$*; in particular* frame *and* unframe *are injective.*

Proof. Put $\nu := n_0^2 + n_1^2$. Expanding the definitions gives the following polynomial identities, valid for all values of the variables (we checked each symbolically; they are the `linear_combination` certificates of the Lean proofs):

$$\begin{aligned}
\mathrm{frame}(\mathrm{lineRefl}\,p) - \mathrm{mir}(\mathrm{frame}\,p) &= \big(0,\ -2(n_0p_0 + n_1p_1)(\nu - 1)\big),\\
\mathrm{orient}(\mathrm{frame}\,a, \mathrm{frame}\,b, \mathrm{frame}\,d) - \mathrm{orient}(a, b, d) &= (\nu - 1)\ \mathrm{orient}(a, b, d),\\
\mathrm{unframe}(\mathrm{frame}\,p) - p &= (\nu - 1)\,(p_0 - cn_0,\ p_1 - cn_1),\\
\mathrm{frame}(\mathrm{unframe}\,q) - q &= (\nu - 1)\,(q_0,\ q_1).
\end{aligned}$$

With $\nu = 1$ all right-hand sides vanish, which proves (a), (b) and the two equalities in (c); a map with a left inverse is injective, and frame and unframe are left inverses of each other. (Conceptually, $\mathrm{frame}(p) = M(p - c\,n)$ and $\mathrm{unframe}(q) = M^{\mathsf{T}}q + c\,n$ with $n = (n_0, n_1)$ and $M = \left(\begin{smallmatrix} n_1 & -n_0 \\ n_0 & n_1 \end{smallmatrix}\right)$, $\det M = \nu$, $M^{\mathsf{T}}M = MM^{\mathsf{T}} = \nu I$; (b) is then Lemma 1.17, and the second coordinate of frame(p) is $n_0p_0 + n_1p_1 - c\nu$, which for $\nu = 1$ is the signed distance from p to the line.) □

Lemma 10.29 (convex polygons under injective affine maps;
Lean: `MirrorOdd.hasConvexNGon_affineMap`, `MIRRORETV/lean/MirrorMain.lean:1859`). *Let $S \subset \mathbb{R}^2$ be finite, $n \in \mathbb{N}$, and $g\colon \mathbb{R}^2 \to \mathbb{R}^2$ an injective affine map. If S contains n points in convex position, then so does $g(S)$.*

Proof. Let $T \subseteq S$ with $|T| = n$ be in convex position. Then $g(T) \subseteq g(S)$, $|g(T)| = n$ because g is injective, and $g(T)$ is in convex position by Lemma 1.27. (Lean argues directly: if $g(p) \in \mathrm{conv}(g(T) \setminus \{g(p)\}) = \mathrm{conv}(g(T \setminus \{p\})) = g(\mathrm{conv}(T \setminus \{p\}))$, injectivity gives $p \in \mathrm{conv}(T \setminus \{p\})$.) □

Lemma 10.30 (transport to the x-axis;
Lean: `MirrorOdd.transport`, `MIRRORETV/lean/MirrorMain.lean:1937`). *Let $n_0, n_1, c \in \mathbb{R}$ with $n_0^2 + n_1^2 = 1$, let $S \subset \mathbb{R}^2$ be finite and $m \in \mathbb{N}$, and assume that S is in general position, that* $\mathrm{lineRefl}_{n_0,n_1,c}(p) \in S$ *for all $p \in S$, and that S does not contain m points in convex position. Put $S' := \mathrm{frame}_{n_0,n_1,c}(S)$. Then S' is in general position, S' is* mir*-symmetric, S' does not contain m points in convex position, and $|S'| = |S|$.*

Proof. The same statement is part (a) of Theorem 9.23; we include the short proof to keep this section self-contained. We use Lemma 10.28 throughout. $|S'| = |S|$ because frame is injective. General position: let $q_1, q_2, q_3 \in S'$ be pairwise distinct, $q_k = \mathrm{frame}(p_k)$ with $p_k \in S$; the p_k are pairwise distinct, so $\mathrm{orient}(q_1, q_2, q_3) = \mathrm{orient}(p_1, p_2, p_3) \neq 0$ by Lemma 10.28(b) and Lemma 1.20; by Lemma 1.20 again S' is in general position. Symmetry: for $q = \mathrm{frame}(p) \in S'$, $\mathrm{mir}(q) = \mathrm{frame}(\mathrm{lineRefl}(p)) \in S'$ because $\mathrm{lineRefl}(p) \in S$. No m points

in convex position: if S' contained m points in convex position, then so would unframe(S') by Lemma 10.29 (unframe is affine and injective); but unframe(S') $= S$ by Lemma 10.28(c). □

Theorem 10.31 (mirror-symmetric sets without convex heptagons;
Lean: `MirrorOdd.mirror_seven`, `MIRRORETV/lean/MirrorMain.lean:1803`,
Lean: `MirrorOdd.mirror_seven_line`, `MIRRORETV/lean/MirrorMain.lean:1969`). *The following two statements hold, (a) for the reflection in the x-axis and (b) for the reflection in an arbitrary line.*

(a) Let $S \subset \mathbb{R}^2$ be finite, in general position and mir*-symmetric. If S does not contain 7 points in convex position, then $|S| \le 32$.*

(b) Let $n_0, n_1, c \in \mathbb{R}$ with $n_0^2+n_1^2 = 1$, and let $S \subset \mathbb{R}^2$ be finite, in general position, with $\mathrm{lineRefl}_{n_0,n_1,c}(p) \in S$ *for every $p \in S$. If S does not contain 7 points in convex position, then $|S| \le 32$.*

Status: Lean-checked (`MIRRORETV/lean/MirrorMain.out`, lines 848 and 850; also `SYMLEAN/mirror/axioms.out`). The proof below is a written proof (not refereed); it rests on the cited Theorem 10.4 (through Proposition 10.17) and on Theorems 9.8 and 9.10.

Proof. (a) By Lemma 10.24(a), (c), (d), the statements (Hcap), (HL), (HR) of Lemma 10.26 hold; that lemma gives $|S| \le 32$. (Lean: `mirror_seven_of cap_mirror_heptagon axis_left_heptagon axis_right_heptagon`.)

(b) By Lemma 10.30 with $m = 7$, $S' = \mathrm{frame}_{n_0,n_1,c}(S)$ is in general position, mir-symmetric, does not contain 7 points in convex position, and $|S'| = |S|$. By (a), $|S| = |S'| \le 32$. □

Remark 10.32. Lean also contains `MirrorOdd.mirror_seven'` (`MIRRORETV/lean/MirrorMain.lean:2631`), which proves the statement of Theorem 10.31(a) again as the instance $j = 3$, $M = 16$ of Theorem 9.17 (`mirror_odd_general`), with the hypothesis on M supplied by Proposition 10.17. This is a second Lean proof: in place of `mirror_seven_of` and the generated polygon lemmas of Lemma 10.24 it uses the general reduction `mirror_odd_general`, which rests on the general mirror lemma (Theorems 9.8 and 9.10).

10.6. **Eight points: the bound 46.**

Theorem 10.33 (mirror-symmetric sets without convex octagons;
Lean: `MirrorOdd.mirror_eight`, `MIRRORETV/lean/MirrorMain.lean:1820`,
Lean: `MirrorOdd.mirror_eight_line`, `MIRRORETV/lean/MirrorMain.lean:1977`). *The following two statements hold, (a) for the reflection in the x-axis and (b) for the reflection in an arbitrary line.*

(a) Let $S \subset \mathbb{R}^2$ be finite, in general position and mir*-symmetric. If S does not contain 8 points in convex position, then $|S| \le 46$.*

(b) Let $n_0, n_1, c \in \mathbb{R}$ with $n_0^2+n_1^2 = 1$, and let $S \subset \mathbb{R}^2$ be finite, in general position, with $\mathrm{lineRefl}_{n_0,n_1,c}(p) \in S$ *for every $p \in S$. If S does not contain 8 points in convex position, then $|S| \le 46$.*

Status: Lean-checked (`MIRRORETV/lean/MirrorMain.out`, lines 849 and 851). The proof below is a written proof (not refereed); it rests on the cited Theorem 10.4 (through Proposition 10.19) and on Theorem 9.8.

Proof. (a) Let $R = R(S)$ and $A = A(S)$. Then $R \subseteq S$ and $y > 0$ on R, so x is injective on R by Lemma 10.21(c) (the condition on axis points is vacuous), and R is in general position (Lemma 1.23(b)). Let $c := \mathrm{cfg}(R)$ (Lemma 10.22), so $\mathcal{P}(c) = R$. We apply Proposition 10.19 to c. Its second hypothesis holds since $\mathcal{P}(c) = R \subseteq S$ (Lemma 1.23(c)). For the first, suppose $c.\mathsf{HasCap}(4)$ with $f\colon [4] \to [|R|]$ and put $d_i := c_{f(i)}$: then $x(d_0) < \cdots < x(d_3)$, all four ordered triples have negative orientation, $y(d_i) > 0$ (as $d_i \in R$), and $d_i, \mathrm{mir}(d_i) \in S$. Lemma 10.24(b) with $P = S$ shows that S contains 8 points in convex position: a contradiction. Hence $|R| \le 22$, and by Lemma 10.21(a),(b) $|S| = 2|R| + |A| \le 44 + 2 = 46$.

(b) By Lemma 10.30 with $m = 8$ and part (a), $|S| = |\mathrm{frame}(S)| \le 46$. □

Remark 10.34. Lean also contains `MirrorOdd.mirror_eight'` (`MIRRORETV/lean/MirrorMain.lean:2636`), which re-derives Theorem 10.33(a) as the instance $j = 4$, $M = 22$ of Theorem 9.18 (`mirror_even_general`) with Proposition 10.19. Only the points strictly above the axis are used in (a); the two possible axis points are simply counted.

10.7. **Nine points: the bound 226.**

Theorem 10.35 (the capacity bound 113; mirror-symmetric sets without convex nonagons;
Lean: `C12Check.config_capacity113`, `MIRRORETV/lean/MirrorNine_append.lean:24`,
Lean: `C12Check.mirror_nine`, `MIRRORETV/lean/MirrorNine_append.lean:131`,
Lean: `C12Check.mirror_nine_line`, `MIRRORETV/lean/MirrorNine_append.lean:136`). *The following three*

statements hold: (a) is a capacity bound in Baek's model of Definition 10.1 (no 5*-cap and no* 9*-gon), (b) and (c) are the mirror bounds for the reflection in the x-axis and in an arbitrary line.*

(a) Let $(\alpha, <)$ *be a linearly ordered set,* C *a configuration on* α *and* $S \subseteq \alpha$ *finite with* $\neg\mathsf{HasNCap}_C(5, S)$ *and* $\neg\mathsf{HasNGon}_C(9, S)$*. Then* $|S| \le 113$*.*

(b) Let $S \subset \mathbb{R}^2$ *be finite, in general position and* mir*-symmetric. If* S *does not contain* 9 *points in convex position, then* $|S| \le 226$*.*

(c) Let $n_0, n_1, c \in \mathbb{R}$ *with* $n_0^2 + n_1^2 = 1$*, and let* $S \subset \mathbb{R}^2$ *be finite, in general position, with* $\mathrm{lineRefl}_{n_0,n_1,c}(p) \in S$ *for every* $p \in S$*. If* S *does not contain* 9 *points in convex position, then* $|S| \le 226$*.*

Status: Lean-checked (`MIRRORETV/lean/MirrorNine.out`, lines 879, 881 and 882). The proofs below are written proofs (not refereed); they rest on the cited Theorems 10.4 and 10.6 and, for (b) and (c), on Theorems 9.17 and 9.23. Part (a) is proved first; Proposition 10.36 transfers it to point configurations; (b) and (c) follow.

Proof of Theorem 10.35(a). We follow the Lean proof. Let T be the set of $p \in S$ that are the first entry of some C-cap of length 4 with all entries in S. Then $T \subseteq S$ and $|S| = |S \setminus T| + |T|$.

Claim 1: $|S \setminus T| \le 29$. Suppose $|S \setminus T| \ge 30 = \binom{6+2}{2} + 2$ (with $\binom{8}{2} = 28$). Theorem 10.4 with $n = 6$ gives $\mathsf{HasNCap}_C(4, S \setminus T)$ or $\mathsf{HasNGon}_C(9, S \setminus T)$. In the first case let $(h, \dots)$ be a C-cap of length 4 with entries in $S \setminus T$; its entries lie in S, so its first entry h lies in T by definition of T, while $h \in S \setminus T$: a contradiction. In the second case Lemma 10.3(a) gives $\mathsf{HasNGon}_C(9, S)$, contrary to the hypothesis.

Claim 2: $|T| \le 84$. Suppose $|T| > 84 = \binom{3+6}{3}$. Theorem 10.6 with $a = 3$, $b = 6$ gives $\mathsf{HasNCap}_C(5, T)$ or $\mathsf{HasNCup}_C(8, T)$. The first is impossible, since a 5-cap in T is one in S (Lemma 10.3(a)). In the second case let $(z_0, \dots, z_7)$ be a C-cup with entries in T, and put $p := z_6$, $q := z_7$. As $q \in T$, there is a C-cap (q, r, r_2, r_3) with entries in S. Note $p < q < r$, since cups and caps are strictly increasing.

- If $\mathrm{Cup3}_C(p, q, r)$, then $(z_0, \dots, z_7, r)$ is strictly increasing and all its consecutive triples satisfy $\mathrm{Cup3}_C$ (those of the cup, and $(z_6, z_7, r) = (p, q, r)$), so it is a C-cup of length 9 with entries in S (as $T \subseteq S$). By Lemma 10.3(b) (with $9 \ge 2$), $\mathsf{HasNGon}_C(9, S)$: a contradiction.
- If $\mathrm{Cap3}_C(p, q, r)$, then (p, q, r, r_2, r_3) is strictly increasing and all its consecutive triples satisfy $\mathrm{Cap3}_C$ (the triple (p, q, r) and those of the cap), so it is a C-cap of length 5 with entries in S ($p \in T \subseteq S$): a contradiction with $\neg\mathsf{HasNCap}_C(5, S)$.

Hence $|S| = |S \setminus T| + |T| \le 29 + 84 = 113$. □

Proposition 10.36 (the capacity bound 113 for point configurations; Lean: `C12Check.capacity113`, `MIRRORETV/lean/MirrorNine_append.lean:116`)**.** *Let* c *be a configuration of* n *points with* $\neg\, c.\mathsf{HasCap}(5)$ *such that* $\mathcal{P}(c)$ *does not contain* 9 *points in convex position. Then* $n \le 113$*.*

Proof. Apply Theorem 10.35(a) (proved above) to $C = C_c$ and $S = [n]$. A C_c-cap of length 5 in $[n]$ would give $c.\mathsf{HasCap}(5)$ (Lemma 10.13(c)), and a 9-gon of C_c in $[n]$ would give 9 points of $\mathcal{P}(c)$ in convex position (Proposition 10.16). Hence $n = |[n]| \le 113$. □

Proof of Theorem 10.35(b) and (c). Theorem 9.17 (`mirror_odd_general`, `MIRRORETV/lean/MirrorMain.lean:2498`) states: let $j \ge 2$ and M be natural numbers such that every configuration e of r points (for every $r \in \mathbb{N}$) with $\neg\, e.\mathsf{HasCap}(j+1)$ and without $2j+1$ points of $\mathcal{P}(e)$ in convex position has $r \le M$; then every finite mir-symmetric set in general position without $2j+1$ points in convex position has at most $2M$ points. Theorem 9.23 contains the same statement for $\mathrm{lineRefl}_{n_0,n_1,c}$-symmetric sets with $n_0^2 + n_1^2 = 1$ (`mirror_odd_general_line`, line 2641).

Take $j = 4$ and $M = 113$. Then $j \ge 2$, $j + 1 = 5$, $2j + 1 = 9$, and the hypothesis on M is exactly Proposition 10.36. So (b) follows from `mirror_odd_general` and (c) from `mirror_odd_general_line`, with $2M = 226$. (Alternatively, (c) follows from (b) by Lemma 10.30.) □

10.8. **Reflections given by a point and a direction; the three-class reduction.** The case classes of Definition 7.7 use a second parametrisation of reflections.

Definition 10.37 (reflection through a point with a unit direction; Lean: `ErdosSzekeres.SymReduction.reflLine`, `SYMLEAN/mirror/ReductionThreeClasses.lean:449`)**.** For $a, u, p \in \mathbb{R}^2$ put $d := (p_0 - a_0)u_0 + (p_1 - a_1)u_1$ and

$$\mathrm{reflLine}_{a,u}(p) := \big(a_0 + 2du_0 - (p_0 - a_0),\ a_1 + 2du_1 - (p_1 - a_1)\big).$$

For $u_0^2 + u_1^2 = 1$ this is the reflection in the line $a + \mathbb{R}u$. (The same definition is `SYMLEAN/SymmetricReduction.lean:423`; the class statements use it with the hypothesis $u_0^2 + u_1^2 = 1$.)

Theorem 10.38 (the two reflections agree; the bound 32 for reflLine;
Lean: `ErdosSzekeres.SymReduction.reflLine_eq_lineRefl`, `SYMLEAN/mirror/ReductionThreeClasses.lean:4253`,
Lean: `ErdosSzekeres.SymReduction.card_le_32_of_reflLine`, `SYMLEAN/mirror/ReductionThreeClasses.lean:4264`). *Let $a, u \in \mathbb{R}^2$ with $u_0^2 + u_1^2 = 1$.*

(a) $\mathrm{reflLine}_{a,u} = \mathrm{lineRefl}_{n_0,n_1,c}$ *as maps $\mathbb{R}^2 \to \mathbb{R}^2$, where $n_0 = -u_1$, $n_1 = u_0$ and $c = -u_1a_0 + u_0a_1$.*

(b) Let $S \subset \mathbb{R}^2$ be finite and in general position with $\mathrm{reflLine}_{a,u}(p) \in S$ for every $p \in S$ (Lean `Invariant (reflLine a u) S`, Definition 1.10). If S does not contain 7 points in convex position, then $|S| \le 32$.

Status: Lean-checked (`SYMLEAN/mirror/axioms.out`). Part (b) puts no condition on the number of points of S on the mirror. The written proof of (b) rests on Theorem 10.31(b), hence on the cited Theorem 10.4 and on Theorems 9.8 and 9.10.

Proof. (a) Fix p and put $w := p - a$. In the first coordinate,

$$\mathrm{reflLine}_{a,u}(p)_0 - \mathrm{lineRefl}_{n_0,n_1,c}(p)_0 = (2a_0 - p_0 + 2du_0) - \big(p_0 + 2(-u_1p_0 + u_0p_1 - c)u_1\big) = 2w_0\,(u_0^2 + u_1^2 - 1),$$

using $d = w_0u_0 + w_1u_1$ and $-u_1p_0 + u_0p_1 - c = -u_1w_0 + u_0w_1$; in the second coordinate the difference is $(2a_1 - p_1 + 2du_1) - \big(p_1 - 2(-u_1w_0 + u_0w_1)u_0\big) = 2w_1\,(u_0^2 + u_1^2 - 1)$. (We checked both identities symbolically; they are the Lean certificates `linear_combination (2 * (p i - a i)) * hu`.) Both vanish since $u_0^2 + u_1^2 = 1$. Geometrically: $n = (-u_1, u_0)$ is u turned by 90°, (u, n) is an orthonormal basis, $w = \langle w, u\rangle u + \langle w, n\rangle n$, $\langle n, p\rangle - c = \langle n, w\rangle$, and both maps send p to $a + \langle w, u\rangle u - \langle w, n\rangle n$.

(b) $n_0^2 + n_1^2 = u_1^2 + u_0^2 = 1$, and for $p \in S$, $\mathrm{lineRefl}_{n_0,n_1,c}(p) = \mathrm{reflLine}_{a,u}(p) \in S$ by (a). Theorem 10.31(b) gives $|S| \le 32$. □

Theorem 10.39 (the two mirror case classes hold;
Lean: `ErdosSzekeres.SymReduction.mirrorOneAxis33_proved`, `SYMLEAN/mirror/ReductionThreeClasses.lean:4274`,
Lean: `ErdosSzekeres.SymReduction.mirrorNoAxis34_proved`, `SYMLEAN/mirror/ReductionThreeClasses.lean:4282`). *The propositions* `MirrorOneAxis33` *and* `MirrorNoAxis34` *of Definition 7.7 hold. Explicitly:*

(a) Let $S \subset \mathbb{R}^2$ be finite and $a, u \in \mathbb{R}^2$ with $u_0^2 + u_1^2 = 1$. If S is in general position, $|S| = 33$, $\mathrm{reflLine}_{a,u}(p) \in S$ for all $p \in S$, and there is $x \in S$ with $\mathrm{reflLine}_{a,u}(x) = x$ such that every $y \in S$ with $\mathrm{reflLine}_{a,u}(y) = y$ equals x, then S contains 7 points in convex position.

(b) Let $S \subset \mathbb{R}^2$ be finite and $a, u \in \mathbb{R}^2$ with $u_0^2 + u_1^2 = 1$. If S is in general position, $|S| = 34$, $\mathrm{reflLine}_{a,u}(p) \in S$ for all $p \in S$, and $\mathrm{reflLine}_{a,u}(p) \neq p$ for all $p \in S$, then S contains 7 points in convex position.

Status: Lean-checked (`SYMLEAN/mirror/axioms.out`); no SAT certificate and no unproved hypothesis is involved; the written proof rests, through Theorems 10.38(b) and 10.31(b), on the cited Theorem 10.4 and on Theorems 9.8 and 9.10.

Proof. In both cases, if S did not contain 7 points in convex position, Theorem 10.38(b) would give $|S| \le 32$, contradicting $|S| = 33$, resp. $|S| = 34$. The last hypothesis of (a) (exactly one point on the mirror) and of (b) (no point on the mirror) is not used. □

Theorem 10.40 (the reduction with three case classes; *conditional*;
Lean: `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_isometry_3`, `SYMLEAN/mirror/ReductionThreeClasses.lean:4292`,
Lean: `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affineEquiv_3`, `SYMLEAN/mirror/ReductionThreeClasses.lean:4303`). *Assume the three propositions* `HalfTurn34`, `Threefold33` *and* `Fivefold35` *of Definition 7.7 (they are explicit hypotheses of both statements).*

(a) Let $S \subset \mathbb{R}^2$ be finite and in general position with $33 \le |S|$, and let $g\colon \mathbb{R}^2 \to \mathbb{R}^2$ be a map with $\mathrm{eDist2}(g(p), g(q)) = \mathrm{eDist2}(p, q)$ for all $p, q \in \mathbb{R}^2$ (`IsEucIsometry g`, Definition 7.1; eDist2 *is the squared Euclidean distance; no continuity or surjectivity is assumed), $g \neq \mathrm{id}$, and $g(p) \in S$ for all $p \in S$. Then S contains 7 points in convex position.*

(b) Let $S \subset \mathbb{R}^2$ be finite and in general position with $33 \le |S|$, and let g be an affine bijection of $\mathbb{R}^2$ (Lean: g is an `AffineEquiv` *of* `Point`*) with $g \neq \mathrm{id}$ (Lean: g differs from* `AffineEquiv.refl`*; two affine equivalences are equal iff they agree at every point) and $g(p) \in S$ for all $p \in S$ (the invariance uses g as a map). Then S contains 7 points in convex position.*

Status: Lean-checked *as a conditional statement* (`SYMLEAN/mirror/axioms.out`: only the three standard axioms; the three case classes are binders of the theorems, not axioms). The theorem does not establish its conclusion unconditionally; see Remark 10.41.

Proof. (a) Theorem 7.39 (`heptagon_of_nontrivial_isometry`) gives the conclusion under the five hypotheses `HalfTurn34`, `Threefold33`, `Fivefold35`, `MirrorOneAxis33`, `MirrorNoAxis34`, for exactly the S and g of (a). The last two hold by Theorem 10.39. (Lean: `heptagon_of_nontrivial_isometry h2 h3 h5 mirrorOneAxis33_proved mirrorNoAxis34_proved S g hgp hcard hg hne hinv`.)

(b) Likewise from Theorem 8.28(a) (`heptagon_of_nontrivial_affineEquiv`) and Theorem 10.39.

The theorems used are the copies in RTC (lines 1023 and 1527), which are byte-identical with `SYMLEAN/SymmetricReduction.lean:997` and `SYMLEAN/AffineReduction.lean:1501` (see the paragraph on copies at the beginning of this section), so the written proofs of Theorems 7.39 and 8.28 apply verbatim. □

Remark 10.41 (status of the hypotheses of Theorem 10.40). `HalfTurn34`, `Threefold33` and `Fivefold35` are not proved in Lean. In the project they are computer-assisted claims, supplied by the following records (`SYMLEAN/mirror/README.md`, §4, and the table in `SYMLEAN/README.md`, §5). The project trusts one proof checker, `cake_lpr`: where a proof is said below to have been accepted by both proof checkers, `cake_lpr` accepted it, and that is the verdict that counts; `lrat-check` is only a secondary check, whose verdicts the project does not count (`WRITEUP/SYMMETRIC.md`, §3; an unpatched build of it accepts invalid proofs, `LRATCHECK20261105`).

- `HalfTurn34`: claimed in `C2FINAL`. The hull cover and the 120 hexagonal hulls of `C2C34` have retained proofs; most parallelogram pieces (`C2P`–`C2P5`) rest on verdicts recorded at solve time (both proof checkers accepted the proof, which was then deleted), with re-solved samples and re-checks (status paragraph of `C2FINAL/README.md`; `C2RECHECK`). The bookkeeping of the subcase orbits, namely that the listed subcases of the eight canonical parallelogram hull cases $P_1, \dots, P_8$ (to which reduction A reduces the others) meet every orbit of second-layer subcases, is Lean-checked according to the record (`C2FINAL/README.md`, §6), by kernel evaluation, so computer-assisted (Theorem 11.98); that the listed subcases are refuted rests on the records. The encoder and the two reductions of the case split have written proofs.
- `Threefold33`: `C3FINAL`. Its 116,883 piece records carry verdicts recorded at solve time (both proof checkers accepted the proof); the proofs were not kept (`C3LEAN/README.md`, §4), and 2,120 randomly chosen pieces were re-solved (`C3SPOT`). `C3LEAN` Lean-checks, by kernel evaluation and so computer-assisted (Theorem 11.82), only that the leaves named in it cover the 910 roots of the reduced case split (that every leaf has such a record is checked outside Lean); the normal form `C3-S/soundness.md`, the encoding `C3a`, the exhaustiveness of the splits and the reductions of the case split are written proofs.
- `Fivefold35`: `SYM5`, two normal forms of one encoder, each with a retained proof accepted by both proof checkers; the encoder has a written proof. The angle normal form (proof P1) needs no layer lemma; the layer normal form (proof L1) uses the layer lemma of `SYM5/README.md`, §2, whose statement that every convex layer is one orbit, and the strict nesting of the layers, are Lean-checked (Theorems 2.22 and 3.16).

`Threefold33` and `Fivefold35` also rest on the written centre-perturbation lemma `C2ENC/perturbation.md`. For none of the three is the step from a certificate to the Lean proposition formal: that each class states what its certificate proves was checked by reading (`SYMLEAN/README.md`, §6 and §9). So the three hypotheses are not Lean-checked, and Theorem 10.40 reduces the statement "every finite set of at least 33 points in general position with a non-identity isometric (or affine) symmetry contains 7 points in convex position" to them and proves it only under them. Only the isometric form and the `AffineEquiv` form have three-class versions in RTC; the other affine variants `heptagon_of_nontrivial_affine`, `heptagon_of_nontrivial_affineMap`, `heptagon_of_nontrivial_affineEquiv_image`, `heptagon_of_nontrivial_affine_coords` (RTC lines 1503, 1515, 1536, 1547) and `halfTurn_of_symmetric_counterexample` (RTC line 1050) still take all five classes (the last one with `Fourfold32` in place of `HalfTurn34`) as hypotheses.

Remark 10.42 (the axis hypotheses). Theorem 10.38(b) shows more than Theorem 10.39: for $u_0^2 + u_1^2 = 1$, every finite $\mathrm{reflLine}_{a,u}$-invariant set in general position with at least 33 points, whatever the number of its points on the mirror, contains 7 points in convex position. In particular the proposition `MirrorExclusionLineCopy` (RTC line 1078, a copy of the text of the statement `Mir.MirrorExclusionLine` of `H3`: for all S, a, u with $u_0^2 + u_1^2 = 1$, S in general position, $|S| = 33$ and $\mathrm{reflLine}_{a,u}(q) \in S$ for all $q \in S$, the set S contains 7 points in convex position) follows at once from Theorem 10.38(b); RTC does not state this implication as a theorem.

Coverage. Covered, by label: the cited vendored theorems `main` (Theorem 10.4) and `Config.cap_cup` (Theorem 10.6); the definitions `Config`, `Config.Cap3`, `Config.Cap`, `Config.Cup`, `Config.Gon`, `Config.NGon`,

`Config.HasNCap`, `Config.HasNCup`, `Config.HasNGon` (Definition 10.1, with `List.Chain3'` and `List.In` in Remark 10.2); `hasNCap_supset`, `hasNGon_supset`, `ncup_is_ngon` (Lemma 10.3); `Baek.four_cap_or_eight_gon`, `Baek.capacity22` (Lemma 10.8); `Configuration` and `Configuration.HasCap` (described at the beginning of Subsection 10.2 and stated as Definition 9.3); `Configuration.cup`, `colorConfig`, `pointAt` (Definition 10.9); `cup_true_iff`, `cup_sign`, `cap_sign` (Lemma 10.10); `four_point_identity_left`, `four_point_identity_right`, `positive_four`, `negative_four` (Lemma 10.11); `chain3_all` (Lemma 10.12); `cup_list_all`, `cap_list_all`, `hasCap_of_list` (Lemma 10.13); `boundary_polygon`, `Boundary.card`, `boundary_convex_position` with the helpers named in its proof (Lemma 10.14); `hasNGon_heptagon`, `hasNGon_octagon`, `hasNGon_nonagon` with `boundaryOfGon`, `boundaryOfGon_inside` (Proposition 10.16); `capacity16` (Proposition 10.17); `geometric_capacity22` (Lemma 10.18); `capacity22g` (Proposition 10.19); `config_capacity113` (Theorem 10.35(a)); `capacity113` (Proposition 10.36); `mir`, `Rset`, `Aset` (Definition 10.20); `card_split`, `card_A_le`, `injOn_x` (Lemma 10.21); `cfg`, `sortPt_image` (Lemma 10.22); `cap_mirror_heptagon`, `cap_mirror_octagon`, `axis_left_heptagon`, `axis_right_heptagon` (Lemma 10.24, by a different proof); `mirror_seven_of` (Lemma 10.26); `lineRefl`, `frame`, `unframe` (Definition 10.27); `frame_lineRefl`, `orient_frame`, `unframe_frame`, `frame_unframe`, `frame_injective`, `unframe_injective` (Lemma 10.28); `hasConvexNGon_affineMap` (Lemma 10.29); `transport` (Lemma 10.30); `mirror_seven`, `mirror_seven_line` (Theorem 10.31); `mirror_eight`, `mirror_eight_line` (Theorem 10.33); `mirror_nine`, `mirror_nine_line` (Theorem 10.35(b),(c)); `reflLine` (Definition 10.37); `reflLine_eq_lineRefl`, `card_le_32_of_reflLine` (Theorem 10.38); `mirrorOneAxis33_proved`, `mirrorNoAxis34_proved` (Theorem 10.39); `heptagon_of_nontrivial_isometry_3`, `heptagon_of_nontrivial_affineEquiv_3` (Theorem 10.40, conditional on `HalfTurn34`, `Threefold33` and `Fivefold35`). The declarations stated in Subsection 10.4, in Definition 10.27 and in Lemmas 10.28–10.30 are also stated in the section of Theorem 9.17 (see the beginning of Subsection 10.4).

Not covered here, deliberately:

- the Lean proofs of `main` and `Config.cap_cup` (cited published theorems);
- the generated Lean proofs of the four polygon lemmas and the wrappers `cyc7`, `cyc8` that they use (replaced by the derivation from Theorems 9.8 and 9.10, Remark 10.25), and the rest of the generated block (the wrappers `cyc4`–`cyc6` and the smaller mirror polygons, which serve the cases $k = 4, 5, 6$);
- `mirror_seven'` and `mirror_eight'` (consistency duplicates, Remarks 10.32 and 10.34);
- the general mirror lemma `cap_mirror_polygon`, `axis_left_polygon`, `axis_right_polygon`, `cap_below_lines`, the general reductions `mirror_odd_general`, `mirror_even_general` and their `_line` versions, and `mirror_four`, `mirror_five`, `mirror_six` with their `_line` versions and their helpers `hasCap3_of`, `capacity_cup` (they are the subject of Lemma 9.7 and of Theorems 9.8, 9.10, 9.17, 9.18, 9.23 and 9.25, or are used in their proofs; Part B of RTC is `MirrorMain.lean` verbatim after its two import lines);
- the Part A content of RTC (the subject of Definitions 7.1, 7.7 and Theorems 7.39, 8.28 and the statements around them);
- helpers absorbed into the proofs above, among them: `mir_zero`, `mir_one`, `mir_mir`, `xset`, `xset_card`, `emb`, `emb_mem`, `sortPt`, `sortPt_mem`, `sortPt_x`, `sortPt_strictMono`, `sortPt_injective`, `unframeLin`, `unframe_apply`; `pointAt_eq`, `pointAt_order`, `OrderedTriples`, `arcOfList`, `cupArc`, `capArc`, `same_first`, `same_last`; from `BoundaryGeometry.lean`, besides the lemmas named in the proof of Lemma 10.14, `CupArc`, `first`, `last`, `reflect`, `reflect_reflect`, `orient_reflect`, `Boundary`, `Boundary.points`, `Boundary.flip`, `reflect_mem_flip`, `Boundary.finite`, `Boundary.finite_coe`, `affine`, `affine_mix`, `affine_positive_convex`, `supporting_edges_expose`, `point_x_bounds`, `first_x_strict`, `last_x_strict`; and, from the vendored library, `Config.NCap`, `Config.NCup` (caps and cups of a given length), the decidability instance `Config.DecidableCap3`, and list lemmas such as `List.takeHead`, `List.takeHead2`, `List.takeLast2`, `List.in_superset`, `List.cons_in`;
- the other declarations of the bridge files, which these proofs do not use: `Configuration.cup_false_iff`, `hasCup_of_list`, `hasNCup_octagon`, `chain3_of_all`, `listCup_of_hasCup`, `listCap_of_hasCap`, `PlanarChains.cup_all`, and the declarations of `Realization.lean` and `ActualRanks.lean` not named in this section (they belong to the project's `ErdosSzekeres/Omission` development and are outside this section);
- the `#print axioms` and `#check` lines.

11. Case-split bookkeeping: the threefold cover and the central 34-point split

11.1. **Overview.** This section is about two Lean files that check the *bookkeeping* of two case splits of the project, that is, the purely combinatorial statement that an explicitly named family of nodes covers the split.

(1) The file `C3LEAN/C3Cover.lean` models the split of threefold rotationally symmetric sets of 33 points (11 orbits of the rotation by 120°, indexed by the residues $0, \dots, 10$) by their convex layers; such sets are the subject of the case class `Threefold33` of Definition 7.7. Its headline is Theorem 11.82 (`C3Cover.all_roots_covered`): each of the 910 depth-two *roots* of the reduced split is *covered*, through layer splits and σ-splits, by the nodes named by the keys of 910 generated certificates. The proof combines a soundness lemma for a certificate checker (Lemma 11.53, with a complete written proof here) with a kernel evaluation of the checker on the 910 certificates (Fact 11.65), the computer-assisted match of the codes of the roots with the roots of the certificates (Fact 11.72), and the decoding of each root from its code (Fact 11.71, which also has a written proof here).

(2) The file `C2FINAL/Central34Orbits.lean` models the second-layer split of the parallelogram hull cases of centrally symmetric sets of 34 points (17 antipodal classes $0, \dots, 16$; the case class `HalfTurn34` of Definition 7.7). Its headline is Theorem 11.98 (`Central34.all_cases_closed`): for each hull case P_a, $1 \le a \le 8$, each of the 560 second-layer subcases, or its image under the reflection that sends i to the residue of $a - i$ modulo 17, belongs (as a set) to an explicit generated list.

What is not established here. Both headline theorems are statements about finite lists of natural numbers. They mention no points, no convexity, no CNF formula and no SAT solver, and they take no hypotheses. The following are *not* established by them, and are not proved in this document:

(i) the geometric meaning of the splits: that every set of the class lies below some root (the top cases, and the reduction of the second layer by Lemma 5 and Lemma H6M of the project's written arguments `C3-S/soundness.md` and `C3H6M/soundness_h6m.md`; for the central split, the reductions A and B reviewed in `C2P/review`), and that a layer split or a σ-split loses no set (Lemma 4 of `C3-S/soundness.md`, rules 2 and 3 of `C3U2/sigma_split.md`; the second-layer covers of the central split);

(ii) the CNF encodings of the nodes;

(iii) the SAT verdicts of the leaves, and even the fact that the leaves are nodes for which the project holds a run record in which both proof checkers, `cake_lpr` and `lrat-check`, are recorded as accepting the proof (the project's term for such a record: "certified"). In Theorem 11.82 the set of leaves is, by definition, the set of nodes named by the keys of the certificates themselves, and in Theorem 11.98 the lists $\mathsf{closedP}_a$ are just lists. Which nodes were put there was decided outside Lean from the run records (by `C3LEAN/tools/gen_c3lean.py`, audited by `C3LEAN/tools/audit_keys.py` and `C3LEAN/review/indep_audit.py`; by `C2FINAL/gen_lean.py` with `C2FINAL/hashmatch.py`).

So neither theorem implies `Threefold33` or `HalfTurn34`. They check one step, "the named nodes form a cut of the split", which the project combines with (i)–(iii); see Remarks 11.83 and 11.100.

Sources and toolchain. Paths are relative to the results directory `cloud/es7-2026-10-19/results` of the repository.

- `C3LEAN/C3Cover.lean` (2,327 lines, SHA-256 `4b64293df2ea...`). Its lines 29–329 (the definitions of the model and of the checker, and the conceptual proofs) are byte-identical to lines 1–301 of `C3LEAN/tools/prelude.lean` (SHA-256 `cdcf96de7739...`), from which the generator `C3LEAN/tools/gen_c3lean.py` copies them. Lines 330–2299 hold the auxiliary definitions `beqLayer`, `isOption`, `distinctCodes`, `codeSet` and `reflectionFacts` (lines 335–365), the facts about the roots (lines 367–409 and 1720–1733), the generated data (lines 413, 423–1717 and 1737), the batch theorems (lines 1740–2245), the theorem `all_checked` (line 2247), the main theorem (line 2251) and seven negative controls (lines 2266–2297); lines 2301–2327 are `#print axioms` commands.
- `C2FINAL/Central34Orbits.lean` (298 lines, SHA-256 `b396a26b2430...`), written by `C2FINAL/gen_lean.py`.

Both files are checked with Lean 4.24.0 according to their READMEs; for `C3LEAN/C3Cover.lean` a compile log is saved, for `C2FINAL/Central34Orbits.lean` no Lean output (paragraph "Axioms" below). Neither has an `import`: only the core library is used, and the repository's Mathlib (pinned at `f897ebcf`) plays no role. Every finite computation is closed by the tactic `decide +kernel`: the Lean kernel itself evaluates the Boolean expression to `true` by definitional unfolding, using its built-in arbitrary-precision arithmetic on natural-number literals (part of the kernel, not an axiom). No `native_decide` is used, so the axiom `Lean.ofReduceBool` does not occur. The option `Elab.async false` set at line 27 of `C3LEAN/C3Cover.lean` only affects elaboration.

Axioms. For `C3LEAN/C3Cover.lean`, the compile log `C3LEAN/logs/compile.log` (and the identical axiom lines of the review's recompilation, `C3LEAN/review/logs/main_compile.log`) records: `all_roots_covered`, `run_sound` and `checkRoot_sound` depend on `propext` and `Quot.sound`; the eleven theorems `reflection_facts_H3`, `reflection_facts_H6_1_0`, ..., `reflection_facts_H6_5_1`, and `roots_eq_tsv` and `all_checked` on `propext` only; `topCases_eq`, `strata`, `roots_match` and `roots_codes_match` on no axiom. The lemmas `Covered.mono`, `beqList_eq`, `beqTok_eq`, `beqNode_eq`, `all_append_of` and the 127 batch theorems are not printed separately; each is used by a printed theorem (`run_sound`, `all_checked` or `all_roots_covered`), so its axioms are among `propext` and `Quot.sound`. No theorem of the file depends on `Classical.choice`, `sorryAx` or `Lean.ofReduceBool`. For `C2FINAL/Central34Orbits.lean` no compile or axioms log is committed: the file ends with three `#print axioms` commands (lines 296–298) whose output is not in the repository, and `C2FINAL/README.md` (§6) records "axioms `propext` only, about 6 s". As no Lean output of any kind (compile or build log, axiom output, pinned message) is saved for this file, the labels of its declarations rest on that record alone: its three theorems and its definitions are *Lean-checked according to the record*, the qualified label defined with the rigor labels at the beginning of this document.

Rigor labels and dependencies. The Lean theorems of `C3LEAN/C3Cover.lean` keep the label *Lean-checked*, and the three theorems of `C2FINAL/Central34Orbits.lean` (Theorem 11.96, Fact 11.97 and Theorem 11.98) carry the label *Lean-checked according to the record*, as do its definitions (paragraph "Axioms" above); the proofs written here are *written proofs (not refereed)*. Every statement that Lean proves by kernel evaluation is marked *computer-assisted* in its statement, and so is every statement whose proof uses one of the computer-assisted statements on generated data (Lemma 11.66, Corollary 11.81, Theorem 11.82 and Corollary 11.99). For the kernel evaluations on explicit generated data (Facts 11.65, 11.72, 11.73 and 11.97, and Theorem 11.98) we state exactly what is evaluated, and we give no hand proof. For the kernel evaluations of the definitions alone (Facts 11.67 to 11.71 and Theorem 11.96) we also give complete written proofs that do not use the computation; the statement names the written proof. Each Lean declaration of the two files other than the seven negative controls of `C3LEAN/C3Cover.lean` (tests, listed in §11.11), or each uniform family of generated declarations (`certs0`, ..., `certs126`; `batch0`, ..., `batch126`; the ten theorems `reflection_facts_H6_1_0`, ..., `reflection_facts_H6_5_1`; `closedP1`, ..., `closedP8`), has a statement below whose head carries its Lean reference; for a family the head cites the first member, and the statement locates the others. The head of Lemma 11.53 carries the two declarations `run_sound` and `checkRoot_sound`. The proofs of this section use no other section.

11.2. Notation.

Numbers. $\mathbb{N} = \{0, 1, 2, \dots\}$. For $x, y \in \mathbb{N}$ with $y > 0$, $\lfloor x/y \rfloor$ and $x \bmod y$ are the quotient and the remainder of Euclidean division (Lean's division and remainder on $\mathbb{N}$), and $x \dot{-} y = \max(x - y, 0)$ is Lean's truncated subtraction on $\mathbb{N}$. For an integer x and $y \in \mathbb{N}$ with $y > 0$, $x \bmod y$ denotes the least nonnegative residue of x modulo y; for $x \in \mathbb{N}$ this is the remainder above. For $\sigma, f \in \{0, 1\}$, $\sigma \oplus f = (\sigma + f) \bmod 2$.

Lists. We write $[x_0, \dots, x_{k-1}]$ for a list, $[]$ for the empty list, $x :: l$ for the list with head x and tail l, $l_1 +\!\!+ l_2$ for concatenation, $|l|$ for the length, and $x \in l$ if x is an entry of l; $\mathrm{set}(l)$ is the set of entries of l. A list is *duplicate-free* if entries at different positions are different; then $|l| = |\mathrm{set}(l)|$. The list functions are those of Lean's core library: $\mathsf{range}(n) = [0, 1, \dots, n-1]$; $\mathsf{map}(g, l) = [g(x_0), \dots, g(x_{k-1})]$; $\mathsf{flatMap}(g, l) = g(x_0) +\!\!+ \cdots +\!\!+ g(x_{k-1})$; $\mathsf{filter}(p, l)$ is the list of the entries x of l with $p(x) = \mathsf{true}$, in their order; $\mathsf{take}(K, l)$ is the list of the first $\min(K, |l|)$ entries; $\mathsf{zip}(l, l') = [(x_0, x'_0), (x_1, x'_1), \dots]$ has length $\min(|l|, |l'|)$; $\mathsf{foldl}(g, z, l) = g(\cdots g(g(z, x_0), x_1) \cdots, x_{k-1})$; $\mathsf{replicate}(k, x)$ is the list of k copies of x. The Boolean-valued functions are: $\mathsf{contains}(l, x) = \mathsf{true}$ iff $x \in l$; $\mathsf{all}(l, p) = \mathsf{true}$ iff $p(x) = \mathsf{true}$ for every $x \in l$; $\mathsf{any}(l, p) = \mathsf{true}$ iff $p(x) = \mathsf{true}$ for some $x \in l$; $\mathsf{isEmpty}(l) = \mathsf{true}$ iff $l = []$. We use without comment the following facts, each immediate by induction on l from the defining equations:

(L1) $\mathsf{all}(l_1 +\!\!+ l_2, p) = \mathsf{all}(l_1, p) \wedge \mathsf{all}(l_2, p)$;
(L2) $y \in \mathsf{flatMap}(g, l)$ iff $y \in g(x)$ for some $x \in l$;
(L3) $|\mathsf{map}(g, l)| = |l|$, and $|\mathsf{flatMap}(g, l)|$ is the sum of $|g(x)|$ over the positions of l;
(L4) $y \in \mathsf{map}(g, l)$ iff $y = g(x)$ for some $x \in l$.

Booleans. The Booleans are true and false. In Boolean expressions $\wedge$, $\vee$, $\neg$ denote Lean's Boolean conjunction, disjunction and negation, and a decidable proposition P used where a Boolean is expected stands for $\mathsf{decide}(P)$, the Boolean that is true iff P holds (Lean inserts this coercion). For $x, y \in \mathbb{N}$, Lean's Boolean equality tests (written `x == y` or `Nat.beq x y`) are true iff $x = y$, and `x != y` is the negation of `x == y`. A statement "$b = \mathsf{true}$" is a Lean proposition about a Boolean b.

Binary digits. For $n \in \mathbb{N}$ let $\beta(n) = \{i \in \mathbb{N} : \lfloor n/2^i \rfloor \text{ is odd}\}$, the set of positions of the digit 1 in the binary expansion of n. Then $n = \sum_{i \in \beta(n)} 2^i$, and $n \mapsto \beta(n)$ is a bijection from $\mathbb{N}$ onto the set of finite subsets of $\mathbb{N}$ (uniqueness of the binary expansion). In Lean, shifting 1 left by c places gives 2^c, so $\beta(2^c) = \{c\}$; the bitwise or of x and y, written here $x \,\mathsf{or}\, y$, is the number with $\beta(x \,\mathsf{or}\, y) = \beta(x) \cup \beta(y)$; and $\mathsf{testBit}(n, i)$ (Lean's `Nat.testBit n i`) is true iff $i \in \beta(n)$.

Lemma 11.1 (Or-folds; written step)**.** *Let $z \in \mathbb{N}$, let l be a list of elements of a set X, and let $g : X \to \mathbb{N}$. Then*

$$\beta\big(\mathsf{foldl}((acc, x) \mapsto acc \,\mathsf{or}\, 2^{g(x)},\ z,\ l)\big) = \beta(z) \cup \{g(x) : x \in l\}.$$

Proof. Induction on l, for all z at once. For $l = []$ the fold is z. For $l = x :: l'$ the fold equals $\mathsf{foldl}((acc, y) \mapsto acc \,\mathsf{or}\, 2^{g(y)},\ z \,\mathsf{or}\, 2^{g(x)},\ l')$, whose set of binary digits is, by induction, $\beta(z) \cup \{g(x)\} \cup \{g(y) : y \in l'\}$. □

11.3. **The threefold split: the model.** Every Lean reference in a statement head of §§11.3–11.9 is to `C3LEAN/C3Cover.lean`, namespace `C3Cover`; the statements tagged "written step" have no Lean counterpart. The definitions transcribe the project's Python implementation of the split (`C3U2/tools/hullcase.py`); that the transcription agrees with the Python code is tested in the project (`C3LEAN/README.md`, §3), not proved, and plays no role in the theorems.

Definition 11.2 (Orbits; Lean: `C3Cover.m`, `C3LEAN/C3Cover.lean:34`)**.** $m = 11$. Below we write 11 for m wherever it occurs.

Definition 11.3 (Tokens; Lean: `C3Cover.Tok`, `C3LEAN/C3Cover.lean:38`)**.** A *token* is a pair $t = (r, s) \in \mathbb{N} \times \mathbb{N}$; we write $t.1 = r$ and $t.2 = s$.

Definition 11.4 (Layers; Lean: `C3Cover.Layer`, `C3LEAN/C3Cover.lean:41`)**.** A *layer* is a list of tokens.

Definition 11.5 (Nodes; Lean: `C3Cover.Node`, `C3LEAN/C3Cover.lean:45`)**.** A *node* $n = \langle \Lambda, C \rangle$ is a pair consisting of a list $n.\mathsf{layers} = \Lambda$ of layers and a list $n.\mathsf{cube} = C$ of pairs $(r, v) \in \mathbb{N} \times \mathbb{N}$. Two nodes are equal iff both components are equal.

Definition 11.6 (Used residues; Lean: `C3Cover.Node.used`, `C3LEAN/C3Cover.lean:51`)**.** For a node n,

$$\mathsf{used}(n) = \mathsf{flatMap}\big(L \mapsto \mathsf{map}(t \mapsto t.1,\ L),\ n.\mathsf{layers}\big),$$

the first components of the tokens of all layers, in order.

Remark 11.7 (Intended reading). In the project, a token (r, s) stands for the orbit with residue $r \in \{0, \dots, 10\}$ together with its σ value $s \in \{0, 1\}$ (the side of a fixed line through the centre on which the point with label r lies), and $s = 2$ stands for "no σ", which is used for a layer of one orbit other than the top layer (the one-orbit top layer $[(0, 0)]$ carries $s = 0$). A layer of one or two tokens is a convex layer of one or two orbits; the layers of a node are listed from the convex hull inwards, and the cube fixes the σ values v of some further residues r. The top layer is $\mathsf{H3} = [(0, 0)]$ (a hull of one orbit) or $\mathsf{H6}(a, s) = [(0, 0), (a, s)]$ with $1 \le a \le 5$ and $s \in \{0, 1\}$ (a hull of the orbits 0 and a, with $\sigma_a = s$); see Lemma 2 of `C3-S/soundness.md`. Nothing in the Lean types restricts $r \le 10$ or $s \le 2$: all such bounds used below are consequences of how particular lists are built. This reading is not used in any proof.

Definition 11.8 (Pair options; Lean: `C3Cover.pairOptions`, `C3LEAN/C3Cover.lean:54`)**.** For a list l of naturals,

$$\mathsf{pairOptions}([]) = [], \qquad \mathsf{pairOptions}(b :: l) = \mathsf{flatMap}\big(c \mapsto Q_{b,c},\ l\big) +\!\!+ \mathsf{pairOptions}(l),$$

where $Q_{b,c} = \big[\,[(b, 0), (c, 0)],\ [(b, 0), (c, 1)],\ [(b, 1), (c, 0)],\ [(b, 1), (c, 1)]\,\big]$.

Definition 11.9 (Layer options; Lean: `C3Cover.layerOptions`, `C3LEAN/C3Cover.lean:61`)**.** For a list U of naturals let $\mathsf{rem}(U) = \mathsf{filter}\big(r \mapsto \neg\mathsf{contains}(U, r),\ \mathsf{range}(11)\big)$ (a local abbreviation inside `layerOptions`), and

$$\mathsf{layerOptions}(U) = \mathsf{map}\big(r \mapsto [(r, 2)],\ \mathsf{rem}(U)\big) +\!\!+ \mathsf{pairOptions}(\mathsf{rem}(U)).$$

Definition 11.10 (Insertion; Lean: `C3Cover.insTok`, `C3LEAN/C3Cover.lean:66`)**.** For a pair $t \in \mathbb{N} \times \mathbb{N}$ and a list of such pairs: $\mathsf{insTok}(t, []) = [t]$; $\mathsf{insTok}(t, u :: us) = t :: u :: us$ if $t.1 \le u.1$, and $\mathsf{insTok}(t, u :: us) = u :: \mathsf{insTok}(t, us)$ if $t.1 > u.1$.

Definition 11.11 (Sorting; Lean: `C3Cover.sortTok`, `C3LEAN/C3Cover.lean:71`)**.** $\mathsf{sortTok}([]) = []$ and $\mathsf{sortTok}(t :: ts) = \mathsf{insTok}(t, \mathsf{sortTok}(ts))$.

Definition 11.12 (Top cases; Lean: `C3Cover.topCases`, `C3LEAN/C3Cover.lean:77`)**.** The list of top cases is

$$\mathsf{topCases} = [(0,0)] :: \mathsf{flatMap}\big(i \mapsto \big[\,[(0,0),(i+1,0)],\ [(0,0),(i+1,1)]\,\big],\ \mathsf{range}(5)\big).$$

We write $\mathsf{H3} = [(0,0)]$ and $\mathsf{H6}(a,s) = [(0,0),(a,s)]$.

Definition 11.13 (Option keys; Lean: `C3Cover.okey`, `C3LEAN/C3Cover.lean:82`)**.** For a layer L, $\mathsf{okey}(L) = \mathsf{map}(\kappa, \mathsf{sortTok}(L))$, where $\kappa(r,s) = 3r$ if $s = 2$ and $\kappa(r,s) = 3r + s + 1$ if $s \neq 2$.

Definition 11.14 (Lexicographic order; Lean: `C3Cover.lexLe`, `C3LEAN/C3Cover.lean:85`)**.** For lists of naturals: $\mathsf{lexLe}([], y) = \mathsf{true}$; $\mathsf{lexLe}(x :: xs, []) = \mathsf{false}$; and $\mathsf{lexLe}(x :: xs,\ y :: ys) = (x < y) \vee \big((x = y) \wedge \mathsf{lexLe}(xs, ys)\big)$.

Definition 11.15 (Mirror; Lean: `C3Cover.mirror`, `C3LEAN/C3Cover.lean:91`)**.** For a layer L,

$$\mathsf{mirror}(L) = \mathsf{sortTok}(\mathsf{map}(\mu, L)), \qquad \mu(r,s) = \big((11 \mathbin{\dot-} r) \bmod 11,\ s'\big), \quad s' = \begin{cases} 2 & \text{if } s = 2, \\ 1 \mathbin{\dot-} s & \text{if } s \neq 2. \end{cases}$$

Definition 11.16 (The map phi; Lean: `C3Cover.phi`, `C3LEAN/C3Cover.lean:95`)**.** For $a, s \in \mathbb{N}$ and a layer L,

$$\mathsf{phi}(a,s,L) = \mathsf{sortTok}(\mathsf{map}(\varphi_{a,s}, L)), \qquad \varphi_{a,s}(r,\sigma) = \big(((a+11) \mathbin{\dot-} r) \bmod 11,\ \sigma'\big),$$

where $\sigma' = 2$ if $\sigma = 2$; $\sigma' = 1 \mathbin{\dot-} \sigma$ if $\sigma \neq 2$ and exactly one of the conditions $r < a$, $s = 0$ holds; and $\sigma' = \sigma$ if $\sigma \neq 2$ and both or neither of them hold.

Definition 11.17 (Reflection; Lean: `C3Cover.refl`, `C3LEAN/C3Cover.lean:101`)**.** For layers *top* and o: $\mathsf{refl}(top, o) = \mathsf{mirror}(o)$ if $top = [(0,0)]$; $\mathsf{refl}(top, o) = \mathsf{phi}(a,s,o)$ if $top = [(0,0),(a,s)]$ for some $a, s \in \mathbb{N}$; and $\mathsf{refl}(top, o) = o$ for every other list *top*.

Definition 11.18 (Kept options; Lean: `C3Cover.keptOptions`, `C3LEAN/C3Cover.lean:109`)**.** For a layer *top*,

$$\mathsf{keptOptions}(top) = \mathsf{filter}\big(\mathsf{kept}_{top},\ \mathsf{layerOptions}(\mathsf{map}(u \mapsto u.1,\ top))\big), \quad \text{where}$$

$$\mathsf{kept}_{top}(o) = \mathsf{lexLe}\big(\mathsf{okey}(o),\ \mathsf{okey}(\mathsf{refl}(top, o))\big).$$

Definition 11.19 (Roots; Lean: `C3Cover.roots`, `C3LEAN/C3Cover.lean:113`)**.** The list of roots is

$$\mathsf{roots} = \mathsf{flatMap}\big(t \mapsto \mathsf{map}(o \mapsto \langle [t, o], [] \rangle,\ \mathsf{keptOptions}(t)),\ \mathsf{topCases}\big).$$

Definition 11.20 (Free residues; Lean: `C3Cover.Node.free`, `C3LEAN/C3Cover.lean:116`)**.** For a node n,

$$\mathsf{free}(n) = \mathsf{filter}\big(r \mapsto (r \neq 0) \wedge \neg\mathsf{contains}(\mathsf{used}(n), r) \wedge \neg\mathsf{contains}(\mathsf{map}(c \mapsto c.1,\ n.\mathsf{cube}), r),\ \mathsf{range}(11)\big).$$

Definition 11.21 (Bits; Lean: `C3Cover.bitsOf`, `C3LEAN/C3Cover.lean:120`)**.** For $K, i \in \mathbb{N}$,

$$\mathsf{bitsOf}(K,i) = \mathsf{map}\big(j \mapsto \lfloor i / 2^{(K \mathbin{\dot-} 1) \mathbin{\dot-} j} \rfloor \bmod 2,\ \mathsf{range}(K)\big).$$

Definition 11.22 (Adding bits; Lean: `C3Cover.addBits`, `C3LEAN/C3Cover.lean:123`)**.** For lists C and *new* of pairs of naturals,

$$\mathsf{addBits}(C, new) = \mathsf{foldl}\big((c,t) \mapsto \mathsf{insTok}(t,c),\ C,\ new\big).$$

Definition 11.23 (Sigma children; Lean: `C3Cover.Node.sigmaChildren`, `C3LEAN/C3Cover.lean:127`)**.** For a node n and $K \in \mathbb{N}$, with the local abbreviation $bits = \mathsf{take}(K, \mathsf{free}(n))$,

$$\mathsf{sigmaChildren}(n,K) = \mathsf{map}\Big(i \mapsto \big\langle n.\mathsf{layers},\ \mathsf{addBits}\big(n.\mathsf{cube},\ \mathsf{zip}(bits, \mathsf{bitsOf}(K,i))\big)\big\rangle,\ \mathsf{range}(2^K)\Big).$$

Definition 11.24 (Conflicts; Lean: `C3Cover.conflicts`, `C3LEAN/C3Cover.lean:132`)**.** For a layer L and a list C of pairs of naturals,

$$\mathsf{conflicts}(L, C) = \mathsf{any}\big(L,\ t \mapsto (t.2 \neq 2) \wedge \mathsf{any}(C,\ c \mapsto (c.1 = t.1) \wedge (c.2 \neq t.2))\big).$$

Definition 11.25 (Layer children; Lean: `C3Cover.Node.layerChildren`, `C3LEAN/C3Cover.lean:137`)**.** For a node n,

$$\mathsf{layerChildren}(n) = \mathsf{map}\Big(o \mapsto \big\langle n.\mathsf{layers} +\!\!+ [o],\ n.\mathsf{cube}\big\rangle,\ \mathsf{filter}\big(o \mapsto \neg\mathsf{conflicts}(o, n.\mathsf{cube}),\ \mathsf{layerOptions}(\mathsf{used}(n))\big)\Big).$$

Remark 11.26 (Reading of the children). If $K \leq |\mathsf{free}(n)|$ and $b_0 < \cdots < b_{K-1}$ are the first K entries of $\mathsf{free}(n)$ (the K smallest residues in $\{1,\dots,10\}$ that occur neither in a layer nor in the cube of n), then the i-th σ-child, $0 \leq i < 2^K$, has the layers of n and the cube of n with the pairs (b_j, v_j) inserted, where $v_j = \lfloor i / 2^{K-1-j} \rfloor \bmod 2$ is the binary digit of i at position $K-1-j$; as i runs through $0,\dots,2^K-1$, $(v_0,\dots,v_{K-1})$ runs through $\{0,1\}^K$. A layer child of n appends to the layers of n one layer option on the residues not used by n, except the options that give some residue of the cube an explicit σ value (not 2) different from the cube's value. No reduction by a reflection is applied to layer children. This reading is not used in any proof.

11.4. **The cover predicate.**

Definition 11.27 (Covered nodes; Lean: `C3Cover.Covered`, `C3LEAN/C3Cover.lean:146`)**.** Let S be a predicate on nodes. A predicate P on nodes is *S-closed* if for every node n the following three rules hold.
(cert): If $S(n)$, then $P(n)$.
(layer): If $2 \le |n.\mathsf{layers}|$, $\mathsf{layerChildren}(n) \neq []$, and $P(c)$ for every $c \in \mathsf{layerChildren}(n)$, then $P(n)$.
(sigma): If $K \in \mathbb{N}$, $2 \le |n.\mathsf{layers}|$, $0 < K$, $K \le |\mathsf{free}(n)|$, and $P(c)$ for every $c \in \mathsf{sigmaChildren}(n, K)$, then $P(n)$.
$\mathsf{Covered}(S)$ is the least S-closed predicate: $\mathsf{Covered}(S)(n)$ holds iff $P(n)$ holds for every S-closed predicate P.

In each rule the predicate occurs in the premises only positively, so an intersection of S-closed predicates is S-closed; hence $\mathsf{Covered}(S)$ is itself S-closed. We use the two resulting principles: *closure* ($\mathsf{Covered}(S)$ satisfies (cert), (layer), (sigma)) and *induction* (if P is S-closed, then $\mathsf{Covered}(S)(n)$ implies $P(n)$ for every node n). In Lean, `Covered S` is an inductive predicate whose three constructors `Covered.cert`, `Covered.layer`, `Covered.sigma` are exactly these three rules, with these premises in this order, and whose recursor yields the induction principle. The Lean predicate is therefore S-closed and contained in every S-closed predicate, so it coincides with the predicate defined above.

Lemma 11.28 (Monotonicity; Lean: `C3Cover.Covered.mono`, `C3LEAN/C3Cover.lean:153`)**.** *Let S, T be predicates on nodes such that $S(n)$ implies $T(n)$ for every node n. Then for every node n, $\mathsf{Covered}(S)(n)$ implies $\mathsf{Covered}(T)(n)$.*

Proof. As in Lean, by induction on the derivation. The predicate $P = \mathsf{Covered}(T)$ is S-closed: for (cert), $S(n)$ gives $T(n)$, hence $P(n)$ by the rule (cert) for T; the rules (layer) and (sigma) have the same side conditions for S and for T, so they hold for P because $\mathsf{Covered}(T)$ satisfies the rules (layer) and (sigma) for T. By the induction principle, $\mathsf{Covered}(S)(n)$ implies $P(n)$. □

11.5. **Keys, codes and Boolean equality.**

Definition 11.29 (Token decoding; Lean: `C3Cover.decodeTok`, `C3LEAN/C3Cover.lean:165`)**.** For $x \in \mathbb{N}$, $\mathsf{decodeTok}(x) = (\lfloor x/3 \rfloor,\ x \bmod 3)$.

Definition 11.30 (Layer decoding; Lean: `C3Cover.decodeLayer`, `C3LEAN/C3Cover.lean:167`)**.** For $c \in \mathbb{N}$,

$$\mathsf{decodeLayer}(c) = \begin{cases} [\mathsf{decodeTok}(c \mathbin{\dot-} 1)] & \text{if } c \le 33, \\ \big[\mathsf{decodeTok}(\lfloor (c \mathbin{\dot-} 34)/33 \rfloor),\ \mathsf{decodeTok}((c \mathbin{\dot-} 34) \bmod 33)\big] & \text{if } c > 33. \end{cases}$$

Definition 11.31 (Decoding layers; Lean: `C3Cover.decodeLayers`, `C3LEAN/C3Cover.lean:172`)**.** For $f, lk \in \mathbb{N}$ and a list acc of layers: $\mathsf{decodeLayers}(0, lk, acc) = acc$; $\mathsf{decodeLayers}(f+1, lk, acc) = acc$ if $lk = 0$; and $\mathsf{decodeLayers}(f+1, lk, acc) = \mathsf{decodeLayers}\big(f,\ \lfloor lk/2048 \rfloor,\ \mathsf{decodeLayer}(lk \bmod 2048) :: acc\big)$ if $lk \neq 0$.

Definition 11.32 (Cube decoding; Lean: `C3Cover.decodeCube`, `C3LEAN/C3Cover.lean:180`)**.** For $f, r, cm, cv \in \mathbb{N}$: $\mathsf{decodeCube}(0, r, cm, cv) = []$; $\mathsf{decodeCube}(f+1, r, cm, cv) = []$ if $cm = 0$; and if $cm \neq 0$, $\mathsf{decodeCube}(f+1, r, cm, cv)$ is $(r,\ cv \bmod 2) :: D$ when $cm \bmod 2 = 1$ and D when $cm \bmod 2 \neq 1$, where $D = \mathsf{decodeCube}(f,\ r+1,\ \lfloor cm/2 \rfloor,\ \lfloor cv/2 \rfloor)$.

Definition 11.33 (Key decoding; Lean: `C3Cover.decodeKey`, `C3LEAN/C3Cover.lean:191`)**.** For $k \in \mathbb{N}$,

$$\mathsf{decodeKey}(k) = \big\langle \mathsf{decodeLayers}(12,\ \lfloor k/2^{22} \rfloor,\ []),\ \mathsf{decodeCube}(11,\ 0,\ \lfloor k/2^{11} \rfloor \bmod 2^{11},\ k \bmod 2^{11}) \big\rangle.$$

(In Lean the constants are written $4194304 = 2^{22}$ and $2048 = 2^{11}$.)

Remark 11.34 (Keys). Read in base 2048, the number $\lfloor k/2^{22} \rfloor$ lists the layer codes of $\mathsf{decodeKey}(k)$, its most significant digit giving the top layer (each step of $\mathsf{decodeLayers}$ puts the layer of the least significant remaining digit in front of the layers decoded so far); the next 11 binary digits of k are the set of cube residues and the last 11 their values. For example (the review's theorem `C3Cover.rev_parent_decodes`, `C3LEAN/review/RevCtrl_pass.lean:315`, whose Lean proof is `decide +kernel`; its file compiles with exit status 0, `C3LEAN/review/logs/revctrl_pass.log`, but the axioms of this theorem are not printed), $\mathsf{decodeKey}(8615616512) = \langle [[(0,0)], [(1,2)]],\ [(2,0), (3,0), (4,0), (5,0), (6,0), (7,0)] \rangle$. The map $\mathsf{decodeKey}$ is not injective: value digits outside the residue mask are ignored, and a zero digit inside the layer part decodes to the layer $[(0,0)]$. Nothing below needs injectivity: the leaves are defined as decoded keys, and the checker compares nodes, not keys.

Definition 11.35 (Token code; Lean: `C3Cover.tokCode`, `C3LEAN/C3Cover.lean:223`)**.** For a token t, $\mathsf{tokCode}(t) = 3\,t.1 + t.2$.

Definition 11.36 (Layer code; Lean: `C3Cover.layerCode`, `C3LEAN/C3Cover.lean:226`)**.** For tokens t, u: $\mathsf{layerCode}([t]) = 1 + \mathsf{tokCode}(t)$, $\mathsf{layerCode}([t,u]) = 34 + 33\,\mathsf{tokCode}(t) + \mathsf{tokCode}(u)$, and $\mathsf{layerCode}(L) = 0$ for every layer L with no or more than two tokens.

Definition 11.37 (Root code; Lean: `C3Cover.rootCode`, `C3LEAN/C3Cover.lean:232`)**.** For a node n: $\mathsf{rootCode}(n) = \mathsf{layerCode}(A) \cdot 1123 + \mathsf{layerCode}(B)$ if $n.\mathsf{layers} = [A, B]$, and $\mathsf{rootCode}(n) = 0$ otherwise.

Definition 11.38 (Root decoding; Lean: `C3Cover.decodeRoot`, `C3LEAN/C3Cover.lean:238`)**.** For $c \in \mathbb{N}$, $\mathsf{decodeRoot}(c) = \langle [\mathsf{decodeLayer}(\lfloor c/1123 \rfloor),\ \mathsf{decodeLayer}(c \bmod 1123)],\ [] \rangle$.

Definition 11.39 (List equality; Lean: `C3Cover.beqList`, `C3LEAN/C3Cover.lean:194`)**.** For a type α and a Boolean-valued function eq of two arguments in α: $\mathsf{beqList}(eq, [], []) = \mathsf{true}$; $\mathsf{beqList}(eq,\ a :: as,\ b :: bs) = eq(a,b) \wedge \mathsf{beqList}(eq, as, bs)$; and $\mathsf{beqList}(eq, l_1, l_2) = \mathsf{false}$ if exactly one of l_1, l_2 is empty.

Definition 11.40 (Token equality; Lean: `C3Cover.beqTok`, `C3LEAN/C3Cover.lean:199`)**.** For pairs a, b of naturals, $\mathsf{beqTok}(a,b) = \mathsf{Nat.beq}(a.1, b.1) \wedge \mathsf{Nat.beq}(a.2, b.2)$.

Definition 11.41 (Node equality; Lean: `C3Cover.beqNode`, `C3LEAN/C3Cover.lean:202`)**.** For nodes a, b,

$$\mathsf{beqNode}(a,b) = \mathsf{beqList}\big(\mathsf{beqList}(\mathsf{beqTok}, \cdot, \cdot),\ a.\mathsf{layers},\ b.\mathsf{layers}\big) \wedge \mathsf{beqList}(\mathsf{beqTok},\ a.\mathsf{cube},\ b.\mathsf{cube}).$$

Definition 11.42 (Layer equality; Lean: `C3Cover.beqLayer`, `C3LEAN/C3Cover.lean:335`)**.** For layers L, L', $\mathsf{beqLayer}(L, L') = \mathsf{beqList}(\mathsf{beqTok}, L, L')$.

Lemma 11.43 (List equality; Lean: `C3Cover.beqList_eq`, `C3LEAN/C3Cover.lean:204`)**.** *Let α be a type and eq a Boolean-valued function on $\alpha \times \alpha$ such that $eq(a,b) = \mathsf{true}$ implies $a = b$ for all $a, b \in \alpha$. Then for all lists l_1, l_2 over α, $\mathsf{beqList}(eq, l_1, l_2) = \mathsf{true}$ implies $l_1 = l_2$.*

Proof. As in Lean, by induction on l_1, for all l_2. If both lists are empty they are equal. If exactly one is empty, beqList is false, contrary to the assumption. If $l_1 = a :: as$ and $l_2 = b :: bs$, the assumption gives $eq(a,b) = \mathsf{true}$ and $\mathsf{beqList}(eq, as, bs) = \mathsf{true}$; so $a = b$ by the hypothesis on eq, and $as = bs$ by induction; hence $l_1 = l_2$. □

Lemma 11.44 (Token equality; Lean: `C3Cover.beqTok_eq`, `C3LEAN/C3Cover.lean:213`)**.** *For all tokens a, b: $\mathsf{beqTok}(a,b) = \mathsf{true}$ implies $a = b$.*

Proof. $\mathsf{Nat.beq}(x,y) = \mathsf{true}$ implies $x = y$, so both components of a and b agree. □

Lemma 11.45 (Node equality; Lean: `C3Cover.beqNode_eq`, `C3LEAN/C3Cover.lean:218`)**.** *For all nodes a, b: $\mathsf{beqNode}(a,b) = \mathsf{true}$ implies $a = b$.*

Proof. By Lemmas 11.44 and 11.43, $\mathsf{beqList}(\mathsf{beqTok}, \cdot, \cdot)$ satisfies the hypothesis of Lemma 11.43 on the type of layers, and beqTok satisfies it on pairs of naturals. Applying Lemma 11.43 to the two conjuncts of $\mathsf{beqNode}(a,b) = \mathsf{true}$ gives $a.\mathsf{layers} = b.\mathsf{layers}$ and $a.\mathsf{cube} = b.\mathsf{cube}$, hence $a = b$. □

Lemma 11.46 (Converses; written step)**.** *$\mathsf{Nat.beq}(x,x)$, $\mathsf{beqTok}(t,t)$, $\mathsf{beqLayer}(L,L)$ and $\mathsf{beqNode}(n,n)$ are* true *for every $x \in \mathbb{N}$, pair t, layer L and node n. Consequently* beqTok, beqLayer, beqNode *and* $\mathsf{beqList}(\mathsf{Nat.beq}, \cdot, \cdot)$ *return* true *exactly on equal arguments.*

Proof. If $eq(x,x) = \mathsf{true}$ for all x, then $\mathsf{beqList}(eq, l, l) = \mathsf{true}$ for every list l, by induction on l. Apply this to $eq = \mathsf{Nat.beq}$ (reflexive), to beqTok (reflexive since Nat.beq is) and to $\mathsf{beqList}(\mathsf{beqTok}, \cdot, \cdot)$; for nodes both conjuncts are then true. The other direction is Lemmas 11.43 (with $eq = \mathsf{Nat.beq}$ or $eq = \mathsf{beqTok}$), 11.44 and 11.45. □

11.6. **Certificates and the checker.**

Definition 11.47 (Certificates; Lean: `C3Cover.Cert`, `C3LEAN/C3Cover.lean:245`)**.** A *certificate* is a triple $c = \langle c.\mathsf{root},\ c.\mathsf{ds},\ c.\mathsf{ks} \rangle$ of natural numbers.

Definition 11.48 (Unpacking; Lean: `C3Cover.unpack`, `C3LEAN/C3Cover.lean:251`)**.** For all $f, x \in \mathbb{N}$ we have $\mathsf{unpack}(0,x) = []$; $\mathsf{unpack}(f+1,x) = []$ if $x = 0$; and $\mathsf{unpack}(f+1,x) = (x \bmod 2^{64}) :: \mathsf{unpack}(f, \lfloor x/2^{64} \rfloor)$ if $x \neq 0$ (in Lean 2^{64} is written 18446744073709551616).

Definition 11.49 (Keys of a certificate; Lean: `C3Cover.Cert.keys`, `C3LEAN/C3Cover.lean:259`)**.** For a certificate c, $\mathsf{keys}(c) = \mathsf{unpack}(100000,\ c.\mathsf{ks})$.

Definition 11.50 (The checker; Lean: `C3Cover.run`, `C3LEAN/C3Cover.lean:262`)**.** For $f, ds \in \mathbb{N}$, a list stk of nodes and a list ks of naturals, the Boolean $\mathsf{run}(f, stk, ds, ks)$ is defined by recursion on f (the first applicable clause is used):

(R1) $\mathsf{run}(f, [], ds, ks) = (ds = 0) \wedge \mathsf{isEmpty}(ks)$, for every f;
(R2) $\mathsf{run}(0,\ n :: rest,\ ds, ks) = \mathsf{false}$;
(R3) $\mathsf{run}(f+1,\ n :: rest,\ ds, ks)$ is computed from $d = ds \bmod 16$ and $ds' = \lfloor ds/16 \rfloor$:

- if $d = 0$: it is false if $ks = []$, and it is $\mathsf{beqNode}(n, \mathsf{decodeKey}(k)) \wedge \mathsf{run}(f, rest, ds', ks')$ if $ks = k :: ks'$;
- if $d = 1$: it is $\mathsf{decide}(2 \le |n.\mathsf{layers}|) \wedge \neg\mathsf{isEmpty}(\mathsf{layerChildren}(n)) \wedge \mathsf{run}\big(f,\ \mathsf{layerChildren}(n) +\!\!+ rest,\ ds',\ ks\big)$;
- if $d \ge 2$: it is $\mathsf{decide}(2 \le |n.\mathsf{layers}|) \wedge \mathsf{decide}(d \mathbin{\dot-} 1 \le |\mathsf{free}(n)|) \wedge \mathsf{run}\big(f,\ \mathsf{sigmaChildren}(n, d \mathbin{\dot-} 1) +\!\!+ rest,\ ds',\ ks\big)$.

Definition 11.51 (Root check; Lean: `C3Cover.checkRoot`, `C3LEAN/C3Cover.lean:277`)**.** For a node ρ and a certificate c, $\mathsf{checkRoot}(\rho, c) = \mathsf{run}(10^6,\ [\rho],\ c.\mathsf{ds},\ \mathsf{keys}(c))$ (in Lean 10^6 is written 1000000).

Remark 11.52 (How a certificate is read). The stack starts as $[\rho]$. At each step the first node n is removed and the least significant base-16 digit d of ds is consumed: $d = 0$ declares n a *leaf*, which must be equal, as a node, to the decoded next key, and that key is consumed; $d = 1$ declares a layer split, and $d = 1 + K$ with $1 \le K \le 14$ a σ-split on the K smallest free residues of n (K σ bits, 2^K children). For a split the checker computes the children itself and puts them in front of the remaining stack. A certificate thus only chooses the kind of split at each node and names the leaves, in depth-first order; it never supplies children. When the stack is empty, the check succeeds iff no nonzero digit and no key is left. This makes the check strict (the negative controls listed in §11.11 exercise it), but it is not used in the soundness proof below. Running out of fuel makes the check fail, so the bound 10^6 is irrelevant for soundness; each step treats one node of the cover tree, and the largest certificate has 849 of them (`C3LEAN/README.md`, §1).

Lemma 11.53 (Checker soundness; Lean: `C3Cover.run_sound`, `C3LEAN/C3Cover.lean:279`, Lean: `C3Cover.checkRoot_sound`, `C3LEAN/C3Cover.lean:323`)**.** *The following hold.*

(a) (`run_sound`*) Let S be a predicate on nodes and $f \in \mathbb{N}$. For every list stk of nodes, every $ds \in \mathbb{N}$ and every list ks of naturals: if $\mathsf{run}(f, stk, ds, ks) = \mathsf{true}$ and $S(\mathsf{decodeKey}(k))$ holds for every $k \in ks$, then $\mathsf{Covered}(S)(n)$ holds for every $n \in stk$.*

(b) (`checkRoot_sound`*) Let ρ be a node and c a certificate with $\mathsf{checkRoot}(\rho, c) = \mathsf{true}$. Then $\mathsf{Covered}(S_c)(\rho)$, where $S_c(n)$ means: there is $k \in \mathsf{keys}(c)$ with $\mathsf{decodeKey}(k) = n$.*

Proof. We follow the Lean proof.

(a) Fix S. We prove by induction on f the statement $\Phi(f)$: *for all stk, ds, ks, if $\mathsf{run}(f, stk, ds, ks) = \mathsf{true}$ and $S(\mathsf{decodeKey}(k))$ for every $k \in ks$, then $\mathsf{Covered}(S)(n)$ for every $n \in stk$.* If $stk = []$ there is nothing to show, for every f; so let $stk = n_0 :: rest$.

Case $f = 0$. By (R2), $\mathsf{run}(0, n_0 :: rest, ds, ks) = \mathsf{false}$; the hypothesis fails, and $\Phi(0)$ holds.

Step from f to $f+1$. Assume $\Phi(f)$, $\mathsf{run}(f+1, n_0 :: rest, ds, ks) = \mathsf{true}$ and $S(\mathsf{decodeKey}(k))$ for every $k \in ks$. Let $d = ds \bmod 16$ and $ds' = \lfloor ds/16 \rfloor$. We show that n_0 and every $n \in rest$ are covered, i.e. satisfy $\mathsf{Covered}(S)$.

- Case $d = 0$. If $ks = []$, the value would be false; so $ks = k :: ks'$, with $\mathsf{beqNode}(n_0, \mathsf{decodeKey}(k)) = \mathsf{true}$ and $\mathsf{run}(f, rest, ds', ks') = \mathsf{true}$. By Lemma 11.45, $n_0 = \mathsf{decodeKey}(k)$; as $k \in ks$, $S(n_0)$ holds, and n_0 is covered by the rule (cert). Every $k' \in ks'$ lies in ks, so $S(\mathsf{decodeKey}(k'))$; hence $\Phi(f)$, applied to $(rest, ds', ks')$, shows that every $n \in rest$ is covered.
- Case $d = 1$. The three conjuncts are true: $2 \le |n_0.\mathsf{layers}|$; $\mathsf{isEmpty}(\mathsf{layerChildren}(n_0)) = \mathsf{false}$, i.e. $\mathsf{layerChildren}(n_0) \ne []$; and $\mathsf{run}(f, \mathsf{layerChildren}(n_0) +\!\!+ rest, ds', ks) = \mathsf{true}$. By $\Phi(f)$, applied to $(\mathsf{layerChildren}(n_0) +\!\!+ rest, ds', ks)$ with the same keys, every entry of $\mathsf{layerChildren}(n_0)$ and every entry of $rest$ is covered. The rule (layer) shows that n_0 is covered.
- Case $d \ge 2$. Put $K = d \mathbin{\dot-} 1 = d - 1$, so $K \ge 1 > 0$. The conjuncts give $2 \le |n_0.\mathsf{layers}|$, $K \le |\mathsf{free}(n_0)|$ and $\mathsf{run}(f, \mathsf{sigmaChildren}(n_0, K) +\!\!+ rest, ds', ks) = \mathsf{true}$. By $\Phi(f)$ every entry of $\mathsf{sigmaChildren}(n_0, K)$ and of $rest$ is covered, and the rule (sigma) with this K shows that n_0 is covered.

This proves $\Phi(f+1)$, and (a) follows. (Keys are consumed only at leaves; this is why the key hypothesis passes unchanged to the recursive call in the two split cases.)

(b) Apply (a) with $S = S_c$, $f = 10^6$, $stk = [\rho]$, $ds = c.\mathsf{ds}$ and $ks = \mathsf{keys}(c)$. The first hypothesis of (a) is $\mathsf{checkRoot}(\rho, c) = \mathsf{true}$, and for every $k \in \mathsf{keys}(c)$ the statement $S_c(\mathsf{decodeKey}(k))$ holds with witness k. Since $\rho \in [\rho]$, $\mathsf{Covered}(S_c)(\rho)$. □

Lemma 11.54 (Concatenation; Lean: `C3Cover.all_append_of`, `C3LEAN/C3Cover.lean:327`)**.** *Let α be a type, f a Boolean-valued function on α, and l_1, l_2 lists over α. If* $\mathsf{all}(l_1, f) = \mathsf{true}$ *and* $\mathsf{all}(l_2, f) = \mathsf{true}$*, then* $\mathsf{all}(l_1 +\!\!+ l_2, f) = \mathsf{true}$.

Proof. By (L1), $\mathsf{all}(l_1 +\!\!+ l_2, f) = \mathsf{all}(l_1, f) \wedge \mathsf{all}(l_2, f) = \mathsf{true}$. (Lean: simplification with `List.all_append`.) □

11.7. **Generated data and the computer-assisted facts.**

Definition 11.55 (TSV root codes; Lean: `C3Cover.tsvRootCodes`, `C3LEAN/C3Cover.lean:413`)**.** tsvRootCodes is an explicit list of natural numbers. According to its docstring and `C3LEAN/README.md` (§4) it is the root column of `C3FINAL/C3FINAL_mirrorh6_roots.tsv`, in file order, written as root codes; this identification is checked by a Python script (`C3LEAN/tools/audit_keys.py`), not in Lean.

Definition 11.56 (Certificate lists; Lean: `C3Cover.certs0`, `C3LEAN/C3Cover.lean:423`)**.** $\mathsf{certs}_0, \dots, \mathsf{certs}_{126}$ (Lean `certs0`, …, `certs126`, at lines 423–1711; the last, `certs126`, at line 1668) are explicit lists of certificates, each certificate written as a triple of numerals.

Definition 11.57 (All certificates; Lean: `C3Cover.certs`, `C3LEAN/C3Cover.lean:1714`)**.** The list of all certificates is, nested to the right,

$$\mathsf{certs} = \mathsf{certs}_0 +\!\!+ \big(\mathsf{certs}_1 +\!\!+ \big(\cdots +\!\!+ (\mathsf{certs}_{125} +\!\!+ \mathsf{certs}_{126})\cdots\big)\big).$$

Definition 11.58 (Batch sizes; Lean: `C3Cover.bounds`, `C3LEAN/C3Cover.lean:1717`)**.** bounds is an explicit list of 127 numbers, the sizes of the lists certs_N (between 1 and 85, with sum 910). No theorem uses it.

Definition 11.59 (Key list; Lean: `C3Cover.certifiedKeys`, `C3LEAN/C3Cover.lean:1737`)**.** $\mathsf{certifiedKeys} = \mathsf{flatMap}(\mathsf{keys}, \mathsf{certs})$, the concatenation of the key lists of all certificates. The *leaf predicate* of the main theorem is

$$S^\star(n) \quad :\Longleftrightarrow \quad \text{there is } k \in \mathsf{certifiedKeys} \text{ with } \mathsf{decodeKey}(k) = n.$$

The data were written by `C3LEAN/tools/gen_c3lean.py` from the run records of the project. Despite the name, certifiedKeys is just the list of the keys of the certificates; nothing in Lean says which records exist.

Definition 11.60 (Options; Lean: `C3Cover.isOption`, `C3LEAN/C3Cover.lean:340`)**.** For a list U of naturals and a layer L: $\mathsf{isOption}(U, [(r,s)]) = (r < 11) \wedge \neg\mathsf{contains}(U, r) \wedge (s = 2)$; $\mathsf{isOption}(U, [(b,\sigma),(c,\tau)]) = (b < c) \wedge (c < 11) \wedge \neg\mathsf{contains}(U, b) \wedge \neg\mathsf{contains}(U, c) \wedge (\sigma < 2) \wedge (\tau < 2)$; and $\mathsf{isOption}(U, L) = \mathsf{false}$ if L has no or more than two tokens.

Definition 11.61 (Distinct codes; Lean: `C3Cover.distinctCodes`, `C3LEAN/C3Cover.lean:346`)**.** For a list l of naturals and $acc \in \mathbb{N}$: $\mathsf{distinctCodes}([], acc) = \mathsf{true}$ and $\mathsf{distinctCodes}(x :: l, acc) = \neg\mathsf{testBit}(acc, x) \wedge \mathsf{distinctCodes}(l,\ acc \,\mathsf{or}\, 2^x)$.

Definition 11.62 (Code sets; Lean: `C3Cover.codeSet`, `C3LEAN/C3Cover.lean:351`)**.** For a list l of naturals, $\mathsf{codeSet}(l) = \mathsf{foldl}((acc, x) \mapsto acc \,\mathsf{or}\, 2^x,\ 0,\ l)$.

Definition 11.63 (Reflection facts; Lean: `C3Cover.reflectionFacts`, `C3LEAN/C3Cover.lean:356`)**.** For a layer t, let $U = \mathsf{map}(u \mapsto u.1, t)$, $O = \mathsf{layerOptions}(U)$, $\iota(o) = \mathsf{refl}(t, o)$, and let kept_t be as in Definition 11.18. Then $\mathsf{reflectionFacts}(t)$ is the 6-tuple of the following values:

(F1) $|O|$;
(F2) $\mathsf{all}(O,\ \mathsf{isOption}(U, \cdot)) \wedge \mathsf{distinctCodes}(\mathsf{map}(\mathsf{layerCode}, O),\ 0)$;
(F3) $\mathsf{all}\big(O,\ o \mapsto \mathsf{isOption}(U, \iota(o)) \wedge \mathsf{beqLayer}(\iota(\iota(o)), o)\big)$;
(F4) $|\mathsf{filter}(o \mapsto \mathsf{beqLayer}(\iota(o), o),\ O)|$;
(F5) $|\mathsf{keptOptions}(t)|$;
(F6) $\mathsf{all}\big(O,\ o \mapsto (\mathsf{kept}_t(o) \vee \mathsf{kept}_t(\iota(o))) \wedge (\mathsf{beqLayer}(\iota(o), o) \vee \neg(\mathsf{kept}_t(o) \wedge \mathsf{kept}_t(\iota(o))))\big)$.

Lemma 11.64 (Distinctness and code sets; written step)**.** *Let l be a list of naturals and $acc \in \mathbb{N}$. Then* $\mathsf{distinctCodes}(l, acc) = \mathsf{true}$ *iff l is duplicate-free and no entry of l lies in $\beta(acc)$. In particular* $\mathsf{distinctCodes}(l, 0) = \mathsf{true}$ *iff l is duplicate-free. Moreover $\beta(\mathsf{codeSet}(l)) = \mathrm{set}(l)$, so for lists l, l' of naturals* $\mathsf{codeSet}(l) = \mathsf{codeSet}(l')$ *iff* $\mathrm{set}(l) = \mathrm{set}(l')$.

Proof. Induction on l, for all acc. For $l = []$ both sides hold. For $l = x :: l'$ the value is true iff $x \notin \beta(acc)$ and $\mathsf{distinctCodes}(l', acc \,\mathsf{or}\, 2^x) = \mathsf{true}$; by induction and $\beta(acc \,\mathsf{or}\, 2^x) = \beta(acc) \cup \{x\}$, this holds iff $x \notin \beta(acc)$, l' is duplicate-free, and no entry of l' lies in $\beta(acc) \cup \{x\}$; that is exactly the statement that $x :: l'$ is duplicate-free with no entry in $\beta(acc)$. For $\beta(0) = \emptyset$ this gives the second sentence. The claim on codeSet is Lemma 11.1 with $z = 0$ and g the identity, together with the injectivity of β. □

Fact 11.65 (Batches; Lean: `C3Cover.batch0`, `C3LEAN/C3Cover.lean:1740`). *(Computer-assisted.) For every* $N \in \{0, \dots, 126\}$,

$$\mathsf{all}\big(\mathsf{certs}_N,\ c \mapsto \mathsf{checkRoot}(\mathsf{decodeRoot}(c.\mathsf{root}), c)\big) = \mathsf{true}.$$

These are the 127 theorems `C3Cover.batch0`, `C3Cover.batch1`, *...,* `C3Cover.batch126`; *the one for* N *is at line* $1740 + 4N$ *(the last,* `batch126`, *at line 2244).*

Each of the 127 theorems is Lean-checked by kernel evaluation (`decide +kernel`); no hand proof is given. What is evaluated: for each of the 910 certificates c in the lists certs_N, the node $\mathsf{decodeRoot}(c.\mathsf{root})$ and the list $\mathsf{keys}(c)$ are computed, and run is evaluated step by step as in Definition 11.50; at every split node the kernel computes $\mathsf{layerChildren}$, or free and $\mathsf{sigmaChildren}$, from the definitions, and at every leaf it compares the node with the decoded key by $\mathsf{beqNode}$. Over all 910 certificates this treats 76,256 nodes: 73,567 leaves, 1,747 σ-splits and 942 layer splits (counts recorded in `C3LEAN/c3cover_summary.json`; the review's independent transcription of the checker into Python obtained the same counts, `C3LEAN/review/logs/indep_audit.json`). For 554 of the 910 roots the decision field $c.\mathsf{ds}$ of the certificate c is the numeral 0 and $\mathsf{keys}(c)$ has one entry, so the certificate consists of the root itself as the only leaf. This number is our own count from the certificate data of `C3LEAN/C3Cover.lean`; it is not a Lean statement, and no project record states it.

Lemma 11.66 (All checked; Lean: `C3Cover.all_checked`, `C3LEAN/C3Cover.lean:2247`). *(Computer-assisted: the proof uses Fact 11.65.) Every certificate passes:*

$$\mathsf{all}\big(\mathsf{certs},\ c \mapsto \mathsf{checkRoot}(\mathsf{decodeRoot}(c.\mathsf{root}), c)\big) = \mathsf{true}.$$

Proof. As in Lean. Let $p(c) = \mathsf{checkRoot}(\mathsf{decodeRoot}(c.\mathsf{root}), c)$, and let $T_{126} = \mathsf{certs}_{126}$ and $T_N = \mathsf{certs}_N +\!\!+ T_{N+1}$ for $N = 125, \dots, 0$, so that $T_0 = \mathsf{certs}$ by Definition 11.57. By Fact 11.65, $\mathsf{all}(T_{126}, p) = \mathsf{true}$; and if $\mathsf{all}(T_{N+1}, p) = \mathsf{true}$, then Lemma 11.54 together with Fact 11.65 for certs_N gives $\mathsf{all}(T_N, p) = \mathsf{true}$. Descending from $N = 125$ to $N = 0$ (126 applications of `all_append_of`, as in the Lean term) gives the claim. □

Fact 11.67 (Reflection facts below H3; Lean: `C3Cover.reflection_facts_H3`, `C3LEAN/C3Cover.lean:367`). *(Computer-assisted in Lean, by kernel evaluation; written proof in Proposition 11.79.)* $\mathsf{reflectionFacts}(\mathsf{H3}) = (190, \mathsf{true}, \mathsf{true}, 10, 100, \mathsf{true})$.

Fact 11.68 (Reflection facts below H6; Lean: `C3Cover.reflection_facts_H6_1_0`, `C3LEAN/C3Cover.lean:370`). *(Computer-assisted in Lean, by kernel evaluation; written proof in Proposition 11.79.) For each* $a \in \{1, \dots, 5\}$ *and* $s \in \{0, 1\}$: $\mathsf{reflectionFacts}(\mathsf{H6}(a, s)) = (153, \mathsf{true}, \mathsf{true}, 9, 81, \mathsf{true})$. *These are the ten theorems* `reflection_facts_H6_1_0`, `reflection_facts_H6_1_1`, `reflection_facts_H6_2_0`, *...,* `reflection_facts_H6_5_1`; *the one for* (a, s) *is at line* $370 + 3(2a + s - 2)$ *(the last at line 397).*

Fact 11.69 (Top cases; Lean: `C3Cover.topCases_eq`, `C3LEAN/C3Cover.lean:401`). *(Computer-assisted in Lean, by kernel evaluation; written proof in Proposition 11.79.) The list of top cases is*

$$\mathsf{topCases} = [\mathsf{H3}, \mathsf{H6}(1,0), \mathsf{H6}(1,1), \mathsf{H6}(2,0), \mathsf{H6}(2,1), \mathsf{H6}(3,0), \mathsf{H6}(3,1), \mathsf{H6}(4,0), \mathsf{H6}(4,1), \mathsf{H6}(5,0), \mathsf{H6}(5,1)].$$

Fact 11.70 (Strata; Lean: `C3Cover.strata`, `C3LEAN/C3Cover.lean:406`). *(Computer-assisted in Lean, by kernel evaluation; written proof in Proposition 11.79.) The numbers of kept options with one and with two tokens are*

$$\mathsf{map}\big(t \mapsto |\mathsf{filter}(o \mapsto (|o| = 1),\ \mathsf{keptOptions}(t))|,\ \mathsf{topCases}\big) = \mathsf{replicate}(11, 5),$$
$$\mathsf{map}\big(t \mapsto |\mathsf{filter}(o \mapsto (|o| = 2),\ \mathsf{keptOptions}(t))|,\ \mathsf{topCases}\big) = 95 :: \mathsf{replicate}(10, 76).$$

Each of the 13 theorems of Facts 11.67–11.70 is Lean-checked by kernel evaluation of the definitions; Proposition 11.79 below gives a complete written proof of all of them. They are not used in the proof of Theorem 11.82; they record that roots has the properties that the reductions of the second layer need (Remark 11.83).

Fact 11.71 (Roots; Lean: `C3Cover.roots_match`, `C3LEAN/C3Cover.lean:1720`). *(Computer-assisted in Lean, by kernel evaluation; written proof in Proposition 11.80.) The list of roots has* 910 *entries, and each is recovered from its code:*

$$|\mathsf{roots}| = 910 \quad \textit{and} \quad \mathsf{all}\big(\mathsf{roots},\ \rho \mapsto \mathsf{beqNode}(\mathsf{decodeRoot}(\mathsf{rootCode}(\rho)), \rho)\big) = \mathsf{true}.$$

Lean-checked by kernel evaluation of the definitions (the kernel computes the list roots); Proposition 11.80 below gives a complete written proof that does not use the computation.

Fact 11.72 (Root codes; Lean: `C3Cover.roots_codes_match`, `C3LEAN/C3Cover.lean:1725`)**.** *(Computer-assisted.) The codes of the roots match the roots of the certificates:*

$$\mathsf{beqList}\big(\mathsf{Nat.beq},\ \mathsf{map}(\mathsf{rootCode},\mathsf{roots}),\ \mathsf{map}(c\mapsto c.\mathsf{root},\mathsf{certs})\big)=\mathsf{true}.$$

Fact 11.73 (TSV codes; Lean: `C3Cover.roots_eq_tsv`, `C3LEAN/C3Cover.lean:1730`)**.** *(Computer-assisted.) The following four statements hold (as one conjunction):*

$$|\mathsf{tsvRootCodes}|=910,\qquad \mathsf{distinctCodes}(\mathsf{tsvRootCodes},0)=\mathsf{true},$$
$$\mathsf{distinctCodes}(\mathsf{map}(c\mapsto c.\mathsf{root},\mathsf{certs}),0)=\mathsf{true},$$
$$\mathsf{codeSet}(\mathsf{map}(c\mapsto c.\mathsf{root},\mathsf{certs}))=\mathsf{codeSet}(\mathsf{tsvRootCodes}).$$

Facts 11.72 and 11.73 are computer-assisted and Lean-checked by kernel evaluation: the kernel compares the 910 codes $\mathsf{rootCode}(\rho)$, computed from the definitions, entry by entry and in order with the 910 generated numerals $c.\mathsf{root}$, and it tests, with the bit-set accumulator of $\mathsf{distinctCodes}$ and the or-fold $\mathsf{codeSet}$, that the latter and the 910 numerals of $\mathsf{tsvRootCodes}$ are duplicate-free lists with equal sets. No hand proof of them is given.

11.8. **Written proofs of the facts about the roots.** Call a token (r,s) *valid* if $r\le 10$ and $s\le 2$, and a layer *valid* if it consists of one or two valid tokens.

Lemma 11.74 (Sorting short lists; written step)**.** $\mathsf{sortTok}([t])=[t]$*;* $\mathsf{sortTok}([t,u])=[t,u]$ *if* $t.1\le u.1$*, and* $\mathsf{sortTok}([t,u])=[u,t]$ *if* $t.1>u.1$*. In particular a two-token list with distinct residues is put into increasing order of residue, and a layer of one token, or of two tokens with increasing residues, is fixed by* $\mathsf{sortTok}$*.*

Proof. $\mathsf{sortTok}([t])=\mathsf{insTok}(t,[])=[t]$, and $\mathsf{sortTok}([t,u])=\mathsf{insTok}(t,[u])$, which is $[t,u]$ if $t.1\le u.1$ and $u::\mathsf{insTok}(t,[])=[u,t]$ otherwise. □

Lemma 11.75 (Layer options; written step)**.** *The following hold.*

(a) Let l be a strictly increasing list of naturals. Then $\mathsf{pairOptions}(l)$ is duplicate-free, has $4\binom{|l|}{2}$ entries, and its entries are exactly the lists $[(b,\sigma),(c,\tau)]$ with $b,c\in l$, $b<c$ and $\sigma,\tau\in\{0,1\}$.

(b) Let U be a list of naturals, $R=\{r\in\{0,\dots,10\}: r\notin U\}$ and $f=|R|$. Then $\mathsf{layerOptions}(U)$ is duplicate-free, has $f+4\binom{f}{2}$ entries, and its set of entries is

$$\mathcal{O}(U)=\{[(r,2)]: r\in R\}\cup\{[(b,\sigma),(c,\tau)]: b,c\in R,\ b<c,\ \sigma,\tau\in\{0,1\}\}.$$

Moreover, for every layer L: $\mathsf{isOption}(U,L)=\mathsf{true}$ iff $L\in\mathcal{O}(U)$.

(c) Every element of $\mathcal{O}(U)$ is a valid layer and is fixed by $\mathsf{sortTok}$.

Proof. (a) Induction on l. For $l=[]$ the list is empty and $\binom{0}{2}=0$. Let $l=b::l'$; then l' is strictly increasing and all its entries exceed b. The first part $\mathsf{flatMap}(c\mapsto Q_{b,c},l')$ consists of the $4|l'|$ lists $[(b,\sigma),(c,\tau)]$ with $c\in l'$ and $\sigma,\tau\in\{0,1\}$; they are pairwise distinct (distinct triples (c,σ,τ), as l' is duplicate-free) and all have first residue b. By induction, $\mathsf{pairOptions}(l')$ is duplicate-free with $4\binom{|l'|}{2}$ entries, all with first residue in l', hence different from b. So $\mathsf{pairOptions}(l)$ is duplicate-free with $4|l'|+4\binom{|l'|}{2}=4\binom{|l'|+1}{2}$ entries, and each entry has the stated form ($b<c$ in the first part since $c\in l'$; by induction in the second). Conversely, let $b',c'\in l$ with $b'<c'$ and $\sigma,\tau\in\{0,1\}$. If $b'=b$, then $c'\ne b$, so $c'\in l'$, and $[(b,\sigma),(c',\tau)]$ is in the first part. If $b'\ne b$, then $b'\in l'$, so $c'>b'>b$ and $c'\in l'$; by induction the list is in $\mathsf{pairOptions}(l')$.

(b) $\mathsf{rem}(U)$ is the list of the elements of R in increasing order (filtering keeps the order of $\mathsf{range}(11)$). The first part $\mathsf{map}(r\mapsto[(r,2)],\mathsf{rem}(U))$ is duplicate-free with f one-token entries; the second part $\mathsf{pairOptions}(\mathsf{rem}(U))$ is described by (a), and its entries have two tokens. So $\mathsf{layerOptions}(U)$ is duplicate-free with $f+4\binom{f}{2}$ entries, and its set of entries is $\mathcal{O}(U)$. By Definition 11.60, $\mathsf{isOption}(U,[(r,s)])=\mathsf{true}$ iff $r\le 10$, $r\notin U$ and $s=2$; $\mathsf{isOption}(U,[(b,\sigma),(c,\tau)])=\mathsf{true}$ iff $b<c\le 10$, $b,c\notin U$ and $\sigma,\tau\in\{0,1\}$; and $\mathsf{isOption}(U,L)=\mathsf{false}$ for the other layers. This is membership in $\mathcal{O}(U)$.

(c) The residues are at most 10, the second components lie in $\{0,1,2\}$, and a two-token element has increasing residues; apply Lemma 11.74. □

Lemma 11.76 (Decoding; written step)**.** *The following hold.*

(a) $\mathsf{decodeTok}(\mathsf{tokCode}(r,s))=(r,s)$ for all $r\in\mathbb{N}$ and $s\le 2$.

(b) For every valid layer L: $1\le\mathsf{layerCode}(L)\le 1122$ and $\mathsf{decodeLayer}(\mathsf{layerCode}(L))=L$. In particular $\mathsf{layerCode}$ is injective on valid layers.

(c) For valid layers A,B: $\mathsf{decodeRoot}\big(\mathsf{rootCode}(\langle[A,B],[]\rangle)\big)=\langle[A,B],[]\rangle$.

Proof. (a) $\mathsf{tokCode}(r,s) = 3r+s$ with $0 \le s < 3$, so $\lfloor (3r+s)/3 \rfloor = r$ and $(3r+s) \bmod 3 = s$.

(b) A valid token $t = (r,s)$ has $\mathsf{tokCode}(t) = 3r+s \le 30+2 = 32$. If $L = [t]$, then $c = \mathsf{layerCode}(L) = 1 + \mathsf{tokCode}(t) \in [1,33]$; as $c \le 33$, $\mathsf{decodeLayer}(c) = [\mathsf{decodeTok}(c \dot{-} 1)] = [\mathsf{decodeTok}(\mathsf{tokCode}(t))] = [t]$ by (a). If $L = [t,u]$, let $x = \mathsf{tokCode}(t)$ and $y = \mathsf{tokCode}(u)$, both in $[0,32]$. Then $c = 34+33x+y \in [34,\ 34+33\cdot 32+32] = [34, 1122]$; as $c > 33$, $\mathsf{decodeLayer}(c)$ is computed from $c \dot{-} 34 = 33x+y$, and since $0 \le y < 33$, $\lfloor (33x+y)/33 \rfloor = x$ and $(33x+y) \bmod 33 = y$; by (a), $\mathsf{decodeLayer}(c) = [\mathsf{decodeTok}(x), \mathsf{decodeTok}(y)] = [t,u]$.

(c) $\mathsf{rootCode}(\langle [A,B], [] \rangle) = q \cdot 1123 + r$ with $q = \mathsf{layerCode}(A)$ and $r = \mathsf{layerCode}(B) \in [1, 1122]$ by (b). As $0 \le r < 1123$, $\lfloor (q \cdot 1123 + r)/1123 \rfloor = q$ and $(q \cdot 1123 + r) \bmod 1123 = r$. Hence $\mathsf{decodeRoot}$ returns $\langle [\mathsf{decodeLayer}(q), \mathsf{decodeLayer}(r)], [] \rangle = \langle [A,B], [] \rangle$ by (b). □

Lemma 11.77 (The order lexLe; written step)**.** *For all lists x, y of naturals: (a) $\mathsf{lexLe}(x,x) = \mathsf{true}$; (b) $\mathsf{lexLe}(x,y) = \mathsf{true}$ or $\mathsf{lexLe}(y,x) = \mathsf{true}$; (c) if $\mathsf{lexLe}(x,y) = \mathsf{lexLe}(y,x) = \mathsf{true}$, then $x = y$.*

Proof. Induction on x, for all y. (a) $\mathsf{lexLe}([],[]) = \mathsf{true}$, and $\mathsf{lexLe}(a :: as, a :: as) = (a < a) \vee ((a = a) \wedge \mathsf{lexLe}(as, as)) = \mathsf{lexLe}(as, as)$, which is true by induction. (b) If $x = []$ or $y = []$, the empty list is lexLe the other one. Let $x = a :: as$ and $y = b :: bs$. If $a < b$, then $\mathsf{lexLe}(x,y) = \mathsf{true}$; if $b < a$, then $\mathsf{lexLe}(y,x) = \mathsf{true}$; if $a = b$, then $\mathsf{lexLe}(x,y) = \mathsf{lexLe}(as, bs)$ and $\mathsf{lexLe}(y,x) = \mathsf{lexLe}(bs, as)$, and one of them is true by induction. (c) If $x = []$, then $\mathsf{lexLe}(y,[]) = \mathsf{true}$ forces $y = []$. If $x = a :: as$, then $\mathsf{lexLe}(x,y) = \mathsf{true}$ forces $y = b :: bs$, and the hypotheses say: $a < b$ or ($a = b$ and $\mathsf{lexLe}(as, bs) = \mathsf{true}$); and $b < a$ or ($b = a$ and $\mathsf{lexLe}(bs, as) = \mathsf{true}$). Since $a < b$ excludes both $b < a$ and $b = a$, and $b < a$ excludes $a = b$, we get $a = b$ and $\mathsf{lexLe}(as, bs) = \mathsf{lexLe}(bs, as) = \mathsf{true}$; so $as = bs$ by induction, and $x = y$. □

Lemma 11.78 (Counting kept options; written step)**.** *Let X be a duplicate-free list of valid layers, each fixed by $\mathsf{sortTok}$, and let ι be a map from layers to layers with $\iota(x) \in X$ and $\iota(\iota(x)) = x$ for every $x \in X$. For $x \in X$ let $k(x) = \mathsf{lexLe}(\mathsf{okey}(x), \mathsf{okey}(\iota(x)))$, and let F be the number of $x \in X$ with $\iota(x) = x$. Then:*

(a) if $\iota(x) = x$, then $k(x) = \mathsf{true}$;
(b) if $\iota(x) \neq x$, then exactly one of $k(x)$, $k(\iota(x))$ is true;
(c) $|\mathsf{filter}(k, X)| = F + (|X| - F)/2$.

Proof. (a) $k(x) = \mathsf{lexLe}(\mathsf{okey}(x), \mathsf{okey}(x)) = \mathsf{true}$ by Lemma 11.77(a).

(b) For $y \in X$ we have $\mathsf{sortTok}(y) = y$, so $\mathsf{okey}(y) = \mathsf{map}(\kappa, y)$. On tokens with second component $s \le 2$, $\kappa(r,s) = 3r + e(s)$ with $e(2) = 0$, $e(0) = 1$, $e(1) = 2$, and e is a bijection of $\{0,1,2\}$; so $r = \lfloor \kappa(r,s)/3 \rfloor$ and $e(s) = \kappa(r,s) \bmod 3$ determine (r,s), and κ is injective on valid tokens. Hence $\mathsf{map}(\kappa, \cdot)$ is injective on lists of valid tokens (equal images have equal lengths and equal entries), and okey is injective on $\mathrm{set}(X)$. Let $\iota(x) \neq x$; both lie in X, so $\mathsf{okey}(x) \neq \mathsf{okey}(\iota(x))$. Moreover $k(\iota(x)) = \mathsf{lexLe}(\mathsf{okey}(\iota(x)), \mathsf{okey}(\iota(\iota(x)))) = \mathsf{lexLe}(\mathsf{okey}(\iota(x)), \mathsf{okey}(x))$. By Lemma 11.77(b) at least one of $k(x)$, $k(\iota(x))$ is true, and by Lemma 11.77(c) not both.

(c) As X is duplicate-free, $|\mathsf{filter}(k, X)|$ is the number of elements x of $\mathrm{set}(X)$ with $k(x) = \mathsf{true}$. The map ι restricted to $\mathrm{set}(X)$ is an involution of $\mathrm{set}(X)$, so $\mathrm{set}(X)$ is the disjoint union of its F fixed points and of $(|X| - F)/2$ two-element orbits $\{x, \iota(x)\}$. By (a) every fixed point is counted, and by (b) exactly one element of each two-element orbit. □

Proposition 11.79 (Proof of the reflection facts; written step)**.** *Facts 11.67, 11.68, 11.69 and 11.70 hold. More precisely, let t be $\mathsf{H3}$, with $R = \{1, \dots, 10\}$, or $\mathsf{H6}(a,s)$ with $1 \le a \le 5$, $s \in \{0,1\}$, with $R = \{1, \dots, 10\} \setminus \{a\}$; let $O = \mathsf{layerOptions}(\mathsf{map}(u \mapsto u.1, t))$ and $\iota = \mathsf{refl}(t, \cdot)$. Then ι maps $\mathrm{set}(O)$ into itself, is an involution of it, preserves the number of tokens, and its fixed points are:*

- *for $\mathsf{H3}$: the ten layers $[(b,\sigma), (11-b, 1-\sigma)]$ with $1 \le b \le 5$ and $\sigma \in \{0,1\}$;*
- *for $\mathsf{H6}(a,s)$: the layer $[(r_0, 2)]$, where $r_0 = a/2$ if a is even and $r_0 = (a+11)/2$ if a is odd (so $r_0 = 6, 1, 7, 2, 8$ for $a = 1, \dots, 5$), and the eight layers $[(b,\sigma), (b^*, \sigma \oplus f_b)]$ with $b \in R$, $b < b^*$ and $\sigma \in \{0,1\}$; here $r^* = ((a+11) \dot{-} r) \bmod 11$ for $r \in R$ (that is, $r^* = a - r$ if $r < a$ and $r^* = a + 11 - r$ if $r > a$), and $f_b = 1$ if exactly one of $b < a$, $s = 0$ holds and $f_b = 0$ otherwise.*

Proof. Lean checks these facts by evaluation; the following proof does not use the evaluation.

The used residues and the options. For $t = \mathsf{H3}$, $\mathsf{map}(u \mapsto u.1, t) = [0]$; for $t = \mathsf{H6}(a,s)$ it is $[0, a]$. So R is the set of Lemma 11.75(b), with $f = |R| = 10$, resp. 9. By that lemma, O is duplicate-free with $\mathrm{set}(O) = \mathcal{O}$ (the set $\mathcal{O}(U)$ of the lemma), and $|O| = 10 + 4\binom{10}{2} = 190$, resp. $9 + 4\binom{9}{2} = 153$. Let X_1 and X_2 be the sublists of one-token and of two-token entries of O; they have 10 and 180, resp. 9 and 144, entries.

The map below $\mathsf{H3}$. Here $\iota = \mathsf{mirror}$ (first case of Definition 11.17). For $r \in \{1,\dots,10\}$, $(11 \dot{-} r) \bmod 11 = 11 - r \in \{1,\dots,10\}$, and for $\sigma \in \{0,1\}$, $1 \dot{-} \sigma = 1 - \sigma$. Hence, by Lemma 11.74,

$$\mathsf{mirror}([(r,2)]) = [(11-r,2)], \qquad \mathsf{mirror}([(b,\sigma),(c,\tau)]) = [(11-c,1-\tau),(11-b,1-\sigma)] \quad (b<c),$$

the second because $11-b > 11-c$. Both results lie in $\mathcal{O}$, and applying the formulas twice returns the original layer ($11-(11-r)=r$, $1-(1-\sigma)=\sigma$). So ι is an involution of $\mathcal{O}$ preserving the number of tokens. A one-token option is never fixed ($11-r=r$ is impossible). A two-token option $[(b,\sigma),(c,\tau)]$, $b<c$, is fixed iff $11-c=b$, $1-\tau=\sigma$, $11-b=c$ and $1-\sigma=\tau$, i.e. iff $c=11-b$ and $\tau=1-\sigma$; with $b<c=11-b$ this means $1\le b\le 5$. So $F=10$: none in X_1 and ten in X_2.

The map below $\mathsf{H6}(a,s)$. Here $\iota = \mathsf{phi}(a,s,\cdot)$ (second case of Definition 11.17, with this a and s). For $r \in R$ let $r^* = ((a+11) \dot{-} r) \bmod 11$, as in the statement. Since $r \le 10 < a+11$, $(a+11) \dot{-} r = a+11-r$. If $r<a$, then $a+11-r \in [12, a+10]$, so $r^* = a-r \in [1,a-1]$; if $r>a$, then $a+11-r \in [a+1,10]$, so $r^* = a+11-r \in [a+1,10]$. In both cases $r^* \in R$, $[r^*<a] = [r<a]$ and $(r^*)^* = r$ (indeed $a-(a-r)=r$, resp. $a+11-(a+11-r)=r$). So $r \mapsto r^*$ is an involution of R. For $\sigma \in \{0,1\}$ the second component of $\varphi_{a,s}(r,\sigma)$ is $\sigma \oplus f_r$, where $f_r = 1$ if exactly one of $r<a$, $s=0$ holds and $f_r = 0$ otherwise (because $1 \dot{-} \sigma = \sigma \oplus 1$); for $\sigma = 2$ it is 2. As $[r^*<a] = [r<a]$, we have $f_{r^*} = f_r$. Hence, by Lemma 11.74,

$$\mathsf{phi}(a,s,[(r,2)]) = [(r^*,2)], \qquad \mathsf{phi}(a,s,[(b,\sigma),(c,\tau)]) = \mathsf{sortTok}\big([(b^*,\sigma\oplus f_b),(c^*,\tau\oplus f_c)]\big),$$

the latter being these two tokens in increasing order of residue ($b^* \ne c^*$ since $r \mapsto r^*$ is injective). Both results lie in $\mathcal{O}$. The token map $h(r,\sigma) = (r^*, \sigma \oplus f_r)$, $h(r,2) = (r^*,2)$, satisfies $h(h(x)) = x$ on the tokens of options, since $(r^*)^* = r$ and $\sigma \oplus f_r \oplus f_{r^*} = \sigma$. For a two-token option o, $\mathsf{sortTok}(\mathsf{map}(h,o))$ is $\mathsf{map}(h,o)$ or its reversal; applying $\mathsf{map}(h,\cdot)$ gives o or its reversal, and $\mathsf{sortTok}$ of that is o, since o has increasing residues. So ι is an involution of $\mathcal{O}$ preserving the number of tokens.

Fixed points: $[(r,2)]$ is fixed iff $r^* = r$, i.e. iff $a = 2r$ (case $r<a$) or $a+11 = 2r$ (case $r>a$). For even a only the first equation has an integer solution, $r_0 = a/2 \in [1,a-1]$; for odd a only the second, $r_0 = (a+11)/2 \in [a+1,10]$ (as $a \le 5$). So $r \mapsto r^*$ has exactly one fixed point r_0 in R, X_1 has exactly one fixed entry, and $R \setminus \{r_0\}$ splits into four two-element orbits $\{b,b^*\}$. Let $o = [(b,\sigma),(c,\tau)]$ with $b<c$. If $\iota(o) = o$, the residues satisfy $\{b^*,c^*\} = \{b,c\}$; $b^* = b$ and $c^* = c$ is impossible (only one fixed residue), so $c = b^*$ and $b = c^*$. In that case $\iota(o) = \mathsf{sortTok}([(c,\sigma\oplus f_b),(b,\tau\oplus f_c)]) = [(b,\tau\oplus f_c),(c,\sigma\oplus f_b)]$, which equals o iff $\tau \oplus f_c = \sigma$ and $\sigma \oplus f_b = \tau$; as $f_c = f_{b^*} = f_b$, both conditions say $\tau = \sigma \oplus f_b$. So each of the four orbits $\{b<c\}$ carries exactly two fixed options (one for each σ), and $F = 9$: one in X_1 and eight in X_2.

Counting. The lists O, X_1, X_2 satisfy the hypotheses of Lemma 11.78 with this ι (Lemma 11.75(c); ι maps $\mathcal{O}$ to itself, preserves the number of tokens and is an involution), and then $k = \mathsf{kept}_t$. Hence the numbers of kept entries are: for $\mathsf{H3}$, $10 + (190-10)/2 = 100$ in O, $0 + 10/2 = 5$ in X_1 and $10 + (180-10)/2 = 95$ in X_2; for $\mathsf{H6}(a,s)$, $9 + (153-9)/2 = 81$ in O, $1 + (9-1)/2 = 5$ in X_1 and $8 + (144-8)/2 = 76$ in X_2.

The six components. (F1) is $|O| = 190$, resp. 153. (F2): $\mathsf{all}(O, \mathsf{isOption}(U,\cdot)) = \mathsf{true}$ by Lemma 11.75(b); the list $\mathsf{map}(\mathsf{layerCode}, O)$ is duplicate-free because O is and $\mathsf{layerCode}$ is injective on valid layers (Lemma 11.76(b)), so $\mathsf{distinctCodes}(\cdot,0) = \mathsf{true}$ by Lemma 11.64. (F3): for $o \in O$, $\iota(o) \in \mathcal{O}$, so $\mathsf{isOption}(U,\iota(o)) = \mathsf{true}$, and $\iota(\iota(o)) = o$, so $\mathsf{beqLayer}(\iota(\iota(o)),o) = \mathsf{true}$ by Lemma 11.46. (F4): by Lemma 11.46, $\mathsf{beqLayer}(\iota(o),o) = \mathsf{true}$ iff $\iota(o) = o$, so the value is $F = 10$, resp. 9. (F5): $\mathsf{keptOptions}(t) = \mathsf{filter}(\mathsf{kept}_t, O)$ has 100, resp. 81, entries. (F6): for $o \in O$, by Lemma 11.78(a),(b) at least one of $\mathsf{kept}_t(o)$, $\mathsf{kept}_t(\iota(o))$ is true; and if $\mathsf{beqLayer}(\iota(o),o) = \mathsf{false}$, then $\iota(o) \ne o$ and not both are true. This proves Facts 11.67 and 11.68.

Top cases and strata. Unfolding Definition 11.12 with $\mathsf{range}(5) = [0,1,2,3,4]$ gives the list of Fact 11.69. In $\mathsf{keptOptions}(t)$ the entries with one token are the kept entries of X_1 and those with two tokens the kept entries of X_2. In the order of $\mathsf{topCases}$ ($\mathsf{H3}$ first, then the ten $\mathsf{H6}(a,s)$), the counts are therefore 5 eleven times, and 95 followed by 76 ten times. This is Fact 11.70. □

Proposition 11.80 (Proof of roots_match; written step)**.** $|\mathsf{roots}| = 910$*, and for every* $\rho \in \mathsf{roots}$*,* $\mathsf{decodeRoot}(\mathsf{rootCode}(\rho)) = \rho$ *and* $\mathsf{beqNode}(\mathsf{decodeRoot}(\mathsf{rootCode}(\rho)),\rho) = \mathsf{true}$*. Hence Fact 11.71 holds.*

Proof. By (L3), $|\mathsf{roots}| = \sum_t |\mathsf{keptOptions}(t)|$ over the 11 entries of $\mathsf{topCases}$, which is $100 + 10 \cdot 81 = 910$ by Proposition 11.79. By (L2) and (L4), every $\rho \in \mathsf{roots}$ is $\langle [t,o], [] \rangle$ with $t \in \mathsf{topCases}$ and $o \in \mathsf{keptOptions}(t)$, so $o \in \mathcal{O}(\mathsf{map}(u \mapsto u.1, t))$ (Lemma 11.75(b)). The layer t is valid ($\mathsf{H3} = [(0,0)]$, or $\mathsf{H6}(a,s) = [(0,0),(a,s)]$ with $a \le 5$ and $s \le 1$, by the unfolding of $\mathsf{topCases}$ in Proposition 11.79), and o is valid by Lemma 11.75(c). Lemma 11.76(c) gives $\mathsf{decodeRoot}(\mathsf{rootCode}(\rho)) = \rho$, and Lemma 11.46 gives $\mathsf{beqNode}(\rho,\rho) = \mathsf{true}$. □

Corollary 11.81 (Roots and certificates; written step)**.** *(Computer-assisted: the proof uses Facts 11.72 and 11.73.) The following hold.*

(a) $\mathsf{map}(\mathsf{rootCode}, \mathsf{roots}) = \mathsf{map}(c \mapsto c.\mathsf{root}, \mathsf{certs})$ *as lists, and* $|\mathsf{certs}| = 910$.

(b) The 910 entries of roots *are pairwise distinct nodes, and* $\{\mathsf{rootCode}(\rho) : \rho \in \mathsf{roots}\} = \mathrm{set}(\mathsf{tsvRootCodes})$.

Proof. (a) Lemma 11.43 with $eq = \mathsf{Nat.beq}$ (whose value true implies equality) turns Fact 11.72 into the equality of lists; then $|\mathsf{certs}| = |\mathsf{map}(\mathsf{rootCode}, \mathsf{roots})| = 910$ by (L3) and Fact 11.71. (b) By Fact 11.73 and Lemma 11.64, $\mathsf{map}(c \mapsto c.\mathsf{root}, \mathsf{certs})$ is duplicate-free, hence so is $\mathsf{map}(\mathsf{rootCode}, \mathsf{roots})$ by (a); two positions of roots holding the same node would give equal codes, so roots is duplicate-free. The equality of the codeSet values and Lemma 11.64 give $\mathrm{set}(\mathsf{map}(c \mapsto c.\mathsf{root}, \mathsf{certs})) = \mathrm{set}(\mathsf{tsvRootCodes})$, and (a) identifies the left side with the set of root codes of the roots. □

The roots are distinct also by a hand argument (distinct top cases and distinct options give distinct nodes), but their identification with tsvRootCodes rests on the computer-assisted Facts 11.72 and 11.73.

11.9. **The threefold cover theorem.**

Theorem 11.82 (Roots covered; Lean: `C3Cover.all_roots_covered`, `C3LEAN/C3Cover.lean:2251`)**.** *(Computer-assisted: the proof uses Facts 11.65 and 11.72.) For every* $\rho \in \mathsf{roots}$, $\mathsf{Covered}(S^\star)(\rho)$, *where* $S^\star(n)$ *means that there is* $k \in \mathsf{certifiedKeys}$ *with* $\mathsf{decodeKey}(k) = n$.

The theorem is Lean-checked (axioms `propext`, `Quot.sound`) and has no hypotheses. Its proof uses the computer-assisted Facts 11.65 (through Lemma 11.66) and 11.72, and Fact 11.71, which also has the written proof of Proposition 11.80.

Proof. We follow the Lean proof. Let $\rho \in \mathsf{roots}$.

(1) By Fact 11.71 (or Proposition 11.80), we have $\mathsf{beqNode}(\mathsf{decodeRoot}(\mathsf{rootCode}(\rho)), \rho) = \mathsf{true}$, so $\mathsf{decodeRoot}(\mathsf{rootCode}(\rho)) = \rho$ by Lemma 11.45.

(2) By Corollary 11.81(a), $\mathsf{map}(\mathsf{rootCode}, \mathsf{roots}) = \mathsf{map}(c \mapsto c.\mathsf{root}, \mathsf{certs})$. As $\mathsf{rootCode}(\rho)$ is an entry of the left side (L4), there is $c \in \mathsf{certs}$ with $c.\mathsf{root} = \mathsf{rootCode}(\rho)$.

(3) By Lemma 11.66, $\mathsf{checkRoot}(\mathsf{decodeRoot}(c.\mathsf{root}), c) = \mathsf{true}$. By (1) and (2), $\mathsf{decodeRoot}(c.\mathsf{root}) = \mathsf{decodeRoot}(\mathsf{rootCode}(\rho)) = \rho$, so $\mathsf{checkRoot}(\rho, c) = \mathsf{true}$.

(4) By Lemma 11.53(b), $\mathsf{Covered}(S_c)(\rho)$, with $S_c(n)$: there is $k \in \mathsf{keys}(c)$ with $\mathsf{decodeKey}(k) = n$.

(5) If $S_c(n)$ holds with witness $k \in \mathsf{keys}(c)$, then $k \in \mathsf{flatMap}(\mathsf{keys}, \mathsf{certs}) = \mathsf{certifiedKeys}$ by (L2), since $c \in \mathsf{certs}$; so $S^\star(n)$. Lemma 11.28 turns $\mathsf{Covered}(S_c)(\rho)$ into $\mathsf{Covered}(S^\star)(\rho)$. □

Remark 11.83 (What the theorem says, and how the project uses it)*. The leaf set.* Theorem 11.82 is only as informative as the predicate $S^\star$, which is read off the certificates themselves. A certificate that names its root as the only leaf satisfies the checker; this is the case for 554 of the 910 roots (our count from the certificate data; see the paragraph after Fact 11.65), and for those roots the statement is immediate. The substantial claim of the project, that every node satisfying $S^\star$ names a run record in which both checkers are recorded as accepting the proof (the criterion of `C3U2/tools/compose.py`; `C3LEAN/README.md`, §4) and whose CNF hash matches the base formula plus the units of the node, is checked outside Lean by the Python scripts `C3LEAN/tools/audit_keys.py` and `C3LEAN/review/indep_audit.py` (73,567 leaves, 73,707 records, no mismatch; `C3LEAN/review/logs/indep_audit.json`). The review demonstrates the gap with a certificate that collapses an inner σ-split of root 0 into one leaf naming a node without such a record and still passes (`C3Cover.rev_uncertified_leaf_passes`, `C3LEAN/review/RevCtrl_pass.lean:307`). The verdicts themselves are recorded fields, and the proofs were not kept (`C3LEAN/README.md`, §4). The project later recorded that acceptance by `lrat-check` alone is not a certificate and that `lrat-check` verdicts are supporting only (lines 111–114 of `WRITEUP/note/symmetric_extremal.tex`; `LRATCHECK20261105`); of the two recorded verdicts, only the `cake_lpr` verdict counts as a certificate (`WRITEUP/SYMMETRIC.md`, §3).

Intended use (not established here). By the induction principle of Definition 11.27, the theorem gives $P(\rho)$ for every root ρ and every $S^\star$-closed predicate P. In the project, $P(n)$ is "no set of the class (threefold symmetric, 33 points, general position, no convex 7-gon, with its canonical labelling) is contained in the node n". That P is $S^\star$-closed amounts to three claims: (i) $P(n)$ for every leaf, which is the record-based fact above together with the soundness of the CNF encoding; (ii) the rule (layer), which is Lemma 4 of `C3-S/soundness.md` and, for nodes with a nonempty cube, rule 3 of `C3U2/sigma_split.md` (Lean's rule (layer) applies to such nodes, whereas the project's composer `C3U2/tools/compose.py` also requires a layer-split record there, `C3LEAN/README.md`, §5,

finding 3); (iii) the rule (sigma), which is rule 2 of `C3U2/sigma_split.md`. The passage from "P holds at every root" to "the class is empty" is the cover theorem with the reduced cut in `C3H6M/soundness_h6m.md`, which uses Lemma 5 of `C3-S/soundness.md` and Lemma H6M; the 90 options below $\mathsf{H3}$ and the 720 options below the ten $\mathsf{H6}(a,s)$ that these reductions drop ($190-100$ and $10\cdot(153-81)$) are not roots of the file, so Theorem 11.82 says nothing about them. Facts 11.67 and 11.68 record that the computed second layers have the properties those reductions need: refl is an involution of the options, and the kept options contain exactly one member of each pair and every fixed option. None of (i)–(iii), and none of the written arguments cited here, is Lean-checked or proved in this document. The project's own list of what the Lean check leaves open, with the status it records for each part, is §6 of `C3LEAN/README.md`; its trust base for the threefold theorem (Theorem 1 of `WRITEUP/SYMMETRIC.md`) is §3 of that file.

11.10. **The central 34-point split.** Every Lean reference in a statement head of this subsection is to `C2FINAL/Central34Orbits.lean`, namespace `Central34`; the statements tagged "written step" have no Lean counterpart.

Definition 11.84 (Inner classes; Lean: `Central34.inner`, `C2FINAL/Central34Orbits.lean:21`)**.** For $a\in\mathbb{N}$,

$$\mathsf{inner}(a)=\mathsf{filter}\big(i\mapsto (i\neq 0)\wedge(i\neq a),\ \mathsf{range}(17)\big).$$

Definition 11.85 (Pairs; Lean: `Central34.choose2`, `C2FINAL/Central34Orbits.lean:24`)**.** For lists of naturals,

$$\mathsf{choose2}([])=[],\qquad \mathsf{choose2}(x::xs)=\mathsf{map}\big(y\mapsto[x,y],\ xs\big)\mathbin{+\!\!+}\mathsf{choose2}(xs).$$

Definition 11.86 (Triples; Lean: `Central34.choose3`, `C2FINAL/Central34Orbits.lean:29`)**.** For lists of naturals,

$$\mathsf{choose3}([])=[],\qquad \mathsf{choose3}(x::xs)=\mathsf{map}\big(W\mapsto x::W,\ \mathsf{choose2}(xs)\big)\mathbin{+\!\!+}\mathsf{choose3}(xs).$$

Definition 11.87 (Subcases; Lean: `Central34.subcases`, `C2FINAL/Central34Orbits.lean:34`)**.** For $a\in\mathbb{N}$, $\mathsf{subcases}(a)=\mathsf{choose2}(\mathsf{inner}(a))\mathbin{+\!\!+}\mathsf{choose3}(\mathsf{inner}(a))$.

Definition 11.88 (Masks; Lean: `Central34.mask`, `C2FINAL/Central34Orbits.lean:37`)**.** For a list V of naturals,

$$\mathsf{mask}(V)=\mathsf{foldl}\big((acc,i)\mapsto acc\ \mathsf{or}\ 2^i,\ 0,\ V\big).$$

Definition 11.89 (Central reflection; Lean: `Central34.refl`, `C2FINAL/Central34Orbits.lean:40`)**.** For $a\in\mathbb{N}$ and a list V of naturals,

$$\mathsf{refl}(a,V)=\mathsf{map}\big(i\mapsto((a+17)\mathbin{\dot-} i)\bmod 17,\ V\big).$$

(This is `Central34.refl`, not to be confused with `C3Cover.refl`.)

Definition 11.90 (Bit sets; Lean: `Central34.bitset`, `C2FINAL/Central34Orbits.lean:43`)**.** For a list L of lists of naturals,

$$\mathsf{bitset}(L)=\mathsf{foldl}\big((acc,W)\mapsto acc\ \mathsf{or}\ 2^{\mathsf{mask}(W)},\ 0,\ L\big).$$

Definition 11.91 (Uncovered subcases; Lean: `Central34.uncovered`, `C2FINAL/Central34Orbits.lean:46`)**.** For $a\in\mathbb{N}$ and a list C of lists of naturals, with the local abbreviation $B=\mathsf{bitset}(C)$,

$$\mathsf{uncovered}(a,C)=\mathsf{filter}\Big(V\mapsto\neg\big(\mathsf{testBit}(B,\mathsf{mask}(V))\vee\mathsf{testBit}(B,\mathsf{mask}(\mathsf{refl}(a,V)))\big),\ \mathsf{subcases}(a)\Big).$$

Definition 11.92 (Closed lists; Lean: `Central34.closedP1`, `C2FINAL/Central34Orbits.lean:51`)**.** The lists $\mathsf{closedP}_1,\dots,\mathsf{closedP}_8$ (Lean `closedP1`, …, `closedP8`, at lines 51, 78, 105, 132, 159, 186, 213, 240; the data end at line 264) are explicit lists of lists of naturals with 287 entries each, except 288 for $\mathsf{closedP}_6$.

Definition 11.93 (Closed subcases; Lean: `Central34.closed`, `C2FINAL/Central34Orbits.lean:267`)**.** For $a\in\mathbb{N}$: $\mathsf{closed}(a)=\mathsf{closedP}_a$ if $1\le a\le 8$, and $\mathsf{closed}(a)=[]$ otherwise.

By Lemma 11.1, $\beta(\mathsf{mask}(V))=\mathrm{set}(V)$, so $\mathsf{mask}(V)=\mathsf{mask}(W)$ iff $\mathrm{set}(V)=\mathrm{set}(W)$; and $\beta(\mathsf{bitset}(L))=\{\mathsf{mask}(W):W\in L\}$, so $\mathsf{testBit}(\mathsf{bitset}(L),\mathsf{mask}(V))=\mathsf{true}$ iff $\mathrm{set}(V)=\mathrm{set}(W)$ for some $W\in L$. For $1\le a\le 16$ put $I_a=\{1,\dots,16\}\setminus\{a\}$.

Remark 11.94 (Intended reading). In the project (header of the file and `C2FINAL/README.md`, §1), a centrally symmetric 34-point set consists of 17 antipodal classes $0,\dots,16$; in the hull case P_a, $1\le a\le 8$, the hull is the parallelogram on the classes 0 and a, and the second convex layer is spanned by 2 or 3 of the other 15 classes, which gives the subcases; the reflection $i\mapsto(a-i)\bmod 17$ is reduction B. The lists $\mathsf{closedP}_a$ were generated by

`C2FINAL/gen_lean.py` as the subcases with a run record in which both checkers, `cake_lpr` and `lrat-check`, are recorded as accepting the proof and whose CNF hash was rebuilt and matched by `C2FINAL/hashmatch.py` (or a re-solve of `C2FINAL/README.md`, §4). This reading is not used in any proof.

Lemma 11.95 (The subcases; written step)**.** *Let l be a strictly increasing list of k naturals. Then* $\mathsf{choose2}(l)$ *is duplicate-free with $\binom{k}{2}$ entries, which are exactly the lists $[x,y]$ with $x,y\in l$, $x<y$; and* $\mathsf{choose3}(l)$ *is duplicate-free with $\binom{k}{3}$ entries, which are exactly the lists $[x,y,z]$ with $x,y,z\in l$, $x<y<z$. Consequently, for $1\le a\le 16$,* $\mathsf{subcases}(a)$ *is a duplicate-free list of $\binom{15}{2}+\binom{15}{3}=105+455=560$ strictly increasing lists, and $V\mapsto \mathrm{set}(V)$ is a bijection from its entries onto the set of subsets of I_a with two or three elements.*

Proof. Induction on l; for $l=[]$ both lists are empty. Let $l=x::xs$, so xs is strictly increasing with $k-1$ entries, all larger than x. The first part of $\mathsf{choose2}(l)$ consists of the $k-1$ distinct lists $[x,y]$, $y\in xs$, all increasing and with head x; by induction $\mathsf{choose2}(xs)$ is duplicate-free with $\binom{k-1}{2}$ increasing entries, with heads in xs, hence different from x. So $\mathsf{choose2}(l)$ is duplicate-free with $k-1+\binom{k-1}{2}=\binom{k}{2}$ entries. Every increasing pair $[u,v]$ of elements of l occurs: if $u=x$, then $v\in xs$ and $[x,v]$ is in the first part; otherwise $u,v\in xs$ (as $v>u>x$) and the pair is in $\mathsf{choose2}(xs)$ by induction. Likewise the first part of $\mathsf{choose3}(l)$ consists of the $\binom{k-1}{2}$ distinct lists $x::W$ with $W\in\mathsf{choose2}(xs)$, increasing with head x, and $\mathsf{choose3}(xs)$ contributes $\binom{k-1}{3}$ increasing triples with heads in xs; the total is $\binom{k-1}{2}+\binom{k-1}{3}=\binom{k}{3}$, without repetition, and every increasing triple of elements of l occurs (the same case distinction on its first element). For $1\le a\le 16$, $\mathsf{inner}(a)$ is the increasing list of the 15 elements of I_a. The pairs and the triples are disjoint by length, and a strictly increasing list is determined by its set of entries; this gives the bijection. □

Theorem 11.96 (Orbits; Lean: `Central34.subcases_reflection`, `C2FINAL/Central34Orbits.lean:273`)**.** *(Computer-assisted in Lean, by kernel evaluation; written proof below.) The statement is*

$$\mathsf{all}\big(\mathsf{range}(8),\ k\mapsto Q(k+1)\big)=\mathsf{true},$$

where for $a\in\mathbb{N}$, with $S_a=\mathsf{bitset}(\mathsf{subcases}(a))$, $Q(a)$ is the Boolean conjunction of

- *(i)* $|\mathsf{subcases}(a)|=560$*;*
- *(ii)* $\mathsf{all}\big(\mathsf{subcases}(a),\ V\mapsto \mathrm{Img}_a(V)\wedge\mathrm{Inv}_a(V)\big)$*, where* $\mathrm{Img}_a(V)=\mathsf{testBit}(S_a,\mathsf{mask}(\mathsf{refl}(a,V)))$ *and* $\mathrm{Inv}_a(V)=\big(\mathsf{mask}(\mathsf{refl}(a,\mathsf{refl}(a,V)))=\mathsf{mask}(V)\big)$*;*
- *(iii)* $|\mathsf{filter}\big(V\mapsto \mathsf{mask}(V)\le\mathsf{mask}(\mathsf{refl}(a,V)),\ \mathsf{subcases}(a)\big)|=287$*;*
- *(iv)* $|\mathsf{filter}\big(V\mapsto \mathsf{mask}(V)=\mathsf{mask}(\mathsf{refl}(a,V)),\ \mathsf{subcases}(a)\big)|=14$.

In words: for every $a\in\{1,\dots,8\}$, the statements (i)–(iv) hold.

The theorem is Lean-checked according to the record (paragraph "Axioms" in §11.1), by kernel evaluation of the definitions. The following written proof does not use the evaluation.

Proof. Since $\mathsf{range}(8)=[0,\dots,7]$, the statement says that $Q(a)=\mathsf{true}$ for $a=1,\dots,8$. Fix such an a and let $\rho_a(i)=((a+17)\mathbin{\dot-} i)\bmod 17$ for $i\in\{0,\dots,16\}$. As $i<a+17$, $\rho_a(i)=(a+17-i)\bmod 17$, so $\rho_a(i)\equiv a-i\pmod{17}$ and $\rho_a(i)\in\{0,\dots,16\}$; hence $\rho_a(\rho_a(i))\equiv i$, and $\rho_a(\rho_a(i))=i$. Also $\rho_a(0)=a$ and $\rho_a(a)=0$. So ρ_a is an involution of $\{0,\dots,16\}$ that maps I_a onto itself. For $V\in\mathsf{subcases}(a)$ the entries of V lie in I_a, $\mathsf{refl}(a,V)=\mathsf{map}(\rho_a,V)$ and $\mathrm{set}(\mathsf{refl}(a,V))=\rho_a(\mathrm{set}(V))$.

By Lemma 11.95 there is, for each $V\in\mathsf{subcases}(a)$, a unique entry V' of $\mathsf{subcases}(a)$ with $\mathrm{set}(V')=\rho_a(\mathrm{set}(V))$, since $\rho_a(\mathrm{set}(V))\subseteq I_a$ has $|\mathrm{set}(V)|\in\{2,3\}$ elements; and $(V')'=V$ because $\rho_a\circ\rho_a$ is the identity. Moreover $\mathsf{mask}(\mathsf{refl}(a,V))=\mathsf{mask}(V')$.

(i) is Lemma 11.95. (ii): $\mathrm{Img}_a(V)=\mathsf{true}$, because the number $\mathsf{mask}(\mathsf{refl}(a,V))=\mathsf{mask}(V')$ lies in $\beta(S_a)$, as $V'\in\mathsf{subcases}(a)$; and $\mathrm{Inv}_a(V)=\mathsf{true}$, because $\mathrm{set}(\mathsf{refl}(a,\mathsf{refl}(a,V)))=\rho_a(\rho_a(\mathrm{set}(V)))=\mathrm{set}(V)$, so the two masks agree.

(iv): $\mathsf{mask}(V)=\mathsf{mask}(\mathsf{refl}(a,V))$ iff $\mathrm{set}(V)=\rho_a(\mathrm{set}(V))$, i.e. iff $\mathrm{set}(V)$ is a union of orbits of ρ_a on I_a. A point $i\in I_a$ is fixed iff $2i\equiv a\pmod{17}$; as $2\cdot 9\equiv 1\pmod{17}$, this means $i\equiv 9a\pmod{17}$, so there is exactly one fixed point $x_0=9a\bmod 17$ in $\{0,\dots,16\}$, and $x_0\in I_a$ because $9a\not\equiv 0$ and $9a\not\equiv a\pmod{17}$ for $1\le a\le 8$ ($x_0=9,1,10,2,11,3,12,4$ for $a=1,\dots,8$). The other 14 elements of I_a form 7 two-element orbits. A union of orbits with two elements is one of these 7 orbits; one with three elements is $\{x_0\}$ together with one of them. So exactly $7+7=14$ entries V satisfy the condition.

(iii): if $\mathsf{mask}(V)=\mathsf{mask}(\mathsf{refl}(a,V))$, the condition $\le$ holds; this accounts for 14 entries. The other 546 entries are paired by $V\leftrightarrow V'$ with $V'\neq V$, and $\mathsf{mask}(V)$, $\mathsf{mask}(V')$ are distinct naturals; as $\mathsf{mask}(\mathsf{refl}(a,V))=\mathsf{mask}(V')$

and $\mathsf{mask}(\mathsf{refl}(a,V')) = \mathsf{mask}(V)$, exactly one member of each pair satisfies the condition. So the count is $14 + 546/2 = 287$. □

Fact 11.97 (Lists; Lean: `Central34.closed_are_subcases`, `C2FINAL/Central34Orbits.lean:284`)**.** *(Computer-assisted.) The statement is*

$$\mathsf{all}\Big(\mathsf{range}(8),\ k \mapsto \mathsf{all}\big(\mathsf{closed}(k+1),\ V \mapsto \mathsf{testBit}(\mathsf{bitset}(\mathsf{subcases}(k+1)), \mathsf{mask}(V))\big)\Big) = \mathsf{true}.$$

Equivalently (by Lemma 11.95 and the remarks after Definition 11.93): for every $a \in \{1,\dots,8\}$ and every $V \in \mathsf{closedP}_a$, $\mathrm{set}(V)$ is a subset of I_a with two or three elements.

Lean-checked according to the record (paragraph "Axioms" in §11.1), by kernel evaluation over the explicit lists: for each a, the kernel computes the masks of the 560 subcases and their or-fold, and tests the mask of each of the 287 (for $a = 6$: 288) entries of $\mathsf{closedP}_a$ against it. No hand proof is given.

Theorem 11.98 (Closure; Lean: `Central34.all_cases_closed`, `C2FINAL/Central34Orbits.lean:291`)**.** *(Computer-assisted.) The statement is*

$$\mathsf{all}\big(\mathsf{range}(8),\ k \mapsto \mathsf{isEmpty}(\mathsf{uncovered}(k+1, \mathsf{closed}(k+1)))\big) = \mathsf{true};$$

that is, $\mathsf{uncovered}(a, \mathsf{closedP}_a) = []$ *for every* $a \in \{1,\dots,8\}$.

The theorem is Lean-checked according to the record (paragraph "Axioms" in §11.1), by kernel evaluation; no hand proof is given, and none is possible without the explicit lists. What is evaluated: for each $a \in \{1,\dots,8\}$ the kernel computes $B_a = \mathsf{bitset}(\mathsf{closedP}_a)$, the or-fold of $2^{\mathsf{mask}(W)}$ over the 287 (for $a = 6$: 288) entries W, and then, for each of the 560 entries V of $\mathsf{subcases}(a)$, the disjunction of the bits of B_a at $\mathsf{mask}(V)$ and at $\mathsf{mask}(\mathsf{refl}(a,V))$ (the second bit is needed only when the first is 0); the disjunction is true for every V, so the filtered list is empty. The two formulations in the statement agree because $\mathsf{range}(8) = [0,\dots,7]$, $\mathsf{closed}(a) = \mathsf{closedP}_a$ for $1 \le a \le 8$, and $\mathsf{isEmpty}(l) = \mathsf{true}$ iff $l = []$. The statement is not vacuous: $\mathsf{uncovered}(a,[]) = \mathsf{subcases}(a)$ has 560 entries.

Corollary 11.99 (Set form; written step)**.** *(Computer-assisted: a consequence of Theorem 11.98.) Let $a \in \{1,\dots,8\}$, and for $i \in \{0,\dots,16\}$ let $\rho_a(i) = ((a+17) \dot{-} i) \bmod 17$, the residue of $a - i$ modulo 17. For every subset $T \subseteq I_a$ with two or three elements there is $W \in \mathsf{closedP}_a$ with $\mathrm{set}(W) = T$ or $\mathrm{set}(W) = \rho_a(T)$.*

Proof. This is a consequence of Theorem 11.98. By Lemma 11.95, $T = \mathrm{set}(V)$ for some $V \in \mathsf{subcases}(a)$. Since V is not an entry of $\mathsf{uncovered}(a, \mathsf{closedP}_a) = []$, $\mathsf{testBit}(B_a, \mathsf{mask}(V))$ or $\mathsf{testBit}(B_a, \mathsf{mask}(\mathsf{refl}(a,V)))$ is true, where $B_a = \mathsf{bitset}(\mathsf{closedP}_a)$. By the remarks after Definition 11.93, there is $W \in \mathsf{closedP}_a$ with $\mathrm{set}(W) = \mathrm{set}(V) = T$ or $\mathrm{set}(W) = \mathrm{set}(\mathsf{refl}(a,V))$; and $\mathrm{set}(\mathsf{refl}(a,V)) = \rho_a(\mathrm{set}(V)) = \rho_a(T)$, because $\mathsf{refl}(a,V) = \mathsf{map}(\rho_a, V)$ by Definition 11.89, the entries of V lying in $I_a \subseteq \{0,\dots,16\}$. □

Remark 11.100 (Scope of the central theorem). Theorem 11.98 and Corollary 11.99 say that the explicit lists meet every orbit $\{T, \rho_a(T)\}$ of second-layer subcases, for each of the eight hull cases. That the listed subcases are refuted is not part of the statement: the lists were selected outside Lean from run records (Remark 11.94), and according to `C2FINAL/README.md` (Status and §9) most parallelogram pieces rest on recorded checker verdicts whose proofs were deleted after checking. As in Remark 11.83, of the two recorded checker verdicts only the `cake_lpr` verdict counts as a certificate; the project's trust base for its central theorem (Theorem 4 of `WRITEUP/SYMMETRIC.md`) is §3 of that file. The second-layer covers of the cases P_a, the 120 hexagonal hull cases, the hull cover, the reductions A ($P_a \sim P_{17-a}$) and B (the reflection), and the soundness of the encoder (`C2ENC`) are written arguments or records outside Lean and outside this document. By inspection of the data, the list $\mathsf{closedP}_6$ contains both members of exactly one orbit, $[1,7]$ and $[5,16]$ (as sets, $\{5,16\} = \rho_6(\{1,7\})$); this accounts for its 288 entries. The header comment of the file mentions a list `pilotUnhashed` that the file does not define (stale text of the generator); it has no effect on the theorems.

11.11. **Coverage.**

Covered. From `C3LEAN/C3Cover.lean`, every definition of the scope is covered by a definition above (the generated family `certs0`, …, `certs126` by a single one): `m`, `Tok`, `Layer`, `Node`, `Node.used`, `pairOptions`, `layerOptions`, `insTok`, `sortTok`, `topCases`, `okey`, `lexLe`, `mirror`, `phi`, `refl`, `keptOptions`, `roots` (Definitions 11.2–11.19); `Node.free`, `bitsOf`, `addBits`, `Node.sigmaChildren`, `conflicts`, `Node.layerChildren` (Definitions 11.20–11.25); `Covered` (Definition 11.27); `decodeTok`, `decodeLayer`, `decodeLayers`, `decodeCube`, `decodeKey`, `tokCode`, `layerCode`, `rootCode`, `decodeRoot`, `beqList`, `beqTok`, `beqNode`, `beqLayer` (Definitions 11.29–11.42); `Cert`, `unpack`, `Cert.keys`, `run`, `checkRoot` (Definitions 11.47–11.51); `tsvRootCodes`,

`certs0`–`certs126`, `certs`, `bounds`, `certifiedKeys` (Definitions 11.55–11.59; the data are described, not reproduced); `isOption`, `distinctCodes`, `codeSet`, `reflectionFacts` (Definitions 11.60–11.63). The theorems: `Covered.mono` (Lemma 11.28); `beqList_eq`, `beqTok_eq`, `beqNode_eq` (Lemmas 11.43–11.45); `run_sound` and `checkRoot_sound` (Lemma 11.53); `all_append_of` (Lemma 11.54); `batch0`–`batch126` (Fact 11.65); `all_checked` (Lemma 11.66); `reflection_facts_H3` (Fact 11.67); the ten `reflection_facts_H6_1_0`, ..., `reflection_facts_H6_5_1` (Fact 11.68); `topCases_eq` (Fact 11.69); `strata` (Fact 11.70); these four with the written proof of Proposition 11.79; `roots_match` (Fact 11.71, with Proposition 11.80); `roots_codes_match` and `roots_eq_tsv` (Facts 11.72 and 11.73, with Corollary 11.81); `all_roots_covered` (Theorem 11.82). From `C2FINAL/Central34Orbits.lean`: `inner`, `choose2`, `choose3`, `subcases`, `mask`, `refl`, `bitset`, `uncovered`, `closedP1`–`closedP8` (by a single definition), `closed` (Definitions 11.84–11.93); `subcases_reflection` (Theorem 11.96), `closed_are_subcases` (Fact 11.97) and `all_cases_closed` (Theorem 11.98, with Corollary 11.99).

Not covered, deliberately.

- The seven negative controls at the end of `C3LEAN/C3Cover.lean`, which are tests, not results: `control_missing_record` (line 2266; root 0 with its fourth key removed is rejected), `control_missing_sigma_child` (2270; root 0 without the subtree of the last child of its first σ-split), `control_swapped_keys` (2274; two adjacent keys swapped), `control_missing_layer_child` (2278; root 100, code 41560, without the key of its last layer child), `control_wrong_root` (2282; the certificate of root 0 does not pass for root 1), `control_strict_reduction` (2286; a strict comparison in the reduction would leave 810 roots; this agrees with Proposition 11.79, as $810 = (100-10)+10\cdot(81-9)$) and `control_phi_mutant` (2292; `phi` without its $[s = 0]$ term would keep options below $\mathsf{H6}(1,0)$ whose root codes are not all in `tsvRootCodes`). The `#print axioms` commands (lines 2301–2327 of `C3LEAN/C3Cover.lean`, lines 296–298 of `C2FINAL/Central34Orbits.lean`) are not statements.
- `C3LEAN/tools/prelude.lean` (SHA-256 `cdcf96de7739...`) is the verbatim source of lines 29–329 of `C3LEAN/C3Cover.lean`; its declarations are covered through that file.
- `C3LEAN/Control_record.lean` (SHA-256 `5f755e897c30...`) and `C3LEAN/Control_sibling.lean` (`fb68b330e6e3...`): must-fail files, each a verbatim copy of the prelude plus one theorem (line 305) asserting that a mutated certificate of root 0 passes; they fail to compile, as intended (`C3LEAN/logs/control_record.log`, `C3LEAN/logs/control_sibling.log`).
- `C3LEAN/review/RevCtrl_mut.lean` (`49bf2cce1b73...`): a copy of the prelude plus nine reviewer controls stating that mutated certificates are rejected (`rev_extra_decision` has the same certificate as `rev_extra_key`). `C3LEAN/review/RevCtrl_pass.lean` (`a135e2b92e6a...`): a copy of the prelude plus two controls stating that two non-genuine certificates pass (`rev_uncertified_leaf_passes`, used in Remark 11.83; `rev_noncanonical_key_passes`) and the evaluation `rev_parent_decodes` (used in Remark 11.34). `C3LEAN/review/RevDiff.lean` (`5aa3f6172026...`): a copy of the prelude plus an `#eval` harness, without theorems of its own. These review files are tests and audits, not results; their prelude copies duplicate covered declarations.
- The Lean package `C3L`: `C3L/C3Encoding.lean` (only imports) with `C3L/C3Encoding/Gen.lean`, `C3L/C3Encoding/Case.lean` and `C3L/C3Encoding/Emit.lean`. It is a Lean program that writes, for parameters m and k, the base CNF of the project's threefold encoding and the node CNFs (the base CNF followed by the unit clauses of a node) as DIMACS files, and it states no theorem (`C3L/README.md`). Its 41 declarations are 39 definitions, a structure and an abbreviation. In `C3L/C3Encoding/Gen.lean`, namespace `C3`: the enumerations `pairs`, `triples`, `quads`, `perms2`, `others`; the helpers `key3`, `paritySort`, `rot`, `eps`; the variables and literal tables `orbits`, the structure `Tables`, `mkTables`, `Tables.s`, `Tables.sig`, `Tables.Cl`, `Tables.Tl`, `Tables.invv`, `Tables.Bvv`, `Tables.B`, `Tables.ccw`, `chainVars`, `numVars`; the clauses `addCl`, `push`, `xor0`, `secSig`, `secSb`, `secPrefix`, `secForced`, `secXorB`, `secXorT`, `secChain` and the base formula `baseClauses`. In `C3L/C3Encoding/Case.lean`, namespace `C3`: the abbreviation `Layer`, `parseLayers`, `layerOrder` and `layerUnits` (the unit clauses of a node). In `C3L/C3Encoding/Emit.lean`, root namespace: `clauseLine`, `body`, `writeCnf` and `main`, the output routines of the executable `c3_dimacs`. That the output agrees with the CNF files of the project's Python generator is recorded in `C3L/logs` (comparisons outside Lean); the soundness theorem planned in `C3L/PLAN.md` (named `c3_base_sound` there) is not in the repository. CNF encodings are outside this document (item (ii) of the paragraph "What is not established here" in §11.1).

Treated in another section. The Lean-checked files `AUDIT3/A1_reduction/SizeArith.lean` and `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean` (the size arithmetic of the reduction of the symmetric $k = 7$ statement to its case classes) and `ES7SPLIT/review/equivalence/BaekSeven.lean` (an instance of Baek's

theorem for abstract configurations) are outside the scope of this section; they are treated in the last section, on size arithmetic.

Audit scripts of results outside Stage 1. The five Lean files of `AUDIT2`, an adversarial review of the project's mirror-class result, state no theorem of their own. `AUDIT2/A1/A1Dump.lean` defines `cube` (line 5) and `main` (line 6), an input–output program that writes the data `V2L.caseTable` (with the leaves of the trees `V2L.treeOf`), `V2L.upTable` and `V1L.hullLeaves` to a file, for comparison with a Python decoding of their source files. `AUDIT2/A1/A1Audit.lean`, `AUDIT2/A2/Ax.lean`, `AUDIT2/A2/Ax2.lean` and `AUDIT2/A6/scripts/A6Check.lean` contain, apart from `import`, `open` and `set_option` lines and comments, only `#check`, `#print`, `#print axioms` and `#eval` commands. The theorems they audit, such as `V2L.mirrorExclusionLine_final`, `V1L.hullVertexLine_of_leaves` and `Mir.mirrorExclusionLine_of_unsat`, belong to the Lean developments in `H3`, `V1-L` and `V2-L`, which lie outside Stage 1; this document does not restate them.

12. Size arithmetic of the symmetric $k = 7$ reduction

This section gives written proofs for three small Lean files of the research records that no other section covers. Two of them check the size arithmetic of the reduction of the symmetric $k = 7$ statement to its case classes; the third instantiates Baek's theorem at 7-gons.

The three files and their role.

(1) `AUDIT3/A1_reduction/SizeArith.lean` (namespace `A1`, 20 theorems). The directory `AUDIT3` records an adversarial audit, run on 1 November 2026 against commit `9cf4949a5`, of the statement "every set of at least 33 points in general position with a nontrivial symmetry, isometric or affine, contains a convex 7-gon"; it records an audit and claims no theorem (`AUDIT3/README.md`). Its component audit A1 (report `AUDIT3/reports/A1_reduction.md`) re-derived by hand the reduction of this statement to the case theorems of the records for half-turns, threefold and fivefold rotations and reflections, to which the case classes of Definition 7.7 correspond. The auditor wrote this file to check in Lean, for all sizes n, the arithmetic of that reduction and the arithmetic behind the upper bounds of the table of the largest symmetric 7-gon-free sizes (trust-table rows 4, 8 and 9 of the report).

(2) `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean` (7 theorems and one anonymous example). Finding A1-F1 of that audit (rated minor) is that the assembled theorem was not written anywhere. The verifier of the finding wrote this file to check, for all n, the bookkeeping of the assembled statement: the half-turn, threefold and fivefold cases, the two mirror cases and the fourfold case through the half-turn, together with a control showing that the half-turn bound fails at 32 points (`AUDIT3/reports/A1_reduction.md`, lines 103–106). The file also states, at its line 24, that an order of at least 7 gives at least one orbit other than the centre (`big_order_orbit`); the report's description of the file, at its line 106, does not mention this lemma.

(3) `ES7SPLIT/review/equivalence/BaekSeven.lean` (one theorem). It instantiates the vendored formalization `main` of Baek's theorem, cited in this document as Theorem 10.4, at 7-gons: 17 elements of an abstract configuration contain a 4-cap or a 7-gon. The records of `ES7SPLIT` use it as the Lean-checked abstract form of the one published theorem that Theorems 5 and 6 and Corollary 7 of `ES7SPLIT/README.md` (written proofs there) use as a black box. It plays no role in the reduction.

What the files do not contain. The statements of the first two files are facts about natural numbers. The geometric modelling that connects them with point sets is not part of them: the header comment of `SizeArith.lean` (lines 4–8) states it and calls it "audited by hand in the report, not formalised here". In that modelling, a rotation of order s about c gives $n = s \cdot k + e$, where k is the number of orbits other than $\{c\}$, $e = 1$ if $c \in S$ and $e = 0$ otherwise, so $e \le 1$, and $e = 0$ for even s; a reflection gives $n = 2k + b$, where k is the number of mirror pairs and $b \le 2$ the number of points on the axis; and deleting whole orbits, the centre or axis points keeps the symmetry and general position. We prove these facts as a written step (Lemma 12.12), which is not a declaration of these files. Related facts are Lean-checked elsewhere in this document, in other forms: Theorem 6.35(a),(b), an abstract form of the count $n = s \cdot k + e$ for a map $z \mapsto \zeta z$ of exact order s in a commutative group with zero; the selection of whole orbits (Lemma 7.16) and the counting of half-turn pairs, rotation orbits and mirror pairs in the Lean proof of Theorem 7.39; the layer results (Theorems 2.22 and 2.24), through verbatim copies of which, and of Theorem 2.26, the Lean proofs of Theorems 6.7 and 6.4 pass from a symmetric set to a subset of exactly 32, respectively 35, points (the file of these proofs repeats lines 1–1147 of `LAYERLEAN/LayerLemma.lean` instead of importing it); and Lemmas 2.16 and 3.18, which the proof of Lemma 12.12 uses. No Lean theorem connects these facts with the statements of the files of this section. Remark 12.13 explains how Lemma 12.12 combines with the arithmetic and with the case classes. The Lean-checked reduction of the symmetric statement

to the case classes is Theorem 7.39 for Euclidean isometries, with Theorem 8.28 for affine symmetries. Both are conditional on the classes, and their Lean proofs do their own arithmetic: a text search of the repository's Lean files finds the names of the declarations of `SizeArith.lean` and `AssembledSizes.lean` only in these two files, so no other Lean file uses them.

Sources and evidence. Paths are relative to the research-records directory `cloud/es7-2026-10-19/results`, except those beginning with `ErdosSzekeres/`, which are relative to the repository root. The table lists the three Lean files and their logs by file name; their full paths are given in the list that follows it. The files are those of commit `7e32ee259`; their line counts and SHA-256 hashes (computed with `shasum -a 256`) are:

file	lines	SHA-256
`SizeArith.lean`	103	`170384d81db76bb5e8d019e886f50b2545dc7581599f077c6431e08edb532418`
`SizeArith.lean.log`	20	`b4bca1cd8542a5e39d2a8560570b573ece40533d9850e8b77f5fc33c41da5770`
`AssembledSizes.lean`	39	`c08bdf200fb59f8353c999460a1069ecd7e3414874c4a79a8030c58dc36f2feb`
`AssembledSizes.lean.log`	11	`1db4174c7c04edd5a20fad31800eaaf076a0af5cb480c08dcc1771018ae377c9`
`BaekSeven.lean`	12	`d6d6636c78dd1a2b271ef8cd1dd09a9c869d5db8f3319f702921319eaf36c2d0`
`BaekSeven.out`	3	`53dced1cd76f1ff0b2afb9ba3ba854ffe0ac1a1dbe895855cb6abfa108f8c747`

- `AUDIT3/A1_reduction/SizeArith.lean`: namespace `A1`, no `import` (core Lean only). Lines 1–9 are the header comment; lines 16–80 hold the 20 theorems; lines 84–103 are one `#print axioms` command for each of them, in the same order. Its log `AUDIT3/A1_reduction/SizeArith.lean.log` is the output of these commands and contains no error and no warning. It reports the axioms `propext`, `Classical.choice`, `Quot.sound` for `A1.c5_reduce`, `A1.c4_reduce`, `A1.c6_reduce`, `A1.c2_reduce`, `A1.c5_gap` and `A1.table_mirror`, and `propext`, `Quot.sound` for the other 14 theorems.
- `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean`: root namespace, no `import`. The seven theorems are declared at lines 6, 9, 12, 15, 18, 21 and 24 (the last one with its proof on lines 25–27), the anonymous `example` at line 30, and lines 33–39 are a `#print axioms` command for each theorem. Its log `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean.log` contains the linter warning "unused variable `hp`" at line 24, column 41, then `propext`, `Classical.choice`, `Quot.sound` for `halfturn_pairs` and `propext`, `Quot.sound` for the other six theorems, and a timing line; it contains no error.
- `ES7SPLIT/review/equivalence/BaekSeven.lean`: imports `ErdosSzekeres.Vendor.ETV.Main.Main` (hence Mathlib). Line 6 is `baek_seven`, line 11 the command `#check @main`, line 12 the command `#print axioms baek_seven`. Its output `ES7SPLIT/review/equivalence/BaekSeven.out` gives in lines 1–2 the type of `main` (Remark 12.11) and in line 3 the axioms `propext`, `Classical.choice`, `Quot.sound` of `baek_seven`. Since `#print axioms` reports every axiom in the dependency closure of a declaration, this line also covers the vendored theorem `main` (`ErdosSzekeres/Vendor/ETV/Main/Main.lean`, SHA-256 prefix `83dd0a214204`) as it is used.

The first two files were checked with Lean 4.24.0 alone (`AUDIT3/README.md`, section "Reproduction"; `AUDIT3/reports/A1_reduction.md`, lines 106 and 262), the third with `lake env lean` from the repository root (`ES7SPLIT/review/README.md`, section "Reproduction"), that is, with the repository's Lean 4.24.0 and Mathlib `f897ebcf`. A text search of the three files finds no `sorry`, `admit`, `native_decide` or `axiom` declaration. The logs were produced by the project and were not re-run for this document.

Rigor. Every statement below whose head names a Lean declaration is *Lean-checked*, with the axioms shown in the logs above (all within `propext`, `Classical.choice`, `Quot.sound`). The proofs given here are *written proofs (not refereed)*. No statement of this section is computer-assisted. None of them is a finite computation: the Lean-checked statements are universally quantified, over natural numbers or, in Theorem 12.10, over linearly ordered types, configurations and finite sets, and the control of Remark 12.9 is the negation of such a statement. The Lean proofs use `decide` only to close side facts inside these proofs, never to evaluate a whole statement: the five closed instances $s = 2, \dots, 6$ of the conclusion of Lemma 12.4, the value $\binom{6}{2} + 2 = 17$ in Theorem 12.10 and $32 \le 32$ in the control; the written proofs check these facts by hand as well. By the convention that accompanies the rigor labels at the beginning of this document, such a use of `decide` does not make a statement computer-assisted. Lemma 12.12 and Remark 12.13 are written steps and are not Lean-checked. Theorem 12.10 rests on Baek's theorem, Theorem 10.4, which is cited and not proved in this document.

Conventions. In Subsections 12.1–12.3 all variables range over $\mathbb{N} = \{0, 1, 2, \dots\}$ (Lean `Nat`), and each statement is universally quantified over its variables in the order in which Lean lists them. No statement subtracts natural numbers. Lean's product `2 * t * k` is $(2t)k$ (multiplication associates to the left). The Lean proofs use the

tactic `omega`, a decision procedure for linear arithmetic over $\mathbb{N}$ and $\mathbb{Z}$ whose proof term is checked by the kernel; the written proofs are direct arguments.

12.1. **Lower bounds for the reduction.** The first eleven theorems of `SizeArith.lean` (its part "Reduction", lines 13–56) turn "at least 33 points" into the numbers of orbits or pairs that the case classes need (Remark 12.13).

Proposition 12.1 (rotations of order 2 to 6;
Lean: `A1.c3_reduce`, `AUDIT3/A1_reduction/SizeArith.lean:16`,
Lean: `A1.c5_reduce`, `AUDIT3/A1_reduction/SizeArith.lean:19`,
Lean: `A1.c4_reduce`, `AUDIT3/A1_reduction/SizeArith.lean:22`,
Lean: `A1.c6_reduce`, `AUDIT3/A1_reduction/SizeArith.lean:25`,
Lean: `A1.c2_reduce`, `AUDIT3/A1_reduction/SizeArith.lean:28`). *All variables range over* $\mathbb{N}$.

(a) (`A1.c3_reduce`) *For all* $k, e \in \mathbb{N}$*: if* $e \le 1$ *and* $33 \le 3k + e$*, then* $11 \le k$.
(b) (`A1.c5_reduce`) *For all* $k, e \in \mathbb{N}$*: if* $e \le 1$ *and* $33 \le 5k + e$*, then* $35 \le 5k + e$ *and* $7 \le k$.
(c) (`A1.c4_reduce`) *For every* $k \in \mathbb{N}$*: if* $33 \le 4k$*, then* $36 \le 4k$*,* $8 \le k$ *and* $17 \le 2k$.
(d) (`A1.c6_reduce`) *For every* $k \in \mathbb{N}$*: if* $33 \le 6k$*, then* $11 \le 2k$ *and* $17 \le 3k$.
(e) (`A1.c2_reduce`) *For every* $k \in \mathbb{N}$*: if* $33 \le 2k$*, then* $34 \le 2k$ *and* $17 \le k$.

Proof. (a) From $33 \le 3k + e$ and $e \le 1$ we get $3k \ge 32$; since $3 \cdot 10 = 30 < 32$, $k \ge 11$. (b) In the same way $5k \ge 32$, and $5 \cdot 6 = 30 < 32$ gives $k \ge 7$; then $5k + e \ge 5k \ge 35$. (c) Since $4 \cdot 8 = 32 < 33 \le 4k$, we have $k \ge 9$; hence $4k \ge 36$, $k \ge 8$ and $2k \ge 18 \ge 17$. (So the statement records the weaker bound $8 \le k$.) (d) Since $6 \cdot 5 = 30 < 33 \le 6k$, we have $k \ge 6$; hence $2k \ge 12 \ge 11$ and $3k \ge 18 \ge 17$. (e) Since $2 \cdot 16 = 32 < 33 \le 2k$, we have $k \ge 17$, and then $2k \ge 34$. Lean proves each part with `omega`. □

Proposition 12.2 (reflections;
Lean: `A1.mirror_reduce_b0`, `AUDIT3/A1_reduction/SizeArith.lean:31`,
Lean: `A1.mirror_reduce_b12`, `AUDIT3/A1_reduction/SizeArith.lean:34`). *All variables range over* $\mathbb{N}$.

(a) (`A1.mirror_reduce_b0`) *For every* $k \in \mathbb{N}$*: if* $33 \le 2k$*, then* $17 \le k$.
(b) (`A1.mirror_reduce_b12`) *For all* $k, b \in \mathbb{N}$*: if* $1 \le b$*,* $b \le 2$ *and* $33 \le 2k + b$*, then* $16 \le k$.

Proof. (a) As in Proposition 12.1(e), $2 \cdot 16 = 32 < 33 \le 2k$ gives $k \ge 17$. (b) From $33 \le 2k + b$ and $b \le 2$ we get $2k \ge 31$; since $2 \cdot 15 = 30 < 31$, $k \ge 16$. The hypothesis $1 \le b$ is not used; it records that the statement is meant for one or two points on the axis. Lean proves both parts with `omega`. □

Proposition 12.3 (orders divisible by 2, 3 or 5;
Lean: `A1.even_order_reduce`, `AUDIT3/A1_reduction/SizeArith.lean:37`,
Lean: `A1.mult3_order_reduce`, `AUDIT3/A1_reduction/SizeArith.lean:42`,
Lean: `A1.mult5_order_reduce`, `AUDIT3/A1_reduction/SizeArith.lean:47`). *All variables range over* $\mathbb{N}$.

(a) (`A1.even_order_reduce`) *For all* $t, k \in \mathbb{N}$*: if* $33 \le (2t)k$*, then* $17 \le tk$.
(b) (`A1.mult3_order_reduce`) *For all* $t, k, e \in \mathbb{N}$*: if* $e \le 1$ *and* $33 \le (3t)k + e$*, then* $11 \le tk$.
(c) (`A1.mult5_order_reduce`) *For all* $t, k, e \in \mathbb{N}$*: if* $e \le 1$ *and* $33 \le (5t)k + e$*, then* $35 \le 5(tk) + e$.

The Lean terms are `2 * t * k`*,* `3 * t * k + e` *and* `5 * t * k + e` *in the hypotheses,* `t * k` *in the conclusions of* (a) *and* (b)*, and* `5 * (t * k) + e` *in the conclusion of* (c).

Proof. By associativity, $(2t)k = 2(tk)$, $(3t)k = 3(tk)$ and $(5t)k = 5(tk)$. Put $m = tk$. Then (a) is Proposition 12.1(e) for m (its conclusion $17 \le m$), (b) is Proposition 12.1(a) for m, and (c) is the first conclusion of Proposition 12.1(b) for m. The Lean proofs are the same: each first proves the associativity identity with `Nat.mul_assoc` and then calls `omega`, which treats the product tk as a single unknown. □

Lemma 12.4 (the order split;
Lean: `A1.order_split`, `AUDIT3/A1_reduction/SizeArith.lean:52`). *For every* $s \in \mathbb{N}$ *with* $2 \le s$*:*
$s \bmod 2 = 0$*, or* $s \bmod 3 = 0$*, or* $s \bmod 5 = 0$*, or* $7 \le s$.

Here $s \bmod q$ is the remainder of s on division by q (Lean `s % q`); the Lean conclusion is the disjunction `s % 2 = 0 ∨ s % 3 = 0 ∨ s % 5 = 0 ∨ 7 ≤ s`.

Proof. Suppose that s is divisible by none of 2, 3, 5. Then $s \notin \{2, 4, 6\}$ (these are divisible by 2), $s \ne 3$ and $s \ne 5$. Since $s \ge 2$, this leaves $s \ge 7$. Lean argues in the same way: it splits on $s < 7$ or $s \ge 7$ (`Nat.lt_or_ge`); in the first case `omega` gives $s \in \{2, 3, 4, 5, 6\}$, and each of the five resulting closed instances of the statement is closed by `decide`, that is, by kernel evaluation of the decision procedure for the whole closed disjunction (for

$s = 5$, for example, this evaluates the remainders of 5 on division by 2, 3 and 5); in the second case the last alternative holds. The five closed instances are side facts inside the proof of a statement about all $s \geq 2$, which is not itself a finite computation, and the written proof establishes each of them by hand, through the disjuncts $2 \mid 2$, $3 \mid 3$, $2 \mid 4$, $5 \mid 5$ and $2 \mid 6$ used above; hence the lemma is Lean-checked and is not computer-assisted (see the paragraph "Rigor" at the beginning of this section). □

12.2. **Upper bounds of the size table.** The second part of `SizeArith.lean` ("Upper bounds of the table, given the case theorems", lines 58–80) turns bounds on the number of orbits or pairs into bounds on the size. The case theorems that give such bounds for sets without 7 points in convex position are not part of the statements: each part of Proposition 12.5 that rests on a case theorem assumes the bound that the theorem gives as a hypothesis, on k in parts (a), (e) and (h), on $2k$ in parts (f) and (g), on $5k + e$ in part (c), and on k, separately for $b = 0$ and for $1 \leq b$, in part (i) (Remark 12.13, item (5)). Parts (b) and (d), `A1.c3_no32` and `A1.c5_gap`, rest on no case theorem: their only hypothesis is $e \leq 1$.

Proposition 12.5 (upper bounds;
Lean: `A1.table_c3`, `AUDIT3/A1_reduction/SizeArith.lean:61`,
Lean: `A1.c3_no32`, `AUDIT3/A1_reduction/SizeArith.lean:62`,
Lean: `A1.table_c5`, `AUDIT3/A1_reduction/SizeArith.lean:65`,
Lean: `A1.c5_gap`, `AUDIT3/A1_reduction/SizeArith.lean:66`,
Lean: `A1.table_c4`, `AUDIT3/A1_reduction/SizeArith.lean:69`,
Lean: `A1.table_c4_only_c2`, `AUDIT3/A1_reduction/SizeArith.lean:70`,
Lean: `A1.table_c6`, `AUDIT3/A1_reduction/SizeArith.lean:73`,
Lean: `A1.table_c2`, `AUDIT3/A1_reduction/SizeArith.lean:76`,
Lean: `A1.table_mirror`, `AUDIT3/A1_reduction/SizeArith.lean:79`). *All variables range over* $\mathbb{N}$.

(a) (`A1.table_c3`) *For all* $k, e \in \mathbb{N}$*: if* $e \leq 1$ *and* $k \leq 10$*, then* $3k + e \leq 31$.
(b) (`A1.c3_no32`) *For all* $k, e \in \mathbb{N}$*: if* $e \leq 1$*, then* $3k + e \neq 32$.
(c) (`A1.table_c5`) *For all* $k, e \in \mathbb{N}$*: if* $e \leq 1$ *and* $5k + e < 35$*, then* $5k + e \leq 31$.
(d) (`A1.c5_gap`) *For all* $k, e \in \mathbb{N}$*: if* $e \leq 1$*, then* $5k + e \neq 32$*,* $5k + e \neq 33$ *and* $5k + e \neq 34$.
(e) (`A1.table_c4`) *For every* $k \in \mathbb{N}$*: if* $k \leq 7$*, then* $4k \leq 28$.
(f) (`A1.table_c4_only_c2`) *For every* $k \in \mathbb{N}$*: if* $2k \leq 16$*, then* $4k \leq 32$.
(g) (`A1.table_c6`) *For every* $k \in \mathbb{N}$*: if* $2k \leq 10$*, then* $6k \leq 30$.
(h) (`A1.table_c2`) *For every* $k \in \mathbb{N}$*: if* $k \leq 16$*, then* $2k \leq 32$.
(i) (`A1.table_mirror`) *For all* $k, b \in \mathbb{N}$*: if* $b \leq 2$*,* $(b = 0 \Rightarrow k \leq 16)$ *and* $(1 \leq b \Rightarrow k \leq 15)$*, then* $2k + b \leq 32$.

Proof. (a) $3k + e \leq 3 \cdot 10 + 1 = 31$. (b) If $3k + e = 32$, then $3k = 32 - e \in \{31, 32\}$; but $31 = 3 \cdot 10 + 1$ and $32 = 3 \cdot 10 + 2$ are not multiples of 3. (c) From $5k \leq 5k + e < 35$ we get $k \leq 6$, so $5k + e \leq 30 + 1 = 31$. (d) If $5k + e = j$ with $j \in \{32, 33, 34\}$, then $5k = j - e$ lies between 31 and 34, strictly between the consecutive multiples 30 and 35 of 5; this is impossible. (e) $4k \leq 4 \cdot 7 = 28$. (f) $4k = 2 \cdot (2k) \leq 2 \cdot 16 = 32$. (g) $6k = 3 \cdot (2k) \leq 3 \cdot 10 = 30$. (h) $2k \leq 2 \cdot 16 = 32$. (i) If $b = 0$, then $k \leq 16$ and $2k + b = 2k \leq 32$. Otherwise $1 \leq b$, so $k \leq 15$, and with $b \leq 2$ we get $2k + b \leq 30 + 2 = 32$. Lean proves each part with `omega`. □

12.3. **Size bookkeeping for the assembled statement (audit finding A1-F1).** The file `AssembledSizes.lean` restates six of the statements above, or parts of them, with the size n as an explicit variable, adds `big_order_orbit` and a negative control. Its declarations are in the root namespace, and the file does not import `SizeArith.lean`.

Proposition 12.6 (assembled sizes;
Lean: `halfturn_pairs`, `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:6`,
Lean: `threefold_orbits`, `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:9`,
Lean: `fivefold_size`, `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:12`,
Lean: `mirror_b0`, `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:15`,
Lean: `mirror_b12`, `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:18`,
Lean: `fourfold_via_halfturn`, `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:21`). *All variables range over* $\mathbb{N}$.

(a) (`halfturn_pairs`) *For all* $n, k \in \mathbb{N}$*: if* $n = 2k$ *and* $33 \leq n$*, then* $17 \leq k$ *and* $34 \leq n$.
(b) (`threefold_orbits`) *For all* $n, k, e \in \mathbb{N}$*: if* $e \leq 1$*,* $n = 3k + e$ *and* $33 \leq n$*, then* $11 \leq k$.
(c) (`fivefold_size`) *For all* $n, k, e \in \mathbb{N}$*: if* $e \leq 1$*,* $n = 5k + e$ *and* $33 \leq n$*, then* $35 \leq n$.
(d) (`mirror_b0`) *For all* $n, k \in \mathbb{N}$*: if* $n = 2k$ *and* $33 \leq n$*, then* $17 \leq k$.

(e) (`mirror_b12`) *For all* $n, k, b \in \mathbb{N}$*: if* $1 \le b$*,* $b \le 2$*,* $n = 2k + b$ *and* $33 \le n$*, then* $16 \le k$*.*
(f) (`fourfold_via_halfturn`) *For all* $n, k \in \mathbb{N}$*: if* $n = 4k$ *and* $33 \le n$*, then* $17 \le 2k$*.*

Proof. Substitute the given value of n into $33 \le n$. (a) Proposition 12.1(e) gives $17 \le k$ and $34 \le 2k = n$. (b) Proposition 12.1(a). (c) Proposition 12.1(b) gives $35 \le 5k+e = n$. (d) Proposition 12.2(a). (e) Proposition 12.2(b); as there, $1 \le b$ is not used. (f) Proposition 12.1(c) gives $17 \le 2k$. Lean proves each part directly with `omega`. □

Lemma 12.7 (an orbit for an order of at least 7;
Lean: `big_order_orbit`, `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:24`)**.** *For all* $n, k, e, p \in \mathbb{N}$*: if* $7 \le p$*,* $e \le 1$*,* $n = pk + e$ *and* $33 \le n$*, then* $1 \le k$*.*

Proof. If $k = 0$, then $n = p \cdot 0 + e = e \le 1 < 33$, which contradicts $33 \le n$. Hence $k \ge 1$. We follow the Lean proof, which splits on k (the product pk is not linear in the unknowns): for $k = 0$, `simp` turns the hypothesis $n = p \cdot 0 + e$ into $n = e$, and `omega` derives a contradiction from $n = e$, $e \le 1$ and $33 \le n$; for $k = k' + 1$ the goal $1 \le k' + 1$ is closed by `omega`. □

Remark 12.8 (the hypothesis $7 \le p$)*.* The proof does not use the hypothesis $7 \le p$ (Lean binder `hp`): the conclusion holds for every $p \in \mathbb{N}$, given $e \le 1$, $n = pk + e$ and $33 \le n$. The compile log records this as the warning "unused variable `hp`" at line 24, column 41. The name of the lemma and the comment above it ("prime order $p \ge 7$") describe the intended use (Remark 12.13, item (1)); the statement does not require p to be prime.

Remark 12.9 (the negative control, `AssembledSizes.lean:30`)*.* Line 30 of `AssembledSizes.lean` is an anonymous `example`, not a named declaration. It states

$$\neg\big(\forall n, k \in \mathbb{N} :\ n = 2k \ \Rightarrow\ (32 \le n \ \Rightarrow\ 17 \le k)\big),$$

that is, Proposition 12.6(a) fails when 33 is replaced by 32. *Proof.* The numbers $n = 32$ and $k = 16$ satisfy $32 = 2 \cdot 16$ and $32 \le 32$, but not $17 \le 16$. The Lean proof is the same: `intro h; have := h 32 16 rfl (by decide); omega`, where `rfl` proves $32 = 2 \cdot 16$ by evaluation, `decide` proves $32 \le 32$, and `omega` refutes $17 \le 16$. As the example has no name, no `#print axioms` line refers to it; the compile log contains no error, so Lean accepted it. The comment above the example (line 29) reads "control: the half-turn line is false below 33 (32 points = 16 pairs; the central 32-point example exists)". The A1 report describes that example, `SYM2/sym7_32points.txt`, as half-turn symmetric about $(0, 0)$, not containing the centre, with no collinear triple, and with largest subset in convex position of 6 points (`AUDIT3/reports/A1_reduction.md`, lines 93 and 107); the set is not examined in this section.

12.4. **Baek's theorem at 7-gons.** We recall the notions of Theorem 10.4. They are defined in Lean in `ErdosSzekeres/Vendor/ETV/Config/Defs.lean`, lines 20–67, with the list predicates `List.In` (line 22) and `List.Chain3'` (line 46, through the inductive predicate `List.Chain3` of line 42) of `ErdosSzekeres/Vendor/ETV/Lib/List/Defs.lean` and the predicate `List.IsChain`, which Mathlib takes from its dependency Batteries. Let $(\gamma, <)$ be a linearly ordered set. A *configuration* C on γ is a ternary relation $\mathrm{Cup3}_C$ on γ (in Lean a structure `Config` γ with this relation as a field and a decidability witness, which plays no mathematical role); $\mathrm{Cap3}_C(a, b, c)$ means $\neg\,\mathrm{Cup3}_C(a, b, c)$. A list $l = (l_0, \dots, l_{r-1})$ of elements of γ is a *C-cap* if $l_0 < l_1 < \cdots < l_{r-1}$ and $\mathrm{Cap3}_C(l_i, l_{i+1}, l_{i+2})$ for every i with $i + 2 < r$, and a *C-cup* if the same holds with $\mathrm{Cup3}_C$ in place of $\mathrm{Cap3}_C$. For a finite set $S \subseteq \gamma$, $\mathsf{HasNCap}_C(4, S)$ means that some C-cap of length 4 has all its entries in S, and $\mathsf{HasNGon}_C(7, S)$ means that there are lists l and l' with all entries in S such that $|l| \ge 2$, l is a C-cap, $|l'| \ge 2$, l' is a C-cup, l and l' have the same first entry and the same last entry, and $|l| + |l'| = 7 + 2$. The relation $\mathrm{Cup3}_C$ is arbitrary, so these notions carry no geometry.

Theorem 12.10 (Baek's theorem at 7-gons;
Lean: `baek_seven`, `ES7SPLIT/review/equivalence/BaekSeven.lean:6`)**.** *Let* γ *be a type (Lean* γ *:* `Type`*, the lowest universe) with a linear order* $<$*, let* C *be a configuration on* γ *(*`C : Config` γ*), and let* S *be a finite subset of* γ *(*`S : Finset` γ*) with* $17 \le |S|$*. Then* $\mathsf{HasNCap}_C(4, S)$ *or* $\mathsf{HasNGon}_C(7, S)$*.*

The theorem is Lean-checked (`BaekSeven.out`, line 3). Its written proof rests on the cited Theorem 10.4.

Proof. We have $\binom{4+2}{2} + 2 = \binom{6}{2} + 2 = \frac{6 \cdot 5}{2} + 2 = 15 + 2 = 17 \le |S|$. Theorem 10.4 with $n = 4$ and the same γ, C and S gives $\mathsf{HasNCap}_C(4, S)$ or $\mathsf{HasNGon}_C(4 + 3, S)$, and $4 + 3 = 7$. The Lean proof is the same. Its first line, `have e : Nat.choose (4 + 2) 2 + 2 = 17 := by decide`, evaluates the binomial coefficient in the kernel (the value $\binom{6}{2} = 15$ computed above by hand). Its second line, `exact main 4 C S (e ▶ h)`, applies `main` with $n = 4$: the term `e ▶ h` (Lean's rewriting operator) turns the hypothesis `h`, which says $17 \le |S|$, into

the hypothesis $\binom{4+2}{2}+2 \le |S|$ of `main` at $n=4$, and Lean accepts the conclusion `C.HasNGon (4 + 3) S` as `C.HasNGon 7 S` because $4+3$ evaluates to 7. □

Remark 12.11 (comparison with Theorem 10.4; scope). (1) Lines 1–2 of `BaekSeven.out` are the output of `#check @main`. They say that `main` takes, in this order, an implicit type γ in an arbitrary universe (`Type u_1`), an instance `LinearOrder` γ, $n \in \mathbb{N}$, `C : Config` γ and `S : Finset` γ, and that it derives from the hypothesis `(n + 2).choose 2 + 2 ≤ S.card`, that is $\binom{n+2}{2}+2 \le |S|$, the conclusion `C.HasNCap 4 S ∨ C.HasNGon (n + 3) S`. This is the statement of Theorem 10.4, with the same notions; it is also the declaration at `ErdosSzekeres/Vendor/ETV/Main/Main.lean:87` together with the variables declared at line 26 of that file. The cap length 4 is fixed; only the gon size $n+3$ and the threshold depend on n.

(2) Theorem 12.10 is the case $n=4$. The only other difference is the universe: `baek_seven` is stated for γ in the lowest universe `Type`, `main` for types in every universe. Read mathematically, both statements are about an arbitrary linearly ordered set.

(3) The theorem concerns abstract configurations; it mentions no points of the plane, and `BaekSeven.lean` contains no bridge from point sets to configurations. The records of `ES7SPLIT` say: "The step from the abstract form to real point sets is not Lean-checked for $n=7$; the repository's bridge covers the octagon case only. Here that step rests on the published paper." (`ES7SPLIT/README.md`, lines 78–79, in the rigor paragraph of its section 2; the status paragraph of `ES7SPLIT/proofs/split_theorem.md`, lines 13–15, says the same.) These records were committed in `9793c06e0` (2 November 2026, 01:03 +0900) and predate `MirrorOdd.capacity16` (`MIRRORETV/lean/MirrorMain.lean:1529`, added in `83c80f38f` at 03:41 the same day), a geometric form of the same instance of Theorem 10.4: a configuration of n points in the sense of Definition 9.3 with no 4-cap and no 7 points in convex position has $n \le 16$. That theorem is Lean-checked: the Lean proof of Theorem 10.31 uses it, and the axioms printed for that theorem cover it. Whether it covers the use of Baek's theorem in the z-split arguments of `ES7SPLIT` is not examined here; the theorems of `ES7SPLIT` remain written proofs. The source of Baek's theorem is J. Baek, *On the Erdős–Tuza–Valtr conjecture*, Eur. J. Combin. 124 (2025) 104085; arXiv:2206.04260.

12.5. **How the arithmetic is used.** The following lemma is the modelling of the header comment of `SizeArith.lean`, proved here as a written step. For $c \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$, $\rho_{c,\theta}$ is the rotation of Definition 1.12; in complex notation it is $z \mapsto c + e^{i\theta}(z-c)$. For a map f we write $f^{[i]}$ for its i-th iterate. The names `HalfTurn34`, `Threefold33`, `Fivefold35`, `MirrorOneAxis33`, `MirrorNoAxis34` and `Fourfold32` denote the case classes of Definition 7.7, in the namespace `ErdosSzekeres.SymReduction`.

Lemma 12.12 (orbit counts; written step)**.** *Let $S \subseteq \mathbb{R}^2$ be a finite set and $n = |S|$.*

(a) Rotations. *Let $c \in \mathbb{R}^2$, let $s \in \mathbb{N}$ with $0 < s$, let $\rho = \rho_{c,2\pi/s}$, and assume* Invariant $\rho\, S$ *(Definition 1.10). For $y \in \mathbb{R}^2$ put $O(y) = \{\rho^{[i]}(y) : 0 \le i < s\}$. For every $y \in S \setminus \{c\}$ the set $O(y)$ is contained in $S \setminus \{c\}$, has exactly s elements and is mapped into itself by ρ; any two of these sets are equal or disjoint; and $n = s \cdot k + e$, where k is the number of distinct sets $O(y)$ with $y \in S \setminus \{c\}$, and $e = 1$ if $c \in S$, $e = 0$ otherwise. If moreover s is even, S is in general position and $S \not\subseteq \{c\}$, then $e = 0$.*

(b) Reflections. *Let $a, u \in \mathbb{R}^2$ with $u_0^2 + u_1^2 = 1$, let $\sigma = \mathrm{reflLine}_{a,u}$, that is*

$$\sigma(p) = \big(a_0 + 2du_0 - (p_0 - a_0),\ a_1 + 2du_1 - (p_1 - a_1)\big), \qquad d = (p_0 - a_0)u_0 + (p_1 - a_1)u_1,$$

the map of the classes `MirrorOneAxis33` *and* `MirrorNoAxis34` *of Definition 7.7, and let $L = \{a + tu : t \in \mathbb{R}\}$. Assume* Invariant $\sigma\, S$ *and that S is in general position. Then $S \setminus L$ is the disjoint union of k two-element sets of the form $\{p, \sigma(p)\}$, each mapped into itself by σ, and $n = 2k + b$ with $b = |S \cap L| \le 2$.*

(c) Deleting. *Let $T \subseteq S$. If S is in general position, so is T; and* HasConvexNGon $7\, T$ *implies* HasConvexNGon $7\, S$. *In* (a)*, a union of sets $O(y)$, together with c or not when $c \in S$, is mapped into itself by ρ; in* (b)*, a union of sets $\{p, \sigma(p)\}$ and of points of $S \cap L$ is mapped into itself by σ.*

Proof. (a) Put $\omega = e^{2\pi i/s}$, so that $\rho(z) = c + \omega(z-c)$. Then $\rho(c) = c$, and ρ is injective because $\omega \ne 0$; so every iterate $\rho^{[i]}$ is injective. Hence ρ maps $S \setminus \{c\}$ into itself: for $y \in S \setminus \{c\}$ we have $\rho(y) \in S$, and $\rho(y) = c = \rho(c)$ would give $y = c$. By induction on i, $\rho^{[i]}(y) \in S \setminus \{c\}$ for all i, so $O(y) \subseteq S \setminus \{c\}$. Let $y \in S \setminus \{c\}$. By Lemma 2.16 (with $m = s$), the minimal period of y under ρ is s: $\rho^{[s]}(y) = y$ and $\rho^{[i]}(y) \ne y$ for $0 < i < s$. The points $\rho^{[i]}(y)$, $0 \le i < s$, are pairwise distinct: if $\rho^{[i]}(y) = \rho^{[j]}(y)$ with $0 \le i < j < s$, then $\rho^{[i]}\big(\rho^{[j-i]}(y)\big) = \rho^{[i]}(y)$, and injectivity of $\rho^{[i]}$ gives $\rho^{[j-i]}(y) = y$ with $0 < j - i < s$, a contradiction. So $|O(y)| = s$. Further, $\rho(\rho^{[i]}(y)) = \rho^{[i+1]}(y) \in O(y)$ for $i + 1 < s$, and $\rho(\rho^{[s-1]}(y)) = \rho^{[s]}(y) = y \in O(y)$; so ρ maps $O(y)$ into itself. Consequently, for $x \in O(y)$ all points $\rho^{[i]}(x)$ lie in $O(y)$, so $O(x) \subseteq O(y)$, and $O(x) = O(y)$

because both sets have s elements. If $z \in O(y) \cap O(y')$, then $O(y) = O(z) = O(y')$. Every $y \in S \setminus \{c\}$ lies in $O(y)$ (take $i = 0$). Therefore the distinct sets $O(y)$ form a partition of $S \setminus \{c\}$ into k blocks of size s, and $n = |S \setminus \{c\}| + e = s \cdot k + e$. If s is even, S is in general position and some $p \in S$ differs from c, then Lemma 3.18 (with $m = s$) gives $c \notin S$, that is, $e = 0$.

(b) Write $\langle x, y\rangle = x_0y_0 + x_1y_1$, so that $d = \langle p - a, u\rangle$ and $\sigma(p) - a = 2du - (p - a)$. *$\sigma$ is an involution:* for $p' = \sigma(p)$ we get $\langle p' - a, u\rangle = 2d\langle u, u\rangle - d = d$, hence $\sigma(p') - a = 2du - (p' - a) = 2du - 2du + (p - a) = p - a$, that is, $\sigma(\sigma(p)) = p$; in particular σ is injective. *The fixed points of σ are the points of L:* $\sigma(p) = p$ means $2du = 2(p - a)$, so $p = a + du \in L$; conversely, for $p = a + tu$ we have $d = t\langle u, u\rangle = t$ and $\sigma(p) = a + 2tu - tu = p$. (This is also shown at the end of Definition 7.7.) Now let $p \in S \setminus L$. Then $\sigma(p) \in S$ and $\sigma(p) \neq p$; and $\sigma(p) \notin L$, since otherwise $\sigma(\sigma(p)) = \sigma(p)$, that is, $p = \sigma(p)$. So $\{p, \sigma(p)\}$ is a two-element subset of $S \setminus L$, mapped into itself by σ because $\sigma(\sigma(p)) = p$. If two such sets $\{p, \sigma(p)\}$ and $\{q, \sigma(q)\}$ meet, then q or $\sigma(q)$ lies in $\{p, \sigma(p)\}$, and applying σ shows that both q and $\sigma(q)$ do; so the two sets are equal. Hence the distinct sets $\{p, \sigma(p)\}$ partition $S \setminus L$, $|S \setminus L| = 2k$, and $n = 2k + b$ with $b = |S \cap L|$. Finally $b \leq 2$: if $x, y, z \in S \cap L$ were pairwise distinct, all differences of two of them would be real multiples of u, so the linear span of these differences would have dimension at most 1 and $\{x, y, z\}$ would be collinear, contrary to general position (Definition 1.5). This is the fact that at most two points of a set in general position lie on a line.

(c) The first two claims are Lemma 1.23(b) and (c). A union of sets each of which is mapped into itself by a map is mapped into itself by that map; $\{c\}$ is mapped into itself by ρ since $\rho(c) = c$, and each point of $S \cap L$ is fixed by σ by (b). □

Remark 12.13 (How the arithmetic is used). This remark is a written step; it is not Lean-checked. It follows the reduction re-derived by the auditor in `AUDIT3/reports/A1_reduction.md` (Summary and trust-table rows 2–9) and shows where each statement of this section enters. Let S be a finite set in general position with $n = |S| \geq 33$.

(1) *Rotations.* Let $c \in \mathbb{R}^2$, $s \geq 2$, $\rho = \rho_{c,2\pi/s}$, and let S be mapped into itself by ρ. By Lemma 12.12(a), $n = sk + e$ with $e \leq 1$, and $e = 0$ for even s (as $n \geq 2$, $S \not\subseteq \{c\}$). By Lemma 12.12(c), a union T of some of the sets $O(y)$ is in general position, mapped into itself by ρ and does not contain c.

- $s = 2$: as $\cos\pi = -1$ and $\sin\pi = 0$, $\rho = \rho_{c,\pi}$ is the half-turn $p \mapsto 2c - p$ (Lean `halfTurn c`), and $n = 2k$. Proposition 12.1(e) (or 12.6(a)) gives $k \geq 17$ and $n \geq 34$. A union T of 17 pairs has 34 points: these are the hypotheses of `HalfTurn34` (Definition 7.7) for T and c.
- $s = 3$: Proposition 12.1(a) (or 12.6(b)) gives $k \geq 11$; 11 orbits give 33 points without c, invariant under $\rho_{c,2\pi/3}$: the hypotheses of `Threefold33`.
- $s = 5$: Proposition 12.1(b) gives $n \geq 35$ and $k \geq 7$ (Proposition 12.6(c) gives $n \geq 35$ only); 7 orbits give 35 points without c: the hypotheses of `Fivefold35`. (The comment of `A1.c5_reduce`, "T_{C5} applies directly", refers to the fivefold case theorem of the records, which admits at least 35 points and the centre; the Lean class asks for exactly 35 points without the centre, which $k \geq 7$ supplies.)
- $s = 4$: Proposition 12.1(c) gives $n = 4k \geq 36$, $k \geq 8$ and $2k \geq 17$ (Proposition 12.6(f) gives $2k \geq 17$ only). Rotations about c compose by adding angles (in complex notation, $c + e^{i\alpha}\big(c + e^{i\beta}(z - c) - c\big) = c + e^{i(\alpha+\beta)}(z - c)$), so $\rho^{[2]} = \rho_{c,\pi}$ is the half-turn; Lemma 12.12(a) for it gives $n = 2k'$, and $k' = 2k \geq 17$, so the case $s = 2$ applies. (Eight of the k orbits would give the 32 points of the class `Fourfold32`. Among the reduction theorems and case lemmas stated in this document, only Theorem 7.41 and the case lemma `SymReduction.case_fourfold` that its proof uses take this class as a hypothesis (so do some checks of that theorem in the review file `SYMLEAN/review/VariantChecks.lean`, among them its group form `SymReduction.VariantChecks.group_form`); a same-text copy, `LayerLemma.Fourfold32`, is the hypothesis of Theorem 6.7 and of its layer form `LayerLemma.fourfold_heptagon_of_erase`, which its proof uses; item (5) uses the class for the size table.)
- $s = 6$: Proposition 12.1(d) gives $2k \geq 11$ and $3k \geq 17$. Here $\rho^{[2]} = \rho_{c,2\pi/3}$ has $2k$ orbits of size 3 and $\rho^{[3]} = \rho_{c,\pi}$ has $3k$ pairs (Lemma 12.12(a) for these rotations, with $n = 6k$), so both the threefold and the half-turn case apply.
- General s: by Lemma 12.4, $2 \mid s$, $3 \mid s$, $5 \mid s$, or $s \geq 7$. If $s = mt$ with $m \in \{2, 3, 5\}$, then $\rho^{[t]} = \rho_{c,2\pi t/s} = \rho_{c,2\pi/m}$ maps S into itself, and Lemma 12.12(a) for it gives $n = mk' + e$ with the same e; from $mk' = n - e = mtk$ we get $k' = tk$. Proposition 12.3 is the arithmetic of this step: $tk \geq 17$ pairs for even s (where $e = 0$), $tk \geq 11$ orbits of size 3 if $3 \mid s$, and $35 \leq 5(tk) + e$ if $5 \mid s$, hence $5tk \geq 34$ and $tk \geq 7$. The cases $s = 2, 3, 5$ then apply to $\rho^{[t]}$. If $s \geq 7$, Lemma 12.7 (with $p = s$) gives $k \geq 1$, so S contains a set $O(y)$ with $s \geq 7$ points. They all have the same distance from c as y, because

$|\omega(z-c)| = |z-c|$, so $O(y)$ is in convex position by Lemma 7.23; any 7 of its points are in convex position (Lemma 1.23(a)), so HasConvexNGon 7 S (Definition 1.7). No case class is needed here.

(2) *Reflections.* Let S be mapped into itself by $\sigma = \mathrm{reflLine}_{a,u}$ with $u_0^2 + u_1^2 = 1$. By Lemma 12.12(b), $n = 2k + b$ with $b \le 2$. If $b = 0$, Proposition 12.2(a) (or 12.6(d)) gives $k \ge 17$, and 17 pairs form a 34-point set that is in general position and mapped into itself by σ (Lemma 12.12(c)) and has no fixed point: the hypotheses of `MirrorNoAxis34`. If $b \in \{1, 2\}$, Proposition 12.2(b) (or 12.6(e)) gives $k \ge 16$, and 16 pairs together with one point $x \in S \cap L$ form a 33-point set that is in general position and mapped into itself by σ (Lemma 12.12(c)) and whose only fixed point is x: the hypotheses of `MirrorOneAxis33`. (The fixed points of σ are the points of L, as shown in the proof of Lemma 12.12(b), and the pairs lie in $S \setminus L$.)

(3) *All symmetries.* By Theorem 7.21(b), a Euclidean isometry $g \ne \mathrm{id}$ that maps S into itself is either a rotation, in complex notation $z \mapsto c + a(z-c)$ with $a \ne 1$ and $a^m = 1$ for some $m > 0$, or a map $\mathrm{reflLine}_{A,u}$ with $u_0^2 + u_1^2 = 1$. In the first case let $s \ge 2$ be the order of a. Then a is a primitive s-th root of unity, so $e^{2\pi i/s} = a^j$ for some $j \in \mathbb{N}$, because a primitive s-th root of unity generates all s-th roots of unity. As $g^{[j]}(z) = c + a^j(z-c)$ (induction on j), $\rho_{c,2\pi/s} = g^{[j]}$, which maps S into itself (here c is read as a point); so item (1) applies. In the second case item (2) applies. Affine symmetries are reduced to Euclidean isometries by Theorem 8.25, which gives Theorem 8.28.

(4) *Status.* Items (1)–(3) derive HasConvexNGon 7 S from the five case classes `HalfTurn34`, `Threefold33`, `Fivefold35`, `MirrorOneAxis33` and `MirrorNoAxis34`, used as hypotheses. Except for the orders $s \ge 7$ in item (1), which need no class, items (1) and (2) exhibit a subset T that satisfies the hypotheses of one class, and the class then gives HasConvexNGon 7 T, hence HasConvexNGon 7 S by Lemma 12.12(c); item (3) reduces every Euclidean isometry other than the identity to items (1) and (2), and affine symmetries to Euclidean isometries. The Lean-checked form of this implication is Theorem 7.39 for Euclidean isometries, with Theorem 8.28 for affine symmetries. The Lean proof of Theorem 7.39 passes to a power of the rotation that is the half-turn or has odd prime order, instead of using Lemma 12.4, and it does its own arithmetic. The two mirror classes are Lean-checked theorems (Theorem 10.39). The classes `HalfTurn34`, `Threefold33` and `Fivefold35` are not proved in Lean or in this document (Theorem 10.40 is conditional on them); the records that supply them are described in the remark that follows Definition 7.7. Of the three ingredients, the modelling (Lemma 12.12), the case classes and the arithmetic, the files of this section check only the arithmetic.

(5) *The size table.* Proposition 12.5 is the arithmetic of the upper bounds in the size table of `WRITEUP/SYMMETRIC.md`, §2 (trust-table row 9 of the A1 report). For a set without 7 points in convex position, the case classes bound k through item (1) or (2), and the parts of Proposition 12.5 turn these bounds into bounds on n. Here S need not have 33 points: if the number of orbits or pairs to which a class is applied below (k, or $2k$ for $\rho^{[2]}$ when $s = 4$ or $s = 6$) reached the number that the class requires, the subset T built in item (1) or (2) would satisfy the hypotheses of the class, which would then give 7 points of S in convex position; for even s, $e = 0$ as soon as $k \ge 1$ (Lemma 12.12(a)), and $k = 0$ gives $n = e \le 1$. For threefold sets, `Threefold33` gives $k \le 10$, so $n \le 31$ by part (a), and $n \ne 32$ in any case by part (b). For fivefold sets, `Fivefold35` gives $k \le 6$, so $n = 5k + e < 35$ and $n \le 31$ by part (c); 32, 33 and 34 are not fivefold sizes by part (d). For fourfold sets, `Fourfold32` gives $k \le 7$, so $n \le 28$ by part (e), while `HalfTurn34` alone gives only $2k \le 16$ and $n \le 32$ by part (f). For sixfold sets, `Threefold33` applied to the $2k$ orbits of size 3 gives $2k \le 10$, so $n \le 30$ by part (g). For half-turns, `HalfTurn34` gives $k \le 16$, so $n \le 32$ by part (h). For reflections, the two mirror classes give $k \le 16$ if $b = 0$ and $k \le 15$ if $b \ge 1$, so $n \le 32$ by part (i). Except for the mirror row, these bounds are conditional on the classes named, which are hypotheses here: the rows for half-turns, threefold and sixfold sets and the bound 32 of part (f) on `HalfTurn34` or `Threefold33`, the fourfold bound 28 on `Fourfold32`, and the fivefold bound 31 on `Fivefold35`. For the last two, Lean-checked conditional forms exist that include the geometric modelling: Theorem 6.7, with the same-text copy `LayerLemma.Fourfold32` as hypothesis, gives HasConvexNGon 7 S for every set S in general position with at least 29 points that is mapped into itself by $\rho_{c,2\pi/4}$, so such a set without 7 points in convex position has at most 28 points; Theorem 6.4, with the same-text copy `LayerLemma.Fivefold35`, does the same with $\rho_{c,2\pi/5}$, at least 32 points and the bound 31. For reflections the bound $n \le 32$ is Lean-checked with no class hypothesis (Theorem 10.38(b) for $\mathrm{reflLine}_{a,u}$, derived from Theorem 10.31(b)), and the two mirror classes are themselves Lean-checked theorems (Theorem 10.39). The matching lower bounds rest on example sets of the records. None of this is part of the files of this section.

Coverage. Covered, by label. In `AUDIT3/A1_reduction/SizeArith.lean`: `A1.c3_reduce`, `A1.c5_reduce`, `A1.c4_reduce`, `A1.c6_reduce`, `A1.c2_reduce` (Proposition 12.1); `A1.mirror_reduce_b0`, `A1.mirror_reduce_b12` (Proposition 12.2); `A1.even_order_reduce`, `A1.mult3_order_reduce`, `A1.mult5_order_reduce`

(Proposition 12.3); `A1.order_split` (Lemma 12.4); `A1.table_c3`, `A1.c3_no32`, `A1.table_c5`, `A1.c5_gap`, `A1.table_c4`, `A1.table_c4_only_c2`, `A1.table_c6`, `A1.table_c2`, `A1.table_mirror` (Proposition 12.5). In `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean`: `halfturn_pairs`, `threefold_orbits`, `fivefold_size`, `mirror_b0`, `mirror_b12`, `fourfold_via_halfturn` (Proposition 12.6); `big_order_orbit` (Lemma 12.7, with Remark 12.8); the anonymous example of line 30 (Remark 12.9). In `ES7SPLIT/review/equivalence/BaekSeven.lean`: `baek_seven` (Theorem 12.10); the vendored theorem `main` that it instantiates is the cited Theorem 10.4. That is all 28 theorems of the three files and the one example.
Not covered, deliberately: the `#print axioms` commands (`SizeArith.lean`, lines 84–103; `AssembledSizes.lean`, lines 33–39; `BaekSeven.lean`, line 12) and the `#check` command (`BaekSeven.lean`, line 11), which produce the logs named in the Sources paragraph and are not results; and the comments of the files, among them the modelling in the header of `SizeArith.lean`, which these files do not formalise and which is treated here as the written step Lemma 12.12. Lemma 12.12 and Remark 12.13 are not declarations of the three files.
Out of scope: the scripts and other logs of `AUDIT3/A1_reduction` and its subdirectories, and the other files of `ES7SPLIT/review/equivalence`; none of them contains a Lean declaration.

Appendix A. Statement map

Each row lists a numbered statement of this document, the Lean declarations that it restates, each with its file and line at commit `7e32ee259`, and a note. For a remark, the Lean column names the declarations that the remark discusses; a remark does not restate a Lean statement. A path that begins with `ErdosSzekeres/`, `docs/`, `experiments/` or `scripts/` is relative to the repository root; every other path is relative to `cloud/es7-2026-10-19/results`. A statement introduced only for the written proofs is marked "written step", one that restates Mathlib definitions or facts is marked "Mathlib", and a remark without a Lean declaration has a dash. Labelled remarks appear where a section lists them. The notes are those of the section maps.

Statement	Lean declaration; file and line	Note
Section 1		
Def. 1.1	`ErdosSzekeres.Point` `ErdosSzekeres/Definitions.lean:9`	Finset vs Set conventions
Rem. 1.2	–	sup norm of the Lean point type; the only topological argument is the compactness step in Lemma 1.40 (Heine-Borel, extreme value theorem; Rudin cited)
Def. 1.3	`ErdosSzekeres.orient` `ErdosSzekeres/Orientation.lean:9`	any commutative ring
Def. 1.4	Mathlib	Mathlib definitions restated: Convex, `convexHull`, `AffineIndependent`, `AffineMap` (map with $f(w+p) =$ `f.linear` $w + fp$), `AffineEquiv`; no repository declaration
Def. 1.5	`ErdosSzekeres.GeneralPosition` `ErdosSzekeres/Definitions.lean:12`	uses Mathlib Collinear (rank of `vectorSpan` at most 1)
Def. 1.6	`ErdosSzekeres.ConvexPosition` `ErdosSzekeres/Definitions.lean:17`	Mathlib `ConvexIndependent` of the inclusion
Def. 1.7	`ErdosSzekeres.HasConvexNGon` `ErdosSzekeres/Definitions.lean:21`	also defines K-free; no emptiness condition
Rem. 1.8	–	`HasConvexNGon` 0 S always holds (no set is 0-free); K-free is upward closed; `HasConvexNGon` n S implies $n \le \lvert S\rvert$
Def. 1.9	`ErdosSzekeres.ForcesConvexNGon` `ErdosSzekeres/Definitions.lean:25`	
Def. 1.10	`ErdosSzekeres.LayerLemma.Invariant` `LAYERLEAN/LayerLemma.lean:105` `ErdosSzekeres.SymReduction.Invariant` `SYMLEAN/SymmetricReduction.lean:271` `ErdosSzekeres.OrbitConsistency.Invariant` `LEANEXT/orbit/OrbitConsistency.lean:47`	maps into; textually identical copies
Lem. 1.11	`ErdosSzekeres.LayerLemma.iterate_mem` `LAYERLEAN/LayerLemma.lean:52`	part (c) is `iterate_mem`, stated as in Lean for an arbitrary set X (Lean: arbitrary type); (a),(b) written step for maps of the plane (the argument of (a) is the step at line 165 of the proof of `LayerLemma.map_mem_hullVerts`, `LAYERLEAN/LayerLemma.lean:162`)

Statement	Lean declaration; file and line	Note
Def. 1.12	`ErdosSzekeres.LayerLemma.rotAbout` `LAYERLEAN/LayerLemma.lean:298` `ErdosSzekeres.SymReduction.rotAbout` `SYMLEAN/SymmetricReduction.lean:400` `ErdosSzekeres.OrbitConsistency.rotAbout` `LEANEXT/orbit/OrbitConsistency.lean:93`	textually identical copies; counterclockwise
Rem. 1.13	–	angle $2\pi/m$ with m natural (cast explicit; $m = 0$ gives the identity); Sixfold66 and Sevenfold70 use real numerals 6 and 7
Lem. 1.14	`ErdosSzekeres.LayerLemma.rotAbout_eq_crot` `LAYERLEAN/LayerLemma.lean:302` `ErdosSzekeres.LayerLemma.rotAff` `LAYERLEAN/LayerLemma.lean:311` `ErdosSzekeres.LayerLemma.rotAbout_iterate` `LAYERLEAN/LayerLemma.lean:465` `ErdosSzekeres.LayerLemma.rotAbout_injective` `LAYERLEAN/LayerLemma.lean:319`	(a) `rotAbout_eq_crot`; (b) `rotAff` (affine packaging: linear part $R_\theta, \rho(p+v) = \rho(p) + R_\theta v$); (c) `rotAbout_iterate`; (d) `rotAbout_injective` (injectivity only); written steps: the matrix form $\rho(p) = c + R_\theta\ (p - c) = R_\theta p + (c - R_\theta c)$ of Definition 1.12 and $\rho(c) = c$ in (b) (the latter proved in Lean inside `invariant_erase_centre`, `LAYERLEAN/LayerLemma.lean:325`), composition law and $\rho_0 = \mathrm{id}$ in (c), bijectivity and inverse in (d), (e), (f) (counterparts named in Coverage)
Lem. 1.15	`ErdosSzekeres.orient_swap` `ErdosSzekeres/Orientation.lean:12` `ErdosSzekeres.orient_cycle` `ErdosSzekeres/Orientation.lean:15` `ErdosSzekeres.orient_translate` `ErdosSzekeres/Orientation.lean:18`	(a) `orient_swap`; (b) `orient_cycle`; (c) `orient_translate`; (d),(e) written step (real case of (d) in Remark 1.16); any commutative ring
Rem. 1.16	`ErdosSzekeres.OrbitConsistency.orient3_perm` `LEANEXT/orbit/OrbitConsistency.lean:178` `ErdosSzekeres.OrbitConsistency.orient3` `LEANEXT/orbit/OrbitConsistency.lean:168`	`orient3_perm` is the case $R =$ real numbers of Lemma 1.15(d) (Lemma 4.14)
Lem. 1.17	`ErdosSzekeres.SymReduction.orient_affine` `SYMLEAN/AffineReduction.lean:1241` `ErdosSzekeres.OrbitConsistency.orient_rotAbout_inv` `LEANEXT/orbit/OrbitConsistency.lean:117`	main formula `orient_affine` (g a map with $g(p) = Lp + w$ for all p; Lean names w as b); verbatim copy at `SYMLEAN/mirror/ReductionThreeClasses.lean:1267`; (i) `orient_rotAbout_inv` (stated for the `OrbitConsistency` copy of `rotAbout`); (ii) written step
Rem. 1.18	`ErdosSzekeres.SymReduction.orient_affine` `SYMLEAN/mirror/ReductionThreeClasses.lean:1267` `ErdosSzekeres.OrbitConsistency.orient_rotAbout_inv` `LEANEXT/orbit/OrbitConsistency.lean:117` `ErdosSzekeres.Suk.orient_reflectY` `ErdosSzekeres/Suk/Reflection.lean:47`	verbatim copy of `orient_affine`; (i) is `orient_rotAbout_inv`, also cited in the head of Lemma 1.17 (Lemma 4.7); the case $o = 0, \phi = 0$ of (ii), the reflection $(p0, p1) \to (p0, -p1)$, is `orient_reflectY`
Lem. 1.19	written step	written step; Mathlib `collinear_iff_of_mem` reproved
Lem. 1.20	`ErdosSzekeres.orient_eq_zero_iff_collinear` `ErdosSzekeres/OrientationGeometry.lean:36` `ErdosSzekeres.generalPosition_iff_orient_ne_zero` `ErdosSzekeres/OrientationGeometry.lean:66`	(a) `orient_eq_zero_iff_collinear`; (b) `generalPosition_iff_orient_ne_zero`; includes degenerate triples
Lem. 1.21	written step	written step (standard facts on convex hulls)
Lem. 1.22	`ErdosSzekeres.convexPosition_iff` `ErdosSzekeres/Basic.lean:15`	Lean proof is Mathlib `convexIndependent_set_iff_notMem_convexHull_diff`; direct proof given
Lem. 1.23	`ErdosSzekeres.ConvexPosition.mono` `ErdosSzekeres/Basic.lean:11` `ErdosSzekeres.GeneralPosition.mono` `ErdosSzekeres/Basic.lean:6` `ErdosSzekeres.HasConvexNGon.mono` `ErdosSzekeres/Basic.lean:20` `MirrorOdd.hasConvexNGon_of_le` `MIRRORETV/lean/MirrorMain.lean:2182`	(a) `ConvexPosition.mono`; (b) `GeneralPosition.mono`; (c) `HasConvexNGon.mono`; (d) `MirrorOdd.hasConvexNGon_of_le`, also verbatim at `SYMLEAN/mirror/ReductionThreeClasses.lean:3741`; axioms of (d) through its use in the proof of `MirrorOdd.mirror_odd_general` (`MIRRORETV/lean/MirrorMain.lean:2498`)

Statement	Lean declaration; file and line	Note
Lem. 1.24	`ErdosSzekeres.HasConvexNGon.card_le` `ErdosSzekeres/Basic.lean:25` `ErdosSzekeres.ConvexPosition.hasConvexNGon` `ErdosSzekeres/Basic.lean:31`	(a) `HasConvexNGon.card_le`; (b) `ConvexPosition.hasConvexNGon`; part (a) is not printed in the axiom audit, and its evidence is of kind (a) of the rigor labels (Section 1): build record `docs/VALIDATION.md:73`, the axiom audit as Lean output of the same library, text search
Lem. 1.25	`ErdosSzekeres.ForcesConvexNGon.mono` `ErdosSzekeres/Basic.lean:47` `ErdosSzekeres.forcesConvexNGon_iff_exact` `ErdosSzekeres/Basic.lean:37`	(a) `ForcesConvexNGon.mono`; (b) `forcesConvexNGon_iff_exact`
Lem. 1.26	`ErdosSzekeres.convexPosition_of_card_three` `ErdosSzekeres/Basic.lean:58`	last assertion of (c) only; (a),(b),(c) otherwise written step
Lem. 1.27	`ErdosSzekeres.SymReduction.convexPosition_image_affine` `SYMLEAN/AffineReduction.lean:1263` `ErdosSzekeres.SymReduction.generalPosition_image_affine` `SYMLEAN/AffineReduction.lean:1278` `ErdosSzekeres.SymReduction.generalPosition_image_affineEquiv` `SYMLEAN/AffineReduction.lean:1290` `ErdosSzekeres.SymReduction.hasConvexNGon_image_affine` `SYMLEAN/AffineReduction.lean:1298` `ErdosSzekeres.SymReduction.injective_of_affine` `SYMLEAN/AffineReduction.lean:1305`	Lean forms of the forward directions: (b') `convexPosition_image_affine`; (c') `generalPosition_image_affine` and `generalPosition_image_affineEquiv`; (d') `hasConvexNGon_image_affine`; (e') `injective_of_affine` (converse of injective $\Rightarrow \det A \neq 0$); (a) is Mathlib `AffineMap.image_convexHull`; reverse directions of (b)-(d) and injective $\Rightarrow \det A \neq 0$ written step; $h(p) = Lp + w$ (Lean names w as b); the affine equivalence of (c') is written ϕ
Rem. 1.28	`ErdosSzekeres.HasConvexNGon.linearEquiv` `ErdosSzekeres/Suk/Reflection.lean:6` `ErdosSzekeres.Suk.reflectY` `ErdosSzekeres/Suk/Reflection.lean:34` `ErdosSzekeres.Suk.reflectY_x` `ErdosSzekeres/Suk/Reflection.lean:42` `ErdosSzekeres.Suk.reflectY_y` `ErdosSzekeres/Suk/Reflection.lean:43` `ErdosSzekeres.Suk.reflectY_reflectY` `ErdosSzekeres/Suk/Reflection.lean:44` `ErdosSzekeres.Suk.generalPosition_reflectY` `ErdosSzekeres/Suk/Reflection.lean:52` `ErdosSzekeres.Suk.hasConvexNGon_of_reflectY` `ErdosSzekeres/Suk/Reflection.lean:64`	main-library special cases of Lemma 1.27 (linear bijections; the reflection `reflectY` in both directions); `reflectY_x`, `reflectY_y` and `reflectY_reflectY` are not printed in the axiom audit: evidence of kind (a) of the rigor labels (Section 1): build record `docs/VALIDATION.md:73`, the axiom audit as Lean output of the same library, text search
Lem. 1.29	written step	written step; Lean uses Mathlib `Caratheodory.minCardFinsetOfMemConvexHull`
Lem. 1.30	`ErdosSzekeres.Suk.convexPosition_of_subsets_card_le_four` `ErdosSzekeres/Suk/FourPoint.lean:10` `ErdosSzekeres.Suk.convexPosition_of_four_subsets` `ErdosSzekeres/Suk/FourPoint.lean:40`	(a) `convexPosition_of_subsets_card_le_four`; (b) `convexPosition_of_four_subsets`; no general position assumed
Lem. 1.31	`ErdosSzekeres.Suk.orient_affine_combination` `ErdosSzekeres/Suk/SupportingLines.lean:6` `ErdosSzekeres.Suk.convex_orient_positive` `ErdosSzekeres/Suk/SupportingLines.lean:11` `ErdosSzekeres.Suk.convex_orient_nonnegative` `ErdosSzekeres/Suk/SupportingLines.lean:21`	(a) `orient_affine_combination`; (b) `convex_orient_positive`; (c) `convex_orient_nonnegative`
Lem. 1.32	`ErdosSzekeres.Suk.convexPosition_of_strict_supports` `ErdosSzekeres/Suk/SupportingLines.lean:28`	
Lem. 1.33	`ErdosSzekeres.Suk.orient_support_sum` `ErdosSzekeres/Suk/SupportingLines.lean:40`	
Lem. 1.34	`ErdosSzekeres.Suk.convexPosition_four_of_positive` `ErdosSzekeres/Suk/SupportingLines.lean:47`	twelve support inequalities as in Lean
Lem. 1.35	`ErdosSzekeres.Suk.convexPosition_of_positive_triples` `ErdosSzekeres/Suk/OrientationClique.lean:9`	requires $4 \leq \operatorname{card} s$ and p injective on all of ι; Lean uses decide only for side facts (order facts among 0,1,2,3 in Fin 4 and `Finset.univ` = {0,1,2,3} in Fin 4) inside the proof of a statement that is not itself a finite computation: not computer-assisted

Statement	Lean declaration; file and line	Note
Rem. 1.36	–	finiteness and injectivity hypotheses; use of decide for side facts (not computer-assisted); variant for card $s \leq 3$ (labelled as a variant, proved from Lemma 1.26)
Lem. 1.37	`ErdosSzekeres.Suk.convexPosition_of_negative_triples` `ErdosSzekeres/Suk/OrientationClique.lean:33`	not printed in the axiom audit: evidence of kind (a) of the rigor labels (Section 1): build record `docs/VALIDATION.md:73`, the axiom audit as Lean output of the same library, text search
Lem. 1.38	`ErdosSzekeres.hasConvexNGon_of_positive_cycle` `ErdosSzekeres/ConvexCycleCuts.lean:11`	all increasing triples of the embedding e
Rem. 1.39	`ErdosSzekeres.ConvexCycleCut` `ErdosSzekeres/ConvexCycleCuts.lean:8` `ErdosSzekeres.convexCycleCut_of_avoids` `ErdosSzekeres/ConvexCycleCuts.lean:31`	clause family of the SMT coordinate-repair searches (`docs/SAT.md`); `convexCycleCut_of_avoids` is the contrapositive of Lemma 1.38
Lem. 1.40	written step	written step; both directions proved (nearest point for the hard direction; Heine-Borel and extreme value theorem cited from Rudin); main-library variant of the easy direction: `Omission.BoundaryGeometry.exposed_not_in_hull` (`ErdosSzekeres/Omission/BoundaryGeometry.lean:30`), named in Coverage
Section 2		
Def. 2.1	`ErdosSzekeres.LayerLemma.orbitFS` `LAYERLEAN/LayerLemma.lean:48`	
Lem. 2.2	`ErdosSzekeres.LayerLemma.iterate_mem` `LAYERLEAN/LayerLemma.lean:52` `ErdosSzekeres.LayerLemma.mem_orbitFS_self` `LAYERLEAN/LayerLemma.lean:59` `ErdosSzekeres.LayerLemma.orbitFS_subset` `LAYERLEAN/LayerLemma.lean:62` `ErdosSzekeres.LayerLemma.orbitFS_invariant` `LAYERLEAN/LayerLemma.lean:68` `ErdosSzekeres.LayerLemma.card_orbitFS_of_minimalPeriod` `LAYERLEAN/LayerLemma.lean:80` `ErdosSzekeres.LayerLemma.mem_orbitFS_of_not_disjoint` `LAYERLEAN/LayerLemma.lean:91`	
Def. 2.3	`ErdosSzekeres.LayerLemma.Invariant` `LAYERLEAN/LayerLemma.lean:105`	
Def. 2.4	`ErdosSzekeres.LayerLemma.hullVerts` `LAYERLEAN/LayerLemma.lean:111`	
Lem. 2.5	`ErdosSzekeres.LayerLemma.mem_hullVerts` `LAYERLEAN/LayerLemma.lean:114` `ErdosSzekeres.LayerLemma.hullVerts_subset` `LAYERLEAN/LayerLemma.lean:118`	
Lem. 2.6	`ErdosSzekeres.LayerLemma.convexPosition_hullVerts` `LAYERLEAN/LayerLemma.lean:121` `ErdosSzekeres.LayerLemma.extreme_mem_hullVerts` `LAYERLEAN/LayerLemma.lean:130` `ErdosSzekeres.LayerLemma.subset_convexHull_hullVerts` `LAYERLEAN/LayerLemma.lean:146` `ErdosSzekeres.LayerLemma.hullVerts_nonempty` `LAYERLEAN/LayerLemma.lean:139`	(b)-(d) proved by elementary arguments, different from the Lean proofs (Lean: (b) by two Mathlib facts on extreme points, (c) by the Krein-Milman theorem, (d) by the Krein-Milman lemma)
Lem. 2.7	`ErdosSzekeres.LayerLemma.map_mem_hullVerts` `LAYERLEAN/LayerLemma.lean:162`	
Thm. 2.8	`ErdosSzekeres.LayerLemma.hullVerts_eq_orbitFS` `LAYERLEAN/LayerLemma.lean:189`	no general position
Lem. 2.9	`ErdosSzekeres.LayerLemma.invariant_sdiff_hullVerts` `LAYERLEAN/LayerLemma.lean:217`	
Rem. 2.10	–	remark on hypotheses and degenerate cases of `hullVerts_eq_orbitFS`

Statement	Lean declaration; file and line	Note

Def. 2.11	`ErdosSzekeres.LayerLemma.toC`
	`LAYERLEAN/LayerLemma.lean:231`
	`ErdosSzekeres.LayerLemma.ofC`
	`LAYERLEAN/LayerLemma.lean:234`
	`ErdosSzekeres.LayerLemma.crot`
	`LAYERLEAN/LayerLemma.lean:252`	

Lem. 2.12	`ErdosSzekeres.LayerLemma.toC_re`
	`LAYERLEAN/LayerLemma.lean:236`
	`ErdosSzekeres.LayerLemma.toC_im`
	`LAYERLEAN/LayerLemma.lean:237`
	`ErdosSzekeres.LayerLemma.ofC_zero`
	`LAYERLEAN/LayerLemma.lean:238`
	`ErdosSzekeres.LayerLemma.ofC_one`
	`LAYERLEAN/LayerLemma.lean:239`
	`ErdosSzekeres.LayerLemma.toC_ofC`
	`LAYERLEAN/LayerLemma.lean:241`
	`ErdosSzekeres.LayerLemma.ofC_toC`
	`LAYERLEAN/LayerLemma.lean:244`
	`ErdosSzekeres.LayerLemma.toC_injective`
	`LAYERLEAN/LayerLemma.lean:248`
	`ErdosSzekeres.LayerLemma.toC_crot`
	`LAYERLEAN/LayerLemma.lean:254`
	`ErdosSzekeres.LayerLemma.crot_one`
	`LAYERLEAN/LayerLemma.lean:257`
	`ErdosSzekeres.LayerLemma.crot_comp`
	`LAYERLEAN/LayerLemma.lean:262`
	`ErdosSzekeres.LayerLemma.crot_iterate`
	`LAYERLEAN/LayerLemma.lean:267`
	`ErdosSzekeres.LayerLemma.crot_eq_self_iff`
	`LAYERLEAN/LayerLemma.lean:273`
	`ErdosSzekeres.LayerLemma.crot_injective`
	`LAYERLEAN/LayerLemma.lean:287`
		includes the seven @[simp] lemmas (`toC_re`, `toC_im`, `ofC_zero`, `ofC_one`, `toC_ofC`, `ofC_toC`, `toC_crot`)

Def. 2.13	`ErdosSzekeres.LayerLemma.rotAbout` `LAYERLEAN/LayerLemma.lean:298`	

Lem. 2.14	`ErdosSzekeres.LayerLemma.rotAbout_eq_crot`
	`LAYERLEAN/LayerLemma.lean:302`
	`ErdosSzekeres.LayerLemma.rotAff`
	`LAYERLEAN/LayerLemma.lean:311`
	`ErdosSzekeres.LayerLemma.rotAbout_injective`
	`LAYERLEAN/LayerLemma.lean:319`
	`ErdosSzekeres.LayerLemma.rotAbout_iterate`
	`LAYERLEAN/LayerLemma.lean:465`
		`rotAff` is a def (affine-map structure); `rotAbout_iterate` is line 465

Lem. 2.15	`ErdosSzekeres.LayerLemma.invariant_erase_centre` `LAYERLEAN/LayerLemma.lean:325`	
Lem. 2.16	`ErdosSzekeres.LayerLemma.minimalPeriod_rotAbout` `LAYERLEAN/LayerLemma.lean:336`	
Thm. 2.17	`ErdosSzekeres.LayerLemma.hullVerts_eq_orbitFS_rotAbout` `LAYERLEAN/LayerLemma.lean:364`	
Def. 2.18	`ErdosSzekeres.LayerLemma.peel` `LAYERLEAN/LayerLemma.lean:374`	

Lem. 2.19	`ErdosSzekeres.LayerLemma.peel_succ`
	`LAYERLEAN/LayerLemma.lean:378`
	`ErdosSzekeres.LayerLemma.peel_subset`
	`LAYERLEAN/LayerLemma.lean:381`	

Lem. 2.20	`ErdosSzekeres.LayerLemma.peel_antitone` `LAYERLEAN/LayerLemma.lean:460`	
Lem. 2.21	`ErdosSzekeres.LayerLemma.invariant_peel` `LAYERLEAN/LayerLemma.lean:386`	
Thm. 2.22	`ErdosSzekeres.LayerLemma.layers_are_orbits` `LAYERLEAN/LayerLemma.lean:402`	
Lem. 2.23	`ErdosSzekeres.LayerLemma.exists_layer` `LAYERLEAN/LayerLemma.lean:419`	
Thm. 2.24	`ErdosSzekeres.LayerLemma.card_peel` `LAYERLEAN/LayerLemma.lean:1037`	

Statement	Lean declaration; file and line	Note
Cor. 2.25	written step	written step (not a Lean declaration): decomposition of a set of exactly $m \cdot t$ points into t nested orbits
Thm. 2.26	`ErdosSzekeres.LayerLemma.convexNGon_of_layers` `LAYERLEAN/LayerLemma.lean:1066`	takes the premise H (every invariant centre-free set of exactly $m \cdot t$ points in general position has a convex K-gon) as an explicit hypothesis
Def. 2.27	`ErdosSzekeres.LayerLemma.Sixfold66` `LAYERLEAN/LayerLemma.lean:1102` `ErdosSzekeres.LayerLemma.Sevenfold70` `LAYERLEAN/LayerLemma.lean:1107`	propositions (Lean def ... : Prop), not proved in Lean; real numerals 6 and 7 in the angle
Thm. 2.28	`ErdosSzekeres.LayerLemma.sixfold_octagon` `LAYERLEAN/LayerLemma.lean:1113`	takes hypotheses: Sixfold66; uses Lemma 3.18 (argument also included)
Thm. 2.29	`ErdosSzekeres.LayerLemma.sevenfold_octagon` `LAYERLEAN/LayerLemma.lean:1136`	takes hypotheses: Sevenfold70
Rem. 2.30	–	status of `Sixfold66/Sevenfold70`: not proved in Lean; supported by `C6UPPER/C7UPPER` `cake_lpr` verdicts (computer-assisted: LRAT proofs checked by `cake_lpr` outside Lean; the project's records call such a result a certificate, a label this document does not use; encoder B's formula alone, encoder A's with a symmetry-breaking unit clause) plus the written soundness proof in `WRITEUP/note/k8_soundness.tex` (not refereed; gaps G1-G4); ENCBLEAN Lean route for $m = 6$ assumes `EncB.CubesUnsat64` (trust items E1-E6)
Rem. 2.31	–	real-cast numerals 6, 7 versus (m : R)
Section 3		
Lem. 3.1	written step	written step; collects the opening steps of the Lean proofs of `strict_nesting`, `lin_pos_of_peel` and `dist2_le_of_later`; uses Theorem 2.17, Lemma 2.21, Lemma 2.6, Lemma 2.20
Lem. 3.2	written step	written step; elementary facts about `rotAbout` (composition, rotation by 0 and 2π, distance to the centre); the Lean proofs obtain the corresponding steps through the complex form of `rotAbout`, by direct computation or through minimal periods
Lem. 3.3	`ErdosSzekeres.LayerLemma.rotAbout_iterate` `LAYERLEAN/LayerLemma.lean:465`	Lean-checked; axioms in `LEANEXT/review/axioms.log`; proof here via the matrix composition rule; Lean uses the complex form
Cor. 3.4	written step	written step; $\rho^m = \mathrm{id}$ for $0 < m$ (Lean uses the minimal period instead)
Lem. 3.5	`ErdosSzekeres.LayerLemma.orient_rotAbout` `LAYERLEAN/LayerLemma.lean:472`	Lean-checked; axioms in `LEANEXT/review/axioms.log`
Lem. 3.6	`ErdosSzekeres.LayerLemma.trig_layer` `LAYERLEAN/LayerLemma.lean:480`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`
Lem. 3.7	`ErdosSzekeres.LayerLemma.orient_regular_nonneg` `LAYERLEAN/LayerLemma.lean:488`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`; its proof uses `trig_layer` (Lemma 3.6, stated just before)
Def. 3.9	`ErdosSzekeres.LayerLemma.lin` `LAYERLEAN/LayerLemma.lean:625`	Lean definition
Lem. 3.10	`ErdosSzekeres.LayerLemma.lin_rotAbout` `LAYERLEAN/LayerLemma.lean:794`	Lean-checked; axioms in `LEANEXT/review/axioms.log`
Lem. 3.11	`ErdosSzekeres.LayerLemma.lin_orbit_pos` `LAYERLEAN/LayerLemma.lean:801`	Lean-checked; axioms in `LEANEXT/review/axioms.log`
Lem. 3.12	`ErdosSzekeres.LayerLemma.orient_next_prev_pos` `LAYERLEAN/LayerLemma.lean:830`	Lean-checked; axioms in `LEANEXT/review/axioms.log`
Lem. 3.13	`ErdosSzekeres.LayerLemma.convex_orient_nonneg` `LAYERLEAN/LayerLemma.lean:516`	Lean-checked; axioms in `LEANEXT/review/axioms.log`

Statement	Lean declaration; file and line	Note
Def. 3.14	`ErdosSzekeres.LayerLemma.dist2` `LAYERLEAN/LayerLemma.lean:999`	Lean definition
Lem. 3.15	`ErdosSzekeres.LayerLemma.convex_disk` `LAYERLEAN/LayerLemma.lean:1002`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`
Thm. 3.16	`ErdosSzekeres.LayerLemma.strict_nesting` `LAYERLEAN/LayerLemma.lean:529`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`; $x \neq \rho x$ proved here via `lin_orbit_pos`; Lean uses the minimal period
Lem. 3.18	`ErdosSzekeres.LayerLemma.centre_not_mem_of_even` `LAYERLEAN/LayerLemma.lean:582`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`
Thm. 3.20	`ErdosSzekeres.LayerLemma.dist2_le_of_later` `LAYERLEAN/LayerLemma.lean:1012`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`; weak inequality only, $n \leq n'$
Lem. 3.22	written step	written step; the planar identity behind `orient_trans_halfplane`, `fan_step` and the key identity of `lin_pos_of_peel`; Lean checks only these three instances, each by ring
Lem. 3.23	`ErdosSzekeres.LayerLemma.orient_trans_halfplane` `LAYERLEAN/LayerLemma.lean:629`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`; key identity derived here from Lemma 3.22; Lean uses ring
Lem. 3.24	`ErdosSzekeres.LayerLemma.fan_step` `LAYERLEAN/LayerLemma.lean:641`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`; key identity derived here from Lemma 3.22; Lean uses ring
Lem. 3.25	`ErdosSzekeres.LayerLemma.orient_eq_lin` `LAYERLEAN/LayerLemma.lean:653`	Lean-checked; axioms in `LEANEXT/review/axioms.log`
Lem. 3.26	`ErdosSzekeres.LayerLemma.fan_positive` `LAYERLEAN/LayerLemma.lean:658`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`
Def. 3.27	`ErdosSzekeres.LayerLemma.fanPt` `LAYERLEAN/LayerLemma.lean:694`	Lean definition
Thm. 3.28	`ErdosSzekeres.LayerLemma.hasConvexNGon_of_fan` `LAYERLEAN/LayerLemma.lean:700`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`; uses `ErdosSzekeres.Suk.convexPosition_of_positive_triples` (Lemma 1.35); no general position
Lem. 3.30	`ErdosSzekeres.LayerLemma.lin_pos_of_peel` `LAYERLEAN/LayerLemma.lean:864`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`; $\rho(p') = p$ proved here via Corollary 3.4 (`rotAbout_iterate` and rotation by 2π); Lean uses the minimal period; key identity derived here from Lemma 3.22; Lean uses ring
Thm. 3.32	`ErdosSzekeres.LayerLemma.chain_clauses_sound` `LAYERLEAN/LayerLemma.lean:914`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`; Lean conclusion False; geometric statement only (that the encoders emit these clauses is not in Lean)
Def. 3.34	`ErdosSzekeres.LayerLemma.polar` `LAYERLEAN/LayerLemma.lean:932`	Lean definition
Lem. 3.35	`ErdosSzekeres.LayerLemma.orient_polar` `LAYERLEAN/LayerLemma.lean:935`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`
Lem. 3.36	`ErdosSzekeres.LayerLemma.orient_polar_centre_inside` `LAYERLEAN/LayerLemma.lean:943`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`
Lem. 3.37	`ErdosSzekeres.LayerLemma.sin_add_lt` `LAYERLEAN/LayerLemma.lean:956`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`
Lem. 3.38	`ErdosSzekeres.LayerLemma.orient_polar_halfplane` `LAYERLEAN/LayerLemma.lean:974`	Lean-checked; axioms in `LAYERLEAN/logs/axioms.log` and `LEANEXT/review/axioms.log`
Section 4		
Def. 4.1	`ErdosSzekeres.OrbitConsistency.Invariant` `LEANEXT/orbit/OrbitConsistency.lean:47` `ErdosSzekeres.OrbitConsistency.rotAbout` `LEANEXT/orbit/OrbitConsistency.lean:93` `ErdosSzekeres.OrbitConsistency.polar` `LEANEXT/orbit/OrbitConsistency.lean:112`	copies of `ErdosSzekeres.LayerLemma.Invariant` (`LAYERLEAN/LayerLemma.lean:105`), `rotAbout` (:298), polar (:932); identical formulas

Statement	Lean declaration; file and line	Note
Rem. 4.2	–	identification of the duplicated definitions
Lem. 4.3	written step	written step: composition of rotations; $R_0, R_{2\pi}, R_\pi$
Lem. 4.4	`ErdosSzekeres.OrbitConsistency.rotAbout_iterate` `LEANEXT/orbit/OrbitConsistency.lean:105` `ErdosSzekeres.LayerLemma.rotAbout_iterate` `LAYERLEAN/LayerLemma.lean:465`	copied declaration (outside the new content); proof by addition formulas instead of complex exponentials; axioms: `OrbitConsistency` copy covered transitively by `LEANEXT/orbit/logs/axioms.log` (used by `antipode_mem_of_even`), `LayerLemma` copy printed in `LEANEXT/review/axioms.log:33`
Cor. 4.5	written step	written step: $\rho^m = \mathrm{id}$; $\rho^{m/2}$ is the half-turn for even m
Lem. 4.6	`ErdosSzekeres.OrbitConsistency.iterate_mem` `LEANEXT/orbit/OrbitConsistency.lean:39` `ErdosSzekeres.LayerLemma.iterate_mem` `LAYERLEAN/LayerLemma.lean:52`	copied declaration (outside the new content); axioms: `OrbitConsistency` copy covered transitively by `LEANEXT/orbit/logs/axioms.log` (used by `antipode_mem_of_even`), `LayerLemma` copy printed in `LEANEXT/review/axioms.log:1` (propext, `Quot.sound` only)
Lem. 4.7	`ErdosSzekeres.OrbitConsistency.orient_rotAbout_inv` `LEANEXT/orbit/OrbitConsistency.lean:117`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log`
Def. 4.8	`ErdosSzekeres.OrbitConsistency.PreservesOrient` `LEANEXT/orbit/OrbitConsistency.lean:124`	
Def. 4.9	`ErdosSzekeres.OrbitConsistency.RotComp` `LEANEXT/orbit/OrbitConsistency.lean:145`	inductive predicate; any m ($2\pi/0 = 0$)
Lem. 4.10	`ErdosSzekeres.OrbitConsistency.preservesOrient_id` `LEANEXT/orbit/OrbitConsistency.lean:127` `ErdosSzekeres.OrbitConsistency.PreservesOrient.comp` `LEANEXT/orbit/OrbitConsistency.lean:129` `ErdosSzekeres.OrbitConsistency.PreservesOrient.iterate` `LEANEXT/orbit/OrbitConsistency.lean:134` `ErdosSzekeres.OrbitConsistency.preservesOrient_rotAbout` `LEANEXT/orbit/OrbitConsistency.lean:141` `ErdosSzekeres.OrbitConsistency.RotComp.preservesOrient` `LEANEXT/orbit/OrbitConsistency.lean:150` `ErdosSzekeres.OrbitConsistency.rotComp_iterate` `LEANEXT/orbit/OrbitConsistency.lean:157`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log` (`RotComp.preservesOrient`, `rotComp_iterate`; helpers transitively); `PreservesOrient.iterate` has no recorded axiom output (compilation of the whole file recorded in `LEANEXT/orbit/logs/build.log`, exit 0)
Def. 4.11	`ErdosSzekeres.OrbitConsistency.orient3` `LEANEXT/orbit/OrbitConsistency.lean:168`	
Fact 4.12	Mathlib	cited standard facts (Mathlib `Equiv.Perm.sign`, `Equiv.Perm.sign_mul`, `Equiv.Perm.sign_swap`, `Equiv.Perm.swap_induction_on`, `Int.units_eq_one_or`)
Lem. 4.13	`ErdosSzekeres.OrbitConsistency.orient3_swap` `LEANEXT/orbit/OrbitConsistency.lean:171`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log`; proof via orient symmetries instead of case expansion
Lem. 4.14	`ErdosSzekeres.OrbitConsistency.orient3_perm` `LEANEXT/orbit/OrbitConsistency.lean:178`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log`
Lem. 4.15	`ErdosSzekeres.OrbitConsistency.orient3_rot_perm` `LEANEXT/orbit/OrbitConsistency.lean:194`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log`
Lem. 4.16	`ErdosSzekeres.OrbitConsistency.sign_eq_of_two_reps` `LEANEXT/orbit/OrbitConsistency.lean:204` `ErdosSzekeres.OrbitConsistency.orient3_of_common_rep` `LEANEXT/orbit/OrbitConsistency.lean:216`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log`
Def. 4.17	`ErdosSzekeres.OrbitConsistency.litVal` `LEANEXT/orbit/OrbitConsistency.lean:223`	
Thm. 4.18	`ErdosSzekeres.OrbitConsistency.literal_sound` `LEANEXT/orbit/OrbitConsistency.lean:228`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log`
Thm. 4.19	`ErdosSzekeres.OrbitConsistency.encoder_literal_sound` `LEANEXT/orbit/OrbitConsistency.lean:253`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log`; encoders compute parity by inversion counting, not modelled in Lean (Remark 4.20)
Rem. 4.20	–	inversion-count parity vs sign: six-case written check; code correspondence outside Lean; Python evidence only

Statement	Lean declaration; file and line	Note
Rem. 4.21	–	how the encoder uses the literal rule
Lem. 4.22	`ErdosSzekeres.OrbitConsistency.antipode_mem_of_even` `LEANEXT/orbit/OrbitConsistency.lean:273` `ErdosSzekeres.LayerLemma.antipode_mem_of_even` `LEANEXT/merged/Merged.lean:1160`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log` (`OrbitConsistency`); Merged copy covered transitively by `LEANEXT/merged/axioms.log`
Lem. 4.23	`ErdosSzekeres.OrbitConsistency.same_ray_false` `LEANEXT/orbit/OrbitConsistency.lean:295` `ErdosSzekeres.LayerLemma.same_ray_false` `LEANEXT/merged/Merged.lean:1181`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log` (`OrbitConsistency`); Merged copy covered transitively by `LEANEXT/merged/axioms.log`
Rem. 4.24	–	$l > 0$ cannot be weakened to $l \neq 0$
Lem. 4.25	`ErdosSzekeres.OrbitConsistency.rotAbout_polar` `LEANEXT/orbit/OrbitConsistency.lean:355` `ErdosSzekeres.OrbitConsistency.polar_add_int` `LEANEXT/orbit/OrbitConsistency.lean:383` `ErdosSzekeres.LayerLemma.rotAbout_polar` `LEANEXT/merged/Merged.lean:1241` `ErdosSzekeres.LayerLemma.polar_add_int` `LEANEXT/merged/Merged.lean:1269`	Lean-checked; axioms covered transitively by `LEANEXT/orbit/logs/axioms.log` and `LEANEXT/merged/axioms.log`
Lem. 4.26	`ErdosSzekeres.OrbitConsistency.eq_iterate_of_phase_eq` `LEANEXT/orbit/OrbitConsistency.lean:363` `ErdosSzekeres.LayerLemma.eq_iterate_of_phase_eq` `LEANEXT/merged/Merged.lean:1249`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log` (`OrbitConsistency`); Merged copy covered transitively by `LEANEXT/merged/axioms.log`
Lem. 4.27	`ErdosSzekeres.OrbitConsistency.eq_iterate_of_phase_eq_int` `LEANEXT/orbit/OrbitConsistency.lean:389` `ErdosSzekeres.LayerLemma.eq_iterate_of_phase_eq_int` `LEANEXT/merged/Merged.lean:1275`	Lean-checked; axioms printed in `LEANEXT/orbit/logs/axioms.log` (`OrbitConsistency`); Merged copy covered transitively by `LEANEXT/merged/axioms.log`
Thm. 4.28	`ErdosSzekeres.OrbitConsistency.phase_ne_of_not_mem_orbit` `LEANEXT/orbit/OrbitConsistency.lean:420` `ErdosSzekeres.LayerLemma.phase_ne_of_not_mem_orbit` `LEANEXT/merged/Merged.lean:1304`	axioms printed in `LEANEXT/orbit/logs/axioms.log` and `LEANEXT/merged/axioms.log`
Rem. 4.29	–	the doc-comment reading (nonzero phases) needs facts outside the statement
Lem. 4.30	`ErdosSzekeres.LayerLemma.not_mem_hullVerts_peel_of_lt` `LEANEXT/merged/Merged.lean:1316` `ErdosSzekeres.LayerLemma.not_mem_hullVerts_peel_of_ne` `LEANEXT/merged/Merged.lean:1322`	Lean-checked; axioms covered transitively by `LEANEXT/merged/axioms.log`
Lem. 4.31	`ErdosSzekeres.LayerLemma.iterate_mem_hullVerts_peel` `LEANEXT/merged/Merged.lean:1330`	Lean-checked; axioms covered transitively by `LEANEXT/merged/axioms.log`; proof uses $\rho^m = \mathrm{id}$ directly instead of `minimalPeriod_rotAbout`
Thm. 4.32	`ErdosSzekeres.LayerLemma.not_same_ray_of_distinct_layers` `LEANEXT/merged/Merged.lean:1350`	Lean-checked; axioms printed in `LEANEXT/merged/axioms.log`
Thm. 4.33	`ErdosSzekeres.LayerLemma.phase_ne_of_distinct_layers` `LEANEXT/merged/Merged.lean:1367`	Lean-checked; axioms printed in `LEANEXT/merged/axioms.log`
Rem. 4.34	–	$c \notin S$ derivable via Lemma 3.18; example showing π/m cannot replace $2\pi/m$
Def. 4.35	Mathlib	Mathlib definitions: `toIcoDiv`/`toIcoMod`, `Int.ediv`/`Int.emod`, lexicographic order (`Prod.Lex`), `StrictMonoOn`
Lem. 4.36	written step	written step (Mathlib `toIcoMod_eq_iff`)
Lem. 4.37	`KeyOrder.cell_lt_iff` `LEANEXT/keys/KeyOrder.lean:23` `KeyOrder.cell_eq_iff` `LEANEXT/keys/KeyOrder.lean:45`	Lean-checked; axioms printed by `KeyOrder.lean:151-158`, output in `ENCBLEAN/logs/m4_build.log:3-10`
Lem. 4.38	`KeyOrder.cell_lt_iff_rank` `LEANEXT/keys/KeyOrder.lean:62` `KeyOrder.cell_lt_iff_rank_finset` `LEANEXT/keys/KeyOrder.lean:71`	Lean-checked; axioms printed by `KeyOrder.lean:151-158`, output in `ENCBLEAN/logs/m4_build.log:3-10`
Lem. 4.39	`KeyOrder.reduce_mod_two_pi` `LEANEXT/keys/KeyOrder.lean:81`	Lean-checked; axioms printed by `KeyOrder.lean:151-158`, output in `ENCBLEAN/logs/m4_build.log:3-10`

Statement	Lean declaration; file and line	Note
Lem. 4.40	`KeyOrder.antipode_odd` `LEANEXT/keys/KeyOrder.lean:100` `KeyOrder.antipode_even` `LEANEXT/keys/KeyOrder.lean:115`	Lean-checked; axioms printed by `KeyOrder.lean:151-158`, output in `ENCBLEAN/logs/m4_build.log:3-10`; oddness of m not assumed in `antipode_odd`
Thm. 4.41	`KeyOrder.reduced_lt_iff_key` `LEANEXT/keys/KeyOrder.lean:139`	Lean-checked; axioms printed by `KeyOrder.lean:151-158`, output in `ENCBLEAN/logs/m4_build.log:3-10`
Rem. 4.42	–	half-open intervals are needed
Rem. 4.43	–	encoder decomposition, local ranks and key-to-circular-order step are outside Lean
Section 5		
Def. 5.1	`PhaseClassification.polar` `LEANEXT/phase/PhaseClassification.lean:41` `ErdosSzekeres.LayerLemma.dist2` `LAYERLEAN/LayerLemma.lean:999`	`PhaseClassification.polar` is a separate constant with the formula of `LayerLemma.polar` (`LAYERLEAN/LayerLemma.lean:932`); dist2 repeated at `LEANEXT/linked/Linked.lean:999` and `LEANEXT/capstone/Capstone2.lean:1007`
Lem. 5.2	`PhaseClassification.orient_polar` `LEANEXT/phase/PhaseClassification.lean:44` `PhaseClassification.orient_polar_centre` `LEANEXT/phase/PhaseClassification.lean:407` `PhaseClassification.orient_polar_centre_inside` `LEANEXT/phase/PhaseClassification.lean:51` `PhaseClassification.sin_add_lt` `LEANEXT/phase/PhaseClassification.lean:64` `PhaseClassification.orient_polar_halfplane` `LEANEXT/phase/PhaseClassification.lean:80`	(a),(c),(d),(e) are copies of `LAYERLEAN/LayerLemma.lean:935-996` (Lemma 3.35, Lemma 3.36, Lemma 3.37, Lemma 3.38); proofs repeated; axioms covered transitively by `LEANEXT/phase/logs/axioms.log`; head cites (a), parts (b)-(e) cite their own declarations
Lem. 5.3	`PhaseClassification.polar_add_int_two_pi` `LEANEXT/phase/PhaseClassification.lean:106` `PhaseClassification.polar_add_two_pi` `LEANEXT/phase/PhaseClassification.lean:111` `PhaseClassification.polar_toIcoMod` `LEANEXT/phase/PhaseClassification.lean:485`	`polar_toIcoMod` in `LEANEXT/phase/logs/axioms.log`; the other two used by logged theorems
Lem. 5.4	`PhaseClassification.exists_polar` `LEANEXT/phase/PhaseClassification.lean:459`	in `LEANEXT/phase/logs/axioms.log`; written proof via arccos (Lean uses `Complex.arg`)
Lem. 5.5	`ErdosSzekeres.Capstone.polar_eq` `LEANEXT/capstone/Capstone.lean:2001` `ErdosSzekeres.Capstone.polar_zero` `LEANEXT/capstone/Capstone2.lean:2051` `ErdosSzekeres.LayerLemma.dist2_polar` `LEANEXT/linked/Linked.lean:1390` `ErdosSzekeres.Capstone.exists_angle` `LEANEXT/capstone/Capstone2.lean:2056` `ErdosSzekeres.Capstone.sqrt_dist2_pos` `LEANEXT/capstone/Capstone2.lean:2064` `ErdosSzekeres.Capstone.ne_of_mem` `LEANEXT/capstone/Capstone2.lean:2073`	no saved axioms output in LEANEXT; used (via the byte-identical Capstone2 copy) in proofs on which `EncB.Geo.labelling_of_counterexample` and `EncB.Geo.modelOfLabelling` depend, whose axioms are pinned at `ENCBLEAN/geo/ES8Geo/Audit.lean:45-49`; the copy is compiled against verbatim copies of the ten repository modules it needs (`ENCBLEAN/geo/VENDOR.sha256`, hashes equal to the repository files)
Def. 5.6	`PhaseClassification.inArc` `LEANEXT/phase/PhaseClassification.lean:117` `PhaseClassification.arcHasAntipode` `LEANEXT/phase/PhaseClassification.lean:120` `PhaseClassification.Alternating` `LEANEXT/phase/PhaseClassification.lean:126` `PhaseClassification.RunMidB` `LEANEXT/phase/PhaseClassification.lean:132` `PhaseClassification.RunMidD` `LEANEXT/phase/PhaseClassification.lean:136` `PhaseClassification.RunMidA` `LEANEXT/phase/PhaseClassification.lean:140`	H_1, H_2, H_3 are notation of the write-up, not Lean names
Rem. 5.7	–	explanatory; gap notation g_1, g_2, g_3
Lem. 5.8	`PhaseClassification.inArc_of` `LEANEXT/phase/PhaseClassification.lean:143` `PhaseClassification.not_inArc` `LEANEXT/phase/PhaseClassification.lean:147`	

Statement	Lean declaration; file and line	Note
Lem. 5.9	written step	written step; makes explicit the have-steps hrun' in the Lean proofs of `orient_pos_of_runMidD/A` (`LEANEXT/phase/PhaseClassification.lean` lines 323-335, 346-362); proves the equivalences
Lem. 5.10	`PhaseClassification.arc1_empty` `LEANEXT/phase/PhaseClassification.lean:161` `PhaseClassification.arc2_empty` `LEANEXT/phase/PhaseClassification.lean:169` `PhaseClassification.arc2_empty'` `LEANEXT/phase/PhaseClassification.lean:177` `PhaseClassification.arc3_empty` `LEANEXT/phase/PhaseClassification.lean:185` `PhaseClassification.arc3_empty'` `LEANEXT/phase/PhaseClassification.lean:193` `PhaseClassification.arc1_empty'` `LEANEXT/phase/PhaseClassification.lean:201`	same lift integers m as the Lean proofs
Lem. 5.11	`PhaseClassification.arc12_nonempty` `LEANEXT/phase/PhaseClassification.lean:229`	written proof uses a coarser three-case split than the Lean proof
Thm. 5.12	`PhaseClassification.alternating_iff_gaps_lt_pi` `LEANEXT/phase/PhaseClassification.lean:210`	no genericity hypothesis
Rem. 5.13	–	written step: a gap equal to π makes alternation fail
Prop. 5.14	`PhaseClassification.alternating_exactly_one` `LEANEXT/phase/PhaseClassification.lean:386`	
Thm. 5.15	`PhaseClassification.runMidB_iff` `LEANEXT/phase/PhaseClassification.lean:251`	hypothesis $\alpha + 2\pi - \delta \neq \pi$ needed for $\Rightarrow$
Rem. 5.16	–	written step: example $\delta - \alpha = \pi$ shows the gap hypothesis is needed
Prop. 5.17	`PhaseClassification.gap_classification` `LEANEXT/phase/PhaseClassification.lean:268`	Lean proves only the disjunction (docstring says exactly one)
Rem. 5.18	–	written steps: counterexample without genericity; disjuncts mutually exclusive (not Lean-checked)
Thm. 5.19	`PhaseClassification.phase_trichotomy` `LEANEXT/phase/PhaseClassification.lean:285`	states the disjunction and Alternating $\rightarrow$ no run; mutual exclusion of runs not stated in Lean
Rem. 5.20	–	written step: runs mutually exclusive under the ordering (not Lean-checked)
Thm. 5.21	`PhaseClassification.orient_pos_of_alternating` `LEANEXT/phase/PhaseClassification.lean:301`	rule A; on the ENCBLEAN route
Thm. 5.22	`PhaseClassification.orient_pos_of_runMidB` `LEANEXT/phase/PhaseClassification.lean:310` `PhaseClassification.orient_pos_of_runMidD` `LEANEXT/phase/PhaseClassification.lean:318` `PhaseClassification.orient_pos_of_runMidA` `LEANEXT/phase/PhaseClassification.lean:342`	rule B; `runMidB` on the ENCBLEAN route
Thm. 5.23	`PhaseClassification.orient_pos_of_phase` `LEANEXT/phase/PhaseClassification.lean:372`	
Prop. 5.24	`PhaseClassification.centre_inside_of_alternating` `LEANEXT/phase/PhaseClassification.lean:414`	
Prop. 5.25	`PhaseClassification.halfplane_of_runMidB` `LEANEXT/phase/PhaseClassification.lean:431`	
Thm. 5.26	`PhaseClassification.orient_sign_of_phase` `LEANEXT/phase/PhaseClassification.lean:496`	case $\beta' = \delta'$ not covered; not combined with the layer theorems in Lean
Rem. 5.27	–	scope remark
Thm. 5.28	`PhaseClassification.word_dichotomy` `LEANEXT/phase/PhaseClassification.lean:539`	computer-assisted in Lean; written proof by hand (8 words) given; Lean proof: decide (evaluated during elaboration and again by the kernel over the 64 functions Fin 6 $\rightarrow$ Bool; the conclusion only for the 8 satisfying the premise), no `native_decide`

Statement	Lean declaration; file and line	Note
Rem. 5.29	–	link between sorted directions and the Boolean word is not in LEANEXT (`LEANEXT/phase/REPORT.md`); Lean-checked encoder-B counterpart of the run link in ENCBLEAN (`EncB.wordSpec`, `EncB.Geo.Arc.arcFrame`, `EncB.Geo.labelling_ruleB`) noted; their axioms: #guard_msgs pin at `ENCBLEAN/EncB/SoundWord.lean:228-229` (with `ENCBLEAN/logs/axioms.log:446`), #print axioms output in `ENCBLEAN/logs/m8_fix_build.log:29,35`, and the `modelOfLabelling` pin at `ENCBLEAN/geo/ES8Geo/Audit.lean:45-46`
Thm. 5.30	`ErdosSzekeres.LayerLemma.orient_pos_of_layers` `LEANEXT/linked/Linked.lean:1398` `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidB` `LEANEXT/capstone/Capstone.lean:1977` `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidD` `LEANEXT/capstone/Capstone2.lean:2011` `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidA` `LEANEXT/capstone/Capstone2.lean:2031` `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidB_intrinsic` `LEANEXT/capstone/Capstone2.lean:2082` `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidD_intrinsic` `LEANEXT/capstone/Capstone2.lean:2098` `ErdosSzekeres.Capstone.orient_pos_of_layers_runMidA_intrinsic` `LEANEXT/capstone/Capstone2.lean:2115`	uses `dist2_le_of_later` (Theorem 3.20); no saved axiom output (reports state standard axioms); polar forms and the angle conditions ((a): $0 < \beta - \alpha, 0 < \delta - \beta, \delta - \alpha < \pi$; (b)-(e): sorted angles, gap condition, pattern) are hypotheses; head cites (a), parts (b)-(e) cite the six Capstone declarations
Cor. 5.31	`ErdosSzekeres.Capstone.orient_pos_of_alternating_polar` `LEANEXT/capstone/Capstone.lean:1992`	no saved axiom output (`LEANEXT/capstone/REPORT.md` states standard axioms)
Rem. 5.32	–	scope remark: what the layer theorems assume
Section 6		
Def. 6.1	`ErdosSzekeres.LayerLemma.Fivefold35` `LEANEXT/es7corr/ES7Corollaries.lean:1157` `ErdosSzekeres.LayerLemma.Fourfold32` `LEANEXT/es7corr/ES7Corollaries.lean:1164`	the case classes: Prop definitions used only as hypotheses; never proved in Lean (supported outside Lean by SYM5 and C4CUT: computer-assisted, with written arguments)
Rem. 6.2	–	status of the case classes `Fivefold35/Fourfold32`: SAT certificates (SYM5, C4CUT; only `cake_lpr` verdicts count, lrat-check is secondary) plus written arguments outside Lean (encoder soundness, normal forms, centre perturbation; the records' reviews and refereeing were internal to the project); layer lemma and even-order ray fact have Lean-checked forms (Theorem 2.22; Lemma 4.23, Theorem 4.28) not linked to the formulas; distinct from the SYMLEAN constants
Thm. 6.3	`ErdosSzekeres.LayerLemma.fivefold_heptagon_of_erase` `LEANEXT/es7corr/ES7Corollaries.lean:1170`	takes hypotheses: Fivefold35; proof via Theorem 2.26
Thm. 6.4	`ErdosSzekeres.LayerLemma.fivefold_heptagon` `LEANEXT/es7corr/ES7Corollaries.lean:1183`	takes hypotheses: Fivefold35
Thm. 6.5	`ErdosSzekeres.LayerLemma.fivefold_heptagon_of_centre_not_mem` `LEANEXT/es7corr/ES7Corollaries.lean:1193`	takes hypotheses: Fivefold35
Thm. 6.6	`ErdosSzekeres.LayerLemma.fourfold_heptagon_of_erase` `LEANEXT/es7corr/ES7Corollaries.lean:1203`	takes hypotheses: Fourfold32; proof via Theorem 2.26
Thm. 6.7	`ErdosSzekeres.LayerLemma.fourfold_heptagon` `LEANEXT/es7corr/ES7Corollaries.lean:1216`	takes hypotheses: Fourfold32; uses Lemma 3.18
Rem. 6.8	–	conditional bounds 31 and 28; SYMMAX examples are exact computation outside Lean

Statement	Lean declaration; file and line	Note
Rem. 6.9	–	alternative route by orbit counting (not the Lean route)
Lem. 6.10	`RefuteC6.sphere_subset_extremePoints` ES8SYM/lean/C6check.lean:24	over C; any real r
Lem. 6.11	`RefuteC6.convexIndependent_of_norm_eq` ES8SYM/lean/C6check.lean:43	any index type
Def. 6.12	`RefuteC6.zeta` ES8SYM/lean/C6check.lean:60	$\exp(2\pi i/m)$, m cast to C
Lem. 6.13	`RefuteC6.zeta_prim` ES8SYM/lean/C6check.lean:62	Mathlib `Complex.isPrimitiveRoot_exp`
Fact 6.14	written step	written step: standard properties of primitive roots (Mathlib lemmas)
Lem. 6.15	`RefuteC6.orbit_injective` ES8SYM/lean/C6check.lean:66 `RefuteC6.orbit_convexIndependent` ES8SYM/lean/C6check.lean:75	
Thm. 6.16	`RefuteC6.part_a` ES8SYM/lean/C6check.lean:85	over C for an arbitrary invariant Set; no finiteness, no general position
Rem. 6.17	–	scope of `part_a`; planar link not in Lean; a rotation of order m is reduced to rotAbout(c,2π/m) by a power
Cor. 6.18	written step	written step: planar form of `part_a` (not in Lean)
Prop. 6.19	`RefuteC6.thr_m4` ES8SYM/lean/C6check.lean:108 `RefuteC6.thr_m5` ES8SYM/lean/C6check.lean:109 `RefuteC6.thr_m6` ES8SYM/lean/C6check.lean:110 `RefuteC6.thr_m7` ES8SYM/lean/C6check.lean:111 `RefuteC6.m4_with_centre_counter` ES8SYM/lean/C6check.lean:115	arithmetic (omega in Lean); `m4_with_centre_counter` has no axiom line
Rem. 6.20	–	geometric part of ES8SYM Lemma 2(b) is not in this file
Def. 6.21	`C7.cross` ES8SYM/lean/PolarOrient.lean:45 `C7.pt` ES8SYM/lean/PolarOrient.lean:48 `C7.coef` ES8SYM/lean/PolarOrient.lean:63 `C7.phase` ES8SYM/lean/PolarOrient.lean:94 `C7.ngonPt` ES8SYM/lean/PolarOrient.lean:97	
Lem. 6.22	`C7.det_mul_eq` ES8SYM/lean/PolarOrient.lean:24	
Lem. 6.23	`C7.orient_polar` ES8SYM/lean/PolarOrient.lean:37	not the radius-form `orient_polar` of LAYERLEAN
Lem. 6.24	`C7.orient_polar_cross` ES8SYM/lean/PolarOrient.lean:50	
Prop. 6.25	`C7.orient_layered` ES8SYM/lean/PolarOrient.lean:68	
Cor. 6.26	`C7.orient_layered_scale` ES8SYM/lean/PolarOrient.lean:80	
Lem. 6.27	`C7.ngonPt_eq_pt` ES8SYM/lean/PolarOrient.lean:100	
Thm. 6.28	`C7.nested_regular_orient` ES8SYM/lean/PolarOrient.lean:108	assumes only $r_o > 0$ (no nesting, no general position)
Lem. 6.29	`C7.orient_centre` ES8SYM/lean/PolarOrient.lean:116	
Rem. 6.30	–	cross versus orient; distinct from Lemma 3.35

Statement	Lean declaration; file and line	Note
Lem. 6.31	`PCompat.ne_zero_of_order` PRIMEFOLD/compat/lean/PCompatCore.lean:36 `PCompat.pow_inj_of_order` PRIMEFOLD/compat/lean/PCompatCore.lean:43 `PCompat.pow_mul_mem` PRIMEFOLD/compat/lean/PCompatCore.lean:58	commutative group with zero
Lem. 6.32	`PCompat.exists_card0` PRIMEFOLD/compat/lean/PCompatCore.lean:129 `PCompat.exists_card1` PRIMEFOLD/compat/lean/PCompatCore.lean:160	takes hypotheses hr0, hdisj on r (not discharged for C in Lean)
Def. 6.33	`PCompat.Compat` PRIMEFOLD/compat/lean/PCompatCore.lean:205	
Prop. 6.34	`PCompat.compat_even` PRIMEFOLD/compat/lean/PCompatCore.lean:208 `PCompat.compat_odd` PRIMEFOLD/compat/lean/PCompatCore.lean:235	
Thm. 6.35	`PCompat.dvd_card` PRIMEFOLD/compat/lean/PCompatCore.lean:67 `PCompat.card_mod` PRIMEFOLD/compat/lean/PCompatCore.lean:112 `PCompat.exists_iff` PRIMEFOLD/compat/lean/PCompatCore.lean:174 `PCompat.classification` PRIMEFOLD/compat/lean/PCompatCore.lean:253	`exists_iff` takes hypotheses hr0, hdisj on r
Prop. 6.36	`PCompat.ex341` PRIMEFOLD/compat/lean/PCompatCore.lean:277	computer-assisted in Lean; written proof by hand (Lean: kernel decide; axioms propext only)
Rem. 6.37	–	Fermat little theorem and general-position existence not formalized
Lem. 6.38	`LOCheck.arith` PRIMEFOLD/compat/lean/LOCheck.lean:21	
Lem. 6.39	`LOCheck.orbit_dvd` PRIMEFOLD/compat/lean/LOCheck.lean:60	written proof derives it from Theorem 6.35(a); Lean re-proves it independently
Thm. 6.40	`LOCheck.L0` PRIMEFOLD/compat/lean/LOCheck.lean:122 `LOCheck.L0_rot` PRIMEFOLD/compat/lean/LOCheck.lean:144	no general position assumed
Rem. 6.41	–	parity excludes the centre for even m; Fermat not used
Section 7		
Def. 7.1	`ErdosSzekeres.SymReduction.eDist2` SYMLEAN/SymmetricReduction.lean:265 `ErdosSzekeres.SymReduction.IsEucIsometry` SYMLEAN/SymmetricReduction.lean:268	
Rem. 7.2	–	remark: Mathlib sup metric on Point; why `eDist2` is used
Def. 7.3	`ErdosSzekeres.SymReduction.Invariant` SYMLEAN/SymmetricReduction.lean:271	same notion as Definition 1.10 (maps into)
Def. 7.4	`ErdosSzekeres.SymReduction.toC` SYMLEAN/SymmetricReduction.lean:274 `ErdosSzekeres.SymReduction.ofC` SYMLEAN/SymmetricReduction.lean:277	

Statement	Lean declaration; file and line	Note
Lem. 7.5	`ErdosSzekeres.SymReduction.toC_re` SYMLEAN/SymmetricReduction.lean:279 `ErdosSzekeres.SymReduction.toC_im` SYMLEAN/SymmetricReduction.lean:280 `ErdosSzekeres.SymReduction.ofC_zero` SYMLEAN/SymmetricReduction.lean:281 `ErdosSzekeres.SymReduction.ofC_one` SYMLEAN/SymmetricReduction.lean:282 `ErdosSzekeres.SymReduction.toC_ofC` SYMLEAN/SymmetricReduction.lean:284 `ErdosSzekeres.SymReduction.ofC_toC` SYMLEAN/SymmetricReduction.lean:287 `ErdosSzekeres.SymReduction.toC_injective` SYMLEAN/SymmetricReduction.lean:291 `ErdosSzekeres.SymReduction.eDist2_eq_normSq` SYMLEAN/SymmetricReduction.lean:294 `ErdosSzekeres.SymReduction.eDist2_pos_of_ne` SYMLEAN/SymmetricReduction.lean:297 `ErdosSzekeres.SymReduction.point_ext` SYMLEAN/SymmetricReduction.lean:502	
Def. 7.6	`ErdosSzekeres.SymReduction.crot` SYMLEAN/SymmetricReduction.lean:349 `ErdosSzekeres.SymReduction.rotAbout` SYMLEAN/SymmetricReduction.lean:400 `ErdosSzekeres.SymReduction.halfTurn` SYMLEAN/SymmetricReduction.lean:413 `ErdosSzekeres.SymReduction.reflLine` SYMLEAN/SymmetricReduction.lean:423	`rotAbout` is the formula of Definition 1.12
Def. 7.7	`ErdosSzekeres.SymReduction.HalfTurn34` SYMLEAN/SymmetricReduction.lean:467 `ErdosSzekeres.SymReduction.Threefold33` SYMLEAN/SymmetricReduction.lean:472 `ErdosSzekeres.SymReduction.Fivefold35` SYMLEAN/SymmetricReduction.lean:477 `ErdosSzekeres.SymReduction.MirrorOneAxis33` SYMLEAN/SymmetricReduction.lean:482 `ErdosSzekeres.SymReduction.MirrorNoAxis34` SYMLEAN/SymmetricReduction.lean:488 `ErdosSzekeres.SymReduction.Fourfold32` SYMLEAN/SymmetricReduction.lean:496	Props used only as hypotheses; none proved in this section
Rem. 7.8	–	remark: project records supplying each class, with the rigor labels that the records claim (C4CUT states Fourfold32 without one)
Lem. 7.9	`ErdosSzekeres.SymReduction.iterate_mem` SYMLEAN/SymmetricReduction.lean:69 `ErdosSzekeres.SymReduction.Invariant.iterate` SYMLEAN/SymmetricReduction.lean:309	
Lem. 7.10	written step	written step (local statement pigeon in `isometry_structure`)
Def. 7.11	written step	written step (Mathlib `Function.minimalPeriod`)
Lem. 7.12	written step	written step (Mathlib minimal-period facts, proved)
Def. 7.13	`ErdosSzekeres.SymReduction.orbitFS` SYMLEAN/SymmetricReduction.lean:65	

Statement	Lean declaration; file and line	Note
Lem. 7.14	`ErdosSzekeres.SymReduction.card_orbitFS_of_minimalPeriod` SYMLEAN/SymmetricReduction.lean:111 `ErdosSzekeres.SymReduction.mem_orbitFS_self` SYMLEAN/SymmetricReduction.lean:90 `ErdosSzekeres.SymReduction.orbitFS_subset` SYMLEAN/SymmetricReduction.lean:93 `ErdosSzekeres.SymReduction.orbitFS_invariant` SYMLEAN/SymmetricReduction.lean:99 `ErdosSzekeres.SymReduction.card_orbitFS` SYMLEAN/SymmetricReduction.lean:76	
Lem. 7.15	written step	written step (the induction shared by `orbit_select` and `orbit_select_minimalPeriod`)
Lem. 7.16	`ErdosSzekeres.SymReduction.orbit_select_minimalPeriod` SYMLEAN/SymmetricReduction.lean:123 `ErdosSzekeres.SymReduction.orbit_select` SYMLEAN/SymmetricReduction.lean:194	
Lem. 7.17	`ErdosSzekeres.SymReduction.toC_crot` SYMLEAN/SymmetricReduction.lean:351 `ErdosSzekeres.SymReduction.crot_one` SYMLEAN/SymmetricReduction.lean:354 `ErdosSzekeres.SymReduction.crot_comp` SYMLEAN/SymmetricReduction.lean:359 `ErdosSzekeres.SymReduction.crot_iterate` SYMLEAN/SymmetricReduction.lean:364 `ErdosSzekeres.SymReduction.crot_eq_self_iff` SYMLEAN/SymmetricReduction.lean:370 `ErdosSzekeres.SymReduction.crot_injective` SYMLEAN/SymmetricReduction.lean:384 `ErdosSzekeres.SymReduction.eDist2_crot_center` SYMLEAN/SymmetricReduction.lean:394	
Lem. 7.18	`ErdosSzekeres.SymReduction.rotAbout_eq_crot` SYMLEAN/SymmetricReduction.lean:404 `ErdosSzekeres.SymReduction.halfTurn_eq_crot` SYMLEAN/SymmetricReduction.lean:415	
Lem. 7.19	`ErdosSzekeres.SymReduction.toC_reflLine` SYMLEAN/SymmetricReduction.lean:427 `ErdosSzekeres.SymReduction.reflLine_involutive` SYMLEAN/SymmetricReduction.lean:435 `ErdosSzekeres.SymReduction.cross_of_reflLine_fixed` SYMLEAN/SymmetricReduction.lean:447 `ErdosSzekeres.SymReduction.orient_eq_zero_of_cross` SYMLEAN/SymmetricReduction.lean:455	
Lem. 7.20	`ErdosSzekeres.SymReduction.IsEucIsometry.injective` SYMLEAN/SymmetricReduction.lean:301 `ErdosSzekeres.SymReduction.isometry_toC_conj` SYMLEAN/SymmetricReduction.lean:313	
Thm. 7.21	`ErdosSzekeres.SymReduction.isometry_complex_cases` SYMLEAN/SymmetricReduction.lean:321 `ErdosSzekeres.SymReduction.isometry_structure` SYMLEAN/SymmetricReduction.lean:869	written proof of part (a) by an elementary argument instead of the Mathlib results used in Lean (a construction and a theorem)
Lem. 7.22	`ErdosSzekeres.Suk.convexPosition_of_strict_supports` ErdosSzekeres/Suk/SupportingLines.lean:28	main-library dependency of `convexPosition_of_concyclic`; same declaration as Lemma 1.32 (proof repeated)
Lem. 7.23	`ErdosSzekeres.SymReduction.convexPosition_of_concyclic` SYMLEAN/SymmetricReduction.lean:509	
Cor. 7.24	`ErdosSzekeres.SymReduction.hasConvexNGon_of_concyclic` SYMLEAN/SymmetricReduction.lean:522	
Lem. 7.25	`ErdosSzekeres.SymReduction.center_not_mem_of_halfTurn` SYMLEAN/SymmetricReduction.lean:528	
Def. 7.26	written step	written step (Mathlib `IsPrimitiveRoot`, `orderOf`)
Lem. 7.27	written step	written step (Mathlib facts on roots of unity, proved)

Statement	Lean declaration; file and line	Note
Lem. 7.28	`ErdosSzekeres.SymReduction.isPrimitiveRoot_rot3` `SYMLEAN/SymmetricReduction.lean:584` `ErdosSzekeres.SymReduction.isPrimitiveRoot_rot5` `SYMLEAN/SymmetricReduction.lean:591` `ErdosSzekeres.SymReduction.isPrimitiveRoot_rot4` `SYMLEAN/SymmetricReduction.lean:598`	
Lem. 7.29	`ErdosSzekeres.SymReduction.root_cases` `SYMLEAN/SymmetricReduction.lean:552`	
Lem. 7.30	`ErdosSzekeres.SymReduction.prime_cases` `SYMLEAN/SymmetricReduction.lean:572`	
Lem. 7.31	`ErdosSzekeres.SymReduction.root_cases2` `SYMLEAN/SymmetricReduction.lean:607`	
Lem. 7.32	`ErdosSzekeres.SymReduction.halfTurn_facts` `SYMLEAN/SymmetricReduction.lean:647`	
Lem. 7.33	`ErdosSzekeres.SymReduction.case_halfTurn` `SYMLEAN/SymmetricReduction.lean:670`	conditional; takes hypotheses: `HalfTurn34`
Lem. 7.34	`ErdosSzekeres.SymReduction.rot_erase_facts` `SYMLEAN/SymmetricReduction.lean:676`	
Lem. 7.35	`ErdosSzekeres.SymReduction.case_rot_big` `SYMLEAN/SymmetricReduction.lean:711`	no general position assumed
Lem. 7.36	`ErdosSzekeres.SymReduction.case_odd_prime` `SYMLEAN/SymmetricReduction.lean:736`	conditional; takes hypotheses: Threefold33, Fivefold35
Lem. 7.37	`ErdosSzekeres.SymReduction.case_fourfold` `SYMLEAN/SymmetricReduction.lean:763`	conditional; takes hypotheses: Fourfold32
Lem. 7.38	`ErdosSzekeres.SymReduction.case_mirror` `SYMLEAN/SymmetricReduction.lean:811`	conditional; takes hypotheses: `MirrorOneAxis33`, `MirrorNoAxis34`
Thm. 7.39	`ErdosSzekeres.SymReduction.heptagon_of_nontrivial_isometry` `SYMLEAN/SymmetricReduction.lean:997`	conditional; takes hypotheses: `HalfTurn34`, Threefold33, Fivefold35, `MirrorOneAxis33`, `MirrorNoAxis34`; axioms log `SYMLEAN/logs/SymmetricReductionAxioms.out`
Rem. 7.40	–	remark: conditional status of the main theorem; the unconditional statement is a project claim (`WRITEUP/SYMMETRIC.md`, Theorem A), established here only as the implication
Thm. 7.41	`ErdosSzekeres.SymReduction.halfTurn_of_symmetric_counterexample` `SYMLEAN/SymmetricReduction.lean:1024`	conditional; takes hypotheses: Threefold33, Fourfold32, Fivefold35, `MirrorOneAxis33`, `MirrorNoAxis34`; axioms log `SYMLEAN/logs/SymmetricReductionAxioms.out`
Rem. 7.42	–	remark: per-symmetry, isometries only; the group form `VariantChecks.group_form` (review file `SYMLEAN/review/VariantChecks.lean:1168`, log `SYMLEAN/logs/VariantChecks.out`) with a written argument, not restated as a numbered statement
Def. 7.43	`ErdosSzekeres.SymReduction.MirrorExclusionLineCopy` `SYMLEAN/SymmetricReduction.lean:1052`	text of `Mir.MirrorExclusionLine` (`H3/MirrorEncoding/Line.lean:94`)
Lem. 7.44	`ErdosSzekeres.SymReduction.mirrorOneAxis33_of_exclusionLine` `SYMLEAN/SymmetricReduction.lean:1056`	conditional; takes hypotheses: `MirrorExclusionLineCopy`
Rem. 7.45	–	remark: the H3 statement is Lean-checked only conditionally on CNF facts (`V2L.mirrorExclusionLine_final`) and is identified with `MirrorExclusionLineCopy` by its text; the converse and the equivalence (review file `SYMLEAN/review/ReviewSanity.lean:1593` and 1629) with a written argument, not restated as numbered statements
Rem. 7.46	–	remark: hypotheses of the classes implied by the others (not used)

Section 8

Def. 8.1	`ErdosSzekeres.SymReduction.nsq` `SYMLEAN/AffineReduction.lean:1088`	

Statement	Lean declaration; file and line	Note
Lem. 8.2	`ErdosSzekeres.SymReduction.mulVec2_fst` SYMLEAN/AffineReduction.lean:1097 `ErdosSzekeres.SymReduction.mulVec2_snd` SYMLEAN/AffineReduction.lean:1101	
Lem. 8.3	`ErdosSzekeres.SymReduction.nsq_nonneg` SYMLEAN/AffineReduction.lean:1090 `ErdosSzekeres.SymReduction.eDist2_eq_nsq` SYMLEAN/AffineReduction.lean:1094	
Def. 8.4	`ErdosSzekeres.SymReduction.gA` SYMLEAN/AffineReduction.lean:1106 `ErdosSzekeres.SymReduction.gB` SYMLEAN/AffineReduction.lean:1107 `ErdosSzekeres.SymReduction.gC` SYMLEAN/AffineReduction.lean:1108	
Lem. 8.5	`ErdosSzekeres.SymReduction.nsq_mulVec` SYMLEAN/AffineReduction.lean:1110	
Def. 8.6	`ErdosSzekeres.SymReduction.avgQ` SYMLEAN/AffineReduction.lean:1116 `ErdosSzekeres.SymReduction.qA` SYMLEAN/AffineReduction.lean:1120 `ErdosSzekeres.SymReduction.qB` SYMLEAN/AffineReduction.lean:1121 `ErdosSzekeres.SymReduction.qC` SYMLEAN/AffineReduction.lean:1122	
Lem. 8.7	`ErdosSzekeres.SymReduction.avgQ_eq` SYMLEAN/AffineReduction.lean:1124	
Lem. 8.8	`ErdosSzekeres.SymReduction.avgQ_ge` SYMLEAN/AffineReduction.lean:1133 `ErdosSzekeres.SymReduction.qA_pos` SYMLEAN/AffineReduction.lean:1152 `ErdosSzekeres.SymReduction.qdisc_pos` SYMLEAN/AffineReduction.lean:1158	each part assumes $0 < n$
Lem. 8.9	`ErdosSzekeres.SymReduction.avgQ_mulVec` SYMLEAN/AffineReduction.lean:1139	assumes $L^n = 1$ only (no positivity of n)
Def. 8.10	`ErdosSzekeres.SymReduction.lin2` SYMLEAN/AffineReduction.lean:1167	bundled linear map
Lem. 8.11	`ErdosSzekeres.SymReduction.lin2_fst` SYMLEAN/AffineReduction.lean:1174 `ErdosSzekeres.SymReduction.lin2_snd` SYMLEAN/AffineReduction.lean:1175 `ErdosSzekeres.SymReduction.orient_lin2` SYMLEAN/AffineReduction.lean:1177	
Lem. 8.12	`ErdosSzekeres.SymReduction.nsq_lin2_chol` SYMLEAN/AffineReduction.lean:1184 `ErdosSzekeres.SymReduction.lin2_chol_inv_left` SYMLEAN/AffineReduction.lean:1191 `ErdosSzekeres.SymReduction.lin2_chol_inv_right` SYMLEAN/AffineReduction.lean:1195	(a) assumes $s \neq 0$; (b),(c) assume $s \neq 0$ and $t \neq 0$
Lem. 8.13	`ErdosSzekeres.SymReduction.iterate_sub_of_affine` SYMLEAN/AffineReduction.lean:1199	
Lem. 8.14	`ErdosSzekeres.SymReduction.exists_periodic_of_injective` SYMLEAN/AffineReduction.lean:1208	
Lem. 8.15	`ErdosSzekeres.SymReduction.matrix_eq_one_of_fix` SYMLEAN/AffineReduction.lean:1217	
Lem. 8.16	`ErdosSzekeres.SymReduction.orient_affine` SYMLEAN/AffineReduction.lean:1241	
Lem. 8.17	`ErdosSzekeres.SymReduction.injective_of_affine` SYMLEAN/AffineReduction.lean:1305	
Lem. 8.18	`ErdosSzekeres.SymReduction.affineMap_eq_mulVec` SYMLEAN/AffineReduction.lean:1284	
Lem. 8.19	`ErdosSzekeres.SymReduction.generalPosition_image_of_orient` SYMLEAN/AffineReduction.lean:1249	injectivity of f not assumed

Statement	Lean declaration; file and line	Note
Lem. 8.20	`ErdosSzekeres.SymReduction.generalPosition_image_affine` `SYMLEAN/AffineReduction.lean:1278` `ErdosSzekeres.SymReduction.generalPosition_image_affineEquiv` `SYMLEAN/AffineReduction.lean:1290`	
Lem. 8.21	`ErdosSzekeres.SymReduction.convexPosition_image_affine` `SYMLEAN/AffineReduction.lean:1263`	
Lem. 8.22	`ErdosSzekeres.SymReduction.hasConvexNGon_image_affine` `SYMLEAN/AffineReduction.lean:1298`	
Def. 8.23	written step	written step: the proposition E(N,n), the type of the hypothesis `hE` (no Lean name)
Prop. 8.24	written step	written step; $L^m = 1$ and parts (b),(c) and $g \neq \mathrm{id} \Rightarrow h \neq \mathrm{id}$ are steps inside the Lean proof of `hasConvexNGon_of_affine_symmetry`; $g^m = \mathrm{id}$ and the rest of (d) (orders) are not in Lean
Thm. 8.25	`ErdosSzekeres.SymReduction.hasConvexNGon_of_affine_symmetry` `SYMLEAN/AffineReduction.lean:1330`	takes hypothesis `hE` (the Euclidean statement E(N,n) for the same N and n, a parameter) and $3 \leq N$
Rem. 8.26	–	role of `hE`; order of g not tracked; order-preserving form not formalized
Cor. 8.27	`ErdosSzekeres.SymReduction.hasConvexNGon_of_affineMap_symmetry` `SYMLEAN/AffineReduction.lean:1423` `ErdosSzekeres.SymReduction.hasConvexNGon_of_affineEquiv_symmetry` `SYMLEAN/AffineReduction.lean:1433` `ErdosSzekeres.SymReduction.hasConvexNGon_of_affine_coords_symmetry` `SYMLEAN/AffineReduction.lean:1444`	takes hypothesis `hE` (E(N,n)) and $3 \leq N$
Thm. 8.28	`ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affineEquiv` `SYMLEAN/AffineReduction.lean:1501` `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affineEquiv_image` `SYMLEAN/AffineReduction.lean:1510` `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affineMap` `SYMLEAN/AffineReduction.lean:1489` `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affine` `SYMLEAN/AffineReduction.lean:1477` `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affine_coords` `SYMLEAN/AffineReduction.lean:1521`	takes hypotheses: `HalfTurn34`, Threefold33, Fivefold35, `MirrorOneAxis33`, `MirrorNoAxis34`
Rem. 8.29	–	what supplies the five class hypotheses and with which rigor label; the mirror theorems concern byte-identical copies in `SYMLEAN/mirror/ReductionThreeClasses.lean`; the unconditional statement is a project claim (`WRITEUP/SYMMETRIC.md`, Theorem A), established here only conditionally
Rem. 8.30	–	what the statements do not say

Section 9

Def. 9.1	`MirrorOdd.mir` `MIRRORETV/lean/MirrorMain.lean:16` `MirrorOdd.Rset` `MIRRORETV/lean/MirrorMain.lean:1598` `MirrorOdd.Aset` `MIRRORETV/lean/MirrorMain.lean:1599`	

Statement	Lean declaration; file and line	Note

Lem. 9.2	`MirrorOdd.mir_zero`
	`MIRRORETV/lean/MirrorMain.lean:18`
	`MirrorOdd.mir_one`
	`MIRRORETV/lean/MirrorMain.lean:19`
	`MirrorOdd.mir_mir`
	`MIRRORETV/lean/MirrorMain.lean:1543`
	`MirrorOdd.orient_mir3`
	`MIRRORETV/lean/MirrorMain.lean:2187`
	`MirrorOdd.orient_mir2`
	`MIRRORETV/lean/MirrorMain.lean:2190`
	`MirrorOdd.orient_cycl`
	`MIRRORETV/lean/MirrorMain.lean:2193`
	`MirrorOdd.orient_swap12`
	`MIRRORETV/lean/MirrorMain.lean:2196`
	`MirrorOdd.orient_rev`
	`MIRRORETV/lean/MirrorMain.lean:2199`
		part (e) is a written step (identity used inside `cap_below_lines`)

Def. 9.3	`ErdosSzekeres.Omission.Configuration`
	`ErdosSzekeres/Omission/Realization.lean:9`
	`ErdosSzekeres.Omission.Configuration.HasCap`
	`ErdosSzekeres/Omission/Realization.lean:64`
	`MirrorOdd.xset`
	`MIRRORETV/lean/MirrorMain.lean:1549`
	`MirrorOdd.emb`
	`MIRRORETV/lean/MirrorMain.lean:1553`
	`MirrorOdd.sortPt`
	`MIRRORETV/lean/MirrorMain.lean:1558`
	`MirrorOdd.cfg`
	`MIRRORETV/lean/MirrorMain.lean:1584`
		`HasCap` is the all-triples cap

Lem. 9.5	`MirrorOdd.xset_card`
	`MIRRORETV/lean/MirrorMain.lean:1551`
	`MirrorOdd.emb_mem`
	`MIRRORETV/lean/MirrorMain.lean:1555`
	`MirrorOdd.sortPt_mem`
	`MIRRORETV/lean/MirrorMain.lean:1560`
	`MirrorOdd.sortPt_x`
	`MIRRORETV/lean/MirrorMain.lean:1563`
	`MirrorOdd.sortPt_strictMono`
	`MIRRORETV/lean/MirrorMain.lean:1566`
	`MirrorOdd.sortPt_injective`
	`MIRRORETV/lean/MirrorMain.lean:1572`
	`MirrorOdd.sortPt_image`
	`MIRRORETV/lean/MirrorMain.lean:1576`
		part (h) is the `generalPosition` field of `MirrorOdd.cfg` (`MIRRORETV/lean/MirrorMain.lean:1587`)

Lem. 9.6	`MirrorOdd.hasConvexNGon_of_le`
	`MIRRORETV/lean/MirrorMain.lean:2182`
	`MirrorOdd.hasConvexNGon_affineMap`
	`MIRRORETV/lean/MirrorMain.lean:1859`	

Lem. 9.7	`MirrorOdd.cap_below_lines` `MIRRORETV/lean/MirrorMain.lean:2204`	part (b) is a written step (inline in `mirror_odd_general` at `MIRRORETV/lean/MirrorMain.lean:2542-2557`; local claim interior at `MIRRORETV/lean/MirrorMain.lean:2060` and `MIRRORETV/lean/MirrorMain.lean:1716`)
Thm. 9.8	`MirrorOdd.cap_mirror_polygon` `MIRRORETV/lean/MirrorMain.lean:2233`	proof follows Lean (every triple of an explicit cyclic order, taken in list order, positive)

Thm. 9.10	`MirrorOdd.axis_left_polygon`
	`MIRRORETV/lean/MirrorMain.lean:2309`
	`MirrorOdd.axis_right_polygon`
	`MIRRORETV/lean/MirrorMain.lean:2403`
		proof follows Lean

Statement	Lean declaration; file and line	Note

Lem. 9.12	`MirrorOdd.cyc4`
	`MIRRORETV/lean/MirrorMain.lean:995`
	`MirrorOdd.cyc5`
	`MIRRORETV/lean/MirrorMain.lean:1026`
	`MirrorOdd.cyc6`
	`MIRRORETV/lean/MirrorMain.lean:1068`
	`MirrorOdd.cyc7`
	`MIRRORETV/lean/MirrorMain.lean:22`
	`MirrorOdd.cyc8`
	`MIRRORETV/lean/MirrorMain.lean:103`
		machine-generated Lean code; instance of Lemma 1.38

Prop. 9.13	`MirrorOdd.cap_mirror_quadrilateral`
	`MIRRORETV/lean/MirrorMain.lean:1126`
	`MirrorOdd.cap_mirror_pentagon`
	`MIRRORETV/lean/MirrorMain.lean:1282`
	`MirrorOdd.cap_mirror_hexagon`
	`MIRRORETV/lean/MirrorMain.lean:1168`
	`MirrorOdd.cap_mirror_heptagon`
	`MIRRORETV/lean/MirrorMain.lean:212`
	`MirrorOdd.cap_mirror_octagon`
	`MIRRORETV/lean/MirrorMain.lean:392`
	`MirrorOdd.axis_left_pentagon`
	`MIRRORETV/lean/MirrorMain.lean:1358`
	`MirrorOdd.axis_right_pentagon`
	`MIRRORETV/lean/MirrorMain.lean:1433`
	`MirrorOdd.axis_left_heptagon`
	`MIRRORETV/lean/MirrorMain.lean:639`
	`MirrorOdd.axis_right_heptagon`
	`MIRRORETV/lean/MirrorMain.lean:816`
		machine-generated Lean code (linarith certificates); written proof differs: derived from Theorem 9.8 and Theorem 9.10

Lem. 9.14	`MirrorOdd.card_split`
	`MIRRORETV/lean/MirrorMain.lean:1601`
	`MirrorOdd.card_A_le`
	`MIRRORETV/lean/MirrorMain.lean:1636`	

Lem. 9.15	`MirrorOdd.injOn_x` `MIRRORETV/lean/MirrorMain.lean:1647`	
Lem. 9.16	written step	written step; isolates the cap argument of the local claims `keyT` and `hR` in `mirror_odd_general`, `mirror_even_general`, `mirror_five`, `mirror_six`
Thm. 9.17	`MirrorOdd.mirror_odd_general` `MIRRORETV/lean/MirrorMain.lean:2498`	takes a parametric hypothesis: `hM` (an assumed bound M for configurations with no (j+1)-cap and no convex (2j+1)-gon, not an unproved claim; instances discharged by Lean-checked theorems, see Remark 9.19); Lean j (used in this section) = `README/PROOFS` j minus 1
Thm. 9.18	`MirrorOdd.mirror_even_general` `MIRRORETV/lean/MirrorMain.lean:2599`	takes a parametric hypothesis: `hM` (an assumed bound M for configurations with no j-cap and no convex 2j-gon, not an unproved claim; instances discharged by Lean-checked theorems, see Remark 9.19)
Rem. 9.19	–	remark (no statement): (`hM`) is a parametric hypothesis (an assumption on M, not an unproved claim); Lean-checked instances (3,16), (4,22), (4,113) via `MirrorOdd.capacity16`, `MirrorOdd.capacity22g`, `C12Check.capacity113` (Baek; cups-caps for (4,113)); small j; relation to N(n,a,b); index shift

Statement	Lean declaration; file and line	Note

Def. 9.20	`MirrorOdd.lineRefl`
	MIRRORETV/lean/MirrorMain.lean:1879
	`MirrorOdd.frame`
	MIRRORETV/lean/MirrorMain.lean:1883
	`MirrorOdd.unframeLin`
	MIRRORETV/lean/MirrorMain.lean:1886
	`MirrorOdd.unframe`
	MIRRORETV/lean/MirrorMain.lean:1892	

Lem. 9.22	`MirrorOdd.unframe_apply`
	MIRRORETV/lean/MirrorMain.lean:1897
	`MirrorOdd.unframe_frame`
	MIRRORETV/lean/MirrorMain.lean:1900
	`MirrorOdd.frame_unframe`
	MIRRORETV/lean/MirrorMain.lean:1907
	`MirrorOdd.frame_lineRefl`
	MIRRORETV/lean/MirrorMain.lean:1914
	`MirrorOdd.orient_frame`
	MIRRORETV/lean/MirrorMain.lean:1920
	`MirrorOdd.frame_injective`
	MIRRORETV/lean/MirrorMain.lean:1925
	`MirrorOdd.unframe_injective`
	MIRRORETV/lean/MirrorMain.lean:1930	

Thm. 9.23	`MirrorOdd.transport`
	MIRRORETV/lean/MirrorMain.lean:1937
	`MirrorOdd.mirror_odd_general_line`
	MIRRORETV/lean/MirrorMain.lean:2641
	`MirrorOdd.mirror_even_general_line`
	MIRRORETV/lean/MirrorMain.lean:2650
		part (a) is transport; parts (b), (c) take the parametric hypothesis `hM` (as in Theorem 9.17 / Theorem 9.18)

Lem. 9.24	`MirrorOdd.hasCap3_of`
	MIRRORETV/lean/MirrorMain.lean:1991
	`MirrorOdd.capacity_cup`
	MIRRORETV/lean/MirrorMain.lean:2002	

Thm. 9.25	`MirrorOdd.mirror_four`
	MIRRORETV/lean/MirrorMain.lean:2019
	`MirrorOdd.mirror_five`
	MIRRORETV/lean/MirrorMain.lean:2040
	`MirrorOdd.mirror_six`
	MIRRORETV/lean/MirrorMain.lean:2129
	`MirrorOdd.mirror_four_line`
	MIRRORETV/lean/MirrorMain.lean:2159
	`MirrorOdd.mirror_five_line`
	MIRRORETV/lean/MirrorMain.lean:2165
	`MirrorOdd.mirror_six_line`
	MIRRORETV/lean/MirrorMain.lean:2171
		unconditional (no hypotheses beyond the stated ones)

Section 10		

Def. 10.1	`Config`
	ErdosSzekeres/Vendor/ETV/Config/Defs.lean:20
	`Config.Cap3`
	ErdosSzekeres/Vendor/ETV/Config/Defs.lean:30
	`Config.Cap`
	ErdosSzekeres/Vendor/ETV/Config/Defs.lean:38
	`Config.Cup`
	ErdosSzekeres/Vendor/ETV/Config/Defs.lean:41
	`Config.Gon`
	ErdosSzekeres/Vendor/ETV/Config/Defs.lean:45
	`Config.NGon`
	ErdosSzekeres/Vendor/ETV/Config/Defs.lean:57
	`Config.HasNCap`
	ErdosSzekeres/Vendor/ETV/Config/Defs.lean:60
	`Config.HasNCup`
	ErdosSzekeres/Vendor/ETV/Config/Defs.lean:63
	`Config.HasNGon`
	ErdosSzekeres/Vendor/ETV/Config/Defs.lean:66
		definitions of Baek's abstract configurations (vendored formalization)

Statement	Lean declaration; file and line	Note
Rem. 10.2	–	remark (no statement): Lean encoding of Definition 10.1

Lem. 10.3	`hasNCap_supset`
	`ErdosSzekeres/Vendor/ETV/Config/Lemmas.lean:293`
	`hasNGon_supset`
	`ErdosSzekeres/Vendor/ETV/Config/Lemmas.lean:305`
	`ncup_is_ngon`
	`ErdosSzekeres/Vendor/ETV/Config/Lemmas.lean:275`
		vendored lemmas; proved here

Thm. 10.4	`main` `ErdosSzekeres/Vendor/ETV/Main/Main.lean:87`	cited (J. Baek, Eur. J. Combin. 124 (2025) 104085; `arXiv:2206.04260`); vendored Lean formalization; not reproved here
Rem. 10.5	–	remark (no statement): source, status, index shift main $n \leftrightarrow$ Baek at $N = n + 3$
Thm. 10.6	`Config.cap_cup` `ErdosSzekeres/Vendor/ETV/Main/CapCup.lean:46`	cited (Erdos-Szekeres 1935 cups-caps theorem); vendored Lean formalization; not reproved here
Rem. 10.7	–	remark (no statement): source of the cups-caps theorem

Lem. 10.8	`ErdosSzekeres.Omission.Baek.four_cap_or_eight_gon`
	`ErdosSzekeres/Omission/Baek.lean:12`
	`ErdosSzekeres.Omission.Baek.capacity22`
	`ErdosSzekeres/Omission/Baek.lean:16`
		instance $n = 5$ of the cited Theorem 10.4

Def. 10.9	`ErdosSzekeres.Omission.Configuration.cup`
	`ErdosSzekeres/Omission/Realization.lean:33`
	`ErdosSzekeres.Omission.ActualRanks.colorConfig`
	`ErdosSzekeres/Omission/ActualRanks.lean:28`
	`ErdosSzekeres.Omission.ListGeometry.pointAt`
	`ErdosSzekeres/Omission/ListGeometry.lean:13`
		definitions (bridge): cup colouring, `colorConfig c.cup` true, `pointAt`

Lem. 10.10	`ErdosSzekeres.Omission.Configuration.cup_true_iff`
	`ErdosSzekeres/Omission/Realization.lean:39`
	`ErdosSzekeres.Omission.ListGeometry.cup_sign`
	`ErdosSzekeres/Omission/ListGeometry.lean:19`
	`ErdosSzekeres.Omission.ListGeometry.cap_sign`
	`ErdosSzekeres/Omission/ListGeometry.lean:24`
		bridge lemma; `cup_true_iff` takes Fin indices, `cup_sign` and `cap_sign` natural-number indices with bounds; `cup_false_iff` is not on the proof paths and not covered

Lem. 10.11	`ErdosSzekeres.Omission.PlanarChains.four_point_identity_left`
	`ErdosSzekeres/Omission/PlanarChains.lean:15`
	`ErdosSzekeres.Omission.PlanarChains.four_point_identity_right`
	`ErdosSzekeres/Omission/PlanarChains.lean:19`
	`ErdosSzekeres.Omission.PlanarChains.positive_four`
	`ErdosSzekeres/Omission/PlanarChains.lean:23`
	`ErdosSzekeres.Omission.PlanarChains.negative_four`
	`ErdosSzekeres/Omission/PlanarChains.lean:35`
		bridge lemma; identities checked symbolically

Lem. 10.12	`ErdosSzekeres.Omission.PlanarChains.chain3_all` `ErdosSzekeres/Omission/PlanarChains.lean:58`	bridge lemma

Lem. 10.13	`ErdosSzekeres.Omission.ListGeometry.cup_list_all`
	`ErdosSzekeres/Omission/ListGeometry.lean:39`
	`ErdosSzekeres.Omission.ListGeometry.cap_list_all`
	`ErdosSzekeres/Omission/ListGeometry.lean:53`
	`ErdosSzekeres.Omission.ListGeometry.hasCap_of_list`
	`ErdosSzekeres/Omission/ListGeometry.lean:85`
		bridge lemma

Lem. 10.14	`ErdosSzekeres.Omission.BoundaryGeometry.boundary_polygon`
	`ErdosSzekeres/Omission/BoundaryGeometry.lean:403`
	`ErdosSzekeres.Omission.BoundaryGeometry.Boundary.card`
	`ErdosSzekeres/Omission/BoundaryGeometry.lean:359`
	`ErdosSzekeres.Omission.BoundaryGeometry.boundary_convex_position`
	`ErdosSzekeres/Omission/BoundaryGeometry.lean:317`
		bridge lemma; stated after the change of variables $v =$ reflect(upperReflected), see Remark 10.15

Rem. 10.15	–	remark (no statement): Lean encoding of Lemma 10.14

Statement	Lean declaration; file and line	Note

Prop. 10.16	`MirrorOdd.hasNGon_heptagon`
	`MIRRORETV/lean/MirrorMain.lean:1512`
	`ErdosSzekeres.Omission.BaekGeometry.hasNGon_octagon`
	`ErdosSzekeres/Omission/BaekGeometry.lean:94`
	`C12Check.hasNGon_nonagon`
	`MIRRORETV/lean/MirrorNine_append.lean:99`
		bridge; three Lean declarations ($m = 7$, 8, 9) whose proofs are identical up to the value of m

Prop. 10.17	`MirrorOdd.capacity16` `MIRRORETV/lean/MirrorMain.lean:1529`	uses the cited Theorem 10.4 ($n = 4$); binomial value $C(6, 2) = 15$ by kernel decide
Lem. 10.18	`ErdosSzekeres.Omission.BaekGeometry.geometric_capacity22` `ErdosSzekeres/Omission/BaekGeometry.lean:111`	uses the cited Theorem 10.4 ($n = 5$)
Prop. 10.19	`MirrorOdd.capacity22g` `MIRRORETV/lean/MirrorMain.lean:1812`	uses the cited Theorem 10.4 ($n = 5$) through Lemma 10.18

Def. 10.20	`MirrorOdd.mir`
	`MIRRORETV/lean/MirrorMain.lean:16`
	`MirrorOdd.Rset`
	`MIRRORETV/lean/MirrorMain.lean:1598`
	`MirrorOdd.Aset`
	`MIRRORETV/lean/MirrorMain.lean:1599`
		definitions

Lem. 10.21	`MirrorOdd.card_split`
	`MIRRORETV/lean/MirrorMain.lean:1601`
	`MirrorOdd.card_A_le`
	`MIRRORETV/lean/MirrorMain.lean:1636`
	`MirrorOdd.injOn_x`
	`MIRRORETV/lean/MirrorMain.lean:1647`
		proved here (also used inside Theorem 9.17 and Theorem 9.18)

Lem. 10.22	`MirrorOdd.cfg`
	`MIRRORETV/lean/MirrorMain.lean:1584`
	`MirrorOdd.sortPt_image`
	`MIRRORETV/lean/MirrorMain.lean:1576`
		the sorting construction of Definition 9.3

Lem. 10.23	written step	written step (local claim interior in the Lean proof of `mirror_seven_of`)

Lem. 10.24	`MirrorOdd.cap_mirror_heptagon`
	`MIRRORETV/lean/MirrorMain.lean:212`
	`MirrorOdd.cap_mirror_octagon`
	`MIRRORETV/lean/MirrorMain.lean:392`
	`MirrorOdd.axis_left_heptagon`
	`MIRRORETV/lean/MirrorMain.lean:639`
	`MirrorOdd.axis_right_heptagon`
	`MIRRORETV/lean/MirrorMain.lean:816`
		Lean proofs are generated code (linarith with hints found offline); written proof derives the statements from Theorem 9.8 and Theorem 9.10 (different proof)

Rem. 10.25	–	remark (no statement): how Lean proves Lemma 10.24
Lem. 10.26	`MirrorOdd.mirror_seven_of` `MIRRORETV/lean/MirrorMain.lean:1676`	takes hypotheses: Hcap, HL, HR (the statements of Lemma 10.24 (a), (c), (d); parametric premises, discharged in Theorem 10.31)

Def. 10.27	`MirrorOdd.lineRefl`
	`MIRRORETV/lean/MirrorMain.lean:1879`
	`MirrorOdd.frame`
	`MIRRORETV/lean/MirrorMain.lean:1883`
	`MirrorOdd.unframe`
	`MIRRORETV/lean/MirrorMain.lean:1892`
		definitions

Lem. 10.28	`MirrorOdd.frame_lineRefl`
	`MIRRORETV/lean/MirrorMain.lean:1914`
	`MirrorOdd.orient_frame`
	`MIRRORETV/lean/MirrorMain.lean:1920`
	`MirrorOdd.unframe_frame`
	`MIRRORETV/lean/MirrorMain.lean:1900`
	`MirrorOdd.frame_unframe`
	`MIRRORETV/lean/MirrorMain.lean:1907`
	`MirrorOdd.frame_injective`
	`MIRRORETV/lean/MirrorMain.lean:1925`
	`MirrorOdd.unframe_injective`
	`MIRRORETV/lean/MirrorMain.lean:1930`
		polynomial identities checked symbolically

Lem. 10.29	`MirrorOdd.hasConvexNGon_affineMap` `MIRRORETV/lean/MirrorMain.lean:1859`	proved here

Statement	Lean declaration; file and line	Note
Lem. 10.30	`MirrorOdd.transport` `MIRRORETV/lean/MirrorMain.lean:1937`	proved here
Thm. 10.31	`MirrorOdd.mirror_seven` `MIRRORETV/lean/MirrorMain.lean:1803` `MirrorOdd.mirror_seven_line` `MIRRORETV/lean/MirrorMain.lean:1969`	rests on the cited Theorem 10.4 and on Theorem 9.8, Theorem 9.10
Rem. 10.32	–	remark (no statement): `mirror_seven'` as a second derivation
Thm. 10.33	`MirrorOdd.mirror_eight` `MIRRORETV/lean/MirrorMain.lean:1820` `MirrorOdd.mirror_eight_line` `MIRRORETV/lean/MirrorMain.lean:1977`	rests on the cited Theorem 10.4 and on Theorem 9.8
Rem. 10.34	–	remark (no statement): `mirror_eight'` as a second derivation
Thm. 10.35	`C12Check.config_capacity113` `MIRRORETV/lean/MirrorNine_append.lean:24` `C12Check.mirror_nine` `MIRRORETV/lean/MirrorNine_append.lean:131` `C12Check.mirror_nine_line` `MIRRORETV/lean/MirrorNine_append.lean:136`	with `config_capacity113` as part (a); (a) uses the cited Theorem 10.4 ($n = 6$) and Theorem 10.6 ($a = 3, b = 6$), $C(8,2) = 28$ and $C(9,3) = 84$ by kernel decide; (b),(c) via Theorem 9.17 and Theorem 9.23 ($j = 4, M = 113$)
Prop. 10.36	`C12Check.capacity113` `MIRRORETV/lean/MirrorNine_append.lean:116`	uses Theorem 10.35(a) (`config_capacity113`); stated and proved between the proofs of Theorem 10.35(a) and (b),(c)
Def. 10.37	`ErdosSzekeres.SymReduction.reflLine` `SYMLEAN/mirror/ReductionThreeClasses.lean:449`	definition (same text as `SYMLEAN/SymmetricReduction.lean:423`)
Thm. 10.38	`ErdosSzekeres.SymReduction.reflLine_eq_lineRefl` `SYMLEAN/mirror/ReductionThreeClasses.lean:4253` `ErdosSzekeres.SymReduction.card_le_32_of_reflLine` `SYMLEAN/mirror/ReductionThreeClasses.lean:4264`	with `reflLine_eq_lineRefl`
Thm. 10.39	`ErdosSzekeres.SymReduction.mirrorOneAxis33_proved` `SYMLEAN/mirror/ReductionThreeClasses.lean:4274` `ErdosSzekeres.SymReduction.mirrorNoAxis34_proved` `SYMLEAN/mirror/ReductionThreeClasses.lean:4282`	the axis hypotheses of both classes are unused
Thm. 10.40	`ErdosSzekeres.SymReduction.heptagon_of_nontrivial_isometry_3` `SYMLEAN/mirror/ReductionThreeClasses.lean:4292` `ErdosSzekeres.SymReduction.heptagon_of_nontrivial_affineEquiv_3` `SYMLEAN/mirror/ReductionThreeClasses.lean:4303`	takes hypotheses: `HalfTurn34`, Threefold33, Fivefold35 (not proved in Lean; computer-assisted claims supplied by the records `C2FINAL/C2C34`, `C3FINAL/C3LEAN`, SYM5, resting also on written encoder-soundness and perturbation proofs and in part on verdicts recorded at solve time; see Remark 10.41); conditional
Rem. 10.41	–	remark (no statement): status of the three hypotheses of Theorem 10.40, with the records and rigor labels of their evidence; the case-split bookkeeping is Lean-checked according to the record (Theorem 11.98) or Lean-checked (Theorem 11.82), in both cases by kernel evaluation, so computer-assisted
Rem. 10.42	–	remark (no statement): `MirrorExclusionLineCopy` follows from Theorem 10.38(b)

Section 11

Lem. 11.1	written step	written step; binary digits of an or-fold
Def. 11.2	`C3Cover.m` `C3LEAN/C3Cover.lean:34`	definition
Def. 11.3	`C3Cover.Tok` `C3LEAN/C3Cover.lean:38`	definition
Def. 11.4	`C3Cover.Layer` `C3LEAN/C3Cover.lean:41`	definition
Def. 11.5	`C3Cover.Node` `C3LEAN/C3Cover.lean:45`	definition
Def. 11.6	`C3Cover.Node.used` `C3LEAN/C3Cover.lean:51`	definition

Statement	Lean declaration; file and line	Note
Rem. 11.7	–	commentary; intended meaning, not used in proofs
Def. 11.8	`C3Cover.pairOptions` `C3LEAN/C3Cover.lean:54`	definition
Def. 11.9	`C3Cover.layerOptions` `C3LEAN/C3Cover.lean:61`	definition; rem is a local abbreviation inside `layerOptions`
Def. 11.10	`C3Cover.insTok` `C3LEAN/C3Cover.lean:66`	definition
Def. 11.11	`C3Cover.sortTok` `C3LEAN/C3Cover.lean:71`	definition
Def. 11.12	`C3Cover.topCases` `C3LEAN/C3Cover.lean:77`	definition
Def. 11.13	`C3Cover.okey` `C3LEAN/C3Cover.lean:82`	definition
Def. 11.14	`C3Cover.lexLe` `C3LEAN/C3Cover.lean:85`	definition
Def. 11.15	`C3Cover.mirror` `C3LEAN/C3Cover.lean:91`	definition
Def. 11.16	`C3Cover.phi` `C3LEAN/C3Cover.lean:95`	definition
Def. 11.17	`C3Cover.refl` `C3LEAN/C3Cover.lean:101`	definition
Def. 11.18	`C3Cover.keptOptions` `C3LEAN/C3Cover.lean:109`	definition
Def. 11.19	`C3Cover.roots` `C3LEAN/C3Cover.lean:113`	definition
Def. 11.20	`C3Cover.Node.free` `C3LEAN/C3Cover.lean:116`	definition
Def. 11.21	`C3Cover.bitsOf` `C3LEAN/C3Cover.lean:120`	definition
Def. 11.22	`C3Cover.addBits` `C3LEAN/C3Cover.lean:123`	definition
Def. 11.23	`C3Cover.Node.sigmaChildren` `C3LEAN/C3Cover.lean:127`	definition; bits is a local abbreviation
Def. 11.24	`C3Cover.conflicts` `C3LEAN/C3Cover.lean:132`	definition
Def. 11.25	`C3Cover.Node.layerChildren` `C3LEAN/C3Cover.lean:137`	definition
Rem. 11.26	–	commentary; reading of the children, not used in proofs
Def. 11.27	`C3Cover.Covered` `C3LEAN/C3Cover.lean:146`	inductive predicate; least predicate closed under the three constructors
Lem. 11.28	`C3Cover.Covered.mono` `C3LEAN/C3Cover.lean:153`	Lean-checked (dependency of `all_roots_covered`); written proof by rule induction
Def. 11.29	`C3Cover.decodeTok` `C3LEAN/C3Cover.lean:165`	definition
Def. 11.30	`C3Cover.decodeLayer` `C3LEAN/C3Cover.lean:167`	definition
Def. 11.31	`C3Cover.decodeLayers` `C3LEAN/C3Cover.lean:172`	definition
Def. 11.32	`C3Cover.decodeCube` `C3LEAN/C3Cover.lean:180`	definition
Def. 11.33	`C3Cover.decodeKey` `C3LEAN/C3Cover.lean:191`	definition
Rem. 11.34	–	commentary; uses the control `rev_parent_decodes` (`C3LEAN/review/RevCtrl_pass.lean:315`); `decodeKey` not injective
Def. 11.35	`C3Cover.tokCode` `C3LEAN/C3Cover.lean:223`	definition
Def. 11.36	`C3Cover.layerCode` `C3LEAN/C3Cover.lean:226`	definition
Def. 11.37	`C3Cover.rootCode` `C3LEAN/C3Cover.lean:232`	definition
Def. 11.38	`C3Cover.decodeRoot` `C3LEAN/C3Cover.lean:238`	definition

Statement	Lean declaration; file and line	Note
Def. 11.39	`C3Cover.beqList` `C3LEAN/C3Cover.lean:194`	definition
Def. 11.40	`C3Cover.beqTok` `C3LEAN/C3Cover.lean:199`	definition
Def. 11.41	`C3Cover.beqNode` `C3LEAN/C3Cover.lean:202`	definition
Def. 11.42	`C3Cover.beqLayer` `C3LEAN/C3Cover.lean:335`	definition
Lem. 11.43	`C3Cover.beqList_eq` `C3LEAN/C3Cover.lean:204`	Lean-checked (dependency of printed theorems); written proof
Lem. 11.44	`C3Cover.beqTok_eq` `C3LEAN/C3Cover.lean:213`	Lean-checked (dependency of printed theorems); written proof
Lem. 11.45	`C3Cover.beqNode_eq` `C3LEAN/C3Cover.lean:218`	Lean-checked (dependency of printed theorems); written proof
Lem. 11.46	written step	written step; reflexivity of the Boolean equalities
Def. 11.47	`C3Cover.Cert` `C3LEAN/C3Cover.lean:245`	definition
Def. 11.48	`C3Cover.unpack` `C3LEAN/C3Cover.lean:251`	definition
Def. 11.49	`C3Cover.Cert.keys` `C3LEAN/C3Cover.lean:259`	definition
Def. 11.50	`C3Cover.run` `C3LEAN/C3Cover.lean:262`	definition
Def. 11.51	`C3Cover.checkRoot` `C3LEAN/C3Cover.lean:277`	definition
Rem. 11.52	–	commentary; how a certificate is read; final test and fuel not needed for soundness
Lem. 11.53	`C3Cover.run_sound` `C3LEAN/C3Cover.lean:279` `C3Cover.checkRoot_sound` `C3LEAN/C3Cover.lean:323`	Lean-checked, axioms propext and `Quot.sound`; written proof by induction on fuel
Lem. 11.54	`C3Cover.all_append_of` `C3LEAN/C3Cover.lean:327`	Lean-checked (dependency of `all_checked`); written proof
Def. 11.55	`C3Cover.tsvRootCodes` `C3LEAN/C3Cover.lean:413`	generated data (910 root codes); identification with the TSV file checked in Python only
Def. 11.56	`C3Cover.certs0` `C3LEAN/C3Cover.lean:423`	generated data; family certs0–certs126 at lines 423-1711 (certs126 at 1668)
Def. 11.57	`C3Cover.certs` `C3LEAN/C3Cover.lean:1714`	generated data; right-nested append
Def. 11.58	`C3Cover.bounds` `C3LEAN/C3Cover.lean:1717`	generated data; unused by any theorem
Def. 11.59	`C3Cover.certifiedKeys` `C3LEAN/C3Cover.lean:1737`	generated data; also defines the leaf predicate S-star of the main theorem
Def. 11.60	`C3Cover.isOption` `C3LEAN/C3Cover.lean:340`	definition
Def. 11.61	`C3Cover.distinctCodes` `C3LEAN/C3Cover.lean:346`	definition
Def. 11.62	`C3Cover.codeSet` `C3LEAN/C3Cover.lean:351`	definition
Def. 11.63	`C3Cover.reflectionFacts` `C3LEAN/C3Cover.lean:356`	definition
Lem. 11.64	written step	written step; characterization of `distinctCodes` and `codeSet`
Fact 11.65	`C3Cover.batch0` `C3LEAN/C3Cover.lean:1740`	computer-assisted (decide +kernel); family batch0–batch126 at lines 1740+4N; no hand proof
Lem. 11.66	`C3Cover.all_checked` `C3LEAN/C3Cover.lean:2247`	Lean-checked, axioms propext; computer-assisted (joins the 127 computer-assisted batch facts by `all_append_of`)
Fact 11.67	`C3Cover.reflection_facts_H3` `C3LEAN/C3Cover.lean:367`	Lean-checked, axioms propext; computer-assisted in Lean (decide +kernel); written proof in Proposition 11.79

Statement	Lean declaration; file and line	Note
Fact 11.68	`C3Cover.reflection_facts_H6_1_0` `C3LEAN/C3Cover.lean:370`	Lean-checked, axioms propext; computer-assisted in Lean (decide +kernel); ten theorems `reflection_facts_H6_1_0`–`H6_5_1` at lines 370+3(2a+s-2); written proof in Proposition 11.79
Fact 11.69	`C3Cover.topCases_eq` `C3LEAN/C3Cover.lean:401`	Lean-checked, no axioms; computer-assisted in Lean (decide +kernel); written proof in Proposition 11.79
Fact 11.70	`C3Cover.strata` `C3LEAN/C3Cover.lean:406`	Lean-checked, no axioms; computer-assisted in Lean (decide +kernel); written proof in Proposition 11.79
Fact 11.71	`C3Cover.roots_match` `C3LEAN/C3Cover.lean:1720`	Lean-checked, no axioms; computer-assisted in Lean (decide +kernel); written proof in Proposition 11.80
Fact 11.72	`C3Cover.roots_codes_match` `C3LEAN/C3Cover.lean:1725`	computer-assisted (decide +kernel on generated certificate data), no axioms; no hand proof
Fact 11.73	`C3Cover.roots_eq_tsv` `C3LEAN/C3Cover.lean:1730`	computer-assisted (decide +kernel on generated data), axioms propext; no hand proof
Lem. 11.74	written step	written step
Lem. 11.75	written step	written step; structure and count of `layerOptions`
Lem. 11.76	written step	written step; decoding lemma (`decodeRoot` after `rootCode` is the identity on valid depth-2 nodes)
Lem. 11.77	written step	written step; `lexLe` is a total order
Lem. 11.78	written step	written step; one kept member per orbit of the reflection
Prop. 11.79	written step	written step; hand proof of reflection_facts_*, `topCases_eq`, strata
Prop. 11.80	written step	written step; hand proof of `roots_match`
Cor. 11.81	written step	written step; computer-assisted (uses `roots_codes_match` and `roots_eq_tsv`)
Thm. 11.82	`C3Cover.all_roots_covered` `C3LEAN/C3Cover.lean:2251`	Lean-checked, axioms propext and `Quot.sound`; no hypotheses; computer-assisted (uses the computer-assisted batch facts and `roots_codes_match`); leaf set is the certificates' own key list (the run records of the leaves are checked outside Lean)
Rem. 11.83	–	commentary; scope and intended use (geometry, SAT verdicts, run records of the leaves not established)
Def. 11.84	`Central34.inner` `C2FINAL/Central34Orbits.lean:21`	definition
Def. 11.85	`Central34.choose2` `C2FINAL/Central34Orbits.lean:24`	definition
Def. 11.86	`Central34.choose3` `C2FINAL/Central34Orbits.lean:29`	definition
Def. 11.87	`Central34.subcases` `C2FINAL/Central34Orbits.lean:34`	definition
Def. 11.88	`Central34.mask` `C2FINAL/Central34Orbits.lean:37`	definition
Def. 11.89	`Central34.refl` `C2FINAL/Central34Orbits.lean:40`	definition
Def. 11.90	`Central34.bitset` `C2FINAL/Central34Orbits.lean:43`	definition
Def. 11.91	`Central34.uncovered` `C2FINAL/Central34Orbits.lean:46`	definition
Def. 11.92	`Central34.closedP1` `C2FINAL/Central34Orbits.lean:51`	generated data; family `closedP1`–`closedP8` at lines 51-264 (287 entries each, 288 for `closedP6`)
Def. 11.93	`Central34.closed` `C2FINAL/Central34Orbits.lean:267`	definition

Statement	Lean declaration; file and line	Note
Rem. 11.94	–	commentary; intended meaning, not used in proofs
Lem. 11.95	written step	written step; subcases enumerate the 2- and 3-subsets
Thm. 11.96	`Central34.subcases_reflection` `C2FINAL/Central34Orbits.lean:273`	Lean-checked according to the record (no saved Lean output; `C2FINAL/README.md` section 6 records propext only); computer-assisted in Lean (decide +kernel); written proof given
Fact 11.97	`Central34.closed_are_subcases` `C2FINAL/Central34Orbits.lean:284`	Lean-checked according to the record (no saved Lean output; `C2FINAL/README.md` section 6 records propext only); computer-assisted (decide +kernel on generated lists); no hand proof
Thm. 11.98	`Central34.all_cases_closed` `C2FINAL/Central34Orbits.lean:291`	Lean-checked according to the record (no saved Lean output; `C2FINAL/README.md` section 6 records propext only); computer-assisted (decide +kernel on generated lists `closedP1`–`closedP8`); no hypotheses
Cor. 11.99	written step	written step; computer-assisted (a consequence of Theorem 11.98)
Rem. 11.100	–	commentary; scope (refutation of the listed subcases, covers, reductions A and B not established)
Section 12		
Prop. 12.1	`A1.c3_reduce` `AUDIT3/A1_reduction/SizeArith.lean:16` `A1.c5_reduce` `AUDIT3/A1_reduction/SizeArith.lean:19` `A1.c4_reduce` `AUDIT3/A1_reduction/SizeArith.lean:22` `A1.c6_reduce` `AUDIT3/A1_reduction/SizeArith.lean:25` `A1.c2_reduce` `AUDIT3/A1_reduction/SizeArith.lean:28`	linear arithmetic over N (omega in Lean); core Lean, no Mathlib
Prop. 12.2	`A1.mirror_reduce_b0` `AUDIT3/A1_reduction/SizeArith.lean:31` `A1.mirror_reduce_b12` `AUDIT3/A1_reduction/SizeArith.lean:34`	linear arithmetic over N (omega in Lean); the hypothesis $1 \le b$ of `mirror_reduce_b12` is not needed
Prop. 12.3	`A1.even_order_reduce` `AUDIT3/A1_reduction/SizeArith.lean:37` `A1.mult3_order_reduce` `AUDIT3/A1_reduction/SizeArith.lean:42` `A1.mult5_order_reduce` `AUDIT3/A1_reduction/SizeArith.lean:47`	`Nat.mul_assoc`, then omega with $t \cdot k$ as one unknown
Lem. 12.4	`A1.order_split` `AUDIT3/A1_reduction/SizeArith.lean:52`	omega, then decide on the five closed cases $s = 2$–6: side facts inside the proof of a statement over all $s \ge 2$, which is not a finite computation, and checked by hand in the written proof; not computer-assisted

Statement	Lean declaration; file and line	Note

Prop. 12.5	`A1.table_c3`
	`AUDIT3/A1_reduction/SizeArith.lean:61`
	`A1.c3_no32`
	`AUDIT3/A1_reduction/SizeArith.lean:62`
	`A1.table_c5`
	`AUDIT3/A1_reduction/SizeArith.lean:65`
	`A1.c5_gap`
	`AUDIT3/A1_reduction/SizeArith.lean:66`
	`A1.table_c4`
	`AUDIT3/A1_reduction/SizeArith.lean:69`
	`A1.table_c4_only_c2`
	`AUDIT3/A1_reduction/SizeArith.lean:70`
	`A1.table_c6`
	`AUDIT3/A1_reduction/SizeArith.lean:73`
	`A1.table_c2`
	`AUDIT3/A1_reduction/SizeArith.lean:76`
	`A1.table_mirror`
	`AUDIT3/A1_reduction/SizeArith.lean:79`
		linear arithmetic over N (omega in Lean); the bounds that the case theorems give are hypotheses, not part of the statements: on k in `table_c3`, `table_c4`, `table_c2`; on 2k in `table_c4_only_c2`, `table_c6`; on 5k+e in `table_c5`; on k for $b = 0$ and for $1 \le b$ in `table_mirror`; `c3_no32` and `c5_gap` assume only $e \le 1$

Prop. 12.6	`halfturn_pairs`
	`AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:6`
	`threefold_orbits`
	`AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:9`
	`fivefold_size`
	`AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:12`
	`mirror_b0`
	`AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:15`
	`mirror_b12`
	`AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:18`
	`fourfold_via_halfturn`
	`AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:21`
		verifier file of finding A1-F1; restatements of six `SizeArith.lean` statements, or parts of them (`fivefold_size`: first conclusion of `c5_reduce`; `fourfold_via_halfturn`: third conclusion of `c4_reduce`), with explicit n (omega in Lean)

Lem. 12.7	`big_order_orbit` `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:24`	hypothesis hp ($7 \le p$) unused (linter warning in the log); case split on k, then omega
Rem. 12.8	–	the hypothesis $7 \le p$ (binder hp) of `big_order_orbit` is unused (linter warning in `AssembledSizes.lean.log`); the statement does not require p to be prime
Rem. 12.9	– `AUDIT3/A1_reduction/verify_A1-F1/AssembledSizes.lean:30`	anonymous example (no declaration name): the half-turn bound fails at 32 points ($n = 32$, $k = 16$); accepted by Lean (no error in `AssembledSizes.lean.log`) but not covered by #print axioms
Thm. 12.10	`baek_seven` `ES7SPLIT/review/equivalence/BaekSeven.lean:6`	instance $n = 4$ of the cited Theorem 10.4; decide only for the side fact $C(6, 2) + 2 = 17$, also checked by hand; the statement is not a finite computation, so not computer-assisted; abstract configurations only, no bridge to point sets
Rem. 12.11	–	compares `baek_seven` with the #check @main output (`BaekSeven.out` lines 1-2) and Theorem 10.4 ($n = 4$; universe); abstract configurations only; the ES7SPLIT remark that the step to point sets is not Lean-checked for $n = 7$ predates `MirrorOdd.capacity16` (`MIRRORETV/lean/MirrorMain.lean:1529`), used in Theorem 10.31

Statement	Lean declaration; file and line	Note
Lem. 12.12	written step	written step: orbit counts $n = s \cdot k + e$ ($e \le 1$, $e = 0$ for even s) and $n = 2k + b$ ($b \le 2$), the modelling of the `SizeArith.lean` header (not formalised there); related Lean-checked forms elsewhere: Theorem 6.35 (a), (b); Lemma 7.16; Theorem 2.22 and Theorem 2.24
Rem. 12.13	–	written step: how Lemma 12.12, the arithmetic and the five case classes (hypotheses) combine; `HalfTurn34`, Threefold33, Fivefold35 not proved; mirror classes Lean-checked (Theorem 10.39); Lean-checked forms of the implication Theorem 7.39 and Theorem 8.28; table bounds conditional except the mirror row (Theorem 10.38); conditional forms Theorem 6.7, Theorem 6.4

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